

2011 ANNUAL REPORT

EXTEND CREDIT
SOVEREIGN ASSET MANAGEMENT
REAL ASSETS MULTINATIONALS
INTERNATIONAL EXPANSION
PRIVATE BANKING ADVISORY CORPORATE FINANCE
GLOBAL RESEARCH
HIGH-PERFORMANCE TECHNOLOGY
GLOBAL CUSTODY MARKET MAKER
DISTINCTIVELY J.P. MORGAN
EMERGING MARKETS
MANAGING RISK
CHARACTER
MANAGING WEALTH
INVESTMENT PERFORMANCE

STABILITY
THOUGHT LEADERSHIP GLOBAL REACH
FORTRESS BALANCE SHEET CREATING JOBS
HIRING VETERANS CAPITAL STRENGTH RESPONSIBLE
JPMORGAN CHASE & CO. MAKING A DIFFERENCE
DOING THE RIGHT THING
INVEST IN OUR COMMUNITIES
OPENING NEW BRANCHES INK
DIFFERENTIATED PRODUCTS
BLUEPRINT
AFFORDABLE HOUSING
PROVIDE CREDIT
TRANSPARENT CREDIT COUNSELING
CHASE WHAT MATTERS
LONG-TERM DECISIONS

LIQUIDITY
TRADE FINANCE
100,000 JOBS MISSION
LENDING & INVESTING IN COMMUNITIES
CHASE PERSON-TO-PERSON QUICKPAY
MOBILE SOLUTIONS
QUICKDEPOSIT
HELPING HOMEOWNERS
CLEAR & SIMPLE
LENDING OUR STRENGTH
GROUNDBREAKING SERVICES
CHASE COMMUNITY GIVING
ULTIMATE REWARDS CHASE FREEDOM SAPPHIRE
CASH BACK

JPMORGAN CHASE & Co.

THE WAY FORWARD >>>

Financial Highlights

As of or for the year ended December 31,
(in millions, except per share, ratio data and headcount)

	2011	2010
Reported basis^(a)		
Total net revenue	\$ 97,234	\$ 102,694
Total noninterest expense	62,911	61,196
Pre-provision profit	34,323	41,498
Provision for credit losses	7,574	16,639
Net income	\$ 18,976	\$ 17,370
Per common share data		
Net income per share:		
Basic	\$ 4.50	\$ 3.98
Diluted	4.48	3.96
Cash dividends declared	1.00	0.20
Book value	46.59	43.04
Selected ratios		
Return on common equity	11%	10%
Return on tangible common equity ^(b)	15	15
Tier 1 capital ratio	12.3	12.1
Total capital ratio	15.4	15.5
Tier 1 common capital ratio ^(b)	10.1	9.8
Selected balance sheet data (period-end)		
Total assets	\$2,265,792	\$ 2,117,605
Loans	723,720	692,927
Deposits	1,127,806	930,369
Total stockholders' equity	183,573	176,106
Headcount	260,157	239,831

(a) Results are presented in accordance with accounting principles generally accepted in the United States of America, except where otherwise noted.

(b) Non-GAAP financial measure. For further discussion, see "Explanation and reconciliation of the firm's use of non-GAAP financial measures" and "Regulatory capital" in this Annual Report.

JPMorgan Chase & Co. (NYSE: JPM) is a leading global financial services firm and one of the largest banking institutions in the United States, with operations worldwide; the firm has \$2.3 trillion in assets and \$183.6 billion in stockholders' equity. The firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing, asset management and private equity. A component of the Dow Jones Industrial Average, JPMorgan Chase & Co. serves millions of consumers in the United States and many of the world's most prominent corporate, institutional and government clients under its J.P. Morgan and Chase brands.

Information about J.P. Morgan capabilities can be found at jpmorgan.com and about Chase capabilities at chase.com. Information about the firm is available at jpmorganchase.com.

J.P. Morgan, Jr., spoke these words in 1933 during the heart of the Great Depression. It was those values that guided us through that tremendous challenge. Today, those values continue to guide us through challenges and help us maintain a standing of vitality and strength.

And, as always, our commitment to our clients remains first and foremost.

We raised \$1.8 trillion for businesses and consumers. For small business, we approved more than \$17 billion in credit and maintained our position as the nation's #1 Small Business Administration lender.

We also continued our support of communities. We raised \$68 billion for not-for-profits and public services. And we hired more than 3,000 military veterans as a proud founding member of the 100,000 Jobs Mission.

We began to see some encouraging signs this past year, and our firm helped put more than 17,000 Americans back to work. We saw more businesses and individuals turning to us for loans. We saw credit quality strengthen and confidence return.

We are optimistic about the future. Throughout our 200-year history, our belief in responsible leadership, our dedication to our clients and our fortress balance sheet have carried us through the toughest challenges. These are the core values we maintain day after day and the values that will sustain us into the future.

Dear Fellow Shareholders,

Your company earned a record \$19.0 billion in 2011, up 9% from the record earnings of \$17.4 billion in 2010.

Our return on tangible equity for 2011 was 15% – the same as last year. Relative to our competitors and given the prevailing economic environment, this is a good result. On an absolute and static basis, we believe that our earnings should be \$23 billion–\$24 billion. The main reason for the difference between what we *are* earning and what we *should* be earning continues to be high costs and losses in mortgage and mortgage-related issues. While these losses are increasingly less severe, they will still persist at elevated levels for a while longer. Looking ahead, we believe our earnings power should grow over time, though we always expect volatility in our earnings – it is the nature of the various businesses we operate.

2011 was another year of challenges for JPMorgan Chase, the financial services industry and the economies of many countries around the world. In addition to the ongoing global economic uncertainty, other traumatic events – such as the earthquake and tsunami in Japan, the debt ceiling fiasco in the United States, revolutions in the Middle East and the European debt crisis – have impeded recovery. In the face of these tragic events and unfortunate setbacks, the frustration with – and hostility toward – our industry continues. We acknowledge it and respect people’s right to express themselves. However, we all have an interest in getting the economy and job creation growing again.

In the face of many difficult challenges, JPMorgan Chase is trying to do its part. We have not retrenched. Just the opposite – we have stepped up.

Over the past year, our people demonstrated once again that the work we do matters. We positively impact the lives of millions of people and the communities in which they live. Our duty is to serve them by stepping into the arena each day and putting our resources and our voices to work on their behalf. For us, standing on the sidelines simply is not an option.

Jamie Dimon,
Chairman and
Chief Executive Officer

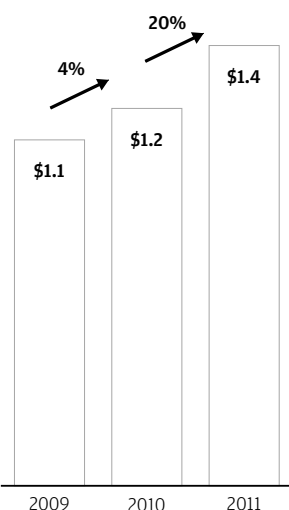


During 2011, the firm raised capital and provided credit of over \$1.8 trillion for our commercial and consumer clients, up 18% from the prior year. We provided more than \$17 billion of credit to U.S. small businesses, up 52% over last year. We raised capital or provided credit of \$68 billion for more than 1,200 not-for-profit and government entities, including states, municipalities, hospitals and universities. We also issued new credit cards to 8.5 million people and originated more than 765,000 mortgages. To help struggling homeowners, we have offered over 1.2 million mortgage modifications since 2009 and completed more than 450,000.

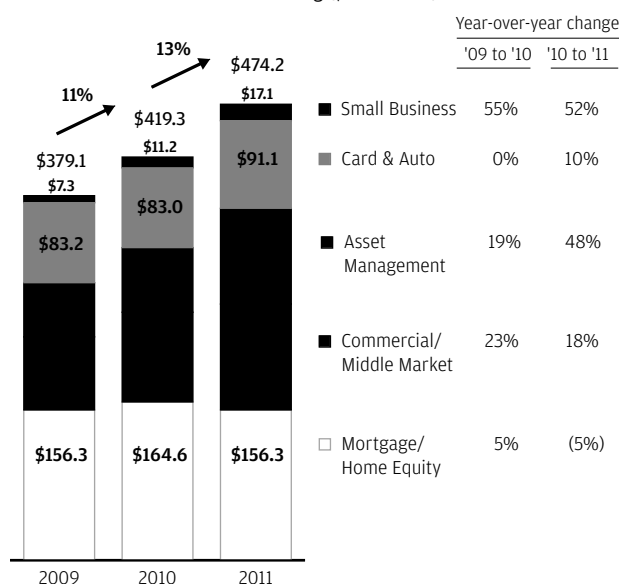
We also bought back \$9 billion of stock and recently received permission to buy back an additional \$15 billion of stock during the remainder of 2012 and the first quarter of 2013. We reinstated our annual dividend to \$1.00 a share in April 2011 and recently announced that we are increasing it to \$1.20 a share in April 2012. And we continued to build our business by heavily investing in infrastructure, systems, technology and new products and by adding bankers and branches around the world.

New and Renewed Capital and Credit for Our Clients

Corporate Clients (\$ in trillions)



Consumer and Commercial Banking (\$ in billions)



The best way to build shareholder value is to build a great company, with exemplary products and services, excellent systems, quality accounting and reporting, effective controls and outstanding people. If you continually build a great company, the stock price will follow. Normally, we don't comment on the stock price. However, we make an exception in Section VIII of this letter because we are buying back a substantial amount of stock and because there are many concerns about investing in bank stocks.

We believe you own an exceptional company. Each of our businesses is among the best in the world, and record earnings were matched by increased market share in most of our businesses. Most importantly, we have outstanding people working at every level in every business across the economic spectrum and around the world. This is no accident - we work hard to bring people with character, integrity and intelligence into this company.

There is always room for improvement, but the strengths that are embedded in this company – our people, client relationships, product capabilities, technology, global presence and fortress balance sheet – provide us with a foundation that is rock solid and an ability to thrive regardless of what the future brings.

In this letter, I will focus my comments on the important issues affecting your company, including some of the regulatory and political issues facing us.

The main sections of the letter are as follows:

- I. Our mission and how we operate to fulfill our role in society
- II. A brief update on our major initiatives
- III. The new One Chase – strengthening the customer experience
- IV. An intense focus in 2012 on adapting our businesses successfully to the new regulatory framework
- V. Comments on global financial reform
- VI. The mortgage business – the good, the bad and the ugly
- VII. Comments on the future of investment banking and the critical role of market making
- VIII. Why would you want to own the stock?
- IX. Closing

We are constantly asked the question of what comes first in your company – customers, employees, shareholder value or being a good corporate citizen – which implies a need to favor one over the other. We disagree with this view. We must serve them all well. If we fail at any one, the whole enterprise suffers.

Our customers, employees, shareholder value and communities *all* come first

Many people seem to think that shareholder value means profit and that a company earns more profit by giving customers or employees less. This has not been our experience. Our job is to build a healthy and vibrant company that satisfies clients, invests in its people – through training, opportunity and compensation – and rewards its shareholders. When this is done well, everyone benefits. At the same time, a company needs to be successful financially because if it isn't, it ultimately will fail. And when a company fails, everyone loses.

How we view our customers – we wouldn't be here without them

There would be no company but for our customers. Without our consumer or corporate clients – and satisfied ones at that – there would be no profits, no bankers, no staff and no CEO.

At JPMorgan Chase, we believe that customers should be treated like we would want to be treated ourselves. Customers usually don't mind paying a fair price for a product or service they need, particularly if it is delivered well and accompanied with a smile. We are constantly looking for better ways to provide, combine and deliver products that meet or exceed our customers' expectations. And we try to listen closely to our customers – even when they complain – because they are doing us a service by telling us how we could do better. It means a lot to a customer when we respond not only by listening but also by actually changing.

How we view our employees – they do it all

Doing a great job starts with great employees. We look for high-quality people with the capability to do a great job and grow with the firm. Then we train and empower them to do the right thing as best they can; to understand and anticipate their customers' needs; and, in effect, to be their advocate. To do this, each employee needs help from the rest of the company.

There are many employees who work behind the scenes that the customers do not see – such as programmers, assistants, network engineers, operations clerks and others. But these are the professionals we depend upon to help us seamlessly deliver integrated and complex products.

And all of our employees drive innovation. They have the knowledge and the deep understanding to find ways – large and small – to improve a system, streamline a process, and save time and money by making things work better for everybody.

How we view our communities – they are our hosts, our customers and our future exceed our customers' expectations and compensate the expectations. In such a global economy, the time source

II. A BRIEF UPDATE ON OUR MAJOR INITIATIVES

The opportunities for JPMorgan Chase over the next 20 years will equal – or maybe even surpass – those of the last 20 years.

In last year's letter, we discussed several specific initiatives we're undertaking in addition to the "normal" growth opportunities that we pursue every day. Each one of these initiatives involves a sustained, full-fledged effort of investment in people, branches and systems over a long period of time. And while we know that these efforts may not turn a profit in the first year, we expect each one to add \$500 million or more in profits annually by the fifth to seventh year.

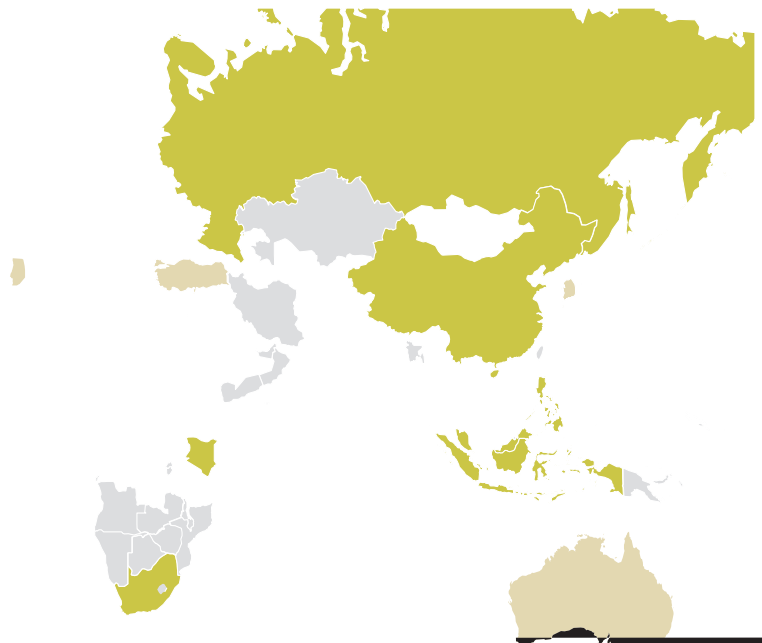
The following segments provide an update on how each of these initiatives is progressing.

The expansion of our international wholesale businesses, including progress in our Global Corporate Bank

Last year, we described our international expansion plan in detail. It involves building out our global presence across our wholesale businesses (Asset Management, the Investment Bank and Treasury & Securities Services) in the rapidly expanding markets of Asia, Latin America, Africa and the Middle East, as well as in emerging and even "frontier" markets.

As our clients – multinational corporations, sovereign wealth funds, public or quasi-public entities – expand globally, we intend to follow them around the world.

We Are Expanding Our Global Platform



We have made good progress:

- Five years ago, we served approximately 200 clients in Brazil, China and India combined. Today, that number has grown to approximately 800 clients. Five years from now, we expect to serve 2,000 clients – including locally headquartered companies (about 50%) and foreign subsidiaries of international companies (about 50%).
- In 2011, we opened offices in the following new locations: Harbin, China; Panama City, Panama; and Doha, Qatar. That's in addition to the offices we opened in 2010 in Bangladesh, Bermuda, Guernsey, Saudi Arabia and the United Arab Emirates. A quick glance at the map on the previous page shows the offices opened over the past two years in new and existing locations and the cities around the world where we plan to add locations in 2012-2013.
- When we started the Global Corporate Bank (GCB), we had 98 bankers. By the end of 2011, we had more than 250 bankers in 35 countries. We plan to have approximately 320 bankers in 40 countries by the end of 2013, who will provide approximately 3,500 multinational corporations with cash management, global custody, foreign exchange, trade finance and other services.
- This strategy has led to a 73% rise in our trade finance loans, a total of \$37 billion in 2011. We also increased other business with these same multinational corporations, including rates, foreign exchange and commodities, by 30%.

Commodities

In 2011, we completed the integration of assets acquired from Sempra. We now are one of the top three firms in commodities – i.e., global sales and trading, as well as advisory services and market making in metals, oil, natural gas, power and others. Our global franchise includes approximately

600 employees and 10 main office locations around the world. Over the course of last year, we grew our client franchise by more than 10% to serve over 2,200 active clients. And we increased the selling of commodities products to already existing clients so that hundreds of clients now come to us for multiple products across different commodity asset classes.

Small business growth

In 2011, we provided more than \$17 billion of new credit to U.S. small businesses in 2011, up 52% from 2010. We are the #1 Small Business Administration (SBA) lender nationwide – for the second year in a row. In 2011, we also became the #1 SBA lender to women-owned and minority-owned businesses.

Since 2009, we have added 1,200 new relationship managers and business bankers, and that includes adding 600 business bankers in the heritage Washington Mutual (WaMu) states of California and Florida. And we plan to continue aggressively hiring bankers who are meeting the needs of small businesses.

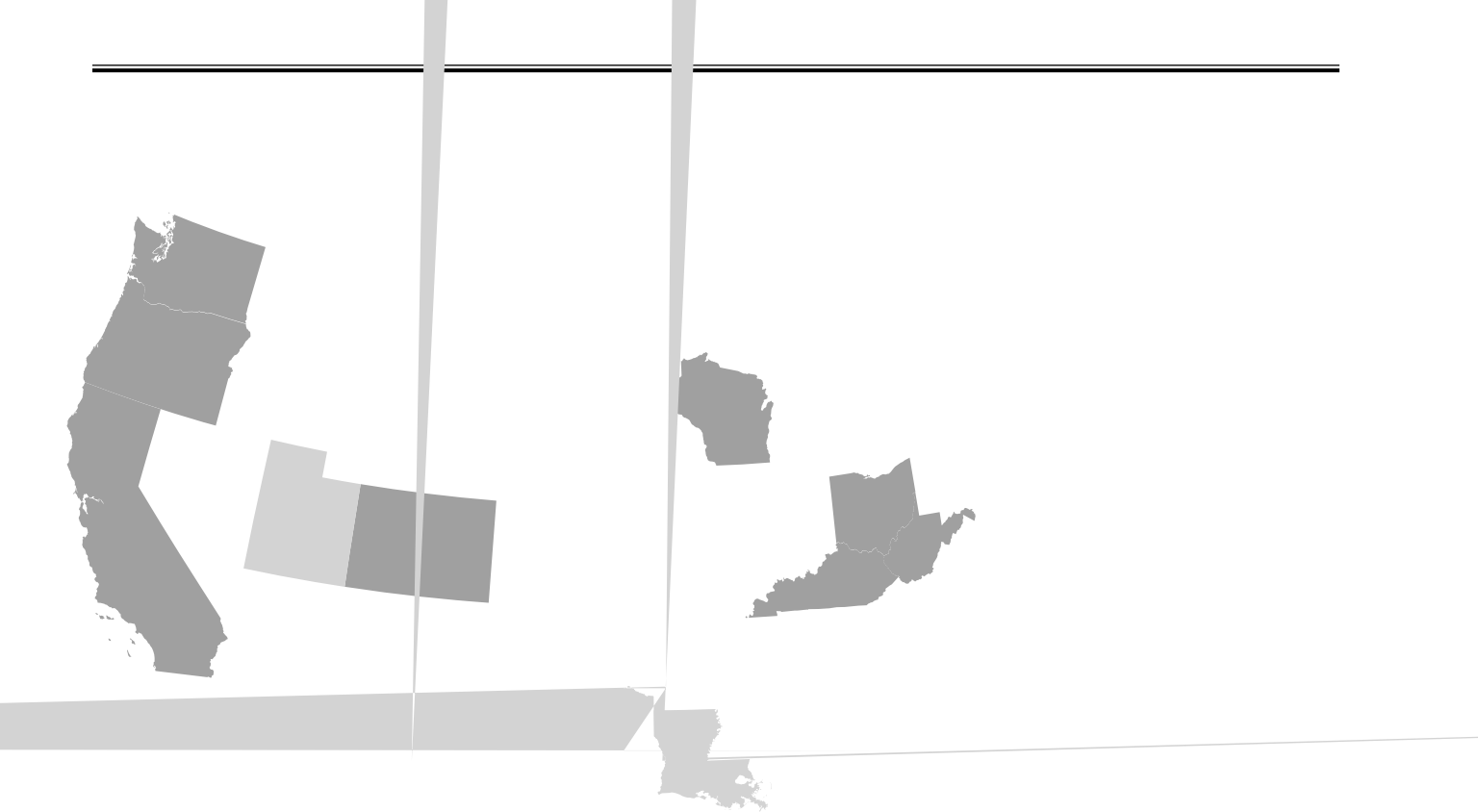
Commercial Banking expansion – particularly in WaMu states

Our Commercial Banking business has performed well in the recession, earning returns of more than 20% during the past two years and over 15% in the most difficult years. We continue to invest in additional bankers and offices to support growth. In 2011, Commercial Banking added 60 new bankers, placing 21 of them in states where WaMu had a presence. Our expansion efforts have made great progress – in California and Florida alone, deposits increased to \$1.8 billion and loans to \$2.0 billion by the end of 2011. Since the WaMu acquisition, our Commercial Banking business has continued to add 200+ new clients a year in the WaMu states.

Commercial Banking's International Banking business unit also has experienced significant growth. In the six years since the unit was

Small Business Growth

	2009	2010	2011	Year-over-year change	
				'09 to '10	'10 to '11
New small business loans (\$ in millions)	\$7,251	\$11,219	\$17,060	55%	52%
Total small business bankers	1,953	2,420	2,886	24%	19%



launched, International Banking has increased the number of U.S. Commercial Banking clients using our international treasury and foreign exchange products – to 2,500 clients – at a rate of approximately 20% per year, and we expect this trend to continue.

As we strive to better and more fully meet the needs of our Commercial Banking clients, we are increasing their access to a broader range of products. Today, our average Commercial Banking client uses more than eight of our products and services, and this number continues to increase.

The growth of our branch network

For years, some have predicted the demise of the physical branch as more customers choose to transact banking business online and on their mobile devices. However, our experience shows that instead of choosing between a branch and a website, customers actively use both. More than 17 million of our customers are paying bills online. But when it's time to take out a mortgage, apply for a credit card or seek personal financial

advice, customers often prefer to meet face to face with a banker. These activities will take place in physical branch locations for the foreseeable future. Our small business and middle market customers also are more comfortable discussing business needs such as cash management in person rather than online. In fact, our middle market business *wouldn't* exist without the branch network. Our branch presence also is a competitive advantage for many of our other businesses:

- For example, when we open a Chase branch, it provides our Card Services and Mortgage Banking businesses with the opportunity to offer more credit cards and retail mortgages. Today, about 45% of our Chase-branded credit cards and about 50% of our retail mortgages are sold through our branches.
- Today, our consumer banking household uses, on average, seven Chase products and services. Increasingly, our customers require and appreciate having the option to transact their business with us virtually and personally. Our network of branches gives consumers that choice.

The map on the preceding page shows our current branch footprint. Since 2009, we have built more than 525 new branches. In 2011, we opened 260 new branches and added more than 3,800 salespeople in the branches. We expect we will add approximately 150-200 branches a year for the next five years, which is fewer than we previously had planned. We are taking a more measured approach because regulatory changes have affected our ability to profitably operate some of our branches.

That said, and despite slight reductions in profit due to an abnormal interest rate environment, our average retail branch still earns approximately \$1 million a year. And the right type of branch in the proper location is profitable not only on its own but is enormously beneficial to the rest of the company. We believe interest rates and spreads will return to normal levels, and we are building our branches accordingly. The map shows we are building branches where we already currently reside. It always has been more valuable to increase your market share in an existing market than it is to go to a new market.

Chase Private Client business continued growth

In 2011, we opened approximately 250 Chase Private Client (CPC) locations – branches dedicated to serving our affluent clients' investment needs – and we plan to open another 750 CPC locations in 2012. Chase Private Client is quickly making an impact in deepening our relationships with the 2 million affluent clients that already bank with Chase. Today, more than 500 Chase Private Client bankers and advisors serve private clients, and we plan to add more than 1,200 private client bankers and advisors in 2012. Since we launched the first phase of CPC expansion in July of 2011, the number of CPC households we serve has nearly quadrupled, and each of those households has grown deposit and investment balances by \$80,000 on average.

At JPMorgan Chase, we are privileged to work with Caterpillar across our markets and services – from community banking in Caterpillar's hometown in

The Chase consumer businesses – Retail Banking, Credit Card, Auto Finance and Mortgage – historically ran as independent companies. Now we are coming together to run all of these companies as one consumer business and one brand – to focus, first and foremost, on serving our customers in the ways they want and with the products they choose. This includes developing common strategies, delivering a consistent customer experience, designing a seamlessly integrated product offering and continually innovating for our customers. We call this effort One Chase.

Doing a better job serving our consumer and small business customers

What does One Chase mean for our customers? It means being known and appreciated for all the business they do with us – across all product lines – and feeling as if they are dealing with one company. It means customers will be treated with consistently great service every time, any way and anywhere they connect with us. It means when customers call Chase, they will get an answer from the Chase representative answering the phone – whether the question is about their mortgage, credit card fees or banking account. It means customers can have more needs met at the Chase branch – including not only being able to get a credit card, mortgage or checking account but also being able to talk with branch professionals about any problems they may be having with any of our products.

Here are some of the things we're doing to serve our consumer and small business customers better:

Making our communications clear and simple

Our customers have told us that the “fine print” on our disclosures was confusing and wordy. Of course, that was not our intent.

When we speak, email or send a letter to a customer, we aim to foster confidence, not confusion. So we have undertaken a number of initiatives designed to simplify the way we communicate with our customers.

At the end of last year, we unveiled a revised summary guide for Chase Total Checking that makes its terms and conditions easier to understand. We developed a simple disclosure form that uses everyday words in a consumer-friendly format. Instead of saying “transaction posting order,” our new disclosure now says “how deposits and withdrawals work,” using words that customers understand. Consumers now can more plainly see a description of fees and services and learn how to avoid certain fees, determine when deposits are available, and track when withdrawals and deposits are processed – on three pages (instead of 40).

In addition to streamlining and clarifying our written disclosures, we also are proactively reaching out to customers with an email or a phone call when we think they should know something about their account. For example, if there are suddenly several unusual transactions in a customer's account that could indicate fraud, we immediately send an email alert or make a phone call to let them know.

Focusing more on customer complaints

Every week, and sometimes every morning, the senior managers in our consumer businesses listen to or read customer complaints to get to the root of problems and to identify options to solve them. These issues are discussed, and the follow-up and feedback are shared with the broader customer support teams.

We know every company makes mistakes. But if you don't acknowledge mistakes, it's unlikely you can fix them. No one should be afraid to make a change because it might imply that something we did in the past was wrong. Instead, every employee at the firm –

including me – should take responsibility for mistakes and take the initiative to fix them and prevent them from occurring in the future. We must continually make changes that make us better.

Empowering our employees to own customer issues

When customers contact Chase, they expect – and deserve – to have us understand and assist them with their entire relationship, regardless of which line of business is involved. To ensure this happens, we increasingly have empowered our front-line employees to better handle customer requests and issues.

For example, we have authorized branch managers to use their judgment in waiving fees for customers they know personally in order to get them a quicker response or expedite a transaction. We are providing real-time information to our bankers and advisors, eliminating the need to transfer many customer calls. These initiatives have helped drive customer complaints down 25% over the last six months.

One Chase means one customer. So when making decisions, we consider the entire relationship our customers have with us. For example, when making a decision about a credit card application, we now more fully consider what type of customer the applicant has been and how long that person has been a customer.

Learning from our bus trips and other feedback

Following a terrific bus trip last summer along the West Coast, we hopped on a bus again in February 2012 and took a week-long, 550-mile journey through the Sunshine State. We visited branches and operations centers throughout Florida, many of which are in off-the-beaten-path locations, like our credit card operations center in Lake Mary. We met face to face with approximately 5,000 employees and hundreds of clients across all our lines of business – from consumer customers to Fortune 500 CEOs. We also met with elected officials and community leaders to talk about how much we're expanding, lending and adding jobs in Florida.

It was an incredible trip that gave us the opportunity to see firsthand how vibrant our business in Florida is: We have become the #1 SBA lender, and our branch count, which was 261 when we bought the WaMu business in 2008, is nearly 300 today – we expect it to grow to 500 in three to five years. Five years ago, we had 6,700 employees in Florida, and, including the 4,500 people we hired last year, we now have 17,550.

One of the most rewarding parts of the trip for us was riding the bus with some of our front-line employees – tellers, branch managers, personal bankers and others. Their perspective and advice on how we could do a better job were invaluable. And, boy, did we get a lot of advice – 160 specific recommendations, which we are in the process of implementing as we speak.

We want to make this drive toward continuous improvement a part of the fiber of every person at our firm.

A new internal tool called “What Do You Think?” is giving our employees throughout the firm a chance to evaluate the products we offer customers, as well as the services we provide internally, from accounts payable to our online benefit enrollment and internal travel services. Some of us predicted these internal services were going to receive the worst ratings – we weren't wrong. But we know that while we won't always like what we learn – in fact, sometimes it is embarrassing – it will help us become better. Providing best-in-class services internally is just as important as providing them to our customers because better services make our colleagues' lives easier so they can spend more time with customers in helping to solve *their* problems.

Continually innovating for our customers

A culture of speed and innovation is imperative. Sometimes people come up with great ideas on their own, but, more often, it happens through informal networking and brainstorming. Also, small improvements, over time, cumulatively may lead to major breakthroughs.

The financial services industry has been highly innovative over the past 20 years, from ATMs to online bill payment and a variety of mobile banking applications. Chase mobile customers increased 57% over the past year to more than 8 million active users at the end of 2011. These customers transact online by paying their bills, checking their balances and transferring money between accounts. Some of our new consumer innovations include:

- Chase QuickDepositSM, part of the Chase Mobile[®] applications that allow customers to make deposits from their smartphones (by taking a picture of the check). Our customers have deposited 10 million checks in 2011. Over the past year, our total deposit volume increased to \$2.6 billion – with \$481 million deposited by QuickDeposit in January 2012 alone.
- We added “pay with points” functionality to our Amazon.com Rewards Visa[®] card, allowing customers to use their rewards instantly as cash.
- We pioneered JotSM, a new mobile application for organizing and tracking expenses, which currently ranks in the top 5% of all financial applications (Apple App StoreSM ranking) and works exclusively for our InkSM from Chase small business cards.
- We continued to partner with some of the world’s best brands, launching new cards with The Ritz-Carlton Hotel Company and United Airlines[®].
- Chase QuickPaySM, our person-to-person payment service that allows our checking customers to use a phone or computer to send or receive money using an email address (money is either taken out or deposited into checking or savings accounts), increased by more than 200% to 2.6 million users in 2011.

- We introduced Chase SapphireSM for the affluent market in late 2009 and generated more than 1.8 million accounts in about two years. In 2011, we launched Chase Sapphire PreferredSM, an enhanced affluent-oriented product that rewards customers with two points for every dollar spent on dining and travel.

We continue to roll out new products. Soon after this letter goes to press, we will be launching an exciting new banking product that will have innovative features and broad appeal. I believe this could be a breakthrough product for consumers in terms of pricing transparency, convenience and simplicity – and we hope you agree when you see it. The management team doesn’t want me to get too excited in case it doesn’t work. I told them that even if it’s a flop, I will be proud of their innovative spirit. You can’t succeed if you don’t try.

The extensive requirements of regulatory reform – which we must meet – demand enormous resources. While we are going to continue the initiatives in all of our businesses in 2012, it is unlikely that we will undertake significant acquisitions due to these regulatory demands and other regulatory constraints. We need to meet these regulatory demands properly while ensuring that our clients are not adversely affected and that we are not creating excessive, stifling bureaucracy. We are totally focused on what is in front of us. It is a new world, and we are going to adjust to it very quickly – whether or not we like it or think it is all needed.

Meeting new regulatory requirements will be a large, costly and complex endeavor – and we must get it right. Therefore, we need to devote enormous attention and resources to it

It has been estimated that there are 14,000 new regulatory requirements that will be implemented over the next few years. Three hundred out of the 400 Dodd-Frank rules still need to be completed. We need to meet the new Basel II, Basel 2.5 and Basel III requirements. We need to meet the new liquidity requirements, the new global systemically important banks (G-SIB) rules, the new requirements due to Resolution Authority and living wills, and any new requirements from two new regulators, the Consumer Financial Protection Bureau and the Office of Financial Research. We need to meet the new derivatives, clearinghouse and Volcker trading rules. We also must complete periodic Comprehensive Capital Analysis and Review (CCAR) stress testing for the Federal Reserve. And, finally, we have major new rules and requirements from Brussels, London and other global jurisdictions.

These new rules will affect virtually every legal entity, system (we have 8,000 of these), banker and client around the world. It will take an enormous amount of resources across all of our disciplines – people, systems, tech-

nology and control functions (finance, risk, legal, audit and compliance) to get it done right. Over the next few years, we estimate that tens of thousands of our people will work on these changes, of whom 3,000 will be devoted full time to the effort, at a cost of close to \$3 billion.

We must not let regulatory reform and requirements create excessive bureaucracy and unnecessary permanent costs

There are so many new rules that they inevitably create more opportunities to build unnecessary bureaucracy within the company. It is incumbent upon us to make sure that we do it right – for the regulators, our clients and our own efficient internal functioning. So we are trying to build streamlined systems to meet the needs of all the regulators in an efficient way. For example, different regulators have asked for different reports on some very complex issues such as global liquidity. We are going to try to build one report that meets all their needs and ours, too – as opposed to preparing three completely different liquidity reports every day or every month. Three reports lead to more mistakes, less understanding and more work.

We must do this in a way that minimizes cost and disruption to our clients

Most clients hope they will not see much change as a result of these new regulations. But for certain clients and certain products, the change will be significant. For example, the cost of credit, in general, will go up modestly, essentially due to the banks' higher capital and liquidity requirements. The cost of credit for some likely will go up substantially – for example, we expect larger increases in trade finance; consumer credit (particularly for consumers with FICO scores below 660); and backup lines of credit

that support commercial paper issuance. Because of the Durbin Amendment, the cost of banking services will go up modestly, but this will likely affect certain clients far more than others – e.g., customers with low account balances.

We also are trying to get ahead of the change and be proactive. We have canceled products and services and will continue to do so when we believe we no longer can adequately provide them, given the new regulatory requirements. We also are exiting products that we think create too much reputational risk for the firm. For example, we no longer bank certain types of clients, we no longer offer tax refund anticipation loans, we essentially have exited the subprime lending business and we no longer offer certain types of complex derivatives. We also have modified our overdraft procedures to be more consumer friendly and are trying to be very responsive to complaints about product disclosures, as we have mentioned previously. We will adjust to all of the new rules very quickly.

We have extensive processes in place to try to do business the right way

We have extensive processes to protect the company and conduct business the right way. We have strong audit, compliance and legal staffs (these groups total more than 3,600 employees). Some of these employees sit in specialized units that cut across the company focusing on the requirements of the Anti-Money Laundering, Bank Secrecy and Privacy acts, and other requirements (these units, which also include dedicated line of business employees, total approximately 1,400 employees). We know we won't always be perfect, but it won't be for lack of trying. Listed below are examples of how each business tries to properly conduct its affairs:

- Our Risk Committees provide general oversight into any and all risk in the business and set overall risk limits from credit extensions to any market-making activities. Risk limits are set by product, by counterparty and by type of specific risk (for example, liquidity risk, interest rate risk, credit risk, country risk, market risk, private equity risk, and legal and fiduciary risk, etc.).

- New Product Committees vet all new products to make sure that we can handle them operationally and, more important, that they meet our ethical standards for conducting business.
- The Capital and Credit Committees review all extensions of credit and uses of capital in the company to make sure we have the right limits, the right structures, the right clients and adequate returns.
- The Commitment Committees review underwritings of stocks, bonds, loans, etc., to ensure that each is properly structured, that we want to do business with the client, that we can meet our commitments and that due diligence is properly done, etc.
- The Operational Risk Committees review the potential errors in processing, legal agreements and others that can lead to any form of operational risk to the company from settlement to clearance, including litigation and processing errors.
- The Reputational Risk Committees review new types of business and out-of-the-ordinary transactions that entail risks relating to the environment, taxes, accounting, disclosures and know-your-customer rules to try to ensure that business is being done appropriately.

We operate in a complex business with high and increasing regulatory demands and risk. Whether or not we agree with all the new rules and business processes, we want you to know that we will strive to meet or exceed every regulatory requirement around the world. This simply is the way we run our business.

We have written extensively about the crisis and the need for financial reform in previous letters. Many of the issues we have discussed have not changed. It is very important, however, that we get this right so I will comment in this section on some of the more critical and recent developments.

We always have acknowledged the need for reform – and we agree with most, but not all, of it. And we all have a huge vested interest in having a strong financial system

Most banks and bankers have acknowledged the need for strong reform. JPMorgan Chase has consistently supported higher capital standards, more liquidity in the system, a Resolution Authority to better manage and unwind large financial firms, better regulation of the mortgage business, the clearing of standardized derivatives through well-structured clearinghouses and even stronger consumer protection (however, we thought this should have been a strengthened department inside the bank regulator). We also supported most of the principles of compensation reform – though you should know that our company, for the most part, had already practiced them.

In addition, we supported the ideas behind the creation of the Financial Stability Oversight Council (FSOC), recognizing that one of the flaws of our financial system was that we did not have strong oversight of the whole system or adequate coordination among many different regulators. We actually believe the FSOC should have even *more* authority than it has been given so that it can *force* coordination among the 11 regulatory authorities of the FSOC, adjudicate where necessary, and properly assign responsibility and authority.

While we agree with much of the reform that has been put in place, we do not agree with all of it. Specifically, we disagree with the Durbin Amendment – which had nothing to do with the crisis and was the adjudication of a dispute between retailers and banks

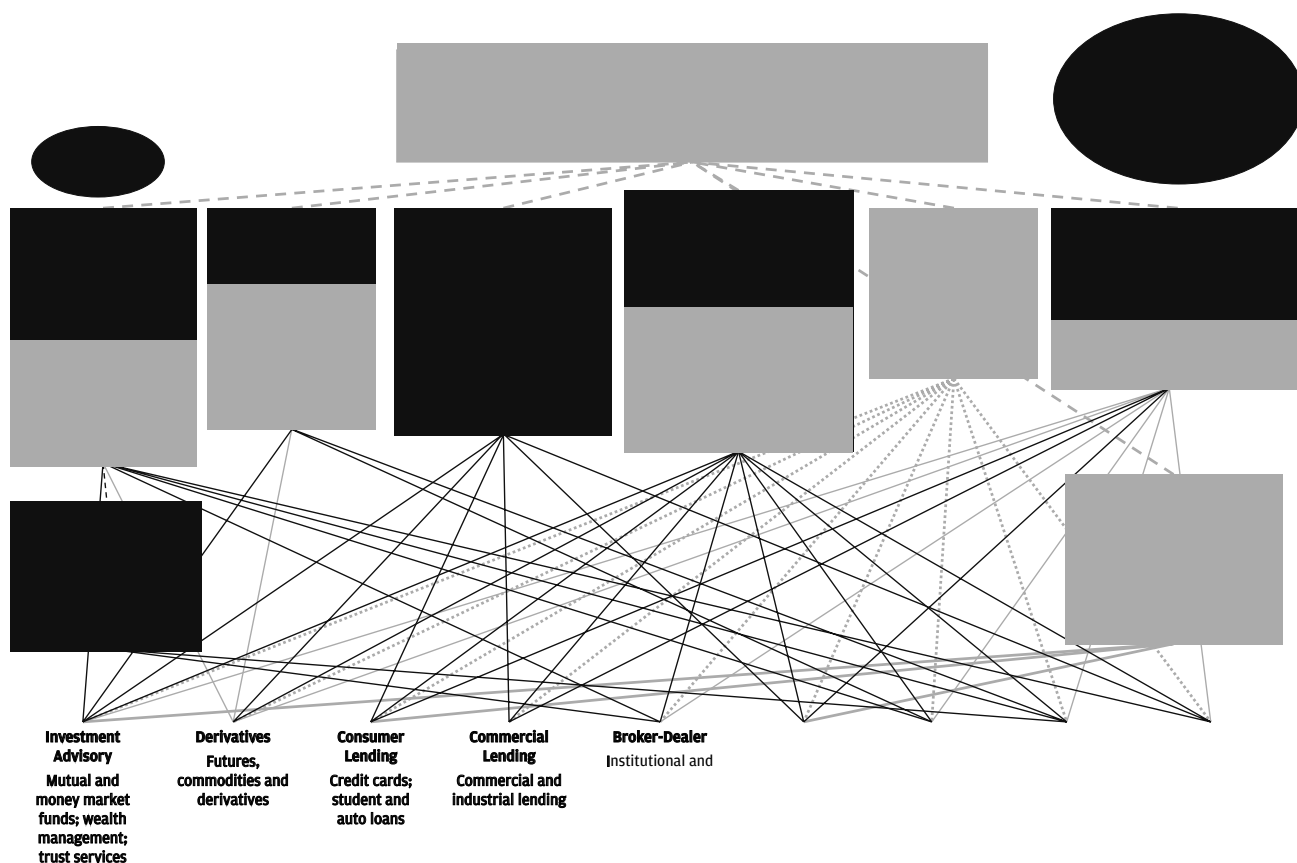
– when the banks were unable to effectively respond. (It essentially is price fixing by the government that will have the unfortunate consequence of leaving millions of Americans unbanked.) Three other specific rules with which we do not completely agree include the G-SIB restrictions and surcharge, the Volcker Rule and some of the derivatives rules. You may be surprised to know that we don't actually disagree with the stated *intent* of these rules. We, however, do disagree with some of the proposed specifics because we think they could have huge negative unintended consequences for American competitiveness and economic growth. As Albert Einstein said, "In theory, theory and practice are the same. In practice, they are not."

The United States has the best financial system on the planet. We have the deepest, widest, most transparent and most innovative capital markets. These markets have helped fuel the great American economic machine – from small businesses to large. And while we need reform, we must be very careful not to throw the baby out with the bathwater. Clear, fair and consistent rules need to be put in place as soon as possible so that our economy, once again, can grow and meet its potential.

But the result of financial reform has not been intelligent design – simplicity, clarity and speed would be better for the system *and* better for the economy

A robust financial system needs coordinated and consistent regulation that is strong, simple and transparent. The regulators should have clear authority and responsibility. Just one look at the chart on the next page shows that this is not what we now have. Complexity and confusion should have been alleviated, not compounded.

As a result of Dodd-Frank, we now have multiple regulatory agencies with overlapping rules and oversight responsibilities. Although the FSOC was created, it is proving to be too



weak to effectively manage the overlap and complexity. We have hundreds of rules, many of which are uncoordinated and inconsistent with each other. While legislation obviously is political, we now have allowed regulation to become politicized, which we believe will likely lead to some bad outcomes.

And we have been very slow in finishing rules that are critical to the health of the system. The rules under which mortgages can be underwritten and securitized still have not been completed – three and a half years after the crisis began. This is unnecessarily keeping the cost of mortgages higher than they otherwise would be, slowing down the recovery. Basel III created additional “capital confusion” as banks did not know what the specific capital rules would be going forward – the banks still don’t know exactly how much capital they will be required to hold, when the regulators would like the banks to get there and how they will be able to use their excess capital when they do get there. The Commodity Futures Trading Commis-

sion (CFTC) and the U.S. Securities and Exchange Commission (SEC), responsible for different parts of the swaps business, have not yet come up with common rules. And the several agencies claiming jurisdiction over the Volcker Rule have proposed regulations of mind-numbing complexity. Even senior regulators now recognize that the current proposed rules are unworkable and will be impossible to implement.

The rules also will create unintended consequences. Nearly 40% of all Americans have FICO scores below 660. Many of the new capital rules make it prohibitively more expensive to lend to this segment (if you are a bank). And the Federal Deposit Insurance Corporation (FDIC) now charges us approximately 10 basis points on all assets (not just the deposits it insures – we now are paying the FDIC approximately \$1.5 billion a year), making all lending more expensive and, in particular, distorting the short-term money markets that lend large sums of money over short periods of time at low interest rates.

No one has considered the cumulative effect of all these changes taking place all at once. And there is little question in my mind that credit contracted globally (particularly in Europe) as a response. Some analysts estimate that even after the European Central Bank's special three-year lending facility to banks, European banks will need to shed another \$3 trillion in assets in the next few years, and that's assuming that banks don't try to meet their new Basel III guidelines ahead of time. This can't possibly help the recovery of an already weakened Europe. With all the new rules, it is unlikely that credit availability will be replaced by new lenders. Even small banks that are exempt from many of the new rules are complaining that these rules will have a substantially negative effect on their businesses – again, not the intended but the

unintended consequence. And certainly the new regulatory burdens for large and small banks have become enormous, but it will be a disproportionate burden on smaller banks.

Recently, we have begun to achieve modest economic growth around the globe, somewhat held back by certain natural disasters such as the tsunami in Japan. But I have no doubt that our own actions – from the debt ceiling fiasco to bad and uncoordinated policy, including the somewhat dramatic restraining of bank leverage in the United States and Europe at precisely the wrong time – made the recovery worse than it otherwise would have been. You cannot prove this in real time, but when economists 20 years from now write the book on the recovery, it may well be entitled, *It Could Have Been Much Better*.

CIVIC ENGAGEMENT AND LOBBYING

You read constantly that banks are lobbying regulators and elected officials as if this is inappropriate. We don't look at it that way. We view it as our responsibility to stay actively engaged in policy debates that will affect our company, our communities and the global economy.

Not only is petitioning the government a *constitutional right*, we have a responsibility as part of our firm's mission to be actively engaged in the political process in the communities and countries where we operate.

Governments are debating issues critical to the financial markets, our company, our shareholders and our customers. It is vital for officials and regulators to have input from people within our businesses who understand the intricacies of how financial markets operate and the consequences of certain policy decisions. Contrary to what you might hear, our input, as often as not, is at the request of government officials who want to draw upon the expertise of our executives who work in the markets every day.

Engagement with government officials and regulators is not only the responsibility of our Government Relations and Regulatory Policy teams, it also has become an important part of the fabric of our entire company. Employees across our company spend time meeting with and briefing government officials and regulators – from Washington to Brussels to Beijing to Sacramento to Albany – about what they are seeing in their local markets, as well as global markets, and how policymaking affects the financial and economic issues of the day.

Our engagement with public officials includes:

- Executives and employees from around the world who visit federal, state and local capitals to provide lawmakers with perspectives on economic conditions in their communities and countries.
- Market participants who respond to requests from policymakers to provide our views on how new regulations or legislation will affect businesses, markets and consumers.
- Small business lenders who offer perspectives on the lending needs of small businesses across the country.
- Analysts and economists who share information on specific industries and economic performance around the world.
- Our Military and Veterans team, which provides policymakers with real-world information on practices that work to employ more veterans and support their financial needs.

Finally, we should recognize that thousands of groups – including unions, veterans, teachers, municipal workers and others – are reasonably engaged in exercising their constitutional rights. We will continue to do so as well.

The United States needs more conversation, collaboration, coordination and confidence

More collaboration would be a good thing. Why should anyone be surprised that financial reform, which is so important to our country, is being rethought and refought (through the courts and otherwise) – since it was passed in a partisan way without sufficient collaboration and without adequate input from experts in the field?

Even with many of the rules and reforms that we support, the details (which are critical) are far from perfect. We're left with hundreds of rules and thousands of pages, that even the regulators are now struggling to make sense of. These are very complex systems that need to be carefully thought through and analyzed, particularly by people who know the subjects best – both academics and practitioners.

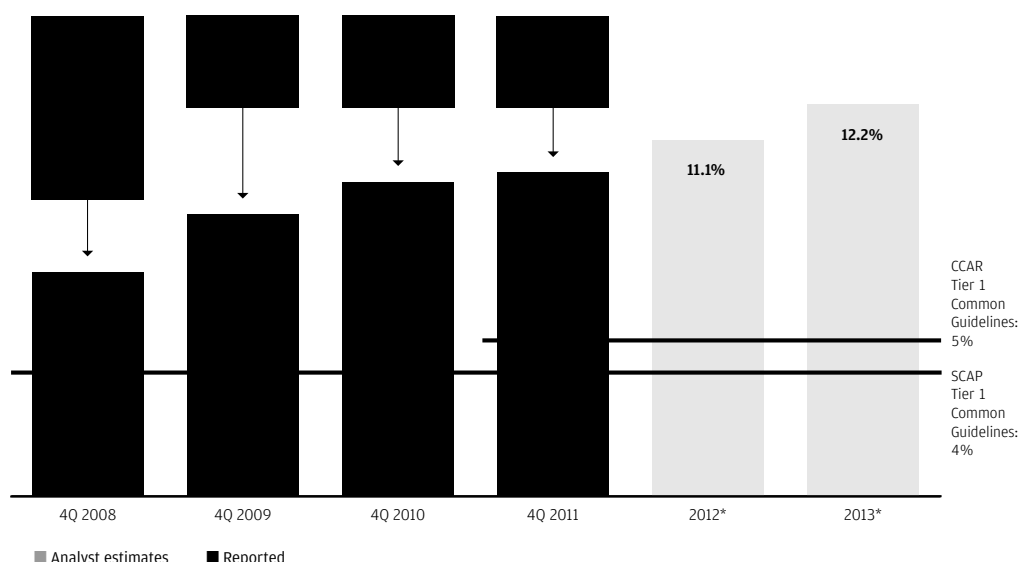
These issues are not Democratic or Republican, and the solution is not political. Many bankers would have loved to support proper reform. But it is hard to support something when you were not involved in the process in a meaningful way. In fact, at a bankers' meeting with 100 bank CEOs in the room, 70%-80% said they were afraid to speak up because of potential retribution from the regulators and examiners. This is not a healthy process for policymaking.

I am struck that so many of our leaders in the United States forget how strong our country can be. The United States of America has the world's best military, and it will have for decades. It has the world's best universities and the best rule of law. We are known for having some of the hardest working, most entrepreneurial and innovative workforces anywhere. The United States has the widest, deepest and most transparent capital markets in the world – and the best businesses on the planet – small to large. These businesses are an essential part of America's strength – they are the engine of the economy. They create the wealth that we have today to enable all of the things we do as a nation. If it weren't for the capital investment, innovation and productivity of American business, we all still would be living in tents and hunting buffalo.

The need for honest dialogue and collaboration goes way beyond the financial system. We need it in fiscal reform, health policy, energy policy, immigration, education and infrastructure. If we don't start working together, we won't get it right. It is critical that we get it right to ensure America has the best possible future.

As Benjamin Franklin said, "We must, indeed, all hang together, or assuredly we shall all hang separately."

JPMorgan Chase Capital Levels (Basel I Tier 1 Common Ratio)



* Assumes analyst estimates for net income and dividends; share repurchases are assumed at the same level as employee issuance to neutralize capital impact

- One category is substitutability – an assessment of how easily clients can replace the important services provided by the bank. One of these measures looks at market share in debt and equity underwriting. We believe this is a flawed measure since any given debt or equity transaction usually involves multiple underwriters – so replacement usually isn't even necessary. And if it were, it could be done easily.
- Another measure looks at “risky” wholesale funding. This clearly is a legitimate risk measure for banks, but the G-SIB calculation treats any funding other than retail deposits as equally risky. Your company, which effectively has no wholesale money market funding, is viewed to be just as risky as a company that mostly is wholesale funded in the notoriously fickle money markets. And no credit is given for deposits from companies (most of which are rather sticky), secure funding sources or long-term funding.
- Another factor in the G-SIB calculation is whether a bank holds assets under custody. This is a business where the assets are completely separated from the rest of the company; i.e., already fully safeguarded. We do not understand why the custody business is in the calculation at all.

We could go on and on – the rule penalizes diversification, it treats liquid securities as being worse than loans, it gives no credit to the newly established Resolution Authority to dismantle a big bank and it is inconsistent with parts of Basel III, particularly around the value of operational deposits.

We don't disagree with all of the intent of the G-SIB – it includes some logical approaches to reducing the complexity of the financial markets and the interconnectedness between financial companies. But the way some of these measures are calculated is contrived and artificial. They are duplicative and completely violate the principles of risk-weighting assets. We believe that while the G-SIB rule will cause bigger banks to hold more capital and give them some incentive to shrink, it will not end up working the way regulators envisioned.

We believe banks will be forced to increase their capital levels in order to “cluster around” their major competitors. Even if a bank could run at 7% capital, it probably will have to run at the higher number to be perceived as strong competitively. Additionally, the rule will create unintended, anti-competitive market-distorting arbitrage. Big banks that have a lot of capital will more easily win certain types of business, such as processing, from smaller competitors. Big banks that need to hold 9.5% capital against mortgages simply will syndicate them out to smaller banks that need to hold only 7% capital against the same specific assets.

Regardless of how we feel about the G-SIB surcharge, we, of course, will meet all the requirements – and currently believe we can do so and still earn adequate returns for our shareholders. We just don't think it is the right way to regulate banks – or operate a financial system.

Resolution Authority – essentially bankruptcy – needs to be made real. We must eliminate “too big to fail”

One of the most important provisions of the Dodd-Frank legislative reforms is the creation of a robust Resolution Authority, which empowers the FDIC to take over a failing systemically important financial institution, including us, and resolve its operations and businesses in an orderly manner, without causing systemic risks to the financial system or excessive risks to the economy as a whole. Shareholders and creditors would bear all the losses (in a predictable and consistent way), with no exposure to taxpayers or damage to innocent bystanders. The management responsible for the failure would be replaced, and prior compensation to directors and senior officers would be clawed back. Ideally, the name of the failed institution also would be buried, memorialized only in the hall of shame of failed institutions.

The FDIC would manage this process, including providing operational liquidity if necessary, so that resolution would occur without a lengthy period of government intervention. Properly executed, there would be minimum value destruction and contagion effects inherent in fire sales or disorderly liquidations (this also would preserve as much

value as possible for unsecured debt holders – just as in an ordinary bankruptcy proceeding). Those responsible for causing the problem would bear the losses. If losses exceeded the amount of shareholder equity and debt, the banking industry, as a whole, would pay for the losses. This essentially is the way the FDIC has operated since its creation in 1933. There would be no cost to taxpayers, and there would be no bailout by the government.

As a result, critical operations that are important to the economy and the functioning of the financial markets would continue uninterrupted. Credit card processing, ATM networks, checking accounts and debit cards would continue to function, but under the control of new owners and management. Similarly, custody services of client assets, payments processing, asset management, and securities and derivatives clearing would continue without economy-damaging interruption.

Although Dodd-Frank calls this process “orderly liquidation,” it really is comparable with a bankruptcy. Implementing this process for financial institutions operating in many jurisdictions around the world brings added complexity. We are working closely with regulators to clearly identify how critical operations in local jurisdictions would continue under a resolution process. Close cooperation is required by multiple regulators. We believe this can best be achieved by actively working together well before any such event occurs and carefully (perhaps legislatively) agreeing on how such an orderly liquidation would be pursued across international borders.

We certainly hope that a large systemically important financial institution never has to go through this process. Certainly, higher capital and liquidity standards, better loan quality and more disciplined underwriting make such a failure significantly less likely. However, the availability of this controlled “bankruptcy” process is critically important for forcing managements and creditors of such institutions to understand that they are NOT too big to fail – and to understand that they are NOT so important that the taxpayers will bail them out and that they are NOT immune to the consequences of excessive risk taking. This type of “bankruptcy” for failed financial institutions is essential for management to maintain market discipline and for risk taking of financial firms.

We need to ensure that America’s large global banks can effectively compete

Many of the new rules potentially affect U.S. global banks more significantly than they affect non-U.S. banks. This is not to say that other countries (for example, the United Kingdom and Switzerland) aren’t doing things to make it harder for their banks to compete. But we need to ensure that the rules, which affect only American banks, don’t hurt – in their *cumulative effect* – American banks’ ability to compete. Following is a list of regulations that are unique to American banks. (Many of these rules did not emanate from Basel but from the U.S. legislative and regulatory process.)

- The Volcker Rule – and we don’t know its final effect yet – will affect only U.S. companies, including, possibly, American banks’ activities outside the United States.
- The derivatives rules – still not complete – may require American banks to follow U.S. regulations outside the United States and effectively could eliminate our ability to offer derivatives to our corporate clients.
- The Collins Amendment eliminates tax-efficient Tier 1 capital, effectively increasing the cost of capital.
- Concentration limits restrict the ability of U.S. banks to acquire institutions outside the United States with no similar limitations on our foreign competitors.
- High Mortgage Servicing Rights capital charges (a uniquely U.S. asset) increase our cost of doing business.
- Proposed accounting changes are more punitive for U.S. banks when they hold marketable debt securities. Foreign banks will be able to hold many of these securities at cost, but American banks will have to deduct any unrealized losses from capital.
- U.S.-specific liquid asset classes are given less credit or excluded. Amazingly, covered bonds in Europe count as 100% liquid assets, but U.S. government-guaranteed mortgage-backed securities count only as 85%.
- The G-SIB capital charge gives no credit for U.S. Resolution Authority in Dodd-Frank.

- U.S. companies that have earned high market shares over time in the investment banking and custody businesses (usually a sign of having a strong business) are specifically penalized with higher capital charges.

Ironically, while the U.S. banking system is far less consolidated than *all* other developed nations (currently only six of the 50 largest financial firms in the world, by market capitalization, are American – they were 44 of the 50 in 1989 – this should give U.S. policymakers pause), the G-SIB charges and some of the other rules penalize American banks more than non-U.S. banks.

Since it to say, the negatives are adding up and bear close watching. While we strongly prefer to have common global rules for everyone, it may not be turning out that way. It is incumbent upon American policymakers to make sure that the final outcome is fair to American banks and that they are *fully free* to compete in the face of increasingly tough global competition.

Basel III, procyclicality, group think and the role of judgment

Quantitative easing may be good policy to help the economy recover, but it does artificially increase the value of government and government-guaranteed securities. The new Liquidity Coverage Ratio gives government and government-guaranteed securities credit only for being liquid – no other assets, including gold, equities or corporate bonds have *any* liquidity value. This also creates higher demand and, therefore, a higher artificial value for government securities. The Volcker Rule, as it currently is written, also allows unimpeded trading and liquidity for government securities and a lot less liquidity for everything else. Pension accounting is forcing pensions to hedge their liabilities by buying fixed-rate securities at precisely the wrong time. Banks hold large available-for-sale securities portfolios to manage their assets and liability risk management. And if rates ever go up (and they will) and there are losses in these portfolios, the losses will have to be deducted in capital – even though the liabilities that they are hedging are not being marked-to-market. All the items we

just mentioned could be looked at as one large “crowded trade.” If things ever start to go wrong, everyone could head to the exit door at the same time. Your company has positioned itself to be protected against rapidly rising rates – in fact, the company would benefit if either short-term or long-term rates went up.

Markets already are naturally procyclical, and Basel III makes it worse. In a crisis, Basel III demands that even more capital be held against risky assets. We estimate that the swing in Tier 1 common capital from benign times to crisis times could be as much as a 20% difference in the capital ratio. We should try to make Basel III countercyclical – but certainly not more procyclical.

Finally, the ultimate goal, with which we mostly agree, is to have Basel III applied fairly and evenly around the world. But this leads to another potential set of issues. Everyone will start to have an increasingly more common view of the risk of a certain type of asset. This is what happened in the United States when everyone thought mortgages were completely safe. Models eventually will replace judgment – and this is a terrible idea. Models always are backward looking and don’t capture true underlying shifts and changes that affect credit or markets; e.g., increasing or reducing liquidity, structural changes in industries that dramatically change the riskiness of an industry (think of what the Internet did to newspapers) or real quality underwriting vs. lax underwriting. And models have a hard time capturing concentration and correlation of risks (think of oil and real estate in oil regions). Many years ago in the United States, there were approximately eight large banks in Texas. Within five years after the oil crisis, only one survived as an independent bank. The others were either sold under duress or went bankrupt – not because of their oil exposure but because of their real estate exposure. Models cannot replace judgment, and judgment helps to balance and diversify the global financial system.

Many of the financial crises of the past hundred years around the world were related to real estate. Real estate was not the only culprit in the recent crisis, but it certainly was at the eye of the storm. I suspect that the mortgage crisis will be the worst financial catastrophe of our lifetime. What the world experienced was almost a collective brain freeze – traditional mortgage underwriting loosened over time (actively supported by the U.S. government) such that we got Alt-A mortgages, subprime mortgages and option-adjustable rate mortgages (option-ARM). These mortgages were packaged into securities (sometimes guaranteed by government entities and insurance companies), and home ownership was going up – it all seemed to be working. But as the process unfolded, unscrupulous mortgage officers were mis-selling mortgages, some borrowers were lying on mortgage documents and speculation was rampant. It was a disaster hidden by rising home prices and false expectations, and once that price bubble burst, we all were in trouble.

We need to write a letter to the next generation that says, “Never forget: 80% loan to value and verify appropriate income.”

Clearly, it was not our finest hour

We were one of the better actors in this situation – but not good enough; we made too many mistakes. We generally were a better underwriter. We did not originate option-ARMs. Many of our problems were inherited from Bear Stearns and WaMu. Even our subprime mortgages outperformed most other subprime mortgages. Early in the crisis, we also stopped dealing with mortgage brokers, some of whom underwrote the worst of the mortgages and probably mis-sold mortgages more than most.

But we did participate in this disaster by originating mortgages that wouldn't have been given a decade earlier (and won't be given a decade later). And when delinquencies and foreclosures grew dramatically, we were ill-prepared operationally to deal with the extraordinary volume of troubled mortgages and upset borrowers. Our servicing operations left a lot to be desired: There were too many paperwork errors, including affidavits that were improperly signed because the signers did not have personal knowledge about what was in the affidavits but, instead, relied on the company's processes. However, the information in the affidavits was largely accurate – i.e., the borrower, in fact, was in default, we did have the mortgage and so on.

Gearing up to deal with this problem meant overcoming the multiple and poor systems we inherited from our acquisitions of Bear Stearns and WaMu. In addition, there were numerous government modification and refinancing programs and multiple changes to these programs to contend with, some of which involved extensive and hard-to-complete paperwork. We now have 23,000 people servicing delinquent loans or dealing with foreclosures – up from 6,800 people in 2008.

These problems, as one might expect, led to a myriad of lawsuits from various U.S. government agencies, attorneys general from the 50 states and private investors.

We have settled with the U.S. government and state attorneys general and implemented strong new policies – for the good of all. In February 2012, JPMorgan Chase and four other top mortgage servicers agreed to a global settlement with the U.S. Department of Justice, the U.S. Department of Housing and Urban Development, the Consumer Financial Protection Bureau and the state attorneys general. The settlement relates to the servicing and origination problems mentioned above.

For us, the settlement will consist of the following:

- Making cash payments of approximately \$1.1 billion (a portion of which will be set aside for payments to borrowers) to 50 states.
- Offering approximately \$500 million of refinancing relief to certain “underwater” borrowers whose loans are owned by the firm.
- Providing approximately \$3.7 billion of additional relief for certain borrowers, including reductions of principal on first and second liens, payments to assist with short sales, deficiency balance waivers on past foreclosures and short sales, and forbearance assistance for unemployed homeowners.
- Agreeing, along with the other banks, to a new set of enhanced nationwide standards for mortgage servicing, including requirements around single point of contact, staffing levels and training, communication with borrowers and document execution in foreclosure cases. The standards also will require banks to offer modifications and other foreclosure alternatives for borrowers before pursuing foreclosure – a practice in which we have and will continue to be actively engaged. We support these new standards – they will help establish a higher level of transparency and clarity for servicer activities and, ultimately, will strengthen the stability of the industry as a whole. (I will talk later in more detail about all the things we are doing, in addition to the things mentioned above, to help homeowners.)

The global settlement releases JPMorgan Chase from further claims related to servicing activities, including foreclosures and loss mitigation activities, certain origination activities and certain bankruptcy activities. Not included in the settlement are claims from investors in private label securities who are making claims both on representations and warranties (i.e., that the underwriting wasn’t done according to the standards in the securities contracts), as well as lawsuits claiming there were misstatements in the underwriting of the securities.

We have substantial reserves for mortgage litigation. One of the challenges our firm continues to face following the economic crisis is litigation relating to mortgage-backed securities issued by JPMorgan Chase, Bear Stearns and WaMu. Investors have brought securities litigation, trustees have demanded loan repurchases and regulators continue to scrutinize these transactions. As I always have said, we will honor our obligations. However, we also will defend against demands that are not reasonable. Securities claims brought by sophisticated investors who understood and accepted the risks associated with their investments – which, in some cases, are current and still paying – face substantial legal hurdles. Likewise, we are going to fight repurchase claims that pretend the steep decline in home prices and unprecedented market conditions had no impact on loan performance or that seek to impose liabilities on us that we believe reside with third-party originators (or, in the case of WaMu securitizations, with the FDIC). These plaintiffs face a long and difficult road, and, as a result, litigation over these issues could take many years. Nonetheless, we have set aside significant reserves to handle these exposures.

How we are trying to properly and fairly deal with delinquencies, modifications and foreclosures

First, some facts: Of 76 million owned homes in America, 24 million do not have a mortgage. Of the remaining 52 million homes with mortgages, approximately 4.7 million have a delinquent mortgage. And approximately half of those that are delinquent are on homes where the value of the home is worth less than the mortgage. Another 10+ million homeowners are current on their mortgages, but their houses are worth less than their mortgages. (We estimate that approximately 25% of these mortgages ultimately will go into default – homeowners for the rest will continue to pay and, it is hoped, will recover the value of their homes.)

Here is where we stand and how we are trying to deal with the situation:

- **If we treated a homeowner improperly, we should make it right.** Anyone who was mis-sold a loan or was foreclosed on improperly deserves redress. Mis-selling a loan is where the borrower was misled about significant loan terms or fees or interest rates that were higher than they should have been. An improper foreclosure is one in which the homeowner did not owe the money or was not in default. If it comes to our attention that we participated in any of these situations, we will fix them immediately. That said, however, many loans were taken out by unscrupulous borrowers, individuals who either lied about their income or lied about their intention to live in the home – they clearly were speculating that they could “flip” the real estate for a profit on rising home prices. These individuals should not receive help for any reason.
 - **If a homeowner can afford to pay the mortgage – whether or not the home is underwater – the mortgage should be paid.** A mortgage is a loan collateralized by the house. It is not a loan that one should feel free to walk away from if the house goes down in value. Most of the people in this situation can, and do, pay their mortgages. Some attempt a “strategic” default – even if they can afford to pay, they just walk away. Even though they still owe the difference, it is hard for the lender to collect. It is hoped, as the housing market recovers, these “underwater” homeowners will get equity back in their homes.
 - **If a homeowner cannot afford the mortgage but can afford a reduced payment, we try to modify the loan.** When a mortgage becomes delinquent, we make a very concerted effort to contact the person. We start reaching out as early as 15 days after a loan becomes delinquent and, for some homeowners, make a hundred or more attempts before foreclosure. We are sympathetic with these borrowers because most of them are unable to make their payments for legitimate reasons – someone lost a job, someone got sick or a person’s income level dropped precipitously. In these cases, we try to modify the mortgage – both under a government initiative called Home Affordable Modification Program (HAMP), which has strict requirements, and through our Chase Home Affordable Modification Program (CHAMP), where we can be more flexible. We often can reduce the interest rate to as low as 2% and, in some cases, reduce the principal. Since 2009, we have offered over 1.2 million modifications and completed more than 450,000. We have reduced payments to borrowers by a current run rate of \$1 billion annually. Ultimately, we expect to reduce payments over the years by more than \$10 billion. For loans owned by JPMorgan Chase, we already have deferred principal of \$1.5 billion, forgiven over \$2.1 billion in principal and reduced interest payments by \$1.2 billion. And by the end of the process, we expect to have forgiven principal of approximately \$4.5 billion and reduced interest payments by a total of \$3.5 billion.
- We treat loans to investors (i.e., loans in private label securities) the same way we treat loans that we own. It is important to note that all modifications are done according to specific contracts. These contracts stipulate that you can modify a mortgage only when it is better for the lender than foreclosing, all things considered (i.e., the net present value of a modified loan is worth more than going through a foreclosure process, with all its expense, and ultimately selling the home at a very distressed price).
- **If a homeowner cannot afford the home, even with the modification, we still try to avoid foreclosure.** If someone can’t afford a mortgage at 2%, even using a reduced valuation on the house, foreclosure is the last option. Since 2009, we have prevented approximately 750,000 foreclosures through our various programs, including modifications – twice as many as have been foreclosed.

Programs designed to prevent foreclosures include short sales or deeds-in-lieu situations in which the homeowner agrees to sell the house or lets us sell the house. In some cases, we pay homeowners to sell their homes, and we waive deficient loan balances (waiving deficient loan balances represents debt forgiveness to these borrowers). These foreclosure programs have cost us \$6 billion so far, including direct payments of \$150 million and balance waivers of \$5.8 billion. When these programs conclude, we expect to have paid a total of \$650 million in direct payments and more than \$12 billion in balance waivers.

- **Foreclosure.** While foreclosure is a terrible option, it sometimes is the only option. While it is awful for the homeowner, it does allow an individual to get a fresh start and more affordable housing – and relief from a crushing debt burden. Foreclosure is the worst option for the bank, too, because the house usually is left in poor condition and sold for substantially less than the outstanding balance on the loan, resulting in a loss. (We even, from time to time, make payments to people to help them leave the home in good condition and be able to afford to relocate.) By the time we actually foreclose on someone, we generally have not received a payment for 17+ months; and in 54% of the cases, the house was either vacant or occupied by someone other than the owner. The loss to the bank, in effect, becomes loan forgiveness to the individual – but this “forgiveness,” it is hoped, is going only to people who really need it: people who truly are unable to pay and really need the debt relief. Since 2007, JPMorgan Chase has recognized losses on first mortgages of more than \$21 billion due to foreclosures and charge-offs. Ultimately, we will have recognized more than \$27 billion in foreclosures and charge-offs.
- **Home equity loans generally are modified if we modify the mortgage loan and almost always are written off if there is a short sale or foreclosure.** We treat home equity loans that we own exactly the same whether we own the first mortgage or service it for someone else. When the first mortgage is modified, the home equity loan generally is modified, and the modification

terms typically are at least as generous to the borrower as the terms of the first mortgage. The home equity loan essentially will pay off only if the first mortgage ultimately pays off. Importantly, if the first mortgage is ever foreclosed on or written down due to a short sale, the second mortgage almost always is written off. Since 2007, we have recognized losses of more than \$16 billion in home equity loans and expect as much as another \$5 billion over the next few years.

This is a miserable situation all around, but we want our shareholders to know that we are trying to treat every borrower fairly and properly based on the individual’s situation and circumstances.

But it also will be the best of JPMorgan Chase

We have brought enormous resources to bear on fixing our mortgage business. Many of our top executives volunteered to help – and we now have some of our best people from finance, risk, technology and operations devoted to this effort. As a result, we are responding rapidly and are improving across the board. For example:

- **In early 2009, Chase opened the first Chase Homeownership Center to help customers under financial stress stay in their homes.** We now have 82 brick-and-mortar centers located in 28 states and the District of Columbia, regions hardest hit by the housing crisis. Six of the 82 are near military bases, and the mortgage counselors at these centers receive special training to understand general military issues, special military programs and the Servicemembers Civil Relief Act. Over the past two years, our Borrower Assistance employees have met with more than 273,000 customers who are behind in their payments or are likely to be, and Chase has held 1,800 outreach events for homeowners who need assistance.
- **On October 4, 2011, our mortgage servicing platform, which, in fact, was three legacy technology systems from Chase, Bear Stearns and WaMu, was consolidated.** This was a huge 13-month effort that resulted in one Chase system, one way to serve customers, and a better and more consistent customer experience.

- **Our customer satisfaction scores in both external and internal surveys have improved considerably.** In the 2011 J.D. Power Mortgage Origination survey, Chase jumped to #5 from #12 in customer satisfaction among lenders nationwide – the largest improvement of any company. (We're still not satisfied with being #5.) At the same time, customer complaints have declined more than 60% from a high point in May 2011.

The mortgage business is important – that's why we are going to stay in it

Providing a mortgage – helping our customers own and stay in their homes – is one of the most important and emotional connections we have with our customers. It also is a product that has the potential to deepen our relationship with customers. Our Retail branch franchise and brand give us an enormous competitive advantage in the mortgage business. There are 5.7 million customers who have an existing Chase mortgage. But with a base of 50 million customers, we think we could double the number of mortgage customers.

Once we finish fixing it, the mortgage business will be a great one for JPMorgan Chase. The winners in the business will be those who have good customer relationships and are good at large-scale servicing and processing – right up our alley. Normalized earnings for this business should be about \$2 billion, with a through-the-cycle return on equity (ROE) of about 15%. We continue to invest in this business by growing our sales force and introducing technology applications to improve the customer experience. Over the past year, we added 700 loan officers – bringing our total to 3,800 – and we are serving more customers as a result. Plus we plan to hire an additional 1,000 loan officers in 2012.

Housing is getting better – there, I said it

There has been a tremendous focus on the fact that housing prices remain depressed and, in fact, are still going down some. The large “shadow inventory” of homes in delinquency or foreclosure that has not yet hit the sale market adds to the fears that this

will continue for a long time. New home construction still is very depressed – so, to most, the future looks bleak. However, if one looks at the leading indicators, all signs are flashing green – the turn is coming if it is not here already. We don't want to be blindly optimistic, but the facts are the facts:

- America has never stopped growing. The United States has added 3 million people a year since the crisis began four years ago. We will add 30 million people in the next 10 years.
- This population growth normally would create a need for 1.2 million additional housing units each year. Household formation has been half of that for the past four years. Our economists believe that there is huge pent-up demand and that household formation will return to 1.2 million a year as job conditions improve.
- Job conditions have been improving, albeit slowly. In the last 24 months, 3.45 million jobs have been created.
- On average, only 845,000 new U.S. housing units were built annually over the last four years – and the destruction of homes from demolition, disaster and dilapidation has averaged 250,000 a year. The growth of new households, even at a reduced rate, has been able to absorb all of this new supply, and more.
- The total inventory of single-family homes and condos for sale currently is 2.7 million units, down from a peak of 4.4 million units in May 2007. It now would take only six months to sell all of the houses for sale at existing sales rates, down from 12 months two years ago. (This low of an inventory number normally would be considered a positive sign for future housing prices.)
- While the shadow inventory mentioned above still is significant, it has shown a visible declining trend since peaking at the end of 2009, when the number of loans delinquent 90+ days or in foreclosure was 5.1 million homes. It now totals 3.9 million, and we estimate it could be 3 million in 12 months. The shadow inventory also may

move more quickly as mortgage servicers get better at packaged sales and short sales and as real money investors start to buy foreclosed homes and rent them out for a good profit. Home prices still are going down a little bit, and they will stay depressed for a while. Distressed sales (short sales, foreclosure sales, real estate-owned sales) still are 25% of all sales, and these sales typically are priced 30% lower than non-distressed sales. As the percentage of distressed sales comes down over the next 12-24 months, their negative effect on housing prices will start to diminish.

- Housing is at an all-time high level of affordability due to both low home prices and low mortgage rates.
- It now is cheaper to buy than to rent in half of the markets in America – this has not been true for more than 15 years. Relatively high rental prices can be a precursor to increasing home prices.
- At the same time, American consumers are finding more solid financial footing relative to their debt. The household debt service ratio, which is the ratio of mortgage plus consumer debt payments to disposable personal income, stands at its lowest level since 1994. This is a result of rapid consumer deleveraging – household mortgage debt now is down \$1 trillion from its 2008 peak. (Reported U.S. mortgage data do not remove mortgage debt from an individual's debt obligations until there is an actual foreclosure. It is estimated that \$600 billion of the \$9 trillion in currently outstanding mortgage debt is not paying interest today and effectively could be removed now from these numbers.)
- Recent senior loan officer surveys by the Federal Reserve show that, while there are not yet clear signs of credit loosening for new mortgages, at least the rush to tighten mortgage lending standards has abated.

- Over the last two years, \$2 trillion of mortgages have been refinanced, substantially aiding homeowner burdens. We expect another \$2 trillion to refinance over the next two years, with approximately 10% coming from recently announced government programs, and, at that point, we estimate that only 15%-20% of Americans will be paying interest rates over 6%.

More jobs, more households, more Americans, good value – it's just a matter of time.

We believe that investment banks provide a critical role in facilitating the flow of capital to meet client needs and that those needs will grow dramatically in the next 10 years

It is important to look at any business from the point of view of the client. Our 5,000 issuer clients and 16,000 investor clients will have large and growing needs in the future.

Corporate clients' need for equity and debt issuance, M&A and other advice, and balance sheet management is projected to almost double over the next 10 years. Global infrastructure investment will more than double over a two-decade period – it is projected to reach \$3.7 trillion by 2030.^(a) Total global financial assets of consumers and businesses, which now total \$198 trillion, are projected to nearly double to \$371 trillion by 2020.^(b) Clearly, these huge capital and investing needs of clients will drive real underlying growth of the investment banking business. And JPMorgan Chase is in the sweet spot because much of the growth will be with our clients – large, often multinational companies, government-related entities and large global investors. And our role as an issuer of securities and as a market maker places us right in the center of key money flows.

Of course, these business volumes, while they will grow over time, frequently have volatile swings within months, quarters and years. Not only can volumes easily move 50% by quarter or year, but spreads and fees also can move dramatically, affecting our revenue. The facts above convince us that the large slowdown we saw in the second half of last year was cyclical, not secular. And volatility does not make the business bad – it simply means you have to manage the business, knowing that it can happen at any time. In 2011, a tough time for many investment banks, your J.P. Morgan Investment Bank earned a 17% ROE.

(a) According to McKinsey Global Institute Study, *Farewell to cheap capital? The implications of long-term shifts in global investment and saving*, December 2010

(b) According to McKinsey Global Institute Study, *The emerging equity gap: Growth and stability in the new investor landscape*, December 2011

Demystifying market making (trading) – why it is so important

While most people understand corporate finance fees are earned for stock or bond issuance or advice, market making is a mystery to most people – it remains a black box. We need to do a better job of describing the important role of market making and explaining how it can be done safely. Before I talk about our market-making business, it is important to recognize that market making is a normal function of any economy. While we make markets in general in financial instruments, others make markets in just about everything, everywhere – farmers markets, all types of food and commodities markets, lumber, paper, ink, advertising, steel, etc. Markets are simply where buyers and sellers meet to exchange products and services, and market makers facilitate the process.

Sixteen thousand investor clients use our market-making services. These clients include mutual funds, corporations, pension plans, states, municipalities, hospitals, universities, etc. The services we provide are research, advice and execution. Clients come to us when they want to buy or sell securities (in this section, when I refer to securities, I mean stocks, bonds and loans of companies, bonds of government entities, mortgage securities of all types, commodities of all types, currencies of all types and derivatives on all of the aforementioned securities, including swaps, options, etc.).

It takes substantial resources to provide these services properly. We have more than 800 professionals carrying out research on 4,300 companies, 1,000 government entities (states, municipalities, etc.) and 80 countries – at a cost of approximately \$600 million a year. We analyze securities, markets and economies around the world. Our job is to educate our investors and issuers and help them accomplish their global financial objectives.

To execute trades, J.P. Morgan has more than 110 trading desks around the world – 2,000 traders – making markets and executing trades in securities, broadly defined. And 2,500 salespeople call on our 16,000 investor clients, offering ideas and advice. Supporting our research, sales and trading are approximately 13,000 technology and operations specialists and 4,000 control, finance and risk management professionals across the Investment Bank. In addition, we hold an average of \$400 billion in inventory (securities, broadly defined), which we turn over constantly, and we provide, on average, more than \$250 billion of securities financing for clients. Our market-making operations also help our issuer clients sell or raise approximately \$430 billion of capital a year.

We trade over a trillion dollars of securities, broadly defined, every day – for example, approximately 90,000 separate trades a day in our fixed income business alone. While we do business with 16,000 clients, the top 1,000 clients account for a large portion of the business. These investors are smart and sophisticated – we want their repeat business, but we have to earn it. Presumably, they keep coming back to us because they value the services we provide; but if we did not give them great value and great prices, we probably would not get their business – they have lots of other options – and there is a lot of competition for their business.

Our aim is simple – to provide our clients with sound investment ideas and value-added, world-class execution at increasingly lower cost.

The cost of these services to clients has been coming down dramatically over time – benefiting both investors and corporate issuers. Thirty years ago, it cost, on average, 15 cents to trade a share of stock, 1% (100 basis points) to buy or sell a corporate single-A bond and \$100,000 to do a \$100,000,000 interest rate swap. Today, it costs, on average, 1.5 cents to trade a share of stock, 10 basis points to buy a corporate single-A bond and \$4,000 to do a \$100,000,000 interest rate swap. Market making creates great liquidity in the market, giving investors confidence that they can buy and sell securities – often at a moment's notice. Market making also

is being done at an increasingly lower cost of execution, which is a benefit to investors and issuers, buyers and sellers. Reducing spreads, or the cost to do a trade, means that the buyer gets to buy at a better price, *and* the seller gets to sell at a better price. This is no different from Wal-Mart Stores, Inc. offering you great products at lower prices. Innovation in products, systems and markets has driven down these costs, and the investor and issuer are the beneficiaries.

Profitability is driven by serving many clients well at a low cost to them – we take on risk, which we manage carefully, to serve our clients. A few examples will suffice. We have huge volumes of business, allowing us to offer good prices. For example, in North America Cash Equities, we buy and sell approximately 160 million shares a day at 1.5 cents per share. In foreign exchange trading, we do approximately 80,000 spot/forward trades a day, netting only \$70 a trade (75% is done electronically). In credit trading, we do 4,000 trades a day (mostly bonds), making \$1,500 per trade. We also trade, on average, approximately 500 interest rate swaps a day. Certain products have higher fees associated with them, but fees generally are consistent with the risk and cost we need to take to execute the trade. In all of these examples, revenue obviously is offset by the cost of operating the business, including the cost of hedging. And when volumes drop or spreads tighten, the business clearly becomes less profitable.

The revenue on 98% of our trades averages \$50,000 or less – per trade. But on a handful of trades, we do make much larger fees because we serve our clients by taking on substantially more risk. Two examples will help explain. In one instance, we executed a multibillion dollar interest rate swap for a leading real estate company. In another trade, we executed a multiyear, half-billion dollar oil hedging program for a leading transportation firm. On some of these large trades, we can make revenue of millions of dollars, but to do so, we take on large risks, which we prudently try to hedge – an undertaking that frequently cannot be completed immediately. On occasion, after all is said and done, we may not make any revenue at all. However, our clients are happy – they have paid us to take on risks that they don't

want. And when we assume the risk, it is our job to manage it so that we are paid fairly, on average, for the risk we took.

In the market-making business, we actively try to hedge our positions to protect the firm from violent price swings. But all hedges are not perfect, and some things simply cannot be hedged. So we do take risk by holding inventory, but that is the cost of doing business – a cost not much different from the inventory a retailer or wholesaler holds in stores to serve their customers. (When they lose money on their inventory, it's called markdowns or sales.) Holding inventory at appropriate levels is a cost of doing business – it is not speculating.

Many clients have a large need for derivatives to manage their exposures. Even more misunderstood than market making in stocks and bonds is derivatives. Ninety percent of the global Fortune 500 companies actively use derivatives. They don't use them because we want them to do so. They use them to manage their own exposures. Ninety percent of what they do, and what we do, is pretty basic – they use interest rate or foreign exchange (FX) derivatives to manage interest rate or FX exposures. In addition, clients use derivatives to manage commodity exposures, credit exposures and other risk exposures. Many companies have huge exposures that they need to hedge so that they are not badly hurt or even bankrupted by violent moves in prices. Farmers have been doing hedging for a long time, and, in the modern world, it also applies to airlines, banks, investors and others who have exposures to oil, interest rates, foreign exchange rates, etc.

We tightly manage our risk in derivatives by limiting our risk to each counterparty, by limiting the type of risk we take within each counterparty and by taking substantial collateral against existing credit exposures. Today, our net credit exposure to all counterparties, net of collateral – in essence, what we are owed by our various counterparties – is approximately \$70 billion. Most of our unsecured exposure is to government entities or corporate clients where we deliberately don't ask for collateral, which essentially is a way to extend credit to them. With all of our major global market counterparties – think

of all the other major financial institutions – we don't leave any material unsecured derivatives exposure at all – we post collateral to each other every day.

One other great fear about derivatives is their “lack of transparency.” If by “transparency” people mean transparent prices, derivatives actually are very transparent. Computer screens provide immediate pricing and very accurate spread information on the majority of derivatives, and many dealers can respond with actual bids, in size and with very tight spreads, to anyone who calls. If by “lack of transparency” people mean that the regulators cannot access the information they need to evaluate the risks, then that is incorrect – they can and do see everything we can see. Finally, if by “transparency” they mean that investors (our shareholders and debtholders) can't see or understand the risks – that's kind of true even though we make extensive disclosures. But you can look at any large company's public disclosures, and there will be some, not deliberate, lack of transparency. For example, it's not transparent what newspaper companies pay for print or paper or how various companies have their inventory marked or what insurance companies' true exposures are. We try to be as transparent as we can meaningfully be, without overwhelming our investors. We welcome any suggestions on how we can get even better at this.

A liquid secondary market is critical to the primary market – where corporate and government-related entities issue securities. Because America has such deep secondary markets, corporate and government-related entities can issue large quantities of securities quickly and at a low cost. When a corporate bond issuer comes to market with a multibillion dollar issue, the world already has been educated on the company, the bonds usually are traded actively and the issue usually can be placed fairly quickly at low cost to the issuer.

This would not be possible if we did not have a high level of efficiency, activity and liquidity in the secondary markets where existing issues constantly are bought and sold. If secondary markets were traded with less frequency, then spreads – or costs –

would increase, thereby making it far more expensive for entities – public and private – to raise capital by issuing new securities. America has the widest, deepest and most transparent capital markets in the world at the lowest prices for both issuer and investor. While we clearly had some issues with parts of these markets and believe reform is needed – let’s not destroy the world’s best capital markets.

We do not disagree with the *intent* of the Volcker Rule. If the intent of the Volcker Rule was to eliminate pure proprietary trading and to ensure that market making is done in a way that won’t jeopardize a financial institution, we agree. And we believe there are many ways to accomplish this: by holding proper capital, by insisting on proper liquidity, by proper marking of positions, by proper reporting of risk, by constantly turning over the risk in inventory positions as appropriate for the type of security – trading in illiquid securities will have less turnover than trading in government securities – and by making sure that most trading is customer driven – much of the trading the Street does with itself is effectively to syndicate out unwanted risk, which is no different from loan syndication. But by its nature, market making requires that traders, in order to facilitate client business, take positions in inventory that they hope to sell later.

The reader should understand that loans, a traditional bank function, are proprietary, illiquid and risky by their nature – but that doesn’t make them bad. And most banks that have gone bankrupt did so by making bad loans – not by trading. Loans and market making both serve a critical function: financing the American business machine.

The Volcker Rule and derivatives rules need to be formulated in such a way as not to severely inhibit American banks’ ability to compete and serve clients. If the Volcker Rule or the derivatives rules are written in a way that constrains our ability to actively make markets or to competitively provide derivatives to our clients, our future will not be as bright as it could be. For both rules, one of the key questions is how they will apply to business conducted outside the United States. We cannot and should not be in a position where the rule affects U.S. banks outside the United States but not our foreign competition. Not only would we be unable to compete effectively in Europe, Asia and Latin America, but much of the business that we currently do in America (with investors or corporations) likely will move to foreign jurisdictions because our competitors will be able to offer a better deal. No matter how much our clients may like us, they will (and should) move their business if they get better pricing elsewhere.

In any case, we are well-positioned to be a winner in the investment banking business. While we do believe that there will be some large-scale changes affecting the business – driven by both regulation and innovation – J.P. Morgan has the breadth – we are one of the top players in almost all of the markets that we deal in – and necessary economies of scale to emerge as a winner.

Why we bought back the stock and how we look at stock value

Our tangible book value per share is a good, very conservative measure of shareholder value. If your assets and liabilities are properly valued, if your accounting is appropriately conservative, if you have real earnings without taking excessive risk and if you have strong franchises with defensible margins, tangible book value should be a very conservative measure of value.

And we have substantial, valuable intangibles. Our brand, our clients, our people, our systems and our capabilities are not replicable – even if I gave you hundreds of billions of dollars to do it. We have many businesses that earn extraordinary returns on equity because there is very little equity involved; e.g., much of our asset management business, our advisory business, parts of our payments businesses and others.

Many of our assets would sell at a substantial premium to what currently is on the books; e.g., credit card loans, consumer branches and others. To be honest, some also would sell at a discount vs. what they're on the books for – though many of these assets or loans will give us the cash flow return we expect and which normally are attached to a client where we earn a lot of non-loan-related, highly profitable revenue (i.e., cash management, etc.). The loan itself might sell at a discount, but the whole relationship would not. And, certainly, most of our businesses, if we sold them whole, would sell at a substantial premium to tangible book value.

Our best and highest use of capital (after the dividend) is always to build our business organically – particularly where we have significant competitive advantages and good returns. We already have described many of those opportunities in this letter, and I won't repeat them here. The second-highest use would be great acquisitions, but, as I also have indicated, it is unlikely that we will do one that requires substantial amounts of capital.

We have huge capital generation. When you look out many years into the future, JPMorgan Chase should generate huge amounts of capital, and much of it will be hard to deploy. Unfortunately, the CCAR test restricts our ability to buy back stock because it looks at just two years of capital generation. So while we have less capital than the 9.5% that we currently believe we will need under Basel III, once we get there, we will be generating extreme amounts of excess capital. And our organic growth and acquisitions unlikely will be able to use it all.

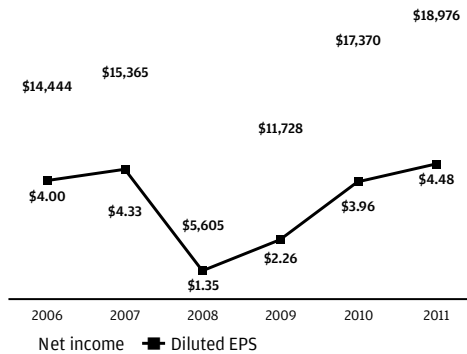
So buying back stock is a great option – you can do the math yourself. Haircut our earnings numbers that analysts project and forecast buying back, say, \$10 billion a year for three years at tangible book value. With these assumptions, after four years, not only would earnings per share be 20% higher than they otherwise would have been, but tangible book value per share would be 15% higher than it otherwise would have been. If you like our businesses, buying back stock at tangible book value is a very good deal. So you can assume that we are a buyer in size around tangible book value. Unfortunately, we were restricted from buying back more stock when it was cheap – below tangible book value – and we did not get permission to buy back stock until it was selling at \$45 a share.

Our appetite for buying back stock is not as great (of course) at higher prices. If you run the same numbers as above, but at \$45 per share, buybacks would be accretive to earnings and approximately break even to tangible book value – still attractive but far less so.

Currently, above \$45 a share, we plan to continue to buy back the amount of stock that we issue every year for employee compensation – we think this is just good discipline. As for the excess capital, we will either find good investments to make or simply use it to more quickly achieve our new Basel III targets. Rest assured, the Board will continuously reevaluate our capital plans and make changes as appropriate but will authorize a buyback of stock only when we think it is a great deal for you, our shareholders.

Earnings and Diluted Earnings per Share 2006-2011

(\$ in millions, except diluted EPS)



Tangible Book Value per Share 2006-2011



The tables above show our earnings per share and tangible book value per share over the last six years. I'd like to make one last comment about our stock and your company. I view it as a great sign of strength that, in the worst financial markets

since the Great Depression, your company could earn money, grow tangible book value, buy Bear Stearns and WaMu and expand our franchise.

Let me close by thanking our 260,000 employees. Day in and day out, they are the people who serve our clients, communities and shareholders with distinction and dedication. They make me very proud, and I am honored to be their partner.

Jamie Dimon
Chairman and Chief Executive Officer

March 30, 2012

From left to right:

Gordon Smith,
CEO, Card Services & Auto

Todd Maclin,
CEO, Consumer & Business Banking

Frank Bisignano,
Chief Administrative Officer and CEO, Mortgage Banking

We will remember 2011 as a turning point. It's the year we united across the Chase businesses to work toward becoming an industry leader in customer service.

Shifting the focus of an entire business, let alone three, isn't easy. But we must do this because we know good products alone aren't enough. We believe that outstanding service is the key to organic growth and long-term success for our franchise.

Why now? Chase has always offered a broad range of financial products and services. In fact, 50 million customers rely on us for their banking needs. There are more than 23 million households with consumer and business banking relationships, and we have 65 million credit card accounts and 8 million mortgage and home equity loans.

But historically, while consumers saw one sign out front – Chase – inside we sometimes operated like three separate businesses. We offer what we believe are the best products in the industry, but we weren't always getting the service part right. Our customer service scores were in the middle of the pack, and that's not nearly good enough.

So in 2011, we began the hard work of moving from a company organized around products to a company focused on our customers first.

We are on a journey to create an outstanding customer experience in everything we do, and we are calling this effort One Chase. What that means is always running Chase as one business for our customers, providing consistently great customer service at every contact.

We are 100% certain that exceptional customer service is the key to growing revenue. We have a tremen-

dous opportunity to earn more business from our current customers. Chase customers who live within our branch network have more than \$10 trillion in deposits and investments with our competitors. And they spend more than \$300 billion annually on non-Chase-issued credit cards.

Customers who say they are completely satisfied are 60% more likely to increase the number of Chase products they use, 26% less likely to switch banks and 61% more likely to recommend us to a friend. Aluent customers who are completely satisfied give us 52% more deposits and investments than those who aren't.

We're proud to say we've already made significant progress. Here's how we have gone about it.

First, we spent more time listening to customers' comments and complaints. Leaders, including our market, district and region managers, gathered for a two-day meeting in May during which they pored over complaint letters and listened to calls. We also launched "Begin Your Day with Our Customer," where the Executive Leadership team starts every day listening to customer calls.

We learned a lot. We found that customers want to interact with people who genuinely care about helping them and are empowered to do so. When customers have issues that need to be resolved, they want to do so quickly and easily. Also, it builds lasting customer loyalty when an employee goes above and beyond what is needed.

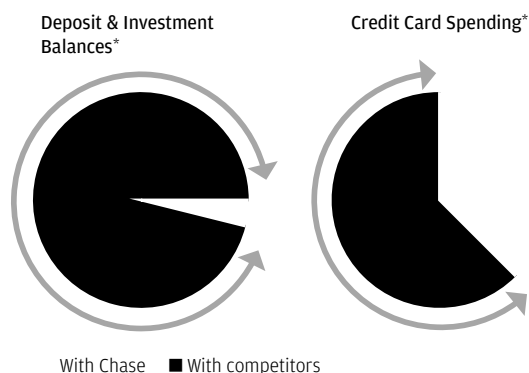
We sought out other companies renowned for service and asked them how they do it. We visited some of the best service providers we know, like The Container Store, The Home Depot, Southwest Airlines, Zappos, and Enterprise Holdings, the parent company of Enterprise Rent-A-Car, many of which are great clients of the firm. Even though their industries and regulatory frameworks are different from ours, we saw a commonality in their approach to customer service that was eye opening.

Every day, our 160,000 Chase employees are working to provide exceptional service to make sure our customers have the products and advice they need.

Estimated Growth Opportunity

The business that customers are not doing with Chase represents a huge opportunity

(\$ in billions)

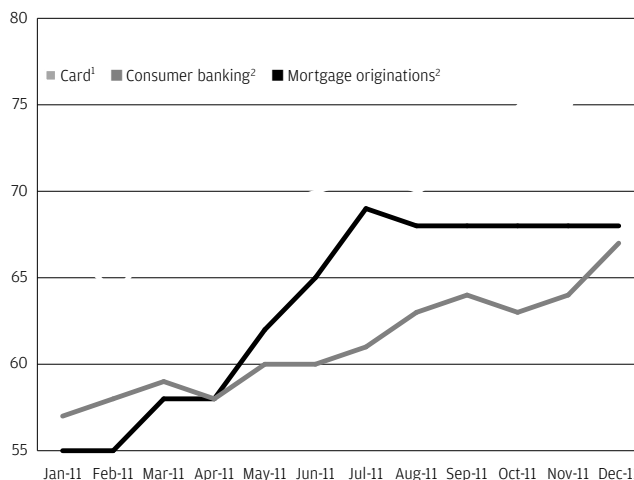


* Numbers are for 23 states with Chase branches

Source: Internal JPMorgan Chase data

Satisfaction Rating for Chase Businesses Increased across the Board

(score in percentage)



1 Satisfaction represents a top ranking of a 4 or 5 on a five-point scale

2 Satisfaction represents a top ranking of a 9 or 10 on a 10-point scale

Source: Based on internal survey process

We learned that great customer service starts with great employees. You have to hire people who have a heart for service and truly care about helping people. Then you have to give them the power to do what they know is right for the customer. We also learned that common policies and processes, while important, aren't the only things that create consistency in large organizations. Having a set of clear values and behaviors lets employees know where they stand and customers know what quality of service they will receive.

From that, we defined a consistent set of behaviors across our businesses that will help every employee interacting with customers, no matter the situation. We are calling these Chase's "Five Keys to a Great Customer Experience." The Five Keys include

things like "exceed expectations" and "own customer issues from start to finish." For the first time, all 160,000 Chase employees understand what's expected of them and how they can provide the best possible experience for all our customers.

Next we hit the road to hear from employees in person. No one knows better what customers are thinking than the people who see and speak with them every day. So we went on bus tours and road shows, holding town halls, barbecues and even rallies to meet as many people as possible across all of Chase. Everywhere we went, we asked employees to tell us what we can do to make the place better. And they did.

We kept a log of everything we heard from employees – the good, the bad and the ugly. That kicked off the most important phase of our work, taking all the suggestions and using them to transform our customer service. We're tracking 160

suggestions that we've gotten from the road. That's in addition to the more than 400 changes we've made to improve the customer experience based on feedback from customers and employees.

While we were on the road, we were inspired by our employees' dedication and integrity and by their heartfelt desire to make their customers' lives better.

We're also working to make sure we continue to get the right people in the door by integrating customer service into our hiring process. In 2011, we created a net 18,000 jobs across Chase. Everyone understands the important role service plays in our business. We've also changed the way we reward people to better align our incentives around customer service.

Chase customers can use more than 5,500 bank branches and more than 17,200 ATMs in 23 states, as well as online and mobile banking services.



And it's not just the three of us who are engaged in this effort. All our senior managers are excited about the changes we're making and are pitching in to get the work done. We've created a combined Chase Executive Committee that meets regularly and two cross-Chase councils to solve problems quickly and put more senior focus on two critical areas: Customer Experience and Brand & Marketing. These two councils aim to take the best practices of each of our businesses and apply them to all.

For example, we're working to make sure experiences in our telephone Customer Service Centers are consistent. In 2011, we simplified our automatic voice menus and the process to reach an agent. We have implemented new training on cultural awareness and communication skills and are hiring people who genuinely want to help others.

We're also simplifying how we talk with customers. Building on work started in Card Services, we adopted an industry-leading standard to create simple, easy-to-understand product disclosures. It sounds simple, and we

should have done it earlier. But we're doing it now, leading the industry and creating happy customers.

Customers are also benefiting from new technology. Our mobile applications are making it easier for our customers to do business with us across channels. Our ATMs now speak 14 languages and accept deposits of multiple checks and cash without an envelope or deposit slip. We've also set up a Twitter feed to help solve customer issues in real time.

We are even more enthusiastic about what's ahead with technology. We are piloting self-serve teller machines with bigger screens and greater functionality. They'll be able to dispense cash in multiple denominations for customers who simply want to get in and out of a branch quickly. We also plan to upgrade chase.com, incorporating feedback from customers on what they want to see.

We're only at the beginning of this journey, but we've already made remarkable progress. Overall, customer satisfaction scores are up, in some cases significantly, across Chase. Turnover is down, and the number of customer letters we receive commending our employees has increased dramatically.

The next great frontier in our industry is creating an outstanding customer experience, and no bank has really conquered it. We plan to do so. And we will. As a firm, when we set our minds to doing something, we do it. We're all consumers. We know what a great customer experience feels like and the loyalty it inspires in us. If we think like customers and focus on delivering the kind of experience we would like to have ourselves, we will build life-long relationships. And stronger relationships will lead to more revenue and future earnings.

So stop by a branch, give us a call or log on to chase.com. We think you'll be excited by the changes you see and the outstanding service you'll receive.

Gordon

Todd

Frank

It's an exciting time to be a part of Chase. I became CEO of Consumer & Business Banking (CBB) this past July after almost 10 years as CEO of Commercial Banking. There we worked closely with the consumer side of the business and relied on our outstanding branch network and terrific consumer bankers to serve our commercial clients. After more than eight months in this new role learning about the operations and products and meeting the dedicated people in this business, I more fully appreciate the power of our network. Our talented, caring and hard-working people, together with great products and channels, make Consumer & Business Banking a truly special part of the firm.

Even in a difficult year for the industry, Consumer & Business Banking produced a strong return on equity of 40% in 2011. We had net income of \$3.8 billion, a 4% increase from 2010 on revenue of \$18.0 billion, up 2% from 2010. Our total average deposits increased 6% to \$360.7 billion.

Last year brought many changes to our business. One of the biggest was the implementation of the Durbin Amendment in the fourth quarter. This legislation, part of the broader Dodd-Frank financial reform bill, caps the amount of money banks can collect from merchants who accept debit cards. We expect these changes to lower net income by \$600 million on an annualized basis. While that's a big hit to our bottom line, I am not worried. I know we have a strong plan to grow the business by

focusing on serving our customers exceptionally well and providing great products that meet all of their financial needs. Over the long term, we'll gain a larger share of their business by serving them better than our competitors, becoming the most trusted advisor to many. We will capture an increased share of our customers' banking activities and continue to grow our business.

A key to our progress has been our continued investment in branches, people, products and technology. And we never stopped investing, even during the darkest days of the financial crisis. For this reason, we are well-positioned today. CBB's strong results in 2011 allow us to commit more resources to serve our customers. We hired more than 6,500 people in 2011, bringing our total number of employees to 88,540. We also promoted nearly 14,000 of our colleagues, giving them new skills and long-term career opportunities.

We added 260 Chase branches, mostly in California and Florida. These new locations allow us to increase our lending to small businesses, offer more mortgages and refinancings, and help more people manage their money through savings and investments. We also do our part to create economic growth by hiring local architects, contractors, builders and staff to assist us.

Our Business Banking expansion is another way we're supporting our communities with more loans and banking services. In 2011, the firm made \$17 billion in new loans to small businesses, 52% more than the previous year. We were the #1 Small Business Administration lender for the second year in a row. Our average business deposits grew 12%, to \$63

billion. Since the start of 2009, we have hired more than 1,200 Business Bankers to serve our more than 2.2 million small business customers.

We set ourselves apart from the competition with strong leadership, careful risk management and continuous investment in our businesses. Our plan in 2012 is to excel at customer service, putting us further ahead of our competitors. We intend to be the first national bank to be known for exceptional customer service. Our 160,000 Chase employees are fully committed to this goal and are already working hard to get us there.

In my experience, good service always leads to more customers and revenue growth. If you are happy with the service you receive, it stands to reason that you will do more business with that company. In fact, the most profitable hotels, airlines and retail stores are usually those that have a higher standard of service integrated into their culture, creating both satisfied employees and loyal customers who seek them out again and again.

In a short time, we've made dramatic progress on providing customers with a great experience. Across CBB, customer satisfaction is up, complaints are down and our customers are moving more money to Chase. This is all great news, but it is a journey, and we still have a long road to travel.

In addition to providing better service, we are developing more customized products that meet the different needs of our customers.

- Chase Private Client:

Opened 246 Chase Private Client locations for a total of 262 nationwide



more than 500 CPC bankers and advisors now serve nearly 22,000 clients

- More custo incre more
- Overa impro durin

Some people are looking for a lower-cost product for basic banking, while others are looking for complex investment advice or help with their businesses. Delivering an experience that “wows,” no matter what type of product or service, means getting to know each of our customers individually and learning what’s important to them. For some people, it means making it easy to do transactions through technology and mobile devices. For others, it means offering experienced and proven investment advice to help grow savings.

We’ll do this across a broad spectrum of products and channels. One example is our accelerated expansion of Chase Private Client (CPC), our banking and investment platform for affluent customers. Since we launched the first phase of CPC expansion in July of 2011, the number of CPC households we serve has nearly quadrupled, and those households have grown their deposit and investment balances by \$80,000 on average.

In 2012, we will introduce a more affordable banking alternative designed for low- and middle-income

customers. We’re also looking to expand our presence in the low- and middle-income markets.

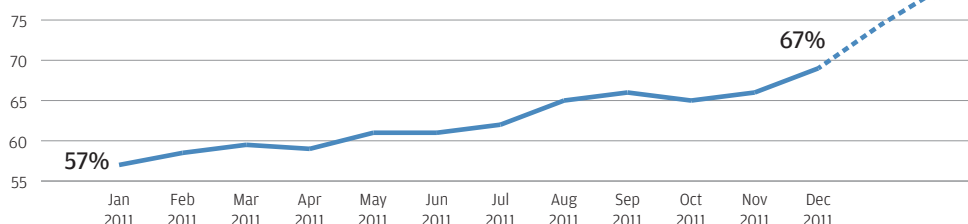
- #3 in deposit market share
- #3 in branches

Todd Maclin
CEO, Consumer & Business Banking

- Firmwide, we made \$17 billion in new loans to small businesses in 2011, up 52% from the previous year. We were the #1 Small Business Administration lender for the second year in a row

Overall Satisfaction with Chase

Customers who rate Chase a 9 or 10 on a 10-point scale (score in percentage)



Source: Chase Relationship Survey

We view 2011 as a defining year for the mortgage business. There is no question that the past several years have been extremely challenging for the industry and Chase, but I couldn't be more proud of the progress we're making. In Mortgage Banking, we remind ourselves every day that every mortgage represents a customer and a home.

While market conditions remained challenging for the mortgage industry, we made progress in several areas of our businesses and continued to tackle regulatory issues.

Although credit losses and higher expenses continue to weigh on earnings, our new Production business made money for the year on strong refinancing activity and lower repurchase losses. We also increased market share, becoming the #2 originator at the end of 2011, up from #3. Core Servicing (excluding legacy portfolio) was firmly profitable.

Real Estate Portfolio performance was better than the past two years as credit improved. This countered lower revenue from portfolio run-off.

Our first priority for 2011 was getting the best team we could on the field. We recruited top talent from across the firm and the industry to make sure we had the right controls, processes, systems and technology. To help manage our portfolios, we tapped experts in risk management

and capital markets from the Investment Bank. We also hired finance industry veterans with deep mortgage experience to ensure we meet all new regulations to the full letter of the law.

With the right team in place, we made improving the customer experience a priority for all areas of Mortgage Banking. Customer complaints declined more than 60% from their May 2011 peak, and overall satisfaction improved to 67% from 58%. Chase was the top-ranked large bank in overall satisfaction in the J.D. Power and Associates 2011 U.S. Primary Mortgage Origination Satisfaction Study.

But 2011 wasn't without its challenges. In April, banking regulators issued Consent Orders to large mortgage servicers, including Chase, requiring changes to how residential mortgage loans and foreclosures are handled. These changes include dedicated borrower assistance when customers are facing foreclosure.

In February 2012, Chase and four other large mortgage servicers entered into a settlement with federal agencies and state regulators, including the Department of Justice and state attorneys general from the 50 states. The agreement provides money directly to states, as well as to borrowers struggling to stay in their homes, and addresses issues related to mortgage servicing and originations.

Because of the regulatory actions and continued stress in the housing market, we spent more on operations, people and legal expenses. Getting

these issues behind us is good for the housing market recovery and good for Chase. We can now redirect the focus and resources consumed by the settlement toward growing our business and serving customers.

The changes we are making this year, which we believe go above and beyond any government requirements, will make us more customer oriented and a better place for employees to work.

The entire Mortgage Banking team is committed to improving our operations, processes, technology, management and controls. Three principles will continue to guide us in this business: helping people stay in their homes, delivering a sustainable profitability model, and managing our business responsibly.

We will continue our work to help people buy and afford homes. We've already made huge strides in our borrower assistance efforts and in preventing foreclosures, which we view as a last resort. We've reduced payments for struggling homeowners by about \$1 billion annually. Since 2009, Chase has prevented about 750,000 foreclosures, which is twice the number of foreclosures that we've acted upon.

During the past two years, we opened 82 Chase Homeownership

Centers across the country, where we meet with customers one-on-one to discuss their options and foreclosure alternatives, including loan modifications. Six of those centers are near military bases, staffed by loan counselors trained in military issues.

We are committed to the mortgage business and committed to returning the business to profitability over the long term for the firm and our shareholders. We believe normalized earnings should be about \$2 billion, and we have the team and a plan to get us there.

We're continuing to hire loan officers and introduce technology that will allow us to more closely monitor the application process, making it easier for our customers to purchase and refinance homes. This year, we expect a modest recovery in the purchase market and continued strong refinancing activity, and I'm confident we'll be able to earn more business.

A mortgage is more than a loan; it's a lasting connection to a customer. Helping customers achieve and maintain homeownership is a responsibility and a great privilege. I am proud of the work we are doing to rebuild this business and I am honored to have this oppor-

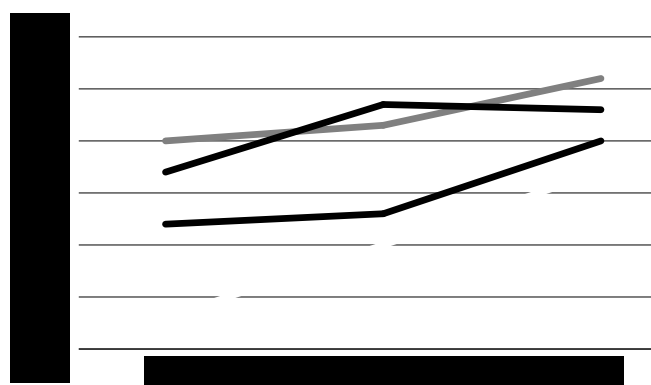
tunity to help Americans affected by the crisis. While we are far from finished, our employees are on a mission to help customers purchase and remain in their homes. It's a mission I know we will accomplish.

Frank Bisignano,
Mortgage Banking CEO,
Chief Administrative Officer,
JPMorgan Chase

- Ranked #2 in originations at year-end, #4 in home equity originations and #3 in servicing:
 - Increased retail market share to 12.6%, up 28% from 2010 and the largest increase among the top 10 lenders
 - Ranked #5 in the J.D. Power and Associates 2011 U.S. Primary Mortgage Origination Study, the highest rating for a large bank and the biggest improvement of any lender
 - Increased loan officers by 23% in 2011
 - Funded approximately 670,000 mortgages for home purchases since 2009
 - Reduced customer complaints by more than 60% from their peak in May 2011
 - Serviced 8 million mortgages and \$1.2 trillion in volume
- Prevented twice as many foreclosures as were acted upon:
 - Met with more than 273,000 struggling homeowners and held 1,800 borrower outreach events
 - Increased foreclosure alternatives 22% year-over-year
 - Completed 452,000 mortgage modifications since 2009
- Completed more than 1 million mortgage refinancings since 2010
 - Increased modifications by 38% per month and short sales by 43% per month
- Consolidated three servicing platforms, providing one Chase system and one way to service customers
- Received approval from the Office of the Comptroller of the Currency on our plan to address Consent Order requirements

Overall Satisfaction with Mortgage Banking

Customers who rate Chase a 9 or 10 on a 10-point scale (score in percentage)



Source: Based on internal survey process

After a difficult few years following the financial crisis, Card Services & Auto is now in an exciting time of growth. We made investments in new products, services and technologies that are paying off in higher market share and an improving customer experience.

The challenging economic environment makes what we do, helping customers manage their financial lives, more important than ever. Since 2009, we increased new accounts by 25%^(a) across our co-brand and Chase-branded products by offering the right products and services to people at the right stage of their lives. Sales from these new accounts were up an impressive 98%^(a) over the same two-year period.

We continued to see encouraging trends in increased consumer spending in 2011, a sign of what we hope is a more sustained economic recovery. Across all merchant categories, fourth quarter 2011 spending was up 14%^(b) from the prior year, with higher-than-average increases in dining (20%)^(b) and travel (15%)^(b).

We also added new features to our products across segments:

- We improved our rewards products, including Chase FreedomSM and Chase SapphireSM, providing even more value. We launched Chase Sapphire PreferredSM, giving customers two points for every dollar they spend on dining and travel, along with other new features and benefits.

- Our co-brand partnerships focused on some of the world's top brands, like Hyatt Hotels Corporation and The Ritz-Carlton Hotel Company. We also added "pay with points" functionality to our Amazon.com Rewards Visa[®] card so customers can use their rewards instantly. We extended our long-standing relationship with United Airlines[®] as the sole issuer of the United MileagePlus[®] co-brand card. Together, we introduced the United MileagePlus[®] Explorer card, which targets rewards-oriented consumers with new travel benefits such as priority boarding, airport lounge passes and a free checked bag.

- As the first U.S. bank to offer credit cards with embedded micro-chips, we made it easier for our customers who travel internationally to use our J.P. Morgan Select, J.P. Morgan Palladium, British Airways and Hyatt cards. Along with a magnetic stripe, the cards feature EMV chip-with-signature technology, making them extremely difficult to copy. The technology is widely used in Europe, Canada, and regions of Africa and Asia, among other locations.

- Chase BlueprintSM, our program that allows cardholders to manage how they pay down balances, continues to attract customers who want more control over their personal finances. With more than 2 million Blueprint plans in place, customers are paying down their balances twice as fast as average consumers.

- We launched JotSM, a new mobile application created exclusively for InkSM from Chase, to help cardholders organize and track business expenses.

Card Services & Auto ended 2011 with improvements across all consumer segments. Total outstanding loan balances are lower than a year ago, but the portfolio has shifted toward customers with a lower risk profile. This is consistent with our strategy to focus on lifelong, quality relationships with engaged customers.

Across our business, net income was \$4.5 billion, up 58% from \$2.9 billion in 2010. Pretax income was \$7.5 billion, also up 58% from \$4.7 billion in 2010.^(c) The improvement was driven by a lower provision for credit losses.

We opened 8.8 million^(d) new credit cards for consumers and businesses in 2011. Card Services sales volume for the year was 13%^(e) higher than 2010, and our market share continues to grow. General purpose credit card (GPCC) share increased 151 basis points^(f) from 2009 to 2011. The higher sales are beginning to translate into higher Card Services loans outstanding: Loans grew 1% quarter-over-quarter in the third quarter of 2011 and 4% in the fourth quarter of 2011.

Chase Paymentech and Auto continued to perform well in 2011. At Paymentech, bank card volume continued to outperform the industry. The number of transactions increased 19% from 20.5 billion to 24.4 billion. Chase branches are acquiring new Paymentech merchants, with more than 34% of new signings coming through branch referrals. Auto, which joined our business in July, had its best year ever.

2012 Priorities: Innovation, Superior Service as Part of One Chase

For 2012, we are once again targeting 20% return on equity.

We will continue our focus on customers, rewards, brand and execution in 2012 with an emphasis

on mobile and online innovation and the One Chase customer experience.

First, innovation continues to be a top priority for everyone, especially in mobile and online. I'm pleased to report that chase.com is the #1 most-visited banking website. Our customers spent more than \$85 billion online during 2011, making Chase one of the largest e-commerce players in the United States. Online is already our most important channel by far.

Second, we are confident that we have the best products in the market. We need to ensure that we also provide the best service in the industry. Delivering a consistently outstanding experience for customers across Chase is the best way to sustain growth. We are making excellent progress, but we still have room to improve. In Card

Services, our overall customer satisfaction increased by 10 percentage points in 2011.

I am extremely encouraged by the success we've had and am even more enthusiastic about the future of this business. As we continue to execute on our strategy, we can deliver strong performance and value for our shareholders over the long term by focusing on the needs of our customers.

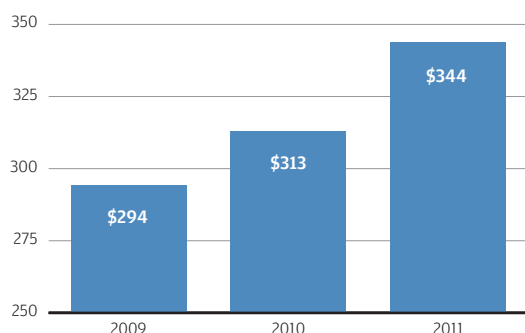


Gordon Smith
CEO, Card Services & Auto

2011 Highlights and Accomplishments

- Credit card loans climbed in the second half of 2011 and rose 4% in the fourth quarter from the previous quarter
- Card Services' sales growth of 13%^(e) year-over-year in 2011 on top of a strong 2010 indicates sustained and robust growth
- Chase Paymentech is the third-largest merchant acquirer in the United States and processed 24.4 billion transactions with 18% year-over-year growth in sales volume
- Credit card net charge-off rate declined from 9.73% in 2010 to 5.44% in 2011. Similarly, delinquency rates also came down
- United MileagePlus® and Southwest Rapid Rewards® were added as Ultimate RewardsSM point transfer partners. Chase Sapphire PreferredSM, Ink PlusSM and Ink BoldSM are the only cards in the marketplace that allow instant point transfers to MileagePlus and Southwest Rapid Rewards, as well as other frequent traveler programs
- Opened 8.8 million^(d) new credit card accounts for consumers and businesses
- \$21 billion in auto originations in 2011

Sales Volume
(\$ in billions)



- (a) Excludes terminated partners
- (b) Based on internal Chase data; excludes WaMu, International and private label portfolios
- (c) Excluding the reduction in the allowance for loan losses, pretax income increased from a loss of \$1.4 billion to income of \$3.6 billion with an upward trajectory each quarter in 2011
- (d) Excludes Commercial Card portfolio
- (e) Excludes Kohl's and Commercial Card portfolios
- (f) GPCC includes consumer, small business and charge card but excludes commercial and private label cards; Chase data excluding WaMu

while delivering very strong financial results despite persistent economic challenges and historically low interest rates.

In 2011, we delivered record revenue of \$6.4 billion and record net income of \$2.4 billion, up 6% and 14%, respectively. Deposits grew by 26% over 2010, and loans were up 13% with all business units generating loan growth. Our credit performance continued to improve with non-performing loans and net charge-offs now trending to pre-crisis levels.

We are proud to have extended \$111 billion in new and renewed financing to our clients in 2011, up from \$92 billion in 2010. In 2011, Corporate Client Banking, which serves Commercial Banking's larger corporate clients, grew loans by 43%, and Middle Market Banking increased loans by 17%. These loans helped our clients, including more than 700 government, not-for-profit, healthcare and educational institutions, achieve their business goals such as purchasing equipment and owner-occupied real estate, refinancing existing debt and funding capital expenditures. Our Community Development Banking efforts brought over \$900 million in capital to underserved communities through New Markets Tax Credit investments and helped create and retain more than 9,500 units of affordable housing in the United States.

We also made significant investments in our business to differentiate our capabilities, deliver exceptional service to our clients, and support our foundation for growth. We hired employees, opened offices

both domestically and abroad, and invested in our technology and infrastructure, all while reducing our overhead ratio to 35% and increasing our return on equity to 30%. In short, 2011 was a terrific year.

We enjoyed growth across Commercial Banking in 2011, but our four key growth areas remain an important focus for 2012.

U.S. Market Expansion – In May 2011, only two years after we began our Middle Market expansion in regions where WaMu had a presence, we achieved positive operating margin in those markets – and we have significantly more revenue potential. California and Florida remain our biggest opportunities, and we continue to gain share in those states. We also are expanding in areas outside our retail branch network with an aim to be a leading commercial bank in 40 of the top 50 metropolitan areas. I am proud of the way we are expanding – we are building strategically, with patience and discipline, while maintaining our culture and credit acumen.

Investment Banking – Our partnership with the Investment Bank remains a tremendous differentiator, generating a record \$1.4 billion in revenue in 2011. With additional dedicated resources in place and a partnership that now is stronger than ever, we are finding new ways to scale the firm's capital markets, risk management and advisory solutions to more of our Commercial Banking clients. I am confident that we are on track to meet our goal of \$2 billion in gross investment banking revenue within the next five years.

When I joined the Commercial Banking team a year and a half ago, I had the good fortune to join an organization with an outstanding track record; a tremendous culture; and a focused, long-term strategy – a legacy of Todd Maclin's nine years of leadership. Above and beyond Commercial Banking's best-in-class franchise, I am continually impressed by the depth of so many of our long-standing client relationships and the difference we make in our communities.

As a business, we are guided by our objectives to expand and deepen client relationships, invest consistently in our franchise, and maintain our risk and expense discipline. As bankers, we operate according to the fundamental principles of the firm, which include putting our clients first and adhering to the highest standards of integrity. This combination helped us add clients and expand geographically in 2011

International Banking – We are observing a powerful trend in overseas activity among U.S.-based small and medium-sized enterprises. We now have more than 2,500 U.S. clients using our international treasury and foreign exchange solutions. This number has grown approximately 20% each year since we launched this business six years ago, and I believe it will continue to grow. With our firm's resources and capabilities, we are one of the few banks able to meet the needs of these companies as they expand overseas.

Commercial Real Estate – Our real estate businesses reported sharp increases in loan production in 2011 with Commercial Term Lending up to \$8 billion from \$2 billion in

2010 and Real Estate Banking up to \$6 billion from \$1 billion in 2010. Market fundamentals are favorable, and our portfolio is in excellent shape. While we expect our Commercial Real Estate businesses to continue performing well, we, as always, are monitoring market conditions carefully to manage the cyclical risks inherent in real estate lending.

We have strong momentum in each of these four areas, and we are confident of our ability to meet our growth targets. Nevertheless, growth should not come at the expense of discipline. We will continue to operate our business responsibly and transparently while relentlessly managing our controls and operational and reputational risks. These are central tenets of our operating philosophy.

As was the case in 2011, the market environment likely will remain challenging in 2012, and competition for the best clients again will be fierce. However, our strong national leadership team, skilled and professional employees, and the scale of the JPMorgan Chase platform paired with our local delivery capabilities give me confidence that Commercial Banking will sustain its outstanding track record. I am proud to be on this team and believe that our best days are ahead.



- Produced record net income of \$2.4 billion, grew deposits 26% year-over-year to a record balance of \$175 billion and increased loans 13% year-over-year with six consecutive quarters of loan growth
- Extended more than \$1.7 billion of new loan and lease commitments to clients through the Lending Our Strength program
- Increased Commercial Banking M&A fees by 57% over 2010
- Won more than 110 bookrunner roles on syndicated lending transactions
- Added more than 1,200 Middle Market clients
- Increased International Banking revenue by 41% over 2010
- Integrated the Citi portfolio acquisition into the Commercial Term Lending business unit
- Achieved the highest return on equity in our peer group^(a) at 30%
- Achieved the lowest overhead ratio in our peer group^(b) at 35% and continued to outperform peers in credit quality with the lowest net charge-off ratio and nonperforming loan ratio
- Maintained our ranking as the nation's #1 multifamily lender^(c) and within the top three middle market syndicated lenders^(d)
- Recognized for our leadership in Middle Market Banking by earning Greenwich Excellence Awards in financial stability, investment banking and international banking^(e)

Steady Growth in Revenue and Profitability Driven by Increase in Loans and Deposits

(\$ in billions)

- (a) Return on equity peer average reflects Commercial Banking equivalent segments at BAC, KEY, PNC, USB
- (b) Peer averages include CB-equivalent segments or wholesale portfolios at BAC, CMA, FITB, KEY, PNC, USB, WFC
- (c) Federal Deposit Insurance Corporation, 12/31/11
- (d) Thomson Reuters, 2011
- (e) Greenwich Associates, 2011

From left to right:

Jes Staley,
CEO, Investment Bank

Mary Callahan Erdoes,
CEO, Asset Management

Mike Cavanagh,
CEO, Treasury & Securities Services

J.P. Morgan is one of the financial industry's outstanding global wholesale franchises. As leaders in each of our businesses – the Investment Bank, Asset Management and Treasury & Securities Services – we serve many of the world's most successful corporations and individuals.

Large multinationals and emerging companies, institutional investors and individuals all turn to J.P. Morgan for capital, insights and solutions to address the opportunities and challenges that arise in today's rapidly evolving global economy.

Over the past decade, despite global crises, world commerce has evolved at a remarkable pace. Today, multinational corporations operate in many large, new markets that, in aggregate, dwarf the revenue potential of the mature economies in Western Europe and the United States; developing market financial assets account for 20% of the global total after years of double-digit increases.

Through sustained investment and strong execution, J.P. Morgan has developed unparalleled scale and capabilities to partner globally with clients to enable them to realize diverse financial and strategic goals.

Last year, while serving more than 25,000 institutional clients headquartered in 170 countries and over 5 million individuals, we cleared 20% of the world's dollar transactions, raised more debt and equity capital than any other firm (\$430 billion), provided custody for nearly \$17 trillion of assets and supervised nearly \$2 trillion of investment assets.

J.P. Morgan's aggregate revenue totaled \$43.5 billion,* roughly half from international sources, mirroring worldwide trends.

Although we already are well positioned for the future, we are adding new dimensions to our capabilities. For example, more Investment Banking clients are using the expertise we've developed in Treasury & Securities Services to streamline their own treasury activities to achieve greater efficiency in diverse operating and regulatory environments around the world.

We are taking bold steps to improve coordination of complementary activities across our lines of business that will grow revenue and strengthen customer relationships. The Global Corporate Bank, a partnership between Treasury & Securities Services and the Investment Bank that began in 2010, targets approximately 3,500 corporate, financial and public sector clients for intensive coverage by a dedicated team of banking and treasury professionals.

The Global Corporate Bank is on track to deliver more than \$1 billion in annual pretax earnings by 2015. Similar ventures are under way involving Asset Management, Commercial Banking and other lines of business. In 2011, revenue synergies attributed to these activities were approximately \$3 billion in the Investment Bank, \$1 billion in

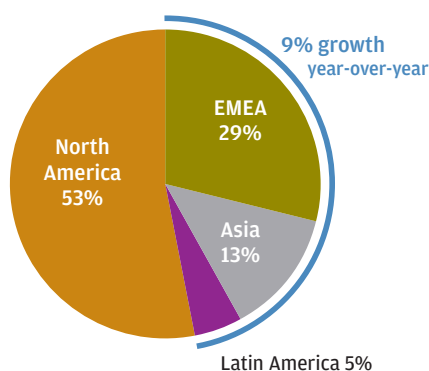
Asset Management and \$3 billion in Treasury & Securities Services, and, in our opinion, we've just scratched the surface of what's achievable.

In addition, we launched another strategically important initiative across our businesses – "Value for Scale" – to eliminate unnecessary complexity, improve communication, and optimize shared and shareable resources. This promises substantial cost savings, further focuses attention and resources where they are most productive, and greatly enhances the quality of our work and client effectiveness.

Although we've really just begun, it's our belief that we're on a path to transform the way global wholesale banking business is conducted, delivering better results for our clients and ultimately more profits for our shareholders. This ambition wouldn't be possible without the efforts and shared vision of the 72,000 extraordinary employees throughout our organization whom we are privileged to lead.


Jes Mary Mike

J.P. Morgan Revenue*



2011 total: \$43.5 billion

Revenue Synergy Examples

Investment Bank	Asset Management	Treasury & Securities Services
- Jointly cover 1,800 clients with Commercial Banking (CB) - Robust client referral to and from Asset Management	- Manage \$120 billion of assets for CB clients - Manage \$90 billion of assets for Treasury & Securities Services clients	78% of top clients shared with Investment Bank 75% of CB clients use Treasury Services products
\$3 billion	\$1 billion	\$3 billion

* Includes Investment Bank, Asset Management and Treasury & Securities Services

In 2011, J.P. Morgan enabled approximately 21,000 issuer and investor clients in over 130 countries to raise capital, invest and implement each client's unique financial and corporate strategies.

Successful execution, risk management and expense discipline produced near record net income for shareholders. We also maintained or improved our leadership position in most major markets and regions. We achieved these strong results even while market sentiment gradually deteriorated from cautious optimism in the first half of 2011 to pronounced anxiety by year-end, affecting transaction volumes and backlogs.

The U.S. debt ceiling impasse, sovereign downgrades, Eurozone instability, a developing markets slowdown and Mideast turmoil are a few examples from the growing list of major issues facing governments – and investors – that undermine confidence in the world recovery. Frenetic financial rulemaking by authorities in diverse venues around the globe also added a special dose of uncertainty to markets.

The Investment Bank (IB) once again made a significant contribution to firmwide results, delivering revenue of \$26 billion with net income of \$6.8 billion – our second best year ever. This 17% annual return on equity (15% excluding Debit Valuation Adjustment (DVA) – the effect of wider JPMorgan Chase credit spreads) is in line with multiyear targets that were set some time ago.

Our client flow business, relative to risk, has never been better. Equities, Fixed Income and Commodities contributed more than 75% of IB total revenue. At \$20 billion, aggregate revenue in these businesses is nearly double pre-crisis levels. Equities, Commodities and Electronic Equities all achieved near or record revenues. In addition, our Fixed Income franchise retained its #1 revenue ranking for the second year in a row.

The acquired Sempra assets were fully integrated into our Commodities platform; we're now one of the top three firms in this profitable and highly competitive global industry.

In addition to strong financial results, we made progress on several strategic objectives. We eliminated regional silos and appointed a global head of Investment Banking to manage industry coverage, capital markets origination, and mergers and acquisitions worldwide. This move accelerates collaboration, streamlines reporting, and greatly improves resource allocation, client service and talent development.

Internationally, wholesale activities across lines of business – Asset Management, Treasury & Securities Services (TSS) and Investment Banking – now are supervised through an International Steering Committee chaired by Mary Erdoes, Mike Cavanagh and myself.

The Global Corporate Bank (GCB), less than two years old, at year-end achieved incremental international revenue exceeding \$600 million. The GCB targets a subset of multinational clients for enhanced coordina-

tion and cross selling of TSS and IB products by a dedicated cadre of 250 bankers in offices around the world.

To better serve the needs of clients in Europe and the Middle East, we opened branches in Poland and Qatar, expanded banking capabilities in Saudi Arabia and launched an EMEA Prime Brokerage platform that created a solid pipeline of new clients.

In Asia, our joint venture with China Securities yielded impressive results in its first year. The joint venture launched its business operations, completed its first sole underwriting and was awarded Foreign Bank of the Year by China Business News.

The IB's technology Strategic Reengineering Program generated significant efficiencies and savings. Since 2008, we've decommissioned 28 overlapping systems, realizing direct run rate savings of roughly \$175 million.

We've retained 98% of our top talent while managing through industry-wide adjustments in the structure and level of compensation. Our scale, sustained investments in career development and franchise strength are strong advantages in the war for talent. The Investment Bank's compensation to revenue ratio is one of the best in our industry.

A vigilant focus on clients' long-term interests always has been our top priority. We are finding new ways to harness the resources at JPMorgan Chase for Investment Bank customers.

We recently launched a Value for Scale initiative that will centralize areas of expertise that support TSS and the Investment Bank. By merging teams and streamlining systems, we improve efficiency and enhance employee specialization. Most important, our clients' strategic objectives gain greater visibility across the firm, allowing professionals to share their knowledge and experience more effectively and to create additional value.

Our partnership with Asset Management's Private Bank improves coverage of private and closely held firms and their owners. This opportunity, similar in concept to the GCB, will lead to

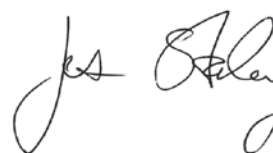
increased activity with thousands of clients worldwide. Close cooperation between the IB and the Private Bank also will strengthen the firm's presence in growth markets where family-owned firms predominate.

As rulemaking moves toward implementation in the United States and Europe, we will add more resources to expand our already considerable engagement with regulators to help them achieve the best outcome for clients and markets.

Our industry will adapt to new rules and capital costs. Markets will recalibrate the pricing of risk, and, together with clients, we will find the most efficient path toward recovery through the thicket of

global challenges. We are fortunate to be able to serve during such an exciting and transformative era.

Few, if any, other global firms have commensurate financial strength, talent and tools. In this environment, our resources will be particularly useful to clients and being a part of J.P. Morgan will be especially satisfying for all who work here.

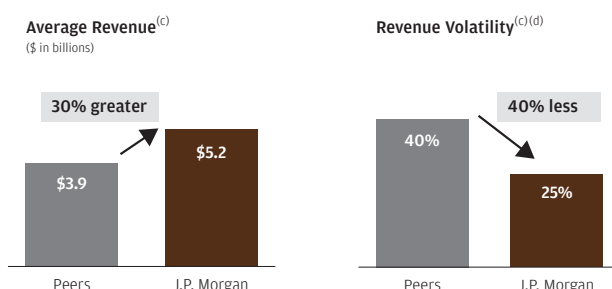


Jes Staley
CEO, Investment Bank

2011 Highlights and Accomplishments

- 2,500 sales professionals, 2,000 traders, 2,000 bankers and 800 research analysts serve clients around the world
- Executed 271 equity transactions,^(a) including lead bookrunner on Mosaic's \$7.5 billion equity offering – largest ever U.S. natural resources equity offering
- Advised clients on 332 announced mergers and acquisitions globally, including six of the year's 10 largest transactions, and achieved 18% market share^(a)
- 110 trading desks around the world execute, in an average quarter^(b):
 - 5 million foreign exchange spot/forward transactions
 - 30,000 interest rate swaps
 - 10 billion North American equity shares
- Global Corporate Bank increasing revenue with multinational corporations^(b):
 - IB Markets: +29% year-over-year
 - Treasury Services: +22% year-over-year
- Market leadership in Banking, Equities and Fixed Income:
 - Top three firm in 13 of 16 major business segments^(b)
 - Retained #1 global IB fee ranking, with 8% market share^(a)
- Helped clients raise \$430 billion of capital globally^(a):
 - More than \$390 billion in debt
 - Around \$43 billion in equity
- Led the market in arranging or lending approximately \$440 billion in 1,204 transactions^(a)
- Raised or provided \$68 billion in capital for U.S. state and local governments, not-for-profits, healthcare organizations and educational institutions^(b)
- For second consecutive year, won U.S. Equity and Fixed Income polls in *Institutional Investor's* All-America Research survey

Over the last 12 quarters, J.P. Morgan's average markets revenue has been 30% greater than its peers, with 40% less volatility over the same period



(a) Dealogic

(b) Internal reporting

(c) Revenue excludes DVA; includes eight IB peers

(d) Volatility equals standard deviation as a percentage of the period average

In 2011, Treasury & Securities Services (TSS) made solid progress toward a long-term performance target that is unchanged – a 25% return on equity (ROE), which equates to a 35% pretax margin. Although the operating environment continues to be a serious challenge for our profitability, I am proud of what we accomplished in 2011, and I could not be more confident about where the business is headed.

J.P. Morgan is a leading global provider of both treasury and securities services – and we have the capacity to continue investing to advance that market leadership as demand for these services grows.

From any angle, this is a great business to be in. We provide essential financial functions like trade finance and securities servicing that keep global commerce and financial markets running. The business generates steady earnings, good margins and high return on capital. These are businesses that are built over decades and require huge scale – making for high barriers to entry.

TSS delivers services that are integral to the development of strong, long-term client relationships across the firm's institutional businesses. In fact, nearly 80% of TSS' top clients are shared with the Investment Bank – underscoring this unique

attribute of the J.P. Morgan franchise. Our expanding partnership with the Investment Bank already is yielding good results, and there's a great deal more potential to be realized. From a firmwide perspective, TSS is a significant source of liquidity, providing an average of roughly \$320 billion in deposit^(a) balances in 2011.

In 2011, TSS reported net income of \$1.2 billion, up 12% from 2010, and a return on equity of 17%. Net revenue was nearly evenly divided between each business: \$3.9 billion in Worldwide Securities Services (WSS) and \$3.8 billion in Treasury Services (TS).

Our investment in global expansion yielded positive results: International revenue rose 17% and comprised 55% of total revenue, up from 49% in 2010. In WSS, 62% of revenue was generated outside North America, and 47% of TS revenue came from outside North America.

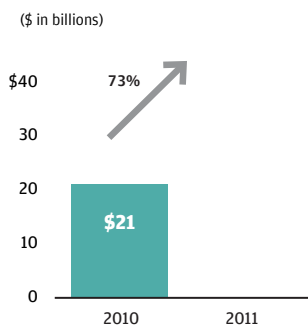
These increases resulted from expanded capabilities in more than 20 countries and the efforts of our 250-strong (and growing) team of Global Corporate Bankers who drive coordinated client planning with the Investment Bank and Asset Management.

I feel good about this performance, especially given our continued investment in capabilities and the low interest rates that compress the spreads we earn on client balances. But these results are still below our potential to earn, given the inherent strengths of the business. I'm confident that we can and will do better as we pursue our growth and efficiency priorities.

Our franchises are strong and well-positioned to capture the opportunities presented by macroeconomic trends. In TS, continued growth in cross-border commerce and trade is driving demand for global cash management and trade finance capabilities. In WSS, continued development of international capital markets and growth in cross-border investment will increase investor client needs for global custody and other securities services.

Our collaboration with the firm's other wholesale businesses distinctively positions J.P. Morgan to support our clients' global objectives. For example, our partnership with the Investment Bank enables us to provide custody for its prime brokerage clients and trade finance

(a) Deposit balances for TSS are shown on an average basis and include deposits, as well as deposits that are swept to on-balance sheet liabilities (e.g., commercial paper, federal funds purchased, time deposits and securities loaned or sold under repurchase agreements) as part of customer cash management programs



For more than 175 years, J.P. Morgan Asset Management has been managing assets for institutions and individuals around the world. While 2011 presented many new challenges for investors, our approach remained the same: Rely on research, incorporate our collective years of experience in developed and emerging markets, and rigorously manage risks.

As global markets and economies continue to become interconnected and clients increasingly require global solutions combined with local expertise, J.P. Morgan is uniquely positioned to be the first call. Our teams in more than 30 countries bring together global macro and region-specific insights to help our clients for the near and long term.

Our Investment Management and Private Banking franchises are built on a fiduciary-minded foundation that puts a relentless focus on highly disciplined investing and generation of long-term investment outperformance. Our time-tested portfolio management skills, combined with our company's capital markets expertise, fortress balance sheet and risk management culture, led clients to invest more with us this year than ever before, resulting in a record 2011.

On the whole, 2011 was a very strong year, but our results were even more gratifying in the context of a challenging geopolitical backdrop and volatile market environment. Some of the highlights include:

Strong and Steady Financial Performance – Asset Management produced record revenue of \$9.5 billion, up 6%. While revenue growth came from almost every region and major asset class, it was particularly strong in our international and alternatives businesses, which were up 12% and 17%, respectively. Net income of \$1.6 billion was down due in large part to continued investments in front office talent and new technology initiatives.

Clients continued to put their confidence in J.P. Morgan Asset Management, entrusting us with \$53 billion in net new long-term assets for the year. We marked our 11th consecutive quarter of positive inflows. We also experienced record loan growth, up 31% to \$58 billion; deposit balances, up 38% to \$127 billion; and mortgage production, up 40% to \$15 billion. We ended the year with record assets under supervision of \$1.9 trillion.

Robust Investment Performance – The foundation of any asset management business is its ability to consistently outperform the benchmark and the competition. I'm proud to report that 78% of our mutual funds are in the first or second performance quartiles over the past five years. This track record has translated into positive flows into virtually every asset class and resulted in industry-leading growth rates in long-term assets under management flows.

Leading Provider of Alternative Solutions – J.P. Morgan has long been a pioneer in providing alternative solutions to clients who want to invest in private equity, real assets and hedge funds. In 2011, our Alter-

natives revenue grew by 17% as the business achieved several successes, including being ranked as the second-largest hedge fund manager by *Absolute Return* magazine. Together with our partners at Gávea Investimentos, we completed the largest-ever private equity fundraising in Brazil with the Gávea Investment Fund IV, making us the largest private equity manager in that country. We also launched the J.P. Morgan Digital Growth Fund; continued building out our Global Real Assets, Private Equity and Highbridge franchises; and maintained our leadership in advising clients on accessing other alternative asset managers.

Invested in Our Future – Our laser-focus on managing expenses and uncovering operating efficiencies enables us to reinvest in our business, create new investment capabilities (we increased our investment professionals by 4%), enter new markets (we opened five new offices) and exploit untapped distribution channels (we increased our client-facing professionals by 8%). Over the past year, we have invested more than \$400 million in new people, systems, technologies and platforms to grow our market-leading positions for years to come.

Improved Market Share – Through the growth of our distribution network, we gained market share in a number of areas, including becoming the first bank-owned asset manager to be among the top 10 (#7 to be exact) in U.S. mutual fund assets under management and ranking #2 in U.S. mutual fund flows. We also generated a compound annual growth rate of 16% for international revenue across our division over the past two years.

Strategic Priorities for 2012: Innovate, Invest and Protect

We remain committed to providing superior investment returns for our clients through active asset management, as we have for decades.

We recently celebrated the 25th anniversary of one of our leading equity products, our Research Enhanced Index portfolio (REI 250). This unique accomplishment is the result of a long-standing dedication to fundamental company research. REI 250 has outperformed the S&P 500 over the 3-, 5-, 10-, 15-, 20- and 25-year periods. Our dedication to local research combined with continuous innovation enables us to constantly adapt our approach and navigate portfolios through time.

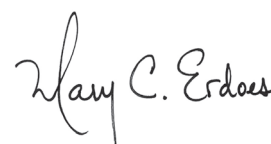
We will continue to invest in providing global and local solutions to our clients around the globe. We have the world's leading Private Banking franchise, and we are committed to gaining market share by adding front-line bankers and client advisors in the locations where clients need us most and by developing solutions that give clients around the world the exposure they seek.

As we have expanded our business internationally, so have our partners in the Investment Bank and Treasury & Securities Services. Working together, we have a tremendous opportunity to provide clients across J.P. Morgan with a complete array of solutions that spans both their corporate and personal balance sheets.

All the energy we direct at searching for opportunities to invest results in an equal amount of energy being focused

on managing risks and protecting clients' interests. Although new regulations will result in many changes in the industry, we anticipate managing through the issues presented while maintaining our client-first approach.

I know I speak for all of my partners in Asset Management when I say we are excited about the opportunities ahead for our business, and we look forward to delivering the returns JPMorgan Chase shareholders expect from us while always doing first-class business in a first-class way.

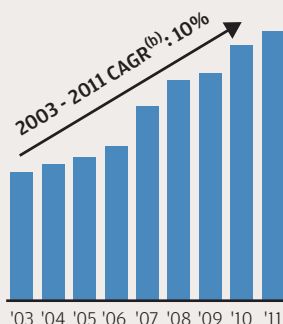


Mary Callahan Erdoes
CEO, Asset Management

2011 Highlights and Accomplishments

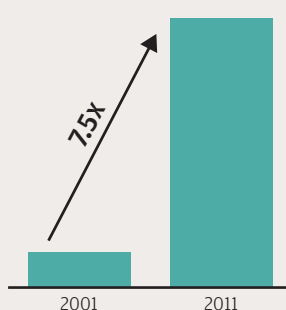
- #1 Ultra-High-Net-Worth Global Private Bank, *Euromoney*
- Leading Pan-European Fund Management Firm, Thomson Reuters
- Best Global Brand in Private Banking, *Financial Times*
- Gold Standard Award for Funds Management in the United Kingdom for nine years in a row, Incisive Media
- Second-largest recipient of U.S. total net mutual fund flows, *Strategic Insight*
- #1 U.S. Real Estate Equity and Infrastructure Money Manager, *Pensions & Investments*
- Asset Management Company of the Year in Asia and Hong Kong, *The Asset*

Global Wealth Management Revenue^(a)



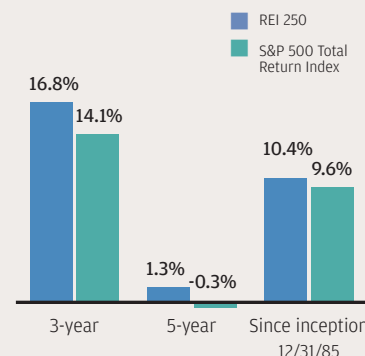
(a) 2003-2004 pro forma for Bank One acquisition
(b) Compound annual growth rate (CAGR)

Alternatives Assets under Management^(c)



(c) Differs from public definition. Excludes currency and includes 130/30 funds

U.S. REI 250



Annualized performance results are as of December 31, 2011, and gross of fees

Growing Our Economy

- Earned the position as #1 Small Business Administration lender by volume in the country. We lent more than \$17 billion to small businesses in 2011, up more than 52% from 2010 and 135% from 2009. We made nearly 400,000 new small business loans in 2011, up 45% from 2010.
- Demonstrated our commitment to investing in the future of the communities where we operate. In the United States, we hired more than 17,000 people in 2011.
- Launched The Brookings JPMorgan Chase Global Cities Initiative with a \$10 million commitment to The Brookings Institution to help the 100 largest U.S. metropolitan areas become more competitive in the global economy.

Giving Consumers New Tools

- Became the first large bank to adopt The Pew Charitable Trusts' model disclosure form that uses everyday words in a consumer-friendly format.

Strengthening Communities

- Invested \$1.4 billion in loans and nearly \$1 billion in equity to build or preserve more than 19,500 units of affordable housing for low- and moderate-income families in over 80 U.S. cities. We also provided more than \$350 million in loans and \$46 million in donations to top-tier Community Development Financial Institutions and other intermediaries to support affordable housing, schools, healthcare clinics and small businesses.
- Donated more than \$200 million in philanthropic support to not-for-profits in our communities, and our employees provided nearly 375,000 hours of volunteer service through our Good Works program in local communities.
- Contributed our time and expertise to digitize the most significant papers of Martin Luther King, Jr., through our Technology for Social Good program. The new website and archive make some of Dr. King's most famous speeches and correspondence available to Internet users for the first time.
- Led the financing of the California FreshWorks Fund, an innovative, \$200 million public-private partnership loan fund created to increase access to healthy, affordable food in underserved communities, spur economic development and inspire innovation in healthy food retailing.
- Created a partnership between our Global Commodities business and the World Bank Group to offer a new product that will provide up to an initial \$4 billion in protection from volatile food prices for farmers, food producers and consumers in developing countries.
- Invested \$8 million through our Social Finance business in a \$25 million initiative with the U.S. Agency for International Development and the Bill & Melinda Gates Foundation to provide private capital to small and medium-sized enterprises in Africa's agriculture sector.
- Donated more than \$10 million to 200 charities in 2011 through the Chase Community Giving crowd-sourcing program on Facebook, for a total of over \$20 million to 500 charities since 2009. The program has more than 3.3 million fans worldwide.

From left to right:

Kimberly Davis,
President, JPMorgan Chase Foundation

Mel Martinez,
Chairman, JPMorgan Chase Foundation

Peter Scher,
Head of Corporate Responsibility

Honoring Military and Veterans

- Expanded our commitment to help transitioning servicemembers and other veterans lead successful lives after their military service. Implemented a firmwide, comprehensive strategy focused on jobs, homeownership and education.
- Launched the 100,000 Jobs Mission along with other major employers with a goal to collectively hire 100,000 veterans by 2020. In 2011, coalition members hired over 6,600 transitioning military servicemembers and veterans. JPMorgan Chase alone hired more than 3,000.
- Awarded 85 mortgage-free homes to deserving veterans and their families and will award 915 more over the next four years.

Promoting Environmental Sustainability

- Renovated our 1.3 million-square-foot global headquarters building in New York City to earn LEED® Platinum certification from the U.S. Green Building Council, making it the world's largest renovation project to achieve Platinum certification.

- (b) Results for 2008 included an accounting conformity loan loss reserve provision related to the acquisition of Washington Mutual Bank's ("Washington Mutual") banking operations.
- (c) On September 25, 2008, JPMorgan Chase acquired the banking operations of Washington Mutual. The acquisition resulted in negative goodwill, and accordingly, the Firm recorded an extraordinary gain. A preliminary gain of \$1.9 billion was recognized at December 31, 2008. The final total extraordinary gain that resulted from the Washington Mutual transaction was \$2.0 billion.
- (d) The calculation of 2009 earnings per share ("EPS") and net income applicable to common equity includes a one-time, noncash reduction of \$1.1 billion, or \$0.27 per share, resulting from repayment of U.S. Troubled Asset Relief Program ("TARP") preferred capital in the second quarter of 2009. Excluding this reduction, the adjusted ROE and ROA were 7% and 11%, respectively, for 2009. The Firm views the adjusted ROE and ROTCE, both non-GAAP financial measures, as meaningful because they enable the comparability to prior periods.
- (e) Share prices shown for JPMorgan Chase's common stock are from the New York Stock Exchange. JPMorgan Chase's common stock is also listed and traded on the London Stock Exchange and the Tokyo Stock Exchange.
- (f) Effective January 1, 2010, the Firm adopted accounting guidance that amended the accounting for the transfer of financial assets and the consolidation of variable interest entities ("VIEs"). Upon adoption of the guidance, the Firm consolidated its Firm-sponsored credit card securitization trusts, Firm-administered multi-seller conduits and certain other consumer loan securitization entities, primarily mortgage-related, adding \$87.7 billion and \$92.2 billion of assets and liabilities, respectively, and decreasing stockholders' equity and the Tier 1 capital ratio by \$4.5 billion and 34 basis points, respectively. The reduction to stockholders' equity was driven by the establishment of an allowance for loan losses of \$7.5 billion (pretax) primarily related to receivables held in credit card securitization trusts that were consolidated at the adoption date.
- (g) Tier 1 common capital ratio ("Tier 1 common ratio") is Tier 1 common capital ("Tier 1 common") divided by risk-weighted assets. The Firm uses Tier 1 common capital along with the other capital measures to assess and monitor its capital position. For further discussion of Tier 1 common capital ratio, see Regulatory capital on pages 119–122 of this Annual Report.
- (h) Effective January 1, 2011, the long-term portion of advances from Federal Home Loan Banks ("FHLBs") was reclassified from other borrowed funds to long-term debt. Prior periods have been revised to conform with the current presentation.
- (i) Excludes the impact of residential real estate purchased credit-impaired ("PCI") loans. For further discussion, see Allowance for credit losses on pages 155–157 of this Annual Report.

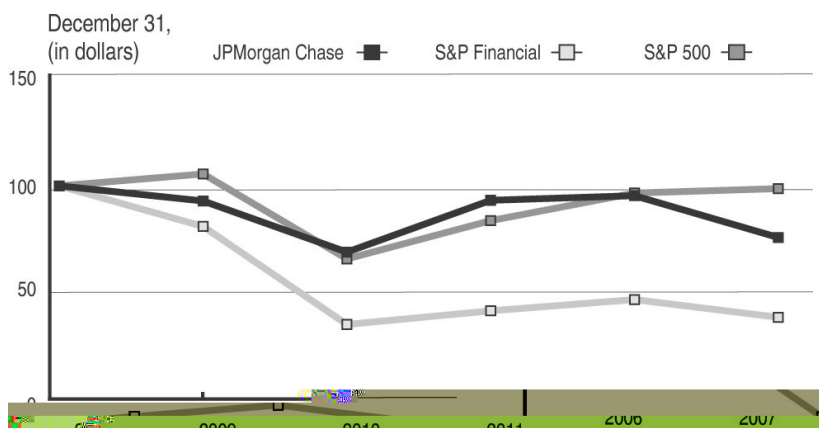
FIVE-YEAR STOCK PERFORMANCE

The following table and graph compare the five-year cumulative total return for JPMorgan Chase & Co. ("JPMorgan Chase" or the "Firm") common stock with the cumulative return of the S&P 500 Index and the S&P Financial Index. The S&P 500 Index is a commonly referenced U.S. equity benchmark consisting of leading

companies from different economic sectors. The S&P Financial Index is an index of 81 financial companies, all of which are components of the S&P 500. The Firm is a component of both industry indices.

The following table and graph assume simultaneous investments of \$100 on December 31, 2006, in JPMorgan Chase common stock and in each of the above S&P indices. The comparison assumes that all dividends are reinvested.

December 31, (in dollars)	2006	2007	2008	2009	2010	2011
JPMorgan Chase	\$ 100.00	\$ 93.07	\$ 69.58	\$ 93.39	\$ 95.50	\$ 76.29
S&P Financial Index	100.00	81.37	36.36	42.62	47.79	39.64
S&P 500 Index	100.00	105.49	66.46	84.05	96.71	98.75



This section of JPMorgan Chase's Annual Report for the year JPMorgan Chase's management and are subject to significant risks and uncertainties. These risks and uncertainties could cause the Firm's actual results to differ materially from those set forth in such forward-looking statements. Certain of such risks and uncertainties are described herein (see Forward-looking Statements on page 175 of this Annual Report) and in JPMorgan Chase's Annual Report on Form 10-K for the year ended December 31, 2011 ("2011 Form 10-K"), in Part I, Item 1A: Risk factors; reference is hereby made to both.

INTRODUCTION

JPMorgan Chase & Co., a financial holding company incorporated under Delaware law in 1968, is a leading global financial services firm and one of the largest banking institutions in the United States of America ("U.S."), with operations worldwide; the Firm has \$2.3 trillion in assets and \$183.6 billion in stockholders' equity as of December 31, 2011. The Firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing, asset management and private equity. Under the J.P. Morgan and Chase brands, the Firm serves millions of customers in the U.S. and many of the world's most prominent corporate, institutional and government clients.

JPMorgan Chase's principal bank subsidiaries are JPMorgan Chase Bank, National Association ("JPMorgan Chase Bank, N.A."), a national bank with U.S. branches in 23 states, and Chase Bank USA, National Association ("Chase Bank USA, N.A."), a national bank that is the Firm's credit card-issuing bank. JPMorgan Chase's principal nonbank subsidiary is J.P. Morgan Securities LLC ("JPMorgan Securities"), the Firm's U.S. investment banking firm. The bank and nonbank subsidiaries of JPMorgan Chase operate nationally as well as through overseas branches and subsidiaries, representative offices and subsidiary foreign banks. One of the Firm's principal operating subsidiaries in the United Kingdom ("U.K.") is J.P. Morgan Securities Ltd., a subsidiary of JPMorgan Chase Bank, N.A.

JPMorgan Chase's activities are organized, for management reporting purposes, into six business segments, as well as Corporate/Private Equity. The Firm's wholesale businesses comprise the Investment Bank, Commercial Banking, Treasury & Securities Services and Asset Management segments. The Firm's consumer businesses comprise the Retail Financial Services and Card Services & Auto segments. A description of the Firm's business segments, and the products and services they provide to their respective client bases, follows.

Investment Bank

J.P. Morgan is one of the world's leading investment banks, with deep client relationships and broad product capabilities. The clients of the Investment Bank ("IB") are corporations, financial institutions, governments and institutional investors. The Firm offers a full range of investment banking products and services in all major capital markets, including advising on corporate strategy and structure, capital-raising in equity and debt markets, sophisticated risk management, market-making in cash securities and derivative instruments, prime brokerage, and research.

Retail Financial Services

Retail Financial Services ("RFS") serves consumers and businesses through personal service at bank branches and through ATMs, online banking and telephone banking. RFS is organized into Consumer & Business Banking and Mortgage Banking (including Mortgage Production and Servicing, and Real Estate Portfolios). Consumer & Business Banking includes branch banking and business banking activities. Mortgage Production and Servicing includes mortgage origination and servicing activities. Real Estate Portfolios comprises residential mortgages and home equity loans, including the PCI portfolio acquired in the Washington Mutual transaction. Customers can use more than 5,500 bank branches (third largest nationally) and more than 17,200 ATMs (second largest nationally), as well as online and mobile banking around the clock. More than 33,500 branch salespeople assist customers with checking and savings accounts, mortgages, home equity and business loans, and investments across the 23-state footprint from New York and Florida to California. As one of the largest mortgage originators in the U.S., Chase helps customers buy or refinance homes resulting in approximately \$150 billion of mortgage originations annually. Chase also services more than 8 million mortgages and home equity loans.

Card Services & Auto

Card Services & Auto ("Card") is one of the nation's largest credit card issuers, with over \$132 billion in credit card loans. Customers have over 65 million open credit card accounts (excluding the commercial card portfolio), and used Chase credit cards to meet over \$343 billion of their spending needs in 2011. Through its Merchant Services business, Chase Paymentech Solutions, Card is a global leader in payment processing and merchant acquiring. Consumers also can obtain loans through more than 17,200 auto dealerships and 2,000 schools and universities nationwide.

Commercial Banking

Commercial Banking ("CB") delivers extensive industry knowledge, local expertise and dedicated service to more than 24,000 clients nationally, including corporations, municipalities, financial institutions and not-for-profit entities with annual revenue generally ranging from \$10 million to \$2 billion, and nearly 35,000 real estate investors/owners. CB partners with the Firm's other businesses to provide comprehensive solutions, including lending, treasury services, investment banking and asset management, to meet its clients' domestic and international financial needs.

Treasury & Securities Services

Treasury & Securities Services ("TSS") is a global leader in transaction, investment and information services. TSS is one of the world's largest cash management providers and a leading global custodian. Treasury Services ("TS") provides cash management, trade, wholesale card and liquidity products and services to small- and mid-sized companies, multinational corporations, financial institutions and government entities. TS partners with IB, CB, RFS and Asset Management businesses to serve clients firmwide. Certain TS revenue is included in other segments' results. Worldwide Securities Services holds, values, clears and services securities, cash and alternative investments for investors and broker-dealers, and manages depository receipt programs globally.

Asset Management

Asset Management ("AM"), with assets under supervision of \$1.9 trillion, is a global leader in investment and wealth management. AM clients include institutions, retail investors and high-net-worth individuals in every major market throughout the world. AM offers global investment management in equities, fixed income, real estate, hedge funds, private equity and liquidity products, including money-market instruments and bank deposits. AM also provides trust and estate, banking and brokerage services to high-net-worth clients, and retirement services for corporations and individuals. The majority of AM's client assets are in actively managed portfolios.

EXECUTIVE OVERVIEW

This executive overview of the MD&A highlights selected information and may not contain all of the information that is important to readers of this Annual Report. For a complete description of events, trends and uncertainties, as well as the capital, liquidity, credit and market risks, and the critical accounting estimates affecting the Firm and its various lines of business, this Annual Report should be read in its entirety.

Economic environment

The global economy lost some momentum during 2011 in the face of several new threats, some transitory and some more deeply entrenched. In the first half of the year, the earthquake and tsunami in Japan represented a significant setback to that country's important economy and probably disrupted activity elsewhere in the world as well, particularly in the global motor vehicle sector. Later in the year, severe floods in Thailand also disrupted motor vehicle supply chains. Furthermore, a sharp rise in oil prices in the spring in the wake of political unrest in the Middle East slowed consumer demand.

Although many of these shocks eased later in the year, Europe's financial crisis posed a new threat. Concerns about sovereign debt in Greece and other Eurozone countries, which raised doubts in the investor community about the viability of the European monetary union, as well as the sovereign debt exposures of the European banking system, were a source of stress in the global financial markets during the second half of 2011. In December 2011, the European Central Bank ("ECB") announced measures to support bank lending and money market activity, offering 36-month, 1 percent loans through two longer-term refinancing operations, known as LTROs. These programs replaced a 12-month lending facility established by the ECB in October 2011 and also allowed banks to use a wider variety of assets as collateral for the loans. The ECB's actions were expected to ease near-term concerns about European bank funding and liquidity.

Despite these headwinds, there were a number of promising developments in the U.S. during 2011. The credit environment improved as consumer and wholesale delinquencies decreased and lending for a broad range of purposes accelerated. Housing prices continued to be largely unchanged and rose in the non-distressed sector, while home builders continued to make good progress working off the excess housing inventory that was built in the last decade. Despite the turmoil in the summer months associated with the debt ceiling crisis and a worsening of the crisis in Europe, the U.S. job market continued to improve, with layoffs easing, employment expanding steadily, and unemployment falling. At the same time the financial health of the business sector, which was already strong, continued to improve. Reflecting these favorable trends, the equity market recovered from the late summer drop.

The Board of Governors of the Federal Reserve System (the "Federal Reserve") took several actions during 2011 to support a stronger economic recovery and to help support conditions in mortgage markets. These actions included extending the average maturity of its holdings of securities, reinvesting principal payments from its holdings of agency debt and U.S. government agency mortgage-backed securities into other agency mortgage-backed securities and maintaining its existing policy of rolling over maturing U.S. Department of the Treasury ("U.S. Treasury") securities at auction. The Federal Reserve maintained the target range for the federal funds rate at zero to one-quarter percent and, in January 2012, provided specific guidance regarding its prediction about policy rates, stating that economic conditions were likely to warrant exceptionally low levels for the federal funds rate at least through late 2014. Also, the Federal Reserve reactivated currency swap lines with the ECB in response to pressures in interbank term funding markets.

Financial performance of JPMorgan Chase

Year ended December 31,

(in millions, except per share data and ratios)

	2011	2010	Change
Selected income statement data			
Total net revenue	\$ 97,234	\$ 102,694	(5)%
Total noninterest expense	62,911	61,196	3
Pre-provision profit	34,323	41,498	(17)
Provision for credit losses	7,574	16,639	(54)
Net income	18,976	17,370	9
Diluted earnings per share	4.48	3.96	13
Return on common equity	11%	10%	
Capital ratios			
Tier 1 capital	12.3	12.1	
Tier 1 common	10.1	9.8	

Business overview

JPMorgan Chase reported full-year 2011 record net income of \$19.0 billion, or \$4.48 per share, on net revenue of \$97.2 billion. Net income increased by \$1.6 billion, or 9%, compared with net income of \$17.4 billion, or \$3.96 per share, in 2010. ROE for the year was 11%, compared with 10% for the prior year.

The increase in net income in 2011 was driven by a lower provision for credit losses, predominantly offset by lower net revenue and higher noninterest expense. The reduction in the provision for credit losses reflected continued improvement in the consumer portfolios. The decline in net revenue from 2010 was driven by lower net interest income, securities gains, mortgage fees and related income, and principal transactions revenue, partially offset by higher asset management, administration and commissions revenue and higher other income. The increase in noninterest expense was driven largely by higher compensation expense, reflecting increased headcount.

During 2011, the credit quality of the Firm's wholesale credit portfolio improved. The delinquency trends in the consumer business modestly improved, though the rate of improvement seen earlier in 2011 slowed somewhat in the latter half of the year. Mortgage net charge-offs and delinquencies modestly improved, but both remained at elevated levels. These positive consumer credit trends resulted in reductions in the allowance for loan losses in Card Services & Auto and in Retail Financial Services (excluding purchased credit-impaired loans). The allowance for loan losses associated with the Washington Mutual purchased credit-impaired loan portfolio in Retail Financial Services increased, reflecting higher than expected loss frequency relative to modeled lifetime loss estimates. Firmwide, net charge-offs were \$12.2 billion for the year, down \$11.4 billion, or 48%, from 2010, and nonperforming assets at year-end were \$11.0 billion, down \$5.5 billion, or 33%. Total firmwide credit reserves were \$28.3 billion, resulting in a loan loss coverage ratio of 3.35% of total loans, excluding the purchased credit-impaired portfolio.

Net income performance varied among JPMorgan Chase's lines of business, but underlying metrics in each business showed positive trends. The second half of 2011 reflected a challenging investment banking and capital markets environment which contributed to lower revenue for the year in the Investment Bank (excluding debit valuation adjustment ("DVA") gains). However, the Investment Bank maintained its #1 ranking in Global Investment Banking Fees for the year. Consumer & Business Banking within Retail Financial Services opened 260 new branches and increased deposits by 8% in 2011. In the Card business, credit card sales volume (excluding Commercial Card) was up 10% for the year. Treasury & Securities Services reported record average liability balances, up 28% for 2011, and a 73% increase in trade loans. Commercial Banking also reported record average liability balances, up 26% for the year, and record revenue and net income for the year. The fourth quarter of 2011 also marked CB's sixth consecutive quarter of loan growth, including a 17% increase in middle-market loans over the prior year end. Asset Management reported record revenue for the year and achieved eleven consecutive quarters of positive long-term flows into assets under management.

JPMorgan Chase ended the year with a Basel I Tier 1 common ratio of 10.1%, compared with 9.8% at year-end 2010. This strong capital position enabled the Firm to repurchase \$8.95 billion of common stock and warrants during 2011. The Firm estimated that its Basel III Tier 1 common ratio was approximately 7.9% at December 31, 2011. Total deposits increased to \$1.1 trillion, up 21% from the prior year. Total stockholders' equity at December 31, 2011, was \$183.6 billion. The Basel I and III Tier 1 common ratios are non-GAAP financial measures, which the Firm uses along with the other capital measures, to assess and monitor its capital position. For further

discussion of the Tier 1 common capital ratios, see Regulatory capital on pages 119–123 of this Annual Report.

During 2011, the Firm worked to help its individual customers, corporate clients and the communities in which it does business. The Firm provided credit to and raised capital of more than \$1.8 trillion for its clients during 2011, up 18% from 2010; this included \$17 billion lent to small businesses, up 52%, and \$68 billion to more than 1,200 not-for-profit and government entities, including states, municipalities, hospitals and universities. The Firm also originated more than 765,000 mortgages, and provided credit cards to approximately 8.5 million people. The Firm remains committed to helping homeowners and preventing foreclosures. Since the beginning of 2009, the Firm has offered more than 1.2 million mortgage modifications, of which approximately 452,000 have achieved permanent modification as of December 31, 2011.

The discussion that follows highlights the performance of each business segment compared with the prior year and presents results on a managed basis. Managed basis starts with the reported results under the accounting principles generally accepted in the United States of America ("U.S. GAAP") and, for each line of business and the Firm as a whole, includes certain reclassifications to present total net revenue on a tax-equivalent basis. Prior to January 1, 2010, the Firm's managed-basis presentation also included certain reclassification adjustments that assumed credit card loans securitized by Card remained on the Consolidated Balance Sheets. For more information about managed basis, as well as other non-GAAP financial measures used by management to evaluate the performance of each line of business, see pages 76–78 of this Annual Report.

Investment Banknet income increased modestly from the prior year as lower noninterest expense was predominantly offset by a lower benefit from the provision for credit losses. Net revenue for the year was approximately flat compared with 2010 and included a \$1.4 billion gain from DVA on certain structured and derivative liabilities, compared with a DVA gain of \$509 million in 2010. In 2011, this was partially offset by a \$769 million loss, net of hedges, from credit valuation adjustments ("CVA") on derivative assets within Credit Portfolio, due to the widening of credit spreads for the Firm's counterparties. In 2010, net revenue was partially offset by a \$403 million loss, net of hedges, from CVA. Fixed Income and Equity Markets revenue increased compared with the prior year partially due to the DVA gain. In addition, results in Fixed Income and Equity Markets reflected solid client revenue across most products. Investment banking fees decreased for the year as the impact of lower volumes in the second half of 2011 more than offset the strong level of fees reported in the first half of the year. The decrease in noninterest expense from the prior-year level was largely driven by lower compensation expense and the absence of

Management's discussion and analysis

the U.K. Bank Payroll Tax. Return on equity for the year was 17% on \$40.0 billion of average allocated capital.

Retail Financial Services net income decreased modestly compared with the prior year driven by higher noninterest expense and lower net revenue, predominantly offset by a lower provision for credit losses. The decline in net revenue was driven by lower mortgage fees and related income and lower net interest income, which reflected the impact of lower loan balances due to portfolio runoff, and narrower loan spreads. Higher investment sales revenue and deposit-related fees partially offset the decline in revenue. A modest improvement in delinquency trends and a decline in net charge-offs compared with 2010 resulted in the lower provision for credit losses; however, the provision continued to reflect elevated losses in the mortgage and home equity portfolios. Additionally, the provision for credit losses in 2011 reflected a lower addition to the allowance for loan losses for the purchased credit-impaired portfolio compared with the prior year. The increase in noninterest expense from the prior year was driven by investment in sales force and new branch builds as well as elevated foreclosure- and default-related costs, including \$1.7 billion of expense for fees and assessments, as well as other costs of foreclosure-related matters. Return on equity for the year was 7% on \$25.0 billion of average allocated capital.

Card Services & Auto net income increased in 2011 compared with the prior year driven by a lower provision for credit losses partially offset by lower net revenue and higher noninterest expense. The decrease in net revenue was driven by a decline in net interest income, reflecting lower average loan balances, the impact of legislative changes and a decreased level of fees. These decreases were largely offset by lower revenue reversals associated with lower net charge-offs. Credit card sales volume, excluding the Commercial Card portfolio, was up 10% from 2010. The lower provision for credit losses reflected lower net charge-offs partially offset by a lower reduction in the allowance for loan losses. The increase in noninterest expense was due to higher marketing expense and the inclusion of the Commercial Card business. Return on equity for the year was 28% on \$16.0 billion of average allocated capital.

Commercial Banking reported record net revenue and net income for the second consecutive year. The increase in revenue was driven by higher net interest income resulting from growth in liability and loan balances, partially offset by spread compression on liability products. Average liability balances reached a record level in 2011, up 26% from 2010. End-of-period loan balances increased in each quarter of 2011 and were up 13% from year-end 2010. The provision for credit losses declined compared with the prior year. Noninterest expense increased from the level in 2010, primarily reflecting higher headcount-related expense. Return on equity for the year was 30% on \$8.0 billion of average allocated capital.

Treasury & Securities Services net income increased from the prior year, driven by higher net revenue reflecting record deposit balances and a benefit from the Global Corporate Bank ("GCB") credit allocation, predominantly offset by higher noninterest expense. Worldwide Securities Services net revenue increased compared to 2010, driven by higher net interest income due to higher deposit balances and net inflows of assets under custody. Assets under custody of \$16.9 trillion were up 5% from 2010. Treasury Services net revenue increased, driven by higher deposit balances and higher trade loan volumes, partially offset by the transfer of the Commercial Card business to Card in the first quarter of 2011. Higher noninterest expense was mainly driven by continued expansion into new markets and expenses related to exiting unprofitable business, partially offset by the transfer of the Commercial Card business to Card. Return on equity for the year was 17% on \$7.0 billion of average allocated capital.

Asset Management net income decreased, reflecting higher noninterest expense, largely offset by record net revenue. The growth in net revenue was due to net inflows to products with higher margins, higher deposit and loan balances, and the effect of higher average market levels. This growth was partially offset by lower performance fees, narrower deposit spreads and lower loan-related revenue. Assets under supervision of \$1.9 trillion increased 4% from the prior year, and assets under management of \$1.3 trillion were up 3%. Both increases were due to net inflows to long-term and liquidity products, partially offset by the effect of lower market levels. In addition, deposit and custody inflows contributed to the increase in assets under supervision. The increase in noninterest expense was due to higher headcount-related expense and non-client-related litigation, partially offset by lower performance-based compensation. Return on equity for the year was 25% on \$6.5 billion of average allocated capital.

Corporate/Private Equity net income decreased in 2011 as income in both Private Equity and Corporate declined. Lower private equity gains were primarily the result of net write-downs on privately-held investments and the absence of prior-year gains from sales in the Private Equity portfolio. In Corporate, lower net interest income was primarily driven by repositioning of the investment securities portfolio and lower funding benefits from financing portfolio positions. Lower securities gains also drove the decline in net income. In 2011, noninterest expense included \$3.2 billion of litigation expense, predominantly for mortgage-related matters, compared with \$5.7 billion of litigation expense in 2010.

2012 Business outlook

The following forward-looking statements are based on the current beliefs and expectations of JPMorgan Chase's management and are subject to significant risks and uncertainties. These risks and uncertainties could cause the Firm's actual results to differ materially from those set forth in such forward-looking statements. See Forward-Looking

Statements on page 175 and Risk Factors section of the 2011 Form 10-K.

JPMorgan Chase's outlook for the full-year 2012 should be viewed against the backdrop of the global and U.S. economies, financial markets activity, the geopolitical environment, the competitive environment, client activity levels, and regulatory and legislative developments in the U.S. and other countries where the Firm does business. Each of these linked factors will affect the performance of the Firm and its lines of business.

In the Consumer & Business Banking business within RFS, the Firm estimates that, given the current low interest rate environment, spread compression will likely negatively affect 2012 net income by approximately \$400 million. In addition, the effect of the Durbin Amendment will likely reduce annualized net income by approximately \$600 million.

In the Mortgage Production and Servicing business within RFS, revenue in 2012 could be negatively affected by continued elevated levels of repurchases of mortgages previously sold, predominantly to U.S. government-sponsored entities ("GSEs"). Management estimates that realized mortgage repurchase losses could be approximately \$350 million per quarter in 2012. Also for Mortgage Production and Servicing, management expects the business to continue to incur elevated default management and foreclosure-related costs including additional costs associated with the Firm's mortgage servicing processes, particularly its loan modification and foreclosure procedures. (See Enhancements to Mortgage Servicing on pages 152-153 and Note 17 on pages 267-271 of this Annual Report.)

For the Real Estate Portfolios within RFS, management believes that quarterly net charge-offs could be approximately \$900 million. Given management's current estimate of portfolio runoff levels, the existing residential real estate portfolio is expected to decline by approximately 10% to 15% in 2012 from year-end 2011 levels. This reduction in the residential real estate portfolio is expected to reduce net interest income by approximately \$500 million in 2012. However, over time, the reduction in net interest income is expected to be more than offset by an improvement in credit costs and lower expenses. In addition, as the portfolio continues to run off, management anticipates that approximately \$1 billion of capital may become available for redeployment each year, subject to the capital requirements associated with the remaining portfolio.

In Card, the net charge-off rate for the combined Chase and Washington Mutual credit card portfolios (excluding Commercial Card) could increase in the first quarter of 2012 to approximately 4.50% from the 4.33% reported in the fourth quarter, reflecting normal seasonality.

The currently anticipated results of RFS and Card described above could be adversely affected by further declines in

U.S. housing prices or increases in the unemployment rate. Given ongoing weak economic conditions, combined with a high level of uncertainty concerning the residential real estate markets, management continues to closely monitor the portfolios in these businesses.

In IB, TSS, CB and AM, revenue will be affected by market levels, volumes and volatility, which will influence client flows and assets under management, supervision and custody. CB and TSS will continue to experience low net interest margins as long as market interest rates remain low. In addition, the wholesale credit environment will influence levels of charge-offs, repayments and provision for credit losses for IB, CB, TSS and AM.

In Private Equity, within the Corporate/Private Equity segment, earnings will likely continue to be volatile and be influenced by capital markets activity, market levels, the performance of the broader economy and investment-specific issues. Corporate's net interest income levels will generally trend with the size and duration of the investment securities portfolio. Corporate quarterly net income (excluding Private Equity results, significant nonrecurring items and litigation expense) could be approximately \$200 million, though these results will depend on the decisions that the Firm makes over the course of the year with respect to repositioning of the investment securities portfolio.

The Firm faces a variety of litigation, including in its various roles as issuer and/or underwriter in mortgage-backed securities ("MBS") offerings, primarily related to offerings involving third parties other than the GSEs. It is possible that these matters will take a number of years to resolve; their ultimate resolution is inherently uncertain and reserves for such litigation matters may need to be increased in the future.

Management and the Firm's Board of Directors continually evaluate ways to deploy the Firm's strong capital base in order to enhance shareholder value. Such alternatives could include the repurchase of common stock and warrants, increasing the common stock dividend and pursuing alternative investment opportunities. Certain of such capital actions, such as increasing dividends, implementing common equity repurchase programs, or redeeming or repurchasing capital instruments, are subject to the Federal Reserve's Comprehensive Capital Analysis and Review ("CCAR") process. The Federal Reserve requires the Firm to submit a capital plan on an annual basis. The Firm submitted its 2012 capital plan on January 9, 2012. The Federal Reserve has indicated that it expects to provide notification of either its objection or non-objection to the Firm's capital plan by March 15, 2012.

Regulatory developments

JPMorgan Chase is subject to regulation under state and federal laws in the U.S., as well as the applicable laws of each of the various other jurisdictions outside the U.S. in which the Firm does business. The Firm is currently

Management's discussion and analysis

experiencing a period of unprecedented change in regulation and such changes could have a significant impact on how the Firm conducts business. The Firm continues to work diligently in assessing and understanding the implications of the regulatory changes it is facing, and is devoting substantial resources to implementing all the new rules and regulations while meeting the needs and expectations of its clients. While the Firm has made a preliminary assessment of the likely impact of certain of the anticipated changes, the Firm cannot, given the current status of the regulatory developments, quantify the possible effects on its business and operations of all of the significant changes that are currently underway. For further discussion of regulatory developments, see Supervision and regulation on pages 1-7 and Risk factors on pages 7-17 of the 2011 Form 10-K.

Subsequent events

Global settlement on servicing and origination of mortgages

On February 9, 2012, the Firm announced that it agreed to a settlement in principle (the "global settlement") with a number of federal and state government agencies, relating to the servicing and origination of mortgages. The global settlement, which is subject to the execution of a definitive agreement and court approval, calls for the Firm to, among other things: (i) make cash payments of approximately \$1.1 billion; (ii) provide approximately \$500 million of refinancing relief to certain "underwater" borrowers whose loans are owned by the Firm; and (iii) provide approximately \$3.7 billion of additional relief for certain borrowers, including reductions of principal, payments to

assist with short sales, deficiency balance waivers on past foreclosures and short sales, and forbearance assistance for unemployed homeowners. While the Firm expects to incur additional operating costs to comply with portions of the global settlement, the Firm's prior period results of operations have reflected the estimated costs of the global settlement. Accordingly, the Firm expects that the financial impact of the global settlement on the Firm's financial condition and results of operations for the first quarter of 2012 and future periods will not be material. For further information on this settlement, see "Subsequent events" in Note 2, and "Mortgage Foreclosure Investigations and Litigation" in Note 31 on pages 183-184 and 295-296, respectively, of this Annual Report.

Washington Mutual, Inc. bankruptcy plan confirmation
On February 17, 2012, a bankruptcy court confirmed the joint plan containing the global settlement agreement resolving numerous disputes among Washington Mutual, Inc. ("WMI"), JPMorgan Chase and the Federal Deposit Insurance Corporation ("FDIC") as well as significant creditor groups (the "WaMu Global Settlement"). Pursuant to this agreement, the Firm expects to recognize additional assets, including certain pension-related assets, as well as tax refunds, in future periods as the settlement is executed and various state and federal tax matters are resolved. For additional information related to the WaMu Global Settlement, see "Subsequent events" in Note 2, and "Washington Mutual Litigations" in Note 31 on page 183-184 and 298, respectively, of this Annual Report.

CONSOLIDATED RESULTS OF OPERATIONS

The following section provides a comparative discussion of JPMorgan Chase's Consolidated Results of Operations on a reported basis for the three-year period ended December 31, 2011. Factors that relate primarily to a single business segment are discussed in more detail within that business segment. For a discussion of the Critical Accounting Estimates Used by the Firm that affect the Consolidated Results of Operations, see pages 168–172 of this Annual Report.

Revenue

Year ended December 31,

(in millions)	2011	2010	2009
Investment banking fees	\$ 5,911	\$ 6,190	\$ 7,087
Principal transactions	10,005	10,894	9,796
Lending- and deposit-related fees	6,458	6,340	7,045
Asset management, administration and commissions	14,094	13,499	12,540
Securities gains	1,593	2,965	1,110
Mortgage fees and related income	2,721	3,870	3,678
Credit card income	6,158	5,891	7,110
Other income	2,605	2,044	916
Noninterest revenue	49,545	51,693	49,282
Net interest income	47,689	51,001	51,152
Total net revenue	\$ 97,234	\$ 102,694	\$ 100,434

2011 compared with 2010

Total net revenue for 2011 was \$97.2 billion, a decrease of \$5.5 billion, or 5%, from 2010. Results for 2011 were driven by lower net interest income in several businesses, lower securities gains in Corporate/Private Equity, lower mortgage fees and related income in RFS, and lower principal transactions revenue in Corporate/Private Equity. These declines were partially offset by higher asset management fees, largely in AM.

Investment banking fees decreased from 2010, predominantly due to declines in equity and debt underwriting fees. The impact from lower industry-wide volumes in the second half of 2011 more than offset the Firm's record level of debt underwriting fees in the first six months of the year. Advisory fees increased for the year, reflecting higher industry-wide completed M&A volumes relative to the 2010 level. For additional information on investment banking fees, which are primarily recorded in IB, see IB segment results on pages 81–84, and Note 7 on pages 211–212 of this Annual Report.

Principal transactions revenue, which consists of revenue from the Firm's market-making and private equity investing activities, decreased compared with 2010. This was driven by lower trading revenue and lower private equity gains. Trading revenue included a \$1.4 billion gain from DVA on certain structured notes and derivative liabilities, resulting from the widening of the Firm's credit spreads, partially

offset by a \$769 million loss, net of hedges, from CVA on derivative assets within Credit Portfolio in IB, due to the widening of credit spreads of the Firm's counterparties. The prior year included a \$509 million gain from DVA, partially offset by a \$403 million loss, net of hedges, from CVA. Excluding DVA and CVA, lower trading revenue reflected the impact of the second half of 2011's challenging market conditions on Corporate and IB. Lower private equity gains were primarily due to net write-downs on privately-held investments and the absence of prior-year gains from sales in the Private Equity portfolio. For additional information on principal transactions revenue, see IB and Corporate/Private Equity segment results on pages 81–84 and 107–108, respectively, and Note 7 on pages 211–212 of this Annual Report.

Lending- and deposit-related fees increased modestly in 2011 compared with the prior year. The increase was primarily driven by the introduction in the first quarter of 2011 of a new checking account product offering in RFS, and the subsequent conversion of certain existing accounts into the new product. The increase was offset partly by the impact of regulatory and policy changes affecting nonsufficient fund/overdraft fees in RFS. For additional information on lending- and deposit-related fees, which are mostly recorded in RFS, CB, TSS and IB, see RFS on pages 85–93, CB on pages 98–100, TSS on pages 101–103 and IB on pages 81–84 of this Annual Report.

Asset management, administration and commissions revenue increased from 2010, reflecting higher asset management fees in AM and RFS, driven by net inflows to products with higher margins and the effect of higher market levels; and higher administration fees in TSS, reflecting net inflows of assets under custody. For additional information on these fees and commissions, see the segment discussions for AM on pages 104–106, RFS on pages 85–93 and TSS on pages 101–103, and Note 7 on pages 211–212 of this Annual Report.

Securities gains decreased compared with the 2010 level, primarily due to the repositioning of the investment securities portfolio in response to changes in the current market environment and to rebalancing exposures. For additional information on securities gains, which are mostly recorded in the Firm's Corporate/Private Equity segment, see the Corporate/Private Equity segment discussion on pages 107–108, and Note 12 on pages 225–230 of this Annual Report.

Mortgage fees and related income decreased in 2011 compared with 2010, reflecting a MSR risk management loss of \$1.6 billion for 2011, compared with income of \$1.1 billion for 2010, largely offset by lower repurchase losses in 2011. The \$1.6 billion loss was driven by a \$7.1 billion loss due to a decrease in the fair value of the mortgage servicing rights ("MSRs") asset, which was predominantly offset by a \$5.6 billion gain on the derivatives used to hedge the MSR asset. For additional information on

Management's discussion and analysis

mortgage fees and related income, which is recorded primarily in RFS, see RFS's Mortgage Production and Servicing discussion on pages 89–91, and Note 17 on pages 267–271 of this Annual Report. For additional information on repurchase losses, see the Mortgage repurchase liability discussion on pages 115–118 and Note 29 on pages 283–289 of this Annual Report.

Credit card income increased during 2011, largely reflecting higher net interchange income associated with higher customer transaction volume on credit and debit cards, as well as lower partner revenue-sharing due to the impact of the Kohl's portfolio sale. These increases were partially offset by lower revenue from fee-based products, as well as the impact of the Durbin Amendment. For additional information on credit card income, see the Card and RFS segment results on pages 94–97, and pages 85–93, respectively, of this Annual Report.

Other income increased in 2011, driven by valuation adjustments on certain assets and incremental revenue from recent acquisitions in IB, and higher auto operating lease income in Card, resulting from growth in lease volume. Also contributing to the increase was a gain on the sale of an investment in AM.

Net interest income decreased in 2011 compared with the prior year, driven by lower average loan balances and yields in Card and RFS, reflecting the expected runoff of credit card balances and residential real estate loans; lower fees on credit card receivables, reflecting the impact of legislative changes; higher average interest-bearing deposit balances and related yields; and lower yields on securities, reflecting portfolio repositioning in anticipation of an increasing interest rate environment. The decrease was offset partially by lower revenue reversals associated with lower credit card charge-offs, and higher trading asset balances. The Firm's average interest-earning assets were \$1.8 trillion for the 2011 full year, and the net yield on those assets, on a fully taxable-equivalent ("FTE") basis, was 2.74%, a decrease of 32 basis points from 2010. For further information on the impact of the legislative changes on the Consolidated Statements of Income, see Card discussion on credit card legislation on page 94 of this Annual Report.

2010 compared with 2009

Total net revenue for 2010 was \$102.7 billion, up by \$2.3 billion, or 2%, from 2009. Results for 2010 were driven by a higher level of securities gains and private equity gains in Corporate/Private Equity, higher asset management fees in AM and administration fees in TSS, and higher other income in several businesses, partially offset by lower credit card income.

Investment banking fees decreased from 2009 due to lower equity underwriting and advisory fees, partially offset by higher debt underwriting fees. Competitive markets combined with flat industry-wide equity underwriting and completed M&A volumes, resulted in lower equity underwriting and advisory fees; while strong industry-wide

loan syndication and high-yield bond volumes drove record debt underwriting fees in IB. For additional information on investment banking fees, which are primarily recorded in IB, see IB segment results on pages 81–84, and Note 7 on pages 211–212 of this Annual Report.

Principal transactions revenue increased compared with 2009. This was driven by the Private Equity business, which had significant private equity gains in 2010, compared with a small loss in 2009, reflecting improvements in market conditions. Trading revenue decreased, reflecting lower results in Corporate, offset by higher revenue in IB primarily reflecting DVA gains. For additional information on principal transactions revenue, see IB and Corporate/Private Equity segment results on pages 81–84 and 107–108, respectively, and Note 7 on pages 211–212 of this Annual Report.

Lending- and deposit-related fees decreased in 2010 from 2009 levels, reflecting lower deposit-related fees in RFS associated, in part, with newly-enacted legislation related to non-sufficient funds and overdraft fees; this was partially offset by higher lending-related service fees in IB, primarily from growth in business volume, and in CB, primarily from higher commitment and letter-of-credit fees. For additional information on lending- and deposit-related fees, which are mostly recorded in IB, RFS, CB and TSS, see segment results for IB on pages 81–84, RFS on pages 85–93, CB on pages 98–100 and TSS on pages 101–103 of this Annual Report.

Asset management, administration and commissions revenue increased from 2009. The increase largely reflected higher asset management fees in AM, driven by the effect of higher market levels, net inflows to products with higher margins and higher performance fees; and higher administration fees in TSS, reflecting the effects of higher market levels and net inflows of assets under custody. This increase was partially offset by lower brokerage commissions in IB, as a result of lower market volumes. For additional information on these fees and commissions, see the segment discussions for AM on pages 104–106 and TSS on pages 101–103, and Note 7 on pages 211–212 of this Annual Report.

Securities gains were significantly higher in 2010 compared with 2009, resulting primarily from the repositioning of the portfolio in response to changes in the interest rate environment and to rebalance exposure. For additional information on securities gains, which are mostly recorded in the Firm's Corporate segment, see the Corporate/Private Equity segment discussion on pages 107–108, and Note 12 on pages 225–230 of this Annual Report.

Mortgage fees and related income increased in 2010 compared with 2009, driven by higher mortgage production revenue, reflecting increased mortgage origination volumes in RFS and AM, and wider margins, particularly in RFS. This increase was largely offset by higher repurchase losses in RFS (recorded as contra-revenue), which were attributable to higher estimated losses related to repurchase demands, predominantly from

GSEs. For additional information on mortgage fees and related income, which is recorded primarily in RFS, see RFS's Mortgage Production and Servicing discussion on pages 89–91, and Note 17 on pages 267–271 of this Annual Report. For additional information on repurchase losses, see the mortgage repurchase liability discussion on pages 115–118 and Note 30 on page 289 of this Annual Report.

Credit card income decreased during 2010, predominantly due to the impact of the accounting guidance related to VIEs, effective January 1, 2010, that required the Firm to consolidate the assets and liabilities of its Firm-sponsored credit card securitization trusts. Adoption of this guidance resulted in the elimination of all servicing fees received from Firm-sponsored credit card securitization trusts, which was offset by related increases in net interest income and provision for credit losses. Lower income from other fee-based products also contributed to the decrease in credit card income. Excluding the impact of the adoption of the accounting guidance, credit card income increased in 2010, reflecting higher customer charge volume on credit and debit cards. For a more detailed discussion of the impact of the adoption of the accounting guidance on the Consolidated Statements of Income, see Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 76–78 of this Annual Report. For additional information on credit card income, see the Card and RFS segment results on pages 94–97, and pages 85–93, respectively, of this Annual Report.

Other income increased in 2010, largely due to the write-down of securitization interests during 2009 and higher auto operating lease income in Card.

Net interest income was relatively flat in 2010 compared with 2009. The effect of lower loan balances was predominantly offset by the effect of the adoption of the new accounting guidance related to VIEs (which increased net interest income by approximately \$5.8 billion in 2010). Excluding the impact of the adoption of the new accounting guidance, net interest income decreased, driven by lower average loan balances, primarily in Card, RFS and IB, reflecting the continued runoff of the credit card balances and residential real estate loans, and net repayments and loan sales; lower yields and fees on credit card receivables, reflecting the impact of legislative changes; and lower yields on securities in Corporate resulting from investment portfolio repositioning. The Firm's average interest-earning assets were \$1.7 trillion in 2010, and the net yield on those assets, on a FTE basis, was 3.06%, a decrease of 6 basis points from 2009. For a more detailed discussion of the impact of the adoption of the new accounting guidance related to VIEs on the Consolidated Statements of Income, see Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 76–78 of this Annual Report. For further information on the impact of the legislative changes on the Consolidated Statements of Income, see Card discussion on credit card legislation on page 94 of this Annual Report.

Provision for credit losses

Year ended December 31,

(in millions)	2011	2010	2009
Wholesale	\$ (23)	\$ (850)	\$ 3,974
Consumer, excluding credit card	4,672	9,452	16,022
Credit card	2,925	8,037	12,019
Total consumer	7,597	17,489	28,041
Total provision for credit losses	\$ 7,574	\$ 16,639	\$ 32,015

2011 compared with 2010

The provision for credit losses declined by \$9.1 billion compared with 2010. The consumer, excluding credit card, provision was down, reflecting improved delinquency and charge-off trends across most portfolios, partially offset by an increase of \$770 million, reflecting additional impairment of the Washington Mutual PCI loans portfolio. The credit card provision was down, driven primarily by improved delinquency trends and net credit losses. The benefit from the wholesale provision was lower in 2011 than in 2010, primarily reflecting loan growth and other portfolio activity. For a more detailed discussion of the loan portfolio and the allowance for credit losses, see the segment discussions for RFS on pages 85–93, Card on pages 94–97, IB on pages 81–84 and CB on pages 98–100, and the Allowance for credit losses section on pages 155–157 of this Annual Report.

2010 compared with 2009

The provision for credit losses declined by \$15.4 billion compared with 2009, due to decreases in both the consumer and wholesale provisions. The decreases in the consumer provisions reflected reductions in the allowance for credit losses for mortgages and credit cards as a result of improved delinquency trends and lower estimated losses. This was partially offset by an increase in the allowance for credit losses associated with the Washington Mutual PCI loans portfolio, resulting from increased estimated future credit losses. The decrease in the wholesale provision in 2010 reflected a reduction in the allowance for credit losses, predominantly as a result of continued improvement in the credit quality of the commercial and industrial loan portfolio, reduced net charge-offs, and net repayments and loan sales. For a more detailed discussion of the loan portfolio and the allowance for credit losses, see the segment discussions for RFS on pages 85–93, Card on pages 94–97, IB on pages 81–84 and CB on pages 98–100, and the Allowance for Credit Losses section on pages 155–157 of this Annual Report.

Management's discussion and analysis

Noninterest expense

Year ended December 31,

(in millions)	2011	2010	2009
Compensation expense	\$ 29,037	\$ 28,124	\$ 26,928
Noncompensation expense:			
Occupancy	3,895	3,681	3,666
Technology, communications and equipment	4,947	4,684	4,624
Professional and outside services	7,482	6,767	6,232
Marketing	3,143	2,446	1,777
Other ^{(a)(b)}	13,559	14,558	7,594
Amortization of intangibles	848	936	1,050
Total noncompensation expense	33,874	33,072	24,943
Merger costs	—	—	481
Total noninterest expense	\$ 62,911	\$ 61,196	\$ 52,352

(a) Included litigation expense of \$4.9 billion, \$7.4 billion and \$161 million for the years ended December 31, 2011, 2010 and 2009, respectively.

(b) Included foreclosed property expense of \$718 million, \$1.0 billion and \$1.4 billion for the years ended December 31, 2011, 2010 and 2009, respectively.

2011 compared with 2010

Total noninterest expense for 2011 was \$62.9 billion, up by \$1.7 billion, or 3%, from 2010. The increase was driven by higher compensation expense and noncompensation expense.

Compensation expense increased from the prior year, due to investments in branch and mortgage production sales and support staff in RFS and increased headcount in AM, largely offset by lower performance-based compensation expense and the absence of the 2010 U.K. Bank Payroll Tax in IB.

The increase in noncompensation expense in 2011 was due to elevated foreclosure- and default-related costs in RFS, including \$1.7 billion of expense for fees and assessments, as well as other costs of foreclosure-related matters, higher marketing expense in Card, higher FDIC assessments across businesses, non-client-related litigation expense in AM, and the impact of continued investments in the businesses, including new branches in RFS. These were offset partially by lower litigation expense in 2011 in Corporate and IB. Effective April 1, 2011, the FDIC changed its methodology for calculating the deposit insurance assessment rate for large banks. The new rule changed the assessment base from insured deposits to average consolidated total assets less average tangible equity, and changed the assessment rate calculation. For a further discussion of litigation expense, see Note 31 on pages 290–299 of this Annual Report. For a discussion of amortization of intangibles, refer to the Balance Sheet Analysis on pages 110–112, and Note 17 on pages 267–271 of this Annual Report.

2010 compared with 2009

Total noninterest expense for 2010 was \$61.2 billion, up by \$8.8 billion, or 17%, from 2009. The increase was driven by higher noncompensation expense, largely due to higher litigation expense, and the effect of investments in the businesses.

Compensation expense increased from the prior year, predominantly due to higher salary expense related to investments in the businesses, including additional sales staff in RFS and client advisors in AM, and the impact of the U.K. Bank Payroll Tax.

In addition to the aforementioned higher litigation expense, which was largely for mortgage-related matters in Corporate and IB, the increase in noncompensation expense was driven by higher marketing expense in Card; higher professional services expense, due to continued investments in new product platforms in the businesses, including those related to international expansion; higher default-related expense, including costs associated with foreclosure affidavit-related suspensions (recorded in other expense), for the serviced portfolio in RFS; and higher brokerage, clearing and exchange transaction processing expense in IB. Partially offsetting these increases was the absence of a \$675 million FDIC special assessment recognized in 2009. For a further discussion of litigation expense, see Note 31 pages 290–299 of this Annual Report. For a discussion of amortization of intangibles, refer to Note 17 on pages 267–271 of this Annual Report.

There were no merger costs recorded in 2010, compared with merger costs of \$481 million in 2009. For additional information on merger costs, refer to Note 11 on page 224 of this Annual Report.

Income tax expense

Year ended December 31, (in millions, except rate)	2011	2010	2009
Income before income tax expense and extraordinary gain	\$ 26,749	\$ 24,859	\$ 16,067
Income tax expense	7,773	7,489	4,415
Effective tax rate	29.1%	30.1%	27.5%

2011 compared with 2010

The decrease in the effective tax rate compared with the prior year was predominantly the result of tax benefits associated with state and local income taxes. This was partially offset by higher reported pretax income and changes in the proportion of income subject to U.S. federal tax. In addition, the current year included tax benefits associated with the disposition of certain investments; the prior year included tax benefits associated with the resolution of tax audits. For additional information on income taxes, see Critical Accounting Estimates Used by the Firm on pages 168–172 and Note 26 on pages 279–281 of this Annual Report.

2010 compared with 2009

The increase in the effective tax rate compared with the prior year was predominantly the result of higher reported pretax book income, as well as changes in the proportion of income subject to U.S. federal and state and local taxes. These increases were partially offset by increased benefits associated with the undistributed earnings of certain non-U.S. subsidiaries that were deemed to be reinvested indefinitely, as well as tax benefits recognized upon the resolution of tax audits in 2010. For additional information on income taxes, see Critical Accounting Estimates Used by the Firm on pages 168–172 and Note 26 on pages 279–281 of this Annual Report.

EXPLANATION AND RECONCILIATION OF THE FIRM'S USE OF NON-GAAP FINANCIAL MEASURES

The Firm prepares its consolidated financial statements using U.S. GAAP; these financial statements appear on pages 178–181 of this Annual Report. That presentation, which is referred to as “reported” basis, provides the reader with an understanding of the Firm’s results that can be tracked consistently from year to year and enables a comparison of the Firm’s performance with other companies’ U.S. GAAP financial statements.

In addition to analyzing the Firm’s results on a reported basis, management reviews the Firm’s results and the results of the lines of business on a “managed” basis, which is a non-GAAP financial measure. The Firm’s definition of managed basis starts with the reported U.S. GAAP results and includes certain reclassifications to present total net revenue for the Firm (and each of the business segments) on a FTE basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in the managed results on a basis comparable to taxable investments and securities. This non-GAAP financial measure allows management to assess the comparability of revenue arising from both taxable and tax-exempt sources. The corresponding income tax impact related to tax-exempt items is recorded within income tax expense. These adjustments have no impact on net income as reported by the Firm as a whole or by the lines of business.

Prior to January 1, 2010, the Firm’s managed-basis presentation also included certain reclassification adjustments that assumed credit card loans securitized by Card remained on the Consolidated Balance Sheets. Effective January 1, 2010, the Firm adopted accounting guidance that required the Firm to consolidate its Firm-sponsored credit card securitization trusts. As a result of the consolidation of the credit card securitization trusts, reported and managed basis relating to credit card securitizations are equivalent for periods beginning after January 1, 2010. The income, expense and credit costs associated with these securitization activities were recorded in the 2011 and 2010 Consolidated Statements of Income in the same classifications that were previously used to report such items on a managed basis. For additional information on the accounting guidance, see Note 16 on pages 256–267 of this Annual Report.

The presentation in 2009 of Card’s results on a managed basis assumed that credit card loans that had been securitized and sold in accordance with U.S. GAAP remained on the Consolidated Balance Sheets, and that the earnings on the securitized loans were classified in the same manner as earnings on retained loans recorded on the Consolidated Balance Sheets. JPMorgan Chase had used this managed-basis information to evaluate the credit performance and overall financial performance of the entire managed credit card portfolio. JPMorgan Chase believed that this managed-basis information was useful to investors, as it enabled them to understand both the credit risks associated with the

loans reported on the Consolidated Balance Sheets and the Firm’s retained interests in securitized loans. For a reconciliation of 2009 reported to managed basis results for Card, see Card’s segment results on pages 94–97 of this Annual Report. For information regarding the securitization process, and loans and residual interests sold and securitized, see Note 16 on pages 256–267 of this Annual Report.

Tangible common equity (“TCE”), a non-GAAP financial measure, represents common stockholders’ equity (i.e., total stockholders’ equity less preferred stock) less goodwill and identifiable intangible assets (other than MSRs), net of related deferred tax liabilities. ROTCE, a non-GAAP financial ratio, measures the Firm’s earnings as a percentage of TCE. Tier 1 common under Basel I and III rules, a non-GAAP financial measure, is used by management to assess the Firm’s capital position in conjunction with its capital ratios under Basel I and III requirements. For additional information on Tier 1 common under Basel I and III, see Regulatory capital on pages 119–124 of this Annual Report. In management’s view, these measures are meaningful to the Firm, as well as analysts and investors, in assessing the Firm’s use of equity and in facilitating comparisons with competitors.

Management also uses certain non-GAAP financial measures at the business-segment level, because it believes these other non-GAAP financial measures provide information to investors about the underlying operational performance and trends of the particular business segment and, therefore, facilitate a comparison of the business segment with the performance of its competitors. Non-GAAP financial measures used by the Firm may not be comparable to similarly named non-GAAP financial measures used by other companies.

The following summary table provides a reconciliation from the Firm's reported U.S. GAAP results to managed basis.

Year ended December 31, (in millions, except per share and ratios)	2011			2010			2009			
	Reported Results	Fully tax- equivalent adjustments	Managed basis	Reported Results	Fully tax- equivalent adjustments	Managed basis	Reported Results	Credit card ^(b)	Fully tax- equivalent adjustments	Managed basis
Revenue										
Investment banking fees	\$ 5,911	\$ —	\$ 5,911	\$ 6,190	\$ —	\$ 6,190	\$ 7,087	\$ —	\$ —	\$ 7,087
Principal transactions	10,005	—	10,005	10,894	—	10,894	9,796	—	—	9,796
Lending- and deposit-related fees	6,458	—	6,458	6,340	—	6,340	7,045	—	—	7,045
Asset management, administration and commissions	14,094	—	14,094	13,499	—	13,499	12,540	—	—	12,540
Securities gains	1,593	—	1,593	2,965	—	2,965	1,110	—	—	1,110
Mortgage fees and related income	2,721	—	2,721	3,870	—	3,870	3,678	—	—	3,678
Credit card income	6,158	—	6,158	5,891	—	5,891	7,110	(1,494)	—	5,616
Other income	2,605	2,003	4,608	2,044	1,745	3,789	916	—	1,440	2,356
Noninterest revenue	49,545	2,003	51,548	51,693	1,745	53,438	49,282	(1,494)	1,440	49,228
Net interest income	47,689	530	48,219	51,001	403	51,404	51,152	7,937	330	59,419
Total net revenue	97,234	2,533	99,767	102,694	2,148	104,842	100,434	6,443	1,770	108,647
Noninterest expense	62,911	—	62,911	61,196	—	61,196	52,352	—	—	52,352
Pre-provision profit	34,323	2,533	36,856	41,498	2,148	43,646	48,082	6,443	1,770	56,295
Provision for credit losses	7,574	—	7,574	16,639	—	16,639	32,015	6,443	—	38,458
Income before income tax expense and extraordinary gain	26,749	2,533	29,282	24,859	2,148	27,007	16,067	—	1,770	17,837
Income tax expense	7,773	2,533	10,306	7,489	2,148	9,637	4,415	—	1,770	6,185
Income before extraordinary gain	18,976	—	18,976	17,370	—	17,370	11,652	—	—	11,652
Extraordinary gain	—	—	—	—	—	—	76	—	—	76
Net income	\$ 18,976	\$ —	\$ 18,976	\$ 17,370	\$ —	\$ 17,370	\$ 11,728	\$ —	\$ —	\$ 11,728
Diluted earnings per share ^(a)	\$ 4.48	\$ —	\$ 4.48	\$ 3.96	\$ —	\$ 3.96	\$ 2.24	\$ —	\$ —	\$ 2.24
Return on assets	0.86%	NM	0.86%	0.85%	NM	0.85%	0.58%	NM	NM	0.55%
Overhead ratio	65	NM	63	60	NM	58	52	NM	NM	48
Loans – period-end	\$ 723,720	\$ —	\$ 723,720	\$ 692,927	\$ —	\$ 692,927	\$ 633,458	\$ 84,626	\$ —	\$ 718,084
Total assets – average	2,198,198	—	2,198,198	2,053,251	—	2,053,251	2,024,201	82,233	—	2,106,434

(a) Based on income before extraordinary gain.

(b) See pages 94–97 of this Annual Report for a discussion of the effect of credit card securitizations on Card's results.

Calculation of certain U.S. GAAP and non-GAAP metrics

The table below reflects the formulas used to calculate both the following U.S. GAAP and non-GAAP measures.

Return on common equity
Net income* / Average common stockholders' equity

Return on tangible common equity
Net income* / Average tangible common equity

Return on assets
Reported net income / Total average assets
Managed net income / Total average managed assets

Overhead ratio
Total noninterest expense / Total net revenue

* Represents net income applicable to common equity

(c) The Firm uses ROTCE, a non-GAAP financial measure, to evaluate its use of equity and to facilitate comparisons with competitors. Refer to the following table for the calculation of average tangible common equity.

(d) The Firm uses return on managed assets, a non-GAAP financial measure, to evaluate the overall performance of the managed credit card portfolio, including securitized credit card loans.

Average tangible common equity

Year ended December 31, (in millions)	2011	2010	2009
Common stockholders' equity	\$ 173,266	\$ 161,520	\$ 145,903
Less: Goodwill	48,632	48,618	48,254
Less: Certain identifiable intangible assets	3,632	4,178	5,095
Add: Deferred tax liabilities ^(a)	2,635	2,587	2,547
Tangible common equity	\$ 123,637	\$ 111,311	\$ 95,101

(a) Represents deferred tax liabilities related to tax-deductible goodwill and to identifiable intangibles created in nontaxable transactions, which are netted against goodwill and other intangibles when calculating TCE.

Management's discussion and analysis

Core net interest income

In addition to reviewing JPMorgan Chase's net interest income on a managed basis, management also reviews core net interest income to assess the performance of its core lending, investing (including asset/liability management) and deposit-raising activities, excluding the impact of IB's market-based activities. The table below presents an analysis of core net interest income, core average interest-earning assets, and the core net interest yield on core average interest-earning assets, on a managed basis. Each of these amounts is a non-GAAP financial measure due to the exclusion of IB's market-based net interest income and the related assets. Management believes the exclusion of IB's market-based activities provides investors and analysts a more meaningful measure to analyze non-market related business trends of the Firm and can be used as a comparable measure to other financial institutions primarily focused on core lending, investing and deposit-raising activities.

Core net interest income data^(a)

Year ended December 31, (in millions, except rates)	2011	2010	2009
Net interest income - managed basis	\$ 48,219	\$ 51,404	\$ 59,419
Impact of market-based net interest income	7,329	7,112	8,238
Core net interest income	\$ 40,890	\$ 44,292	\$ 51,181
Average interest-earning assets - managed basis	\$ 1,761,355	\$ 1,677,521	\$ 1,735,866
Impact of market-based earning assets	519,655	470,927	428,471
Core average interest-earning assets	\$ 1,241,700	\$ 1,206,594	\$ 1,307,395
Net interest yield on interest-earning assets - managed basis	2.74%	3.06%	3.42%
Net interest yield on market-based activity	1.41	1.51	1.92
Core net interest yield on interest-earning assets	3.29%	3.67%	3.91%

(a) Includes core lending activities, investing and deposit-raising activities on a managed basis, across RFS, Card, CB, TSS, AM and Corporate/Private Equity, as well as IB credit portfolio loans.

2011 compared with 2010

Core net interest income decreased by \$3.4 billion to \$40.9 billion for 2011. The decrease was primarily driven by lower loan levels and yields in RFS and Card compared with 2010 levels. Core average interest-earning assets increased by \$35.1 billion in 2011 to \$1,241.7 billion. The increase was driven by higher levels of deposits with banks and securities borrowed due to wholesale and retail client deposit growth. The core net interest yield decreased by 38 basis points in 2011 driven by lower loan yields and higher deposit balances, and lower yields on investment securities due to portfolio mix and lower long-term interest rates.

2010 compared with 2009

Core net interest income decreased by \$6.9 billion to \$44.3 billion in 2010. The decrease was primarily driven by lower loan levels and yields in RFS, Card and IB compared with

2009 levels. Core average interest-earning assets decreased by \$100.8 billion in 2010 to \$1,206.6 billion. The decrease was primarily driven by lower loan balances and deposits with banks due to a decline in wholesale and retail deposits. The core net interest yield decreased by 24 basis points in 2010 driven by lower yields on loans and investment securities.

Impact of redemption of TARP preferred stock issued to the U.S. Treasury

The calculation of 2009 net income applicable to common equity included a one-time, noncash reduction of \$1.1 billion resulting from the redemption of TARP preferred capital. Excluding this reduction, ROE would have been 7% for 2009. The Firm views adjusted ROE, a non-GAAP financial measure, as meaningful because it enables the comparability to the other periods reported.

Year ended December 31, 2009 (in millions, except ratios)	As reported	Excluding the TARP redemption
Return on equity		
Net income	\$ 11,728	\$ 11,728
Less: Preferred stock dividends	1,327	1,327
Less: Accelerated amortization from redemption of preferred stock issued to the U.S. Treasury	1,112	—
Net income applicable to common equity	\$ 9,289	\$ 10,401
Average common stockholders' equity	\$ 145,903	\$ 145,903
ROE	6%	7%

In addition, the calculation of diluted earnings per share ("EPS") for the year ended December 31, 2009, was also affected by the TARP repayment, as presented below.

Year ended December 31, 2009 (in millions, except per share)	As reported	Effect of TARP redemption
Diluted earnings per share		
Net income	\$ 11,728	\$ —
Less: Preferred stock dividends	1,327	—
Less: Accelerated amortization from redemption of preferred stock issued to the U.S. Treasury	1,112	1,112
Net income applicable to common equity	9,289	(1,112)
Less: Dividends and undistributed earnings allocated to participating securities	515	(62)
Net income applicable to common stockholders	8,774	(1,050)
Total weighted average diluted shares outstanding	3,879.7	3,879.7
Net income per share	\$ 2.26	\$ (0.27)

Other financial measures

The Firm also discloses the allowance for loan losses to total retained loans, excluding residential real estate purchased credit-impaired loans. For a further discussion of this credit metric, see Allowance for Credit Losses on pages 155–157 of this Annual Report.

BUSINESS SEGMENT RESULTS

The Firm is managed on a line-of-business basis. The business segment financial results presented reflect the current organization of JPMorgan Chase. There are six major reportable business segments: the Investment Bank, Retail Financial Services, Card Services & Auto, Commercial Banking, Treasury & Securities Services and Asset Management, as well as a Corporate/Private Equity segment.

The business segments are determined based on the products and services provided, or the type of customer served, and reflect the manner in which financial information is currently evaluated by management. Results of the lines of business are presented on a managed basis. For a definition of managed basis, see Explanation and Reconciliation of the Firm's use of non-GAAP financial measures, on pages 76–78 of this Annual Report.

Business segment changes

Commencing July 1, 2011, the Firm's business segments were reorganized as follows:

Auto and Student Lending transferred from the RFS segment and are reported with Card in a single segment. Retail Financial Services continues as a segment, organized in two components: Consumer & Business Banking (formerly Retail Banking) and Mortgage Banking (which includes Mortgage Production and Servicing, and Real Estate Portfolios).

The business segment information associated with RFS and Card have been revised to reflect the business reorganization retroactive to January 1, 2009.

JPMorgan Chase

Investment Bank	Retail Financial Services	Card Services & Auto	Commercial Banking	Treasury & Securities Services	Asset Management
Businesses: •Investment Banking – Advisory – Debt and equity underwriting •Market-making – Fixed income – Commodities – Equities •Prime Services •Research •Corporate Lending •Credit Portfolio Management	Businesses: •Consumer & Business Banking •Mortgage Production and Servicing •Real Estate Portfolios – Residential mortgage loans – Home equity loans and originations	Businesses: •Card Services – Credit Card – Merchant Services •Auto •Student	Businesses: •Middle Market Banking •Commercial Term Lending •Corporate Client Banking •Real Estate Banking	Businesses: •Treasury Services •Worldwide Securities Services	Businesses: •Private Banking •Investment Management: – Institutional – Retail •Highbridge

Description of business segment reporting methodology

Results of the business segments are intended to reflect each segment as if it were essentially a stand-alone business. The management reporting process that derives business segment results allocates income and expense using market-based methodologies. The Firm continues to assess the assumptions, methodologies and reporting classifications used for segment reporting, and further refinements may be implemented in future periods.

Revenue sharing

When business segments join efforts to sell products and services to the Firm's clients, the participating business segments agree to share revenue from those transactions. The segment results reflect these revenue-sharing agreements.

Funds transfer pricing

Funds transfer pricing is used to allocate interest income and expense to each business and transfer the primary interest rate risk exposures to the Treasury group within the Corporate/Private Equity business segment. The allocation process is unique to each business segment and considers the interest rate risk, liquidity risk and regulatory requirements of that segment as if it were operating independently, and as compared with its stand-alone peers. This process is overseen by senior management and reviewed by the Firm's Asset-Liability Committee ("ALCO"). Business segments may be permitted to retain certain interest rate exposures subject to management approval.

Capital allocation

Each line of business is allocated an amount of capital the Firm believes the business would require if it were operating independently, incorporating sufficient capital to

Management's discussion and analysis

address regulatory capital requirements (including Basel III Tier 1 common capital requirements), economic risk measures and capital levels for similarly rated peers. For a further discussion on capital allocation, see Capital Management – Line of business equity on page 123 of this Annual Report.

Expense allocation

Where business segments use services provided by support units within the Firm, the costs of those support units are allocated to the business segments. The expense is allocated based on their actual cost or the lower of actual

cost or market, as well as upon usage of the services provided. In contrast, certain other expense related to certain corporate functions, or to certain technology and operations, are not allocated to the business segments and are retained in Corporate. Retained expense includes: parent company costs that would not be incurred if the segments were stand-alone businesses; adjustments to align certain corporate staff, technology and operations allocations with market prices; and other one-time items not aligned with a particular business segment.

Segment Results – Managed Basis

The following table summarizes the business segment results for the periods indicated.

Year ended December 31, (in millions)	Total net revenue			Noninterest expense			Pre-provision profit ^(a)		
	2011	2010	2009	2011	2010	2009	2011	2010	2009
Investment Bank ^(b)	\$ 26,274	\$ 26,217	\$ 28,109	\$ 16,116	\$ 17,265	\$ 15,401	\$ 10,158	\$ 8,952	\$ 12,708
Retail Financial Services	26,538	28,447	29,797	19,458	16,483	15,512	7,080	11,964	14,285
Card Services & Auto	19,141	20,472	23,199	8,045	7,178	6,617	11,096	13,294	16,582
Commercial Banking	6,418	6,040	5,720	2,278	2,199	2,176	4,140	3,841	3,544
Treasury & Securities Services	7,702	7,381	7,344	5,863	5,604	5,278	1,839	1,777	2,066
Asset Management	9,543	8,984	7,965	7,002	6,112	5,473	2,541	2,872	2,492
Corporate/Private Equity	4,151	7,301	6,513	4,149	6,355	1,895	2	946	4,618
Total	\$ 99,767	\$ 104,842	\$ 108,647	\$ 62,911	\$ 61,196	\$ 52,352	\$ 36,856	\$ 43,646	\$ 56,295

Year ended December 31, (in millions, except ratios)	Provision for credit losses			Net income/(loss)			Return on equity		
	2011	2010	2009	2011	2010	2009	2011	2010	2009
Investment Bank ^(b)	\$ (286)	\$ (1,200)	\$ 2,279	\$ 6,789	\$ 6,639	\$ 6,899	17%	17%	21%
Retail Financial Services	3,999	8,919	14,754	1,678	1,728	(335)	7	7	(1)
Card Services & Auto	3,621	8,570	19,648	4,544	2,872	(1,793)	28	16	(10)
Commercial Banking	208	297	1,454	2,367	2,084	1,271	30	26	16
Treasury & Securities Services	1	(47)	55	1,204	1,079	1,226	17	17	25
Asset Management	67	86	188	1,592	1,710	1,430	25	26	20
Corporate/Private Equity	(36)	14	80	802	1,258	3,030	NM	NM	NM
Total	\$ 7,574	\$ 16,639	\$ 38,458	\$ 18,976	\$ 17,370	\$ 11,728	11%	10%	6%

(a) Corporate/Private Equity includes an adjustment to offset IB's inclusion of a credit allocation income/(expense) to TSS in total net revenue; TSS reports the credit allocation as a separate line item on its income statement (not within total net revenue).

(b) Pre-provision profit is total net revenue less noninterest expense. The Firm believes that this financial measure is useful in assessing the ability of a lending institution to generate income in excess of its provision for credit losses.

INVESTMENT BANK

J.P. Morgan is one of the world's leading investment banks, with deep client relationships and broad product capabilities. The clients of IB are corporations, financial institutions, governments and institutional investors. The Firm offers a full range of investment banking products and services in all major capital markets, including advising on corporate strategy and structure, capital-raising in equity and debt markets, sophisticated risk management, market-making in cash securities and derivative instruments, prime brokerage, and research.

Selected income statement data

Year ended December 31,

(in millions, except ratios)

Revenue

Investment banking fees

Principal transactions

Lending- and deposit-related fees

Asset management, administration
and commissions

All other income^(e)

Noninterest revenue

Net interest income

Total net revenue^(e) AITD 4.8 TF (b3 /)

Management's discussion and analysis

gains of \$181 million in the prior year. Credit Portfolio revenue was \$246 million as net interest income and fees on retained loans, as well as DVA gains of \$528 million were predominantly offset by a \$769 million loss, net of hedges, from CVA on derivative assets. Results were approximately flat to the prior year, which included net CVA losses of \$403 million.

The provision for credit losses was a benefit of \$286 million, compared with a benefit of \$1.2 billion in the prior year. The current-year provision reflected a net reduction in the allowance for loan losses largely driven by portfolio activity, partially offset by new loan growth. Net charge-offs were \$161 million, compared with \$735 million in the prior year.

Noninterest expense was \$16.1 billion, down 7% driven primarily by lower compensation expense compared with the prior period which included the impact of the U.K. Bank Payroll Tax. Noncompensation expense was also lower compared with the prior year, which included higher litigation reserves. This decrease was partially offset by additional operating expense related to growth in business activities in 2011.

Return on Equity was 17% on \$40.0 billion of average allocated capital.

2010 compared with 2009

Net income was \$6.6 billion, down 4% compared with the prior year. These results primarily reflected lower net revenue as well as higher noninterest expense, largely offset by a benefit from the provision for credit losses, compared with an expense in the prior year.

Net revenue was \$26.2 billion, compared with \$28.1 billion in the prior year. Investment banking fees were \$6.2 billion, down 14% from the prior year; these consisted of record debt underwriting fees of \$3.1 billion (up 18%), equity underwriting fees of \$1.6 billion (down 40%), and advisory fees of \$1.5 billion (down 21%). Fixed Income Markets revenue was \$15.0 billion, compared with \$17.6 billion in the prior year. The decrease from the prior year largely reflected lower results in rates and credit markets, partially offset by DVA gains of \$287 million from the widening of the Firm's credit spread on certain structured liabilities, compared with DVA losses of \$1.1 billion in the prior year. Equity Markets revenue was \$4.8 billion, compared with \$4.4 billion in the prior year, reflecting solid client revenue, as well as DVA gains of \$181 million, compared with DVA losses of \$596 million in the prior year. Credit Portfolio revenue was \$243 million, primarily reflecting net interest income and fees on loans, partially offset by net CVA losses on derivative assets and mark-to-market losses on hedges of retained loans.

The provision for credit losses was a benefit of \$1.2 billion, compared with an expense of \$2.3 billion in the prior year. The current-year provision reflected a reduction in the allowance for loan losses, largely related to net repayments and loan sales. Net charge-offs were \$735 million, compared with \$1.9 billion in the prior year.

Noninterest expense was \$17.3 billion, up \$1.9 billion from the prior year, driven by higher noncompensation expense, which included increased litigation reserves, and higher compensation expense which included the impact of the U.K. Bank Payroll Tax.

Return on Equity was 17% on \$40.0 billion of average allocated capital.

Selected metrics

As of or for the year ended December 31,

(in millions, except headcount)	2011	2010	2009
Selected balance sheet data (period-end)			
Total assets	\$ 776,430	\$ 825,150	\$ 706,944
Loans:			
Loans retained ^(a)	68,208	53,145	45,544
Loans held-for-sale and loans at fair value	2,915	3,746	3,567
Total loans	71,123	56,891	49,111
Equity	40,000	40,000	33,000
Selected balance sheet data (average)			
Total assets	\$ 812,779	\$ 731,801	\$ 699,039
Trading assets-debt and equity instruments	346,461	307,061	273,624
Trading assets-derivative receivables	73,201	70,289	96,042
Loans:			
Loans retained ^(a)	57,007	54,402	62,722
Loans held-for-sale and loans at fair value	3,119	3,215	7,589
Total loans	60,126	57,617	70,311
Adjusted assets ^(b)	600,160	540,449	538,724
Equity	40,000	40,000	33,000
Headcount	25,999	26,314	24,654

- (a) Loans retained included credit portfolio loans, leveraged leases and other held-for-investment loans, and excluded loans held-for-sale and loans at fair value.
- (b) Adjusted assets, a non-GAAP financial measure, equals total assets minus: (1) securities purchased under resale agreements and securities borrowed less securities sold, not yet purchased; (2) assets of consolidated VIEs; (3) cash and securities segregated and on deposit for regulatory and other purposes; (4) goodwill and intangibles; and (5) securities received as collateral. The amount of adjusted assets is presented to assist the reader in comparing IB's asset and capital levels to other investment banks in the securities industry. Asset-to-equity leverage ratios are commonly used as one measure to assess a company's capital adequacy. IB believes an adjusted asset amount that excludes the assets discussed above, which were considered to have a low risk profile, provides a more meaningful measure of balance sheet leverage in the securities industry.

Selected metrics

As of or for the year ended
December 31,

(in millions, except ratios)	2011	2010	2009
Credit data and quality statistics			
Net charge-offs	\$ 161	\$ 735	\$ 1,904
Nonperforming assets:			
Nonaccrual loans:			
Nonaccrual loans retained ^(a)	1,035	3,159	3,196
Nonaccrual loans held-for-sale and loans at fair value	166	460	308
Total nonaccrual loans	1,201	3,619	3,504
Derivative receivables	14	34	529
Assets acquired in loan satisfactions	79	117	203
Total nonperforming assets	1,294	3,770	4,236
Allowance for credit losses:			
Allowance for loan losses	1,436	1,863	3,756
Allowance for lending-related commitments	418	447	485
Total allowance for credit losses	1,854	2,310	4,241
Net charge-off rate ^{(a)(c)}	0.28%	1.35%	3.04%
Allowance for loan losses to period-end loans retained ^(c)	2.11	3.51	8.25
Allowance for loan losses to nonaccrual loans retained ^{(b)(c)}	139	59	118
Nonaccrual loans to period-end loans	1.69	6.36	7.13
Market risk-average trading and credit portfolio VaR – 95% confidence level			
Trading activities:			
Fixed income	\$ 50	\$ 65	\$ 160
Foreign exchange	11	11	18
Equities	23	22	47
Commodities and other	16	16	20
Diversification ^(d)	(42)	(43)	(91)
Total trading VaR ^(e)	58	71	154
Credit portfolio VaR	33	26	52
Diversification ^(d)	(15)	(10)	(42)
Total trading and credit portfolio VaR	\$ 76	\$ 87	\$ 164

- (a) Loans retained included credit portfolio loans, leveraged leases and other held-for-investment loans, and excluded loans held-for-sale and loans at fair value.
- (b) Allowance for loan losses of \$263 million, \$1.1 billion and \$1.3 billion were held against these nonaccrual loans at December 31, 2011, 2010 and 2009, respectively.
- (c) Loans held-for-sale and loans at fair value were excluded when calculating the allowance coverage ratio and net charge-off rate.
- (d) Average value-at-risk ("VaR") was less than the sum of the VaR of the components described above, due to portfolio diversification. The diversification effect reflects the fact that the risks were not perfectly correlated. The risk of a portfolio of positions is therefore usually less than the sum of the risks of the positions themselves.
- (e) Trading VaR includes substantially all market-making and client-driven activities as well as certain risk management activities in IB, including the credit spread sensitivities of certain mortgage products and syndicated lending facilities that the Firm intends to distribute; however, particular risk parameters of certain products are not fully captured, for example, correlation risk. Trading VaR does not include the DVA on derivative and structured liabilities to reflect the credit

quality of the Firm. See VaR discussion on pages 158–160 and the DVA sensitivity table on page 161 of this Annual Report for further details.

- (f) Credit portfolio VaR includes the derivative CVA, hedges of the CVA and mark-to-market ("MTM") hedges of the retained loan portfolio, which are all reported in principal transactions revenue. This VaR does not include the retained loan portfolio, which is not MTM.

Market shares and rankings^(a)

Year ended December 31,	2011		2010		2009	
	Market Share	Rankings	Market Share	Rankings	Market Share	Rankings
Global investment banking fees ^(b)	8.1%	#1	7.6%	#1	9.0%	#1
Debt, equity and equity-related						
Global	6.8	1	7.2	1	8.8	1
U.S.	11.1	1	11.1	1	14.8	1
Syndicated loans						
Global	11.0	1	8.5	2	8.1	1
U.S.	21.4	1	19.1	2	21.8	1
Long-term debt^(c)						
Global	6.7	1	7.2	2	8.4	1
U.S.	11.2	1	10.9	2	14.2	1
Equity and equity-related						
Global ^(d)	6.8	3	7.3	3	11.6	1
U.S.	12.5	1	13.1	2	15.5	2
Announced M&A^(e)						
Global	18.6	2	15.9	4	23.7	3
U.S.	27.5	2	21.9	3	35.6	2

(a) Source: Dealogic. Global Investment Banking fees reflects ranking of fees and market share. Remainder of rankings reflects transaction volume rank and market share. Global announced M&A is based on transaction value at announcement; because of joint M&A assignments, M&A market share of all participants will add up to more than 100%. All other transaction volume-based rankings are based on proceeds, with full credit to each book manager/equal if joint.

(b) Global Investment Banking fees rankings exclude money market, short-term debt and shelf deals.

(c) Long-term debt rankings include investment-grade, high-yield, supranationals, sovereigns, agencies, covered bonds, asset-backed securities ("ABS") and mortgage-backed securities; and exclude money market, short-term debt, and U.S. municipal securities.

(d) Global Equity and equity-related ranking includes rights offerings and Chinese A-Shares.

(e) Announced M&A reflects the removal of any withdrawn transactions. U.S. announced M&A represents any U.S. involvement ranking.

According to Dealogic, the Firm was ranked #1 in Global Investment Banking Fees generated during 2011, based on revenue; #1 in Global Debt, Equity and Equity-related; #1 in Global Syndicated Loans; #1 in Global Long-Term Debt; #3 in Global Equity and Equity-related; and #2 in Global Announced M&A, based on volume.

Management's discussion and analysis

International metrics

Year ended December 31,

(in millions)	2011	2010	2009
Total net revenue ^(a)			
Europe/Middle East/Africa	\$ 8,418	\$ 7,380	\$ 9,164
Asia/Pacific	3,334	3,809	3,470
Latin America/Caribbean	1,079	897	1,157
North America	13,443	14,131	14,318
Total net revenue	\$ 26,274	\$ 26,217	\$ 28,109
Loans retained (period-end) ^(b)			
Europe/Middle East/Africa	\$ 15,905	\$ 13,961	\$ 13,079
Asia/Pacific	7,889	5,924	4,542
Latin America/Caribbean	3,148	2,200	2,523
North America	41,266	31,060	25,400
Total loans	\$ 68,208	\$ 53,145	\$ 45,544

- (a) Regional revenue is based primarily on the domicile of the client and/or location of the trading desk.
- (b) Includes retained loans based on the domicile of the customer. Excludes loans held-for-sale and loans at fair value.

RETAIL FINANCIAL SERVICES

Retail Financial Services serves consumers and businesses through personal service at bank branches and through ATMs, online banking and telephone banking. RFS is organized into Consumer & Business Banking and Mortgage Banking (including Mortgage Production and Servicing, and Real Estate Portfolios). Consumer & Business Banking includes branch banking and business banking activities. Mortgage Production and Servicing includes mortgage origination and servicing activities. Real Estate Portfolios comprises residential mortgages and home equity loans, including the PCI portfolio acquired in the Washington Mutual transaction. Customers can use more than 5,500 bank branches (third largest nationally) and more than 17,200 ATMs (second largest nationally), as well as online and mobile banking around the clock. More than 33,500 branch salespeople assist customers with checking and savings accounts, mortgages, home equity and business loans, and investments across the 23-state footprint from New York and Florida to California. As one of the largest mortgage originators in the U.S., Chase helps customers buy or refinance homes resulting in approximately \$150 billion of mortgage originations annually. Chase also services more than 8 million mortgages and home equity loans.

Effective July 1, 2011, RFS was organized into two components: (1) Consumer & Business Banking (formerly Retail Banking) and (2) Mortgage Banking (including Mortgage Production and Servicing, and Real Estate Portfolios). For a further discussion of the business segment reorganization, see Business segment changes on page 79, and Note 33 on pages 300–303 of this Annual Report.

Selected income statement data

Year ended December 31,
(in millions, except ratios)

	2011	2010	2009
Revenue			
Lending- and deposit-related fees	\$ 3,190	\$ 3,061	\$ 3,897
Asset management, administration and commissions	1,991	1,776	1,665
Mortgage fees and related income	2,714	3,855	3,794
Credit card income	2,025	1,955	1,634
Other income	485	580	424
Noninterest revenue	10,405	11,227	11,414
Net interest income	16,133	17,220	18,383
Total net revenue ^(a)	26,538	28,447	29,797
Provision for credit losses	3,999	8,919	14,754
Noninterest expense			
Compensation expense	8,044	7,072	6,349
Noncompensation expense	11,176	9,135	8,834
Amortization of intangibles	238	276	329
Total noninterest expense	19,458	16,483	15,512
Income/(loss) before income tax expense/(benefit)	3,081	3,045	(469)
Income tax expense/(benefit)	1,403	1,317	(134)
Net income/(loss)	\$ 1,678	\$ 1,728	\$ (335)
Financial ratios			
Return on common equity	7%	7%	(1)%
Overhead ratio	73	58	52
Overhead ratio excluding core deposit intangibles ^(b)	72	57	51

- (a) Total net revenue included tax-equivalent adjustments associated with tax-exempt loans to municipalities and other qualified entities of \$7 million, \$8 million and \$9 million for the years ended December 31, 2011, 2010 and 2009, respectively.
- (b) RFS uses the overhead ratio (excluding the amortization of core deposit intangibles ("CDI")), a non-GAAP financial measure, to evaluate the underlying expense trends of the business. Including CDI amortization expense in the overhead ratio calculation would result in a higher overhead ratio in the earlier years and a lower overhead ratio in later years; this method would therefore result in an improving overhead ratio over time, all things remaining equal. This non-GAAP ratio excluded Consumer & Business Banking's CDI amortization expense related to prior business combination transactions of \$238 million, \$276 million and \$328 million for the years ended December 31, 2011, 2010 and 2009, respectively.

2011 compared with 2010

Retail Financial Services reported net income of \$1.7 billion, down 3% when compared with the prior year.

Net revenue was \$26.5 billion, a decrease of \$1.9 billion, or 7%, compared with the prior year. Net interest income was \$16.1 billion, down by \$1.1 billion, or 6%, reflecting the impact of lower loan balances, due to portfolio runoff, and narrower loan spreads. Noninterest revenue was \$10.4 billion, down by \$822 million, or 7%, driven by lower mortgage fees and related income partially offset by higher investment sales revenue and higher deposit-related fees.

The provision for credit losses was \$4.0 billion, a decrease of \$4.9 billion from the prior year. While delinquency trends and net charge-offs improved compared with the prior year, the current-year provision continued to reflect elevated losses in the mortgage and home equity portfolios. The current year provision also included a \$230 million net reduction in the allowance for loan losses which reflects a reduction of \$1.0 billion in the allowance related to the non-credit-impaired portfolio, as estimated losses in the portfolio have declined, predominantly offset by an increase of \$770 million reflecting additional impairment of the Washington Mutual PCI portfolio due to higher-than-expected default frequency relative to modeled lifetime loss estimates. The prior-year provision reflected a higher impairment on the PCI portfolio and higher net charge-offs. See Consumer Credit Portfolio on pages 145–154 of this Annual Report for the net charge-off amounts and rates.

Noninterest expense was \$19.5 billion, an increase of \$3.0 billion, or 18%, from the prior year driven by elevated foreclosure- and default-related costs, including \$1.7 billion for fees and assessments, as well as other costs of foreclosure-related matters during 2011, compared with \$350 million in 2010.

2010 compared with 2009

Net income was \$1.7 billion, compared with a net loss of \$335 million in the prior year.

Net revenue was \$28.4 billion, a decrease of \$1.4 billion, or 5%, compared with the prior year. Net interest income was \$17.2 billion, down by \$1.2 billion, or 6%, reflecting the impact of lower loan and deposit balances and narrower

Management's discussion and analysis

loan spreads, partially offset by a shift to wider-spread deposit products. Noninterest revenue was \$11.2 billion, a decrease of \$187 million, or 2%, compared with the prior year, as lower deposit-related fees were partially offset by higher debit card income.

The provision for credit losses was \$8.9 billion, compared with \$14.8 billion in the prior year. The current-year provision reflected an addition to the allowance for loan losses of \$3.4 billion for the PCI portfolio and a reduction in the allowance for loan losses of \$1.7 billion, predominantly for the mortgage loan portfolios. In comparison, the prior-year provision reflected an addition to the allowance for loan losses of \$5.5 billion, predominantly for the home equity and mortgage portfolios, and also included an addition of \$1.6 billion for the PCI portfolio. While delinquency trends and net charge-offs improved compared with the prior year, the provision continued to reflect elevated losses for the mortgage and home equity portfolios. See Consumer Credit Portfolio on page 145–154 of this Annual Report for the net charge-off amounts and rates.

Noninterest expense was \$16.5 billion, an increase of \$971 million, or 6%, from the prior year, reflecting higher default-related expense.

Selected metrics

As of or for the year ended December 31,

(in millions, except headcount and ratios)

	2011	2010	2009
Selected balance sheet data (period-end)			
Total assets	\$ 274,795	\$ 299,950	\$ 322,185
Loans:			
Loans retained	232,555	253,904	280,246
Loans held-for-sale and loans at fair value ^(e)	12,694	14,863	12,920
Total loans	245,249	268,767	293,166
Deposits	395,797	369,925	356,614
Equity	25,000	24,600	22,457
Selected balance sheet data (average)			
Total assets	\$ 286,716	\$ 314,046	\$ 344,727
Loans:			
Loans retained	241,621	268,902	296,959
Loans held-for-sale and loans at fair value ^(e)	16,354	15,395	16,236
Total loans	257,975	284,297	313,195
Deposits	380,663	361,525	366,996
Equity	25,000	24,600	22,457
Headcount	133,075	116,882	103,733

As of or for the year ended December 31,

(in millions, except ratios)	2011	2010	2009
Credit data and quality statistics			
Net charge-offs	\$ 4,304	\$ 7,221	\$ 9,233
Nonaccrual loans:			
Nonaccrual loans retained	7,170	8,568	10,373
Nonaccrual loans held-for-sale and loans at fair value	103	145	234
Total nonaccrual loans ^{(b)(c)(d)}	7,273	8,713	10,607
Nonperforming assets ^{(b)(c)(d)}	8,064	9,999	11,761
Allowance for loan losses	15,247	15,554	13,734
Net charge-off rate ^(e)	1.78%	2.69%	3.11%
Net charge-off rate excluding PCI loans ^{(e)(f)}	2.49	3.76	4.36
Allowance for loan losses to ending loans retained	6.56	6.13	4.90
Allowance for loan losses to ending loans retained excluding PCI loans ^(e)	5.71	5.86	6.11
Allowance for loan losses to nonaccrual loans retained ^{(e)(f)}	133	124	117
Nonaccrual loans to total loans	2.97	3.24	3.62
Nonaccrual loans to total loans excluding PCI loans ^(e)	4.05	4.45	5.01

- (a) Predominantly consists of prime mortgages originated with the intent to sell that are accounted for at fair value and classified as trading assets on the Consolidated Balance Sheets.
- (b) Excludes PCI loans that were acquired as part of the Washington Mutual transaction, which are accounted for on a pool basis. Since each pool is accounted for as a single asset with a single composite interest rate and an aggregate expectation of cash flows, the past-due status of the pools, or that of the individual loans within the pools, is not meaningful. Because the Firm is recognizing interest income on each pool of loans, they are all considered to be performing.
- (c) Certain of these loans are classified as trading assets on the Consolidated Balance Sheets.
- (d) At December 31, 2011, 2010 and 2009, nonperforming assets excluded: (1) mortgage loans insured by U.S. government agencies of \$11.5 billion, \$9.4 billion and \$9.0 billion, respectively, that are 90 or more days past due; and (2) real estate owned insured by U.S. government agencies of \$954 million, \$1.9 billion and \$579 million, respectively. These amounts were excluded as reimbursement of insured amounts is proceeding normally. For further discussion, see Note 14 on pages 231–252 of this Annual Report which summarizes loan delinquency information.
- (e) Loans held-for-sale and loans accounted for at fair value were excluded when calculating the net charge-off rate.
- (f) Excludes the impact of PCI loans that were acquired as part of the Washington Mutual transaction. These loans were accounted for at fair value on the acquisition date, which incorporated management's estimate, as of that date, of credit losses over the remaining life of the portfolio. An allowance for loan losses of \$5.7 billion, \$4.9 billion and \$1.6 billion was recorded for these loans at December 31, 2011, 2010 and 2009, respectively; these amounts were also excluded from the applicable ratios. To date, no charge-offs have been recorded for these loans.

Consumer & Business Banking

Selected income statement data

Year ended December 31,

(in millions, except ratios)	2011	2010	2009
Noninterest revenue	\$ 7,201	\$ 6,844	\$ 7,204
Net interest income	10,809	10,884	10,864
Total net revenue	18,010	17,728	18,068
Provision for credit losses	419	630	1,176
Noninterest expense	11,202	10,717	10,421
Income before income tax expense	6,389	6,381	6,471
Net income	\$ 3,816	\$ 3,652	\$ 3,915
Overhead ratio	62%	60%	58%
Overhead ratio excluding core deposit intangibles ^(a)	61	59	56

(a) Consumer & Business Banking uses the overhead ratio (excluding the amortization of CDI), a non-GAAP financial measure, to evaluate the underlying expense trends of the business. Including CDI amortization expense in the overhead ratio calculation would result in a higher overhead ratio in the earlier years and a lower overhead ratio in later years; this method would therefore result in an improving overhead ratio over time, all things remaining equal. This non-GAAP ratio excluded Consumer & Business Banking's CDI amortization expense related to prior business combination transactions of \$238 million and \$276 million and \$328 million for the years ended December 31, 2011, 2010 and 2009, respectively.

2011 compared with 2010

Consumer & Business Banking reported net income of \$3.8 billion, an increase of \$164 million, or 4%, compared with the prior year.

Net revenue was \$18.0 billion, up 2%, from the prior year. Net interest income was \$10.8 billion, relatively flat compared with the prior year, as the impact from higher deposit balances was offset predominantly by the effect of lower deposit spreads. Noninterest revenue was \$7.2 billion, an increase of 5%, driven by higher investment sales revenue and higher deposit-related fees.

The provision for credit losses was \$419 million, compared with \$630 million in the prior year. Net charge-offs were \$494 million, compared with \$730 million in the prior year.

Noninterest expense was \$11.2 billion, up 5%, from the prior year resulting from investment in sales force and new branch builds.

2010 compared with 2009

Consumer & Business Banking reported net income of \$3.7 billion, a decrease of \$263 million, or 7%, compared with the prior year.

Total net revenue was \$17.7 billion, down 2% compared with the prior year. The decrease was driven by lower deposit-related fees, largely offset by higher debit card income and a shift to wider-spread deposit products.

The provision for credit losses was \$630 million, down \$546 million compared with the prior year. The current-year provision reflected lower net charge-offs and a reduction of \$100 million to the allowance for loan losses due to lower estimated losses, compared with a \$300 million addition to the allowance for loan losses in the prior year. Net charge-offs were \$730 million, compared with \$876 million in the prior year.

Noninterest expense was \$10.7 billion, up 3% compared with the prior year, resulting from sales force increases in Business Banking and bank branches.

Management's discussion and analysis

Selected metrics

As of or for the year ended December 31,

(in millions, except ratios)

	2011	2010	2009
Business metrics			
Business banking origination volume	\$ 5,827	\$ 4,688	\$ 2,299
End-of-period loans	17,652	16,812	16,974
End-of-period deposits:			
Checking	147,779	131,702	123,220
Savings	191,891	170,604	156,140
Time and other	36,743	45,967	58,185
Total end-of-period deposits	376,413	348,273	337,545
Average loans	17,121	16,863	17,991
Average deposits:			
Checking	136,579	123,490	116,568
Savings	182,587	166,112	151,909
Time and other	41,574	51,149	76,550
Total average deposits	360,740	340,751	345,027
Deposit margin	2.82%	3.00%	2.92%
Average assets	\$ 29,729	\$ 29,307	\$ 29,791

Selected metrics

As of or for the year ended December 31,

(in millions, except ratios and where otherwise noted)

	2011	2010	2009
Credit data and quality statistics			
Net charge-offs	\$ 494	\$ 730	\$ 876
Net charge-off rate	2.89%	4.32%	4.87%
Allowance for loan losses	\$ 798	\$ 875	\$ 977
Nonperforming assets	710	846	839
Retail branch business metrics			
Investment sales volume	\$ 22,716	\$ 23,579	\$ 21,784
Client investment assets	137,853	133,114	120,507
% managed accounts	24%	20%	13%
Number of:			
Branches	5,508	5,268	5,154
Chase Private Client branch locations	262	16	16
ATMs	17,235	16,145	15,406
Personal bankers ^(a)	24,308	21,735	18,009
Sales specialists ^(a)	6,017	4,876	3,915
Client advisors	3,201	3,066	2,731
Active online customers (in thousands) ^(a)	17,334	16,855	14,627
Active mobile customers (in thousands) ^(a)	8,391	5,337	1,249
Chase Private Clients	21,723	4,242	2,933
Checking accounts (in thousands)	26,626	27,252	25,712

(a) In 2011, the classification of personal bankers, sales specialists, and active online and mobile customers was refined; as such, prior periods have been revised to conform with the current presentation.

Mortgage Production and Servicing

Selected income statement data

Year ended December 31,

(in millions, except ratios)	2011	2010	2009
Mortgage fees and related income	\$ 2,714	\$ 3,855	\$ 3,794
Other noninterest revenue	452	413	442
Net interest income	770	904	973
Total net revenue	3,936	5,172	5,209
Provision for credit losses	5	58	15
Noninterest expense	6,735	4,139	3,244
Income/(loss) before income tax expense/(benefit)	(2,804)	975	1,950
Net income/(loss)	\$ (1,832)	\$ 569	\$ 1,199
Overhead ratio	171%	80%	62%

Functional results

Production

Production revenue	\$ 3,395	\$ 3,440	\$ 2,115
Production-related net interest & other income	840	869	1,079
Production-related revenue, excluding repurchase losses	4,235	4,309	3,194
Production expense	1,895	1,613	1,575
Income, excluding repurchase losses	2,340	2,696	1,619
Repurchase losses	(1,347)	(2,912)	(1,612)
Income/(loss) before income tax expense/(benefit)	993	(216)	7

Servicing

Loan servicing revenue	4,134	4,575	4,942
Servicing-related net interest & other income	390	433	240
Servicing-related revenue	4,524	5,008	5,182
MSR asset modeled amortization	(1,904)	(2,384)	(3,279)
Default servicing expense	3,814	1,747	1,002
Core servicing expense	1,031	837	682
Income/(loss), excluding MSR risk management	(2,225)	40	219
MSR risk management, including related net interest income/(expense)	(1,572)	1,151	1,724
Income/(loss) before income tax expense/(benefit)	(3,797)	1,191	1,943
Net income/(loss)	\$ (1,832)	\$ 569	\$ 1,199

Selected income statement data

Year ended December 31,

(in millions)	2011	2010	2009
Supplemental mortgage fees and related income details			
Net production revenue:			
Production revenue	\$ 3,395	\$ 3,440	\$ 2,115
Repurchase losses	(1,347)	(2,912)	(1,612)
Net production revenue	2,048	528	503
Net mortgage servicing revenue:			
Operating revenue:			
Loan servicing revenue	4,134	4,575	4,942
Changes in MSR asset fair value due to modeled amortization	(1,904)	(2,384)	(3,279)
Total operating revenue	2,230	2,191	1,663
Risk management:			
Changes in MSR asset fair value due to inputs or assumptions in model	(7,117)	(2,268)	5,804
Derivative valuation adjustments and other	5,553	3,404	(4,176)
Total risk management ^(a)	(1,564)	1,136	1,628
Total net mortgage servicing revenue	666	3,327	3,291
Mortgage fees and related income	\$ 2,714	\$ 3,855	\$ 3,794

(a) Includes \$1.7 billion of fees and assessments, as well as other costs of foreclosure-related matters for the year ended December 31, 2011, and \$350 million for foreclosure-related matters for the year ended December 31, 2010.

(b) Predominantly includes: (1) changes in the MSR asset fair value due to changes in market interest rates and other modeled inputs and assumptions, and (2) changes in the value of the derivatives used to hedge the MSR asset. See Note 17 on pages 267–271 of this Annual Report for further information regarding changes in value of the MSR asset and related hedges.

2011 compared with 2010

Mortgage Production and Servicing reported a net loss of \$1.8 billion, compared with net income of \$569 million in the prior year.

Mortgage production pretax income was \$993 million, compared with a pretax loss of \$216 million in the prior year. Production-related revenue, excluding repurchase losses, was \$4.2 billion, a decrease of 2% from the prior year reflecting lower volumes and narrower margins when compared with the prior year. Production expense was \$1.9 billion, an increase of \$282 million, or 17%, reflecting a strategic shift to higher-cost retail originations both through the branch network and direct to the consumer. Repurchase losses were \$1.3 billion, compared with prior-year repurchase losses of \$2.9 billion, which included a \$1.6 billion increase in the repurchase reserve.

Management's discussion and analysis

Mortgage servicing, including MSR risk management, resulted in a pretax loss of \$3.8 billion, compared with pretax income of \$1.2 billion in the prior year. Servicing-related revenue was \$4.5 billion, a decline of 10% from the prior year, as a result of the decline in third-party loans serviced. MSR asset amortization was \$1.9 billion, compared with \$2.4 billion in the prior year; this reflected reduced amortization as a result of a lower MSR asset value. Servicing expense was \$4.8 billion, an increase of \$2.3 billion, driven by \$1.7 billion recorded for fees and assessments, and other costs of foreclosures-related matters, as well as higher core and default servicing costs. MSR risk management was a loss of \$1.6 billion, compared with income of \$1.2 billion in the prior year, driven by refinements to the valuation model and related inputs. See Note 17 on pages 267-271 of this Annual Report for further information regarding changes in value of the MSR asset and related hedges.

2010 compared with 2009

Mortgage Production and Servicing reported net income of \$569 million, a decrease of \$630 million, or 53%, from the prior year.

Mortgage production pretax loss was \$216 million, compared with pretax income of \$7 million in the prior year. Production-related revenue, excluding repurchase losses, was \$4.3 billion, an increase of 35% from the prior year reflecting wider mortgage margins and higher origination volumes when compared with the prior year. Production expense was \$1.6 billion, an increase of \$38 million, due to increased volumes. Repurchase losses were \$2.9 billion, compared with prior-year repurchase losses of \$1.6 billion. The current year losses included a \$1.6 billion increase in the repurchase reserve, reflecting higher estimated future repurchase demands.

Mortgage servicing, including MSR risk management, resulted in pretax income of \$1.2 billion, compared with pretax income of \$1.9 billion in the prior year. Servicing-related revenue was \$5.0 billion, a decline of 3% from the prior year, as a result of the decline in third-party loans serviced. MSR asset amortization was \$2.4 billion compared with \$3.3 billion in the prior year, reflecting reduced amortization as a result of a lower MSR asset value. Servicing expense was \$2.6 billion, an increase of \$900 million, driven by higher core and default servicing costs, including \$350 million for foreclosure-related matters. MSR risk management income was \$1.2 billion, compared with income of \$1.7 billion in the prior year.

Selected metrics

As of or for the year ended
December 31,

(in millions, except ratios and
where otherwise noted)

	2011	2010	2009
Selected balance sheet data			
End-of-period loans:			
Prime mortgage, including option ARMs ^(a)	\$16,891	\$14,186	\$11,964
Loans held-for-sale and loans at fair value ^(b)	12,694	14,863	12,920
Average loans:			
Prime mortgage, including option ARMs ^(a)	14,580	13,422	8,894
Loans held-for-sale and loans at fair value ^(b)	16,354	15,395	16,236
Average assets	59,891	57,778	51,317
Repurchase reserve (ending)	3,213	3,000	1,448
Credit data and quality statistics			
Net charge-offs:			
Prime mortgage, including option ARMs	5	41	14
Net charge-off rate:			
Prime mortgage, including option ARMs	0.03%	0.31%	0.17%
30+ day delinquency rate ^(c)	3.15	3.44	2.89
Nonperforming assets ^(d)	\$ 716	\$ 729	\$ 575
Business metrics (in billions)			
Origination volume by channel			
Retail	\$ 87.2	\$ 68.8	\$ 53.9
Wholesale ^(e)	0.5	1.3	3.6
Correspondent ^(f)	52.1	75.3	81.0
CNT (negotiated transactions)	5.8	10.2	12.2
Total origination volume	\$ 145.6	\$ 155.6	\$ 150.7
Application volume by channel			
Retail	\$ 137.2	\$ 115.1	\$ 90.9
Wholesale ^(e)	1.0	2.4	4.9
Correspondent ^(f)	66.5	97.3	110.8
Total application volume	\$ 204.7	\$ 214.8	\$ 206.6
Third-party mortgage loans serviced (ending)	\$ 902.2	\$ 967.5	\$1,082.1
Third-party mortgage loans serviced (average)	937.6	1,037.6	1,119.1
MSR net carrying value (ending)	7.2	13.6	15.5
Ratio of MSR net carrying value (ending) to third-party mortgage loans serviced (ending)	0.80%	1.41%	1.43%
Ratio of loan servicing revenue to third-party mortgage loans serviced (average)	0.44	0.44	0.44
MSR revenue multiple	1.82x	3.20x	3.25x

(a) Predominantly represents prime loans repurchased from Government National Mortgage Association ("Ginnie Mae") pools, which are insured by U.S. government agencies. See further discussion of loans repurchased from Ginnie Mae pools in Mortgage repurchase liability on pages 115-118 of this Annual Report.

(b) Loans at fair value consist of prime mortgages originated with the intent to sell that are accounted for at fair value and classified as trading assets on the Consolidated Balance Sheets. These loans

totaled \$12.7 billion, \$14.7 billion and \$12.5 billion at December 31, 2011, 2010 and 2009, respectively. Average balances of these loans totaled \$16.3 billion, \$15.2 billion and \$15.8 billion for the years ended December 31, 2011, 2010 and 2009, respectively.

- (c) At December 31, 2011, 2010 and 2009, excluded mortgage loans insured by U.S. government agencies of \$12.6 billion, \$10.3 billion and \$9.7 billion, respectively, that are 30 or more days past due. These amounts were excluded as reimbursement of insured amounts is proceeding normally. For further discussion, see Note 14 on pages 231–252 of this Annual Report which summarizes loan delinquency information.
- (d) At December 31, 2011, 2010 and 2009, nonperforming assets excluded: (1) mortgage loans insured by U.S. government agencies of \$11.5 billion, \$9.4 billion and \$9.0 billion, respectively, that are 90 or more days past due; and (2) real estate owned insured by U.S. government agencies of \$954 million, \$1.9 billion and \$579 million, respectively. These amounts were excluded as reimbursement of insured amounts is proceeding normally. For further discussion, see Note 14 on pages 231–252 of this Annual Report which summarizes loan delinquency information.
- (e) Includes rural housing loans sourced through brokers and correspondents, which are underwritten and closed in conjunction with the U.S. Department of Agriculture Rural Development, who acts as the guarantor in the transaction.
- (f) Represents the ratio of MSR net carrying value (ending) to third-party mortgage loans serviced (ending) divided by the ratio of loan servicing revenue to third-party mortgage loans serviced (average).

Mortgage Production and Servicing revenue comprises the following:

Net production revenue– Includes net gains or losses on originations and sales of prime and subprime mortgage loans, other production-related fees and losses related to the repurchase of previously-sold loans.

Net mortgage servicing revenue includes the following components:

(a) Operating revenue comprises:

- all gross income earned from servicing third-party mortgage loans including stated service fees, excess service fees, late fees and other ancillary fees; and
- modeled MSR asset amortization (or time decay).

(b) Risk management comprises:

- changes in MSR asset fair value due to market-based inputs such as interest rates, as well as updates to assumptions used in the MSR valuation model; and
- derivative valuation adjustments and other, which represents changes in the fair value of derivative instruments used to offset the impact of changes in interest rates to the MSR valuation model.

Mortgage origination channels comprise the following:

Retail – Borrowers buy or refinance a home through direct contact with a mortgage banker employed by the Firm using a branch office, the Internet or by phone. Borrowers are frequently referred to a mortgage banker by a banker in a Chase branch, real estate brokers, home builders or other third parties.

Wholesale – Third-party mortgage brokers refer loan application packages to the Firm. The Firm then underwrites and funds the loan. Brokers are independent loan originators that specialize in counseling applicants on available home financing options, but do not provide funding for loans. Chase materially eliminated broker-originated loans in 2008, with the exception of a small number of loans guaranteed by the U.S. Department of Agriculture under its Section 502 Guaranteed Loan program that serves low-and-moderate income families in small rural communities.

Correspondent– Banks, thrifts, other mortgage banks and other financial institutions sell closed loans to the Firm.

Correspondent negotiated transactions (“CNTs”) Mid-to-large-sized mortgage lenders, banks and bank-owned mortgage companies sell servicing to the Firm on an as-originated basis (excluding sales of bulk servicing). These transactions supplement traditional production channels and provide growth opportunities in the servicing portfolio in periods of stable and rising interest rates.

Real Estate Portfolios

Selected income statement data

Year ended December 31, (in millions, except ratios)	2011	2010	2009
Noninterest revenue	\$ 38	\$ 115	\$ (26)
Net interest income	4,554	5,432	6,546
Total net revenue	4,592	5,547	6,520
Provision for credit losses	3,575	8,231	13,563
Noninterest expense	1,521	1,627	1,847
Income/(loss) before income tax expense/(benefit)	(504)	(4,311)	(8,890)
Net income/(loss)	\$ (306)	\$ (2,493)	\$ (5,449)
Overhead ratio	33%	29%	28%

2011 compared with 2010

Real Estate Portfolios reported a net loss of \$306 million, compared with a net loss of \$2.5 billion in the prior year. The improvement was driven by a lower provision for credit losses, partially offset by lower net revenue.

Net revenue was \$4.6 billion, down by \$955 million, or 17%, from the prior year. The decrease was driven by a decline in net interest income as a result of lower loan balances due to portfolio runoff and narrower loan spreads.

The provision for credit losses was \$3.6 billion, compared with \$8.2 billion in the prior year, reflecting an improvement in charge-off trends and a net reduction of the allowance for loan losses of \$230 million. The net change in the allowance reflected a \$1.0 billion reduction related to the non-credit-impaired portfolios as estimated losses declined, predominately offset by an increase of \$770 million reflecting additional impairment of the Washington Mutual PCI portfolio due to higher-than-expected default frequency relative to modeled lifetime loss estimates. The prior-year provision reflected a higher impairment of the PCI portfolio and higher net charge-offs. See Consumer Credit Portfolio on pages 145–154 of this Annual Report for the net charge-off amounts and rates.

Noninterest expense was \$1.5 billion, down by \$106 million, or 7%, from the prior year, reflecting a decrease in foreclosed asset expense due to temporary delays in foreclosure activity.

2010 compared with 2009

Real Estate Portfolios reported a net loss of \$2.5 billion, compared with a net loss of \$5.4 billion in the prior year. The improvement was driven by a lower provision for credit losses, partially offset by lower net interest income.

Net revenue was \$5.5 billion, down by \$973 million, or 15%, from the prior year. The decrease was driven by a decline in net interest income as a result of lower loan balances, reflecting net portfolio runoff.

The provision for credit losses was \$8.2 billion, compared with \$13.6 billion in the prior year. The current-year provision reflected a \$1.9 billion reduction in net charge-

offs and a \$1.6 billion reduction in the allowance for the mortgage loan portfolios. This reduction in the allowance for loan losses included the effect of \$632 million of charge-offs related to an adjustment of the estimated net realizable value of the collateral underlying delinquent residential home loans. The remaining reduction of the allowance of approximately \$950 million was a result of an improvement in delinquencies and lower estimated losses, compared with prior year additions of \$3.6 billion for the home equity and mortgage portfolios. Additionally, the current-year provision reflected an addition to the allowance for loan losses of \$3.4 billion for the PCI portfolio, compared with a prior year addition of \$1.6 billion for this portfolio. See Consumer Credit Portfolio on pages 145–154 of this Annual Report for the net charge-off amounts and rates.

Noninterest expense was \$1.6 billion, down by \$220 million, or 12%, from the prior year, reflecting lower default-related expense.

PCI Loans

Included within Real Estate Portfolios are PCI loans that the Firm acquired in the Washington Mutual transaction. For PCI loans, the excess of the undiscounted gross cash flows expected to be collected over the carrying value of the loans (the “accretable yield”) is accreted into interest income at a level rate of return over the expected life of the loans.

The net spread between the PCI loans and the related liabilities are expected to be relatively constant over time, except for any basis risk or other residual interest rate risk that remains and for certain changes in the accretable yield percentage (e.g., from extended loan liquidation periods and from prepayments). As of December 31, 2011, the remaining weighted-average life of the PCI loan portfolio is expected to be 7.5 years. The loan balances are expected to decline more rapidly in the earlier years as the most troubled loans are liquidated, and more slowly thereafter as the remaining troubled borrowers have limited refinancing opportunities. Similarly, default and servicing expense are expected to be higher in the earlier years and decline over time as liquidations slow down.

To date the impact of the PCI loans on Real Estate Portfolios' net income has been negative. This is due to the current net spread of the portfolio, the provision for loan losses recognized subsequent to its acquisition, and the higher level of default and servicing expense associated with the portfolio. Over time, the Firm expects that this portfolio will contribute positively to net income.

For further information, see Note 14, PCI loans, on pages 248–249 of this Annual Report.

Selected metrics

As of or for the year ended
December 31, (in millions)

	2011	2010	2009
Loans excluding PCI			
End-of-period loans owned:			
Home equity	\$ 77,800	\$ 88,385	\$ 101,425
Prime mortgage, including option ARMs	44,284	49,768	55,891
Subprime mortgage	9,664	11,287	12,526
Other	718	857	671
Total end-of-period loans owned	\$ 132,466	\$ 150,297	\$ 170,513
Average loans owned:			
Home equity	\$ 82,886	\$ 94,835	\$ 108,333
Prime mortgage, including option ARMs	46,971	53,431	62,155
Subprime mortgage	10,471	12,729	13,901
Other	773	954	841
Total average loans owned	\$ 141,101	\$ 161,949	\$ 185,230
PCI loans^(a)			
End-of-period loans owned:			
Home equity	\$ 22,697	\$ 24,459	\$ 26,520
Prime mortgage	15,180	17,322	19,693
Subprime mortgage	4,976	5,398	5,993
Option ARMs	22,693	25,584	29,039
Total end-of-period loans owned	\$ 65,546	\$ 72,763	\$ 81,245
Average loans owned:			
Home equity	\$ 23,514	\$ 25,455	\$ 27,627
Prime mortgage	16,181	18,526	20,791
Subprime mortgage	5,170	5,671	6,350
Option ARMs	24,045	27,220	30,464
Total average loans owned	\$ 68,910	\$ 76,872	\$ 85,232
Total Real Estate Portfolios			
End-of-period loans owned:			
Home equity	\$ 100,497	\$ 112,844	\$ 127,945
Prime mortgage, including option ARMs	82,157	92,674	104,623
Subprime mortgage	14,640	16,685	18,519
Other	718	857	671
Total end-of-period loans owned	\$ 198,012	\$ 223,060	\$ 251,758
Average loans owned:			
Home equity	\$ 106,400	\$ 120,290	\$ 135,960
Prime mortgage, including option ARMs	87,197	99,177	113,410
Subprime mortgage	15,641	18,400	20,251
Other	773	954	841
Total average loans owned	\$ 210,011	\$ 238,821	\$ 270,462
Average assets	\$ 197,096	\$ 226,961	\$ 263,619
Home equity origination volume	1,127	1,203	2,479

(a) PCI loans represent loans acquired in the Washington Mutual transaction for which a deterioration in credit quality occurred between the origination date and JPMorgan Chase's acquisition date.

These loans were initially recorded at fair value and accrete interest income over the estimated lives of the loans as long as cash flows are reasonably estimable, even if the underlying loans are contractually past due.

Credit data and quality statistics

As of or for the year ended

December 31, (in millions, except ratios)	2011	2010	2009
Net charge-offs excluding PCI loans^(a)			
Home equity	\$ 2,472	\$ 3,444	\$ 4,682
Prime mortgage, including option ARMs	682	1,573	1,935
Subprime mortgage	626	1,374	1,648
Other	25	59	78
Total net charge-offs	\$ 3,805	\$ 6,450	\$ 8,343
Net charge-off rate excluding PCI loans^(a)			
Home equity	2.98%	3.63%	4.32%
Prime mortgage, including option ARMs	1.45	2.95	3.11
Subprime mortgage	5.98	10.82	11.86
Other	3.23	5.90	9.75
Total net charge-off rate excluding PCI loans	2.70	3.98	4.50
Net charge-off rate – reported:			
Home equity	2.32%	2.86%	3.45%
Prime mortgage, including option ARMs	0.78	1.59	1.70
Subprime mortgage	4.00	7.47	8.16
Other	3.23	5.90	9.75
Total net charge-off rate – reported	1.81	2.70	3.08
30+ day delinquency rate excluding PCI loans ^(b)	5.69%	6.45%	7.73%
Allowance for loan losses	\$ 14,429	\$ 14,659	\$ 12,752
Nonperforming assets ^(c)	6,638	8,424	10,347
Allowance for loan losses to ending loans retained	7.29%	6.57%	5.06%
Allowance for loan losses to ending loans retained excluding PCI loans ^(b)	6.58	6.47	6.55

- (a) Excludes the impact of PCI loans that were acquired as part of the Washington Mutual transaction. These loans were accounted for at fair value on the acquisition date, which incorporated management's estimate, as of that date, of credit losses over the remaining life of the portfolio. An allowance for loan losses of \$5.7 billion, \$4.9 billion and \$1.6 billion was recorded for these loans at December 31, 2011, 2010 and 2009, respectively; these amounts were also excluded from the applicable ratios. To date, no charge-offs have been recorded for these loans.
- (b) The delinquency rate for PCI loans was 23.30%, 28.20% and 27.62% at December 31, 2011, 2010 and 2009, respectively.
- (c) Excludes PCI loans that were acquired as part of the Washington Mutual transaction, which are accounted for on a pool basis. Since each pool is accounted for as a single asset with a single composite interest rate and an aggregate expectation of cash flows, the past-due status of the pools, or that of the individual loans within the pools, is not meaningful. Because the Firm is recognizing interest income on each pool of loans, they are all considered to be performing.

CARD SERVICES & AUTO

Card Services & Auto is one of the nation's largest credit card issuers, with over \$132 billion in credit card loans. Customers have over 65 million open credit card accounts (excluding the commercial card portfolio), and used Chase credit cards to meet over \$343 billion of their spending needs in 2011. Through its Merchant Services business, Chase Paymentech Solutions, Card is a global leader in payment processing and merchant acquiring. Consumers also can obtain loans through more than 17,200 auto dealerships and 2,000 schools and universities nationwide.

Effective July 1, 2011, Card includes Auto and Student Lending. For a further discussion of the business segment reorganization, see Business segment changes on page 79, and Note 33 on pages 300–303 of this Annual Report.

Selected income statement data – managed basis^(a)

Year ended December 31, (in millions, except ratios)	2011	2010	2009
Revenue			
Credit card income	\$ 4,127	\$ 3,514	\$ 3,613
All other income	765	764	93
Noninterest revenue ^(c)	4,892	4,278	3,706
Net interest income	14,249	16,194	19,493
Total net revenue ^(d)	19,141	20,472	23,199
Provision for credit losses	3,621	8,570	19,648
Noninterest expense			
Compensation expense	1,826	1,651	1,739
Noncompensation expense	5,818	5,060	4,362
Amortization of intangibles	401	467	516
Total noninterest expense ^(e)	8,045	7,178	6,617
Income/(loss) before income tax expense/ (benefit)	7,475	4,724	(3,066)
Income tax expense/ (benefit)	2,931	1,852	(1,273)
Net income/(loss)	\$ 4,544	\$ 2,872	\$ (1,793)
Memo: Net securitization income/(loss)	NA	NA	(474)
Financial ratios ^(a)			
Return on common equity	28%	16%	(10)%
Overhead ratio	42	35	29

- (a) Effective January 1, 2011, the commercial card business that was previously in TSS was transferred to Card. There is no material impact on the financial data; prior-year periods were not revised.
- (b) Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. As a result of the consolidation of the securitization trusts, reported and managed basis are equivalent for periods beginning after January 1, 2010. See Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 76–78 of this Annual Report for additional information. Also, for further details regarding the Firm's application and impact of the VIE guidance, see Note 16 on pages 256–267 of this Annual Report.
- (c) Included Commercial Card noninterest revenue of \$290 million for the year ended December 31, 2011.
- (d) Total net revenue included tax-equivalent adjustments associated with tax-exempt loans to certain qualified entities of \$2 million, \$7 million and \$13 million for the years ended December 31, 2011,

2010 and 2009, respectively.

- (e) Included Commercial Card noninterest expense of \$298 million for the year ended December 31, 2011.

NA: Not applicable

2011 compared with 2010

Net income was \$4.5 billion, compared with \$2.9 billion in the prior year. The increase was driven primarily by lower net charge-offs, partially offset by a lower reduction in the allowance for loan losses compared with the prior year.

Net revenue was \$19.1 billion, a decrease of \$1.3 billion, or 7%, from the prior year. Net interest income was \$14.2 billion, down by \$1.9 billion, or 12%. The decrease was driven by lower average loan balances, the impact of legislative changes, and a decreased level of fees. These decreases were largely offset by lower revenue reversals associated with lower charge-offs. Noninterest revenue was \$4.9 billion, an increase of \$614 million, or 14%, from the prior year. The increase was driven by the transfer of the Commercial Card business to Card from Treasury & Securities Services in the first quarter of 2011, higher net interchange income, and lower partner revenue-sharing due to the impact of the Kohl's portfolio sale. These increases were partially offset by lower revenue from fee-based products. Excluding the impact of the Commercial Card business, noninterest revenue increased 8%.

The provision for credit losses was \$3.6 billion, compared with \$8.6 billion in the prior year. The current-year provision reflected lower net charge-offs and an improvement in delinquency rates, as well as a reduction of \$3.9 billion to the allowance for loan losses due to lower estimated losses. The prior-year provision included a reduction of \$6.2 billion to the allowance for loan losses. The net charge-off rate was 3.99%, down from 7.12% in the prior year; the 30+ day delinquency rate was 2.32%, down from 3.23% in the prior year. Excluding the Washington Mutual and Commercial Card portfolios, the Credit Card net charge-off rate was 4.93%, down from 8.72% in the prior year; and the 30+ day delinquency rate was 2.54%, down from 3.66% in the prior year. The Auto net charge-off rate was 0.32%, down from 0.63% in the prior year. The Student net charge-off rate was 3.10%, up from 2.61% in the prior year.

Noninterest expense was \$8.0 billion, an increase of \$867 million, or 12%, from the prior year, due to higher marketing expense and the inclusion of the Commercial Card business. Excluding the impact of the Commercial Card business, noninterest expense increased 8%.

In May 2009, the CARD Act was enacted. The changes required by the CARD Act were fully implemented by the end of the fourth quarter of 2010. The total estimated reduction in net income resulting from the CARD Act was approximately \$750 million and \$300 million in 2011 and 2010, respectively.

COMMERCIAL BANKING

Commercial Banking delivers extensive industry knowledge, local expertise and dedicated service to more than 24,000 clients nationally, including corporations, municipalities, financial institutions and not-for-profit entities with annual revenue generally ranging from \$10 million to \$2 billion, and nearly 35,000 real estate investors/owners. CB partners with the Firm's other businesses to provide comprehensive solutions, including lending, treasury services, investment banking and asset management to meet its clients' domestic and international financial needs.

Commercial Banking is divided into four primary client segments: Middle Market Banking, Commercial Term Lending, Corporate Client Banking, and Real Estate Banking. Middle Market Banking covers corporate, municipal, financial institution and not-for-profit clients, with annual revenue generally ranging between \$10 million and \$500 million. Commercial Term Lending primarily provides term financing to real estate investors/owners for multifamily properties as well as financing office, retail and industrial properties. Corporate Client Banking, known as Mid-Corporate Banking prior to 2011, covers clients with annual revenue generally ranging between \$500 million and \$2 billion and focuses on clients that have broader investment banking needs. Real Estate Banking provides full-service banking to investors and developers of institutional-grade real estate properties. Lending and investment activity within the Community Development Banking and Chase Capital segments are included in other.

Selected income statement data

Year ended December 31, (in millions, except ratios)	2011	2010	2009
Revenue			
Lending- and deposit-related fees	\$1,081	\$1,099	\$1,081
Asset management, administration and commissions	136	144	140
All other income ^(a)	978	957	596
Noninterest revenue	2,195	2,200	1,817
Net interest income	4,223	3,840	3,903
Total net revenue ^(b)	6,418	6,040	5,720
Provision for credit losses	208	297	1,454
Noninterest expense			
Compensation expense	886	820	776
Noncompensation expense	1,361	1,344	1,359
Amortization of intangibles	31	35	41
Total noninterest expense	2,278	2,199	2,176
Income before income tax expense	3,932	3,544	2,090
Income tax expense	1,565	1,460	819
Net income	\$2,367	\$2,084	\$1,271
Revenue by product			
Lending ^(c)	\$3,455	\$2,749	\$2,663
Treasury services ^(d)	2,270	2,632	2,642
Investment banking	498	466	394
Other	195	193	21
Total Commercial Banking revenue	\$6,418	\$6,040	\$5,720
IB revenue, gross ^(e)	\$1,421	\$1,335	\$1,163
Revenue by client segment			
Middle Market Banking	\$3,145	\$3,060	\$3,055
Commercial Term Lending	1,168	1,023	875
Corporate Client Banking	1,261	1,154	1,102
Real Estate Banking	416	460	461
Other	428	343	227
Total Commercial Banking revenue	\$6,418	\$6,040	\$5,720
Financial ratios			
Return on common equity	30%	26%	16%
Overhead ratio	35	36	38

- (a) CB client revenue from investment banking products and commercial card transactions is included in all other income.
- (b) Total net revenue included tax-equivalent adjustments from income tax credits related to equity investments in designated community development entities that provide loans to qualified businesses in low-income communities, as well as tax-exempt income from municipal bond activity, totaling \$345 million, \$238 million, and \$170 million for the years ended December 31, 2011, 2010 and 2009, respectively.
- (c) Effective January 1, 2011, product revenue from commercial card and standby letters of credit transactions was included in lending. For the year ended December 31, 2011, the impact of the change was \$438 million. In prior-year periods, it was reported in treasury services.
- (d) Represents the total revenue related to investment banking products sold to CB clients.
- (e) Corporate Client Banking was known as Mid-Corporate Banking prior to January 1, 2011.

2011 compared with 2010

Record net income was \$2.4 billion, an increase of \$283 million, or 14%, from the prior year. The improvement was driven by higher net revenue and a reduction in the provision for credit losses, partially offset by an increase in noninterest expense.

Net revenue was a record \$6.4 billion, up by \$378 million, or 6%, compared with the prior year. Net interest income was \$4.2 billion, up by \$383 million, or 10%, driven by growth in liability and loan balances partially offset by spread compression on liability products. Noninterest revenue was \$2.2 billion, flat compared with the prior year.

On a client segment basis, revenue from Middle Market Banking was \$3.1 billion, an increase of \$85 million, or 3%, from the prior year due to higher liability and loan balances offset by spread compression on liability products and lower lending- and deposit-related fees. Revenue from Commercial Term Lending was \$1.2 billion, an increase of \$145 million, or 14%, and includes the full year impact of the purchase of a \$3.5 billion loan portfolio during the third quarter of 2010. Revenue from Corporate Client Banking was \$1.3 billion, an increase of \$107 million, or 9% due to growth in liability and loan balances and higher lending- and deposit-related fees, partially offset by spread compression on liability products. Revenue from Real Estate Banking was \$416 million, a decrease of \$44 million, or 10%, driven by a reduction in loan balances and lower gains on sales of loans and other real estate owned, partially offset by wider loan spreads.

The provision for credit losses was \$208 million, compared with \$297 million in the prior year. Net charge-offs were \$187 million (0.18% net charge-off rate) compared with \$909 million (0.94% net charge-off rate) in the prior year. The reduction was largely related to commercial real estate. The allowance for loan losses to period-end loans retained was 2.34%, down from 2.61% in the prior year. Nonaccrual loans were \$1.1 billion, down by \$947 million, or 47% from the prior year, largely as a result of commercial real estate repayments and loans sales.

Noninterest expense was \$2.3 billion, an increase of \$79 million, or 4% from the prior year, reflecting higher headcount-related expense.

2010 compared with 2009

Record net income was \$2.1 billion, an increase of \$813 million, or 64%, from the prior year. The increase was driven by a reduction in the provision for credit losses and higher net revenue.

Net revenue was a record \$6.0 billion, up by \$320 million, or 6%, compared with the prior year. Net interest income was \$3.8 billion, down by \$63 million, or 2%, driven by spread compression on liability products and lower loan balances, predominantly offset by growth in liability balances and wider loan spreads. Noninterest revenue was \$2.2 billion, an increase of \$383 million, or 21%, from the prior year, reflecting higher net gains from asset sales, higher lending- and deposit-related fees, an improvement in the market conditions impacting the value of investments held at fair value, higher investment banking fees and increased community development investment-related revenue.

On a client segment basis, revenue from Middle Market Banking was \$3.1 billion, flat compared with the prior year. Revenue from Commercial Term Lending was \$1.0 billion, an increase of \$148 million, or 17%, and included the impact of the purchase of a \$3.5 billion loan portfolio during the third quarter of 2010 and higher net gains from asset sales. Corporate Client Banking revenue was \$1.2 billion, an increase of \$52 million, or 5%, compared with the prior year due to wider loan spreads, higher lending- and deposit-related fees and higher investment banking fees offset partially by reduced loan balances. Real Estate Banking revenue was \$460 million, flat compared with the prior year.

The provision for credit losses was \$297 million, compared with \$1.5 billion in the prior year. The decline was mainly due to stabilization in the credit quality of the loan portfolio and refinements to credit loss estimates. Net charge-offs were \$909 million (0.94% net charge-off rate), compared with \$1.1 billion (1.02% net charge-off rate) in the prior year. The allowance for loan losses to period-end loans retained was 2.61%, down from 3.12% in the prior year. Nonaccrual loans were \$2.0 billion, a decrease of \$801 million, or 29%, from the prior year.

Noninterest expense was \$2.2 billion, an increase of \$23 million, or 1%, compared with the prior year reflecting higher headcount-related expense partially offset by lower volume-related expense.

Management's discussion and analysis

Selected metrics

Year ended December 31,
(in millions, except headcount
and ratios)

	2011	2010	2009
Selected balance sheet data (period-end)			
Total assets	\$ 158,040	\$ 142,646	\$ 130,280
Loans:			
Loans retained	111,162	97,900	97,108
Loans held-for-sale and loans at fair value	840	1,018	324
Total loans	\$ 112,002	\$ 98,918	\$ 97,432
Equity	8,000	8,000	8,000
Period-end loans by client segment			
Middle Market Banking	\$ 44,437	\$ 37,942	\$ 34,170
Commercial Term Lending	38,583	37,928	36,201
Corporate Client Banking	16,747	11,678	12,500
Real Estate Banking	8,211	7,591	10,619
Other	4,024	3,779	3,942
Total Commercial Banking loans	\$ 112,002	\$ 98,918	\$ 97,432
Selected balance sheet data (average)			
Total assets	\$ 146,230	\$ 133,654	\$ 135,408
Loans:			
Loans retained	103,462	96,584	106,421
Loans held-for-sale and loans at fair value	745	422	317
Total loans	\$ 104,207	\$ 97,006	\$ 106,738
Liability balance ^(b)	174,729	138,862	113,152
Equity	8,000	8,000	8,000
Average loans by client segment			
Middle Market Banking	\$ 40,759	\$ 35,059	\$ 37,459
Commercial Term Lending	38,107	36,978	36,806
Corporate Client Banking	13,993	11,926	15,951
Real Estate Banking	7,619	9,344	12,066
Other	3,729	3,699	4,456
Total Commercial Banking loans	\$ 104,207	\$ 97,006	\$ 106,738
Headcount	5,520	4,881	4,151

Year ended December 31,
(in millions, except headcount
and ratios)

	2011	2010	2009
Credit data and quality statistics			
Net charge-offs	\$ 187	\$ 909	\$ 1,089
Nonperforming assets			
Nonaccrual loans:			
Nonaccrual loans retained	1,036	1,964	2,764
Nonaccrual loans held-for-sale and loans held at fair value	17	36	37
Total nonaccrual loans	1,053	2,000	2,801
Assets acquired in loan satisfactions	85	197	188
Total nonperforming assets	1,138	2,197	2,989
Allowance for credit losses:			
Allowance for loan losses	2,603	2,552	3,025
Allowance for lending-related commitments	189	209	349
Total allowance for credit losses	2,792	2,761	3,374
Net charge-off rate ^(c)	0.18%	0.94%	1.02%
Allowance for loan losses to period-end loans retained	2.34	2.61	3.12
Allowance for loan losses to nonaccrual loans retained	251	130	109
Nonaccrual loans to total period- end loans	0.94	2.02	2.87

- (a) Corporate Client Banking was known as Mid-Corporate Banking prior to January 1, 2011.
- (b) Liability balances include deposits, as well as deposits that are swept to on-balance sheet liabilities (e.g., commercial paper, federal funds purchased, time deposits and securities loaned or sold under repurchase agreements) as part of customer cash management programs.
- (c) Allowance for loan losses of \$176 million, \$340 million and \$581 million was held against nonaccrual loans retained at December 31, 2011, 2010 and 2009, respectively.
- (d) Loans held-for-sale and loans at fair value were excluded when calculating the net charge-off rate.

TREASURY & SECURITIES SERVICES

Treasury & Securities Services is a global leader in transaction, investment and information services. TSS is one of the world's largest cash management providers and a leading global custodian. Treasury Services provides cash management, trade, wholesale card and liquidity products and services to small- and mid-sized companies, multinational corporations, financial institutions and government entities. TS partners with IB, CB, RFS and AM businesses to serve clients firmwide. Certain TS revenue is included in other segments' results. Worldwide Securities Services holds, values, clears and services securities, cash and alternative investments for investors and broker-dealers, and manages depositary receipt programs globally.

Year ended December 31, (in millions)	2011	2010	2009
Revenue by business			
Worldwide Securities Services ("WSS")			
Investor Services	\$ 3,019	\$ 2,869	\$ 2,836
Clearance, Collateral Management and Depository Receipts	842	814	806
Total WSS revenue	\$ 3,861	\$ 3,683	\$ 3,642
Treasury Services ("TS")			
Transaction Services	\$ 3,240	\$ 3,233	\$ 3,312
Trade Finance	601	465	390
Total TS revenue	\$ 3,841	\$ 3,698	\$ 3,702

Selected income statement data

Year ended December 31, (in millions, except ratio data)	2011	2010	2009
Revenue			
Lending- and deposit-related fees	\$ 1,240	\$ 1,256	\$ 1,285
Asset management, administration and commissions	2,748	2,697	2,631
All other income	556	804	831
Noninterest revenue	4,544	4,757	4,747
Net interest income	3,158	2,624	2,597
Total net revenue	7,702	7,381	7,344
Provision for credit losses	1	(47)	55
Credit allocation income/ (expense) ^(a)	8	(121)	(121)
Noninterest expense			
Compensation expense	2,824	2,734	2,544
Noncompensation expense	2,971	2,790	2,658
Amortization of intangibles	68	80	76
Total noninterest expense	5,863	5,604	5,278
Income before income tax expense	1,846	1,703	1,890
Income tax expense	642	624	664
Net income	\$ 1,204	\$ 1,079	\$ 1,226
Financial ratios			
Return on common equity	17%	17%	25%
Pretax margin ratio	24	23	26
Overhead ratio	76	76	72
Pre-provision profit ratio ^(b)	24	24	28

(a) IB manages traditional credit exposures related to GCB on behalf of IB and TSS. Effective January 1, 2011, IB and TSS share the economics related to the Firm's GCB clients. Included within this allocation are net revenue, provision for credit losses and expenses. The prior years reflected a reimbursement to IB for a portion of the total costs of managing the credit portfolio. IB recognizes this credit allocation as a component of all other income.

(b) Pre-provision profit ratio represents total net revenue less total noninterest expense divided by total net revenue. This reflects the operating performance before the impact of credit, and is another measure of performance for TSS against the performance of competitors.

2011 compared with 2010

Net income was \$1.2 billion, an increase of \$125 million, or 12%, from the prior year.

Net revenue was \$7.7 billion, an increase of \$321 million, or 4%, from the prior year. Excluding the impact of the Commercial Card business, net revenue was up 7%. Worldwide Securities Services net revenue was \$3.9 billion, an increase of \$178 million, or 5%. The increase was driven mainly by higher net interest income due to higher deposit balances and net inflows of assets under custody. Treasury Services net revenue was \$3.8 billion, an increase of \$143 million, or 4%. The increase was driven by higher deposit balances as well as higher trade loan volumes, partially offset by the transfer of the Commercial Card business to Card in the first quarter of 2011. Excluding the impact of the Commercial Card business, TS net revenue increased 10%.

TSS generated firmwide net revenue of \$10.2 billion, including \$6.4 billion by Treasury Services; of that amount, \$3.8 billion was recorded in Treasury Services, \$2.3 billion in Commercial Banking and \$265 million in other lines of business. The remaining \$3.9 billion of firmwide net revenue was recorded in Worldwide Securities Services.

The provision for credit losses was an expense of \$1 million, compared with a benefit of \$47 million in the prior year.

Noninterest expense was \$5.9 billion, an increase of \$259 million, or 5%, from the prior year. The increase was mainly driven by continued expansion into new markets and expenses related to exiting unprofitable business, partially offset by the transfer of the Commercial Card business to Card. Excluding the impact of the Commercial Card business, TSS noninterest expense increased 10%.

Results for 2011 included an \$8 million pretax benefit related to the traditional credit portfolio for GCB clients that are managed jointly by IB and TSS.

Management's discussion and analysis

2010 compared with 2009

Net income was \$1.1 billion, a decrease of \$147 million, or 12%, from the prior year. These results reflected higher noninterest expense partially offset by the benefit from the provision for credit losses and higher net revenue.

Net revenue was \$7.4 billion, an increase of \$37 million, or 1%, from the prior year. Treasury Services net revenue was \$3.7 billion, relatively flat compared with the prior year as lower spreads on liability products were offset by higher trade loan and card product volumes. Worldwide Securities Services net revenue was \$3.7 billion, relatively flat compared with the prior year as higher market levels and net inflows of assets under custody were offset by lower spreads in securities lending, lower volatility on foreign exchange, and lower balances on liability products.

TSS generated firmwide net revenue of \$10.3 billion, including \$6.6 billion by Treasury Services; of that amount, \$3.7 billion was recorded in Treasury Services, \$2.6 billion in Commercial Banking and \$247 million in other lines of business. The remaining \$3.7 billion of firmwide net revenue was recorded in Worldwide Securities Services.

The provision for credit losses was a benefit of \$47 million, compared with an expense of \$55 million in the prior year. The decrease in the provision expense was primarily due to an improvement in credit quality.

Noninterest expense was \$5.6 billion, up \$326 million, or 6%, from the prior year. The increase was driven by continued investment in new product platforms, primarily related to international expansion and higher performance-based compensation.

Selected metrics

Year ended December 31, (in millions, except headcount data)	2011	2010	2009
Selected balance sheet data (period-end)			
Total assets	\$ 68,665	\$ 45,481	\$ 38,054
Loans ^(a)	42,992	27,168	18,972
Equity	7,000	6,500	5,000
Selected balance sheet data (average)			
Total assets	\$ 56,151	\$ 42,494	\$ 35,963
Loans ^(a)	34,268	23,271	18,397
Liability balances	318,802	248,451	248,095
Equity	7,000	6,500	5,000
Headcount	27,825	29,073	26,609

Year ended December 31, (in millions, except ratio data, and where otherwise noted)	2011	2010	2009
Credit data and quality statistics			
Net charge-offs	\$ —	\$ 1	\$ 19
Nonaccrual loans	4	12	14
Allowance for credit losses:			
Allowance for loan losses	65	65	88
Allowance for lending-related commitments	49	51	84
Total allowance for credit losses	114	116	172
Net charge-off rate	—%	—%	0.10%
Allowance for loan losses to period-end loans	0.15	0.24	0.46
Allowance for loan losses to nonaccrual loans	NM	NM	NM
Nonaccrual loans to period-end loans	0.01	0.04	0.07
WSS business metrics			
Assets under custody ("AUC") by assets class (period-end) (in billions)			
Fixed income	\$ 10,926	\$ 10,364	\$ 10,073
Equity	4,878	4,850	4,090
Other ^(b)	1,066	906	722
Total AUC	\$ 16,870	\$ 16,120	\$ 14,885
Liability balances (average)	100,660	79,457	86,936
TS business metrics			
TS liability balances (average)	218,142	168,994	161,159
Trade finance loans (period-end)	36,696	21,156	10,227

- (a) Loan balances include trade finance loans, wholesale overdrafts and commercial card. Effective January 1, 2011, the commercial card loan business (of approximately \$1.2 billion) that was previously in TSS was transferred to Card. There is no material impact on the financial data; the prior years were not revised.
- (b) Consists of mutual funds, unit investment trusts, currencies, annuities, insurance contracts, options and nonsecurities contracts.

Selected metrics

Year ended December 31,
(in millions, except where
otherwise noted)

	2011	2010	2009
International metrics			
Net revenue by geographic region^(a)			
Asia/Pacific	\$ 1,235	\$ 978	\$ 845
Latin America/Caribbean	329	257	221
Europe/Middle East/Africa	2,658	2,389	2,462
North America	3,480	3,757	3,816
Total net revenue	\$ 7,702	\$ 7,381	\$ 7,344
Average liability balances^(b)			
Asia/Pacific	\$ 43,524	\$ 32,862	\$ 28,501
Latin America/Caribbean	12,625	11,558	8,231
Europe/Middle East/Africa	123,920	102,014	101,683
North America	138,733	102,017	109,680
Total average liability balances	\$ 318,802	\$ 248,451	\$ 248,095
Trade finance loans (period-end)^(a)			
Asia/Pacific	\$ 19,280	\$ 11,834	\$ 4,519
Latin America/Caribbean	6,254	3,628	2,458
Europe/Middle East/Africa	9,726	4,874	2,171
North America	1,436	820	1,079
Total trade finance loans	\$ 36,696	\$ 21,156	\$ 10,227
AUC (period-end)(in billions)^(b)			
North America	\$ 9,735	\$ 9,836	\$ 9,391
All other regions	7,135	6,284	5,494
Total AUC	\$ 16,870	\$ 16,120	\$ 14,885
TSS firmwide disclosures^(b)			
TS revenue – reported	\$ 3,841	\$ 3,698	\$ 3,702
TS revenue reported in CB	2,270	2,632	2,642
TS revenue reported in other lines of business	265	247	245
TS firmwide revenue ^(b)	6,376	6,577	6,589
WSS revenue	3,861	3,683	3,642
TSS firmwide revenue^(b)	\$ 10,237	\$ 10,260	\$ 10,231
TSS total foreign exchange ("FX") revenue^(d)			
	658	636	661
TS firmwide liability balances (average)^(e)			
	393,022	308,028	274,472
TSS firmwide liability balances (average)^(e)			
	493,531	387,313	361,247
Number of:			
U.S.\$ ACH transactions originated	3,906	3,892	3,896
Total U.S.\$ clearing volume (in thousands)	129,417	122,123	113,476
International electronic funds transfer volume (in thousands)	250,537	232,453	193,348
Wholesale check volume	2,333	2,060	2,184
Wholesale cards issued (in thousands) ^(f)	25,187	29,785	27,138

(a) Total net revenue, average liability balances, trade finance loans and AUC are based on the domicile of the client.

(b) TSS firmwide metrics include revenue recorded in CB, Consumer & Business Banking and AM lines of business and net TSS FX revenue (it excludes TSS FX revenue recorded in IB). In order to capture the firmwide impact of TS and TSS products and revenue, management reviews firmwide metrics in assessing financial performance of TSS.

Firmwide metrics are necessary in order to understand the aggregate TSS business.

- (c) Effective January 1, 2011, certain CB revenues were excluded in the TS firmwide metrics; they are instead directly captured within CB's lending revenue by product. The impact of this change was \$438 million for the year ended December 31, 2011. In previous years, these revenues were included in CB's treasury services revenue by product.
- (d) IB executes FX transactions on behalf of TSS customers under revenue sharing agreements. FX revenue generated by TSS customers is recorded in TSS and IB. TSS Total FX revenue reported above is the gross (pre-split) FX revenue generated by TSS customers. However, TSS firmwide revenue includes only the FX revenue booked in TSS, i.e., it does not include the portion of TSS FX revenue recorded in IB.
- (e) Firmwide liability balances include liability balances recorded in CB.
- (f) International electronic funds transfer includes non-U.S. dollar Automated Clearing House ("ACH") and clearing volume.
- (g) Wholesale cards issued and outstanding include commercial, stored value, prepaid and government electronic benefit card products. Effective January 1, 2011, the commercial card portfolio was transferred from TSS to Card.

Description of a business metric within TSS:

Liability balances include deposits, as well as deposits that are swept to on-balance sheet liabilities (e.g., commercial paper, federal funds purchased, time deposits and securities loaned or sold under repurchase agreements) as part of customer cash management programs.

Description of selected products and services within TSS:

Investor Services includes primarily custody, fund accounting and administration, and securities lending products sold principally to asset managers, insurance companies and public and private investment funds.

Clearance, Collateral Management & Depositary Receipts primarily includes broker-dealer clearing and custody services, including tri-party repo transactions, collateral management products, and depositary bank services for American and global depositary receipt programs.

Transaction Services includes a broad range of products that enable clients to manage payments and receipts, as well as invest and manage funds. Products include U.S. dollar and multi-currency clearing, ACH, lockbox, disbursement and reconciliation services, check deposits, and currency related services.

Trade Finance enables the management of cross-border trade for bank and corporate clients. Products include loans directly tied to goods crossing borders, export/import loans, commercial letters of credit, standby letters of credit, and supply chain finance.

ASSET MANAGEMENT

Asset Management, with assets under supervision of \$1.9 trillion, is a global leader in investment and wealth management. AM clients include institutions, retail investors and high-net-worth individuals in every major market throughout the world. AM offers global investment management in equities, fixed income, real estate, hedge funds, private equity and liquidity products, including money market instruments and bank deposits. AM also provides trust and estate, banking and brokerage services to high-net-worth clients, and retirement services for corporations and individuals. The majority of AM's client assets are in actively managed portfolios.

Selected income statement data

Year ended December 31, (in millions, except ratios)	2011	2010	2009
Revenue			
Asset management, administration and commissions	\$ 6,748	\$ 6,374	\$ 5,621
All other income	1,147	1,111	751
Noninterest revenue	7,895	7,485	6,372
Net interest income	1,648	1,499	1,593
Total net revenue	9,543	8,984	7,965
Provision for credit losses	67	86	188
Noninterest expense			
Compensation expense	4,152	3,763	3,375
Noncompensation expense	2,752	2,277	2,021
Amortization of intangibles	98	72	77
Total noninterest expense	7,002	6,112	5,473
Income before income tax expense	2,474	2,786	2,304
Income tax expense	882	1,076	874
Net income	\$ 1,592	\$ 1,710	\$ 1,430
Revenue by client segment			
Private Banking	\$ 5,116	\$ 4,860	\$ 4,320
Institutional	2,273	2,180	2,065
Retail	2,154	1,944	1,580
Total net revenue	\$ 9,543	\$ 8,984	\$ 7,965
Financial ratios			
Return on common equity	25%	26%	20%
Overhead ratio	73	68	69
Pretax margin ratio	26	31	29

2011 compared with 2010

Net income was \$1.6 billion, a decrease of \$118 million, or 7%, from the prior year. These results reflected higher noninterest expense, largely offset by higher net revenue and a lower provision for credit losses.

Net revenue was \$9.5 billion, an increase of \$559 million, or 6%, from the prior year. Noninterest revenue was \$7.9 billion, up by \$410 million, or 5%, due to net inflows to products with higher margins and the effect of higher market levels, partially offset by lower performance fees

and lower loan-related revenue. Net interest income was \$1.6 billion, up by \$149 million, or 10%, due to higher deposit and loan balances, partially offset by narrower deposit spreads.

Revenue from Private Banking was \$5.1 billion, up 5% from the prior year due to higher deposit and loan balances and higher brokerage revenue, partially offset by narrower deposit spreads and lower loan-related revenue. Revenue from Institutional was \$2.3 billion, up 4% due to net inflows to products with higher margins and the effect of higher market levels. Revenue from Retail was \$2.2 billion, up 11% due to net inflows to products with higher margins and the effect of higher market levels.

The provision for credit losses was \$67 million, compared with \$86 million in the prior year.

Noninterest expense was \$7.0 billion, an increase of \$890 million, or 15%, from the prior year, due to higher headcount-related expense and non-client-related litigation, partially offset by lower performance-based compensation.

2010 compared with 2009

Net income was \$1.7 billion, an increase of \$280 million, or 20%, from the prior year, due to higher net revenue and a lower provision for credit losses, largely offset by higher noninterest expense.

Net revenue was a record \$9.0 billion, an increase of \$1.0 billion, or 13%, from the prior year. Noninterest revenue was \$7.5 billion, an increase of \$1.1 billion, or 17%, due to the effect of higher market levels, net inflows to products with higher margins, higher loan originations, and higher performance fees. Net interest income was \$1.5 billion, down by \$94 million, or 6%, from the prior year, due to narrower deposit spreads, largely offset by higher deposit and loan balances.

Revenue from Private Banking was \$4.9 billion, up 13% from the prior year due to higher loan originations, higher deposit and loan balances, the effect of higher market levels and net inflows to products with higher margins, partially offset by narrower deposit spreads. Revenue from Institutional was \$2.2 billion, up 6% due to the effect of higher market levels, partially offset by liquidity outflows. Revenue from Retail was \$1.9 billion, up 23% due to the effect of higher market levels and net inflows to products with higher margins, partially offset by lower valuations of seed capital investments.

The provision for credit losses was \$86 million, compared with \$188 million in the prior year, reflecting an improving credit environment.

Noninterest expense was \$6.1 billion, an increase of \$639 million, or 12%, from the prior year, resulting from increased headcount and higher performance-based compensation.

Selected metrics

Business metrics

As of or for the year ended
December 31, (in millions,
except headcount, ranking data
and where otherwise noted)

	2011	2010	2009
Number of:			
Client advisors ^(a)	2,444	2,281	1,936
Retirement planning services participants (in thousands)	1,798	1,580	1,628
JPMorgan Securities brokers	439	415	376
% of customer assets in 4 & 5 Star Fund ^(b)	43%	49%	42%
% of AUM in 1 st and 2 nd quartiles ^(c)			
1 year	48	67	57
3 years	72	72	62
5 years	78	80	74

Selected balance sheet data (period-end)

Total assets	\$ 86,242	\$68,997	\$64,502
Loans	57,573	44,084	37,755
Equity	6,500	6,500	7,000

Selected balance sheet data (average)

Total assets	\$ 76,141	\$65,056	\$60,249
Loans	50,315	38,948	34,963
Deposits	106,421	86,096	77,005
Equity	6,500	6,500	7,000

Headcount	18,036	16,918	15,136
-----------	--------	--------	--------

Credit data and quality statistics

Net charge-offs	\$ 92	\$ 76	\$ 117
Nonaccrual loans	317	375	580

Allowance for credit losses:

Allowance for loan losses	209	267	269
Allowance for lending-related commitments	10	4	9

Total allowance for credit losses	219	271	278
-----------------------------------	-----	-----	-----

Net charge-off rate	0.18%	0.20%	0.33%
---------------------	-------	-------	-------

Allowance for loan losses to period-end loans	0.36	0.61	0.71
---	------	------	------

Allowance for loan losses to nonaccrual loans	66	71	46
---	----	----	----

Nonaccrual loans to period-end loans	0.55	0.85	1.54
--------------------------------------	------	------	------

- (a) Effective January 1, 2011, the methodology used to determine client advisors was revised. Prior periods have been revised.
(b) Derived from Morningstar for the U.S., the U.K., Luxembourg, France, Hong Kong and Taiwan; and Nomura for Japan.
(c) Quartile ranking sourced from: Lipper for the U.S. and Taiwan; Morningstar for the U.K., Luxembourg, France and Hong Kong; and Nomura for Japan.

AM's client segments comprise the following:

Private Banking offers investment advice and wealth management services to high- and ultra-high-net-worth individuals, families, money managers, business owners and small corporations worldwide, including investment management, capital markets and risk management, tax and estate planning, banking, capital raising and specialty-wealth advisory services.

Institutional brings comprehensive global investment services – including asset management, pension analytics, asset-liability management and active risk-budgeting strategies – to corporate and public institutions, endowments, foundations, not-for-profit organizations and governments worldwide.

Retail provides worldwide investment management services and retirement planning and administration, through third-party and direct distribution of a full range of investment vehicles.

J.P. Morgan Asset Management has two high-level measures of its overall fund performance.

- Percentage of assets under management in funds rated 4- and 5-stars (three years). Mutual fund rating services rank funds based on their risk-adjusted performance over various periods. A 5-star rating is the best and represents the top 10% of industry wide ranked funds. A 4-star rating represents the next 22% of industry wide ranked funds. The worst rating is a 1-star rating.
- Percentage of assets under management in first- or second- quartile funds (one, three and five years). Mutual fund rating services rank funds according to a peer-based performance system, which measures returns according to specific time and fund classification (small-, mid-, multi- and large-cap).

Assets under supervision

2011 compared with 2010

Assets under supervision were \$1.9 trillion at December 31, 2011, an increase of \$81 billion, or 4%, from the prior year. Assets under management were \$1.3 trillion, an increase of \$38 billion, or 3%. Both increases were due to net inflows to long-term and liquidity products, partially offset by the impact of lower market levels. Custody, brokerage, administration and deposit balances were \$585 billion, up by \$43 billion, or 8%, due to deposit and custody inflows.

2010 compared with 2009

Assets under supervision were \$1.8 trillion at December 31, 2010, an increase of \$139 billion, or 8%, from the prior year. Assets under management were \$1.3 trillion, an increase of \$49 billion, or 4%, due to the effect of higher market levels and net inflows in long-term products, largely offset by net outflows in liquidity products. Custody, brokerage, administration and deposit balances were \$542 billion, up by \$90 billion, or 20%, due to custody and brokerage inflows and the effect of higher market levels.

Management's discussion and analysis

Assets under supervision^(a)

As of or the year ended December 31, (in billions)	2011	2010	2009
Assets by asset class			
Liquidity	\$ 515	\$ 497	\$ 591
Fixed income	336	289	226
Equity and multi-asset	372	404	339
Alternatives	113	108	93
Total assets under management	1,336	1,298	1,249
Custody/brokerage/ administration/deposits	585	542	452
Total assets under supervision	\$ 1,921	\$ 1,840	\$ 1,701
Assets by client segment			
Private Banking	\$ 291	\$ 284	\$ 270
Institutional ^(b)	722	703	731
Retail ^(b)	323	311	248
Total assets under management	\$ 1,336	\$ 1,298	\$ 1,249
Private Banking	\$ 781	\$ 731	\$ 636
Institutional ^(b)	723	703	731
Retail ^(b)	417	406	334
Total assets under supervision	\$ 1,921	\$ 1,840	\$ 1,701
Mutual fund assets by asset class			
Liquidity	\$ 458	\$ 446	\$ 539
Fixed income	107	92	67
Equity and multi-asset	147	169	143
Alternatives	8	7	9
Total mutual fund assets	\$ 720	\$ 714	\$ 758

(a) Excludes assets under management of American Century Companies, Inc., in which the Firm sold its ownership interest on August 31, 2011. The Firm previously had an ownership interest of 41% and 42% in American Century Companies, Inc., whose AUM is not included in the table above, at December 31, 2010 and 2009, respectively.

(b) In 2011, the client hierarchy used to determine asset classification was revised, and the prior-year periods have been revised.

Year ended December 31, (in billions)	2011	2010	2009
Assets under management rollforward			
Beginning balance	\$ 1,298	\$ 1,249	\$ 1,133
Net asset flows:			
Liquidity	18	(89)	(23)
Fixed income	40	50	34
Equity, multi-asset and alternatives	13	19	17
Market/performance/other impacts	(33)	69	88
Ending balance, December 31	\$ 1,336	\$ 1,298	\$ 1,249
Assets under supervision rollforward			
Beginning balance	\$ 1,840	\$ 1,701	\$ 1,496
Net asset flows	123	28	50
Market/performance/other impacts	(42)	111	155
Ending balance, December 31	\$ 1,921	\$ 1,840	\$ 1,701

International metrics

Year ended December 31, (in billions, except where otherwise noted)	2011	2010	2009
Total net revenue^(a)(in millions)^(a)			
Europe/Middle East/Africa	\$ 1,704	\$ 1,642	\$ 1,380
Asia/Pacific	971	925	752
Latin America/Caribbean	808	541	426
North America	6,060	5,876	5,407
Total net revenue	\$ 9,543	\$ 8,984	\$ 7,965
Assets under management			
Europe/Middle East/Africa	\$ 278	\$ 282	\$ 293
Asia/Pacific	105	111	99
Latin America/Caribbean	34	35	19
North America	919	870	838
Total assets under management	\$ 1,336	\$ 1,298	\$ 1,249
Assets under supervision			
Europe/Middle East/Africa	\$ 329	\$ 331	\$ 338
Asia/Pacific	139	147	125
Latin America/Caribbean	89	84	55
North America	1,364	1,278	1,183
Total assets under supervision	\$ 1,921	\$ 1,840	\$ 1,701

(a) Regional revenue is based on the domicile of the client.

CORPORATE/PRIVATE EQUITY

The Corporate/Private Equity sector comprises Private Equity, Treasury, the Chief Investment Office ("CIO"), corporate staff units and expense that is centrally managed. Treasury and CIO manage capital, liquidity and structural risks of the Firm. The corporate staff units include Central Technology and Operations, Internal Audit, Executive Office, Finance, Human Resources, Marketing & Communications, Legal & Compliance, Corporate Real Estate and General Services, Risk Management, Corporate Responsibility and Strategy & Development. Other centrally managed expense includes the Firm's occupancy and pension-related expense, net of allocations to the business.

Selected income statement data

Year ended December 31, (in millions, except headcount)	2011	2010	2009
Revenue			
Principal transactions	\$ 1,434	\$ 2,208	\$ 1,574
Securities gains	1,600	2,898	1,139
All other income	604	253	58
Noninterest revenue	3,638	5,359	2,771
Net interest income	505	2,063	3,863
Total net revenue ^(a)	4,143	7,422	6,634
Provision for credit losses	(36)	14	80
Noninterest expense			
Compensation expense	2,425	2,357	2,811
Noncompensation expense	6,884	8,788	3,597
Merger costs	—	—	481
Subtotal	9,309	11,145	6,889
Net expense allocated to other businesses	(5,160)	(4,790)	(4,994)
Total noninterest expense	4,149	6,355	1,895
Income before income tax expense/(benefit) and extraordinary gain	30	1,053	4,659
Income tax expense/(benefit)	(772)	(205)	1,705
Income before extraordinary gain	802	1,258	2,954
Extraordinary gain ^(b)	—	—	76
Net income	\$ 802	\$ 1,258	\$ 3,030
Total net revenue			
Private equity	\$ 836	\$ 1,239	\$ 18
Corporate	3,307	6,183	6,616
Total net revenue	\$ 4,143	\$ 7,422	\$ 6,634
Net income			
Private equity	\$ 391	\$ 588	\$ (78)
Corporate ^(c)	411	670	3,108
Total net income	\$ 802	\$ 1,258	\$ 3,030
Total assets (period-end)	\$693,153	\$ 526,588	\$ 595,877
Headcount	22,117	20,030	20,119

(a) Total net revenue included tax-equivalent adjustments, predominantly due to tax-exempt income from municipal bond investments of \$298 million, \$226 million and \$151 million for the years ended December 31, 2011, 2010 and 2009, respectively.

- (b) Included litigation expense of \$3.2 billion and \$5.7 billion for the years ended December 31, 2011 and 2010, respectively, compared with net benefits of \$0.3 billion for the year ended December 31, 2009.
- (c) Includes tax benefits recognized upon the resolution of tax audits.
- (d) On September 25, 2008, JPMorgan Chase acquired the banking operations of Washington Mutual from the FDIC for \$1.9 billion. The acquisition resulted in negative goodwill, and accordingly, the Firm recorded an extraordinary gain. A preliminary gain of \$1.9 billion was recognized at December 31, 2008. As a result of the final refinement of the purchase price allocation in 2009, the Firm recognized a \$76 million increase in the extraordinary gain. The final total extraordinary gain that resulted from the Washington Mutual transaction was \$2.0 billion.
- (e) 2009 included merger costs and the extraordinary gain related to the Washington Mutual transaction, as well as items related to the Bear Stearns merger, including merger costs, asset management liquidation costs and JPMorgan Securities broker retention expense.

2011 compared with 2010

Net income was \$802 million, compared with \$1.3 billion in the prior year.

Private Equity net income was \$391 million, compared with \$588 million in the prior year. Net revenue was \$836 million, a decrease of \$403 million, primarily related to net write-downs on privately-held investments and the absence of prior-year gains from sales. Noninterest expense was \$238 million, a decrease of \$85 million from the prior year.

Corporate reported net income of \$411 million, compared with net income of \$670 million in the prior year. Net revenue was \$3.3 billion, including \$1.6 billion of securities gains. Net interest income in 2011 was lower compared with 2010, primarily driven by repositioning of the investment securities portfolio and lower funding benefits from financing the portfolio.

Noninterest expense was \$4.1 billion which included \$3.2 billion of litigation expense, predominantly for mortgage-related matters. Noninterest expense in the prior year was \$6.4 billion, which included \$5.7 billion of litigation expense.

2010 compared with 2009

Net income was \$1.3 billion compared with \$3.0 billion in the prior year. The decrease was driven by higher litigation expense, partially offset by higher net revenue.

Net income for Private Equity was \$588 million, compared with a net loss of \$78 million in the prior year, reflecting the impact of improved market conditions on certain investments in the portfolio. Net revenue was \$1.2 billion compared with \$18 million in the prior year, reflecting private equity gains of \$1.3 billion compared with losses of \$54 million in 2009. Noninterest expense was \$323 million, an increase of \$182 million, driven by higher compensation expense.

Net income for Corporate was \$670 million, compared with \$3.1 billion in the prior year. Results for 2010 reflect after-tax litigation expense of \$3.5 billion, lower net interest

Management's discussion and analysis

income and trading gains, partially offset by a higher level of securities gains, primarily driven by repositioning of the investment securities portfolio in response to changes in the interest rate environment and to rebalance exposure. The prior year included merger-related net loss of \$635 million and a \$419 million FDIC assessment.

Treasury and CIO

Selected income statement and balance sheet data

Year ended December 31, (in millions)	2011	2010	2009
Securities gain ^(a)	\$ 1,385	\$ 2,897	\$ 1,147
Investment securities portfolio (average)	330,885	323,673	324,037
Investment securities portfolio (ending)	355,605	310,801	340,163
Mortgage loans (average)	13,006	9,004	7,427
Mortgage loans (ending)	13,375	10,739	8,023

(a) Reflects repositioning of the Corporate investment securities portfolio.

For further information on the investment securities portfolio, see Note 3 and Note 12 on pages 184–198 and 225–230, respectively, of this Annual Report. For further information on CIO VaR and the Firm's nontrading interest rate-sensitive revenue at risk, see the Market Risk Management section on pages 158–163 of this Annual Report.

Private Equity Portfolio

Selected income statement and balance sheet data

Year ended December 31, (in millions)	2011	2010	2009
Private equity gains/(losses)			
Realized gains	\$ 1,842	\$ 1,409	\$ 109
Unrealized gains/(losses) ^(b)	(1,305)	(302)	(81)
Total direct investments	537	1,107	28
Third-party fund investments	417	241	(82)
Total private equity gains/ (losses) ^(c)	\$ 954	\$ 1,348	\$ (54)

Private equity portfolio information^(c)

Direct investments

December 31, (in millions)	2011	2010	2009
Publicly held securities			
Carrying value	\$ 805	\$ 875	\$ 762
Cost	573	732	743
Quoted public value	896	935	791
Privately held direct securities			
Carrying value	4,597	5,882	5,104
Cost	6,793	6,887	5,959
Third-party fund investments ^(d)			
Carrying value	2,283	1,980	1,459
Cost	2,452	2,404	2,079
Total private equity portfolio			
Carrying value	\$ 7,685	\$ 8,737	\$ 7,325
Cost	\$ 9,818	\$ 10,023	\$ 8,781

- (a) Unrealized gains/(losses) contain reversals of unrealized gains and losses that were recognized in prior periods and have now been realized.
- (b) Included in principal transactions revenue in the Consolidated Statements of Income.
- (c) For more information on the Firm's policies regarding the valuation of the private equity portfolio, see Note 3 on pages 184–198 of this Annual Report.
- (d) Unfunded commitments to third-party private equity funds were \$789 million, \$1.0 billion and \$1.5 billion at December 31, 2011, 2010 and 2009, respectively.

2011 compared with 2010

The carrying value of the private equity portfolio at December 31, 2011, was \$7.7 billion, down from \$8.7 billion at December 31, 2010. The decrease in the portfolio is predominantly driven by sales of investments, partially offset by new investments. The portfolio represented 5.7% of the Firm's stockholders' equity less goodwill at December 31, 2011, down from 6.9% at December 31, 2010.

physical commodities inventories generally carried at the lower of cost or fair value. Trading assets – debt and equity instruments decreased, driven by client market-making activity in IB; this resulted in lower levels of equity securities, U.S. government and agency mortgage-backed securities, and non-U.S. government securities. For additional information, refer to Note 3 on pages 184–198 of this Annual Report.

Trading assets and liabilities- derivative receivables and payables

The Firm uses derivative instruments predominantly for market-making activities. Derivatives enable customers and the Firm to manage their exposure to fluctuations in interest rates, currencies and other markets. The Firm also uses derivative instruments to manage its market and credit exposure. Derivative receivables and payables increased, predominantly due to increases in interest rate derivative balances driven by declining interest rates, and higher commodity derivative balances driven by price movements in base metals and energy. For additional information, refer to Derivative contracts on pages 141–144, and Note 3 and Note 6 on pages 184–198 and 202–210, respectively, of this Annual Report.

Securities

Substantially all of the securities portfolio is classified as available-for-sale (“AFS”) and used primarily to manage the Firm’s exposure to interest rate movements and to invest cash resulting from excess liquidity. Securities increased, largely due to repositioning of the portfolio in Corporate in response to changes in the market environment. This repositioning increased the levels of non-U.S. government debt and residential mortgage-backed securities, as well as collateralized loan obligations and commercial mortgage-backed securities, and reduced the levels of U.S. government agency securities. For additional information related to securities, refer to the discussion in the Corporate/Private Equity segment on pages 107–108, and Note 3 and Note 12 on pages 184–198 and 225–230, respectively, of this Annual Report.

Loans and allowance for loan losses

The Firm provides loans to a variety of customers, from large corporate and institutional clients to individual consumers and small businesses. Loans increased, reflecting continued growth in client activity across all of the Firm’s wholesale businesses and regions. This increase was offset by a decline in consumer, excluding credit card loan balances, due to paydowns, portfolio run-off and charge-offs, and in credit card loans, due to higher repayment rates, run-off of the Washington Mutual portfolio and the Firm’s sale of the Kohl’s portfolio.

The allowance for loan losses decreased predominantly due to lower estimated losses in the credit card loan portfolio, reflecting improved delinquency trends and lower levels of credit card outstandings, and the impact of loan sales in the wholesale portfolio. For a more detailed discussion of the loan portfolio and the allowance for loan losses, refer to

Credit Risk Management on pages 132–157, and Notes 3, 4, 14 and 15 on pages 184–198, 198–200, 231–252 and 252–255, respectively, of this Annual Report.

Accrued interest and accounts receivable

This caption consists of accrued interest receivables from interest-earning assets; receivables from customers; receivables from brokers, dealers and clearing organizations; and receivables from failed securities sales. Accrued interest and accounts receivable decreased, primarily in IB, driven by a large reduction in customer margin receivables due to changes in client activity.

Premises and Equipment

The Firm’s premises and equipment consist of land, buildings, leasehold improvements, furniture and fixtures, hardware and software, and other equipment. The increase in premises and equipment was predominantly due to renovation of JPMorgan Chase’s headquarters in New York City; the purchase of a building in London; retail branch expansion in the U.S.; and investments in technology hardware and software, as well as other equipment. The increase was partially offset by depreciation and amortization.

Goodwill

Goodwill arises from business combinations and represents the excess of the purchase price of an acquired entity or business over the fair values assigned to the assets acquired and liabilities assumed. The decrease in goodwill was predominantly due to AM’s sale of its investment in an asset manager. For additional information on goodwill, see Note 17 on pages 267–271 of this Annual Report.

Mortgage servicing rights

MSRs represent the fair value of net cash flows expected to be received for performing specified mortgage-servicing activities for others. MSRs decreased, predominantly as a result of a decline in market interest rates, amortization and other changes in valuation inputs and assumptions, including increased cost to service assumptions, partially offset by new MSR originations. For additional information on MSRs, see Note 17 on pages 267–271 of this Annual Report.

Other intangible assets

Other intangible assets consist of purchased credit card relationships, other credit card-related intangibles, core deposit intangibles and other intangibles. The decrease in other intangible assets was due to amortization. For additional information on other intangible assets, see Note 17 on pages 267–271 of this Annual Report.

Other assets

Other assets consist of private equity and other instruments, cash collateral pledged, corporate- and bank-owned life insurance policies, assets acquired in loan satisfactions (including real estate owned), and all other assets. Other assets remained relatively flat in 2011.

Management's discussion and analysis

Deposits

Deposits represent a liability to customers, both retail and wholesale, related to non-brokerage funds held on their behalf. Deposits provide a stable and consistent source of funding for the Firm. Deposits increased significantly, predominantly due to an overall growth in wholesale client balances and, to a lesser extent, growth in consumer deposit balances. The increase in wholesale client balances, particularly in TSS and CB, was primarily driven by lower returns on other available alternative investments and low interest rates during 2011, and in AM, driven by growth in the number of clients and level of deposits. For more information on deposits, refer to the RFS and AM segment discussions on pages 85–93 and 104–106, respectively; the Liquidity Risk Management discussion on pages 127–132; and Notes 3 and 19 on pages 184–198 and 272, respectively, of this Annual Report. For more information on wholesale liability balances, which includes deposits, refer to the CB and TSS segment discussions on pages 98–100 and 101–103, respectively, of this Annual Report.

Federal funds purchased and securities loaned or sold under repurchase agreements

The Firm uses these instruments as part of its liquidity management activities and to support its client-driven market-making activities. In particular, federal funds purchased and securities loaned or sold under repurchase agreements are used by the Firm as short-term funding sources and to provide securities to clients for their short-term liquidity purposes. Securities sold under repurchase agreements decreased, predominantly in IB, reflecting the lower funding requirements of the Firm based on lower trading inventory levels, and change in the mix of funding sources. For additional information on the Firm's Liquidity Risk Management, see pages 127–132 of this Annual Report.

Commercial paper and other borrowed funds

The Firm uses commercial paper and other borrowed funds in its liquidity management activities to meet short-term funding needs, and in connection with a TSS liquidity management product, whereby excess client funds are transferred into commercial paper overnight sweep accounts. Commercial paper increased due to growth in the volume of liability balances in sweep accounts related to TSS's cash management product. Other borrowed funds, which includes short-term advances from FHLBs decreased, predominantly driven by maturities of short-term secured borrowings, unsecured bank notes and short-term FHLB advances. For additional information on the Firm's Liquidity Risk Management and other borrowed funds, see pages 127–132 of this Annual Report.

Accounts payable and other liabilities

Accounts payable and other liabilities consist of payables to customers; payables to brokers, dealers and clearing organizations; payables from failed securities purchases; accrued expense, including interest-bearing liabilities; and all other liabilities, including litigation reserves and obligations to return securities received as collateral.

Accounts payable and other liabilities increased predominantly due to higher IB customer balances. For additional information on the Firm's accounts payable and other liabilities, see Note 20 on page 272 of this Annual Report.

Beneficial interests issued by consolidated VIEs

Beneficial interests issued by consolidated VIEs represent interest-bearing beneficial-interest liabilities, which decreased, predominantly due to maturities of Firm-sponsored credit card securitization transactions. For additional information on Firm-sponsored VIEs and loan securitization trusts, see Off-Balance Sheet Arrangements, and Note 16 on pages 256–267 of this Annual Report.

Long-term debt

The Firm uses long-term debt (including trust-preferred capital debt securities and long-term FHLB advances) to provide cost-effective and diversified sources of funds and as critical components of the Firm's liquidity and capital management activities. Long-term debt decreased, predominantly due to net redemptions and maturities of long-term borrowings. For additional information on the Firm's long-term debt activities, see the Liquidity Risk Management discussion on pages 127–132 of this Annual Report.

Stockholders' equity

Total stockholders' equity increased, predominantly due to net income, as well as net issuances and commitments to issue under the Firm's employee stock-based compensation plans. The increase was partially offset by repurchases of common equity; and the declaration of cash dividends on common and preferred stock.

OFF-BALANCE SHEET ARRANGEMENTS AND CONTRACTUAL CASH OBLIGATIONS

JPMorgan Chase is involved with several types of off-balance sheet arrangements, including through unconsolidated special-purpose entities ("SPEs"), which are a type of VIE, and through lending-related financial instruments (e.g., commitments and guarantees).

Special-purpose entities

The most common type of VIE is a special purpose entity ("SPE"). SPEs are commonly used in securitization transactions in order to isolate certain assets and distribute the cash flows from those assets to investors. SPEs are an important part of the financial markets, including the mortgage- and asset-backed securities and commercial paper markets, as they provide market liquidity by facilitating investors' access to specific portfolios of assets and risks. SPEs may be organized as trusts, partnerships or corporations and are typically established for a single, discrete purpose. SPEs are not typically operating entities and usually have a limited life and no employees. The basic SPE structure involves a company selling assets to the SPE; the SPE funds the purchase of those assets by issuing securities to investors.

JPMorgan Chase uses SPEs as a source of liquidity for itself and its clients by securitizing financial assets, and by creating investment products for clients. The Firm is involved with SPEs through multi-seller conduits, investor intermediation activities, and loan securitizations. As a result of changes in the accounting guidance, certain VIEs were consolidated on the Firm's Consolidated Balance Sheets effective January 1, 2010. For further information on the types of SPEs and the impact of the change in the accounting guidance, see Note 16 on pages 256–267 for further information on these types of SPEs.

The Firm holds capital, as deemed appropriate, against all SPE-related transactions and related exposures, such as derivative transactions and lending-related commitments and guarantees.

The Firm has no commitments to issue its own stock to support any SPE transaction, and its policies require that transactions with SPEs be conducted at arm's length and reflect market pricing. Consistent with this policy, no JPMorgan Chase employee is permitted to invest in SPEs with which the Firm is involved where such investment would violate the Firm's Code of Conduct. These rules prohibit employees from self-dealing and acting on behalf of the Firm in transactions with which they or their family have any significant financial interest.

Implications of a credit rating downgrade to JPMorgan Chase Bank, N.A.

For certain liquidity commitments to SPEs, JPMorgan Chase Bank, N.A., could be required to provide funding if its short-term credit rating were downgraded below specific levels, primarily "P-1," "A-1" and "F1" for Moody's, Standard & Poor's and Fitch, respectively. These liquidity commitments support the issuance of asset-backed commercial paper by

both Firm-administered consolidated and third party sponsored nonconsolidated SPEs. In the event of a short-term credit rating downgrade, JPMorgan Chase Bank, N.A., absent other solutions, would be required to provide funding to the SPE, if the commercial paper could not be reissued as it matured. The aggregate amounts of commercial paper outstanding, issued by both Firm-administered and third-party-sponsored SPEs, that are held by third parties as of December 31, 2011 and 2010, was \$19.7 billion and \$23.1 billion, respectively. In addition, the aggregate amounts of commercial paper outstanding could increase in future periods should clients of the Firm-administered consolidated or third party sponsored nonconsolidated SPEs draw down on certain unfunded lending-related commitments. JPMorgan Chase Bank, N.A. had unfunded lending-related commitments to clients to fund an incremental \$11.0 billion and \$10.5 billion at December 31, 2011 and 2010, respectively. The Firm could facilitate the refinancing of some of the clients' assets in order to reduce the funding obligation. For further information, see the discussion of Firm-administered multi-seller conduits in Note 16 on page 260 of this Annual Report.

The Firm also acts as liquidity provider for certain municipal bond vehicles. The liquidity provider's obligation to perform is conditional and is limited by certain termination events, which include bankruptcy or failure to pay by the municipal bond issuer or credit enhancement provider, an event of taxability on the municipal bonds or the immediate downgrade of the municipal bond to below investment grade. See Note 16 on pages 260–261 of this Annual Report for additional information.

Off-balance sheet lending-related financial instruments, guarantees, and other commitments

JPMorgan Chase provides lending-related financial instruments (e.g., commitments and guarantees) to meet the financing needs of its customers. For further discussion of lending-related commitments and guarantees and the Firm's accounting for them, see Lending-related commitments on page 144, and Note 29 (including a table that presents, as of December 31, 2011, the amounts, by contractual maturity, of off-balance sheet lending-related financial instruments, guarantees and other commitments) on pages 283–289, of this Annual Report. For a discussion of loan repurchase liabilities, see Mortgage repurchase liability on pages 115–118 and Note 29 on pages 283–289, respectively, of this Annual Report.

Contractual cash obligations

In the normal course of business, the Firm enters into various contractual obligations that may require future cash payments. Certain obligations are recognized on-balance sheet, while others are off-balance sheet under U.S. GAAP. The accompanying table summarizes, by remaining maturity, JPMorgan Chase's significant contractual cash obligations at December 31, 2011. The contractual cash obligations included in the table below reflect the minimum contractual obligation under legally enforceable contracts

with terms that are both fixed and determinable. The carrying amount of on-balance sheet obligations on the Consolidated Balance Sheets may differ from the minimum contractual amount of the obligations reported below. For a discussion of mortgage loan repurchase liabilities, see Mortgage repurchase liability on pages 115–118 of this Annual Report. For further discussion of other obligations, see the Notes to Consolidated Financial Statements in this Annual Report.

Contractual cash obligations

By remaining maturity at December 31, (in millions)	2011					2010
	2012	2013-2014	2015-2016	After 2016	Total	Total
On-balance sheet obligations						
Deposits ^(a)	\$ 1,108,154	\$ 9,681	\$ 5,570	\$ 2,065	\$ 1,125,470	\$ 927,682
Federal funds purchased and securities loaned or sold under repurchase agreements	200,049	11,271	875	1,337	213,532	276,644
Commercial paper	51,631	—	—	—	51,631	35,363
Other borrowed funds	12,450	—	—	—	12,450	24,611
Beneficial interests issued by consolidated VIEs	39,729	14,317	3,464	8,467	65,977	77,649
Long-term debt ^(b)	50,077	59,749	43,464	83,615	236,905	249,434
Other ^(c)	1,355	1,136	924	2,617	6,032	7,329
Total on-balance sheet obligations	1,463,445	96,154	54,297	98,101	1,711,997	1,598,712
Off-balance sheet obligations						
Unsettled reverse repurchase and securities borrowing agreements	39,939	—	—	—	39,939	39,927
Contractual interest payments	9,551	13,006	9,669	44,192	76,418	78,454
Operating leases	1,753	3,335	2,738	7,188	15,014	16,000
Equity investment commitments	933	4	7	1,346	2,290	2,468
Contractual purchases and capital expenditures	1,244	713	288	415	2,660	2,822
Obligations under affinity and co-brand programs	1,197	1,996	1,875	325	5,393	5,801
Other	115	108	48	13	284	567
Total off-balance sheet obligations	54,732	19,162	14,625	53,479	141,998	146,039
Total contractual cash obligations	\$ 1,518,177	\$ 115,316	\$ 68,922	\$ 151,580	\$ 1,853,995	\$ 1,744,751

(a) Excludes structured notes where the Firm is not obligated to return a stated amount of principal at the maturity of the notes, but is obligated to return an amount based on the performance of the structured notes.

(b) Primarily includes deferred annuity contracts, pension and postretirement obligations and insurance liabilities.

(c) For further information, refer to unsettled reverse repurchase and securities borrowing agreements in Note 29 on page 286 of this Annual Report.

(d) Includes accrued interest and future contractual interest obligations. Excludes interest related to structured notes where the Firm's payment obligation is based on the performance of certain benchmarks.

(e) Includes noncancelable operating leases for premises and equipment used primarily for banking purposes and for energy-related tolling service agreements. Excludes the benefit of noncancelable sublease rentals of \$1.5 billion and \$1.8 billion at December 31, 2011 and 2010, respectively.

(f) At December 31, 2011 and 2010, included unfunded commitments of \$789 million and \$1.0 billion, respectively, to third-party private equity funds that are generally valued as discussed in Note 3 on pages 184–198 of this Annual Report; and \$1.5 billion and \$1.4 billion of unfunded commitments, respectively, to other equity investments.

Mortgage repurchase liability

In connection with the Firm's mortgage loan sale and securitization activities with Fannie Mae and Freddie Mac (the "GSEs") and other mortgage loan sale and private-label securitization transactions, the Firm has made representations and warranties that the loans sold meet certain requirements. For transactions with the GSEs, these representations relate to type of collateral, underwriting standards, validity of certain borrower representations made in connection with the loan, primary mortgage insurance being in force for any mortgage loan with a loan-to-value ("LTV") ratio greater than 80% at the loan's origination date, and the use of the GSEs' standard legal documentation. The Firm may be, and has been, required to repurchase loans and/or indemnify the GSEs and other investors for losses due to material breaches of these representations and warranties. To the extent that repurchase demands that are received relate to loans that the Firm purchased from third parties that remain viable, the Firm typically will have the right to seek a recovery of related repurchase losses from the related third party.

To date, the repurchase demands the Firm has received from the GSEs primarily relate to loans originated from 2005 to 2008. Demands against pre-2005 and post-2008 vintages have not been significant; the Firm attributes this to the comparatively favorable credit performance of these vintages and to the enhanced underwriting and loan qualification standards implemented progressively during 2007 and 2008. From 2005 to 2008, excluding Washington Mutual, the principal amount of loans sold to the GSEs subject to certain representations and warranties for which the Firm may be liable was approximately \$380 billion; this amount has not been adjusted for subsequent activity, such as borrower repayments of principal or repurchases completed to date. See the discussion below for information concerning the process the Firm uses to evaluate repurchase demands for breaches of representations and warranties, and the Firm's estimate of probable losses related to such exposure.

From 2005 to 2008, Washington Mutual sold approximately \$150 billion principal amount of loans to the GSEs subject to certain representations and warranties. Subsequent to the Firm's acquisition of certain assets and liabilities of Washington Mutual from the FDIC in September 2008, the Firm resolved and/or limited certain current and future repurchase demands for loans sold to the GSEs by Washington Mutual, although it remains the Firm's position that such obligations remain with the FDIC receivership. The Firm will continue to evaluate and may pay (subject to reserving its rights for indemnification by the FDIC) certain future repurchase demands related to individual loans, subject to certain limitations, and has considered such potential repurchase demands in its repurchase liability. The Firm believes that the remaining GSE repurchase exposure related to Washington Mutual presents minimal future risk to the Firm's financial results.

The Firm also sells loans in securitization transactions with Ginnie Mae; these loans are typically insured or guaranteed by another government agency. The Firm, in its role as servicer, may elect, but is not required, to repurchase delinquent loans securitized by Ginnie Mae, including those that have been sold back to Ginnie Mae subsequent to modification. Principal amounts due under the terms of these repurchased loans continue to be insured and the reimbursement of insured amounts is proceeding normally. Accordingly, the Firm has not recorded any mortgage repurchase liability related to these loans.

From 2005 to 2008, the Firm and certain acquired entities made certain loan level representations and warranties in connection with approximately \$450 billion of residential mortgage loans that were sold or deposited into private-label securitizations. While the terms of the securitization transactions vary, they generally differ from loan sales to the GSEs in that, among other things: (i) in order to direct the trustee to investigate potential claims, the security holders must make a formal request for the trustee to do so, and typically, this requires agreement of the holders of a specified percentage of the outstanding securities; (ii) generally, the mortgage loans are not required to meet all GSE eligibility criteria; and (iii) in many cases, the party demanding repurchase is required to demonstrate that a loan-level breach of a representation or warranty has materially and adversely affected the value of the loan. Of the \$450 billion originally sold or deposited (including \$165 billion by Washington Mutual, as to which the Firm maintains that certain of the repurchase obligations remain with the FDIC receivership), approximately \$191 billion of principal has been repaid (including \$71 billion related to Washington Mutual). In addition, approximately \$97 billion of the principal amount of loans has been liquidated (including \$35 billion related to Washington Mutual), with an average loss severity of 58%. Accordingly, the remaining outstanding principal balance of these loans (including Washington Mutual) was, as of December 31, 2011, approximately \$162 billion, of which \$55 billion was 60 days or more past due. The remaining outstanding principal balance of loans related to Washington Mutual was approximately \$59 billion, of which \$20 billion were 60 days or more past due.

Although there have been generalized allegations, as well as specific demands, that the Firm should repurchase loans sold or deposited into private-label securitizations, these claims for repurchases of loans sold or deposited into private-label securitizations (including claims from insurers that have guaranteed certain obligations of the securitization trusts) have, thus far, generally manifested themselves through threatened or pending litigation. Accordingly, the Firm does not consider these claims in estimating its mortgage repurchase liability; rather, the Firm separately evaluates such exposures in establishing its litigation reserves. For additional information regarding litigation, see Note 31 on pages 290–299 of this Annual Report.

Management's discussion and analysis

With respect to repurchase claims from private-label securitizations other than those considered in the Firm's litigation reserves, the Firm experienced an increase in the number of requests for loan files ("file requests") in the latter part of 2011; however, loan-level repurchase demands and repurchases from private-label securitizations have been limited to date. While it is possible that the volume of repurchases may increase in the future, the Firm cannot at the current time offer a reasonable estimate of probable future repurchases from such private-label securitizations. As a result, the Firm's mortgage repurchase liability primarily relates to loan sales to the GSEs and is calculated predominantly based on the Firm's repurchase activity experience with the GSEs.

Repurchase demand process

The Firm first becomes aware that a GSE is evaluating a particular loan for repurchase when the Firm receives a file request from the GSE. Upon completing its review, the GSE may submit a repurchase demand to the Firm; historically, most file requests have not resulted in repurchase demands.

The primary reasons for repurchase demands from the GSEs relate to alleged misrepresentations primarily arising from: (i) credit quality and/or undisclosed debt of the borrower; (ii) income level and/or employment status of the borrower; and (iii) appraised value of collateral. Ineligibility of the borrower for the particular product, mortgage insurance rescissions and missing documentation are other reasons for repurchase demands. The successful rescission of mortgage insurance typically results in a violation of representations and warranties made to the GSEs and, therefore, has been a significant cause of repurchase demands from the GSEs. The Firm actively reviews all rescission notices from mortgage insurers and contests them when appropriate.

As soon as practicable after receiving a repurchase demand from a GSE, the Firm evaluates the request and takes appropriate actions based on the nature of the repurchase demand. Loan-level appeals with the GSEs are typical and the Firm seeks to resolve the repurchase demand (i.e., either repurchase the loan or have the repurchase demand

rescinded) within three to four months of the date of receipt. In many cases, the Firm ultimately is not required to repurchase a loan because it is able to resolve the purported defect. Although repurchase demands may be made until the loan is paid in full, most repurchase demands from the GSEs historically have related to loans that became delinquent in the first 24 months following origination.

When the Firm accepts a repurchase demand from one of the GSEs, the Firm may either (i) repurchase the loan or the underlying collateral from the GSE at the unpaid principal balance of the loan plus accrued interest, or (ii) reimburse the GSE for its realized loss on a liquidated property (a "make-whole" payment).

Estimated mortgage repurchase liability

To estimate the Firm's mortgage repurchase liability arising from breaches of representations and warranties, the Firm considers the following factors, which are predominantly based on the Firm's historical repurchase activity with the GSEs:

- (i) the level of outstanding unresolved repurchase demands,
- (ii) estimated probable future repurchase demands, considering information about file requests, delinquent and liquidated loans, resolved and unresolved mortgage insurance rescission notices and the Firm's historical experience,
- (iii) the potential ability of the Firm to cure the defects identified in the repurchase demands ("cure rate"),
- (iv) the estimated severity of loss upon repurchase of the loan or collateral, make-whole settlement, or indemnification,
- (v) the Firm's potential ability to recover its losses from third-party originators, and
- (vi) the terms of agreements with certain mortgage insurers and other parties.

Based on these factors, the Firm has recognized a mortgage repurchase liability of \$3.6 billion and \$3.3 billion as of December 31, 2011 and 2010, respectively.

The following table provides information about outstanding repurchase demands and unresolved mortgage insurance rescission notices, excluding those related to Washington Mutual, at each of the past five quarter-end dates.

Outstanding repurchase demands and unresolved mortgage insurance rescission notices by counterparty^(a) type

(in millions)	December 31, 2011	September 30, 2011	June 30, 2011	March 31, 2011	December 31, 2010
GSEs and other ^(b)	\$ 2,345	\$ 2,133	\$ 1,826	\$ 1,321	\$ 1,251
Mortgage insurers	1,034	1,112	1,093	1,240	1,121
Overlapping population ^(c)	(113)	(155)	(145)	(127)	(104)
Total	\$ 3,266	\$ 3,090	\$ 2,774	\$ 2,434	\$ 2,268

- (a) Mortgage repurchase demands associated with pending or threatened litigation are not reported in this table because the Firm separately evaluates its exposure to such repurchase demands in establishing its litigation reserves.
- (b) The Firm's outstanding repurchase demands are predominantly from the GSEs. Other represents repurchase demands received from parties other than GSEs that have been presented in accordance with the terms of the underlying sale or securitization agreement.
- (c) Because the GSEs may make repurchase demands based on mortgage insurance rescission notices that remain unresolved, certain loans may be subject to both an unresolved mortgage insurance rescission notice and an outstanding repurchase demand.

The following tables show the trend in repurchase demands and mortgage insurance rescission notices received by loan origination vintage, excluding those related to Washington Mutual, for the past five quarters. The Firm expects repurchase demands to remain at elevated levels or to increase if there is a significant increase in private label repurchase demands outside of litigation.

Quarterly mortgage repurchase demands received by loan origination vintage

(in millions)	December 31, 2011	September 30, 2011	June 30, 2011	March 31, 2011	December 31, 2010
Pre-2005	\$ 39	\$ 34	\$ 32	\$ 15	\$ 39
2005	55	200	57	45	73
2006	315	232	363	158	198
2007	804	602	510	381	539
2008	291	323	301	249	254
Post-2008	81	153	89	94	65
Total repurchase demands received	\$ 1,585	\$ 1,544	\$ 1,352	\$ 942	\$ 1,168

(a) Mortgage repurchase demands associated with pending or threatened litigation are not reported in this table because the Firm separately evaluates its exposure to such repurchase demands in establishing its litigation reserves.

Quarterly mortgage insurance rescission notices received by loan origination vintage

(in millions)	December 31, 2011	September 30, 2011	June 30, 2011	March 31, 2011	December 31, 2010
Pre-2005	\$ 4	\$ 3	\$ 3	\$ 5	\$ 3
2005	12	15	24	32	9
2006	19	31	39	65	53
2007	48	63	72	144	142
2008	26	30	31	49	50
Post-2008	2	1	1	1	1
Total mortgage insurance rescissions received	\$ 111	\$ 143	\$ 170	\$ 296	\$ 258

(a) Mortgage insurance rescissions typically result in a repurchase demand from the GSEs. This table includes mortgage insurance rescission notices for the GSEs also have issued a repurchase demand.

Since the beginning of 2010, the Firm's overall cure rate, excluding Washington Mutual, has been approximately 50%. Repurchases that have resulted from mortgage insurance rescissions are reflected in the Firm's overall cure rate. While the actual cure rate may vary from quarter to quarter, the Firm expects that the overall cure rate will remain in the 40-50% range for the foreseeable future.

The Firm has not observed a direct relationship between the type of defect that causes the breach of representations and warranties and the severity of the realized loss. Therefore, the loss severity assumption is estimated using the Firm's historical experience and projections regarding changes in home prices. Actual principal loss severities on finalized repurchases and "make-whole" settlements to date, excluding Washington Mutual, currently average approximately 50%, but may vary from quarter to quarter based on the characteristics of the underlying loans and changes in home prices.

When a loan was originated by a third-party originator, the Firm typically has the right to seek a recovery of related repurchase losses from the third-party originator.

Estimated and actual third-party recovery rates may vary from quarter to quarter based upon the underlying mix of correspondents (e.g., active, inactive, out-of-business originators) from which recoveries are being sought.

The Firm has entered into agreements with two mortgage insurers to resolve their claims on certain portfolios for which the Firm is a servicer. These two agreements cover and have resolved approximately one-third of the Firm's total mortgage insurance rescission risk exposure, both in terms of the unpaid principal balance of serviced loans covered by mortgage insurance and the amount of mortgage insurance coverage. The impact of these agreements is reflected in the mortgage repurchase liability and the outstanding mortgage insurance rescission notices as of December 31, 2011 disclosed above. The Firm has considered its remaining unresolved mortgage insurance rescission risk exposure in estimating the mortgage repurchase liability as of December 31, 2011.

Substantially all of the estimates and assumptions underlying the Firm's established methodology for computing its recorded mortgage repurchase liability — including the amount of probable future demands from purchasers, trustees or investors (which is in part based on historical experience), the ability of the Firm to cure identified defects, the severity of loss upon repurchase or foreclosure and recoveries from third parties — require application of a significant level of management judgment. Estimating the mortgage repurchase liability is further complicated by historical data that is not necessarily indicative of future expectations and uncertainty

Management's discussion and analysis

surrounding numerous external factors, including: (i) economic factors (for example, further declines in home prices and changes in borrower behavior may lead to increases in the number of defaults, the severity of losses, or both), and (ii) the level of future demands, which is dependent, in part, on actions taken by third parties, such as the GSEs, mortgage insurers, trustees and investors. While the Firm uses the best information available to it in estimating its mortgage repurchase liability, the estimation process is inherently uncertain, imprecise and potentially volatile as additional information is obtained and external factors continue to evolve.

The following table summarizes the change in the mortgage repurchase liability for each of the periods presented.

Summary of changes in mortgage repurchase liability

Year ended December 31, (in millions)	2011	2010	2009
Repurchase liability at beginning of period	\$ 3,285	\$ 1,705	\$ 1,093
Realized losses	(1,263)	(1,423)	(1,253) ^(d)
Provision for repurchase losses	1,535	3,003	1,865
Repurchase liability at end of period	\$ 3,557 ^(c)	\$ 3,285	1,705

- (a) Mortgage repurchase liabilities associated with pending or threatened litigation are not reported in this table because the Firm separately evaluates its exposure to such repurchases in establishing its litigation reserves.
- (b) Includes principal losses and accrued interest on repurchased loans, "make-whole" settlements, settlements with claimants, and certain related expense. For the years ended 2011, 2010 and 2009, make-whole settlements were \$640 million, \$632 million and \$277 million, respectively.
- (c) Includes \$173 million at December 31, 2011, related to future demands on loans sold by Washington Mutual to the GSEs.
- (d) Includes the Firm's resolution with the GSEs of certain current and future repurchase demands for certain loans sold by Washington Mutual. The unpaid principal balance of loans related to this resolution is not included in the table below, which summarizes the unpaid principal balance of repurchased loans.

The following table summarizes the total unpaid principal balance of repurchases during the periods indicated.

Unpaid principal balance of mortgage loan repurchases

Year ended December 31, (in millions)	2011	2010	2009
Ginnie Mae	\$ 5,981	\$ 8,717	\$ 6,966
GSEs and other ^(d)	1,334	1,773	1,019
Total	\$ 7,315	\$ 10,490	\$ 7,985

- (a) This table includes (i) repurchases of mortgage loans due to breaches of representations and warranties, and (ii) loans repurchased from Ginnie Mae loan pools as described in (b) below. This table does not include mortgage insurance rescissions; while the rescission of mortgage insurance typically results in a repurchase demand from the GSEs, the mortgage insurers themselves do not present repurchase demands to the Firm. This table also excludes mortgage loan repurchases associated with pending or threatened litigation because the Firm separately evaluates its exposure to such repurchases in establishing its litigation reserves.
- (b) In substantially all cases, these repurchases represent the Firm's voluntary repurchase of certain delinquent loans from loan pools as permitted by Ginnie Mae guidelines (i.e., they do not result from repurchase demands due to breaches of representations and warranties). The Firm typically elects to repurchase these delinquent loans as it continues to service them and/or manage the foreclosure process in accordance with applicable requirements of Ginnie Mae, the Federal Housing Administration ("FHA"), Rural Housing Services ("RHS") and/or the U.S. Department of Veterans Affairs ("VA").
- (c) Predominantly all of the repurchases related to demands by GSEs.
- (d) Nonaccrual loans held-for-investment included \$477 million, \$354 million and \$218 million at December 31, 2011, 2010 and 2009, respectively, of loans repurchased as a result of breaches of representations and warranties.

For additional information regarding the mortgage repurchase liability, see Note 29 on pages 283-289 of this Annual Report.

CAPITAL MANAGEMENT

A strong capital position is essential to the Firm's business strategy and competitive position. The Firm's capital strategy focuses on long-term stability, which enables the Firm to build and invest in market-leading businesses, even in a highly stressed environment. Senior management considers the implications on the Firm's capital strength prior to making any decision on future business activities. Capital and earnings are inextricably linked, as earnings directly affect capital generation for the Firm. In addition to considering the Firm's earnings outlook, senior management evaluates all sources and uses of capital and makes decisions to vary sources or uses to preserve the Firm's capital strength.

The Firm's capital management objectives are to hold capital sufficient to:

- Cover all material risks underlying the Firm's business activities;
- Maintain "well-capitalized" status under regulatory requirements;
- Maintain debt ratings, which will enable the Firm to optimize its funding mix and liquidity sources while minimizing costs;
- Retain flexibility to take advantage of future investment opportunities; and
- Build and invest in businesses, even in a highly stressed environment.

To meet these objectives, the Firm maintains a robust and disciplined capital adequacy assessment process, which is performed regularly, and is intended to enable the Firm to remain well-capitalized and fund ongoing operations under adverse conditions. The process assesses the potential impact of alternative economic and business scenarios on earnings and capital for the Firm's businesses individually and in the aggregate over a rolling three-year period. Economic scenarios, and the parameters underlying those scenarios, are defined centrally and applied uniformly across the businesses. These scenarios are articulated in terms of macroeconomic factors, which are key drivers of business results; global market shocks, which generate short-term but severe trading losses; and operational risk events, which generate significant losses. However, when defining a broad range of scenarios, realized events can always be worse. Accordingly, management considers additional stresses outside these scenarios as necessary.

The Firm utilized this capital adequacy process in completing the Federal Reserve Comprehensive Capital Analysis and Review ("CCAR"). The Federal Reserve requires the Firm to submit a capital plan on an annual basis. The Firm submitted its 2012 capital plan on January 9, 2012. The Federal Reserve has indicated that it expects to provide notification of either its objection or non-objection to the Firm's capital plan by March 15, 2012.

Capital adequacy is also evaluated with the Firm's liquidity

risk management processes. For further information on the Firm's Liquidity Risk Management, see pages 127–132 of this Annual Report.

The quality and composition of capital are key factors in senior management's evaluation of the Firm's capital adequacy. Accordingly, the Firm holds a significant amount of its capital in the form of common equity. The Firm uses three capital measurements in assessing its levels of capital:

- **Regulatory capital** The capital required according to standards stipulated by U.S. bank regulatory agencies.
- **Economic risk capital** The capital required as a result of a bottom-up assessment of the underlying risks of the Firm's business activities, utilizing internal risk-assessment methodologies.
- **Line of business equity** The amount of equity the Firm believes each business segment would require if it were operating independently, which incorporates sufficient capital to address economic risk measures, regulatory capital requirements and capital levels for similarly rated peers.

Regulatory capital

The Federal Reserve establishes capital requirements, including well-capitalized standards, for the consolidated financial holding company. The Office of the Comptroller of the Currency ("OCC") establishes similar capital requirements and standards for the Firm's national banks, including JPMorgan Chase Bank, N.A. and Chase Bank USA, N.A. As of December 31, 2011 and 2010, JPMorgan Chase and all of its banking subsidiaries were well-capitalized and each met all capital requirements to which it was subject.

In connection with the U.S. Government's Supervisory Capital Assessment Program in 2009, U.S. banking regulators developed a new measure of capital, Tier 1 common, which is defined as Tier 1 capital less elements of Tier 1 capital not in the form of common equity — such as perpetual preferred stock, noncontrolling interests in subsidiaries and trust preferred capital debt securities. Tier 1 common, a non-GAAP financial measure, is used by banking regulators, investors and analysts to assess and compare the quality and composition of the Firm's capital with the capital of other financial services companies. The Firm uses Tier 1 common along with the other capital measures to assess and monitor its capital position.

At December 31, 2011 and 2010, JPMorgan Chase maintained Tier 1 and Total capital ratios in excess of the well-capitalized standards established by the Federal Reserve, as indicated in the tables below. In addition, the Firm's Tier 1 common ratio was significantly above the 4% well-capitalized standard established at the time of the Supervisory Capital Assessment Program. For more information, see Note 28 on pages 281–283 of this Annual Report.

Management's discussion and analysis

The following table presents the regulatory capital, assets and risk-based capital ratios for JPMorgan Chase at December 31, 2011 and 2010. These amounts are determined in accordance with regulations issued by the Federal Reserve and OCC.

Risk-based capital ratios

December 31,	2011	2010
Capital ratios		
Tier 1 capital	12.3%	12.1%
Total capital	15.4	15.5
Tier 1 leverage	6.8	7.0
Tier 1 common ^(a)	10.1	9.8

(a) The Tier 1 common ratio is Tier 1 common capital divided by RWA.

A reconciliation of total stockholders' equity to Tier 1 common, Tier 1 capital and Total qualifying capital is presented in the table below.

Risk-based capital components and assets

December 31, (in millions)	2011	2010
Total stockholders' equity	\$ 183,573	\$ 176,106
Less: Preferred stock	7,800	7,800
Common stockholders' equity	175,773	168,306
Effect of certain items in accumulated other comprehensive income/(loss) excluded from Tier 1 common	(970)	(748)
Less: Goodwill	45,873	46,915
Fair value DVA on derivative and structured note liabilities related to the Firm's credit quality	2,150	1,261
Investments in certain subsidiaries and other	993	1,032
Other intangible assets	2,871	3,587
Tier 1 common	122,916	114,763
Preferred stock	7,800	7,800
Qualifying hybrid securities and noncontrolling interests ^(b)	19,668	19,887
Total Tier 1 capital	150,384	142,450
Long-term debt and other instruments qualifying as Tier 2	22,275	25,018
Qualifying allowance for credit losses	15,504	14,959
Adjustment for investments in certain subsidiaries and other	(75)	(211)
Total Tier 2 capital	37,704	39,766
Total qualifying capital	\$ 188,088	\$ 182,216
Risk-weighted assets	\$ 1,221,198	\$ 1,174,978
Total adjusted average assets	\$ 2,202,087	\$ 2,024,515

(a) Goodwill and other intangible assets are net of any associated deferred tax liabilities.

(b) Primarily includes trust preferred capital debt securities of certain business trusts.

The Firm's Tier 1 common was \$122.9 billion at December 31, 2011, an increase of \$8.2 billion from December 31, 2010. The increase was predominantly due to net income (adjusted for DVA) of \$18.1 billion, lower deductions related to goodwill and other intangibles of \$1.8 billion, and net issuances and commitments to issue common stock under the Firm's employee stock-based

compensation plans of \$2.1 billion. The increase was partially offset by \$8.95 billion (on a trade-date basis) of repurchases of common stock and warrants and \$4.7 billion of dividends on common and preferred stock. The Firm's Tier 1 capital was \$150.4 billion at December 31, 2011, an increase of \$7.9 billion from December 31, 2010. The increase in Tier 1 capital reflected the increase in Tier 1 common.

Additional information regarding the Firm's capital ratios and the federal regulatory capital standards to which it is subject is presented in Supervision and regulation and Part I, Item 1A, Risk Factors, on pages 1–7 and 7–17, respectively, of the 2011 Form 10-K, and Note 28 on pages 281–283 of this Annual Report.

Basel II

The minimum risk-based capital requirements adopted by the U.S. federal banking agencies follow the Capital Accord of the Basel Committee on Banking Supervision ("Basel I"). In 2004, the Basel Committee published a revision to the Accord ("Basel II"). The goal of the Basel II Framework is to provide more risk-sensitive regulatory capital calculations and promote enhanced risk management practices among large, internationally active banking organizations. U.S. banking regulators published a final Basel II rule in December 2007, which requires JPMorgan Chase to implement Basel II at the holding company level, as well as at certain of its key U.S. bank subsidiaries.

Prior to full implementation of the new Basel II Framework, JPMorgan Chase is required to complete a qualification period of four consecutive quarters during which it needs to demonstrate that it meets the requirements of the rule to the satisfaction of its U.S. banking regulators. JPMorgan Chase is currently in the qualification period and expects to be in compliance with all relevant Basel II rules within the established timelines. In addition, the Firm has adopted, and will continue to adopt, based on various established timelines, Basel II rules in certain non-U.S. jurisdictions, as required.

"Basel 2.5"

During 2011, the U.S. federal banking agencies issued proposals for industry comment to revise the market risk capital rules of Basel II that would result in additional capital requirements for trading positions and securitizations. The Firm anticipates these rules will be finalized and implemented in 2012. It is currently estimated that implementation of these rules could result in approximately a 100 basis point decrease in the Firm's Basel I Tier 1 common ratio, but the actual impact upon implementation on the Firm's capital ratios could differ depending on the outcome of the final U.S. rules and regulatory approval of the Firm's internal models.

Basel III

In addition to the Basel II Framework, on December 16, 2010, the Basel Committee issued the final version of the Capital Accord, commonly referred to as "Basel III," which revised Basel II by, among other things, narrowing the definition of capital, increasing capital requirements for specific exposures, introducing minimum standards for short-term liquidity coverage – the liquidity coverage ratio (the "LCR") – and term funding – the net stable funding ratio (the "NSFR"), and establishing an international leverage ratio. The LCR is a short-term liquidity measure which identifies a firm's unencumbered, high-quality liquid assets that can be converted into cash to meet net cash outflows during a 30-day severe stress scenario. The NSFR measures the amount of longer-term, stable sources of funding available to support the portion of all assets (on- and off-balance sheet) that cannot be monetized over a one-year period of extended stress. The Basel Committee also announced higher capital ratio requirements under Basel III, which provide that the common equity requirement will be increased to 7%, comprised of a minimum ratio of 4.5% plus a 2.5% capital conservation buffer.

On June 25, 2011, the Basel Committee announced an agreement to require global systemically important banks ("GSIBs") to maintain Tier 1 common requirements above the 7% minimum in amounts ranging from an additional 1% to an additional 2.5%. The Basel Committee also stated it intended to require certain GSIBs to maintain a further Tier 1 common requirement of an additional 1% under certain circumstances, to act as a disincentive for the GSIB from taking actions that would further increase its systemic importance. On July 19, 2011, the Basel Committee published a proposal on the GSIB assessment methodology, which reflects an approach based on five broad categories: size; interconnectedness; lack of substitutability; cross-jurisdictional activity; and complexity. In late September, the Basel Committee finalized the GSIB assessment methodology and Tier 1 common requirements.

In addition, the U.S. federal banking agencies have published proposed risk-based capital floors pursuant to the requirements of the Dodd-Frank Wall Street Reform and Consumer Protection Act (the "Dodd-Frank Act") to establish a permanent Basel I floor under Basel II and Basel III capital calculations.

Estimated Tier 1 common under Basel III rules

The following table presents a comparison of the Firm's Tier 1 common under Basel I rules to its estimated Tier 1 common under Basel III rules, along with the Firm's estimated risk-weighted assets and the Tier 1 common ratio under Basel III rules, all of which are non-GAAP financial measures. Tier 1 common under Basel III includes additional adjustments and deductions not included in Basel I Tier 1 common, such as the inclusion of accumulated other comprehensive income ("AOCI") related to AFS securities and defined benefit pension and other postretirement employee benefit plans, and the deduction of the Firm's

defined benefit pension fund assets.

The Firm estimates that its Tier 1 common ratio under Basel III rules would be 7.9% as of December 31, 2011.

Management considers this estimate as a key measure to assess the Firm's capital position in conjunction with its capital ratios under Basel I requirements, in order to enable management, investors and analysts to compare the Firm's capital under the Basel III capital standards with similar estimates provided by other financial services companies.

December 31, 2011
(in millions, except ratios)

Tier 1 common under Basel I rules	\$	122,916
Adjustments related to AOCI for AFS securities and defined benefit pension and other postretirement employee benefit plans		919
Deduction for net defined benefit pension asset		(1,430)
All other adjustments		(534)
Estimated Tier 1 common under Basel III rules	\$	121,871
Estimated risk-weighted assets under Basel III rules	\$	1,545,801
Estimated Tier 1 common ratio under Basel III rules		7.9%

- (a) Key differences in the calculation of risk-weighted assets between Basel I and Basel III include: (a) Basel III credit risk risk-weighted assets ("RWA") is based on risk-sensitive approaches which largely rely on the use of internal credit models and parameters, whereas Basel I RWA is based on fixed supervisory risk weightings which vary only by counterparty type and asset class; (b) Basel III market risk RWA reflects the new capital requirements related to trading assets and securitizations, which include incremental capital requirements for stress VaR, correlation trading, and re-securitization positions; and (c) Basel III includes RWA for operational risk, whereas Basel I does not.
- (b) The Tier 1 common ratio is Tier 1 common divided by RWA.

The Firm's estimate of its Tier 1 common ratio under Basel III reflects its current understanding of the Basel III rules and the application of such rules to its businesses as currently conducted, and therefore excludes the impact of any changes the Firm may make in the future to its businesses as a result of implementing the Basel III rules. The Firm's understanding of the Basel III rules is based on information currently published by the Basel Committee and U.S. federal banking agencies.

The Firm intends to maintain its strong liquidity position in the future as the short-term liquidity coverage (LCR) and term funding (NSFR) standards of the Basel III rules are implemented, in 2015 and 2018, respectively. In order to do so the Firm believes it may need to modify the liquidity profile of certain of its assets and liabilities. Implementation of the Basel III rules may also cause the Firm to increase prices on, or alter the types of, products it offers to its customers and clients.

The Basel III revisions governing liquidity and capital requirements are subject to prolonged observation and transition periods. The observation periods for both the LCR and NSFR began in 2011, with implementation in 2015 and 2018, respectively. The transition period for banks to meet the revised Tier 1 common requirement will begin in 2013, with implementation on January 1, 2019. The Firm fully expects to be in compliance with the higher Basel III capital

Management's discussion and analysis

standards, as well as any additional Dodd-Frank Act capital requirements, as they become effective. The additional capital requirements for GSIBs will be phased-in starting January 1, 2016, with full implementation on January 1, 2019.

The Firm will continue to monitor the ongoing rule-making process to assess both the timing and the impact of Basel III on its businesses and financial condition.

Broker-dealer regulatory capital

JPMorgan Chase's principal U.S. broker-dealer subsidiaries are J.P. Morgan Securities LLC ("JPMorgan Securities") and J.P. Morgan Clearing Corp. ("JPMorgan Clearing"). JPMorgan Clearing is a subsidiary of JPMorgan Securities and provides clearing and settlement services. JPMorgan Securities and JPMorgan Clearing are each subject to Rule 15c3-1 under the Securities Exchange Act of 1934 (the "Net Capital Rule"). JPMorgan Securities and JPMorgan Clearing are also each registered as futures commission merchants and subject to Rule 1.17 of the Commodity Futures Trading Commission ("CFTC"). Effective June 1, 2011, J.P. Morgan Futures Inc., a registered Futures Commission Merchant and a wholly owned subsidiary of JPMorgan Chase, merged with and into JPMorgan Securities. The merger created a combined Broker-Dealer/Futures Commission Merchant entity that provides capital and operational efficiencies.

JPMorgan Securities and JPMorgan Clearing have elected to compute their minimum net capital requirements in accordance with the "Alternative Net Capital Requirements" of the Net Capital Rule. At December 31, 2011, JPMorgan Securities' net capital, as defined by the Net Capital Rule, was \$11.1 billion, exceeding the minimum requirement by \$9.5 billion, and JPMorgan Clearing's net capital was \$7.4 billion, exceeding the minimum requirement by \$5.5 billion.

In addition to its minimum net capital requirement, JPMorgan Securities is required to hold tentative net capital in excess of \$1.0 billion and is also required to notify the SEC in the event that tentative net capital is less than \$5.0 billion, in accordance with the market and credit risk standards of Appendix E of the Net Capital Rule. As of December 31, 2011, JPMorgan Securities had tentative net capital in excess of the minimum and notification requirements.

Economic risk capital

JPMorgan Chase assesses its capital adequacy relative to the risks underlying its business activities using internal risk-assessment methodologies. The Firm measures economic capital primarily based on four risk factors: credit, market, operational and private equity risk.

Year ended December 31, (in billions)	Yearly Average		
	2011	2010	2009
Credit risk	\$ 48.2	\$ 49.7	\$ 51.3
Market risk	14.5	15.1	15.4
Operational risk	8.5	7.4	8.5
Private equity risk	6.9	6.2	4.7
Economic risk capital	78.1	78.4	79.9
Goodwill	48.6	48.6	48.3
Other ^(a)	46.6	34.5	17.7
Total common stockholders' equity	\$ 173.3	\$ 161.5	\$ 145.9

(a) Reflects additional capital required, in the Firm's view, to meet its regulatory and debt rating objectives.

Credit risk capital

Credit risk capital is estimated separately for the wholesale businesses (IB, CB, TSS and AM) and consumer businesses (RFS and Card).

Credit risk capital for the overall wholesale credit portfolio is defined in terms of unexpected credit losses, both from defaults and from declines in the portfolio value due to credit deterioration, measured over a one-year period at a confidence level consistent with an "AA" credit rating standard. Unexpected losses are losses in excess of those for which allowances for credit losses are maintained. The capital methodology is based on several principal drivers of credit risk: exposure at default (or loan-equivalent amount), default likelihood, credit spreads, loss severity and portfolio correlation.

Credit risk capital for the consumer portfolio is based on product and other relevant risk segmentation. Actual segment-level default and severity experience are used to estimate unexpected losses for a one-year horizon at a confidence level consistent with an "AA" credit rating standard. See Credit Risk Management on pages 132–157 of this Annual Report for more information about these credit risk measures.

Market risk capital

The Firm calculates market risk capital guided by the principle that capital should reflect the risk of loss in the value of portfolios and financial instruments caused by adverse movements in market variables, such as interest and foreign exchange rates, credit spreads, and securities and commodities prices, taking into account the liquidity of the financial instruments. Results from daily VaR, biweekly stress-tests, issuer credit spreads and default risk calculations, as well as other factors, are used to determine appropriate capital levels. Market risk capital is allocated to each business segment based on its risk assessment. See Market Risk Management on pages 158–163 of this Annual Report for more information about these market risk measures.

Operational risk capital

Capital is allocated to the lines of business for operational risk using a risk-based capital allocation methodology which estimates operational risk on a bottom-up basis. The operational risk capital model is based on actual losses and potential scenario-based stress losses, with adjustments to the capital calculation to reflect changes in the quality of the control environment or the use of risk-transfer products. The Firm believes its model is consistent with the Basel II Framework. See Operational Risk Management on pages 166–167 of this Annual Report for more information about operational risk.

Private equity risk capital

Capital is allocated to privately- and publicly-held securities, third-party fund investments, and commitments in the private equity portfolio to cover the potential loss associated with a decline in equity markets and related asset devaluations. In addition to negative market fluctuations, potential losses in private equity investment portfolios can be magnified by liquidity risk. Capital allocation for the private equity portfolio is based on measurement of the loss experience suffered by the Firm and other market participants over a prolonged period of adverse equity market conditions.

Line of business equity

The Firm's framework for allocating capital is based on the following objectives:

- Integrate firmwide and line of business capital management activities;
- Measure performance consistently across all lines of business; and
- Provide comparability with peer firms for each of the lines of business

Equity for a line of business represents the amount the Firm believes the business would require if it were operating independently, incorporating sufficient capital to address regulatory capital requirements (including Basel III Tier 1 common capital requirements), economic risk measures and capital levels for similarly rated peers. Capital is also allocated to each line of business for, among other things, goodwill and other intangibles associated with acquisitions effected by the line of business. ROE is measured and internal targets for expected returns are established as key measures of a business segment's performance.

Line of business equity

December 31, (in billions)	2011	2010
Investment Bank	\$ 40.0	\$ 40.0
Retail Financial Services	25.0	24.6
Card Services & Auto	16.0	18.4
Commercial Banking	8.0	8.0
Treasury & Securities Services	7.0	6.5
Asset Management	6.5	6.5
Corporate/Private Equity	73.3	64.3
Total common stockholders' equity	\$ 175.8	\$ 168.3

Line of business equity

Year ended December 31, (in billions)	Yearly Average		
	2011	2010	2009
Investment Bank	\$ 40.0	\$ 40.0	\$ 33.0
Retail Financial Services	25.0	24.6	22.5
Card Services & Auto	16.0	18.4	17.5
Commercial Banking	8.0	8.0	8.0
Treasury & Securities Services	7.0	6.5	5.0
Asset Management	6.5	6.5	7.0
Corporate/Private Equity	70.8	57.5	52.9
Total common stockholders' equity	\$ 173.3	\$ 161.5	\$ 145.9

Effective January 1, 2010, the Firm enhanced its line of business equity framework to better align equity assigned to the lines of business with changes anticipated to occur in each line of business, and to reflect the competitive and regulatory landscape. The lines of business are now capitalized based on the Tier 1 common standard, rather than the Tier 1 capital standard. Effective January 1, 2011, capital allocated to Card was reduced by \$2.4 billion to \$16.0 billion, largely reflecting portfolio runoff and the improving risk profile of the business; capital allocated to TSS was increased by \$500 million, to \$7.0 billion, reflecting growth in the underlying business.

Effective January 1, 2012, the Firm further revised the capital allocated to certain businesses, reflecting additional refinement of each segment's Basel III Tier 1 common capital requirements. The Firm continues to assess the level of capital required for each line of business, as well as the assumptions and methodologies used to allocate capital to the business segments, and further refinements may be implemented in future periods.

Capital actions

Dividends

On February 23, 2009, the Board of Directors reduced the Firm's quarterly common stock dividend from \$0.38 to \$0.05 per share, effective with the dividend paid on April 30, 2009, to shareholders of record on April 6, 2009. The action enabled the Firm to retain approximately \$5.5 billion in common equity in each of 2010 and 2009, and was taken to ensure the Firm had sufficient capital strength in the event the very weak economic conditions that existed at the beginning of 2009 deteriorated further. JPMorgan Chase declared quarterly cash dividends on its common stock in the amount of \$0.05 per share for each quarter of 2010 and 2009.

On March 18, 2011, the Board of Directors increased the Firm's quarterly common stock dividend from \$0.05 to \$0.25 per share, effective with the dividend paid on April 30, 2011, to shareholders of record on April 6, 2011. The Firm's common stock dividend policy reflects JPMorgan Chase's earnings outlook; desired dividend payout ratio; capital objectives; and alternative investment opportunities. The Firm's current expectation is to return to a payout ratio of approximately 30% of normalized earnings over time.

For information regarding dividend restrictions, see Note 22 and Note 27 on page 276 and 281, respectively, of this Annual Report.

The following table shows the common dividend payout ratio based on reported net income.

Year ended December 31,	2011	2010	2009
Common dividend payout ratio	22%	5%	9%

Common equity repurchases

On March 18, 2011, the Board of Directors approved a \$15.0 billion common equity (i.e., common stock and warrants) repurchase program, of which \$8.95 billion was authorized for repurchase in 2011. The \$15.0 billion repurchase program superseded a \$10.0 billion repurchase program approved in 2007. During 2011 and 2010, the Firm repurchased (on a trade-date basis) an aggregate of 240 million and 78 million shares of common stock and warrants, for \$8.95 billion and \$3.0 billion, at an average price per unit of \$37.35 and \$38.49, respectively. The Firm did not repurchase any of the warrants during 2010, and did not repurchase any shares of its common stock or warrants during 2009.

The Firm may, from time to time, enter into written trading plans under Rule 10b5-1 of the Securities Exchange Act of 1934 to facilitate repurchases in accordance with the repurchase program. A Rule 10b5-1 repurchase plan allows the Firm to repurchase its equity during periods when it would not otherwise be repurchasing common equity — for example, during internal trading “black-out periods.” All purchases under a Rule 10b5-1 plan must be made according to a predefined plan established when the Firm is not aware of material nonpublic information.

The authorization to repurchase common equity will be utilized at management's discretion, and the timing of purchases and the exact amount of common equity that may be repurchased is subject to various factors, including market conditions; legal considerations affecting the amount and timing of repurchase activity; the Firm's capital position (taking into account goodwill and intangibles); internal capital generation; and alternative investment opportunities. The repurchase program does not include specific price targets or timetables; may be executed through open market purchases or privately negotiated transactions, or utilizing Rule 10b5-1 programs; and may be suspended at any time.

For additional information regarding repurchases of the Firm's equity securities, see Part II, Item 5: Market for registrant's common equity, related stockholder matters and issuer purchases of equity securities, on pages 18–20 of JPMorgan Chase's 2011 Form 10-K.

Issuance

Common stock

On June 5, 2009, the Firm issued \$5.8 billion, or 163 million shares, of common stock at \$35.25 per share. The proceeds from these issuances were used for general corporate purposes. For additional information regarding common stock, see Note 23 on pages 276-277 of this Annual Report.

Capital Purchase Program

Pursuant to the U.S. Treasury's Capital Purchase Program, on October 28, 2008, the Firm issued to the U.S. Treasury a Warrant to purchase up to 88,401,697 shares of the Firm's common stock, at an exercise price of \$42.42 per share, subject to certain antidilution and other adjustments. The U.S. Treasury exchanged the Warrant for 88,401,697 warrants, each of which was a warrant to purchase a share of the Firm's common stock at an exercise price of \$42.42 per share and, on December 11, 2009, the U.S. Treasury sold the warrants to the public in a secondary public offering for \$950 million. In 2011, the Firm repurchased 10,167,698 of these warrants as part of the common equity repurchase program discussed above. The warrants are exercisable, in whole or in part, at any time and from time to time until October 28, 2018.

RISK MANAGEMENT

Risk is an inherent part of JPMorgan Chase's business activities. The Firm's risk management framework and governance structure are intended to provide comprehensive controls and ongoing management of the major risks inherent in its business activities. The Firm employs a holistic approach to risk management to ensure the broad spectrum of risk types are considered in managing its business activities. The Firm's risk management framework is intended to create a culture of risk awareness and personal responsibility throughout the Firm where collaboration, discussion, escalation and sharing of information is encouraged.

The Firm's overall risk appetite is established in the context of the Firm's capital, earnings power, and diversified business model. The Firm employs a formalized risk appetite framework to clearly link risk appetite and return targets, controls and capital management. The Firm's CEO is responsible for setting the overall risk appetite of the Firm and the LOB CEOs are responsible for setting the risk appetite for their respective lines of business. The Risk Policy Committee of the Firm's Board of Directors approves the risk appetite policy on behalf of the entire Board of Directors.

Risk governance

The Firm's risk governance structure is based on the principle that each line of business is responsible for managing the risk inherent in its business, albeit with appropriate Corporate oversight. Each line of business risk committee is responsible for decisions regarding the business' risk strategy, policies and controls. There are nine major risk types identified in the business activities of the Firm: liquidity risk, credit risk, market risk, interest rate risk, country risk, private equity risk, operational risk, legal and fiduciary risk, and reputation risk.

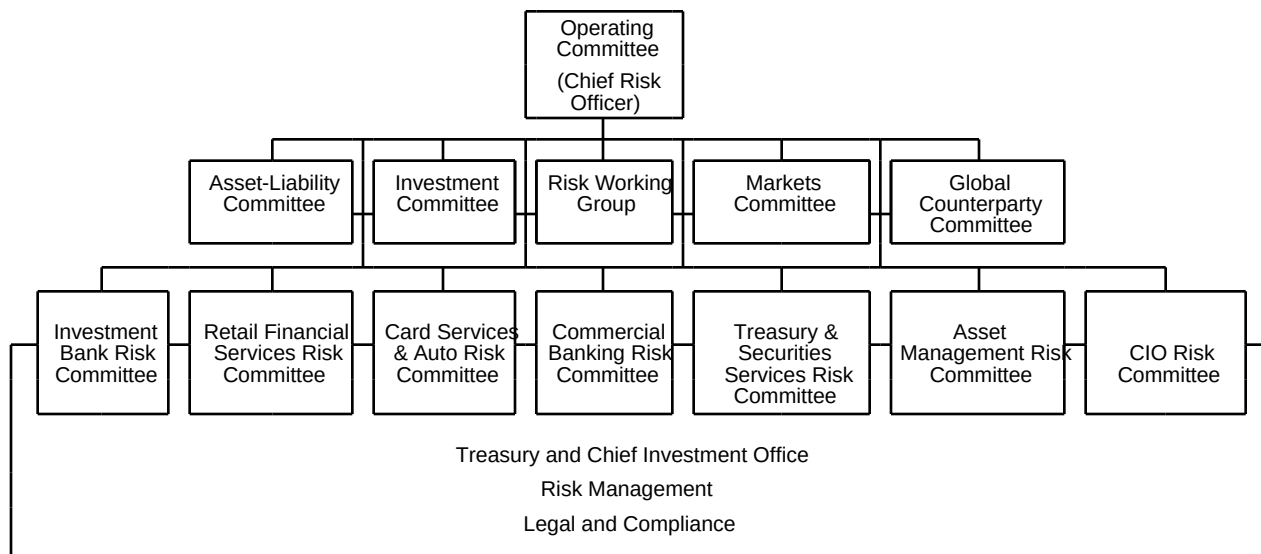
Overlaying line of business risk management are four corporate functions with risk management-related responsibilities: Risk Management, the Chief Investment Office, Corporate Treasury, and Legal and Compliance.

Risk Management operates independently of the lines of businesses to provide oversight of firmwide risk management and controls, and is viewed as a partner in achieving appropriate business objectives. Risk Management coordinates and communicates with each line of business through the line of business risk committees and chief risk officers to manage risk. The Risk Management function is headed by the Firm's Chief Risk Officer, who is a member of the Firm's Operating Committee and who reports to the Chief Executive Officer and is accountable to the Board of Directors, primarily through the Board's Risk Policy Committee. The Chief Risk Officer is also a member of the line of business risk committees. Within the Firm's Risk Management function are units responsible for credit risk, market risk, country risk, private equity risk and operational risk, as well as risk reporting, risk policy and risk technology and operations. Risk technology and operations is responsible for building the information technology infrastructure used to monitor and manage risk.

The Chief Investment Office and Corporate Treasury are responsible for measuring, monitoring, reporting and managing the Firm's liquidity, interest rate and foreign exchange risk, and other structural risks.

Legal and Compliance has oversight for legal risk.

In addition to the risk committees of the lines of business and the above-referenced risk management functions, the Firm also has an Investment Committee, an Asset-Liability Committee and three other risk-related committees – the Risk Working Group, the Global Counterparty Committee and the Markets Committee. All of these committees are accountable to the Operating Committee. The membership of these committees are composed of senior management of the Firm, including representatives of the lines of business, Risk Management, Finance and other senior executives. The committees meet frequently to discuss a broad range of topics including, for example, current market conditions and other external events, risk exposures, and risk concentrations to ensure that the impact of risk factors are considered broadly across the Firm's businesses.



The Asset-Liability Committee (“ALCO”), chaired by the Corporate Treasurer, monitors the Firm’s overall interest rate risk and liquidity risk. ALCO is responsible for reviewing and approving the Firm’s liquidity policy and contingency funding plan. ALCO also reviews the Firm’s funds transfer pricing policy (through which lines of business “transfer” interest rate and foreign exchange risk to Corporate Treasury in the Corporate/Private Equity segment), nontrading interest rate-sensitive revenue-at-risk, overall interest rate position, funding requirements and strategy, and the Firm’s securitization programs (and any required liquidity support by the Firm of such programs).

The Investment Committee, chaired by the Firm’s Chief Financial Officer, oversees global merger and acquisition activities undertaken by JPMorgan Chase for its own account that fall outside the scope of the Firm’s private equity and other principal finance activities.

The Risk Working Group, chaired by the Firm’s Chief Risk Officer, meets monthly to review issues that cross lines of business such as risk policy, risk methodology, risk concentrations, regulatory capital and other regulatory issues, and such other topics referred to it by line of business risk committees.

The Markets Committee, chaired by the Firm’s Chief Risk Officer, meets weekly to review, monitor and discuss significant risk matters, which may include credit, market and operational risk issues; market moving events; large transactions; hedging strategies; transactions that may give rise to reputation risk or conflicts of interest; and other issues.

The Global Counterparty Committee, chaired by the Firm’s Chief Risk Officer, reviews exposures to counterparties when such exposure levels are above portfolio-established thresholds. The Committee meets quarterly to review total exposures with these counterparties, with particular focus

on counterparty trading exposures to ensure that such exposures are deemed appropriate and to direct changes in exposure levels as needed.

The Board of Directors exercises its oversight of risk management, principally through the Board’s Risk Policy Committee and Audit Committee. The Risk Policy Committee oversees senior management risk-related responsibilities, including reviewing management policies and performance against these policies and related benchmarks. The Audit Committee is responsible for oversight of guidelines and policies that govern the process by which risk assessment and management is undertaken. In addition, the Audit Committee reviews with management the system of internal controls that is relied upon to provide reasonable assurance of compliance with the Firm’s operational risk management processes.

Risk monitoring and control

The Firm’s ability to properly identify, measure, monitor and report risk is critical to both its soundness and profitability.

- **Risk identification:** The Firm’s exposure to risk through its daily business dealings, including lending and capital markets activities, is identified and aggregated through the Firm’s risk management infrastructure. In addition, individuals who manage risk positions, particularly those that are complex, are responsible for identifying and estimating potential losses that could arise from specific or unusual events that may not be captured in other models, and for communicating those risks to senior management.
- **Risk measurement:** The Firm measures risk using a variety of methodologies, including calculating probable loss, unexpected loss and value-at-risk, and by conducting stress tests and making comparisons to external benchmarks. Measurement models and related assumptions are routinely subject to internal model

review, empirical validation and benchmarking with the goal of ensuring that the Firm's risk estimates are reasonable and reflective of the risk of the underlying positions.

- Risk monitoring/control: The Firm's risk management policies and procedures incorporate risk mitigation strategies and include approval limits by customer, product, industry, country and business. These limits are monitored on a daily, weekly and monthly basis, as appropriate.

- Risk reporting: The Firm reports risk exposures on both a line of business and a consolidated basis. This information is reported to management on a daily, weekly and monthly basis, as appropriate. There are nine major risk types identified in the business activities of the Firm: liquidity risk, credit risk, market risk, interest rate risk, country risk, private equity risk, operational risk, legal and fiduciary risk, and reputation risk.

LIQUIDITY RISK MANAGEMENT

Liquidity is essential to the ability to operate financial services businesses and, therefore, the ability to maintain surplus levels of liquidity through economic cycles is crucial to financial services companies, particularly during periods of adverse conditions. The Firm relies on external sources to finance a significant portion of its operations, and the Firm's funding strategy is intended to ensure that it will have sufficient liquidity and a diversity of funding sources necessary to enable it to meet actual and contingent liabilities during both normal and stress periods.

JPMorgan Chase's primary sources of liquidity include a diversified deposit base, which was \$1,127.8 billion at December 31, 2011, and access to the equity capital markets and to long-term unsecured and secured funding sources, including through asset securitizations and borrowings from FHLBs. Additionally, JPMorgan Chase maintains significant amounts of highly-liquid unencumbered assets. The Firm actively monitors the availability of funding in the wholesale markets across various geographic regions and in various currencies. The Firm's ability to generate funding from a broad range of sources in a variety of geographic locations and in a range of tenors is intended to enhance financial flexibility and limit funding concentration risk.

Management considers the Firm's liquidity position to be strong, based on its liquidity metrics as of December 31, 2011, and believes that the Firm's unsecured and secured funding capacity is sufficient to meet its on- and off-balance sheet obligations. The Firm was able to access the funding markets as needed during the year ended December 31, 2011, despite increased market volatility.

Governance

The Firm's liquidity risk governance process is designed to ensure that its liquidity position remains strong. The Asset-Liability Committee reviews and approves the Firm's liquidity policy and contingency funding plan. Corporate Treasury is responsible for executing the Firm's liquidity policy and contingency funding plan as well as measuring, monitoring, reporting and managing the Firm's liquidity risk profile. JPMorgan Chase centralizes the management of global funding and liquidity risk within Corporate Treasury. This centralized approach maximizes liquidity access, minimizes funding costs and enhances global identification

and coordination of liquidity risk and involves frequent communication with the business segments, disciplined management of liquidity at the parent holding company, comprehensive market-based pricing of all financial assets and liabilities, continuous balance sheet monitoring, frequent stress testing of liquidity sources, and frequent reporting and communication provided to senior management and the Board of Directors regarding the Firm's liquidity position.

Liquidity monitoring

The Firm employs a variety of metrics to monitor and manage liquidity. One set of analyses used by the Firm relates to the timing of liquidity sources versus liquidity uses (e.g., funding gap analysis and parent holding company funding, as discussed below). A second set of analyses focuses on measurements of the Firm's reliance on short-term unsecured funding as a percentage of total liabilities, as well as the relationship of short-term unsecured funding to highly-liquid assets, the deposits-to-loans ratio and other balance sheet measures.

The Firm performs regular liquidity stress tests as part of its liquidity monitoring activities. The purpose of the liquidity stress tests is intended to ensure sufficient liquidity for the Firm under both idiosyncratic and systemic market stress conditions. These scenarios measure the Firm's liquidity position across a full-year horizon by analyzing the net funding gaps resulting from contractual and contingent cash and collateral outflows versus the Firm's ability to generate additional liquidity by pledging or selling excess collateral and issuing unsecured debt. The scenarios are produced for the parent holding company and major bank subsidiaries as well as the Firm's principal U.S. broker-dealer subsidiary.

The Firm currently has liquidity in excess of its projected full-year liquidity needs under both its idiosyncratic stress scenario (which evaluates the Firm's net funding gap after a short-term ratings downgrade to A-2/P-2), as well as under its systemic market stress scenario (which evaluates the Firm's net funding gap during a period of severe market stress similar to market conditions in 2008 and assumes that the Firm is not uniquely stressed versus its peers).

Parent holding company

Liquidity monitoring of the parent holding company takes

into consideration regulatory restrictions that limit the extent to which bank subsidiaries may extend credit to the parent holding company and other nonbank subsidiaries. Excess cash generated by parent holding company issuance activity is used to purchase liquid collateral through reverse repurchase agreements or is placed with both bank and nonbank subsidiaries in the form of deposits and advances to satisfy a portion of subsidiary funding requirements. The Firm's liquidity management takes into consideration its subsidiaries' ability to generate replacement funding in the event the parent holding company requires repayment of the aforementioned deposits and advances.

The Firm closely monitors the ability of the parent holding company to meet all of its obligations with liquid sources of cash or cash equivalents for an extended period of time without access to the unsecured funding markets. The Firm targets pre-funding of parent holding company obligations for at least 12 months; however, due to conservative liquidity management actions taken by the Firm in the current environment, the current pre-funding of such obligations is significantly greater than target.

Global Liquidity Reserve

In addition to the parent holding company, the Firm maintains a significant amount of liquidity – primarily at its bank subsidiaries, but also at its nonbank subsidiaries. The Global Liquidity Reserve represents consolidated sources of available liquidity to the Firm, including cash on deposit at central banks, and cash proceeds reasonably expected to be received in secured financings of highly liquid, unencumbered securities, such as government-issued debt, government- and FDIC-guaranteed corporate debt, U.S. government agency debt, and agency MBS. The liquidity amount estimated to be realized from secured financings is based on management's current judgment and assessment of the Firm's ability to quickly raise funds from secured financings. The Global Liquidity Reserve also includes the Firm's borrowing capacity at various FHLBs, the Federal Reserve Bank discount window and various other central banks as a result of collateral pledged by the Firm to such banks. Although considered as a source of available liquidity, the Firm does not view borrowing capacity at the Federal Reserve Bank discount window and various other central banks as a primary source of funding.

As of December 31, 2011, the Global Liquidity Reserve was estimated to be approximately \$379 billion, compared with approximately \$262 billion at December 31, 2010. The increase in the Global Liquidity Reserve reflected the placement of funds with various central banks, including Federal Reserve Banks, which was driven by an increase in deposits during the second half of 2011. For further discussion see Sources of funds below.

In addition to the Global Liquidity Reserve, the Firm has significant amounts of other high-quality, marketable securities available to raise liquidity, such as corporate debt and equity securities.

Basel III

On December 16, 2010, the Basel Committee published the final Basel III rules pertaining to capital and liquidity requirements, including minimum standards for short-term liquidity coverage – the liquidity coverage ratio (the "LCR") – and term funding – the net stable funding ratio (the "NSFR"). For more information, see the discussion on Basel III on pages 121–122 of this Annual Report.

Funding

Sources of funds

A key strength of the Firm is its diversified deposit franchise, through the RFS, CB, TSS and AM lines of business, which provides a stable source of funding and decreases reliance on the wholesale markets. As of December 31, 2011, total deposits for the Firm were \$1,127.8 billion, compared with \$930.4 billion at December 31, 2010. The significant increase in deposits was predominantly due to an overall growth in wholesale client balances and, to a lesser extent, consumer deposit balances. The increase in wholesale client balances, particularly in TSS and CB, was primarily driven by lower returns on other available alternative investments and low interest rates during 2011. Also contributing to the increase in deposits was growth in the number of clients and level of deposits in AM and RFS (the RFS deposits were net of attrition related to the conversion of Washington Mutual Free Checking accounts). Average total deposits for the Firm were \$1,012.0 billion and \$881.1 billion for the years ended December 31, 2011 and 2010, respectively.

The Firm typically experiences higher customer deposit inflows at period-ends. A significant portion of the Firm's deposits are retail deposits (35% and 40% at December 31, 2011 and 2010, respectively), which are considered particularly stable as they are less sensitive to interest rate changes or market volatility. A significant portion of the Firm's wholesale deposits are also considered to be stable sources of funding due to the nature of the relationships from which they are generated, particularly customers' operating service relationships with the Firm. As of December 31, 2011, the Firm's deposits-to-loans ratio was 156%, compared with 134% at December 31, 2010. For further discussions of deposit and liability balance trends, see the discussion of the results for the Firm's business segments and the Balance Sheet Analysis on pages 79–80 and 110–112, respectively, of this Annual Report.

Additional sources of funding include a variety of unsecured and secured short-term and long-term instruments. Short-term unsecured funding sources include federal funds and Eurodollars purchased, certificates of deposit, time deposits, commercial paper and other borrowed funds. Long-term unsecured funding sources include long-term debt, preferred stock and common stock.

The Firm's short-term secured sources of funding consist of securities loaned or sold under agreements to repurchase and other short-term secured other borrowed funds. Secured long-term funding sources include asset-backed

securitizations, and borrowings from the Chicago, Pittsburgh and San Francisco FHLBs.

Funding markets are evaluated on an ongoing basis to achieve an appropriate global balance of unsecured and secured funding at favorable rates.

Short-term funding

The Firm's reliance on short-term unsecured funding sources is limited. Short-term unsecured funding sources include federal funds and Eurodollars purchased, which represent overnight funds; certificates of deposit; time deposits; commercial paper, which is generally issued in amounts not less than \$100,000 and with maturities of 270 days or less; and other borrowed funds, which consist of demand notes, term federal funds purchased, and various other borrowings that generally have maturities of one year or less.

Total commercial paper liabilities were \$51.6 billion as of December 31, 2011, compared with \$35.4 billion as of December 31, 2010. However, of those totals, \$47.4 billion and \$29.2 billion as of December 31, 2011 and 2010, respectively, originated from deposits that customers chose to sweep into commercial paper liabilities as a cash management product offered by the Firm. Therefore, commercial paper liabilities sourced from wholesale funding markets were \$4.2 billion as of December 31, 2011, compared with \$6.2 billion as of December 31, 2010; the average balance of commercial paper liabilities sourced from wholesale funding markets were \$6.1 billion and \$9.5 billion for the years ended December 31, 2011 and 2010, respectively.

Securities loaned or sold under agreements to repurchase, which generally mature between one day and three months, are secured predominantly by high-quality securities collateral, including government-issued debt, agency debt and agency MBS. The balances of securities loaned or sold under agreements to repurchase, which constitute a significant portion of the federal funds purchased and securities loaned or sold under repurchase agreements, was \$212.0 billion as of December 31, 2011, compared with \$273.3 billion as of December 31, 2010; the average balance was \$252.6 billion and \$271.5 billion for the years ended December 31, 2011 and 2010, respectively. At December 31, 2011, the decline in the balance, compared with the balance at December 31, 2010, and the average balance for the year ended December 31, 2011, was driven largely by lower financing of the Firm's trading assets and change in the mix of funding sources. The balances associated with securities loaned or sold under agreements to repurchase fluctuate over time due to customers' investment and financing activities; the Firm's demand for financing; the Firm's matched book activity; the ongoing management of the mix of the Firm's liabilities, including its secured and unsecured financing (for both the investment and market-making portfolios); and other market and portfolio factors.

Total other borrowed funds was \$21.9 billion as of December 31, 2011, compared with \$34.3 billion as of December 31, 2010; the average balance of other borrowed funds was \$30.9 billion and \$33.0 billion for the years ended December 31, 2011 and 2010, respectively. At December 31, 2011, the decline in the balance, compared with the balance at December 31, 2010, and the average balances for the year ended December 31, 2011, was predominantly driven by maturities of short-term unsecured bank notes, short-term FHLB advances, and other secured short-term borrowings.

For additional information, see the Balance Sheet Analysis on pages 110–112, Note 13 on page 231 and the table of Short-term and other borrowed funds on page 307 of this Annual Report.

Long-term funding and issuance

During the year ended December 31, 2011, the Firm issued \$49.0 billion of long-term debt, including \$29.0 billion of senior notes issued in the U.S. market, \$5.2 billion of senior notes issued in non-U.S. markets, and \$14.8 billion of IB structured notes. In addition, in January 2012, the Firm issued \$3.3 billion of senior notes in the U.S. market and \$2.1 billion of senior notes in non-U.S. markets. During the year ended December 31, 2010, the Firm issued \$36.1 billion of long-term debt, including \$17.1 billion of senior notes issued in U.S. markets, \$2.9 billion of senior notes issued in non-U.S. markets, \$1.5 billion of trust preferred capital debt securities and \$14.6 billion of IB structured notes. During the year ended December 31, 2011, \$58.5 billion of long-term debt matured or was redeemed, including \$18.7 billion of IB structured notes. During the year ended December 31, 2010, \$53.4 billion of long-term debt matured or was redeemed, including \$907 million of trust preferred capital debt securities and \$22.8 billion of IB structured notes.

In addition to the unsecured long-term funding and issuances discussed above, the Firm securitizes consumer credit card loans, residential mortgages, auto loans and student loans for funding purposes. During the year ended December 31, 2011, the Firm securitized \$1.8 billion of credit card loans; \$14.0 billion of loan securitizations matured or were redeemed, including \$13.6 billion of credit card loan securitizations, \$156 million of residential mortgage loan securitizations and \$322 million of student loan securitizations. During the year ended December 31, 2010, the Firm did not securitize any loans for funding purposes; \$25.8 billion of loan securitizations matured or were redeemed, including \$24.9 billion of credit card loan securitizations, \$294 million of residential mortgage loan securitizations, \$326 million of student loan securitizations, and \$210 million of auto loan securitizations.

In addition, the Firm's wholesale businesses securitize loans for client-driven transactions; those client-driven loan securitizations are not considered to be a source of funding for the Firm.

Management's discussion and analysis

During the year ended December 31, 2011, the Firm borrowed \$4.0 billion in long-term advances from the FHLBs and there were \$9.2 billion of maturities. For the year ended December 31, 2010, the Firm borrowed \$18.7 billion in long-term advances from the FHLBs, which was offset by \$18.6 billion of maturities.

Cash flows

For the years ended December 31, 2011, 2010 and 2009, cash and due from banks increased \$32.0 billion and \$1.4 billion, and decreased \$689 million, respectively. The following discussion highlights the major activities and transactions that affected JPMorgan Chase's cash flows during 2011, 2010 and 2009, respectively.

Cash flows from operating activities

JPMorgan Chase's operating assets and liabilities support the Firm's capital markets and lending activities, including the origination or purchase of loans initially designated as held-for-sale. Operating assets and liabilities can vary significantly in the normal course of business due to the amount and timing of cash flows, which are affected by client-driven and risk management activities, and market conditions. Management believes cash flows from operations, available cash balances and the Firm's ability to generate cash through short- and long-term borrowings are sufficient to fund the Firm's operating liquidity needs.

For the year ended December 31, 2011, net cash provided by operating activities was \$95.9 billion. This resulted from a net decrease in trading assets and liabilities—debt and equity instruments, driven by client market-making activity in IB; an increase in accounts payable and other liabilities predominantly due to higher IB customer balances; and a decrease in accrued interest and accounts receivables, primarily in IB, driven by a large reduction in customer margin receivables due to changes in client activity. Partially offsetting these cash proceeds was an increase in securities borrowed, predominantly in Corporate due to higher excess cash positions at year-end. Net cash generated from operating activities was higher than net income largely as a result of adjustments for noncash items such as the provision for credit losses, depreciation and amortization, and stock-based compensation. Additionally, cash provided by proceeds from sales and paydowns of loans originated or purchased with an initial intent to sell was higher than cash used to acquire such loans, and also reflected a higher level of activity over the prior-year period.

For the year ended December 31, 2010, net cash used by operating activities was \$3.8 billion, mainly driven by an increase primarily in trading assets—debt and equity instruments; principally due to improved market activity primarily in equity securities, foreign debt and physical commodities, partially offset by an increase in trading liabilities due to higher levels of positions taken to facilitate customer-driven activity. Net cash was provided by net income and from adjustments for non-cash items such as the provision for credit losses, depreciation and

amortization and stock-based compensation. Additionally, proceeds from sales and paydowns of loans originated or purchased with an initial intent to sell were higher than cash used to acquire such loans.

For the year ended December 31, 2009, net cash provided by operating activities was \$122.8 billion, reflecting the net decline in trading assets and liabilities affected by the impact of the challenging capital markets environment that existed in 2008, and continued into the first half of 2009. Net cash generated from operating activities was higher than net income, largely as a result of adjustments for non-cash items such as the provision for credit losses. In addition, proceeds from sales, securitizations and paydowns of loans originated or purchased with an initial intent to sell were higher than cash used to acquire such loans, but the cash flows from these loan activities remained at reduced levels as a result of the lower activity in these markets.

Cash flows from investing activities

The Firm's investing activities predominantly include loans originated to be held for investment, the AFS securities portfolio and other short-term interest-earning assets. For the year ended December 31, 2011, net cash of \$170.8 billion was used in investing activities. This resulted from a significant increase in deposits with banks reflecting the placement of funds with various central banks, including Federal Reserve Banks, predominantly resulting from the overall growth in wholesale client deposits; an increase in loans reflecting continued growth in client activity across all of the Firm's wholesale businesses and regions; net purchases of AFS securities, largely due to repositioning of the portfolio in Corporate in response to changes in the market environment; and an increase in securities purchased under resale agreements, predominantly in Corporate due to higher excess cash positions at year-end. Partially offsetting these cash outflows were a decline in consumer, excluding credit card, loan balances due to paydowns and portfolio run-off, and in credit card loans, due to higher repayment rates, run-off of the Washington Mutual portfolio and the Firm's sale of the Kohl's portfolio.

For the year ended December 31, 2010, net cash of \$54.0 billion was provided by investing activities. This resulted from a decrease in deposits with banks largely due to a decline in deposits placed with the Federal Reserve Bank and lower interbank lending as market stress eased since the end of 2009; net proceeds from sales and maturities of AFS securities used in the Firm's interest rate risk management activities in Corporate; and a net decrease in the credit card loan portfolio, driven by the expected runoff of the Washington Mutual portfolio, a decline in lower-yielding promotional credit card balances, continued runoff of loan balances in the consumer, excluding credit card portfolio, primarily related to residential real estate, and repayments and loan sales in the wholesale portfolio, primarily in IB and CB; the decrease was partially offset by higher originations across the wholesale and consumer businesses. Partially offsetting these cash proceeds was an increase in securities purchased under resale agreements,

predominantly due to higher financing volume in IB; and cash used for business acquisitions, primarily RBS Semptra. For the year ended December 31, 2009, net cash of \$29.4 billion was provided by investing activities, primarily from a decrease in deposits with banks reflecting lower demand for inter-bank lending and lower deposits with the Federal Reserve Bank relative to the elevated levels at the end of 2008; a net decrease in the loan portfolio across most businesses, driven by continued lower customer demand and loan sales in the wholesale portfolio, lower charge volume on credit cards, slightly higher credit card securitizations, and paydowns; and the maturity of all asset-backed commercial paper issued by money market mutual funds in connection with the Federal Reserve Bank of Boston's Asset-Backed Commercial Paper Money Market Mutual Fund Liquidity Facility ("AML facility"). Largely offsetting these cash proceeds were net purchases of AFS securities associated with the Firm's management of interest rate risk and investment of cash resulting from an excess funding position.

Cash flows from financing activities

The Firm's financing activities primarily reflect cash flows related to taking customer deposits, and issuing long-term debt as well as preferred and common stock. For the year ended December 31, 2011, net cash provided by financing activities was \$107.7 billion. This was largely driven by a significant increase in deposits, predominantly due to an overall growth in wholesale client balances and, to a lesser extent, consumer deposit balances. The increase in wholesale client balances, particularly in TSS and CB, was primarily driven by lower returns on other available alternative investments and low interest rates during 2011, and in AM, driven by growth in the number of clients and level of deposits. In addition, there was an increase in commercial paper due to growth in the volume of liability balances in sweep accounts related to TSS's cash management product. Cash was used to reduce securities sold under repurchase agreements, predominantly in IB, reflecting the lower funding requirements of the Firm based on lower trading inventory levels, and change in the mix of funding sources; for net repayments of long-term borrowings, including a decrease in long-term debt, predominantly due to net redemptions and maturities, as well as a decline in long-term beneficial interests issued by consolidated VIEs due to maturities of Firm-sponsored credit card securitization transactions; to reduce other borrowed funds, predominantly driven by maturities of short-term secured borrowings, unsecured bank notes and short-term FHLB advances; and for repurchases of common stock and warrants, and payments of cash dividends on common and preferred stock.

In 2010, net cash used in financing activities was \$49.2 billion. This resulted from net repayments of long-term borrowings as new issuances were more than offset by payments primarily reflecting a decline in beneficial interests issued by consolidated VIEs due to maturities related to Firm-sponsored credit card securitization trusts;

a decline in deposits associated with wholesale funding activities due to the Firm's lower funding needs; lower deposit levels in TSS, offset partially by net inflows from existing customers and new business in AM, CB and RFS; a decline in commercial paper and other borrowed funds due to lower funding requirements; payments of cash dividends; and repurchases of common stock. Cash was generated as a result of an increase in securities sold under repurchase agreements largely as a result of an increase in activity levels in IB partially offset by a decrease in CIO reflecting repositioning activities.

In 2009, net cash used in financing activities was \$153.1 billion; this reflected a decline in wholesale deposits, predominantly in TSS, driven by the continued normalization of wholesale deposit levels resulting from the mitigation of credit concerns, compared with the heightened market volatility and credit concerns in the latter part of 2008; a decline in other borrowings, due to the absence of borrowings from the Federal Reserve under the Term Auction Facility program; net repayments of short-term advances from FHLBs and the maturity of the nonrecourse advances under the Federal Reserve Bank of Boston AML Facility; the June 17, 2009, repayment in full of the \$25.0 billion principal amount of Series K Preferred Stock issued to the U.S. Treasury; and the payment of cash dividends on common and preferred stock. Cash was also used for the net repayment of long-term borrowings as issuances of FDIC-guaranteed debt and non-FDIC guaranteed debt in both the U.S. and European markets were more than offset by repayments including long-term advances from FHLBs. Cash proceeds resulted from an increase in securities loaned or sold under repurchase agreements, partly attributable to favorable pricing and to financing the increased size of the Firm's AFS securities portfolio; and the issuance of \$5.8 billion of common stock. There were no repurchases of common stock or the warrants during 2009.

Credit ratings

The cost and availability of financing are influenced by credit ratings. Reductions in these ratings could have an adverse effect on the Firm's access to liquidity sources, increase the cost of funds, trigger additional collateral or funding requirements and decrease the number of investors and counterparties willing to lend to the Firm. Additionally, the Firm's funding requirements for VIEs and other third-party commitments may be adversely affected by a decline in credit ratings. For additional information on the impact of a credit ratings downgrade on the funding requirements for VIEs, and on derivatives and collateral agreements, see Special-purpose entities on page 113, and Note 6 on pages 202–210, respectively, of this Annual Report.

Critical factors in maintaining high credit ratings include a stable and diverse earnings stream, strong capital ratios, strong credit quality and risk management controls, diverse funding sources, and disciplined liquidity monitoring procedures.

Management's discussion and analysis

The credit ratings of the parent holding company and each of the Firm's significant banking subsidiaries as of December 31, 2011, were as follows.

	Short-term debt			Senior long-term debt		
	Moody's	S&P	Fitch	Moody's	S&P	Fitch
JPMorgan Chase & Co.	P-1	A-1	F1+	Aa3	A	AA-
JPMorgan Chase Bank, N.A.	P-1	A-1	F1+	Aa1	A+	AA-
Chase Bank USA, N.A.	P-1	A-1	F1+	Aa1	A+	AA-

On July 18, 2011, Moody's placed the long-term debt ratings of the Firm and its subsidiaries under review for possible downgrade. The Firm's current long-term debt ratings by Moody's reflect "support uplift" above the Firm's stand-alone financial strength due to Moody's assessment of the likelihood of U.S. government support. Moody's action was directly related to Moody's placing the U.S. government's Aaa rating on review for possible downgrade on July 13, 2011. Moody's indicated that the action did not reflect a change to Moody's opinion of the Firm's stand-alone financial strength. The short-term debt ratings of the Firm and its subsidiaries were affirmed and were not affected by the action. Subsequently, on August 3, 2011, Moody's confirmed the long-term debt ratings of the Firm and its subsidiaries at their current levels and assigned a negative outlook on the ratings. The rating confirmation was directly related to Moody's confirmation on August 2, 2011, of the Aaa rating assigned to the U.S. government.

On November 29, 2011, S&P lowered the long-term debt rating of the parent holding company from A+ to A, and the long-term and short-term debt ratings of the Firm's significant banking subsidiaries from AA- to A+ and from A-1+ to A-1, respectively. The action resulted from a review of the Firm along with all other banks rated by S&P under S&P's revised bank rating criteria. The downgrade had no adverse impact on the Firm's ability to fund itself.

The senior unsecured ratings from Moody's and Fitch on JPMorgan Chase and its principal bank subsidiaries remained unchanged at December 31, 2011, from

December 31, 2010. At December 31, 2011, Moody's outlook was negative, while S&P's and Fitch's outlooks were stable.

On February 15, 2012, Moody's announced that it had placed 17 banks and securities firms with global capital markets operations on review for possible downgrade, including JPMorgan Chase. As part of this announcement, the long-term ratings of the Firm and its major operating entities were placed on review for possible downgrade, while all of the Firm's short-term ratings were affirmed.

If the Firm's senior long-term debt ratings were downgraded by one notch or two notches, the Firm believes its cost of funds would increase; however, the Firm's ability to fund itself would not be materially adversely impacted. JPMorgan Chase's unsecured debt does not contain requirements that would call for an acceleration of payments, maturities or changes in the structure of the existing debt, provide any limitations on future borrowings or require additional collateral, based on unfavorable changes in the Firm's credit ratings, financial ratios, earnings, or stock price.

Rating agencies continue to evaluate various ratings factors, such as regulatory reforms, economic uncertainty and sovereign creditworthiness, and their potential impact on ratings of financial institutions. Although the Firm closely monitors and endeavors to manage factors influencing its credit ratings, there is no assurance that its credit ratings will not be changed in the future.

CREDIT RISK MANAGEMENT

Credit risk is the risk of loss from obligor or counterparty default. The Firm provides credit (for example, through loans, lending-related commitments, guarantees and derivatives) to a variety of customers, from large corporate and institutional clients to the individual consumers and small businesses. Loans originated or acquired by the Firm's wholesale businesses are generally retained on the balance sheet. Credit risk management actively monitors the wholesale portfolio to ensure that it is well diversified across industry, geography, risk rating, maturity and individual client categories. Portfolio management for wholesale loans includes, for the Firm's syndicated loan business, distributing originations into the market place and

targeting exposure held in the retained wholesale portfolio at less than 10% of the customer facility. With regard to the consumer credit market, the Firm focuses on creating a portfolio that is diversified from a product, industry and geographic perspective. Loss mitigation strategies are being employed for all residential real estate portfolios. These strategies include interest rate reductions, term or payment extensions, principal and interest deferral and other actions intended to minimize economic loss and avoid foreclosure. In the mortgage business, originated loans are either retained in the mortgage portfolio or securitized and sold to U.S. government agencies and U.S. government-sponsored enterprises.

Credit risk organization

Credit risk management is overseen by the Chief Risk Officer and implemented within the lines of business. The Firm's credit risk management governance consists of the following functions:

- Establishing a comprehensive credit risk policy framework
- Monitoring and managing credit risk across all portfolio segments, including transaction and line approval
- Assigning and managing credit authorities in connection with the approval of all credit exposure
- Managing criticized exposures and delinquent loans
- Determining the allowance for credit losses and ensuring appropriate credit risk-based capital management

Risk identification and measurement

The Firm is exposed to credit risk through lending and capital markets activities. Credit Risk Management works in partnership with the business segments in identifying and aggregating exposures across all lines of business. To measure credit risk, the Firm employs several methodologies for estimating the likelihood of obligor or counterparty default. Methodologies for measuring credit risk vary depending on several factors, including type of asset (e.g., consumer versus wholesale), risk measurement parameters (e.g., delinquency status and borrower's credit score versus wholesale risk-rating) and risk management and collection processes (e.g., retail collection center versus centrally managed workout groups). Credit risk measurement is based on the amount of exposure should the obligor or the counterparty default, the probability of default and the loss severity given a default event. Based on these factors and related market-based inputs, the Firm estimates both probable losses and unexpected losses for the wholesale and consumer portfolios as follows:

- Probable credit losses are based primarily upon statistical estimates of credit losses as a result of obligor or counterparty default. However, probable losses are not the sole indicators of risk.
- Unexpected losses, reflected in the allocation of credit risk capital, represent the potential volatility of actual losses relative to the probable level of incurred losses.

Risk measurement for the wholesale portfolio is assessed primarily on a risk-rated basis; for the consumer portfolio, it is assessed primarily on a credit-scored basis.

Risk-rated exposure

Risk ratings are assigned to differentiate risk within the portfolio and are reviewed on an ongoing basis by Credit Risk Management and revised, if needed, to reflect the borrowers' current financial positions, risk profiles and the related collateral. For portfolios that are risk-rated, probable and unexpected loss calculations are based on estimates of probability of default and loss severity given a default. These risk-rated portfolios are generally held in IB, CB, TSS and AM; they also include approximately \$20.0

billion of certain business banking loans in RFS and certain auto loans in Card that are risk-rated because they have characteristics similar to commercial loans. Probability of default is the likelihood that a loan will default and will not be repaid. Probability of default is calculated for each client who has a risk-rated loan. Loss given default is an estimate of losses given a default event and takes into consideration collateral and structural support for each credit facility. Calculations and assumptions are based on management information systems and methodologies which are under continual review.

Credit-scored exposure

For credit-scored portfolios (generally held in RFS and Card), probable loss is based on a statistical analysis of inherent losses expected to emerge over discrete periods of time for each portfolio. The credit-scored portfolio includes residential real estate loans, credit card loans, certain auto and business banking loans, and student loans. Probable credit losses inherent in the portfolio are estimated using sophisticated portfolio modeling, credit scoring and decision-support tools, which take into account factors such as delinquency, LTV ratios, credit scores and geography. These analyses are applied to the Firm's current portfolios in order to estimate the severity of losses, which determines the amount of probable losses. Other risk characteristics utilized to evaluate probable losses include recent loss experience in the portfolios, changes in origination sources, portfolio seasoning, potential borrower behavior and the macroeconomic environment. These factors and analyses are updated on a quarterly basis or more frequently as market conditions dictate.

Risk monitoring and control

The Firm has developed policies and practices that are designed to preserve the independence and integrity of the approval and decision-making process of extending credit and to ensure credit risks are assessed accurately, approved properly, monitored regularly and managed actively at both the transaction and portfolio levels. The policy framework establishes credit approval authorities, concentration limits, risk-rating methodologies, portfolio review parameters and guidelines for management of distressed exposures. In addition, certain models, assumptions and inputs used in evaluating and monitoring credit risk are independently validated by groups that are separate from the line of businesses.

For consumer credit risk, delinquency and other trends, including any concentrations at the portfolio level, are monitored for potential problems, as certain of these trends can be ameliorated through changes in underwriting policies and portfolio guidelines. Consumer Credit Risk Management evaluates delinquency and other trends against business expectations, current and forecasted economic conditions, and industry benchmarks. Historical and forecasted trends are incorporated into the modeling of estimated consumer credit losses and are part of the monitoring of the credit risk profile of the portfolio. In the Firm's consumer credit portfolio, the Internal Audit

Management's discussion and analysis

department periodically tests the internal controls around the modeling process including the integrity of the data utilized. For further discussion of consumer loans, see Note 14 on pages 231–252 of this Annual Report.

Wholesale credit risk is monitored regularly at an aggregate portfolio, industry and individual counterparty basis with established concentration limits that are reviewed and revised, as deemed appropriate by management, typically on an annual basis. Industry and counterparty limits, as measured in terms of exposure and economic credit risk capital, are subject to stress-based loss constraints.

Management of the Firm's wholesale exposure is accomplished through a number of means including:

- Loan syndications and participations
- Loan sales and securitizations
- Credit derivatives
- Use of master netting agreements
- Collateral and other risk-reduction techniques

In addition to Risk Management, the Firm's Internal Audit department performs periodic exams, as well as continuous review, where appropriate, of the Firm's consumer and wholesale portfolios.

For risk-rated portfolios, a credit review group within the Internal Audit department is responsible for:

- Independently assessing and validating the changing risk grades assigned to exposures; and
- Evaluating the effectiveness of business units' risk ratings, including the accuracy and consistency of risk grades, the timeliness of risk grade changes and the justification of risk grades in credit memoranda

Risk reporting

To enable monitoring of credit risk and decision-making, aggregate credit exposure, credit quality forecasts, concentration levels and risk profile changes are reported regularly to senior Credit Risk Management. Detailed portfolio reporting of industry, customer, product and geographic concentrations occurs monthly, and the appropriateness of the allowance for credit losses is reviewed by senior management at least on a quarterly basis. Through the risk reporting and governance structure, credit risk trends and limit exceptions are provided regularly to, and discussed with, senior management. For further discussion of Risk monitoring and control, see pages 126–127 of this Annual Report.

CREDIT PORTFOLIO

2011 Credit Risk Overview

In the first half of 2011, the credit environment showed signs of improvement compared with 2010. During the second half of the year, macroeconomic conditions became more challenging, with increased market volatility and heightened concerns around the European financial crisis. Over the course of the year, the Firm continued to actively manage its underperforming and nonaccrual loans and reduce such exposures through repayments, loan sales and workouts. The Firm also saw decreased downgrade, default and charge-off activity and improved consumer delinquency trends. At the same time, the Firm increased its overall lending activity driven by the wholesale businesses. The combination of these factors resulted in an improvement in the credit quality of the portfolio compared with 2010 and contributed to the Firm's reduction in the allowance for credit losses, particularly in Card.

The credit quality of the Firm's wholesale portfolio improved in 2011. The rise in commercial client activity resulted in an increase in credit exposure across all businesses, regions and products. Underwriting guidelines across all areas of lending continue to remain in focus, consistent with evolving market conditions and the Firm's risk management activities. The wholesale portfolio continues to be actively managed, in part by conducting ongoing, in-depth reviews of credit quality and of industry, product and client concentrations. During the year, criticized assets, nonperforming assets and charge-offs

decreased from higher levels experienced in 2010, including a reduction in nonaccrual loans by over one half. As a result, the ratio of nonaccrual loans to total loans, the net charge-off rate and the allowance for loan loss coverage ratio all declined. For further discussion of wholesale loans, see Note 14 on pages 231–252 of this Annual Report.

The credit performance of the consumer portfolio across the entire product spectrum has improved, particularly in credit card, with lower levels of delinquent loans and charge-offs. Weak overall economic conditions continued to have a negative impact on the number of real estate loans charged off, while continued weak housing prices have resulted in an elevated severity of loss recognized on these defaulted loans. The Firm has taken proactive steps to assist homeowners most in need of financial assistance throughout the economic downturn. In addition, the Firm has taken actions since the onset of the economic downturn in 2007 to tighten underwriting and loan qualification standards and to eliminate certain products and loan origination channels, which have resulted in the reduction of credit risk and improved credit performance for recent loan vintages. For further discussion of the consumer credit environment and consumer loans, see Consumer Credit Portfolio on pages 145–154 and Note 14 on pages 231–252 of this Annual Report.

The following table presents JPMorgan Chase's credit portfolio as of December 31, 2011 and 2010. Total credit exposure was \$1.8 trillion at December 31, 2011, an

increase of \$44.4 billion from December 31, 2010, reflecting increases in loans of \$30.8 billion, lending related commitments of \$17.0 billion and derivative receivables of \$12.0 billion. These increases were partially offset by a decrease in receivables from customers and interests in purchased receivables of \$15.4 billion. The \$44.4 billion net increase during 2011 in total credit exposure reflected an increase in the wholesale portfolio of \$88.6 billion partially offset by a decrease in the consumer portfolio of \$44.2 billion.

The Firm provided credit to and raised capital of more than \$1.8 trillion for its clients during 2011, up 18% from

2010; this included \$17 billion lent to small businesses, up 52%, and \$68 billion to more than 1,200 not-for-profit and government entities, including states, municipalities, hospitals and universities. The Firm also originated more than 765,000 mortgages, and provided credit cards to approximately 8.5 million consumers. The Firm remains committed to helping homeowners and preventing foreclosures. Since the beginning of 2009, the Firm has offered more than 1.2 million mortgage modifications of which approximately 452,000 have achieved permanent modification as of December 31, 2011.

In the table below, reported loans include loans retained (i.e., held-for-investment); loans held-for-sale (which are carried at the lower of cost or fair value, with changes in value recorded in noninterest revenue); and loans accounted for at fair value. For additional information on the Firm's loans and derivative receivables, including the Firm's accounting policies, see Note 4 and Note 6 on pages 231–252 and 202–210, respectively, of this Annual Report. Average retained loan balances are used for net charge-off rate calculations.

Total credit portfolio

As of or for the year ended December 31, (in millions, except ratios)	Credit exposure		Nonperforming ^{(a)(d)(e)}		Net charge-offs		Average annual net charge-off rate ^(f)	
	2011	2010	2011	2010	2011	2010	2011	2010
Loans retained	\$ 718,997	\$ 685,498	\$ 9,810	\$ 14,345	\$ 12,237	\$ 23,673	1.78%	3.39%
Loans held-for-sale	2,626	5,453	110	341	—	—	—	—
Loans at fair value	2,097	1,976	73	155	—	—	—	—
Total loans – reported	723,720	692,927	9,993	14,841	12,237	23,673	1.78	3.39
Derivative receivables	92,477	80,481	18	34	NA	NA	NA	NA
Receivables from customers and interests in purchased receivables	17,561	32,932	—	—	—	—	—	—
Total credit-related assets	833,758	806,340	10,011	14,875	12,237	23,673	1.78	3.39
Lending-related commitments	975,662	958,709	865	1,005	NA	NA	NA	NA
Assets acquired in loan satisfactions								
Real estate owned	NA	NA	975	1,610	NA	NA	NA	NA
Other	NA	NA	50	72	NA	NA	NA	NA
Total assets acquired in loan satisfactions	NA	NA	1,025	1,682	NA	NA	NA	NA
Total credit portfolio	\$ 1,809,420	\$ 1,765,049	\$ 11,901	\$ 17,562	\$ 12,237	\$ 23,673	1.78%	3.39%
Net credit derivative hedges notional	\$ (26,240)	\$ (23,108)	\$ (38)	\$ (55)	NA	NA	NA	NA
Liquid securities and other cash collateral held against derivatives	(21,807)	(16,486)	NA	NA	NA	NA	NA	NA

(a) The amounts in nonperforming represent commitments that are risk rated as nonaccrual.

(b) Represents the net notional amount of protection purchased and sold of single-name and portfolio credit derivatives used to manage both performing and nonperforming credit exposures; these derivatives do not qualify for hedge accounting under U.S. GAAP. For additional information, see Credit derivatives on pages 143–144 and Note 6 on pages 202–210 of this Annual Report.

(c) At December 31, 2011 and 2010, nonperforming assets excluded: (1) mortgage loans insured by U.S. government agencies of \$11.5 billion and \$9.4 billion, respectively, that are 90 or more days past due; (2) real estate owned insured by U.S. government agencies of \$954 million and \$1.9 billion, respectively; and (3) student loans insured by U.S. government agencies under the FFELP of \$551 million and \$625 million, respectively, that are 90 or more days past due. These amounts were excluded as reimbursement of insured amounts is proceeding normally. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance issued by the Federal Financial Institutions Examination Council ("FFIEC"). Credit card loans are charged-off by the end of the month in which the account becomes 180 days past due within 60 days from receiving notification about a specified event (e.g., bankruptcy of the borrower), whichever is earlier.

(d) Excludes PCI loans acquired as part of the Washington Mutual transaction, which are accounted for on a pool basis. Since each pool is accounted for as a single asset with a single composite interest rate and an aggregate expectation of cash flows, the past due status of the pools, or that of individual loans within the pools, is not meaningful. Because the Firm is recognizing interest income on each pool of loans, they are all considered to be performing.

(e) At December 31, 2011 and 2010, total nonaccrual loans represented 1.38% and 2.14% of total loans.

(f) For the years ended December 31, 2011 and 2010, net charge-off rates were calculated using average retained loans of \$688.2 billion and \$698.2 billion, respectively. These average retained loans include average PCI loans of \$69.0 billion and \$77.0 billion, respectively. Excluding these PCI loans, the Firm's total charge-off rates would have been 1.98% and 3.81%, respectively.

The following table presents summaries of the maturity and ratings profiles of the wholesale portfolio as of December 31, 2011 and 2010. The increase in loans retained was predominately in loans to investment-grade (“IG”) counterparties and was largely loans having a shorter maturity profile. The ratings scale is based on the Firm’s internal risk ratings, which generally correspond to the ratings as defined by S&P and Moody’s. Also included in this table is the notional value of net credit derivative hedges; the counterparties to these hedges are predominantly investment-grade banks and finance companies.

Wholesale credit exposure – maturity and ratings profile

December 31, 2011 (in millions, except ratios)	Maturity profile ^(a)				Ratings profile			
	Due in 1 year or less	Due after 1 year through 5 years	Due after 5 years	Total	Investment-grade	Noninvestment- grade	Total	Total % of IG
					AAA/Aaa to BBB-/Baa3	BB+/Ba1 & below		
Loans retained	\$ 113,222	\$ 101,959	\$ 63,214	\$ 278,395	\$ 197,070	\$ 81,325	\$ 278,395	71%
Derivative receivables				92,477			92,477	
Less: Liquid securities and other cash collateral held against derivatives				(21,807)			(21,807)	
Total derivative receivables, net of all collateral	8,243	29,910	32,517	70,670	57,637	13,033	70,670	82
Lending-related commitments	139,978	233,396	9,365	382,739	310,107	72,632	382,739	81
Subtotal	261,443	365,265	105,096	731,804	564,814	166,990	731,804	77
Loans held-for-sale and loans at fair value				4,621			4,621	
Receivables from customers and interests in purchased receivables				17,461			17,461	
Total exposure – net of liquid securities and other cash collateral held against derivatives				\$ 753,886			\$ 753,886	
Net credit derivative hedges notional	\$ (2,034)	\$ (16,450)	\$ (7,756)	\$ (26,240)	\$ (26,300)	\$ 60	\$ (26,240)	100%

December 31, 2010 (in millions, except ratios)	Maturity profile ^(a)				Ratings profile			
	Due in 1 year or less	Due after 1 year through 5 years	Due after 5 years	Total	Investment-grade	Noninvestment- grade	Total	Total % of IG
					AAA/Aaa to BBB-/Baa3	BB+/Ba1 & below		
Loans retained	\$ 78,017	\$ 85,987	\$ 58,506	\$ 222,510	\$ 146,047	\$ 76,463	\$ 222,510	66%
Derivative receivables				80,481			80,481	
Less: Liquid securities and other cash collateral held against derivatives				(16,486)			(16,486)	
Total derivative receivables, net of all collateral	11,499	24,415	28,081	63,995	47,557	16,438	63,995	74
Lending-related commitments	126,389	209,299	10,391	346,079	276,298	69,781	346,079	80
Subtotal	215,905	319,701	96,978	632,584	469,902	162,682	632,584	74
Loans held-for-sale and loans at fair value				5,123			5,123	
Receivables from customers and interests in purchased receivables				32,932			32,932	
Total exposure – net of liquid securities and other cash collateral held against derivatives				\$ 670,639			\$ 670,639	
Net credit derivative hedges notional	\$ (1,228)	\$ (16,415)	\$ (5,465)	\$ (23,108)	\$ (23,159)	\$ 51	\$ (23,108)	100%

(a) Represents loans held-for-sale primarily related to syndicated loans and loans transferred from the retained portfolio, and loans at fair value.

(b) Represents the net notional amounts of protection purchased and sold of single-name and portfolio credit derivatives used to manage the credit exposures; these derivatives do not qualify for hedge accounting under U.S. GAAP.

(c) The maturity profiles of retained loans and lending-related commitments are based on the remaining contractual maturity. The maturity profiles of derivative receivables are based on the maturity profile of average exposure. For further discussion of average exposure, see Derivative receivables on pages 141–144 of this Annual Report.

Receivables from customers primarily represent margin loans to prime and retail brokerage clients and are collateralized through a pledge of assets maintained in clients’ brokerage accounts that are subject to daily minimum collateral requirements. In the event that the collateral value decreases, a maintenance margin call is made to the client to provide additional collateral into the account. If additional collateral is not provided by the client, the client’s position may be liquidated by the Firm to meet the minimum collateral requirements.

Wholesale credit exposure – selected industry exposures
The Firm focuses on the management and diversification of its industry exposures, with particular attention paid to industries with actual or potential credit concerns. Exposures deemed criticized generally represent a ratings profile similar to a rating of “CCC+”/“Caa1” and lower, as defined by S&P and Moody’s, respectively. The total criticized component of the portfolio, excluding loans held-for-sale and loans at fair value, decreased 29% to \$15.9 billion at December 31, 2011, from \$22.4 billion at December 31, 2010. The decrease was primarily related to net repayments and loan sales.

Management's discussion and analysis

Below are summaries of the top 25 industry exposures as of December 31, 2011 and 2010. For additional information on industry concentrations, see Note 5 on page 201 of this Annual Report.

As of or for the year ended	Noninvestment-grade					30 days or more past due and accruing loans	Full year net charge-offs/(recoveries)	Credit derivative hedges ⁵	Liquid securities and other cash collateral held against derivative receivables
December 31, 2011	Credit exposure ⁽¹⁾	Investment-grade	Noncriticized	Criticized performing	Criticized nonperforming				
(in millions)									
Top 25 industries ⁽³⁾									
Banks and finance companies	\$ 71,440	\$ 59,115	\$ 11,742	\$ 560	\$ 23	\$ 20	\$ (211)	\$ (3,053)	\$ (9,585)
Real estate	67,594	40,921	21,541	4,246	886	411	256	(97)	(359)
Healthcare	42,247	35,147	6,817	247	36	166	—	(304)	(320)
State and municipal governments	41,930	40,565	1,124	225	16	23	—	(185)	(147)
Oil and gas	35,437	25,004	10,337	96	—	3	—	(119)	(88)
Asset managers	33,465	28,835	4,530	99	1	24	—	—	(4,807)
Consumer products	29,637	19,728	9,439	447	23	3	13	(272)	(50)
Utilities	28,650	23,557	4,423	614	56	—	76	(105)	(359)
Retail and consumer services	22,891	14,568	7,796	464	63	15	1	(96)	(1)
Technology	17,898	12,494	5,085	319	—	—	4	(191)	—
Central government	17,138	16,524	488	126	—	—	—	(9,796)	(813)
Machinery and equipment manufacturing	16,498	9,014	7,375	103	6	1	(1)	(19)	—
Transportation	16,305	12,061	4,070	149	25	6	17	(178)	—
Metals/mining	15,254	8,716	6,388	150	—	6	(19)	(423)	—
Insurance	13,092	9,425	3,064	591	12	—	—	(552)	(454)
Business services	12,408	7,093	5,168	113	34	17	22	(20)	(2)
Securities firms and exchanges	12,394	10,799	1,564	30	1	10	73	(395)	(3,738)
Media	11,909	6,853	3,921	720	415	1	18	(188)	—
Building materials/construction	11,770	5,175	5,674	917	4	6	(4)	(213)	—
Chemicals/plastics	11,728	7,867	3,720	140	1	—	—	(95)	(20)
Telecom services	11,552	8,502	2,235	814	1	2	5	(390)	—
Automotive	9,910	5,699	4,188	23	—	9	(11)	(819)	—
Aerospace	8,560	7,646	848	66	—	7	—	(208)	—
Agriculture/paper manufacturing	7,594	4,888	2,586	120	—	9	—	-	—
Leisure	5,650	3,051	1,752	629	218	1	1	(81)	(26)
All other ⁽⁴⁾	180,660	161,568	17,011	1,486	595	1,099	200	(8,441)	(1,038)
Subtotal	\$ 753,611	\$ 584,815	\$ 152,886	\$ 13,494	\$ 2,416	\$ 1,839	\$ 440	\$ (26,240)	\$ (21,807)
Loans held-for-sale and loans at fair value	4,621								
Receivables from customers and interests in purchased receivables	17,461								
Total	\$ 775,693								

Presented below is a discussion of several industries to which the Firm has significant exposure, as well as industries the Firm continues to monitor because of actual or potential credit concerns. For additional information, refer to the tables above and on the next page.

- **Banks and finance companies** Exposure to this industry increased by \$5.6 billion or 8%, and criticized exposure decreased 3%, compared with 2010. The portfolio increased from 2010 and the investment grade portion remained high in proportion to the overall industry increase. As of December 31, 2011, 83% of the portfolio continued to be rated investment-grade, unchanged from 2010.

Real estate Exposure to this sector increased by \$3.2 billion or 5%, in 2011 to \$67.6 billion. The increase was primarily driven by CB, partially offset by decreases in credit exposure in IB. The credit quality of this industry improved as the investment-grade portion of this industry increased by 19% from 2010, while the criticized portion declined by 45% from 2010, primarily as a result of repayments and loans sales. The ratio of nonaccrual loans to total loans decreased to 2% from 5% in line with the decrease in real estate criticized exposure. For further information on commercial real estate loans, see Note 14 on pages 231–252 of this Annual Report.

As of or for the year ended	Noninvestment-grade					30 days or more past due and accruing loans	Full year net charge-offs/ (recoveries)	Credit derivative hedges ⁽⁹⁾	Liquid securities and other cash collateral held against derivative receivables
December 31, 2010	Credit exposure ⁽⁹⁾	Investment-grade	Noncriticized	Criticized performing	Criticized nonperforming				
(in millions)									
Top 25 industries ⁽⁹⁾									
Banks and finance companies	\$ 65,867	\$ 54,839	\$ 10,428	\$ 467	\$ 133	\$ 26	\$ 69	\$ (3,456)	\$ (9,216)
Real estate	64,351	34,440	20,569	6,404	2,938	399	862	(76)	(57)
Healthcare	41,093	33,752	7,019	291	31	85	4	(768)	(161)
State and municipal governments	35,808	34,641	912	231	24	34	3	(186)	(233)
Oil and gas	26,459	18,465	7,850	143	1	24	—	(87)	(50)
Asset managers	29,364	25,533	3,401	427	3	7	—	—	(2,948)
Consumer products	27,508	16,747	10,379	371	11	217	1	(752)	(2)
Utilities	25,911	20,951	4,101	498	361	3	49	(355)	(230)
Retail and consumer services	20,882	12,021	8,316	338	207	8	23	(623)	(3)
Technology	14,348	9,355	4,534	399	60	47	50	(158)	—
Central government	11,173	10,677	496	—	—	—	—	(6,897)	(42)
Machinery and equipment manufacturing	13,311	7,690	5,372	244	5	8	2	(74)	(2)
Transportation	9,652	6,630	2,739	245	38	—	(16)	(132)	—
Metals/mining	11,426	5,260	5,748	362	56	7	35	(296)	—
Insurance	10,918	7,908	2,690	320	—	—	(1)	(805)	(567)
Business services	11,247	6,351	4,735	115	46	11	15	(5)	—
Securities firms and exchanges	9,415	7,678	1,700	37	—	—	5	(38)	(2,358)
Media	10,967	5,808	3,945	672	542	2	92	(212)	(3)
Building materials/construction	12,808	6,557	5,065	1,129	57	9	6	(308)	—
Chemicals/plastics	12,312	8,375	3,656	274	7	—	2	(70)	—
Telecom services	10,709	7,582	2,295	821	11	3	(8)	(820)	—
Automotive	9,011	3,915	4,822	269	5	—	52	(758)	—
Aerospace	5,732	4,903	732	97	—	—	—	(321)	—
Agriculture/paper manufacturing	7,368	4,510	2,614	242	2	8	7	(44)	(2)
Leisure	5,405	2,895	1,367	941	202	—	90	(253)	(21)
All other ^(c)	146,025	128,074	15,648	1,499	804	954	385	(5,614)	(591)
Subtotal	\$ 649,070	\$ 485,557	\$ 141,133	\$ 16,836	\$ 5,544	\$ 1,852	\$ 1,727	\$ (23,108)	\$ (16,486)
Loans held-for-sale and loans at fair value	5,123								
Receivables from customers and interests in purchased receivables	32,932								
Total	\$ 687,125								

- (a) All industry rankings are based on exposure at December 31, 2011. The industry rankings presented in the table as of December 31, 2010, are based on the industry rankings of the corresponding exposures at December 31, 2011, not actual rankings of such exposures at December 31, 2010.
- (b) In addition to the credit risk exposure to states and municipal governments at December 31, 2011 and 2010, noted above, the Firm held \$16.7 billion and \$14.0 billion, respectively, of trading securities and \$16.5 billion and \$11.6 billion, respectively, of AFS securities issued by U.S. state and municipal governments. For further information, see Note 3 and Note 12 on pages 184–198 and 225–230, respectively, of this Annual Report.
- (c) For further information on the All other category refer to the discussion in the following section on page 140 of this Annual Report. All other for credit derivative hedges includes credit default swap (“CDS”) index hedges of CVA.
- (d) Credit exposure is net of risk participations and excludes the benefit of credit derivative hedges and collateral held against derivative receivables or loans.
- (e) Represents the net notional amounts of protection purchased and sold of single-name and portfolio credit derivatives used to manage the credit exposures; these derivatives do not qualify for hedge accounting under U.S. GAAP.

Management's discussion and analysis

- State and municipal government: Exposure to this segment increased by \$6.1 billion or 17% in 2011 to \$41.9 billion. Lending-related commitments comprise approximately 67% of exposure to this sector, generally in the form of bond and commercial paper liquidity and standby letter of credit commitments. Credit quality of the portfolio remains high as 97% of the portfolio was rated investment-grade, unchanged from 2010. Criticized exposure was less than 1% of this industry's exposure. The non-U.S. portion of this industry was less than 5% of the total. The Firm continues to actively monitor and manage this exposure in light of the challenging environment faced by state and municipal governments. For further discussion of commitments for bond liquidity and standby letters of credit, see Note 29 on pages 283–289 of this Annual Report.
- Media: Exposure to this industry increased by 9% to \$11.9 billion in 2011. Criticized exposure of \$1.1 billion decreased by 7% in 2011 from \$1.2 billion, but remains elevated relative to total industry exposure due to

continued pressure on the traditional media business model from expanding digital and online technology.

- All other: All other at December 31, 2011 (excluding loans held-for-sale and loans at fair value), included \$180.7 billion of credit exposure. Concentrations of exposures include: (1) Individuals, Private Education & Civic Organizations, which were 54% of this category and (2) SPEs which were 35% of this category. Each of these categories has high credit quality, and over 90% of each of these categories were rated investment-grade. SPEs provide secured financing (generally backed by receivables, loans or bonds with a diverse group of obligors); the lending in this category was all secured and well-structured. For further discussion of SPEs, see Note 1 on pages 182–183 and Note 16 on pages 256–267 of this Annual Report. The remaining exposure within this category is well-diversified, with no category being more than 6% of its total.

The following table presents the geographic distribution of wholesale credit exposure including nonperforming assets and past due loans as of December 31, 2011 and 2010. The geographic distribution of the wholesale portfolio is determined based predominantly on the domicile of the borrower.

December 31, 2011 (in millions)	Credit exposure				Nonperforming				Assets acquired in loan satisfactions	30 days or more past due and accruing loans
	Loans	Lending- related commitments	Derivative receivables	Total credit exposure	Nonaccrual loans ^(a)	Derivatives	Lending- related commitments	Total non- performing credit exposure		
Europe/Middle East/Africa	\$ 36,637	\$ 60,681	\$ 43,204	\$ 140,522	\$ 44	\$ —	\$ 25	\$ 69	\$ —	\$ 68
Asia/Pacific	31,119	17,194	10,943	59,256	1	13	—	14	—	6
Latin America/Caribbean	25,141	20,859	5,316	51,316	386	—	15	401	3	222
Other North America	2,267	6,680	1,488	10,435	3	—	1	4	—	—
Total non-U.S.	95,164	105,414	60,951	261,529	434	13	41	488	3	296
Total U.S.	183,231	277,325	31,526	492,082	1,964	5	824	2,793	176	1,543
Loans held-for-sale and loans at fair value	4,621	—	—	4,621	183	NA	—	183	NA	—
Receivables from customers and interests in purchased receivables	—	—	—	17,461	—	NA	NA	—	NA	—
Total	\$ 283,016	\$ 382,739	\$ 92,477	\$ 775,693	\$ 2,581	\$ 18	\$ 865	\$ 3,464	\$ 179	\$ 1,839

December 31, 2010 (in millions)	Credit exposure				Nonperforming				Assets acquired in loan satisfactions	30 days or more past due and accruing loans
	Loans	Lending- related commitments	Derivative receivables	Total credit exposure	Nonaccrual loans ^(a)	Derivatives	Lending- related commitments	Total non- performing credit exposure		
Europe/Middle East/Africa	\$ 27,934	\$ 58,418	\$ 35,196	\$ 121,548	\$ 153	\$ 1	\$ 23	\$ 177	\$ —	\$ 127
Asia/Pacific	20,552	15,002	10,991	46,545	579	21	—	600	—	74
Latin America/Caribbean	16,480	12,170	5,634	34,284	649	—	13	662	1	131
Other North America	1,185	6,149	2,039	9,373	6	—	5	11	—	—
Total non-U.S.	66,151	91,739	53,860	211,750	1,387	22	41	1,450	1	332
Total U.S.	156,359	254,340	26,621	437,320	4,123	12	964	5,099	320	1,520
Loans held-for-sale and loans at fair value	5,123	—	—	5,123	496	NA	—	496	NA	—
Receivables from customers and interests in purchased receivables	—	—	—	32,932	—	NA	NA	—	NA	—
Total	\$ 227,633	\$ 346,079	\$ 80,481	\$ 687,125	\$ 6,006	\$ 34	\$ 1,005	\$ 7,045	\$ 321	\$ 1,852

(a) At December 31, 2011 and 2010, the Firm held an allowance for loan losses of \$496 million and \$1.6 billion, respectively, related to nonaccrual retained loans resulting in allowance coverage ratios of 21% and 29%, respectively. Wholesale nonaccrual loans represented 0.91% and 2.64% of total wholesale loans at December 31, 2011 and 2010, respectively.

Loans

In the normal course of business, the Firm provides loans to a variety of wholesale customers, from large corporate and institutional clients to high-net-worth individuals. For further discussion on loans, including information on credit quality indicators, see Note 14 on pages 231–252 of this Annual Report.

The Firm actively manages wholesale credit exposure. One way of managing credit risk is through sales of loans and lending-related commitments. During 2011, the Firm sold \$5.2 billion of loans and commitments, recognizing net gains of \$22 million. During 2010, the Firm sold \$8.3 billion of loans and commitments, recognizing net gains of \$99 million. These results included gains or losses on sales of nonaccrual loans, if any, as discussed below. These sale activities are not related to the Firm's securitization activities. For further discussion of securitization activity, see Liquidity Risk Management and Note 16 on pages 127–132 and 256–267 respectively, of this Annual Report.

The following table presents the change in the nonaccrual loan portfolio for the years ended December 31, 2011 and 2010. Nonaccrual wholesale loans decreased by \$3.4 billion from December 31, 2010, primarily reflecting net repayments and loan sales.

Wholesale nonaccrual loan activity

Year ended December 31, (in millions)	2011	2010
Beginning balance	\$ 6,006	\$ 6,904
Additions	2,519	9,249
Reductions:		
Paydowns and other	2,841	5,540
Gross charge-offs	907	1,854
Returned to performing status	807	364
Sales	1,389	2,389
Total reductions	5,944	10,147
Net additions/(reductions)	(3,425)	(898)
Ending balance	\$ 2,581	\$ 6,006

The following table presents net charge-offs, which are defined as gross charge-offs less recoveries, for the years ended December 31, 2011 and 2010. The amounts in the table below do not include gains or losses from sales of nonaccrual loans.

Wholesale net charge-offs

Year ended December 31, (in millions, except ratios)	2011	2010
Loans – reported		
Average loans retained	\$ 245,111	\$ 213,609
Net charge-offs/(recoveries)	440	1,727
Net charge-off/(recovery) rate	0.18%	0.81%

Derivative contracts

In the normal course of business, the Firm uses derivative instruments predominantly for market-making activity. Derivatives enable customers and the Firm to manage exposures to fluctuations in interest rates, currencies and other markets. The Firm also uses derivative instruments to manage its credit exposure. For further discussion of derivative contracts, see Note 5 and Note 6 on page 201 and 202–210, respectively, of this Annual Report.

The following tables summarize the net derivative receivables for the periods presented

Derivative receivables

December 31, (in millions)	Derivative receivables	
	2011	2010
Interest rate	\$ 46,369	\$ 32,555
Credit derivatives	6,684	7,725
Foreign exchange	17,890	25,858
Equity	6,793	4,204
Commodity	14,741	10,139
Total, net of cash collateral	92,477	80,481
Liquid securities and other cash collateral held against derivative receivables	(21,807)	(16,486)
Total, net of all collateral	\$ 70,670	\$ 63,995

Derivative receivables reported on the Consolidated Balance Sheets were \$92.5 billion and \$80.5 billion at December 31, 2011 and 2010, respectively. These represent the fair value of the derivative contracts after giving effect to legally enforceable master netting agreements, cash collateral held by the Firm and the CVA. However, in management's view, the appropriate measure of current credit risk should take into consideration additional liquid securities (primarily U.S. government and agency securities and other G7 government bonds) and other cash collateral held by the Firm of \$21.8 billion and \$16.5 billion at December 31, 2011 and 2010, respectively that may be used as security when the fair value of the client's exposure is in the Firm's favor, as shown in the table above.

In addition to the collateral described in the preceding paragraph the Firm also holds additional collateral (including cash, U.S. government and agency securities, and other G7 government bonds) delivered by clients at the initiation of transactions, as well as collateral related to contracts that have a non-daily call frequency and collateral that the Firm has agreed to return but has not yet settled as of the reporting date. Though this collateral does not reduce the balances and is not included in the table above, it is available as security against potential exposure that could arise should the fair value of the client's derivative transactions move in the Firm's favor. As of December 31, 2011 and 2010, the Firm held \$17.6 billion and \$18.0 billion, respectively, of this additional collateral. The derivative receivables fair value, net of all collateral, also do not include other credit enhancements, such as letters of credit. For additional information on the Firm's use of

Management's discussion and analysis

collateral agreements, see Note 6 on pages 202–210 of this Annual Report.

While useful as a current view of credit exposure, the net fair value of the derivative receivables does not capture the potential future variability of that credit exposure. To capture the potential future variability of credit exposure, the Firm calculates, on a client-by-client basis, three measures of potential derivatives-related credit loss: Peak, Derivative Risk Equivalent (“DRE”), and Average exposure (“AVG”). These measures all incorporate netting and collateral benefits, where applicable.

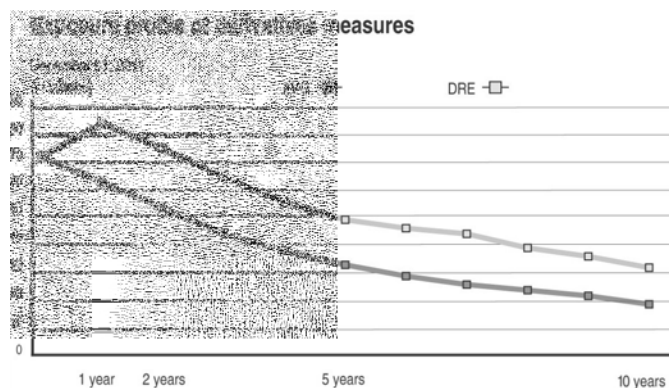
Peak exposure to a counterparty is an extreme measure of exposure calculated at a 97.5% confidence level. DRE exposure is a measure that expresses the risk of derivative exposure on a basis intended to be equivalent to the risk of loan exposures. The measurement is done by equating the unexpected loss in a derivative counterparty exposure (which takes into consideration both the loss volatility and the credit rating of the counterparty) with the unexpected loss in a loan exposure (which takes into consideration only the credit rating of the counterparty). DRE is a less extreme measure of potential credit loss than Peak and is the primary measure used by the Firm for credit approval of derivative transactions.

Finally, AVG is a measure of the expected fair value of the Firm's derivative receivables at future time periods, including the benefit of collateral. AVG exposure over the total life of the derivative contract is used as the primary metric for pricing purposes and is used to calculate credit capital and the CVA, as further described below. AVG exposure was \$53.6 billion and \$45.3 billion at December 31, 2011 and 2010, respectively, compared with derivative receivables, net of all collateral, of \$70.7 billion and \$64.0 billion at December 31, 2011 and 2010, respectively.

The fair value of the Firm's derivative receivables incorporates an adjustment, the CVA, to reflect the credit

quality of counterparties. The CVA is based on the Firm's AVG to a counterparty and the counterparty's credit spread in the credit derivatives market. The primary components of changes in CVA are credit spreads, new deal activity or unwinds, and changes in the underlying market environment. The Firm believes that active risk management is essential to controlling the dynamic credit risk in the derivatives portfolio. In addition, the Firm's risk management process takes into consideration the potential impact of wrong-way risk, which is broadly defined as the potential for increased correlation between the Firm's exposure to a counterparty (AVG) and the counterparty's credit quality. Many factors may influence the nature and magnitude of these correlations over time. To the extent that these correlations are identified, the Firm may adjust the CVA associated with that counterparty's AVG. The Firm risk manages exposure to changes in CVA by entering into credit derivative transactions, as well as interest rate, foreign exchange, equity and commodity derivative transactions.

The accompanying graph shows exposure profiles to derivatives over the next 10 years as calculated by the DRE and AVG metrics. The two measures generally show declining exposure after the first year, if no new trades were added to the portfolio.



The following table summarizes the ratings profile of the Firm's derivative receivables, net of other liquid securities collateral, at the dates indicated.

Ratings profile of derivative receivables

Rating equivalent

December 31,
(in millions, except ratios)

	2011		2010	
	Exposure net of all collateral	% of exposure net of all collateral	Exposure net of all collateral	% of exposure net of all collateral
AAA/Aaa to AA-/Aa3	\$ 25,100	35%	\$ 23,342	36%
A+/A1 to A-/A3	22,942	32	15,812	25
BBB+/Baa1 to BBB-/Baa3	9,595	14	8,403	13
BB+/Ba1 to B-/B3	10,545	15	13,716	22
CCC+/Caa1 and below	2,488	4	2,722	4
Total	\$ 70,670	100%	\$ 63,995	100%

As noted above, the Firm uses collateral agreements to mitigate counterparty credit risk. The percentage of the

Firm's derivatives transactions subject to collateral agreements – excluding foreign exchange spot trades, which

are not typically covered by collateral agreements due to their short maturity – was 88% as of December 31, 2011, unchanged compared with December 31, 2010. The Firm posted \$82.1 billion and \$58.3 billion of collateral at December 31, 2011 and 2010, respectively.

Credit derivatives

Credit derivatives are financial instruments whose value is derived from the credit risk associated with the debt of a third party issuer (the reference entity) and which allow one party (the protection purchaser) to transfer that risk to another party (the protection seller) when the reference entity suffers a credit event. If no credit event has occurred, the protection seller makes no payments to the protection purchaser.

As a purchaser of credit protection, the Firm has risk that the counterparty providing the credit protection will default. As a seller of credit protection, the Firm has risk that the underlying entity referenced in the contract will be subject to a credit event. Upon the occurrence of a credit event, which may include, among other events, the bankruptcy or failure to pay by, or certain restructurings of the debt of, the reference entity, neither party has recourse to the reference entity. The protection purchaser has recourse to the protection seller for the difference between the face value of the credit derivative contract and the fair value of the reference obligation at the time of settling the

credit derivative contract. The determination as to whether a credit event has occurred is made by the relevant ISDA Determination Committee, comprised of 10 sell-side and five buy-side ISDA member firms.

One type of credit derivatives the Firm enters into with counterparties are CDS. The large majority of CDS are subject to collateral arrangements to protect the Firm from counterparty credit risk. The use of collateral to settle against defaulting counterparties has generally performed as designed and has significantly mitigated the Firm's exposure to these counterparties. In 2011 the frequency and size of defaults related to the underlying debt referenced in credit derivatives was lower than 2010. For a more detailed description of credit derivatives, including other types of credit derivatives, see Credit derivatives in Note 6 on pages 202–210 of this Annual Report.

The Firm uses credit derivatives for two primary purposes: first, in its capacity as a market-maker in the dealer/client business to meet the needs of customers; and second, in order to mitigate the Firm's own credit risk associated with its overall derivative receivables and traditional commercial credit lending exposures (loans and unfunded commitments). For further information on the Firm's dealer/client business, see Credit derivatives in Note 6, on pages 202–210 of this Annual Report.

The following table presents the Firm's notional amounts of credit derivatives protection purchased and sold as of December 31, 2011 and 2010, distinguishing between dealer/client activity and credit portfolio activity.

Credit derivative notional amounts

December 31, (in millions)	2011					2010				
	Dealer/client		Credit portfolio			Dealer/client		Credit portfolio		
	Protection purchased ^(a)	Protection sold	Protection purchased	Protection sold	Total	Protection purchased ^(a)	Protection sold	Protection purchased	Protection sold	Total
Credit default swaps	\$ 2,800,975	\$ 2,839,361	\$ 26,371	\$ 131	\$ 5,666,838	\$ 2,661,657	\$ 2,658,825	\$ 23,523	\$ 415	\$ 5,344,420
Other credit derivatives ^(b)	27,246	79,711	—	—	106,957	34,250	93,776	—	—	128,026
Total	\$ 2,828,221	\$ 2,919,072	\$ 26,371	\$ 131	\$ 5,773,795	\$ 2,695,907	\$ 2,752,601	\$ 23,523	\$ 415	\$ 5,472,446

(a) Primarily consists of total return swaps and credit default swap options.

(b) At December 31, 2011 and 2010, included \$2,803 billion and \$2,662 billion, respectively, of notional exposure where the Firm has sold protection on the identical underlying reference instruments.

Dealer/client business

Within the dealer/client business, the Firm actively manages credit derivatives by buying and selling credit protection, predominantly on corporate debt obligations, according to client demand. For further information, see Note 6 on pages 202–210 of this Annual Report. At December 31, 2011, the total notional amount of protection purchased and sold increased by \$298.8 billion from year-end 2010, primarily due to increased activity, particularly in the EMEA region.

Credit portfolio activities

Management of the Firm's wholesale exposure is accomplished through a number of means including loan syndication and participations, loan sales, securitizations, credit derivatives, use of master netting agreements, and

collateral and other risk-reduction techniques. The Firm also manages its wholesale credit exposure by purchasing protection through single-name and portfolio credit derivatives to manage the credit risk associated with loans, lending-related commitments and derivative receivables. Changes in credit risk on the credit derivatives are expected to offset changes in credit risk on the loans, lending-related commitments or derivative receivables. This activity does not reduce the reported level of assets on the Consolidated Balance Sheets or the level of reported off-balance sheet commitments, although it does provide the Firm with credit risk protection.

Management's discussion and analysis

Use of single-name and portfolio credit derivatives

December 31, (in millions)	Notional amount of protection purchased and sold	
	2011	2010
Credit derivatives used to manage:		
Loans and lending-related commitments	\$ 3,488	\$ 6,698
Derivative receivables	22,883	16,825
Total protection purchased	26,371	23,523
Total protection sold	131	415
Credit derivatives hedges notional, net	\$ 26,240	\$ 23,108

The credit derivatives used by JPMorgan Chase for credit portfolio management activities do not qualify for hedge accounting under U.S. GAAP; these derivatives are reported at fair value, with gains and losses recognized in principal transactions revenue. In contrast, the loans and lending-related commitments being risk-managed are accounted for on an accrual basis. This asymmetry in accounting treatment, between loans and lending-related commitments and the credit derivatives used in credit portfolio management activities, causes earnings volatility that is not representative, in the Firm's view, of the true changes in value of the Firm's overall credit exposure. In addition, the effectiveness of the Firm's CDS protection as a hedge of the Firm's exposures may vary depending upon a number of factors, including the contractual terms of the CDS. The fair value related to the Firm's credit derivatives used for managing credit exposure, as well as the fair value related to the CVA (which reflects the credit quality of derivatives counterparty exposure), are included in the gains and losses realized on credit derivatives disclosed in the table below. These results can vary from period to period due to market conditions that affect specific positions in the portfolio. For further information on credit derivative protection purchased in the context of country risk, see Country Risk Management on pages 163–165 of this Annual Report.

Net gains and losses on credit portfolio hedges

Year ended December 31, (in millions)	2011	2010	2009
Hedges of loans and lending-related commitments	\$ (32)	\$ (279)	\$ (3,258)
CVA and hedges of CVA	(769)	(403)	1,920
Net gains/(losses)	\$ (801)	\$ (682)	\$ (1,338)

Lending-related commitments

JPMorgan Chase uses lending-related financial instruments, such as commitments and guarantees, to meet the financing needs of its customers. The contractual amounts of these financial instruments represent the maximum possible credit risk should the counterparties draw down on these commitments or the Firm fulfills its obligations under these guarantees, and the counterparties subsequently fails to perform according to the terms of these contracts.

In the Firm's view, the total contractual amount of these wholesale lending-related commitments is not representative of the Firm's actual credit risk exposure or funding requirements. In determining the amount of credit risk exposure the Firm has to wholesale lending-related commitments, which is used as the basis for allocating credit risk capital to these commitments, the Firm has established a "loan-equivalent" amount for each commitment; this amount represents the portion of the unused commitment or other contingent exposure that is expected, based on average portfolio historical experience, to become drawn upon in an event of a default by an obligor. The loan-equivalent amount of the Firm's lending-related commitments was \$206.5 billion and \$178.9 billion as of December 31, 2011 and 2010, respectively.

CONSUMER CREDIT PORTFOLIO

JPMorgan Chase's consumer portfolio consists primarily of residential real estate loans, credit cards, auto loans, business banking loans, and student loans. The Firm's primary focus is on serving the prime segment of the consumer credit market. For further information on consumer loans, see Note 14 on pages 231–252 of this Annual Report.

A substantial portion of the consumer loans acquired in the Washington Mutual transaction were identified as PCI based on an analysis of high-risk characteristics, including product type, LTV ratios, FICO scores and delinquency status. These PCI loans are accounted for on a pool basis, and the pools are considered to be performing. For further information on PCI loans see Note 14 on pages 231–252 of this Annual Report.

The credit performance of the consumer portfolio across the entire product spectrum has improved, particularly in credit card, but high unemployment and weak overall economic conditions continued to result in an elevated number of residential real estate loans that were charged-off, and weak housing prices continued to negatively affect the severity of loss recognized on residential real estate loans that defaulted. Early-stage residential real estate delinquencies (30–89 days delinquent) declined during the first half of the year, but flattened during the second half of the year, while late-stage delinquencies (150+ days delinquent), excluding government guaranteed loans, have steadily declined in 2011. In spite of the declines, residential real estate loan delinquencies remained elevated. The elevated level of the late-stage delinquent loans is due, in part, to loss-mitigation activities currently being undertaken and to elongated foreclosure processing timelines. Losses related to these loans continued to be recognized in accordance with the Firm's standard charge-off practices, but some delinquent loans that would otherwise have been foreclosed upon remain in the mortgage and home equity loan portfolios. In addition to these elevated levels of delinquencies, ongoing weak economic conditions and housing prices, the estimated effects of the mortgage foreclosure-related settlement with federal and state officials, uncertainties regarding the ultimate success of loan modifications, and the risk attributes of certain loans within the portfolio (e.g., loans with high LTV ratios, junior lien loans behind a delinquent or modified senior lien) have resulted in a high level of uncertainty regarding credit risk in the residential real estate portfolio and have been considered in estimating the allowance for loan losses.

Since the global economic crisis began in mid-2007, the Firm has taken actions to reduce risk exposure to consumer loans by tightening both underwriting and loan qualification standards, as well as eliminating certain products and loan origination channels for residential real estate lending. To manage the risk associated with lending-related commitments, the Firm has reduced or canceled certain lines of credit as permitted by law. For example, the Firm may reduce or close home equity lines of credit when there are significant decreases in the value of the underlying property or when there has been a demonstrable decline in the creditworthiness of the borrower. Also, the Firm typically closes credit card lines when the borrower is 60 days or more past due. The tightening of underwriting criteria for auto loans has resulted in the reduction of both extended-term and high LTV financing. In addition, new originations of private student loans are limited to school-certified loans, the majority of which include a qualified co-borrower.

Management's discussion and analysis

The following table presents managed consumer credit-related information (including RFS, Card Services & Auto, and residential real estate loans reported in Asset Management and the Corporate/Private Equity segment) for the dates indicated. For further information about the Firm's nonaccrual and charge-off accounting policies, see Note 14 on pages 231–252 of this Annual Report.

Consumer credit portfolio

As of or for the year ended December 31, (in millions, except ratios)	Credit exposure		Nonaccrual loans ^(g)		Net charge-offs		Average annual net charge-off rate ^(d)	
	2011	2010	2011	2010	2011	2010	2011	2010
Consumer, excluding credit card								
Loans, excluding PCI loans and loans held-for-sale								
Home equity – senior lien	\$ 21,765	\$ 24,376	\$ 495	\$ 479	\$ 284	\$ 262	1.20%	1.00%
Home equity – junior lien	56,035	64,009	792	784	2,188	3,182	3.69	4.63
Prime mortgage, including option ARMs	76,196	74,539	3,462	4,320	708	1,627	0.95	2.15
Subprime mortgage	9,664	11,287	1,781	2,210	626	1,374	5.98	10.82
Auto ^(a)	47,426	48,367	118	141	152	298	0.32	0.63
Business banking	17,652	16,812	694	832	494	707	2.89	4.23
Student and other	14,143	15,311	69	67	420	459	2.85	2.85
Total loans, excluding PCI loans and loans held-for-sale	242,881	254,701	7,411	8,833	4,872	7,909	1.97	3.00
Loans – PCI								
Home equity	22,697	24,459	NA	NA	NA	NA	NA	NA
Prime mortgage	15,180	17,322	NA	NA	NA	NA	NA	NA
Subprime mortgage	4,976	5,398	NA	NA	NA	NA	NA	NA
Option ARMs	22,693	25,584	NA	NA	NA	NA	NA	NA
Total loans – PCI	65,546	72,763	NA	NA	NA	NA	NA	NA
Total loans – retained	308,427	327,464	7,411	8,833	4,872	7,909	1.54	2.32
Loans held-for-sale	—	154	—	—	—	—	—	—
Total consumer, excluding credit card loans	308,427	327,618	7,411	8,833	4,872	7,909	1.54	2.32
Lending-related commitments								
Home equity – senior lien	16,542	17,662						
Home equity – junior lien	26,408	30,948						
Prime mortgage	1,500	1,266						
Subprime mortgage	—	—						
Auto	6,694	5,246						
Business banking	10,299	9,702						
Student and other	864	579						
Total lending-related commitments	62,307	65,403						
Receivables from customers	100	—						
Total consumer exposure, excluding credit card	370,834	393,021						
Credit Card								
Loans retained	132,175	135,524	1	2	6,925	14,037	5.44	9.73
Loans held-for-sale	102	2,152	—	—	—	—	—	—
Total credit card loans	132,277	137,676	1	2	6,925	14,037	5.44	9.73
Lending-related commitments	530,616	547,227						
Total credit card exposure	662,893	684,903						
Total consumer credit portfolio	\$ 1,033,727	\$ 1,077,924	\$ 7,412	\$ 8,835	\$ 11,797	\$ 21,946	2.66%	4.53%
Memo: Total consumer credit portfolio, excluding PCI	\$ 968,181	\$ 1,005,161	\$ 7,412	\$ 8,835	\$ 11,797	\$ 21,946	3.15%	5.38%

- (a) At December 31, 2011 and 2010, excluded operating lease-related assets of \$4.4 billion and \$3.7 billion, respectively.
- (b) Charge-offs are not recorded on PCI loans until actual losses exceed estimated losses that were recorded as purchase accounting adjustments at the acquisition. To date, no charge-offs have been recorded for these loans.
- (c) Represents prime mortgage loans held-for-sale.
- (d) Credit card and home equity lending-related commitments represent the total available lines of credit for these products. The Firm has not experienced and does not anticipate, that all available lines of credit would be used at the same time. For credit card and home equity commitments (if certain conditions are met), the Firm can reduce or cancel these lines of credit by providing the borrower notice or, in some cases, without notice as permitted by law.
- (e) Receivables from customers primarily represent margin loans to retail brokerage customers, which are included in accrued interest and accounts receivable on the Consolidated Balance Sheets.
- (f) Includes billed finance charges and fees net of an allowance for uncollectible amounts.
- (g) At December 31, 2011 and 2010, nonaccrual loans excluded: (1) mortgage loans insured by U.S. government agencies of \$11.5 billion and \$9.4 billion, respectively.

respectively, that are 90 or more days past due; and (2) student loans insured by U.S. government agencies under the FFELP of \$551 million and \$62 million, respectively, that are 90 or more days past due. These amounts were excluded as reimbursement of insured amounts is proceeding normally. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance. Under guidance issued by the FFIEC, credit card loans are charged off by the end of the month in which the account becomes 180 days past due or within 6 months from receiving notification about a specified event (e.g., bankruptcy of the borrower), whichever is earlier.

- (h) Excludes PCI loans that were acquired as part of the Washington Mutual transaction, which are accounted for on a pool basis. Since each pool is accounted for as a single asset with a single composite interest rate and an aggregate expectation of cash flows, the past-due status of the pools, or that of individual loans within the pools, is not meaningful. Because the Firm is recognizing interest income on each pool of loans, they are all considered to be performing.
- (i) Average consumer loans held-for-sale were \$924 million and \$1.5 billion, respectively, for the years ended December 31, 2011 and 2010. These amounts were excluded when calculating net charge-off rates.
- (j) Net charge-off rates for 2010 reflect the impact of an aggregate \$632 million adjustment related to the Firm's estimate of the net realizable value of the collateral underlying the loans at the charge-off date. Absent this adjustment, net charge-off rates would have been 0.92%, 4.57%, 1.73% and 8.87% for home equity – senior lien; home equity – junior lien; prime mortgage, including option ARMs; and subprime mortgage, respectively. Total consumer, excluding credit card and PCI loans, and total consumer, excluding credit card, net charge-off rates would have been 2.76% and 2.14%, respectively, excluding this adjustment.

Consumer, excluding credit card

Portfolio analysis

Consumer loan balances declined during the year ended December 31, 2011, due to paydowns, portfolio run-off and charge-offs. Credit performance has improved across most portfolios but remains under stress. The following discussion relates to the specific loan and lending-related categories. PCI loans are generally excluded from individual loan product discussions and are addressed separately below. For further information about the Firm's consumer portfolio, including information about delinquencies, loan modifications and other credit quality indicators, see Note 14 on pages 231–252 of this Annual Report.

Home equity: Home equity loans at December 31, 2011, were \$77.8 billion, compared with \$88.4 billion at December 31, 2010. The decrease in this portfolio primarily reflected loan paydowns and charge-offs. Both senior lien and junior lien nonaccrual loans increased slightly from 2010. Senior lien early-stage delinquencies were relatively flat to 2010 and charge-offs increased slightly, but junior lien early-stage delinquencies and charge-offs showed improvement.

Approximately 20% of the Firm's home equity portfolio consists of home equity loans ("HELOANS") and the remainder consists of home equity lines of credit ("HELOCs"). HELOANS are generally fixed-rate, closed-end, amortizing loans, with terms ranging from 3–30 years. Approximately half of the HELOANS are senior liens and the remainder are junior liens. In general, HELOCs are open-ended, revolving loans for a 10-year period, after which time the HELOC converts to a loan with a 20-year amortization period. At the time of origination, the borrower typically selects one of two minimum payment options that will generally remain in effect during the revolving period: a monthly payment of 1% of the outstanding balance, or interest-only payments based on a variable index (typically Prime).

The Firm manages the risk of HELOCs during their revolving period by closing or reducing the undrawn line to the extent permitted by law when borrowers are experiencing financial difficulty or when the collateral does not support the loan amount. Because the majority of the HELOCs were funded in

2005 or later, a fully-amortizing payment is not required until 2015 or later for the most significant portion of the HELOC portfolio. The Firm regularly evaluates both the near-term and longer-term repricing risks inherent in its HELOC portfolio to ensure that the allowance for credit losses and its account management practices are appropriate given the portfolio risk profile.

At December 31, 2011, the Firm estimates that its home equity portfolio contained approximately \$3.7 billion of junior lien loans where the borrower has a first mortgage loan that is either delinquent or has been modified ("high-risk seconds"). Such loans are considered to pose a higher risk of default than that of junior lien loans for which the senior lien is neither delinquent nor modified. Of this estimated \$3.7 billion balance, the Firm owns approximately 5% and services approximately 30% of the related senior lien loans to these same borrowers. The Firm estimates the balance of its total exposure to high-risk seconds on a quarterly basis using summary-level output from a database of information about senior and junior lien mortgage and home equity loans maintained by one of the bank regulatory agencies. This database comprises loan-level data provided by a number of servicers across the industry (including JPMorgan Chase). The performance of the Firm's junior lien loans is generally consistent regardless of whether the Firm owns, services or does not own or service the senior lien. The increased probability of default associated with these higher-risk junior lien loans was considered in estimating the allowance for loan losses.

Mortgage: Mortgage loans at December 31, 2011, including prime, subprime and loans held-for-sale, were \$85.9 billion, compared with \$86.0 billion at December 31, 2010. Balances remained relatively flat as declines resulting from paydowns, portfolio run-off and the charge-off or liquidation of delinquent loans were offset by new prime mortgage originations and Ginnie Mae loans that the Firm elected to repurchase. Net charge-offs decreased from 2010 as a result of improvement in delinquencies, but remained elevated.

Prime mortgages, including option adjustable-rate mortgages ("ARMs") and loans held-for-sale, were \$76.2 billion at December 31, 2011, compared with \$74.7 billion at December 31, 2010. The increase was due primarily to

Management's discussion and analysis

prime mortgage originations and Ginnie Mae loans that the Firm elected to repurchase, partially offset by the charge-off or liquidation of delinquent loans, paydowns, and portfolio run-off of option ARM loans. Excluding loans insured by U.S. government agencies, both early-stage and late-stage delinquencies showed modest improvement during the year but remained elevated. Nonaccrual loans showed improvement, but also remained elevated as a result of ongoing foreclosure processing delays. Net charge-offs declined year-over-year but remained high.

Option ARM loans, which are included in the prime mortgage portfolio, were \$7.4 billion and \$8.1 billion and represented 10% and 11% of the prime mortgage portfolio at December 31, 2011 and 2010, respectively. The decrease in option ARM loans resulted from portfolio run-off partially offset by the purchase of loans previously securitized as the securitization entities were terminated. The Firm's option ARM loans, other than those held in the PCI portfolio, are primarily loans with lower LTV ratios and higher borrower FICO scores. Accordingly, the Firm expects substantially lower losses on this portfolio when compared with the PCI option ARM pool. As of December 31, 2011, approximately 6% of option ARM borrowers were delinquent, 3% were making interest-only or negatively amortizing payments, and 91% were making amortizing payments (such payments are not necessarily fully amortizing). Approximately 85% of borrowers within the portfolio are subject to risk of payment shock due to future payment recast, as only a limited number of these loans have been modified. The cumulative amount of unpaid interest added to the unpaid principal balance due to negative amortization of option ARMs was not material at either December 31, 2011 or 2010. The Firm estimates the following balances of option ARM loans will experience a recast that results in a payment increase: \$160 million in 2012, \$528 million in 2013 and \$636 million in 2014. The Firm did not originate option ARMs and new originations of option ARMs were discontinued by Washington Mutual prior to the date of JPMorgan Chase's acquisition of its banking operations.

Subprime mortgages at December 31, 2011, were \$9.7 billion, compared with \$11.3 billion at December 31, 2010. The decrease was due to portfolio run-off and the charge-off or liquidation of delinquent loans. Both early-stage and late-stage delinquencies improved from December 31, 2010. However, delinquencies and nonaccrual loans remained at elevated levels. Net charge-offs improved from the prior year.

Auto: Auto loans at December 31, 2011, were \$47.4 billion, compared with \$48.4 billion at December 31, 2010. Loan balances declined due to paydowns and payoffs, which were only partially offset by new originations reflecting the impact of increased competition. Delinquent and nonaccrual loans have decreased from December 31, 2010. Net charge-offs declined from the prior year as a result of a decline in loss severity due to a strong used-car market nationwide. The auto loan portfolio reflected a high

concentration of prime-quality credits.

Business bankingBusiness banking loans at December 31, 2011, were \$17.7 billion, compared with \$16.8 billion at December 31, 2010. The increase was due to growth in new loan origination volumes. These loans primarily include loans that are collateralized, often with personal loan guarantees, and may also include Small Business Administration guarantees. Delinquent loans and nonaccrual loans showed some improvement from December 31, 2010, but remain elevated. Net charge-offs declined from the prior year.

Student and otherStudent and other loans at December 31, 2011, were \$14.1 billion, compared with \$15.3 billion at December 31, 2010. The decrease was primarily due to paydowns and charge-offs of student loans. Other loans primarily include other secured and unsecured consumer loans. Delinquencies and nonaccrual loans remained elevated, but charge-offs decreased from 2010.

Purchased credit-impaired loansPCI loans at December 31, 2011, were \$65.5 billion, compared with \$72.8 billion at December 31, 2010. This portfolio represents loans acquired in the Washington Mutual transaction, which were recorded at fair value at the time of acquisition.

During 2011, in connection with the Firm's quarterly review of the PCI portfolios' expected cash flows, management concluded that it was probable that higher expected credit losses would result in a decrease to the expected cash flows in certain portfolios. As a result, the Firm recognized an additional \$770 million of impairment related to the home equity, prime mortgage and subprime mortgage PCI portfolios. As a result of this impairment, the Firm increased the allowance for loan losses for this portfolio. At December 31, 2011, the allowance for loan losses for the home equity, prime mortgage, option ARM and subprime mortgage PCI portfolios was \$1.9 billion, \$1.9 billion, \$1.5 billion and \$380 million, respectively, compared with an allowance for loan losses at December 31, 2010, of \$1.6 billion, \$1.8 billion, \$1.5 billion and \$98 million.

As of December 31, 2011, approximately 31% of the option ARM PCI loans were delinquent and 42% have been modified into fixed-rate, fully amortizing loans. Substantially all of the remaining loans are making amortizing payments, although such payments are not necessarily fully amortizing; in addition, substantially all of these loans are subject to the risk of payment shock due to future payment recast. The cumulative amount of unpaid interest added to the unpaid principal balance of the option ARM PCI pool was \$1.1 billion and \$1.4 billion at December 31, 2011 and 2010, respectively. The Firm estimates the following balances of option ARM PCI loans will experience a recast that results in a payment increase: \$2.1 billion in 2012 and \$361 million in 2013 and \$410 million in 2014.

The following table provides a summary of lifetime principal loss estimates included in both the nonaccretable difference and the allowance for loan losses. Lifetime principal loss estimates, which exclude the effect of foregone interest as a result of loan modifications, were relatively unchanged from December 31, 2010 to December 31, 2011. Although the credit quality of the non-modified PCI loans generally deteriorated during 2011, this was offset by a decrease in estimated principal losses on the modified portion of the PCI portfolio. The impairment recognized in the fourth quarter of 2011 was driven by an increase in estimated principal losses on non-modified PCI loans, as the improvement in estimated principal losses on modified PCI loans was predominately offset by contractual interest cash flows foregone as a result of the modification. Principal charge-offs will not be recorded on these pools until the nonaccretable difference has been fully depleted.

Summary of lifetime principal loss estimates December 31, (in billions)	Lifetime loss estimates		LTD liquidation losses	
	2011	2010	2011	2010
Home equity	\$ 14.9	\$ 14.7	\$ 10.4	\$ 8.8
Prime mortgage	4.6	4.9	2.3	1.5
Subprime mortgage	3.8	3.7	1.7	1.2
Option ARMs	11.5	11.6	6.6	4.9
Total	\$ 34.8	\$ 34.9	\$ 21.0	\$ 16.4

- (a) Includes the original nonaccretable difference established in purchase accounting of \$30.5 billion for principal losses only plus additional principal losses recognized subsequent to acquisition through the provision and allowance for loan losses. The remaining nonaccretable difference for principal losses only was \$9.4 billion and \$14.1 billion at December 31, 2011 and 2010, respectively.
- (b) Life-to-date ("LTD") liquidation losses represent realization of loss upon loan resolution.

Geographic composition and current estimated LTVs of residential real estate loans

The consumer, excluding credit card, loan portfolio is geographically diverse.

At both December 31, 2011 and 2010, California had the greatest concentration of residential real estate loans with 24% of the total retained residential real estate loan portfolio, excluding mortgage loans insured by U.S. government agencies and PCI loans. Of the total retained residential real estate loan portfolio, excluding mortgage loans insured by U.S. government agencies and PCI loans, \$79.5 billion, or 54%, were concentrated in California, New York, Arizona, Florida and Michigan at December 31, 2011, compared with \$86.4 billion, or 54%, at December 31, 2010. The unpaid principal balance of PCI loans concentrated in these five states represented 72% of total PCI loans at both December 31, 2011 and 2010.

The current estimated average LTV ratio for residential real estate loans retained, excluding mortgage loans insured by U.S. government agencies and PCI loans, was 83% at both December 31, 2011 and 2010. Excluding mortgage loans insured by U.S. government agencies and PCI loans, 24% of the retained portfolio had a current estimated LTV ratio greater than 100%, and 10% of the retained portfolio had a current estimated LTV ratio greater than 125% at both December 31, 2011 and 2010. The decline in home prices since 2007 has had a significant impact on the collateral values underlying the Firm's residential real estate loan portfolio. In general, the delinquency rate for loans with high LTV ratios is greater than the delinquency rate for loans in which the borrower has equity in the collateral. While a large portion of the loans with current estimated LTV ratios greater than 100% continue to pay and are current, the continued willingness and ability of these borrowers to pay remains uncertain.

Management's discussion and analysis

The following table for PCI loans presents the current estimated LTV ratio, as well as the ratio of the carrying value of the underlying loans to the current estimated collateral value. Because such loans were initially measured at fair value, the ratio of the carrying value to the current estimated collateral value will be lower than the current

estimated LTV ratio, which is based on the unpaid principal balance. The estimated collateral values used to calculate these ratios do not represent actual appraised loan-level collateral values; as such, the resulting ratios are necessarily imprecise and should therefore be viewed as estimates.

LTV ratios and ratios of carrying values to current estimated collateral values – PCI loans

December 31, (in millions, except ratios)	2011				2010			
	Unpaid principal balance	Current estimated LTV ratio ^(a)	Net carrying value ^(c)	Ratio of net carrying value to current estimated collateral value ^(e)	Unpaid principal balance	Current estimated LTV ratio ^(a)	Net carrying value ^(c)	Ratio of net carrying value to current estimated collateral value ^(e)
Home equity	\$ 25,064	117% ^(b)	\$ 20,789	97%	\$ 28,312	117% ^(b)	\$ 22,876	95%
Prime mortgage	16,060	110	13,251	91	18,928	109	15,556	90
Subprime mortgage	7,229	115	4,596	73	8,042	113	5,300	74
Option ARMs	26,139	109	21,199	89	30,791	111	24,090	87

- (a) Represents the aggregate unpaid principal balance of loans divided by the estimated current property value. Current property values are estimated at least quarterly based on home valuation models that utilize nationally recognized home price index valuation estimates; such models incorporate actual data to the extent available and forecasted data where actual data is not available.
- (b) Represents current estimated combined LTV for junior home equity liens, which considers all available lien positions related to the property. All other products are presented without consideration of subordinate liens on the property.
- (c) Net carrying value includes the effect of fair value adjustments that were applied to the consumer PCI portfolio at the date of acquisition and is also net of the allowance for loan losses at December 31, 2011 and 2010, of \$1.9 billion and \$1.6 billion for home equity, respectively, \$1.9 billion and \$1.8 billion for prime mortgage, respectively, \$1.5 billion and \$1.5 billion for option ARMs, respectively, and \$380 million and \$98 million for subprime mortgage, respectively. Prior-period amounts have been revised to conform to the current-period presentation.

The current estimated average LTV ratios were 117% and 140% for California and Florida PCI loans, respectively, at December 31, 2011, compared with 118% and 135%, respectively, at December 31, 2010. Continued pressure on

Management's discussion and analysis

Foreclosure prevention Foreclosure is a last resort, and the Firm makes significant efforts to help borrowers stay in their homes. Since the third quarter of 2009, the Firm has prevented two foreclosures (through loan modification, short sales, and other foreclosure prevention means) for every foreclosure completed.

The Firm has a well-defined foreclosure prevention process when a borrower fails to pay on his or her loan. Customer contacts are attempted multiple times in various ways to pursue options other than foreclosure. In addition, if the Firm is unable to contact a customer, various reviews are completed of a borrower's facts and circumstances before a foreclosure sale is completed. By the time of a foreclosure sale, borrowers have not made a payment on average for more than 17 months.

Nonperforming assets

The following table presents information as of December 31, 2011 and 2010, about consumer, excluding credit card, nonperforming assets.

Nonperforming assets ^(a)		
December 31, (in millions)	2011	2010
Nonaccrual loans ^(b)		
Home equity – senior lien	\$ 495	\$ 479
Home equity – junior lien	792	784
Prime mortgage, including option ARMs	3,462	4,320
Subprime mortgage	1,781	2,210
Auto	118	141
Business banking	694	832
Student and other	69	67
Total nonaccrual loans	7,411	8,833
Assets acquired in loan satisfactions		
Real estate owned	802	1,294
Other	44	67
Total assets acquired in loan satisfactions	846	1,361
Total nonperforming assets	\$ 8,257	\$ 10,194

- (a) At December 31, 2011 and 2010, nonperforming assets excluded: (1) mortgage loans insured by U.S. government agencies of \$11.5 billion and \$9.4 billion, respectively, that are 90 or more days past due; (2) real estate owned insured by U.S. government agencies of \$954 million and \$1.9 billion, respectively; and (3) student loans insured by U.S. government agencies under the FFELP of \$551 million and \$625 million, respectively, that are 90 or more days past due. These amounts were excluded as reimbursement of insured amounts is proceeding normally.
- (b) Excludes PCI loans that were acquired as part of the Washington Mutual transaction, which are accounted for on a pool basis. Since each pool is accounted for as a single asset with a single composite interest rate and an aggregate expectation of cash flows, the past-due status of the pools, or that of individual loans within the pools, is not meaningful. Because the Firm is recognizing interest income on each pool of loans, they are all considered to be performing.
- (c) At December 31, 2011 and 2010, consumer, excluding credit card nonaccrual loans represented 2.40% and 2.70%, respectively, of total consumer, excluding credit card loans.

Nonaccrual loans Total consumer, excluding credit card, nonaccrual loans were \$7.4 billion at December 31, 2011, compared with \$8.8 billion at December 31, 2010. Nonaccrual loans have declined, but remain at elevated levels. The elongated foreclosure processing timelines is expected to continue to result in elevated levels of nonaccrual loans in the residential real estate portfolios. In addition, modified loans have also contributed to the elevated level of nonaccrual loans, since the Firm's policy requires modified loans that are on nonaccrual to remain on nonaccrual status until payment is reasonably assured and the borrower has made a minimum of six payments under the modified terms. Nonaccrual loans in the residential real estate portfolio totaled \$6.5 billion at December 31, 2011, of which 69% were greater than 150 days past due; this compared with nonaccrual residential real estate loans of \$7.8 billion at December 31, 2010, of which 71% were greater than 150 days past due. At December 31, 2011 and 2010, modified residential real estate loans of \$2.0 billion and \$1.3 billion, respectively, were classified as nonaccrual loans, of which \$886 million and \$580 million, respectively, had yet to make six payments under their modified terms; the remaining nonaccrual modified loans have redefaulted. In the aggregate, the unpaid principal balance of residential real estate loans greater than 150 days past due was charged down by approximately 50% and 46% to estimated collateral value at December 31, 2011 and 2010, respectively.

Real estate owned ("REO") REO assets are managed for prompt sale and disposition at the best possible economic value. REO assets are those individual properties where the Firm gains ownership and possession at the completion of the foreclosure process. REO assets, excluding those insured by U.S. government agencies, decreased by \$492 million from \$1.3 billion at December 31, 2010, to \$802 million at December 31, 2011.

Enhancements to mortgage servicing

During the second quarter of 2011, the Firm entered into Consent Orders with banking regulators relating to its residential mortgage servicing, foreclosure and loss-mitigation activities. In their Orders, the regulators have mandated significant changes to the Firm's servicing and default business and outlined requirements to implement these changes. In accordance with the requirements of the Consent Orders, the Firm submitted comprehensive action plans, the plans have been approved, and the Firm has commenced implementation. The plans sets forth the steps necessary to ensure the Firm's residential mortgage servicing, foreclosure and loss-mitigation activities are conducted in accordance with the requirements of the Orders.

To date, the Firm has implemented a number of corrective actions including the following:

- Established an independent Compliance Committee which meets regularly and monitors progress against the Consent Orders.
- Launched a new Customer Assistance Specialist organization for borrowers to facilitate the single point of contact initiative and ensure effective coordination and communication related to foreclosure, loss-mitigation and loan modification.
- Enhanced its approach to oversight over third-party vendors for foreclosure or other related functions.
- Standardized the processes for maintaining appropriate controls and oversight of the Firm's activities with respect to the Mortgage Electronic Registration system ("MERS") and compliance with MERSCORP's membership rules, terms and conditions.
- Strengthened its compliance program so as to ensure mortgage-servicing and foreclosure operations, including loss-mitigation and loan modification, comply with all applicable legal requirements.
- Enhanced management information systems for loan modification, loss-mitigation and foreclosure activities.
- Developed a comprehensive assessment of risks in servicing operations including, but not limited to, operational, transaction, legal and reputational risks.
- Made technological enhancements to automate and streamline processes for the Firm's document management, training, skills assessment and payment processing initiatives.
- Deployed an internal validation process to monitor progress under the comprehensive action plans.

In addition, pursuant to the Consent Orders, the Firm is required to enhance oversight of its mortgage servicing activities, including oversight by compliance, management and audit personnel and, accordingly, has made and continues to make changes in its organization structure, control oversight and customer service practices.

Pursuant to the Consent Orders, the Firm has retained an independent consultant to conduct a review of its residential foreclosure actions during the period from January 1, 2009, through December 31, 2010 (including foreclosure actions brought in respect of loans being serviced), and to remediate any errors or deficiencies identified by the independent consultant, including, if required, by reimbursing borrowers for any identified financial injury they may have incurred. The borrower outreach process was launched in the fourth quarter of 2011, and the independent consultant has begun its review. For additional information, see "Mortgage Foreclosure Investigations and Litigation" in Note 31 on pages 290–299 of this Annual Report.

In connection with the Firm's February 2012 settlement with the U.S. Department of Justice, other federal agencies, and the State Attorneys General relating to the Firm's residential mortgage servicing, foreclosure, loss mitigation and origination activities, the Firm will make significant further changes to its servicing and default business pursuant to servicing standards agreed upon in the settlement. The servicing standards include, among other items, the following enhancements to the Firm's servicing of loans: a pre-foreclosure notice to all borrowers, which will include account information, holder status, and loss mitigation steps taken; enhancements to payment application and collections processes; strengthening procedures for filings in bankruptcy proceedings; deploying specific restrictions on "dual track" of foreclosure and loss mitigation; standardizing the process for appeal of loss mitigation denials; and implementing certain restrictions on fees, including the waiver of certain fees while a borrower's loss mitigation application is being evaluated.

Credit Card

Total credit card loans were \$132.3 billion at December 31, 2011, a decrease of \$5.4 billion from December 31, 2010, due to higher repayment rates, runoff of the Washington Mutual portfolio and the Firm's sale of the \$3.7 billion Kohl's portfolio on April 1, 2011.

For the retained credit card portfolio, the 30+ day delinquency rate decreased to 2.81% at December 31, 2011, from 4.14% at December 31, 2010. For the years ended December 31, 2011 and 2010, the net charge-off rates were 5.44% and 9.73% respectively. The delinquency trend showed improvement in the first half of the year, but delinquencies flattened during the second half of the year. Charge-offs have improved as a result of lower delinquent loans. The credit card portfolio continues to reflect a well-seasoned, largely rewards-based portfolio that has good U.S. geographic diversification. The greatest geographic concentration of credit card retained loans is in California, which represented 13% of total retained loans at both December 31, 2011 and 2010. Loan concentration for the top five states of California, New York, Texas, Florida and

Illinois consisted of \$53.6 billion in receivables, or 40% of the retained loan portfolio, at December 31, 2011, compared with \$54.4 billion, or 40%, at December 31, 2010.

Total retained credit card loans, excluding the Washington Mutual portfolio, were \$121.1 billion at December 31, 2011, compared with \$121.8 billion at December 31, 2010. The 30+ day delinquency rate was 2.53% at December 31, 2011, down from 3.73% at December 31, 2010. For the years ended December 31, 2011 and 2010, the net charge-off rates were 4.91% and 8.73% respectively.

Retained credit card loans in the Washington Mutual portfolio were \$11.1 billion at December 31, 2011, compared with \$13.7 billion at December 31, 2010. The Washington Mutual portfolio's 30+ day delinquency rate was 5.82% at December 31, 2011, down from 7.74% at December 31, 2010. For the years ended December 31, 2011 and 2010, the net charge-off rates were 10.49% and 17.73% respectively.

Top 5 States Credit Card - Retained (at December 31, 2011)

California
13.3%

Top 5 States Credit Card - Retained (at December 31, 2010)

California
13.4%

Modifications of credit card loans

At December 31, 2011 and 2010, the Firm had \$7.2 billion and \$10.0 billion, respectively, of on-balance sheet credit card loans outstanding that have been modified in TDRs. These balances included both credit card loans with modified payment terms and credit card loans that reverted back to their pre-modification payment terms. The decrease in modified credit card loans outstanding from December 31, 2010, was attributable to a reduction in new modifications as well as ongoing payments and charge-offs on previously modified credit card loans.

Consistent with the Firm's policy, all credit card loans typically remain on accrual status. However, the Firm establishes an allowance, which is reflected as a charge to interest income, for the estimated uncollectible portion of billed and accrued interest and fee income on credit card loans.

For additional information about loan modification programs to borrowers, see Note 14 on pages 231–252 of this Annual Report.

COMMUNITY REINVESTMENT ACT EXPOSURE

The Community Reinvestment Act ("CRA") encourages banks to meet the credit needs of borrowers in all segments of their communities, including neighborhoods with low or moderate incomes. JPMorgan Chase is a national leader in community development by providing loans, investments and community development services in communities across the United States.

At December 31, 2011 and 2010, the Firm's CRA loan portfolio was approximately \$15 billion and \$16 billion,

respectively. At December 31, 2011 and 2010, 63% and 65%, respectively, of the CRA portfolio were residential mortgage loans; 17% and 15%, respectively, were business banking loans; 14%, for both periods, were commercial real estate loans; and 6%, for both periods, were other loans. CRA nonaccrual loans were 6% of the Firm's total nonaccrual loans at both December 31, 2011 and 2010, respectively. For the years ended December 31, 2011 and 2010, net charge-offs in the CRA portfolio were 3% for both periods, of the Firm's net charge-offs.

ALLOWANCE FOR CREDIT LOSSES

JPMorgan Chase's allowance for loan losses covers the wholesale (risk-rated), and consumer, excluding credit card and credit card portfolios (primarily scored). The allowance represents management's estimate of probable credit losses inherent in the Firm's loan portfolio. Management also determines an allowance for wholesale and certain consumer, excluding credit card, lending-related commitments.

For a further discussion of the components of the allowance for credit losses, see Critical Accounting Estimates Used by the Firm on pages 168–169 and Note 15 on pages 252–255 of this Annual Report.

At least quarterly, the allowance for credit losses is reviewed by the Chief Risk Officer, the Chief Financial Officer and the Controller of the Firm, and discussed with the Risk Policy and Audit Committees of the Board of Directors of the Firm. As of December 31, 2011, JPMorgan Chase deemed the allowance for credit losses to be appropriate (i.e., sufficient to absorb probable credit losses inherent in the portfolio).

The allowance for credit losses was \$28.3 billion at December 31, 2011, a decrease of \$4.7 billion from \$33.0

billion at December 31, 2010. The credit card allowance for loan losses decreased by \$4.0 billion from December 31, 2010, primarily as a result of lower estimated losses primarily related to improved delinquency trends as well as lower levels of outstandings. The wholesale allowance for loan losses decreased by \$445 million from December 31, 2010, primarily related to the impact of loan sales. The consumer, excluding credit card, allowance for loan losses decreased \$177 million largely due to a reduction of \$1.0 billion in the allowance related to the non-credit-impaired residential real estate portfolio, as estimated losses in that portfolio declined, predominantly offset by a \$770 million increase related to an increase in estimated lifetime losses in the PCI portfolio.

The allowance for lending-related commitments for both the wholesale and consumer, excluding credit card portfolios, which is reported in other liabilities, totaled \$673 million and \$717 million at December 31, 2011 and 2010, respectively.

The credit ratios in the table below are based on retained loan balances, which exclude loans held-for-sale and loans accounted for at fair value.

Management's discussion and analysis

Summary of changes in the allowance for credit losses

Year ended December 31, (in millions, except ratios)	2011				2010			
	Wholesale	Consumer, excluding credit card	Credit card	Total	Wholesale	Consumer, excluding credit card	Credit card	Total
Allowance for loan losses								
Beginning balance at January 1,	\$ 4,761	\$ 16,471	\$ 11,034	\$ 32,266	\$ 7,145	\$ 14,785	\$ 9,672	\$ 31,602
Cumulative effect of change in accounting principles ^(a)	—	—	—	—	14	127	7,353	7,494
Gross charge-offs	916	5,419	8,168	14,503	1,989	8,383	15,410	25,782
Gross recoveries	(476)	(547)	(1,243)	(2,266)	(262)	(474)	(1,373)	(2,109)
Net charge-offs	440	4,872	6,925	12,237	1,727	7,909	14,037	23,673
Provision for loan losses	17	4,670	2,925	7,612	(673)	9,458	8,037	16,822
Other	(22)	25	(35)	(32)	2	10	9	21
Ending balance at December 31,	\$ 4,316	\$ 16,294	\$ 6,999	\$ 27,609	\$ 4,761	\$ 16,471	\$ 11,034	\$ 32,266
Impairment methodology								
Asset-specific ^(b)	\$ 516	\$ 828	\$ 2,727	\$ 4,071	\$ 1,574	\$ 1,075	\$ 4,069	\$ 6,718
Formula-based	3,800	9,755	4,272	17,827	3,187	10,455	6,965	20,607
PCI	—	5,711	—	5,711	—	4,941	—	4,941
Total allowance for loan losses	\$ 4,316	\$ 16,294	\$ 6,999	\$ 27,609	\$ 4,761	\$ 16,471	\$ 11,034	\$ 32,266
Allowance for lending-related commitments								
Beginning balance at January 1,	\$ 711	\$ 6	\$ —	\$ 717	\$ 927	\$ 12	\$ —	\$ 939
Cumulative effect of change in accounting principles ^(a)	—	—	—	—	(18)	—	—	(18)
Provision for lending-related commitments	(40)	2	—	(38)	(177)	(6)	—	(183)
Other	(5)	(1)	—	(6)	(21)	—	—	(21)
Ending balance at December 31,	\$ 666	\$ 7	\$ —	\$ 673	\$ 711	\$ 6	\$ —	\$ 717
Impairment methodology								
Asset-specific	\$ 150	\$ —	\$ —	\$ 150	\$ 180	\$ —	\$ —	\$ 180
Formula-based	516	7	—	523	531	6	—	537
Total allowance for lending-related commitments	\$ 666	\$ 7	\$ —	\$ 673	\$ 711	\$ 6	\$ —	\$ 717
Total allowance for credit losses	\$ 4,982	\$ 16,301	\$ 6,999	\$ 28,282	\$ 5,472	\$ 16,477	\$ 11,034	\$ 32,983
Memo:								
Retained loans, end of period	\$278,395	\$308,427	\$132,175	\$718,997	\$222,510	\$327,464	\$135,524	\$685,498
Retained loans, average	245,111	315,736	127,334	688,181	213,609	340,334	144,219	698,162
PCI loans, end of period	21	65,546	—	65,567	44	72,763	—	72,807
Credit ratios								
Allowance for loan losses to retained loans	1.55%	5.28%	5.30%	3.84%	2.14%	5.03%	8.14%	4.71%
Allowance for loan losses to retained nonaccrual loans ^(c)	180	220	NM	281	86	186	NM	225
Allowance for loan losses to retained nonaccrual loans excluding credit card	180	220	NM	210	86	186	NM	148
Net charge-off rate ^(d)	0.18	1.54	5.44	1.78	0.81	2.32	9.73	3.39
Credit ratios, excluding residential real estate PCI loans								
Allowance for loan losses to retained loans ^(e)	1.55	4.36	5.30	3.35	2.14	4.53	8.14	4.46
Allowance for loan losses to retained nonaccrual loans ^(e)	180	143	NM	223	86	131	NM	190
Allowance for loan losses to retained nonaccrual loans excluding credit card ^(e)	180	143	NM	152	86	131	NM	114
Net charge-off rate ^(d)	0.18%	1.97%	5.44%	1.98%	0.81%	3.00%	9.73%	3.81%

(a) Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. Upon adoption of the guidance, the Firm consolidated its Firm-sponsored credit card securitization trusts, its Firm-administered multi-seller conduits and certain other consumer loan securitization entities, primarily mortgage-related. As a result, \$7.4 billion,

\$14 million and \$127 million, respectively, of allowance for loan losses were recorded on-balance sheet with the consolidation of these entities. For further discussion, see Note 16 on pages 256–267 of this Annual Report.

- (b) Includes risk-rated loans that have been placed on nonaccrual status and loans that have been modified in a TDR.
- (c) The Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance. Under the guidance issued by the FFIEC, credit card loans are charged off by the end of the month in which the account becomes 180 days past due or within 60 days from receiving notification about a specified event (e.g., bankruptcy of the borrower), whichever is earlier.
- (d) Charge-offs are not recorded on PCI loans until actual losses exceed estimated losses recorded as purchase accounting adjustments at the time of acquisition.
- (e) Excludes the impact of PCI loans acquired as part of the Washington Mutual transaction.

Provision for credit losses

For the year ended December 31, 2011, the provision for credit losses was \$7.6 billion down 54% from 2010. For the year ended December 31, 2011, the consumer, excluding credit card, provision for credit losses was \$4.7 billion, down 51% from 2010, reflecting improved delinquency and net charge-off trends in 2011 across most portfolios, partially offset by an increase of \$770 million reflecting additional impairment of the Washington Mutual PCI loans portfolio. The credit card provision for credit losses was \$2.9 billion, down 64% from the prior year

period, driven primarily by improved delinquency and net charge-offs which led to a reduction in the allowance for loan losses for both the prior and current year periods. For the year ended December 31, 2011, the wholesale provision for credit losses was a benefit of \$23 million, compared with a benefit of \$850 million in the prior-year period. The change in the wholesale provision when compared with the prior year period primarily reflects loan growth and other portfolio activity including the effect of lower net-charge offs on the provision.

Year ended December 31, (in millions)	Provision for loan losses			Provision for lending-related commitments			Total provision for credit losses		
	2011	2010	2009	2011	2010	2009	2011	2010	2009
Wholesale	\$ 17	\$ (673)	\$ 3,684	\$ (40)	\$ (177)	\$ 290	\$ (23)	\$ (850)	\$ 3,974
Consumer, excluding credit card	4,670	9,458	16,032	2	(6)	(10)	4,672	9,452	16,022
Credit card – reported ^(a)	2,925	8,037	12,019	—	—	—	2,925	8,037	12,019
Total provision for credit losses – reported	7,612	16,822	31,735	(38)	(183)	280	7,574	16,639	32,015
Credit card – securitized ^(b)	NA	NA	6,443	NA	NA	—	NA	NA	6,443
Total provision for credit losses – managed	\$ 7,612	\$ 16,822	\$ 38,178	\$ (38)	\$ (183)	\$ 280	\$ 7,574	\$ 16,639	\$ 38,458

- (a) Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. As a result of the consolidation of the credit card securitization trusts reported and managed basis relating to credit card securitizations are equivalent for periods beginning after January 1, 2010. For further discussion regarding the Firm's application and the impact of the new guidance, see Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 76–78 of this Annual Report.
- (b) Loans securitized are defined as loans that were sold to unconsolidated securitization trusts and were not included in reported loans. For further discussion of credit card securitizations, see Note 16 on pages 256–267 of this Annual Report.

MARKET RISK MANAGEMENT

Market risk is the exposure to an adverse change in the market value of portfolios and financial instruments caused by a change in market prices or rates.

Market risk management

Market Risk is an independent risk management function that works in close partnership with the business segments to identify and monitor market risks throughout the Firm and to define market risk policies and procedures. The risk management function is headed by the Firm's Chief Risk Officer.

Market Risk seeks to facilitate efficient risk/return decisions, reduce volatility in operating performance and provide transparency into the Firm's market risk profile for senior management, the Board of Directors and regulators. Market Risk is responsible for the following functions:

- Establishing a market risk policy framework
- Independent measurement, monitoring and control of line-of-business market risk
- Definition, approval and monitoring of limits
- Performance of stress testing and qualitative risk assessments

Risk identification and classification

Each line of business is responsible for the comprehensive identification and verification of market risks within its units. The Firm's market risks arise primarily from the activities in IB, Mortgage Production and Servicing, and CIO in Corporate/Private Equity.

IB makes markets in products across the fixed income, foreign exchange, equities and commodities markets. This trading activity may lead to a potential decline in net income due to adverse changes in market rates. In addition to these risks, there are risks in IB's credit portfolio from retained loans and commitments, derivative credit valuation adjustments, hedges of the credit valuation adjustments and the fair value of hedges of the retained loan portfolio. Additional risk positions result from the debit valuation adjustments taken on certain structured liabilities and derivatives to reflect the credit quality of the Firm.

The Firm's Mortgage Production and Servicing business includes the Firm's mortgage pipeline and warehouse loans, MSR's and all related hedges. These activities give rise to complex interest rate risks, as well as option and basis risk. Option risk arises primarily from prepayment options embedded in mortgages and changes in the probability of newly originated mortgage commitments actually closing. Basis risk results from differences in the relative movements of the rate indices underlying mortgage exposure and other interest rates.

CIO is primarily concerned with managing structural risks which arise out of the various business activities of the Firm. Market Risk measures and monitors the gross structural exposures as well as the net exposures related to these activities.

Risk measurement

Tools used to measure risk

Because no single measure can reflect all aspects of market risk, the Firm uses various metrics, both statistical and nonstatistical, including:

- Value-at-risk
- Economic-value stress testing
- Nonstatistical risk measures
- Loss advisories
- Revenue drawdowns
- Risk identification for large exposures ("RIFLEs")
- Nontrading interest rate-sensitive revenue-at-risk stress testing

Value-at-risk

JPMorgan Chase utilizes VaR, a statistical risk measure, to estimate the potential loss from adverse market moves. Each business day, as part of its risk management activities, the Firm undertakes a comprehensive VaR calculation that includes the majority of its material market risks. VaR provides a consistent cross-business measure of risk profiles and levels of diversification and is used for comparing risks across businesses and monitoring limits. These VaR results are reported to senior management and regulators, and they are utilized in regulatory capital calculations.

The Firm calculates VaR to estimate possible economic outcomes for its current positions using historical simulation, which measures risk across instruments and portfolios in a consistent, comparable way. The simulation is based on data for the previous 12 months. This approach assumes that historical changes in market values are representative of the distribution of potential outcomes in the immediate future. VaR is calculated using a one day time horizon and an expected tail-loss methodology, and approximates a 95% confidence level. This means that, assuming current changes in market values are consistent with the historical changes used in the simulation, the Firm would expect to incur losses greater than that predicted by VaR estimates five times in every 100 trading days, or about 12 to 13 times a year. However, differences between current and historical market price volatility may result in fewer or greater VaR exceptions than the number indicated by the historical simulation. The Firm's VaR calculation is highly granular and incorporates numerous risk factors, which are selected based on the risk profile of each portfolio.

The table below shows the results of the Firm's VaR measure using a 95% confidence level.

Total IB trading VaR by risk type, Credit portfolio VaR and other VaR

As of or for the year ended December 31, (in millions)	2011			2010			At December 31,	
	Avg.	Min	Max	Avg.	Min	Max	2011	2010
IB VaR by risk type								
Fixed income	\$ 50	\$ 31	\$ 68	\$ 65	\$ 33	\$ 95	\$ 49	\$ 52
Foreign exchange	11	6	19	11	6	20	19	16
Equities	23	15	42	22	10	52	19	30
Commodities and other	16	8	24	16	11	32	22	13
Diversification benefit to IB trading VaR	(42) ^(a)	NM ^(b)	NM ^(b)	(43) ^(a)	NM ^(b)	NM ^(b)	(55) ^(a)	(34) ^(a)
IB trading VaR	58	34	80	71	40	107	54	77
Credit portfolio VaR	33	19	55	26	15	40	42	27
Diversification benefit to IB trading and credit portfolio VaR	(15) ^(a)	NM ^(b)	NM ^(b)	(10) ^(a)	NM ^(b)	NM ^(b)	(20) ^(a)	(5) ^(a)
Total IB trading and credit portfolio VaR	76	42	102	87	50	128	76	99
Other VaR								
Mortgage Production and Servicing VaR	30	6	98	23	8	47	16	9
Chief Investment Office ("CIO") VaR	57	30	80	61	44	80	77	56
Diversification benefit to total other VaR	(17) ^(a)	NM ^(b)	NM ^(b)	(13) ^(a)	NM ^(b)	NM ^(b)	(10) ^(a)	(10) ^(a)
Total other VaR	70	46	110	71	48	100	83	55
Diversification benefit to total IB and other VaR	(45) ^(a)	NM ^(b)	NM ^(b)	(59) ^(a)	NM ^(b)	NM ^(b)	(46) ^(a)	(65) ^(a)
Total IB and other VaR	\$ 101	\$ 67	\$ 147	\$ 99	\$ 66	\$ 142	\$ 113	\$ 89

Management's discussion and analysis

in the fixed income risk component.

CIO VaR averaged \$57 million in 2011, compared with \$61 million for 2010. The decrease was also driven by a decline in market volatility in the first half of 2011, as well as position changes.

Mortgage Production and Servicing VaR averaged \$30 million for 2011, compared with \$23 million for 2010. The increase was driven by position changes in the MSR Portfolio.

The Firm's average IB and other VaR diversification benefit

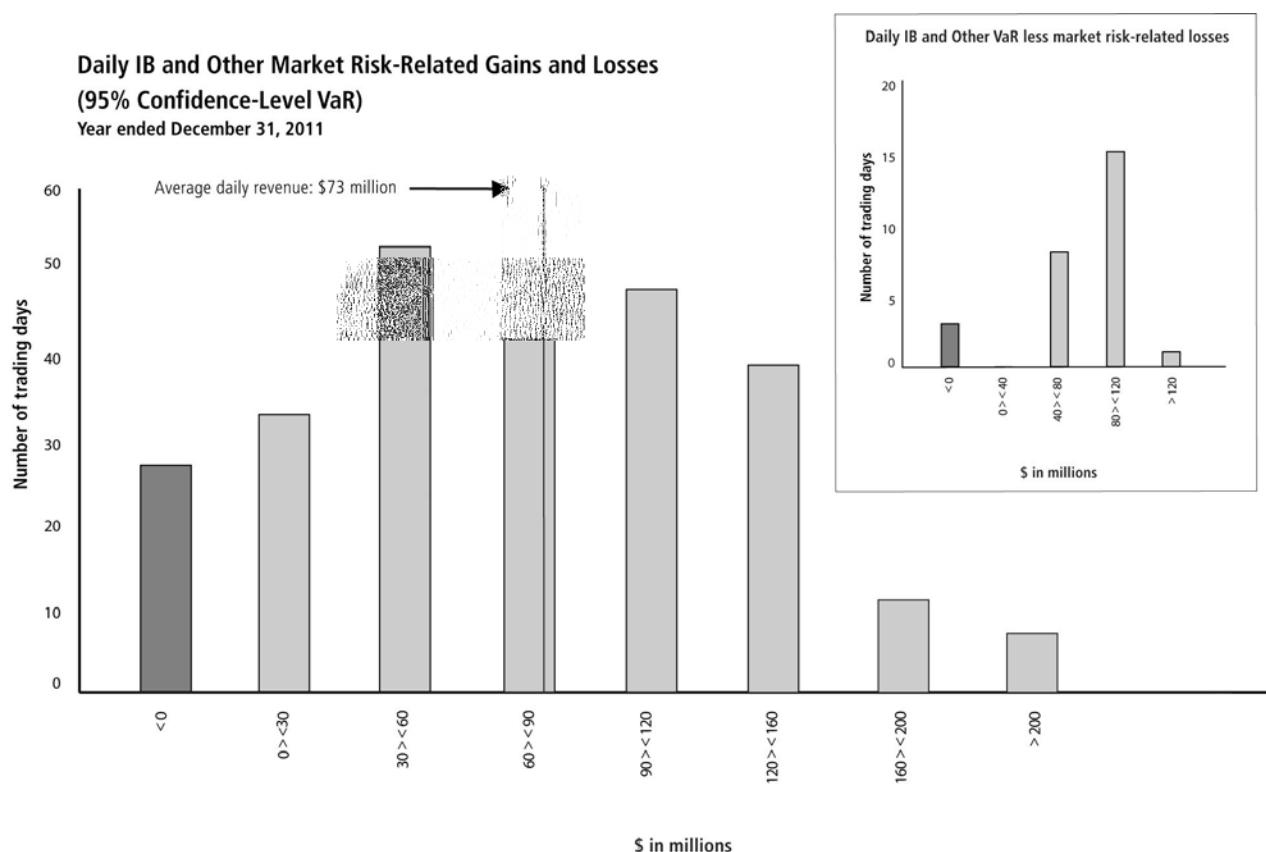
was \$45 million or 31% of the sum for 2011, compared with \$59 million or 37% of the sum for 2010. In general, over the course of the year, VaR exposure can vary significantly as positions change, market volatility fluctuates and diversification benefits change.

VaR back-testing

The Firm conducts daily back-testing of VaR against its market risk related revenue. In the year ended December 31, 2011, losses were sustained on 27 days, of which three days exceeded the VaR measure.

The following histogram illustrates the daily market risk related gains and losses for IB, CIO and Mortgage Production and Servicing positions for 2011. This market risk related revenue is defined as the change in value of: principal transactions revenue for IB and CIO (less Private Equity gains/losses and revenue from longer-term CIO investments); trading-related net interest income for IB, CIO and Mortgage Production and Servicing; IB brokerage commissions, underwriting fees or other revenue; revenue from syndicated lending facilities that the Firm intends to distribute; and mortgage fees and related income for the Firm's mortgage pipeline and warehouse loans, MSRs, and all related hedges. Daily firmwide market risk related revenue excludes gains and losses from DVA.

The chart shows that the Firm posted market risk related gains on 233 of the 260 days in this period, with seven days exceeding \$200 million. The inset graph looks at those days on which the Firm experienced losses and depicts the amount which the VaR exceeded the actual loss on each of those days.



The following table provides information about the gross sensitivity of DVA to a one-basis-point increase in JPMorgan Chase's credit spreads. This sensitivity represents the impact from a one-basis-point parallel shift in JPMorgan Chase's entire credit curve. As credit curves do not typically move in a parallel fashion, the sensitivity multiplied by the change in spreads at a single maturity point may not be representative of the actual revenue recognized.

Debit valuation adjustment sensitivity

December 31, (in millions)	One basis-point increase in JPMorgan Chase's credit spread
2011	\$ 35
2010	35

Economic-value stress testing

While VaR reflects the risk of loss due to adverse changes in markets using recent historical market behavior as an indicator of losses, stress testing captures the Firm's exposure to unlikely but plausible events in abnormal markets using multiple scenarios that assume significant changes in credit spreads, equity prices, interest rates, currency rates or commodity prices. Scenarios are updated dynamically and may be redefined on an ongoing basis to reflect current market conditions. Along with VaR, stress testing is important in measuring and controlling risk; it enhances understanding of the Firm's risk profile and loss potential, as stress losses are monitored against limits. Stress testing is also employed in cross-business risk management. Stress-test results, trends and explanations based on current market risk positions are reported to the Firm's senior management and to the lines of business to allow them to better understand event risk-sensitive positions and manage risks with more transparency.

Nonstatistical risk measures

Nonstatistical risk measures as well as stress testing include sensitivities to variables used to value positions, such as credit spread sensitivities, interest rate basis point values and market values. These measures provide granular information on the Firm's market risk exposure. They are aggregated by line-of-business and by risk type, and are used for tactical control and monitoring limits.

Loss advisories and revenue drawdowns

Loss advisories and net revenue drawdowns are tools used to highlight trading losses above certain levels of risk tolerance. Net revenue drawdown is defined as the decline in net revenue since the year-to-date peak revenue level.

Risk identification for large exposures

Individuals who manage risk positions in IB are responsible for identifying potential losses that could arise from specific, unusual events, such as a potential change in tax legislation, or a particular combination of unusual market moves. This information allows the Firm to monitor further earnings vulnerability not adequately covered by standard risk measures.

Nontrading interest rate-sensitive revenue-at-risk (i.e., "earnings-at-risk")

The VaR and stress-test measures described above illustrate the total economic sensitivity of the Firm's Consolidated Balance Sheets to changes in market variables. The effect of interest rate exposure on reported net income is also important. Interest rate risk represents one of the Firm's significant market risk exposures. This risk arises not only from trading activities but also from the Firm's traditional banking activities which include extension of loans and credit facilities, taking deposits and issuing debt (i.e., asset/liability management positions including accrual loans within IB and CIO, and off—balance sheet positions). ALCO establishes the Firm's interest rate risk policies, sets risk guidelines and limits and reviews the risk profile of the Firm. Treasury, working in partnership with the lines of business, calculates the Firm's interest rate risk profile weekly and reviews it with senior management.

Interest rate risk for nontrading activities can occur due to a variety of factors, including:

- Differences in the timing among the maturity or repricing of assets, liabilities and off—balance sheet instruments. For example, if liabilities reprice more quickly than assets and funding interest rates are declining, earnings will increase initially.
- Differences in the amounts of assets, liabilities and off-balance sheet instruments that are repricing at the same time. For example, if more deposit liabilities are repricing than assets when general interest rates are declining, earnings will increase initially.
- Differences in the amounts by which short-term and long-term market interest rates change (for example, changes in the slope of the yield curve) because the Firm has the ability to lend at long-term fixed rates and borrow at variable or short-term fixed rates. Based on these scenarios, the Firm's earnings would be affected negatively by a sudden and unanticipated increase in short-term rates paid on its liabilities (e.g., deposits) without a corresponding increase in long-term rates received on its assets (e.g., loans). Conversely, higher long-term rates received on assets generally are beneficial to earnings, particularly when the increase is not accompanied by rising short-term rates paid on liabilities.
- The impact of changes in the maturity of various assets, liabilities or off-balance sheet instruments as interest rates change. For example, if more borrowers than forecasted pay down higher-rate loan balances when general interest rates are declining, earnings may decrease initially.

The Firm manages interest rate exposure related to its assets and liabilities on a consolidated, corporate-wide basis. Business units transfer their interest rate risk to Treasury through a transfer-pricing system, which takes into account the elements of interest rate exposure that can be risk-managed in financial markets. These elements

Management's discussion and analysis

include asset and liability balances and contractual rates of interest, contractual principal payment schedules, expected prepayment experience, interest rate reset dates and maturities, rate indices used for repricing, and any interest rate ceilings or floors for adjustable rate products. All transfer-pricing assumptions are dynamically reviewed.

The Firm manages this interest rate risk generally through its investment securities portfolio and related derivatives. The Firm evaluates its nontrading interest rate risk exposure through the stress testing of earnings-at-risk, which measures the extent to which changes in interest rates will affect the Firm's Core net interest income (see page 78 of this Annual Report for further discussion on Core net interest income) and interest rate-sensitive fees ("nontrading interest rate-sensitive revenue"). Earnings-at-risk excludes the impact of trading activities and MSRs as these sensitivities are captured under VaR.

The Firm conducts simulations of changes in nontrading interest rate-sensitive revenue under a variety of interest rate scenarios. Earnings-at-risk tests measure the potential change in this revenue, and the corresponding impact to the Firm's pretax earnings, over the following 12 months. These tests highlight exposures to various interest rate-sensitive factors, such as the rates themselves (e.g., the prime lending rate), pricing strategies on deposits, optionality and changes in product mix. The tests include forecasted balance sheet changes, such as asset sales and securitizations, as well as prepayment and reinvestment behavior. Mortgage prepayment assumptions are based on current interest rates compared with underlying contractual rates, the time since origination, and other factors which are updated periodically based on historical experience and forward market expectations. The amount and pricing assumptions of deposits that have no stated maturity are based on historical performance, the competitive environment, customer behavior, and product mix.

Immediate changes in interest rates present a limited view of risk, and so a number of alternative scenarios are also reviewed. These scenarios include the implied forward curve, nonparallel rate shifts and severe interest rate shocks on selected key rates. These scenarios are intended to provide a comprehensive view of JPMorgan Chase's earnings-at-risk over a wide range of outcomes.

JPMorgan Chase's 12-month pretax earnings sensitivity profiles.
(Excludes the impact of trading activities and MSRs)

December 31, (in millions)	Immediate change in rates			
	+200bp	+100bp	-100bp	-200bp
2011	\$ 4,046	\$ 2,326	NM ^(a)	NM ^(a)
2010	2,465	1,483	NM ^(a)	NM ^(a)

(a) Downward 100- and 200-basis-point parallel shocks result in a Federal Funds target rate of zero and negative three- and six-month treasury rates. The earnings-at-risk results of such a low-probability scenario are not meaningful.

The change in earnings at risk from December 31, 2010, resulted from investment portfolio repositioning and an

assumed higher level of deposit balances. The Firm's risk to rising rates was largely the result of widening deposit margins, which are currently compressed due to very low short-term interest rates.

Additionally, another interest rate scenario used by the Firm — involving a steeper yield curve with long-term rates rising by 100 basis points and short-term rates staying at current levels — results in a 12-month pretax earnings benefit of \$669 million. The increase in earnings under this scenario is due to reinvestment of maturing assets at the higher long-term rates, with funding costs remaining unchanged.

Risk monitoring and control

Limits

Market risk is controlled primarily through a series of limits. Limits reflect the Firm's risk appetite in the context of the market environment and business strategy. In setting limits, the Firm takes into consideration factors such as senior management risk appetite, market volatility, product liquidity, accommodation of client business and management experience.

Market risk management regularly reviews and updates risk limits. Senior management, including the Firm's Chief Executive Officer and Chief Risk Officer, is responsible for reviewing and approving certain risk limits on an ongoing basis.

The Firm maintains different levels of limits. Corporate-level limits include VaR and stress limits. Similarly, line-of-business limits include VaR and stress limits and may be supplemented by loss advisories, nonstatistical measurements and profit and loss drawdowns. Businesses are responsible for adhering to established limits, against which exposures are monitored and reported. Limit breaches are reported in a timely manner to senior management, and the affected line-of-business is required to reduce trading positions or consult with senior management on the appropriate action.

Model review

Some of the Firm's financial instruments cannot be valued based on quoted market prices but are instead valued using pricing models. These pricing models and VaR models are used for management of risk positions, such as reporting against limits, as well as for valuation. The Model Risk Group, which is independent of the businesses and market risk management, reviews the models the Firm uses and assesses model appropriateness and consistency. The model reviews consider a number of factors about the model's suitability for valuation and risk management of a particular product. These factors include whether the model accurately reflects the characteristics of the transaction and its significant risks, the suitability and convergence properties of numerical algorithms, reliability of data sources, consistency of the treatment with models for similar products, and sensitivity to input parameters and assumptions that cannot be priced from the market. Reviews are conducted of new or changed models, as well

as previously accepted models, to assess whether there have been any changes in the product or market that may affect the model's validity and whether there are theoretical or competitive developments that may require reassessment of the model's adequacy. For a summary of valuations based on models, see Critical Accounting Estimates Used by the Firm on pages 168–172 and Note 3 on pages 184–198 of this Annual Report.

Risk reporting

Nonstatistical risk measures, VaR, loss advisories and limit excesses are reported daily to the lines of business and to senior management. Market risk exposure trends, VaR trends, profit-and-loss changes and portfolio concentrations are reported weekly. Stress-test results are also reported weekly to the lines of business and to senior management.

COUNTRY RISK MANAGEMENT

Country risk is the risk that a sovereign event or action alters the value or terms of contractual obligations of counterparties and issuers related to a country. The Firm has a comprehensive country risk management framework for assessing country risks, determining risk tolerance, and measuring and monitoring direct country exposures in the Firm's wholesale lines of business, including CIO. The Country Risk Management group is responsible for developing guidelines and policy for managing country risk in both emerging and developed countries. The Country Risk Management group actively monitors the wholesale portfolio to ensure the Firm's country risk exposures are diversified and that exposure levels are appropriate given the Firm's strategy and risk tolerance relative to a country.

Country risk organization

The Country Risk Management group is an independent risk management function which works in close partnership with other risk functions and across wholesale lines of business, including CIO. The Country Risk Management governance consists of the following functions:

- Developing guidelines and policies consistent with a comprehensive country risk framework
- Assigning sovereign ratings and assessing country risks
- Measuring and monitoring country risk exposure across the Firm
- Managing country limits and reporting utilization to senior management
- Developing surveillance tools for early identification of potential country risk concerns
- Providing country risk scenario analysis

Country risk identification and measurement

The Firm is exposed to country risk through its wholesale lending, investing, and market-making activities, whether cross-border or locally funded. Country exposure includes activity with both government and private-sector entities in a country. Under the Firm's internal risk management approach, country exposure is reported based on the country where the majority of the assets of the obligor, counterparty, issuer or guarantor are located or where the majority of its revenue is derived, which may be different than the domicile (legal residence) of the obligor, counterparty, issuer or guarantor. Exposures are generally measured by considering the Firm's risk to an immediate

default of the counterparty or obligor, with zero recovery. For example:

- Lending exposures are measured at the total committed amount (funded and unfunded), net of the allowance for credit losses and cash and marketable securities collateral received
- AFS securities are measured at par value
- Securities financing exposures are measured at their receivable balance, net of collateral received
- Debt and equity securities in market-making and investing activities are measured at the fair value of all positions, both long and short positions
- Counterparty exposure on derivative receivables, including credit derivative receivables, is measured at the derivative's fair value, net of the fair value of the related collateral
- Credit derivatives protection purchased and sold are reported based on the underlying reference entity and is measured at the notional amount of protection purchased or sold, net of the fair value of the recognized derivative receivable or payable. Credit derivatives protection purchased and sold in the Firm's market-making activities are presented on a net basis, as such activities often result in selling and purchasing protection related to the same underlying reference entity, and which reflects the manner in which the Firm manages these exposures

In addition, the Firm also has indirect exposures to country risk (for example, related to the collateral received on securities financing receivables or related to client clearing activities). These indirect exposures are managed in the normal course of business through the Firm's credit, market, and operational risk governance, rather than through the country risk governance.

The Firm's internal risk management approach differs from the reporting provided under FFIEC bank regulatory requirements. There are significant reporting differences in reporting methodology, including with respect to the treatment of collateral received and the benefit of credit derivative protection. For further information on the FFIEC's reporting methodology, see Cross-border outstandings on page 322 of the 2011 Form 10-K.

Management's discussion and analysis

Country risk monitoring and control

The Country Risk Policy Group establishes guidelines for sovereign ratings reviews and limit management. In addition, the Country Risk Management group uses surveillance tools for early identification of potential country risk concerns, such as signaling models and ratings indicators. The limit framework includes a risk-tier approach and stress testing procedures for assessing the potential risk of loss associated with a significant sovereign crisis. Country ratings and limits activity are actively monitored and reported on a regular basis. Country limit requirements are reviewed and approved by senior management as often as necessary, but at least annually. For further information on market-risk stress testing the Firm performs in the normal course of business, see Market Risk Management on pages 161–162 of this Annual Report. For further information on credit loss estimates, see Critical Accounting Estimates – Allowance for credit losses on pages 168–169 of this Annual Report.

Country risk reporting

The following table presents the Firm's top 20 country exposures (excluding U.S.) based on its internal measurements of exposure. The selection of countries is based solely on the Firm's largest total exposures by country and does not represent its view of any actual or potentially adverse credit conditions.

Top 20 country exposures

December 31, 2011 (in billions)	Lending ^(a)	Trading and investing ^(b)	Other ^(c)	Total exposure
United Kingdom	\$ 23.6	\$ 58.4	\$ 12.1	\$ 94.1
Switzerland	41.4	1.1	0.5	43.0
Netherlands	4.7	34.5	2.9	42.1
France	16.8	13.9	—	30.7
Germany	13.6	16.0	—	29.6
Australia	7.6	20.4	—	28.0
Brazil	5.3	14.1	—	19.4
Canada	9.1	5.9	0.2	15.2
India	7.8	7.1	—	14.9
Korea	7.7	5.7	—	13.4
China	7.0	4.4	0.2	11.6
Japan	3.5	5.4	—	8.9
Hong Kong	3.5	4.2	—	7.7
Mexico	3.2	4.5	—	7.7
Belgium	2.1	5.2	0.1	7.4
Spain	3.3	3.8	0.1	7.2
Italy	3.1	3.4	0.1	6.6
Singapore	3.0	2.2	1.0	6.2
Sweden	1.6	3.6	0.5	5.7
Taiwan	2.8	2.5	—	5.3

(a) Lending includes loans and accrued interest receivable, net of the allowance for loan losses, deposits with banks, acceptances, other monetary assets, issued letters of credit net of participations, and undrawn commitments to extend credit.

(b) Includes market-making inventory, securities held in AFS accounts and hedging.

(c) Includes capital invested in local entities and physical commodity storage.

Selected European exposure

Several European countries, including Spain, Italy, Ireland, Portugal and Greece, have been subject to credit deterioration due to weaknesses in their economic and fiscal situations. The Firm believes its exposure to these five countries is modest relative to the Firm's overall risk exposures and is manageable given the size and types of exposures to each of the countries and the diversification of the aggregate exposure. The Firm continues to conduct business and support client activity in these countries and, therefore, the Firm's aggregate net exposures and sector distribution may vary over time. In addition, the net exposures may be affected by changes in market conditions, including the effects of interest rates and credit spreads on market valuations. The Firm is closely monitoring its exposures in these countries. The following table presents the Firm's direct exposure to these five countries at December 31, 2011, as measured under the Firm's internal risk management approach.

December 31, 2011 (in billions)	Lending ^(a)	AFS securities ^(b)	Trading ^(c)	Derivative collateral ^(d)	Portfolio hedging ^(e)	Total exposure
Spain						
Sovereign	\$ —	\$ 2.0	\$ —	\$ —	(0.1)	\$ 1.9
Non-sovereign	3.3	0.2	4.4	(2.3)	(0.3)	5.3
Total Spain exposure	\$ 3.3	\$ 2.2	\$ 4.4	\$ (2.3)	\$ (0.4)	\$ 7.2
Italy						
Sovereign	\$ —	\$ —	\$ 6.4	\$ (1.1)	\$ (2.8)	\$ 2.5
Non-sovereign	3.1	0.1	2.9	(1.5)	(0.5)	4.1
Total Italy exposure	\$ 3.1	\$ 0.1	\$ 9.3	\$ (2.6)	\$ (3.3)	\$ 6.6
Other (Ireland, Portugal and Greece)						
Sovereign	\$ —	\$ 1.0	\$ 0.1	\$ —	(0.9)	\$ 0.2
Non-sovereign	1.4	—	2.1	(1.4)	(0.1)	2.0
Total other exposure	\$ 1.4	\$ 1.0	\$ 2.2	\$ (1.4)	\$ (1.0)	\$ 2.2
Total exposure	\$ 7.8	\$ 3.3	\$ 15.9	\$ (6.3)	\$ (4.7)	\$ 16.0

- (a) Lending includes loans and accrued interest receivable, net of the allowance for loan losses, deposits with banks, acceptances, other monetary assets issued letters of credit net of participations, and undrawn commitments to extend credit. Includes \$2.2 billion of unfunded lending exposure at December 31, 2011. These exposures consist typically of committed, but unused corporate credit agreements, with market-based lending terms and covenants.
- (b) The fair value of AFS securities was \$3.1 billion at December 31, 2011.
- (c) Includes: (1) \$1.2 billion of issuer exposure on debt and equity securities held in trading, as well as market-making CDS exposure and (2) \$14.5 billion derivative and securities financing counterparty exposure. As of December 31, 2011, there were approximately \$18.4 billion of securities financing receivables, which were collateralized with approximately \$21.5 billion of marketable securities.
- (d) Includes cash and marketable securities pledged to the Firm, of which approximately 98% of the collateral was cash as of December 31, 2011.
- (e) Reflects net CDS protection purchased through the Firm's credit portfolio management activities, which are managed separately from its market-making activities.

Corporate clients represent approximately 77% of the Firm's non-sovereign net exposure in these five countries, and substantially all of the remaining 23% of the non-sovereign exposure is to the banking sector.

The table above includes single-name CDS protection sold and purchased, as well as portfolio and tranche CDS for which one or more of the underlying reference entities is in one of the named European countries. As of December 31, 2011, the notional amount of single-name CDS protection sold and purchased related to these countries was \$142.4 billion and \$147.3 billion, respectively, on a gross basis, before consideration of counterparty master netting agreements or collateral arrangements. In each of the five countries, the aggregate gross notional amount of single-name protection sold was more than 97% offset by the aggregate gross notional amount of single-name protection purchased on the same reference entities on which the Firm sold protection. The notional amount of single-name CDS protection sold and purchased related to these countries, after consideration of counterparty master netting agreements (which is a measure used by certain market peers and therefore presented for comparative purposes), was \$13.7 billion and \$18.5 billion, respectively.

The fair value of the single-name CDS protection sold and purchased in the five named European countries as of December 31, 2011 was \$22.9 billion and \$24.1 billion, respectively, prior to consideration of collateral and master netting agreements, and was \$2.7 billion and \$3.9 billion, respectively, after consideration of counterparty master netting agreements for single-name credit derivatives within the selected European countries.

The Firm's credit derivative activity is presented on a net basis, as market-making activities often result in selling and purchasing protection related to the same underlying reference entity. This presentation reflects the manner in which this exposure is managed, and reflects, in the Firm's view, the substantial mitigation of counterparty credit and market risk in its credit derivative activities. The Firm believes that the counterparty credit risk on credit derivative purchased protection has been substantially mitigated based on the following characteristics, by notional amount, as of December 31, 2011:

- 99% is purchased under contracts that require posting of cash collateral;
- 83% is purchased from investment-grade counterparties domiciled outside of the select European countries;
- 75% of the protection purchased offsets protection sold on the identical reference entity, with the identical counterparty subject to master netting agreements.

The Firm generally seeks to purchase credit protection with the same or similar maturity date on its exposures for which the protection was purchased. However, there are instances where the purchased protection has a shorter maturity date than the maturity date on the exposure for which the protection was purchased. These exposures are actively monitored and managed by the Firm.

The effectiveness of the Firm's CDS protection as a hedge of the Firm's exposures may vary depending upon a number of factors, including the contractual terms of the CDS. For further information about credit derivatives (otecae(ess)fp thuv

indicative of future loss levels. Management applies judgment in making this adjustment, taking into account uncertainties associated with current macroeconomic and political conditions, quality of underwriting standards, borrower behavior, the estimated effects of the mortgage foreclosure-related settlement with federal and state officials, uncertainties regarding the ultimate success of loan modifications, and other relevant internal and external factors affecting the credit quality of the portfolio. For junior lien products, management considers the delinquency and/or modification status of any senior liens in determining the adjustment. The application of different inputs into the statistical calculation, and the assumptions used by management to adjust the statistical calculation, are subject to management judgment, and emphasizing one input or assumption over another, or considering other inputs or assumptions, could affect the estimate of the allowance for loan losses for the consumer credit portfolio.

The allowance for credit losses for the consumer portfolio, including credit card, is sensitive to changes in the economic environment, delinquency status, the realizable value of collateral, FICO scores, borrower behavior and other risk factors. Significant judgment is required to estimate the duration of current weak overall economic conditions, as well as the impact on housing prices and the labor market. The allowance for credit losses is highly sensitive to both home prices and unemployment rates, and in the current market it is difficult to estimate how potential changes in one or both of these factors might affect the allowance for credit losses. For example, while both factors are important determinants of overall allowance levels, changes in one factor or the other may not occur at the same rate, or changes may be directionally inconsistent such that improvement in one factor may offset deterioration in the other. In addition, changes in these factors would not necessarily be consistent across all geographies or product types. Finally, it is difficult to predict the extent to which changes in both or either of these factors would ultimately affect the frequency of losses, the severity of losses or both.

PCI loans

In connection with the Washington Mutual transaction, JPMorgan Chase acquired certain PCI loans, which are accounted for as described in Note 14 on pages 231–252 of this Annual Report. The allowance for loan losses for the PCI portfolio is based on quarterly estimates of the amount of principal and interest cash flows expected to be collected over the estimated remaining lives of the loans.

These cash flow projections are based on estimates regarding default rates, loss severities, the amounts and timing of prepayments and other factors that are reflective of current and expected future market conditions. These estimates are dependent on assumptions regarding the level of future home price declines, and the duration of current weak overall economic conditions, among other factors. These estimates and assumptions require

significant management judgment and certain assumptions are highly subjective.

Allowance for credit losses sensitivity

As noted above, the Firm's allowance for credit losses is sensitive to numerous factors, depending on the portfolio. Changes in economic conditions or in the Firm's assumptions could affect the Firm's estimate of probable credit losses inherent in the portfolio at the balance sheet date. For example, deterioration in the following inputs would have the following effects on the Firm's modeled loss estimates as of December 31, 2011, without consideration of any offsetting or correlated effects of other inputs in the Firm's allowance for loan losses:

- A one-notch downgrade in the Firm's internal risk ratings for its entire wholesale loan portfolio could imply an increase in the Firm's modeled loss estimates of approximately \$1.9 billion.
- An adverse national home price scenario (reflecting an additional 8% decline in housing prices when geographically weighted for the PCI portfolio), could result in an increase in credit loss estimates for PCI loans of approximately \$1.5 billion.
- The same adverse scenario, weighted for the residential real estate portfolio, excluding PCI loans, could result in an increase to modeled annual loss estimates of approximately \$600 million.
- A 50 basis point deterioration in forecasted credit card loss rates could imply an increase to modeled annualized credit card loan loss estimates of approximately \$800 million.

The purpose of these sensitivity analyses is to provide an indication of the isolated impacts of hypothetical alternative assumptions on credit loss estimates. The changes in the inputs presented above are not intended to imply management's expectation of future deterioration of those risk factors.

It is difficult to estimate how potential changes in specific factors might affect the allowance for credit losses because management considers a variety of factors and inputs in estimating the allowance for credit losses. Changes in these factors and inputs may not occur at the same rate and may not be consistent across all geographies or product types, and changes in factors may be directionally inconsistent, such that improvement in one factor may offset deterioration in other factors. In addition, it is difficult to predict how changes in specific economic conditions or assumptions could affect borrower behavior or other factors considered by management in estimating the allowance for credit losses. Given the process the Firm follows in evaluating the risk factors related to its loans, including risk ratings, home price assumptions, and credit card loss estimates, management believes that its current estimate of the allowance for credit loss is appropriate.

Management's discussion and analysis

Fair value of financial instruments, MSRs and commodities inventory

JPMorgan Chase carries a portion of its assets and liabilities at fair value. The majority of such assets and liabilities are measured at fair value on a recurring basis. Certain assets and liabilities are measured at fair value on a nonrecurring basis, including loans accounted for at the lower of cost or fair value that are only subject to fair value adjustments under certain circumstances.

Under U.S. GAAP there is a three-level valuation hierarchy for disclosure of fair value measurements. An instrument's categorization within the hierarchy is based on the lowest level of input that is significant to the fair value measurement. Therefore, for instruments classified in levels 1 and 2 of the hierarchy, where inputs are principally based on observable market data, there is less judgment applied in arriving at a fair value measurement. For instruments classified within level 3 of the hierarchy, judgments are more significant. The Firm reviews and updates the fair value hierarchy classifications on a quarterly basis. Changes from one quarter to the next related to the observability of inputs to a fair value measurement may result in a reclassification between hierarchy levels.

Assets measured at fair value

The following table includes the Firm's assets measured at fair value and the portion of such assets that are classified within level 3 of the valuation hierarchy. For further information, see Note 3 on pages 184–198 of this Annual Report.

December 31, (in billions, except ratio data)	2011	
	Total assets at fair value	Total level 3 assets
Trading debt and equity instruments	\$ 351.5	\$ 33.0
Derivative receivables – gross	1,884.5	35.0
Netting adjustment	(1,792.0)	—
Derivative receivables – net	92.5	35.0
AFS securities	364.8	25.5
Loans	2.1	1.6
MSRs	7.2	7.2
Private equity investments	7.6	6.8
Other	49.1	4.4
Total assets measured at fair value on a recurring basis	874.8	113.5
Total assets measured at fair value on a nonrecurring basis	5.3	4.9
Total assets measured at fair value	\$ 880.1	\$ 118.4 ^(a)
Total Firm assets	\$ 2,265.8	
Level 3 assets as a percentage of total Firm assets		5.2%
Level 3 assets as a percentage of total Firm assets at fair value		13.5%

(a) At December 31, 2011, included \$63.0 billion of level 3 assets, consisting of recurring and nonrecurring assets carried by IB.

Valuation

The Firm has an established and well-documented process for determining fair value. Fair value is based on quoted market prices, where available. If listed prices or quotes are not available, fair value is based on internally developed models that consider relevant transaction data such as maturity and use as inputs market-based or independently

sourced market parameters. For further information on the Firm's valuation process, see Note 3 on pages 184–198 of this Annual Report.

For instruments classified within level 3 of the hierarchy, judgments used to estimate fair value may be significant. In arriving at an estimate of fair value for an instrument within level 3, management must first determine the appropriate model to use. Second, due to the lack of observability of significant inputs, management must assess all relevant empirical data in deriving valuation inputs – including, but not limited to, transaction details, yield curves, interest rates, volatilities, equity or debt prices, valuations of comparable instruments, foreign exchange rates and credit curves. Finally, management judgment must be applied to assess the appropriate level of valuation adjustments to reflect counterparty credit quality, the Firm's creditworthiness, constraints on liquidity and unobservable parameters, where relevant. The judgments made are typically affected by the type of product and its specific contractual terms, and the level of liquidity for the product or within the market as a whole.

The Firm has numerous controls in place to ensure that its valuations are appropriate. An independent model review group reviews the Firm's valuation models and approves them for use for specific products. All valuation models of the Firm are subject to this review process. A price verification group, independent from the risk-taking functions, ensures observable market prices and market-based parameters are used for valuation whenever possible. For those products with material parameter risk for which observable market levels do not exist, an independent review of the assumptions made on pricing is performed. Additional review includes deconstruction of the model valuations for certain structured instruments into their components; benchmarking valuations, where possible, to similar products; validating valuation estimates through actual cash settlement; and detailed review and explanation of recorded gains and losses, which are analyzed daily and over time. Valuation adjustments, which are also determined by the independent price verification group, are based on established policies and applied consistently over time. Any changes to the valuation methodology are reviewed by management to confirm the changes are justified. As markets and products develop and the pricing for certain products becomes more transparent, the Firm continues to refine its valuation methodologies.

Imprecision in estimating unobservable market inputs can affect the amount of revenue or loss recorded for a particular position. Furthermore, while the Firm believes its valuation methods are appropriate and consistent with those of other market participants, the use of different methodologies or assumptions to determine the fair value of certain financial instruments could result in a different estimate of fair value at the reporting date. For a detailed discussion of the determination of fair value for individual financial instruments, see Note 3 on pages 184–198 of this Annual Report.

Goodwill impairment

Under U.S. GAAP, goodwill must be allocated to reporting units and tested for impairment at least annually. The Firm's process and methodology used to conduct goodwill impairment testing is described in Note 17 on pages 267–271 of this Annual Report.

Management applies significant judgment when estimating the fair value of its reporting units. Estimates of fair value are dependent upon estimates of (a) the future earnings potential of the Firm's reporting units, including the estimated effects of regulatory and legislative changes, such as the Dodd-Frank Act, the CARD Act, and limitations on non-sufficient funds and overdraft fees and (b) the relevant cost of equity and long-term growth rates. Imprecision in estimating these factors can affect the estimated fair value of the reporting units.

Based upon the updated valuations for all of its reporting units, the Firm concluded that goodwill allocated to its reporting units was not impaired at December 31, 2011 nor was any goodwill written off during 2011. The fair values of a significant majority of the Firm's reporting units exceeded their carrying values by substantial amounts (excess fair value as a percent of carrying value ranged from approximately 20% to 200%) and did not indicate a significant risk of goodwill impairment based on current projections and valuations.

However, the fair value of the Firm's consumer lending businesses in RFS and Card each exceeded their carrying values by less than 15% and the associated goodwill remains at an elevated risk for goodwill impairment due to their exposure to U.S. consumer credit risk and the effects of regulatory and legislative changes. The assumptions used in the valuation of these businesses include (a) estimates of future cash flows for the business (which are dependent on portfolio outstanding balances, net interest margin, operating expense, credit losses and the amount of capital necessary given the risk of business activities), and (b) the cost of equity used to discount those cash flows to a present value. Each of these factors requires significant judgment and the assumptions used are based on management's best estimate and most current projections, derived from the Firm's business forecasting process reviewed with senior management. These projections are consistent with the short-term assumptions discussed in the Business Outlook on pages 68–69 of this Annual Report, and, in the longer term, incorporate a set of macroeconomic assumptions and the Firm's best estimates of long-term growth and returns of its businesses. Where possible, the Firm uses third-party and peer data to benchmark its assumptions and estimates.

Deterioration in economic market conditions, increased estimates of the effects of recent regulatory or legislative changes, or additional regulatory or legislative changes may result in declines in projected business performance beyond management's current expectations. For example, in RFS, such declines could result from increases in costs to resolve foreclosure-related matters or from deterioration in

economic conditions that result in increased credit losses, including decreases in home prices beyond management's current expectations. In Card, declines in business performance could result from deterioration in economic conditions such as increased unemployment claims or bankruptcy filings that result in increased credit losses or changes in customer behavior that cause decreased account activity or receivable balances. In addition, the earnings or estimated cost of equity of the Firm's capital markets businesses could also be affected by regulatory or legislative changes. Declines in business performance, increases in equity capital requirements, or increases in the estimated cost of equity, could cause the estimated fair values of the Firm's reporting units or their associated goodwill to decline, which could result in a material impairment charge to earnings in a future period related to some portion of the associated goodwill.

For additional information on goodwill, see Note 17 on pages 267–271 of this Annual Report.

Income taxes

JPMorgan Chase is subject to the income tax laws of the various jurisdictions in which it operates, including U.S. federal, state and local and non-U.S. jurisdictions. These laws are often complex and may be subject to different interpretations. To determine the financial statement impact of accounting for income taxes, including the provision for income tax expense and unrecognized tax benefits, JPMorgan Chase must make assumptions and judgments about how to interpret and apply these complex tax laws to numerous transactions and business events, as well as make judgments regarding the timing of when certain items may affect taxable income in the U.S. and non-U.S. tax jurisdictions.

JPMorgan Chase's interpretations of tax laws around the world are subject to review and examination by the various taxing authorities in the jurisdictions where the Firm operates, and disputes may occur regarding its view on a tax position. These disputes over interpretations with the various taxing authorities may be settled by audit, administrative appeals or adjudication in the court systems of the tax jurisdictions in which the Firm operates. JPMorgan Chase regularly reviews whether it may be assessed additional income taxes as a result of the resolution of these matters, and the Firm records additional reserves as appropriate. In addition, the Firm may revise its estimate of income taxes due to changes in income tax laws, legal interpretations and tax planning strategies. It is possible that revisions in the Firm's estimate of income taxes may materially affect the Firm's results of operations in any reporting period.

The Firm's provision for income taxes is composed of current and deferred taxes. Deferred taxes arise from differences between assets and liabilities measured for financial reporting versus income tax return purposes. Deferred tax assets are recognized if, in management's judgment, their realizability is determined to be more likely

than not. The Firm has also recognized deferred tax assets in connection with certain net operating losses. The Firm performs regular reviews to ascertain whether deferred tax assets are realizable. These reviews include management's estimates and assumptions regarding future taxable income, which also incorporates various tax planning strategies, including strategies that may be available to utilize net operating losses before they expire. In connection with these reviews, if it is determined that a deferred tax asset is not realizable, a valuation allowance is established. The valuation allowance may be reversed in a subsequent reporting period if the Firm determines that, based on revised estimates of future taxable income or changes in tax planning strategies, it is more likely than not that all or part of the deferred tax asset will become realizable. As of December 31, 2011, management has determined it is more likely than not that the Firm will realize its deferred tax assets, net of the existing valuation allowance.

JPMorgan Chase does not provide U.S. federal income taxes on the undistributed earnings of certain non-U.S. subsidiaries, to the extent that such earnings have been reinvested abroad for an indefinite period of time. Changes

to the income tax rates applicable to these non-U.S. subsidiaries may have a material impact on the effective tax rate in a future period if such changes were to occur.

The Firm adjusts its unrecognized tax benefits as necessary when additional information becomes available. Uncertain tax positions that meet the more-likely-than-not recognition threshold are measured to determine the amount of benefit to recognize. An uncertain tax position is measured at the largest amount of benefit that management believes is more likely than not to be realized upon settlement. It is possible that the reassessment of JPMorgan Chase's unrecognized tax benefits may have a material impact on its effective tax rate in the period in which the reassessment occurs.

For additional information on income taxes, see Note 26 on pages 279–281 of this Annual Report.

Litigation reserves

For a description of the significant estimates and judgments associated with establishing litigation reserves, see Note 31 on pages 290–299 of this Annual Report.

ACCOUNTING AND REPORTING DEVELOPMENTS

Fair value measurement and disclosures

In January 2010, the FASB issued guidance that requires new disclosures, and clarifies existing disclosure requirements, about fair value measurements. The clarifications and the requirement to separately disclose transfers of instruments between level 1 and level 2 of the fair value hierarchy was effective for interim reporting periods beginning after December 15, 2009; the Firm adopted this guidance in the first quarter of 2010. In addition, a new requirement to provide purchases, sales, issuances and settlements in the level 3 rollforward on a gross basis was effective for fiscal years beginning after December 15, 2010. The Firm adopted the new guidance, effective January 1, 2011. For information about fair value measurements, see Note 3 on pages 184–198 of this Annual Report.

In May 2011, the FASB issued guidance that amends the requirements for fair value measurement and disclosure. The guidance changes and clarifies certain existing requirements related to portfolios of financial instruments and valuation adjustments, requires additional disclosures for fair value measurements categorized in level 3 of the fair value hierarchy (including disclosure of the range of inputs used in certain valuations), and requires additional disclosures for certain financial instruments that are not carried at fair value. The guidance is effective in the first quarter of 2012. The application of this guidance is not expected to have a material effect on the Firm's Consolidated Balance Sheets or results of operations.

Determining whether a restructuring is a troubled debt restructuring

In April 2011, the FASB issued guidance to clarify existing standards for determining whether a modification represents a TDR from the perspective of the creditor. In addition, the guidance established an effective date for enhanced disclosures related to TDRs. The guidance and new disclosures became effective in the third quarter of 2011 and were applied retrospectively to January 1, 2011. For information regarding the Firm's TDRs, see Note 14 on pages 231–252 of this Annual Report. The application of this guidance did not have a material effect on the Firm's Consolidated Balance Sheets or results of operations.

Accounting for repurchase and similar agreements

In April 2011, the FASB issued guidance that amends the criteria used to assess whether repurchase and similar agreements should be accounted for as financings or sales (purchases) with forward agreements to repurchase (resell). Specifically, the guidance eliminates circumstances in which the lack of adequate collateral maintenance requirements could result in a repurchase agreement being accounted for as a sale. The guidance is effective for new transactions or existing transactions that are modified beginning January 1, 2012. The Firm has accounted for its repurchase and similar agreements as secured financings, and therefore, the Firm does not expect the application of this guidance will have an impact on the Firm's Consolidated Balance Sheets or results of operations.

Presentation of other comprehensive income

In June 2011, the FASB issued guidance that modifies the presentation of other comprehensive income in the Consolidated Financial Statements. The guidance requires that items of net income, items of other comprehensive income, and total comprehensive income be presented in one continuous statement or in two separate but consecutive statements. For public companies the guidance is effective for interim and annual reporting periods beginning after December 15, 2011. However, in December 2011, the FASB issued guidance that deferred the presentation requirements relating to reclassifications of items out of accumulated other comprehensive income and into the income statement. The application of this guidance will only affect the presentation of the Consolidated Financial Statements and will have no impact on the Firm's Consolidated Balance Sheets or results of operations.

Balance sheet netting

In December 2011, the FASB issued guidance that requires enhanced disclosures about derivatives and securities financing agreements that are subject to legally enforceable master netting or similar agreements, or that have otherwise been offset on the balance sheet under certain specific conditions that permit net presentation. The guidance is effective in the first quarter of 2013. The application of this guidance will only affect the disclosure of these instruments and will have no impact on the Firm's Consolidated Balance Sheets or results of operations.

NONEXCHANGE TRADED COMMODITY DERIVATIVE CONTRACTS AT FAIR VALUE

In the normal course of business, JPMorgan Chase trades nonexchange-traded commodity derivative contracts. To determine the fair value of these contracts, the Firm uses various fair value estimation techniques, primarily based on internal models with significant observable market parameters. The Firm's nonexchange-traded commodity derivative contracts are primarily energy-related.

The following table summarizes the changes in fair value for nonexchange-traded commodity derivative contracts for the year ended December 31, 2011.

Year ended December 31, 2011 (in millions)	Asset position	Liability position
Net fair value of contracts outstanding at January 1, 2011	\$ 8,166	\$ 7,184
Effect of legally enforceable master netting agreements	41,284	41,919
Gross fair value of contracts outstanding at January 1, 2011	49,450	49,103
Contracts realized or otherwise settled	(22,855)	(20,826)
Fair value of new contracts	21,517	23,195
Changes in fair values attributable to changes in valuation techniques and assumptions	—	—
Other changes in fair value	(1,495)	(2,260)
Gross fair value of contracts outstanding at December 31, 2011	46,617	49,212
Effect of legally enforceable master netting agreements	(33,495)	(35,695)
Net fair value of contracts outstanding at December 31, 2011	\$ 13,122	\$ 13,517

The following table indicates the maturities of nonexchange-traded commodity derivative contracts at December 31, 2011.

December 31, 2011 (in millions)	Asset position	Liability position
Maturity less than 1 year	\$ 20,876	\$ 18,993
Maturity 1–3 years	16,564	16,949
Maturity 4–5 years	7,745	7,593
Maturity in excess of 5 years	1,432	5,677
Gross fair value of contracts outstanding at December 31, 2011	46,617	49,212
Effect of legally enforceable master netting agreements	(33,495)	(35,695)
Net fair value of contracts outstanding at December 31, 2011	\$ 13,122	\$ 13,517

FORWARD-LOOKING STATEMENTS

From time to time, the Firm has made and will make forward-looking statements. These statements can be identified by the fact that they do not relate strictly to historical or current facts. Forward-looking statements often use words such as “anticipate,” “target,” “expect,” “estimate,” “intend,” “plan,” “goal,” “believe,” or other words of similar meaning. Forward-looking statements provide JPMorgan Chase's current expectations or forecasts of future events, circumstances, results or aspirations. JPMorgan Chase's disclosures in this Annual Report contain forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. The Firm also may make forward-looking statements in its other documents filed or furnished with the Securities and Exchange Commission. In addition, the Firm's senior management may make forward-looking statements orally to analysts, investors, representatives of the media and others.

All forward-looking statements are, by their nature, subject to risks and uncertainties, many of which are beyond the Firm's control. JPMorgan Chase's actual future results may differ materially from those set forth in its forward-looking statements. While there is no assurance that any list of risks and uncertainties or risk factors is complete, below are certain factors which could cause actual results to differ from those in the forward-looking statements:

- Local, regional and international business, economic and political conditions and geopolitical events;
- Changes in laws and regulatory requirements, including as a result of recent financial services legislation;
- Changes in trade, monetary and fiscal policies and laws;
- Securities and capital markets behavior, including changes in market liquidity and volatility;
- Changes in investor sentiment or consumer spending or savings behavior;
- Ability of the Firm to manage effectively its liquidity;
- Changes in credit ratings assigned to the Firm or its subsidiaries;
- Damage to the Firm's reputation;
- Ability of the Firm to deal effectively with an economic slowdown or other economic or market disruption;
- Technology changes instituted by the Firm, its counterparties or competitors;
- Mergers and acquisitions, including the Firm's ability to integrate acquisitions;
- Ability of the Firm to develop new products and services, and the extent to which products or services previously sold by the Firm (including but not limited to mortgages and asset-backed securities) require the Firm to incur liabilities or absorb losses not contemplated at their initiation or origination;
- Ability of the Firm to address enhanced regulatory requirements affecting its mortgage business;
- Acceptance of the Firm's new and existing products and services by the marketplace and the ability of the Firm to increase market share;
- Ability of the Firm to attract and retain employees;
- Ability of the Firm to control expense;
- Competitive pressures;
- Changes in the credit quality of the Firm's customers and counterparties;
- Adequacy of the Firm's risk management framework;
- Adverse judicial or regulatory proceedings;
- Changes in applicable accounting policies;
- Ability of the Firm to determine accurate values of certain assets and liabilities;
- Occurrence of natural or man-made disasters or calamities or conflicts, including any effect of any such disasters, calamities or conflicts on the Firm's power generation facilities and the Firm's other commodity-related activities;
- Ability of the Firm to maintain the security of its financial, accounting, technology, data processing and other operating systems and facilities;
- The other risks and uncertainties detailed in Part I, Item 1A: Risk Factors in the Firm's Annual Report on Form 10-K for the year ended December 31, 2011.

Any forward-looking statements made by or on behalf of the Firm speak only as of the date they are made, and JPMorgan Chase does not undertake to update forward-looking statements to reflect the impact of circumstances or events that arise after the date the forward-looking statements were made. The reader should, however, consult any further disclosures of a forward-looking nature the Firm may make in any subsequent Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, or Current Reports on Form 8-K.

Management's report on internal control over financial reporting

Management of JPMorgan Chase & Co. ("JPMorgan Chase" or the "Firm") is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed by, or under the supervision of, the Firm's principal executive and principal financial officers, or persons performing similar functions, and effected by JPMorgan Chase's Board of Directors, management and other personnel, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with accounting principles generally accepted in the United States of America.

JPMorgan Chase's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records, that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the Firm's assets; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the Firm are being made only in accordance with authorizations of JPMorgan Chase's management and directors; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of the Firm's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Management has completed an assessment of the effectiveness of the Firm's internal control over financial reporting as of December 31, 2011. In making the assessment, management used the framework in "Internal Control — Integrated Framework" promulgated by the Committee of Sponsoring Organizations of the Treadway Commission, commonly referred to as the "COSO" criteria.

Based upon the assessment performed, management concluded that as of December 31, 2011, JPMorgan Chase's internal control over financial reporting was effective based upon the COSO criteria. Additionally, based upon management's assessment, the Firm determined that there were no material weaknesses in its internal control over financial reporting as of December 31, 2011.

The effectiveness of the Firm's internal control over financial reporting as of December 31, 2011, has been audited by PricewaterhouseCoopers LLP, an independent registered public accounting firm, as stated in their report which appears herein.

A handwritten signature in black ink, appearing to read "James Dimon". The signature is stylized with a large, sweeping initial "J" and "D".

James Dimon
Chairman and Chief Executive Officer

Douglas L. Braunstein
Executive Vice President and Chief Financial Officer

February 29, 2012



To the Board of Directors and Stockholders of JPMorgan Chase & Co.:

In our opinion, the accompanying consolidated balance sheets and the related consolidated statements of income, changes in stockholders' equity and comprehensive income and cash flows present fairly, in all material respects, the financial position of JPMorgan Chase & Co. and its subsidiaries (the "Firm") at December 31, 2011 and 2010, and the results of their operations and their cash flows for each of the three years in the period ended December 31, 2011, in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Firm maintained, in all material respects, effective internal control over financial reporting as of December 31, 2011, based on criteria established in Internal Control - Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO). The Firm's management is responsible for these financial statements, for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying "Management's report on internal control over financial reporting." Our responsibility is to express opinions on these financial statements and on the Firm's internal control over financial reporting based on our integrated audits. We conducted our audits in accordance with the standards of the Public Company Accounting Oversight Board (United States). Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the financial statements are free of material misstatement and whether effective internal control over financial reporting was maintained in all material respects. Our audits of the financial statements included examining, on a test basis, evidence supporting the amounts and disclosures in the financial statements, assessing the accounting principles used and significant estimates made by management, and evaluating the overall financial statement presentation. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk

that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

A handwritten signature in dark ink that reads "PricewaterhouseCoopers LLP". The signature is written in a cursive, flowing style.

February 29, 2012

Consolidated statements of income

Year ended December 31, (in millions, except per share data)	2011	2010	2009
Revenue			
Investment banking fees	\$ 5,911	\$ 6,190	\$ 7,087
Principal transactions	10,005	10,894	9,796
Lending- and deposit-related fees	6,458	6,340	7,045
Asset management, administration and commissions	14,094	13,499	12,540
Securities gain [§]	1,593	2,965	1,110
Mortgage fees and related income	2,721	3,870	3,678
Credit card income	6,158	5,891	7,110
Other income	2,605	2,044	916
Noninterest revenue	49,545	51,693	49,282
Interest income	61,293	63,782	66,350
Interest expense	13,604	12,781	15,198
Net interest income	47,689	51,001	51,152
Total net revenue	97,234	102,694	100,434
Provision for credit losses	7,574	16,639	32,015
Noninterest expense			
Compensation expense	29,037	28,124	26,928
Occupancy expense	3,895	3,681	3,666
Technology, communications and equipment expense	4,947	4,684	4,624
Professional and outside services	7,482	6,767	6,232
Marketing	3,143	2,446	1,777
Other expense	13,559	14,558	7,594
Amortization of intangibles	848	936	1,050
Merger costs	—	—	481
Total noninterest expense	62,911	61,196	52,352
Income before income tax expense and extraordinary gain	26,749	24,859	16,067
Income tax expense	7,773	7,489	4,415
Income before extraordinary gain	18,976	17,370	11,652
Extraordinary gain	—	—	76
Net income	\$ 18,976	\$ 17,370	\$ 11,728
Net income applicable to common stockholders	\$ 17,568	\$ 15,764	\$ 8,774
Per common share data			
Basic earnings per share			
Income before extraordinary gain	\$ 4.50	\$ 3.98	\$ 2.25
Net income	4.50	3.98	2.27
Diluted earnings per share			
Income before extraordinary gain	4.48	3.96	2.24
Net income	4.48	3.96	2.26
Weighted-average basic shares	3,900.4	3,956.3	3,862.8
Weighted-average diluted shares	3,920.3	3,976.9	3,879.7
Cash dividends declared per common share	\$ 1.00	\$ 0.20	\$ 0.20
(a) The following other-than-temporary impairment losses are included in securities gains for the periods presented			
Year ended December 31, (in millions)	2011	2010	2009
Total other-than-temporary impairment losses	\$ (27)	\$ (94)	\$ (946)
Losses recorded in/(reclassified from) other comprehensive income	(49)	(6)	368
Total credit losses recognized in income	\$ (76)	\$ (100)	\$ (578)

The Notes to Consolidated Financial Statements are an integral part of these statements.

Consolidated balance sheets

December 31, (in millions, except share data)	2011	2010
Assets		
Cash and due from banks	\$ 59,602	\$ 27,567
Deposits with banks	85,279	21,673
Federal funds sold and securities purchased under resale agreements (included \$15,308 and \$13,961 at fair value)	235,314	222,554
Securities borrowed (included \$1,530 and \$1,961 at fair value)	142,462	123,587
Trading assets (included assets pledged \$19,856 and \$73,056)	443,963	489,892
Securities (included \$364,781 and \$316,318 at fair value and assets pledged \$94,691 and \$86,891)	364,793	316,336
Loans (included \$2,097 and \$1,976 at fair value)	723,720	692,927
Allowance for loan losses	(27,609)	(32,266)
Loans, net of allowance for loan losses	696,111	660,661
Accrued interest and accounts receivable	61,478	70,147
Premises and equipment	14,041	13,355
Goodwill	48,188	48,854
Mortgage servicing rights	7,223	13,649
Other intangible assets	3,207	4,039
Other assets (included \$16,499 and \$18,201 at fair value and assets pledged \$1,316 and \$1,485)	104,131	105,291
Total assets^(a)	\$ 2,265,792	\$ 2,117,605
Liabilities		
Deposits (included \$4,933 and \$4,369 at fair value)	\$ 1,127,806	\$ 930,369
Federal funds purchased and securities loaned or sold under repurchase agreements (included \$2,517 and \$4,060 at fair value)	213,532	276,644
Commercial paper	51,631	35,363
Other borrowed funds (included \$9,576 and \$9,931 at fair value)	21,908	34,325
Trading liabilities	141,695	146,166
Accounts payable and other liabilities (included \$51 and \$236 at fair value)	202,895	170,330
Beneficial interests issued by consolidated variable interest entities (included \$1,250 and \$1,495 at fair value)	65,977	77,649
Long-term debt (included \$34,720 and \$38,839 at fair value)	256,775	270,653
Total liabilities^(a)	2,082,219	1,941,499
Commitments and contingencies (see Notes 29, 30 and 31 of this Annual Report)		
Stockholders' equity		
Preferred stock (\$1 par value; authorized 200,000,000 shares; issued 7,800,000 shares)	7,800	7,800
Common stock (\$1 par value; authorized 9,000,000,000 shares; issued 4,104,933,895 shares)	4,105	4,105
Capital surplus	95,602	97,415
Retained earnings	88,315	73,998
Accumulated other comprehensive income/(loss)	944	1,001
Shares held in RSU Trust, at cost (\$2,906 and 1,192,712 shares)	(38)	(53)
Treasury stock, at cost (\$2,243,180 and 194,639,785 shares)	(13,155)	(8,160)
Total stockholders' equity	183,573	176,106
Total liabilities and stockholders' equity	\$ 2,265,792	\$ 2,117,605

(a) The following table presents information on assets and liabilities related to VIEs that are consolidated by the Firm at December 31, 2011 and 2010. The difference between the VIE assets and liabilities represents the Firm's interests in those entities, which were eliminated in consolidation.

December 31, (in millions)	2011	2010
Assets		
Trading assets	\$ 12,079	\$ 9,837
Loans	86,754	95,587
All other assets	2,638	3,494
Total assets	\$ 101,471	\$ 108,918
Liabilities		
Beneficial interests issued by consolidated variable interest entities	\$ 65,977	\$ 77,649
All other liabilities	1,487	1,922
Total liabilities	\$ 67,464	\$ 79,571

The assets of the consolidated VIEs are used to settle the liabilities of those entities. The holders of the beneficial interests do not have recourse to the general credit of JPMorgan Chase. At December 31, 2011 and 2010, the Firm provided limited program-wide credit enhancement of \$3.1 billion and \$2.0 billion, respectively, related to its Firm-administered multi-seller conduits, which are eliminated in consolidation. For further discussion, see Note 16 on pages 256–267 of this Annual Report.

The Notes to Consolidated Financial Statements are an integral part of these statements.

Consolidated statements of changes in stockholders' equity and comprehensive income

Year ended December 31, (in millions, except per share data)	2011	2010	2009
Preferred stock			
Balance at January 1	\$ 7,800	\$ 8,152	\$ 31,939
Accretion of preferred stock discount on issuance to the U.S. Treasury	—	—	1,213
Redemption of preferred stock issued to the U.S. Treasury	—	—	(25,000)
Redemption of other preferred stock	—	(352)	—
Balance at December 31	7,800	7,800	8,152
Common stock			
Balance at January 1	4,105	4,105	3,942
Issuance of common stock	—	—	163
Balance at December 31	4,105	4,105	4,105
Capital surplus			
Balance at January 1	97,415	97,982	92,143
Issuance of common stock	—	—	5,593
Shares issued and commitments to issue common stock for employee stock-based compensation awards, and related tax effects	(1,688)	706	474
Other	(125)	(1,273)	(228)
Balance at December 31	95,602	97,415	97,982
Retained earnings			
Balance at January 1	73,998	62,481	54,013
Cumulative effect of changes in accounting principles	—	(4,376)	—
Net income	18,976	17,370	11,728
Dividends declared:			
Preferred stock	(629)	(642)	(1,328)
Accelerated amortization from redemption of preferred stock issued to the U.S. Treasury	—	—	(1,112)
Common stock (\$1.00, \$0.20 and \$0.20 per share for 2011, 2010 and 2009, respectively)	(4,030)	(835)	(820)
Balance at December 31	88,315	73,998	62,481
Accumulated other comprehensive income/(loss)			
Balance at January 1	1,001	(91)	(5,687)
Cumulative effect of changes in accounting principles	—	(144)	—
Other comprehensive (loss)/income	(57)	1,236	5,596
Balance at December 31	944	1,001	(91)
Shares held in RSU Trust, at cost			
Balance at January 1	(53)	(68)	(217)
Reissuance from RSU Trust	15	15	149
Balance at December 31	(38)	(53)	(68)
Treasury stock, at cost			
Balance at January 1	(8,160)	(7,196)	(9,249)
Purchase of treasury stock	(8,741)	(2,999)	—
Reissuance from treasury stock	3,750	2,040	2,079
Share repurchases related to employee stock-based compensation awards	(4)	(5)	(26)
Balance at December 31	(13,155)	(8,160)	(7,196)
Total stockholders' equity	\$ 183,573	\$ 176,106	\$ 165,365
Comprehensive income			
Net income	\$ 18,976	\$ 17,370	\$ 11,728
Other comprehensive (loss)/income	(57)	1,236	5,596
Comprehensive income	\$ 18,919	\$ 18,606	\$ 17,324

The Notes to Consolidated Financial Statements are an integral part of these statements.

Consolidated statements of cash flows

Year ended December 31, (in millions)	2011	2010	2009
Operating activities			
Net income	\$ 18,976	\$ 17,370	\$ 11,728
Adjustments to reconcile net income to net cash provided by/(used in) operating activities:			
Provision for credit losses	7,574	16,639	32,015
Depreciation and amortization	4,257	4,029	3,308
Amortization of intangibles	848	936	1,050
Deferred tax expense/(benefit)	1,693	(968)	(3,622)
Investment securities gains	(1,593)	(2,965)	(1,110)
Stock-based compensation	2,675	3,251	3,355
Originations and purchases of loans held-for-sale	(52,561)	(37,085)	(22,417)
Proceeds from sales, securitizations and paydowns of loans held-for-sale	54,092	40,155	33,902
Net change in:			
Trading assets	36,443	(72,082)	133,488
Securities borrowed	(18,936)	(3,926)	4,452
Accrued interest and accounts receivable	8,655	443	(6,312)
Other assets	(15,456)	(12,452)	32,557
Trading liabilities	7,905	19,344	(79,314)
Accounts payable and other liabilities	35,203	17,325	(26,450)
Other operating adjustments	6,157	6,234	6,167
Net cash provided by/(used in) operating activities	95,932	(3,752)	122,797
Investing activities			
Net change in:			
Deposits with banks	(63,592)	41,625	74,829
Federal funds sold and securities purchased under resale agreements	(12,490)	(26,957)	7,082
Held-to-maturity securities:			
Proceeds	6	7	9
Available-for-sale securities:			
Proceeds from maturities	86,850	92,740	87,712
Proceeds from sales	68,631	118,600	114,041
Purchases	(202,309)	(179,487)	(346,372)
Proceeds from sales and securitizations of loans held-for-investment	10,478	9,476	31,034
Other changes in loans, net	(58,365)	3,022	50,651
Net cash received from/(used in) business acquisitions or dispositions	102	(4,910)	(97)
Net maturities of asset-backed commercial paper guaranteed by the FRBB	—	—	11,228
All other investing activities, net	(63)	(114)	(762)
Net cash (used in)/provided by investing activities	(170,752)	54,002	29,355
Financing activities			
Net change in:			
Deposits	203,420	(9,637)	(107,700)
Federal funds purchased and securities loaned or sold under repurchase agreements	(63,116)	15,202	67,785
Commercial paper and other borrowed funds	7,230	(6,869)	(67,198)
Beneficial interests issued by consolidated variable interest entities	1,165	2,426	(4,076)
Proceeds from long-term borrowings and trust preferred capital debt securities	54,844	55,181	51,324
Payments of long-term borrowings and trust preferred capital debt securities	(82,078)	(99,043)	(68,441)
Excess tax benefits related to stock-based compensation	867	26	17
Redemption of preferred stock issued to the U.S. Treasury	—	—	(25,000)
Redemption of other preferred stock	—	(352)	—
Proceeds from issuance of common stock	—	—	5,756
Treasury stock and warrants repurchased	(8,863)	(2,999)	—
Dividends paid	(3,895)	(1,486)	(3,422)
All other financing activities, net	(1,868)	(1,666)	(2,124)
Net cash provided by/(used in) financing activities	107,706	(49,217)	(153,079)
Effect of exchange rate changes on cash and due from banks	(851)	328	238
Net increase/(decrease) in cash and due from banks	32,035	1,361	(689)
Cash and due from banks at the beginning of the period	27,567	26,206	26,895
Cash and due from banks at the end of the period	\$ 59,602	\$ 27,567	\$ 26,206
Cash interest paid	\$ 13,725	\$ 12,404	\$ 16,875
Cash income taxes paid, net	8,153	9,747	5,434

Note: Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. Upon adoption of the guidance, the Firm consolidated noncash assets and liabilities of \$87.7 billion and \$92.2 billion, respectively.

The Notes to Consolidated Financial Statements are an integral part of these statements.

Note 1 – Basis of presentation

JPMorgan Chase & Co. (“JPMorgan Chase” or the “Firm”), a financial holding company incorporated under Delaware law in 1968, is a leading global financial services firm and one of the largest banking institutions in the United States of America (“U.S.”), with operations worldwide. The Firm is a leader in investment banking, financial services for consumers and small business, commercial banking, financial transaction processing, asset management and private equity. For a discussion of the Firm’s business segments, see Note 33 on pages 300–303 of this Annual Report.

The accounting and financial reporting policies of JPMorgan Chase and its subsidiaries conform to accounting principles generally accepted in the U.S. (“U.S. GAAP”). Additionally, where applicable, the policies conform to the accounting and reporting guidelines prescribed by regulatory authorities.

Certain amounts reported in prior periods have been reclassified to conform to the current presentation.

Consolidation

The Consolidated Financial Statements include the accounts of JPMorgan Chase and other entities in which the Firm has a controlling financial interest. All material intercompany balances and transactions have been eliminated. The Firm determines whether it has a controlling financial interest in an entity by first evaluating whether the entity is a voting interest entity or a variable interest entity (“VIE”).

Voting Interest Entities

Voting interest entities are entities that have sufficient equity and provide the equity investors voting rights that enable them to make significant decisions relating to the entity’s operations. For these types of entities, the Firm’s determination of whether it has a controlling interest is primarily based on the amount of voting equity interests held. Entities in which the Firm has a controlling financial interest, through ownership of the majority of the entities’ voting equity interests, or through other contractual rights that give the Firm control, are consolidated by the Firm.

Investments in companies in which the Firm has significant influence over operating and financing decisions (but does not own a majority of the voting equity interests) are accounted for (i) in accordance with the equity method of accounting (which requires the Firm to recognize its proportionate share of the entity’s net earnings), or (ii) at fair value if the fair value option was elected at the inception of the Firm’s investment. These investments are generally included in other assets, with income or loss included in other income.

Certain Firm-sponsored asset management funds are structured as limited partnerships or limited liability companies. For many of these entities, the Firm is the general partner or managing member, but the non-affiliated partners or members have the ability to remove the Firm as

the general partner or managing member without cause (i.e., kick-out rights), based on a simple majority vote, or the non-affiliated partners or members have rights to participate in important decisions. Accordingly, the Firm does not consolidate these funds. In the limited cases where the non-affiliated partners or members do not have substantive kick-out or participating rights, the Firm consolidates the funds.

The Firm’s investment companies make investments in both publicly-held and privately-held entities, including investments in buyouts, growth equity and venture opportunities. These investments are accounted for under investment company guidelines and accordingly, irrespective of the percentage of equity ownership interests held, are carried on the Consolidated Balance Sheets at fair value, and are recorded in other assets.

Variable Interest Entities

VIEs are entities that, by design, either (1) lack sufficient equity to permit the entity to finance its activities without additional subordinated financial support from other parties, or (2) have equity investors that do not have the ability to make significant decisions relating to the entity’s operations through voting rights, or do not have the obligation to absorb the expected losses, or do not have the right to receive the residual returns of the entity.

The most common type of VIE is a special purpose entity (“SPE”). SPEs are commonly used in securitization transactions in order to isolate certain assets and distribute the cash flows from those assets to investors. The basic SPE structure involves a company selling assets to the SPE; the SPE funds the purchase of those assets by issuing securities to investors. The legal documents that govern the transaction specify how the cash earned on the assets must be allocated to the SPE’s investors and other parties that have rights to those cash flows. SPEs are generally structured to insulate investors from claims on the SPE’s assets by creditors of other entities, including the creditors of the seller of the assets.

The primary beneficiary of a VIE (i.e., the party that has a controlling financial interest) is required to consolidate the assets and liabilities of the VIE. The primary beneficiary is the party that has both (1) the power to direct the activities of an entity that most significantly impact the VIE’s economic performance; and (2) through its interests in the VIE, the obligation to absorb losses or the right to receive benefits from the VIE that could potentially be significant to the VIE.

To assess whether the Firm has the power to direct the activities of a VIE that most significantly impact the VIE's economic performance, the Firm considers all the facts and circumstances, including its role in establishing the VIE and its ongoing rights and responsibilities. This assessment includes, first, identifying the activities that most significantly impact the VIE's economic performance; and second, identifying which party, if any, has power over those activities. In general, the parties that make the most significant decisions affecting the VIE (such as asset managers, collateral managers, servicers, or owners of call options or liquidation rights over the VIE's assets) or have the right to unilaterally remove those decision-makers are deemed to have the power to direct the activities of a VIE.

To assess whether the Firm has the obligation to absorb losses of the VIE or the right to receive benefits from the VIE that could potentially be significant to the VIE, the Firm considers all of its economic interests, including debt and equity investments, servicing fees, and derivative or other arrangements deemed to be variable interests in the VIE. This assessment requires that the Firm apply judgment in determining whether these interests, in the aggregate, are considered potentially significant to the VIE. Factors considered in assessing significance include: the design of the VIE, including its capitalization structure; subordination of interests; payment priority; relative share of interests held across various classes within the VIE's capital structure; and the reasons why the interests are held by the Firm.

The Firm performs on-going reassessments of: (1) whether entities previously evaluated under the majority voting-interest framework have become VIEs, based on certain events, and therefore subject to the VIE consolidation framework; and (2) whether changes in the facts and circumstances regarding the Firm's involvement with a VIE cause the Firm's consolidation conclusion to change.

In January 2010, the Financial Accounting Standards Board ("FASB") issued an amendment which deferred the requirements of the accounting guidance for VIEs for certain investment funds, including mutual funds, private equity funds and hedge funds. For the funds to which the deferral applies, the Firm continues to apply other existing authoritative accounting guidance to determine whether such funds should be consolidated.

Assets held for clients in an agency or fiduciary capacity by the Firm are not assets of JPMorgan Chase and are not included in the Consolidated Balance Sheets.

Use of estimates in the preparation of consolidated financial statements

The preparation of the Consolidated Financial Statements requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities, revenue and expense, and disclosures of contingent assets and liabilities. Actual results could be different from these estimates.

Foreign currency translation

JPMorgan Chase revalues assets, liabilities, revenue and expense denominated in non-U.S. currencies into U.S. dollars using applicable exchange rates.

Gains and losses relating to translating functional currency financial statements for U.S. reporting are included in other comprehensive income/(loss) ("OCI") within stockholders' equity. Gains and losses relating to nonfunctional currency transactions, including non-U.S. operations where the functional currency is the U.S. dollar, are reported in the Consolidated Statements of Income.

Statements of cash flows

For JPMorgan Chase's Consolidated Statements of Cash Flows, cash is defined as those amounts included in cash and due from banks.

Significant accounting policies

The following table identifies JPMorgan Chase's other significant accounting policies and the Note and page where a detailed description of each policy can be found.

Business changes and developments	Note 2	Page 183
Fair value measurement	Note 3	Page 184
Fair value option	Note 4	Page 198
Derivative instruments	Note 6	Page 202
Noninterest revenue	Note 7	Page 211
Interest income and interest expense	Note 8	Page 212
Pension and other postretirement employee benefit plans	Note 9	Page 213
Employee stock-based incentives	Note 10	Page 222
Securities	Note 12	Page 225
Securities financing activities	Note 13	Page 231
Loans	Note 14	Page 231
Allowance for credit losses	Note 15	Page 252
Variable interest entities	Note 16	Page 256
Goodwill and other intangible assets	Note 17	Page 267
Premises and equipment	Note 18	Page 272
Long-term debt	Note 21	Page 273
Income taxes	Note 26	Page 279
Off-balance sheet lending-related financial instruments, guarantees and other commitments	Note 29	Page 283
Litigation	Note 31	Page 290

Note 2 – Business changes and developments

Changes in common stock dividend

On February 23, 2009, the Board of Directors reduced the Firm's quarterly common stock dividend from \$0.38 to \$0.05 per share, effective with the dividend paid on April 30, 2009, to shareholders of record on April 6, 2009. On March 18, 2011, the Board of Directors raised the Firm's quarterly common stock dividend from \$0.05 to \$0.25 per share, effective with the dividend paid on April 30, 2011, to shareholders of record on April 6, 2011.

Other business events

RBS Sempra transaction

On July 1, 2010, JPMorgan Chase completed the acquisition of RBS Sempra Commodities' global oil, global metals and European power and gas businesses. The Firm acquired approximately \$1.7 billion of net assets which included \$3.3 billion of debt which was immediately repaid. This acquisition almost doubled the number of clients the Firm's commodities business can serve and has enabled the Firm to offer clients more products in more regions of the world.

Purchase of remaining interest in J.P. Morgan Cazenove

On January 4, 2010, JPMorgan Chase purchased the remaining interest in J.P. Morgan Cazenove, an investment banking business partnership formed in 2005, which resulted in an adjustment to the Firm's capital surplus of approximately \$1.3 billion.

Purchase of remaining interest in Highbridge Capital Management

In July 2009, JPMorgan Chase completed its purchase of the remaining interest in Highbridge, which resulted in a \$228 million adjustment to capital surplus.

Subsequent events

Global settlement on servicing and origination of mortgages

On February 9, 2012, the Firm announced that it agreed to a settlement in principle (the "global settlement") with a number of federal and state government agencies, including the U.S. Department of Justice, the U.S. Department of Housing and Urban Development, the Consumer Financial Protection Bureau and the State Attorneys General, relating to the servicing and origination of mortgages. The global settlement, which is subject to the execution of a definitive agreement and court approval, calls for the Firm to, among other things: (i) make cash payments of approximately \$1.1 billion (a portion of which will be set aside for payments to borrowers); (ii) provide approximately \$500 million of refinancing relief to certain "underwater" borrowers whose loans are owned by the Firm; and (iii) provide approximately \$3.7 billion of additional relief for certain borrowers, including reductions of principal on first and second liens, payments to assist with short sales, deficiency balance waivers on past foreclosures and short sales, and forbearance assistance for unemployed homeowners. (If the Firm does not meet certain targets for provision of the refinancing or other borrower relief within certain prescribed time periods, the Firm will instead make cash payments.) In addition, under the global settlement the Firm will be required to adhere to certain enhanced mortgage servicing standards.

The global settlement releases the Firm from further claims related to servicing activities, including foreclosures and loss mitigation activities; certain origination activities; and certain bankruptcy-related activities. Not included in the global settlement are any claims arising out of securitization activities, including representations made to investors respecting mortgage-backed securities; criminal

claims; and repurchase demands from the GSEs, among other items.

Also on February 9, 2012, the Firm entered into agreements in principle with the Federal Reserve and the Office of the Comptroller of the Currency for the payment of civil money penalties related to conduct that was the subject of consent orders entered into with the banking regulators in April 2011. The Firm's payment obligations under those agreements will be deemed satisfied by the Firm's payments and provisions of relief under the global settlement.

While the Firm expects to incur additional operating costs to comply with portions of the global settlement, including the enhanced servicing standards, the Firm's prior period results of operations have reflected the estimated costs of the global settlement. Accordingly, the Firm expects that the financial impact of the global settlement on the Firm's financial condition and results of operations for the first quarter of 2012 and future periods will not be material. For further information on this settlement, see "Mortgage Foreclosure Investigations and Litigation" in Note 31 on pages 290–299 of this Annual Report.

Washington Mutual, Inc. bankruptcy plan confirmation

On February 17, 2012, a bankruptcy court confirmed the joint plan containing the global settlement agreement resolving numerous disputes among Washington Mutual, Inc. ("WMI"), JPMorgan Chase and the Federal Deposit Insurance Corporation ("FDIC") as well as significant creditor groups (the "WaMu Global Settlement"). Pursuant to this agreement, the Firm expects to recognize additional assets, including certain pension-related assets, as well as tax refunds, in future periods as the settlement is executed and various state and federal tax matters are resolved. For additional information related to the WaMu Global Settlement, see "Washington Mutual Litigations" in Note 31 on pages 290–299 of this Annual Report.

Note 3 – Fair value measurement

JPMorgan Chase carries a portion of its assets and liabilities at fair value. These assets and liabilities are predominantly carried at fair value on a recurring basis. Certain assets and liabilities are carried at fair value on a nonrecurring basis, including mortgage, home equity and other loans, where the carrying value is based on the fair value of the underlying collateral.

The Firm has an established and well-documented process for determining fair values. Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. Fair value is based on quoted market prices, where available. If listed prices or quotes are not available, fair value is based on internally developed models that consider relevant transaction data such as maturity and use as inputs, market-based or independently sourced market parameters, including but

not limited to yield curves, interest rates, volatilities, equity or debt prices, foreign exchange rates and credit curves. Valuation adjustments may be made to ensure that financial instruments are recorded at fair value. These adjustments include amounts to reflect counterparty credit quality, the Firm's creditworthiness, constraints on liquidity and unobservable parameters. Valuation adjustments are applied consistently over time.

- Credit valuation adjustments ("CVA") are necessary when the market price (or parameter) is not indicative of the credit quality of the counterparty. As few classes of derivative contracts are listed on an exchange, derivative positions are predominantly valued using internally developed models that use as their basis observable market parameters. An adjustment is necessary to reflect the credit quality of each derivative counterparty to arrive at fair value. The adjustment also takes into account contractual factors designed to reduce the Firm's credit exposure to each counterparty, such as collateral and legal rights of offset.
- Debit valuation adjustments ("DVA") are taken to reflect the credit quality of the Firm in the valuation of liabilities measured at fair value. The methodology to determine the adjustment is consistent with CVA and incorporates JPMorgan Chase's credit spread as observed through the credit default swap market.
- Liquidity valuation adjustments are necessary when the Firm may not be able to observe a recent market price for a financial instrument that trades in inactive (or less active) markets or to reflect the cost of exiting larger-than-normal market-size risk positions (liquidity adjustments are not taken for positions classified within level 1 of the fair value hierarchy; see below). The Firm estimates the amount of uncertainty in the initial valuation based on the degree of liquidity in the market in which the financial instrument trades and makes liquidity adjustments to the carrying value of the financial instrument. The Firm measures the liquidity adjustment based on the following factors: (1) the amount of time since the last relevant pricing point; (2) whether there was an actual trade or relevant external quote; and (3) the volatility of the principal risk component of the financial instrument. Costs to exit larger-than-normal market-size risk positions are determined based on the size of the adverse market move that is likely to occur during the period required to bring a position down to a nonconcentrated level.
- Unobservable parameter valuation adjustments are necessary when positions are valued using internally developed models that use as their basis unobservable parameters – that is, parameters that must be estimated and are, therefore, subject to management judgment. Unobservable parameter valuation adjustments are applied to mitigate the possibility of error and revision in the estimate of the market price provided by the model.

The Firm has numerous controls in place intended to ensure that its fair values are appropriate. An independent model review group reviews the Firm's valuation models and approves them for use for specific products. All valuation models within the Firm are subject to this review process. A price verification group, independent from the risk-taking function, ensures observable market prices and market-based parameters are used for valuation wherever possible. For those products with material parameter risk for which observable market levels do not exist, an independent review of the assumptions made on pricing is performed. Additional review includes deconstruction of the model valuations for certain structured instruments into their components and benchmarking valuations, where possible, to similar products; validating valuation estimates through actual cash settlement; and detailed review and explanation of recorded gains and losses, which are analyzed daily and over time. Valuation adjustments, which are also determined by the independent price verification group, are based on established policies and applied consistently over time. Any changes to the valuation methodology are reviewed by management to confirm that the changes are justified. As markets and products develop and the pricing for certain products becomes more or less transparent, the Firm continues to refine its valuation methodologies.

The methods described above to estimate fair value may produce a fair value calculation that may not be indicative of net realizable value or reflective of future fair values. Furthermore, while the Firm believes its valuation methods are appropriate and consistent with other market participants, the use of different methodologies or assumptions to determine the fair value of certain financial instruments could result in a different estimate of fair value at the reporting date.

Valuation Hierarchy

A three-level valuation hierarchy has been established under U.S. GAAP for disclosure of fair value measurements. The valuation hierarchy is based on the transparency of inputs to the valuation of an asset or liability as of the measurement date. The three levels are defined as follows.

- Level 1 – inputs to the valuation methodology are quoted prices (unadjusted) for identical assets or liabilities in active markets.
- Level 2 – inputs to the valuation methodology include quoted prices for similar assets and liabilities in active markets, and inputs that are observable for the asset or liability, either directly or indirectly, for substantially the full term of the financial instrument.
- Level 3 – one or more inputs to the valuation methodology are unobservable and significant to the fair value measurement.

A financial instrument's categorization within the valuation hierarchy is based on the lowest level of input that is significant to the fair value measurement.

Notes to consolidated financial statements

The following table is a description of the valuation methodologies used by the Firm to measure its more significant product instruments at fair value, including the general classification of such instruments pursuant to the valuation hierarchy.

Product/instrument	Valuation methodology, inputs and assumptions
Securities financing agreements	Valuations are based on discounted cash flows, which consider: • Derivative features • Market rates for respective maturity • Collateral
Loans and lending-related commitments - wholesale	
Trading portfolio	Where observable market data is available, valuations are based on: • Observed market prices (circumstances are limited) • Relevant broker quotes • Observed market prices for similar instruments Where observable market data is unavailable or limited, valuations are based on discounted cash flows, which consider the following: • Discount rate • Expected credit losses • Loss severity rates • Prepayment rates • Servicing costs
Loans held for investment and associated lending related commitments	Valuations are based on discounted cash flows, which consider: • Credit spreads, derived from the cost of credit default swaps ("CDS"); or benchmark credit curves developed by the Firm by industry and credit rating, and which take into account the
Loans - consumer	
Held for investment consumer loans, excluding credit card	
Credit card receivables	
Conforming residential mortgage loans expected to be sold	

The following table presents the asset and liabilities measured at fair value as of December 31, 2011 and 2010 by major product category and fair value hierarchy.

Assets and liabilities measured at fair value on a recurring basis

December 31, 2011 (in millions)	Fair value hierarchy			Netting adjustments	Total fair value
	Level 1 ⁽¹⁾	Level 2 ⁽²⁾	Level 3 ⁽³⁾		
Federal funds sold and securities purchased under resale agreements	\$ —	\$ 24,891	\$ —	\$ —	\$ 24,891
Securities borrowed	—	15,308	—	—	15,308
Trading assets:					
Debt instruments:					
Mortgage-backed securities:					
U.S. government agencies	27,082	7,801	86	—	34,969
Residential – nonagency	—	2,956	796	—	3,752
Commercial – nonagency	—	870	1,758	—	2,628
Total mortgage-backed securities	27,082	11,627	2,640	—	41,349
U.S. Treasury and government agencies	11,508	8,391	—	—	19,899
Obligations of U.S. states and municipalities	—	15,117	1,619	—	16,736
Certificates of deposit, bankers' acceptances and commercial paper	—	2,615	—	—	2,615
Non-U.S. government debt securities	18,618	40,080	104	—	58,802
Corporate debt securities	—	33,938	6,373	—	40,311
Loan ⁽⁴⁾	—	21,589	12,209	—	33,798
Asset-backed securities	—	2,406	7,965	—	10,371
Total debt instruments	57,208	135,763	30,910	—	223,881
Equity securities	93,799	3,502	1,177	—	98,478
Physical commodities	21,066	4,898	—	—	25,964
Other	—	2,283	880	—	3,163
Total debt and equity instrument ⁽⁵⁾	172,073	146,446	32,967	—	351,486
Derivative receivables:					
Interest rate	1,324	1,433,469	6,728	(1,395,152)	46,369
Credit	—	152,569	17,081	(162,966)	6,684
Foreign exchange	833	162,689	4,641	(150,273)	17,890
Equity	—	43,604	4,132	(40,943)	6,793
Commodity	4,561	50,409	2,459	(42,688)	14,741
Total derivative receivable ⁽⁶⁾	6,718	1,842,740	35,041	(1,792,022)	92,477
Total trading assets	178,791	1,989,186	68,008	(1,792,022)	443,963
Available-for-sale securities:					
Mortgage-backed securities:					
U.S. government agencies	92,426	14,681	—	—	107,107
Residential – nonagency	—	67,554	3	—	67,557
Commercial – nonagency	—	10,962	267	—	11,229
Total mortgage-backed securities	92,426	93,197	270	—	185,893
U.S. Treasury and government agencies	3,837	4,514	—	—	8,351
Obligations of U.S. states and municipalities	36	16,246	258	—	16,540
Certificates of deposit	—	3,017	—	—	3,017
Non-U.S. government debt securities	25,381	19,884	—	—	45,265
Corporate debt securities	—	62,176	—	—	62,176
Asset-backed securities:					
Credit card receivables	—	4,655	—	—	4,655
Collateralized loan obligations	—	116	24,745	—	24,861
Other	—	11,105	213	—	11,318
Equity securities	2,667	38	—	—	2,705
Total available-for-sale securities	124,347	214,948	25,486	—	364,781
Loans	—	450	1,647	—	2,097
Mortgage servicing rights	—	—	7,223	—	7,223
Other assets:					
Private equity investments	99	706	6,751	—	7,556
All other	4,336	233	4,374	—	8,943
Total other assets	4,435	939	11,125	—	16,499
Total assets measured at fair value on a recurring basis	\$ 307,573	\$ 2,245,722	\$ 113,489	\$ (1,792,022)	\$ 874,762
Deposits	\$ —	\$ 3,515	\$ 1,418	\$ —	\$ 4,933
Federal funds purchased and securities loaned or sold under repurchase agreements	—	9,517	—	—	9,517
Other borrowed funds	—	8,069	1,507	—	9,576
Trading liabilities:					
Debt and equity instrument ⁽⁵⁾	50,830	15,677	211	—	66,718
Derivative payables:					
Interest rate	1,537	1,395,113	3,167	(1,371,807)	28,010
Credit	—	155,772	9,349	(159,511)	5,610
Foreign exchange	846	159,258	5,904	(148,573)	17,435
Equity	—	39,129	7,237	(36,711)	9,655
Commodity	3,114	53,684	3,146	(45,677)	14,267
Total derivative payable ⁽⁶⁾	5,497	1,802,956	28,803	(1,762,279)	74,977
Total trading liabilities	56,327	1,818,633	29,014	(1,762,279)	141,695
Accounts payable and other liabilities	—	—	51	—	51
Beneficial interests issued by consolidated VIEs	—	459	791	—	1,250
Long-term debt	—	24,410	10,310	—	34,720
Total liabilities measured at fair value on a recurring basis	\$ 56,327	\$ 1,864,603	\$ 43,091	\$ (1,762,279)	\$ 201,742

Notes to consolidated financial statements

December 31, 2010 (in millions)	Fair value hierarchy			Netting adjustments	Total fair value
	Level 1 ^(a)	Level 2 ^(b)	Level 3 ^(c)		
Federal funds sold and securities purchased under resale agreements	\$ —	\$ 20,299	\$ —	\$ —	\$ 20,299
Securities borrowed	—	13,961	—	—	13,961
Trading assets:					
Debt instruments:					
Mortgage-backed securities:					
U.S. government agencies	36,813	10,738	174	—	47,725
Residential – nonagency	—	2,807	687	—	3,494
Commercial – nonagency	—	1,093	2,069	—	3,162
Total mortgage-backed securities	36,813	14,638	2,930	—	54,381
U.S. Treasury and government agencies	12,863	9,026	—	—	21,889
Obligations of U.S. states and municipalities	—	11,715	2,257	—	13,972
Certificates of deposit, bankers' acceptances and commercial paper	—	3,248	—	—	3,248
Non-U.S. government debt securities	31,127	38,482	202	—	69,811
Corporate debt securities	—	42,280	4,946	—	47,226
Loans ^(d)	—	21,736	13,144	—	34,880
Asset-backed securities	—	2,743	8,460	—	11,203
Total debt instruments	80,803	143,868	31,939	—	256,610
Equity securities	124,400	3,153	1,685	—	129,238
Physical commodities	18,327	2,708	—	—	21,035
Other	—	1,598	930	—	2,528
Total debt and equity instruments ^(e)	223,530	151,327	34,554	—	409,411
Derivative receivables:					
Interest rate	2,278	1,120,282	5,422	(1,095,427)	32,555
Credit	—	111,827	17,902	(122,004)	7,725
Foreign exchange	1,121	163,114	4,236	(142,613)	25,858
Equity	30	38,718	4,885	(39,429)	4,204
Commodity	1,324	56,076	2,197	(49,458)	10,139
Total derivative receivable ^(f)	4,753	1,490,017	34,642	(1,448,931)	80,481
Total trading assets	228,283	1,641,344	69,196	(1,448,931)	489,892
Available-for-sale securities:					
Mortgage-backed securities:					
U.S. government agencies	104,736	15,490	—	—	120,226
Residential – nonagency	1	48,969	5	—	48,975
Commercial – nonagency	—	5,403	251	—	5,654
Total mortgage-backed securities	104,737	69,862	256	—	174,855
U.S. Treasury and government agencies	522	10,826	—	—	11,348
Obligations of U.S. states and municipalities	31	11,272	256	—	11,559
Certificates of deposit	6	3,641	—	—	3,647
Non-U.S. government debt securities	13,107	7,670	—	—	20,777
Corporate debt securities	—	61,793	—	—	61,793
Asset-backed securities:					
Credit card receivables	—	7,608	—	—	7,608
Collateralized loan obligations	—	128	13,470	—	13,598
Other	—	8,777	305	—	9,082
Equity securities	1,998	53	—	—	2,051
Total available-for-sale securities	120,401	181,630	14,287	—	316,318
Loans	—	510	1,466	—	1,976
Mortgage servicing rights	—	—	13,649	—	13,649
Other assets:					
Private equity investments	49	826	7,862	—	8,737
All other	5,093	192	4,179	—	9,464
Total other assets	5,142	1,018	12,041	—	18,201
Total assets measured at fair value on a recurring basis	\$ 353,826	\$ 1,858,762	\$ 110,639	\$ (1,448,931)	\$ 874,296
Deposits	\$ —	\$ 3,596	\$ 773	\$ —	\$ 4,369
Federal funds purchased and securities loaned or sold under repurchase agreements	—	4,060	—	—	4,060
Other borrowed funds	—	8,547	1,384	—	9,931
Trading liabilities:					
Debt and equity instruments	58,468	18,425	54	—	76,947
Derivative payables:					
Interest rate	2,625	1,085,233	2,586	(1,070,057)	20,387
Credit	—	112,545	12,516	(119,923)	5,138
Foreign exchange	972	158,908	4,850	(139,715)	25,015
Equity	22	39,046	7,331	(35,949)	10,450
Commodity	862	54,611	3,002	(50,246)	8,229
Total derivative payable ^(f)	4,481	1,450,343	30,285	(1,415,890)	69,219
Total trading liabilities	62,949	1,468,768	30,339	(1,415,890)	146,166
Accounts payable and other liabilities	—	—	236	—	236
Beneficial interests issued by consolidated VIEs	—	622	873	—	1,495
Long-term debt	—	25,795	13,044	—	38,839
Total liabilities measured at fair value on a recurring basis	\$ 62,949	\$ 1,511,388	\$ 46,649	\$ (1,415,890)	\$ 205,096

- (a) At December 31, 2011 and 2010, included total U.S. government-sponsored enterprise obligations of \$122.4 billion and \$137.3 billion respectively, which were predominantly mortgage-related.
- (b) At December 31, 2011 and 2010, included within trading loans were \$20.1 billion and \$22.7 billion, respectively, of residential first-lien mortgages, and \$2.0 billion and \$2.6 billion, respectively, of commercial first-lien mortgages. Residential mortgage loans include conforming mortgage loans originated with the intent to sell to U.S. government agencies of \$11.0 billion and \$13.1 billion, respectively, and reverse mortgages of \$4.0 billion and \$4.0 billion, respectively.
- (c) Physical commodities inventories are generally accounted for at the lower of cost or fair value.

- (d) Balances reflect the reduction of securities owned (long positions) by the amount of securities sold but not yet purchased (short positions) when the long and short positions have identical Committee on Uniform Security Identification Procedures numbers ("CUSIPs").
- (e) As permitted under U.S. GAAP, the Firm has elected to net derivative receivables and derivative payables and the related cash collateral received and paid when a legally enforceable master netting agreement exists. For purposes of the tables above, the Firm does not reduce derivative receivables and derivative payables balances for this netting adjustment, either within or across the levels of the fair value hierarchy, as such netting is not relevant to a presentation based on the transparency of inputs to the valuation of an asset or liability. Therefore, the balances reported in the fair value hierarchy table are gross of any counterparty netting adjustments. However, if the Firm were to net such balances within level 3, the reduction in the level 3 derivative receivable and payable balances would be \$11.7 billion and \$12.7 billion at December 31, 2011 and 2010, respectively; this is exclusive of the netting benefit associated with cash collateral, which would further reduce the level 3 balances.
- (f) Private equity instruments represent investments within the Corporate/Private Equity line of business. The cost basis of the private equity investment portfolio totaled \$9.5 billion and \$10.0 billion at December 31, 2011 and 2010, respectively.
- (g) At December 31, 2011 and 2010, balances included investments valued at net asset values of \$10.8 billion and \$12.1 billion, respectively, of which \$5.3 billion and \$5.9 billion, respectively, were classified in level 1, \$1.2 billion and \$2.0 billion, respectively, in level 2, and \$4.3 billion and \$4.2 billion, respectively, in level 3.
- (h) For the years ended December 31, 2011 and 2010, there were no significant transfers between levels 1 and 2. For the year ended December 31, 2011, transfers from level 1 into level 2 included \$2.6 billion of long-term debt due to a decrease in valuation uncertainty of certain structured notes. For the year ended December 31, 2010, transfers from level 3 into level 2 included \$1.2 billion of trading loans due to increased price transparency. There were no significant transfers into level 3 for the years ended December 31, 2011 and 2010. All transfers are assumed to occur at the beginning of the reporting period.

Changes in level 3 recurring fair value measurements

The following tables include a rollforward of the Consolidated Balance Sheet amounts (including changes in fair value) for financial instruments classified by the Firm within level 3 of the fair value hierarchy for the years ended December 31, 2011, 2010 and 2009. When a determination is made to classify a financial instrument within level 3, the determination is based on the significance of the unobservable parameters to the overall fair value measurement. However, level 3 financial instruments typically include, in addition to the unobservable or level 3 components, observable

components (that is, components that are actively quoted and can be validated to external sources); accordingly, the gains and losses in the table below include changes in fair value due in part to observable factors that are part of the valuation methodology. Also, the Firm risk-manages the observable components of level 3 financial instruments using securities and derivative positions that are classified within level 1 or 2 of the fair value hierarchy; as these level 1 and level 2 risk management instruments are not included below, the gains or losses in the following tables do not reflect the effect of the Firm's risk management activities related to such level 3 instruments.

Notes to consolidated financial statements

Fair value measurements using significant unobservable inputs

Year ended December 31, 2011 (in millions)	Fair value at January 1, 2011	Total realized/ unrealized gains/(losses)	Purchases [§]	Sales	Issuances	Settlements	Transfers into and/ or out of level 3 ^{g)}	Fair value at Dec. 31, 2011	Change in unrealized gains/ (losses) related to financial instruments held at Dec. 31, 2011
Assets:									
Trading assets:									
Debt instruments:									
Mortgage-backed securities:									
U.S. government agencies	\$ 174	\$ 24	\$ 28	\$ (39)	\$ —	\$ (43)	\$ (58)	\$ 86	\$ (51)
Residential – nonagency	687	109	708	(432)	—	(221)	(55)	796	(9)
Commercial – nonagency	2,069	37	796	(973)	—	(171)	—	1,758	33
Total mortgage-backed securities	2,930	170	1,532	(1,444)	—	(435)	(113)	2,640	(27)
Obligations of U.S. states and municipalities	2,257	9	807	(1,465)	—	(1)	12	1,619	(11)
Non-U.S. government debt securities	202	35	552	(531)	—	(80)	(74)	104	38
Corporate debt securities	4,946	32	8,080	(5,939)	—	(1,005)	259	6,373	26
Loans	13,144	329	5,532	(3,873)	—	(2,691)	(232)	12,209	142
Asset-backed securities	8,460	90	4,185	(4,368)	—	(424)	22	7,965	(217)
Total debt instruments	31,939	665	20,688	(17,620)	—	(4,636)	(126)	30,910	(49)
Equity securities	1,685	267	180	(541)	—	(352)	(62)	1,177	278
Other	930	48	36	(39)	—	(95)	—	880	79
Total trading assets – debt and equity instruments	34,554	980 ^(b)	20,904	(18,200)	—	(5,083)	(188)	32,967	308 ^(b)
Net derivative receivables:									
Interest rate	2,836	5,205	511	(219)	—	(4,534)	(238)	3,561	1,497
Credit	5,386	2,240	22	(13)	—	116	(19)	7,732	2,744
Foreign exchange	(614)	(1,913)	191	(20)	—	886	207	(1,263)	(1,878)
Equity	(2,446)	(60)	715	(1,449)	—	37	98	(3,105)	(132)
Commodity	(805)	596	328	(350)	—	(294)	(162)	(687)	208
Total net derivative receivables	4,357	6,068 ^(b)	1,767	(2,051)	—	(3,789)	(114)	6,238	2,439 ^(b)
Available-for-sale securities:									
Asset-backed securities	13,775	(95)	15,268	(1,461)	—	(2,529)	—	24,958	(106)
Other	512	—	57	(15)	—	(26)	—	528	8
Total available-for-sale securities	14,287	(95) ^(c)	15,325	(1,476)	—	(2,555)	—	25,486	(98) ^(c)
Loans	1,466	504 ^(b)	326	(9)	—	(639)	(1)	1,647	484 ^(b)
Mortgage servicing rights	13,649	(7,119) ^(d)	2,603	—	—	(1,910)	—	7,223	(7,119) ^(d)
Other assets:									
Private equity investments	7,862	943 ^(b)	1,452	(2,746)	—	(594)	(166)	6,751	(242) ^(b)
All other	4,179	(54) ^(e)	938	(139)	—	(521)	(29)	4,374	(83) ^(e)

Fair value measurements using significant unobservable inputs

Year ended December 31, 2011 (in millions)	Fair value at January 1, 2011	Total realized/ unrealized (gains)/losses	Purchases [§]	Sales	Issuances	Settlements	Transfers into and/ or out of level 3 ^{g)}	Fair value at Dec. 31, 2011	Change in unrealized (gains)/losses related to financial instruments held at Dec. 31, 2011
Liabilities:^(a)									
Deposits	\$ 773	\$ 15 ^(b)	\$ —	\$ —	\$ 433	\$ (386)	\$ 583	\$ 1,418	\$ 4 ^(b)
Other borrowed funds	1,384	(244) ^(b)	—	—	1,597	(834)	(396)	1,507	(85) ^(b)
Trading liabilities – debt and equity instruments	54	17 ^(b)	(533)	778	—	(109)	4	211	(7) ^(b)
Accounts payable and other liabilities	236	(61) ^(e)	—	—	—	(124)	—	51	5 ^(e)
Beneficial interests issued by consolidated VIEs	873	17 ^(b)	—	—	580	(679)	—	791	(15) ^(b)
Long-term debt	13,044	60 ^(b)	—	—	2,564	(3,218)	(2,140)	10,310	288 ^(b)

Fair value measurements using significant unobservable inputs

Year ended December 31, 2010 (in millions)	Fair value at January 1, 2010	Total realized/ unrealized gains/ (losses)	Purchases, issuances, settlements, net	Transfers into and/or out of level 3 ^(a)	Fair value at Dec. 31, 2010	Change in unrealized gains/ (losses) related to financial instruments held at Dec. 31, 2010
Assets:						
Trading assets:						
Debt instruments:						
Mortgage-backed securities:						
U.S. government agencies	\$ 260	\$ 24	\$ (107)	\$ (3)	\$ 174	\$ (31)
Residential – nonagency	1,115	178	(564)	(42)	687	110
Commercial – nonagency	1,770	230	(33)	102	2,069	130
Total mortgage-backed securities	3,145	432	(704)	57	2,930	209
Obligations of U.S. states and municipalities	1,971	2	142	142	2,257	(30)
Non-U.S. government debt securities	89	(36)	194	(45)	202	(8)
Corporate debt securities	5,241	(325)	115	(85)	4,946	28
Loans	13,218	(40)	1,296	(1,330)	13,144	(385)
Asset-backed securities	8,620	237	(408)	11	8,460	195
Total debt instruments	32,284	270	635	(1,250)	31,939	9
Equity securities	1,956	133	(351)	(53)	1,685	199
Other	1,441	211	(801)	79	930	299
Total trading assets – debt and equity instruments	35,681	614 ^(b)	(517)	(1,224)	34,554	507 ^(b)
Net derivative receivables:						
Interest rate	2,040	3,057	(2,520)	259	2,836	487
Credit	10,350	(1,757)	(3,102)	(105)	5,386	(1,048)
Foreign exchange	1,082	(913)	(434)	(349)	(614)	(464)
Equity	(2,306)	(194)	(82)	136	(2,446)	(212)
Commodity	(329)	(700)	134	90	(805)	(76)
Total net derivative receivables	10,837	(507) ^(b)	(6,004)	31	4,357	(1,313) ^(b)
Available-for-sale securities:						
Asset-backed securities	12,732	(146)	1,189	—	13,775	(129)
Other	461	(49)	37	63	512	18
Total available-for-sale securities	13,193	(195) ^(c)	1,226	63	14,287	(111) ^(c)
Loans	990	145 ^(b)	323	8	1,466	37 ^(b)
Mortgage servicing rights	15,531	(2,268) ^(d)	386	—	13,649	(2,268) ^(d)
Other assets:						
Private equity investments	6,563	1,038 ^(b)	715	(454)	7,862	688 ^(b)
All other	9,521	(113) ^(e)	(5,132)	(97)	4,179	37 ^(e)

Fair value measurements using significant unobservable inputs

Year ended December 31, 2010 (in millions)	Fair value at January 1, 2010	Total realized/ unrealized (gains)/losses	Purchases, issuances, settlements, net	Transfers into and/or out of level 3 ^(a)	Fair value at Dec. 31, 2010	Change in unrealized (gains)/losses related to financial instruments held at Dec. 31, 2010
Liabilities:^(a)						
Deposits	\$ 476	\$ 54 ^(b)	\$ (86)	\$ 329	\$ 773	\$ (77) ^(b)
Other borrowed funds	542	(242) ^(b)	1,326	(242)	1,384	445 ^(b)
Trading liabilities – debt and equity instruments	10	2 ^(b)	19	23	54	—
Accounts payable and other liabilities	355	(138) ^(e)	19	—	236	37 ^(e)
Beneficial interests issued by consolidated VIEs	625	(7) ^(b)	87	168	873	(76) ^(b)
Long-term debt	18,287	(532) ^(b)	(4,796)	85	13,044	662 ^(b)

Notes to consolidated financial statements

	Fair value measurements using significant unobservable inputs					Change in unrealized gains/ (losses) related to financial instruments held at Dec. 31, 2009
Year ended December 31, 2009 (in millions)	Fair Value at January 1, 2009	Total realized/ unrealized gains/ (losses)	Purchases, issuances settlements, net	Transfers into and/or out of level 3 ⁽ⁱ⁾	Fair value at Dec. 31, 2009	
Assets:						
Trading assets:						
Debt instruments:						
Mortgage-backed securities:						
U.S. government agencies	\$ 163	\$ (38)	\$ 62	\$ 73	\$ 260	\$ (38)
Residential – nonagency	3,339	(782)	(245)	(1,197)	1,115	(871)
Commercial – nonagency	2,487	(242)	(325)	(150)	1,770	(313)
Total mortgage-backed securities	5,989	(1,062)	(508)	(1,274)	3,145	(1,222)
Obligations of U.S. states and municipalities	2,641	(22)	(648)	—	1,971	(123)
Non-U.S. government debt securities	11	36	(22)	64	89	32
Corporate debt securities	5,280	38	(3,416)	3,339	5,241	(72)
Loans	17,091	(871)	(3,497)	495	13,218	(1,167)
Asset-backed securities	7,802	1,438	(431)	(189)	8,620	736
Total debt instruments	38,814	(443)	(8,522)	2,435	32,284	(1,816)
Equity securities	1,380	(149)	(512)	1,237	1,956	(51)
Other	1,694	(12)	(273)	32	1,441	(52)
Total trading assets – debt and equity instruments	41,888	(604) ^(b)	(9,307)	3,704	35,681	(1,919) ^(b)
Total net derivative receivables	9,039	(11,473) ^(b)	(3,428)	16,699	10,837	(10,902) ^(b)
Available-for-sale securities:						
Asset-backed securities	11,447	(2)	1,112	175	12,732	(48)
Other	944	(269)	302	(516)	461	43
Total available-for-sale securities	12,391	(271) ^(c)	1,414	(341)	13,193	(5) ^(c)
Loans	2,667	(448) ^(b)	(1,906)	677	990	(488) ^(b)
Mortgage servicing rights	9,403	5,807 ^(d)	321	—	15,531	5,807 ^(d)
Other assets:						
Private equity investments	6,369	(407) ^(b)	582	19	6,563	(369) ^(b)
All other	8,114	(676) ^(e)	2,439	(356)	9,521	(612) ^(e)

	Fair value measurements using significant unobservable inputs					Change in unrealized (gains)/losses related to financial instruments held at Dec. 31, 2009
Year ended December 31, 2009 (in millions)	Fair value at January 1, 2009	Total realized/unrealized (gains)/losses	Purchases, issuances settlements, net	Transfers into and/or out of level 3 ^(e)	Fair value at Dec.31, 2009	
Liabilities: ^(a)						
Deposits	\$ 1,235	\$ 47 ^(b)	\$ (870)	\$ 64	\$ 476	\$ (36) ^(b)
Other borrowed funds	101	(73) ^(b)	621	(107)	542	9 ^(b)
Trading liabilities:						
Debt and equity instruments	288	64 ^(b)	(339)	(3)	10	12 ^(b)
Accounts payable and other liabilities	—	(55) ^(b)	410	—	355	(29) ^(b)
Beneficial interests issued by consolidated VIEs	—	344 ^(b)	(598)	879	625	327 ^(b)
Long-term debt	16,548	1,367 ^(b)	(2,738)	3,110	18,287	1,728 ^(b)

- (a) Level 3 liabilities as a percentage of total Firm liabilities accounted for at fair value (including liabilities measured at fair value on a nonrecurring basis) were 21%, 23% and 29% at December 31, 2011, 2010 and 2009, respectively.
- (b) Predominantly reported in principal transactions revenue, except for changes in fair value for Retail Financial Services ("RFS") mortgage loans and lending-related commitments originated with the intent to sell, which are reported in mortgage fees and related income.
- (c) Realized gains/(losses) on available-for-sale ("AFS") securities, as well as other-than-temporary impairment losses that are recorded in earnings, are reported in securities gains. Unrealized gains/(losses) are reported in OCI. Realized gains/(losses) and foreign exchange remeasurement adjustments recorded in income on AFS securities were \$(240) million, \$(66) million, and \$(345) million for the years ended December 31, 2011, 2010 and 2009, respectively. Unrealized gains/(losses) recorded on AFS securities in OCI were \$145 million, \$(129) million and \$74 million for the years ended December 31, 2011, 2010 and 2009, respectively.
- (d) Changes in fair value for RFS mortgage servicing rights are reported in mortgage fees and related income.
- (e) Largely reported in other income.
- (f) Loan originations are included in purchases.
- (g) All transfers into and/or out of level 3 are assumed to occur at the beginning of the reporting period.

Assets and liabilities measured at fair value on a nonrecurring basis

Certain assets, liabilities and unfunded lending-related commitments are measured at fair value on a nonrecurring basis; that is, they are not measured at fair value on an ongoing basis but are subject to fair value adjustments only in certain circumstances (for example, when there is evidence of impairment). At December 31, 2011 and 2010, assets measured at fair value on a nonrecurring basis were \$5.3 billion and \$9.9 billion, respectively, comprised predominantly of loans. At December 31, 2011, \$369 million and \$4.9 billion of these assets were classified in levels 2 and 3 of the fair value hierarchy, respectively. At December 31, 2010, \$312 million and \$9.6 billion of these assets were classified in levels 2 and 3 of the fair value hierarchy, respectively. Liabilities measured at fair value on a nonrecurring basis were not significant at December 31, 2011 and 2010. For the years ended December 31, 2011 and 2010, there were no significant transfers between levels 1, 2, and 3. The total change in the value of assets and liabilities for which a fair value adjustment has been included in the Consolidated Statements of Income for the years ended December 31, 2011, 2010 and 2009, related to financial instruments held at those dates were losses of \$2.2 billion, \$3.6 billion and \$4.7 billion, respectively; these losses were predominantly associated with loans.

For further information about the measurement of impaired collateral-dependent loans, and other loans where the carrying value is based on the fair value of the underlying collateral (e.g., residential mortgage loans charged off in accordance with regulatory guidance), see Note 14 on pages 231–252 of this Annual Report.

Level 3 analysis

Level 3 assets at December 31, 2011, predominantly included derivative receivables, MSRs, CLOs held within the available-for-sale and trading portfolios, loans within the trading portfolio and private equity investments.

- Derivative receivables included \$35.0 billion related to interest rate, credit, foreign exchange, equity and commodity contracts. Credit derivative receivables of \$17.1 billion included \$12.1 billion of structured credit derivatives with corporate debt underlying and \$3.4 billion of CDS largely on commercial mortgages where the risks are partially mitigated by similar and offsetting derivative payables. Interest rate derivative receivables of \$6.7 billion include long-dated structured interest rate derivatives which are dependent on the correlation between different interest rate curves. Foreign exchange derivative receivables of \$4.6 billion included long-dated foreign exchange derivatives which are dependent on the correlation between foreign exchange and interest rates. Equity derivative receivables of \$4.1 billion principally included long-dated contracts where the volatility levels are unobservable. Commodity derivative receivables of \$2.5 billion largely included long-dated oil contracts
- CLOs totaling \$30.9 billion are securities backed by

corporate loans. At December 31, 2011, \$24.7 billion of CLOs were held in the AFS securities portfolio and \$6.2 billion were included in asset-backed securities held in the trading portfolio. Substantially all of the securities are rated “AAA,” “AA” and “A” and had an average credit enhancement of 30%. Credit enhancement in CLOs is primarily in the form of subordination, which is a form of structural credit enhancement where realized losses associated with assets held by the issuing vehicle are allocated to the various tranches of securities issued by the vehicle considering their relative seniority. For a further discussion of CLOs held in the AFS securities portfolio, see Note 12 on pages 225–230 of this Annual Report.

- Trading loans totaling \$12.2 billion included \$6.0 billion of residential mortgage whole loans and commercial mortgage loans for which there is limited price transparency; and \$4.0 billion of reverse mortgages for which the principal risk sensitivities are mortality risk and home prices. The fair value of the commercial and residential mortgage loans is estimated by projecting expected cash flows, considering relevant borrower-specific and market factors, and discounting those cash flows at a rate reflecting current market liquidity. Loans are partially hedged by level 2 instruments, including credit default swaps and interest rate derivatives, for which valuation inputs are observable and liquid.
- MSRs represent the fair value of future cash flows for performing specified mortgage servicing activities for others (predominantly with respect to residential mortgage loans). For a further discussion of the MSR asset, the interest rate risk management and valuation methodology used for MSRs, including valuation assumptions and sensitivities, and a summary of the changes in the MSR asset, see Note 17 on pages 267–271 of this Annual Report.

Consolidated Balance Sheets changes

Level 3 assets (including assets measured at fair value on a nonrecurring basis) were 5.2% of total Firm assets at December 31, 2011. The following describes significant changes to level 3 assets since December 31, 2010.

For the year ended December 31, 2011

Level 3 assets decreased by \$1.8 billion during 2011, due to the following:

- \$11.2 billion increase in asset-backed AFS securities, predominantly driven by purchases of CLOs;
- \$6.4 billion decrease in MSRs. For further discussion of the change, refer to Note 17 on pages 267–271 of this Annual Report;
- \$2.3 billion decrease in nonrecurring loans held-for-sale, predominantly driven by sales in the loan portfolios;
- \$2.2 billion decrease in nonrecurring retained loans predominantly due to portfolio runoff;
- \$1.6 billion decrease in trading assets – debt and equity instruments, largely driven by sales and settlements of certain securities, partially offset by purchases of corporate debt; and

Notes to consolidated financial statements

- \$1.1 billion decrease in private equity investments, predominantly driven by sales of investments, partially offset by new investments.

Gains and Losses

Gains and losses included in the tables for 2011, 2010 and 2009 included:

2011

Included in the tables for the year ended December 31, 2011

- \$7.1 billion of losses on MSRs. For further discussion of the change, refer to Note 17 on pages 267–271 of this Annual Report; and
- \$6.1 billion of net gains on derivatives, related to declining interest rates and tightening of credit spreads, partially offset by losses due to fluctuation in foreign exchange rates.

2010

Included in the tables for the year ended December 31, 2010

- \$2.3 billion of losses on MSRs; and
- \$1.0 billion gain in private equity largely driven by gains on investments in the portfolio.

2009

Included in the tables for the year ended December 31, 2009

- \$11.5 billion of net losses on derivatives, primarily related to the tightening of credit spreads;
- Net losses on trading – debt and equity instruments of \$604 million, consisting of \$2.1 billion of losses, primarily related to residential and commercial loans and mortgage-backed securities (“MBS”), principally driven by markdowns and sales, partially offset by gains of \$1.4 billion, reflecting increases in the fair value of other asset-backed securities (“ABS”);
- \$5.8 billion of gains on MSRs; and
- \$1.4 billion of losses related to structured note liabilities, predominantly due to volatility in the equity markets.

Credit adjustments

When determining the fair value of an instrument, it may be necessary to record a valuation adjustment to arrive at an exit price under U.S. GAAP. Valuation adjustments include, but are not limited to, amounts to reflect counterparty credit quality and the Firm’s own creditworthiness. The market’s view of the Firm’s credit quality is reflected in credit spreads observed in the credit default swap market. For a detailed discussion of the valuation adjustments the Firm considers, see the valuation discussion at the beginning of this Note.

The following table provides the credit adjustments, excluding the effect of any hedging activity, reflected within the Consolidated Balance Sheets as of the dates indicated.

December 31, (in millions)	2011	2010
Derivative receivables balance (net of derivatives CVA)	\$ 92,477	\$ 80,481
Derivatives CVA	(6,936)	(4,362)
Derivative payables balance (net of derivatives DVA)	74,977	69,219
Derivatives DVA	(1,420)	(882)
Structured notes balance (net of structured notes DVA) ^(c)	49,229	53,139
Structured notes DVA	(2,052)	(1,153)

- (a) Derivatives CVA, gross of hedges, includes results managed by the Credit Portfolio and other lines of business within the Investment Bank (“IB”).
- (b) Structured notes are recorded within long-term debt, other borrowed funds or deposits on the Consolidated Balance Sheets, depending upon the tenor and legal form of the note.
- (c) Structured notes are measured at fair value based on the Firm’s election under the fair value option. For further information on these elections, see Note 4 on pages 198–200 of this Annual Report.

The following table provides the impact of credit adjustments on earnings in the respective periods, excluding the effect of any hedging activity.

Year ended December 31, (in millions)	2011	2010	2009
Credit adjustments:			
Derivative CVA	\$ (2,574)	\$ (665)	\$ 5,869
Derivative DVA	538	41	(548)
Structured note DVA	899	468	(1,748)

- (a) Derivatives CVA, gross of hedges, includes results managed by the Credit Portfolio and other lines of business within IB.
- (b) Structured notes are measured at fair value based on the Firm’s election under the fair value option. For further information on these elections, see Note 4 on pages 198–200 of this Annual Report.

Additional disclosures about the fair value of financial instruments (including financial instruments not carried at fair value)

U.S. GAAP requires disclosure of the estimated fair value of certain financial instruments, and the methods and significant assumptions used to estimate their fair value. Financial instruments within the scope of these disclosure requirements are included in the following table. However, certain financial instruments and all nonfinancial instruments are excluded from the scope of these disclosure requirements. Accordingly, the fair value disclosures provided in the following table include only a partial estimate of the fair value of JPMorgan Chase’s assets and liabilities. For example, the Firm has developed long-term relationships with its customers through its deposit base and credit card accounts, commonly referred to as core deposit intangibles and credit card relationships. In the opinion of management, these items, in the aggregate, add significant value to JPMorgan Chase, but their fair value is not disclosed in this Note.

Financial instruments for which carrying value approximates fair value

Certain financial instruments that are not carried at fair value on the Consolidated Balance Sheets are carried at amounts that approximate fair value, due to their short-term nature and generally negligible credit risk. These instruments include cash and due from banks; deposits with banks; federal funds sold; securities purchased under resale agreements and securities borrowed with short-dated maturities; short-term receivables and accrued interest receivable; commercial paper; federal funds purchased; securities loaned and sold under repurchase agreements with short-dated maturities; other borrowed funds (excluding advances from the Federal Home Loan Banks ("FHLBs")); accounts payable; and accrued liabilities. In addition, U.S. GAAP requires that the fair value for deposit liabilities with no stated maturity (i.e., demand, savings and certain money market deposits) be equal to their carrying value; recognition of the inherent funding value of these instruments is not permitted.

The following table presents the carrying values and estimated fair values of financial assets and liabilities.

December 31, (in billions)	2011		2010	
	Carrying value	Estimated fair value	Carrying value	Estimated fair value
Financial assets				
Assets for which fair value approximates carrying value	\$ 144.9	\$ 144.9	\$ 49.2	\$ 49.2
Accrued interest and accounts receivable	61.5	61.5	70.1	70.1
Federal funds sold and securities purchased under resale agreements (included at fair value)	235.3	235.3	222.6	222.6
Securities borrowed (included at fair value)	142.5	142.5	123.6	123.6
Trading assets	444.0	444.0	489.9	489.9
Securities (included at fair value)	364.8	364.8	316.3	316.3
Loans (included at fair value)	696.1	695.8	660.7	663.5
Mortgage servicing rights at fair value	7.2	7.2	13.6	13.6
Other (included at fair value)	66.3	66.8	64.9	65.0
Financial liabilities				
Deposits (included at fair value)	\$ 1,127.8	\$ 1,128.3	\$ 930.4	\$ 931.5
Federal funds purchased and securities loaned or sold under repurchase agreements (included at fair value)	213.5	213.5	276.6	276.6
Commercial paper	51.6	51.6	35.4	35.4
Other borrowed funds (included at fair value)	21.9	21.9	34.3	34.3
Trading liabilities	141.7	141.7	146.2	146.2
Accounts payable and other liabilities (included at fair value)	167.0	166.9	138.2	138.2
Beneficial interests issued by consolidated VIEs (included at fair value)	66.0	66.2	77.6	77.9
Long-term debt and junior subordinated deferrable interest debentures (included at fair value)	256.8	254.2	270.7	271.9

- (a) Fair value is typically estimated using a discounted cash flow model that incorporates the characteristics of the underlying loans (including principal, contractual interest rate and contractual fees) and other key inputs, including expected lifetime credit losses, interest rates, prepayment rates, and primary origination or secondary market spreads. For certain loans, the fair value is measured based on the value of the underlying collateral. The difference between the estimated fair value and carrying value of a financial asset or liability is the result of the different methodologies used to determine fair value as compared with carrying value. For example, credit losses are estimated for a financial asset's remaining life in a fair value calculation but are estimated for a loss emergence period in a loan loss reserve calculation; future loan income (interest and fees) is incorporated in a fair value calculation but is generally not considered in a loan loss reserve calculation. For a further discussion of the Firm's methodologies for estimating the fair value of loans and lending-related commitments, see pages 186–188 of this Note.
- (b) Effective January 1, 2011, \$23.0 billion of long-term advances from FHLBs were reclassified from other borrowed funds to long-term debt. The prior-year period has been revised to conform with the current presentation.

Notes to consolidated financial statements

The majority of the Firm's lending-related commitments are not carried at fair value on a recurring basis on the Consolidated Balance Sheets, nor are they actively traded. The carrying value and estimated fair value of the Firm's wholesale lending-related commitments were as follows for the periods indicated.

December 31, (in billions)	2011		2010	
	Carrying value ^(a)	Estimated fair value	Carrying value ^(a)	Estimated fair value
Wholesale lending-related commitments	\$ 0.7	\$ 3.4	\$ 0.7	\$ 0.9

(a) Represents the allowance for wholesale lending-related commitments. Excludes the current carrying values of the guarantee liability and the offsetting asset, each of which are recognized at fair value at the inception of guarantees.

The Firm does not estimate the fair value of consumer lending-related commitments. In many cases, the Firm can reduce or cancel these commitments by providing the borrower notice or, in some cases, without notice as permitted by law. For a further discussion of lending-related commitments, see Note 29 on pages 283–289 of this Annual Report; for further information on the valuation of lending-related commitments, see pages 186–188 of this Note.

Trading assets and liabilities

Trading assets include debt and equity instruments owned by JPMorgan Chase ("long" positions) that are held for client market-making and client-driven activities, as well as for certain risk management activities, certain loans managed on a fair value basis and for which the Firm has elected the fair value option, and physical commodities inventories that are generally accounted for at the lower of cost or fair value. Trading liabilities include debt and equity instruments that the Firm has sold to other parties but does not own ("short" positions). The Firm is obligated to

purchase instruments at a future date to cover the short positions. Included in trading assets and trading liabilities are the reported receivables (unrealized gains) and payables (unrealized losses) related to derivatives. Trading assets and liabilities are carried at fair value on the Consolidated Balance Sheets. Balances reflect the reduction of securities owned (long positions) by the amount of securities sold but not yet purchased (short positions) when the long and short positions have identical Committee on Uniform Security Identification Procedures numbers ("CUSIPs").

Trading assets and liabilities – average balances

Average trading assets and liabilities were as follows for the periods indicated.

Year ended December 31, (in millions)	2011	2010	2009
Trading assets – debt and equity instruments	\$ 393,890	\$ 354,441	\$ 318,063
Trading assets – derivative receivables	90,003	84,676	110,457
Trading liabilities – debt and equity instruments	81,916	78,159	60,224
Trading liabilities – derivative payables	71,539	65,714	77,901

(a) Balances reflect the reduction of securities owned (long positions) by the amount of securities sold, but not yet purchased (short positions) when the long and short positions have identical CUSIP numbers.

(b) Primarily represent securities sold, not yet purchased.

Note 4 – Fair value option

The fair value option provides an option to elect fair value as an alternative measurement for selected financial assets, financial liabilities, unrecognized firm commitments, and written loan commitments not previously carried at fair value.

Elections

Elections were made by the Firm to:

- Mitigate income statement volatility caused by the differences in the measurement basis of elected instruments (for example, certain instruments elected were previously accounted for on an accrual basis) while the associated risk management arrangements are accounted for on a fair value basis;
- Eliminate the complexities of applying certain accounting models (e.g., hedge accounting or bifurcation accounting for hybrid instruments); and/or
- Better reflect those instruments that are managed on a fair value basis.

Elections include the following:

- Loans purchased or originated as part of securitization warehousing activity, subject to bifurcation accounting, or managed on a fair value basis.
- Securities financing arrangements with an embedded derivative and/or a maturity of greater than one year.
- Owned beneficial interests in securitized financial assets that contain embedded credit derivatives, which would otherwise be required to be separately accounted for as a derivative instrument.
- Certain investments that receive tax credits and other equity investments acquired as part of the Washington Mutual transaction.
- Structured notes issued as part of IB's client-driven activities. (Structured notes are financial instruments that contain embedded derivatives.)
- Long-term beneficial interests issued by IB's consolidated securitization trusts where the underlying assets are carried at fair value.

Changes in fair value under the fair value option election

The following table presents the changes in fair value included in the Consolidated Statements of Income for the years ended December 31, 2011, 2010 and 2009, for items for which the fair value option was elected. The profit and loss information presented below only includes the financial instruments that were elected to be measured at fair value; related risk management instruments, which are required to be measured at fair value, are not included in the table.

December 31, (in millions)	2011			2010			2009		
	Principal transactions	Other income	Total changes in fair value recorded	Principal transactions	Other income	Total changes in fair value recorded	Principal transactions	Other income	Total changes in fair value recorded
Federal funds sold and securities purchased under resale agreements	\$ 270	\$ —	\$ 270	\$ 173	\$ —	\$ 173	\$ (553)	\$ —	\$ (553)
Securities borrowed	(61)	—	(61)	31	—	31	82	—	82
Trading assets:									
Debt and equity instruments, excluding loans	53	(6) ^(c)	47	556	(2) ^(c)	554	619	25 ^(c)	644
Loans reported as trading assets:									
Changes in instrument-specific credit risk	934	(174) ^(c)	760	1,279	(6) ^(c)	1,273	(300)	(177) ^(c)	(477)
Other changes in fair value	127	5,263 ^(c)	5,390	(312)	4,449 ^(c)	4,137	1,132	3,119 ^(c)	4,251
Loans:									
Changes in instrument-specific credit risk	2	—	2	95	—	95	(78)	—	(78)
Other changes in fair value	535	—	535	90	—	90	(343)	—	(343)
Other assets	(49)	(19) ^(d)	(68)	—	(263) ^(d)	(263)	—	(731) ^(d)	(731)
Deposits ^(b)	(237)	—	(237)	(564)	—	(564)	(770)	—	(770)
Federal funds purchased and securities loaned or sold under repurchase agreements	(4)	—	(4)	(29)	—	(29)	116	—	116
Other borrowed funds	2,986	—	2,986	123	—	123	(1,287)	—	(1,287)
Trading liabilities	(57)	—	(57)	(23)	—	(23)	(3)	—	(3)
Beneficial interests issued by consolidated VIEs	(83)	—	(83)	(12)	—	(12)	(351)	—	(351)
Other liabilities	(3)	(5) ^(d)	(8)	(9)	8 ^(d)	(1)	64	—	64
Long-term debt:									
Changes in instrument-specific credit risk ^(a)	927	—	927	400	—	400	(1,704)	—	(1,704)
Other changes in fair value	322	—	322	1,297	—	1,297	(2,393)	—	(2,393)

(a) Total changes in instrument-specific credit risk related to structured notes were \$899 million, \$468 million, and \$(1.7) billion for the years ended December 31, 2011, 2010 and 2009, respectively. These totals include adjustments for structured notes classified within deposits and other borrowed funds, as well as long-term debt.

(b) Structured notes are debt instruments with embedded derivatives that are tailored to meet a client's need. The embedded derivative is the primary driver of risk. Although the risk associated with the structured notes is actively managed, the gains reported in this table do not include the income statement impact of such risk management instruments.

(c) Reported in mortgage fees and related income.

(d) Reported in other income.

Notes to consolidated financial statements

Determination of instrument-specific credit risk for items for which a fair value election was made

The following describes how the gains and losses included in earnings during 2011, 2010 and 2009, which were attributable to changes in instrument-specific credit risk, were determined.

- Loans and lending-related commitments: For floating-rate instruments, all changes in value are attributed to instrument-specific credit risk. For fixed-rate instruments, an allocation of the changes in value for the period is made between those changes in value that are interest rate-related and changes in value that are credit-related. Allocations are generally based on an analysis of borrower-specific credit spread and

recovery information, where available, or benchmarking to similar entities or industries.

- Long-term debt: Changes in value attributable to instrument-specific credit risk were derived principally from observable changes in the Firm's credit spread.
- Resale and repurchase agreements, securities borrowed agreements and securities lending agreements: Generally, for these types of agreements, there is a requirement that collateral be maintained with a market value equal to or in excess of the principal amount loaned; as a result, there would be no adjustment or an immaterial adjustment for instrument-specific credit risk related to these agreements.

Difference between aggregate fair value and aggregate remaining contractual principal balance outstanding

The following table reflects the difference between the aggregate fair value and the aggregate remaining contractual principal balance outstanding as of December 31, 2011 and 2010, for loans, long-term debt and long-term beneficial interests for which the fair value option has been elected.

December 31, (in millions)	2011			2010		
	Contractual principal outstanding	Fair value	Fair value over/ (under) contractual principal outstanding	Contractual principal outstanding	Fair value	Fair value over/ (under) contractual principal outstanding
Loans ^(a)						
Nonaccrual loans						
Loans reported as trading assets	\$ 4,875	\$ 1,141	\$ (3,734)	\$ 5,246	\$ 1,239	\$ (4,007)
Loans	820	56	(764)	927	132	(795)
Subtotal	5,695	1,197	(4,498)	6,173	1,371	(4,802)
All other performing loans						
Loans reported as trading assets	37,481	32,657	(4,824)	39,490	33,641	(5,849)
Loans	2,136	1,601	(535)	2,496	1,434	(1,062)
Total loans	\$ 45,312	\$ 35,455	\$ (9,857)	\$ 48,159	\$ 36,446	\$ (11,713)
Long-term debt						
Principal-protected debt	\$ 19,417 ^(c)	\$ 19,890	\$ 473	\$ 20,761 ^(c)	\$ 21,315	\$ 554
Nonprincipal-protected debt	NA	14,830	NA	NA	17,524	NA
Total long-term debt	NA	\$ 34,720	NA	NA	\$ 38,839	NA
Long-term beneficial interests						
Principal-protected debt	\$ —	\$ —	\$ —	\$ 49	\$ 49	\$ —
Nonprincipal-protected debt	NA	1,250	NA	NA	1,446	NA
Total long-term beneficial interests	NA	\$ 1,250	NA	NA	\$ 1,495	NA

(a) There were no performing loans which were ninety days or more past due as of December 31, 2011 and 2010, respectively.

(b) Remaining contractual principal is not applicable to nonprincipal-protected notes. Unlike principal-protected structured notes, for which the Firm is obligated to return a stated amount of principal at the maturity of the note, nonprincipal-protected structured notes do not obligate the Firm to return a stated amount of principal at maturity, but to return an amount based on the performance of an underlying variable or derivative feature embedded in the note.

(c) Where the Firm issues principal-protected zero-coupon or discount notes, the balance reflected as the remaining contractual principal is the final principal payment at maturity.

At December 31, 2011 and 2010, the contractual amount of letters of credit for which the fair value option was elected was \$3.9 billion and \$3.8 billion, respectively, with a corresponding fair value of \$(5) million and \$(6) million, respectively. For further information regarding off-balance sheet lending-related financial instruments, see Note 29 on pages 283–289 of this Annual Report.

Note 5 – Credit risk concentrations

Concentrations of credit risk arise when a number of customers are engaged in similar business activities or activities in the same geographic region, or when they have similar economic features that would cause their ability to meet contractual obligations to be similarly affected by changes in economic conditions.

JPMorgan Chase regularly monitors various segments of its credit portfolio to assess potential concentration risks and to obtain collateral when deemed necessary. Senior management is significantly involved in the credit approval and review process, and risk levels are adjusted as needed to reflect the Firm's risk appetite.

In the Firm's wholesale portfolio, risk concentrations are evaluated primarily by industry and monitored regularly on both an aggregate portfolio level and on an individual customer basis. Management of the Firm's wholesale exposure is accomplished through loan syndication and participation, loan sales, securitizations, credit derivatives, use of master netting agreements, and collateral and other risk-reduction techniques. In the consumer portfolio, concentrations are evaluated primarily by product and by U.S. geographic region, with a key focus on trends and concentrations at the portfolio level, where potential risk concentrations can be remedied through changes in underwriting policies and portfolio guidelines.

The Firm does not believe that its exposure to any particular loan product (e.g., option adjustable rate mortgages ("ARMs")), industry segment (e.g., commercial

real estate) or its exposure to residential real estate loans with high loan-to-value ratios results in a significant concentration of credit risk. Terms of loan products and collateral coverage are included in the Firm's assessment when extending credit and establishing its allowance for

prospectively and retrospectively. To assess effectiveness, the Firm uses statistical methods such as regression analysis, as well as nonstatistical methods including dollar-value comparisons of the change in the fair value of the derivative to the change in the fair value or cash flows of the hedged item. The extent to which a derivative has been, and is expected to continue to be, effective at offsetting changes in the fair value or cash flows of the hedged item must be assessed and documented at least quarterly. Any hedge ineffectiveness (i.e., the amount by which the gain or loss on the designated derivative instrument does not exactly offset the change in the hedged item attributable to the hedged risk) must be reported in current-period earnings. If it is determined that a derivative is not highly effective at hedging the designated exposure, hedge accounting is discontinued.

There are three types of hedge accounting designations: fair value hedges, cash flow hedges and net investment hedges. JPMorgan Chase uses fair value hedges primarily to hedge fixed-rate long-term debt, AFS securities and certain commodities inventories. For qualifying fair value hedges, the changes in the fair value of the derivative, and in the value of the hedged item, for the risk being hedged, are recognized in earnings. If the hedge relationship is terminated, then the fair value adjustment to the hedged item continues to be reported as part of the basis of the hedged item and for interest-bearing instruments is amortized to earnings as a yield adjustment. Derivative amounts affecting earnings are recognized consistent with the classification of the hedged item – primarily net interest income and principal transactions revenue.

JPMorgan Chase uses cash flow hedges to hedge the exposure to variability in cash flows from floating-rate financial instruments and forecasted transactions, primarily the rollover of short-term assets and liabilities, and foreign currency-denominated revenue and expense. For qualifying cash flow hedges, the effective portion of the change in the fair value of the derivative is recorded in OCI and recognized in the Consolidated Statements of Income when the hedged cash flows affect earnings. Derivative amounts affecting earnings are recognized consistent with the classification of the hedged item – primarily interest income, interest expense, noninterest revenue and compensation expense. The ineffective portions of cash flow hedges are immediately recognized in earnings. If the hedge relationship is terminated, then the value of the derivative recorded in accumulated other comprehensive income/(loss) ("AOCI") is recognized in earnings when the cash flows that were hedged affect earnings. For hedge relationships that are discontinued because a forecasted transaction is not expected to occur according to the original hedge forecast, any related derivative values recorded in AOCI are immediately recognized in earnings.

JPMorgan Chase uses foreign currency hedges to protect the value of the Firm's net investments in certain non-U.S. subsidiaries or branches whose functional currencies are not the U.S. dollar. For foreign currency qualifying net investment hedges, changes in the fair value of the derivatives are recorded in the translation adjustments account within AOCI.

Notional amount of derivative contracts

The following table summarizes the notional amount of derivative contracts outstanding as of December 31, 2011 and 2010.

December 31, (in billions)	Notional amount ^(a)	
	2011	2010
Interest rate contracts		
Swaps	\$ 38,704	\$ 46,299
Futures and forwards	7,888	9,298
Written options	3,842	4,075
Purchased options	4,026	3,968
Total interest rate contracts	54,460	63,640
Credit derivatives	5,774	5,472
Foreign exchange contracts		
Cross-currency swaps	2,931	2,568
Spot, futures and forwards	4,512	3,893
Written options	674	674
Purchased options	670	649
Total foreign exchange contracts	8,787	7,784
Equity contracts		
Swaps	119	116
Futures and forwards	38	49
Written options	460	430
Purchased options	405	377
Total equity contracts	1,022	972
Commodity contracts		
Swaps	341	349
Spot, futures and forwards	188	170
Written options	310	264
Purchased options	274	254
Total commodity contracts	1,113	1,037
Total derivative notional amounts	\$ 71,156	\$ 78,905

(a) Represents the sum of gross long and gross short third-party notional derivative contracts.

While the notional amounts disclosed above give an indication of the volume of the Firm's derivatives activity, the notional amounts significantly exceed, in the Firm's view, the possible losses that could arise from such transactions. For most derivative transactions, the notional amount is not exchanged; it is used simply as a reference to calculate payments.

Notes to consolidated financial statements

Impact of derivatives on the Consolidated Balance Sheets

The following table summarizes information on derivative receivables and payables (before and after netting adjustments) that are reflected on the Firm's Consolidated Balance Sheets as of December 31, 2011 and 2010, by accounting designation (e.g., whether the derivatives were designated as hedges or not) and contract type.

Free-standing derivative receivables and payables

December 31, 2011 (in millions)	Gross derivative receivables				Gross derivative payables			
	Not designated as hedges	Designated as hedges	Total derivative receivables	Net derivative receivables	Not designated as hedges	Designated as hedges	Total derivative payables	Net derivative payables
Trading assets and liabilities								
Interest rate	\$ 1,433,900	\$ 7,621	\$ 1,441,521	\$ 46,369	\$ 1,397,625	\$ 2,192	\$ 1,399,817	\$ 28,010
Credit	169,650	—	169,650	6,684	165,121	—	165,121	5,610
Foreign exchange ^(a)	163,497	4,666	168,163	17,890	165,353	655	166,008	17,435
Equity	47,736	—	47,736	6,793	46,366	—	46,366	9,655
Commodity	53,894	3,535	57,429	14,741	58,836	1,108	59,944	14,267
Total fair value of trading assets and liabilities	\$ 1,868,677	\$ 15,822	\$ 1,884,499	\$ 92,477	\$ 1,833,301	\$ 3,955	\$ 1,837,256	\$ 74,977

December 31, 2010 (in millions)	Gross derivative receivables				Gross derivative payables			
	Not designated as hedges	Designated as hedges	Total derivative receivables	Net derivative receivables	Not designated as hedges	Designated as hedges	Total derivative payables	Net derivative payables
Trading assets and liabilities								
Interest rate	\$ 1,121,703	\$ 6,279	\$ 1,127,982	\$ 32,555	\$ 1,089,604	\$ 840	\$ 1,090,444	\$ 20,387
Credit	129,729	—	129,729	7,725	125,061	—	125,061	5,138
Foreign exchange ^(a)	165,240	3,231	168,471	25,858	163,671	1,059	164,730	25,015
Equity	43,633	—	43,633	4,204	46,399	—	46,399	10,450
Commodity	59,573	24	59,597	10,139	56,397	2,078 ^(c)	58,475	8,229
Total fair value of trading assets and liabilities	\$ 1,519,878	\$ 9,534	\$ 1,529,412	\$ 80,481	\$ 1,481,132	\$ 3,977	\$ 1,485,109	\$ 69,219

(a) Excludes structured notes for which the fair value option has been elected. See Note 4 on pages 198–200 of this Annual Report for further information.

(b) Excludes \$11 million and \$21 million of foreign currency-denominated debt designated as a net investment hedge at December 31, 2011 and 2010, respectively.

(c) Excludes \$1.0 billion related to commodity derivatives that were embedded in a debt instrument and used as fair value hedging instruments that were recorded in the line item of the host contract (other borrowed funds) at December 31, 2010.

Impact of derivatives on the Consolidated Statements of Income

Fair value hedge gains and losses

The following tables present derivative instruments, by contract type, used in fair value hedge accounting relationships, as well as pretax gains/(losses) recorded on such derivatives and the related hedged items for the years ended December 31, 2011, 2010 and 2009, respectively. The Firm includes gains/(losses) on the hedging derivative and the related hedged item in the same line item in the Consolidated Statements of Income.

Year ended December 31, 2011 (in millions)	Gains/(losses) recorded in income			Income statement impact due to:	
	Derivatives	Hedged items	Total income statement impact	Hedge ineffectiveness ^(e)	Excluded components ^(f)
Contract type					
Interest rate ^(a)	\$ 558	\$ 6	\$ 564	\$ 104	\$ 460
Foreign exchange ^(b)	5,684 ^(d)	(3,761)	1,923	—	1,923
Commodity ^(c)	1,784	(2,880)	(1,096)	(10)	(1,086)
Total	\$ 8,026	\$ (6,635)	\$ 1,391	\$ 94	\$ 1,297

Year ended December 31, 2010 (in millions)	Gains/(losses) recorded in income			Income statement impact due to:	
	Derivatives	Hedged items	Total income statement impact	Hedge ineffectiveness ^(e)	Excluded components ^(f)
Contract type					
Interest rate ^(a)	\$ 1,066	\$ (454)	\$ 612	\$ 172	\$ 440
Foreign exchange ^(b)	1,357 ^(d)	(1,812)	(455)	—	(455)
Commodity ^(c)	(1,354)	1,882	528	—	528
Total	\$ 1,069	\$ (384)	\$ 685	\$ 172	\$ 513

Year ended December 31, 2009 (in millions)	Gains/(losses) recorded in income			Income statement impact due to:	
	Derivatives	Hedged items	Total income statement impact	Hedge ineffectiveness ^(e)	Excluded components ^(f)
Contract type					
Interest rate ^(a)	\$ (3,830)	\$ 4,638	\$ 808	\$ (466)	\$ 1,274
Foreign exchange ^(b)	(1,421) ^(d)	1,445	24	—	24
Commodity ^(c)	(430)	399	(31)	—	(31)
Total	\$ (5,681)	\$ 6,482	\$ 801	\$ (466)	\$ 1,267

- (a) Primarily consists of hedges of the benchmark (e.g., London Interbank Offered Rate ("LIBOR")) interest rate risk of fixed-rate long-term debt and AFS securities. Gains and losses were recorded in net interest income.
- (b) Primarily consists of hedges of the foreign currency risk of long-term debt and AFS securities for changes in spot foreign currency rates. Gains and losses related to the derivatives and the hedged items, due to changes in spot foreign currency rates, were recorded in principal transactions revenue.
- (c) Consists of overall fair value hedges of certain commodities inventories. Gains and losses were recorded in principal transactions revenue.
- (d) Included \$4.9 billion, \$278 million and \$(1.6) billion for the years ended December 31, 2011, 2010 and 2009, respectively, of revenue related to certain foreign exchange trading derivatives designated as fair value hedging instruments.
- (e) Hedge ineffectiveness is the amount by which the gain or loss on the designated derivative instrument does not exactly offset the gain or loss on the hedged item attributable to the hedged risk.
- (f) Certain components of hedging derivatives are permitted to be excluded from the assessment of hedge effectiveness, such as forward points on foreign exchange forward contracts. Amounts related to excluded components are recorded in current-period income.

Notes to consolidated financial statements

Cash flow hedge gains and losses

The following tables present derivative instruments, by contract type, used in cash flow hedge accounting relationships, and the pretax gains/(losses) recorded on such derivatives, for the years ended December 31, 2011, 2010 and 2009, respectively. The Firm includes the gain/(loss) on the hedging derivative in the same line item as the offsetting change in cash flows on the hedged item in the Consolidated Statements of Income.

Year ended December 31, 2011 (in millions)	Gains/(losses) recorded in income and other comprehensive income/(loss)				
	Derivatives – effective portion reclassified from AOCI to income	Hedge ineffectiveness recorded directly in income ^(d)	Total income statement impact	Derivatives – effective portion recorded in OCI	Total change in OCI for period
Contract type					
Interest rate ^(a)	\$ 310	\$ 19	\$ 329	\$ 107	\$ (203)
Foreign exchange ^(b)	(9)	—	(9)	(57)	(48)
Total	\$ 301	\$ 19	\$ 320	\$ 50	\$ (251)

Year ended December 31, 2010 (in millions)	Gains/(losses) recorded in income and other comprehensive income/(loss)				
	Derivatives – effective portion reclassified from AOCI to income	Hedge ineffectiveness recorded directly in income ^(d)	Total income statement impact	Derivatives – effective portion recorded in OCI	Total change in OCI for period
Contract type					
Interest rate ^(a)	\$ 288	\$ 20	\$ 308	\$ 388	\$ 100
Foreign exchange ^(b)	(82)	(3)	(85)	(141)	(59)
Total	\$ 206	\$ 17	\$ 223	\$ 247	\$ 41

Year ended December 31, 2009 (in millions)	Gains/(losses) recorded in income and other comprehensive income/(loss)				
	Derivatives – effective portion reclassified from AOCI to income	Hedge ineffectiveness recorded directly in income ^(d)	Total income statement impact	Derivatives – effective portion recorded in OCI	Total change in OCI for period
Contract type					
Interest rate ^(a)	\$ (158)	\$ (62)	\$ (220)	\$ 61	\$ 219
Foreign exchange ^(b)	282	—	282	706	424
Total	\$ 124	\$ (62)	\$ 62	\$ 767	\$ 643

- (a) Primarily consists of benchmark interest rate hedges of LIBOR-indexed floating-rate assets and floating-rate liabilities. Gains and losses were recorded in net interest income.
- (b) Primarily consists of hedges of the foreign currency risk of non-U.S. dollar-denominated revenue and expense. The income statement classification of gains and losses follows the hedged item – primarily net interest income, noninterest revenue and compensation expense.
- (c) The Firm did not experience any forecasted transactions that failed to occur for the years ended December 31, 2011 and 2009. In 2010, the Firm reclassified a \$25 million loss from AOCI to earnings because the Firm determined that it was probable that forecasted interest payment cash flows related to certain wholesale deposits would not occur.
- (d) Hedge ineffectiveness is the amount by which the cumulative gain or loss on the designated derivative instrument exceeds the present value of the cumulative expected change in cash flows on the hedged item attributable to the hedged risk.

Over the next 12 months, the Firm expects that \$26 million (after-tax) of net gains recorded in AOCI at December 31, 2011, related to cash flow hedges will be recognized in income. The maximum length of time over which forecasted transactions on the hedged item is 10 years, and such transactions primarily relate to core lending and borrowing activities.

Net investment hedge gains and losses

The following tables present hedging instruments, by contract type, that were used in net investment hedge accounting relationships, and the pretax gains/(losses) recorded on such instruments for the years ended December 31, 2011, 2010 and 2009.

Year ended December 31, (in millions)	Gains/(losses) recorded in income and other comprehensive income/(loss)					
	2011		2010		2009	
	Excluded components recorded directly in income ^(a)	Effective portion recorded in OCI	Excluded components recorded directly in income ^(a)	Effective portion recorded in OCI	Excluded components recorded directly in income ^(a)	Effective portion recorded in OCI
Contract type						
Foreign exchange derivatives	\$ (251)	\$ 225	\$ (139)	\$ (30)	\$ (112)	\$ (259)
Foreign currency denominated debt	—	1	—	41	NA	NA
Total	\$ (251)	\$ 226	\$ (139)	\$ 11	\$ (112)	\$ (259)

(a) Certain components of hedging derivatives are permitted to be excluded from the assessment of hedge effectiveness, such as forward points on foreign exchange forward contracts. Amounts related to excluded components are recorded in current-period income. The Firm measures the ineffectiveness in net investment hedge accounting relationships based on changes in spot foreign currency rates, and therefore there was no ineffectiveness for net investment hedge accounting relationships during 2011, 2010 and 2009.

Risk management derivatives gains and losses (not designated as hedging instruments)

The following table presents nontrading derivatives, by contract type, that were not designated in hedge relationships, and the pretax gains/(losses) recorded on such derivatives for the years ended December 31, 2011, 2010 and 2009. These derivatives are risk management instruments used to mitigate or transform market risk exposures arising from banking activities other than trading activities, which are discussed separately below.

Year ended December 31, (in millions)	Derivatives gains/(losses) recorded in income		
	2011	2010	2009
Contract type			
Interest rate ^(a)	\$ 8,084	\$ 4,987	\$ (3,113)
Credit ^(b)	(52)	(237)	(3,222)
Foreign exchange ^(c)	(157)	(64)	(197)
Equity ^(b)	—	—	(8)
Commodity ^(c)	41	(48)	(50)
Total	\$ 7,916	\$ 4,638	\$ (6,590)

(a) Gains and losses were recorded in principal transactions revenue, mortgage fees and related income, and net interest income.

(b) Gains and losses were recorded in principal transactions revenue.

(c) Gains and losses were recorded in principal transactions revenue and net interest income.

Trading derivative gains and losses

The Firm has elected to present derivative gains and losses related to its trading activities together with the nonderivative instruments with which they are risk managed. All amounts are recorded in principal transactions revenue in the Consolidated Statements of Income for the years ended December 31, 2011, 2010 and 2009. The amounts below do not represent a comprehensive view of the Firm's trading activities because they do not include certain revenue associated with those activities, including net interest income earned on cash instruments used in trading activities and gains and losses on cash instruments that are risk managed without derivative instruments.

Year ended December 31, (in millions)	Gains/(losses) recorded in principal transactions revenue		
	2011	2010	2009
Type of instrument			
Interest rate	\$ (1,531)	\$ (683)	\$ 4,375
Credit	3,346	4,636	5,022
Foreign exchange	1,216	1,854	2,583
Equity	1,956	1,827	1,475
Commodity	3,697	243	1,329
Total	\$ 8,684	\$ 7,877	\$ 14,784

Credit risk, liquidity risk and credit-related contingent features

In addition to the specific market risks introduced by each derivative contract type, derivatives expose JPMorgan Chase to credit risk — the risk that derivative counterparties may fail to meet their payment obligations under the derivative contracts and the collateral, if any, held by the Firm proves to be of insufficient value to cover the payment obligation. It is the policy of JPMorgan Chase to actively pursue the use of legally enforceable master netting arrangements and collateral agreements to mitigate derivative counterparty credit risk. The amount of derivative receivables reported on the Consolidated Balance Sheets is the fair value of the derivative contracts after giving effect to legally enforceable master netting agreements and cash collateral held by the Firm. These amounts represent the cost to the Firm to replace the contracts at then-current market rates should the counterparty default.

Notes to consolidated financial statements

While derivative receivables expose the Firm to credit risk, derivative payables expose the Firm to liquidity risk, as the derivative contracts typically require the Firm to post cash or securities collateral with counterparties as the mark-to-market ("MTM") of the contracts moves in the counterparties' favor, or upon specified downgrades in the Firm's and its subsidiaries' respective credit ratings. Certain derivative contracts also provide for termination of the contract, generally upon a downgrade of either the Firm or

the counterparty, at the fair value of the derivative contracts. The following table shows the aggregate fair value of net derivative payables that contain contingent collateral or termination features that may be triggered upon a downgrade and the associated collateral the Firm has posted in the normal course of business at December 31, 2011 and 2010.

Derivative payables containing downgrade triggers

December 31, (in millions)	2011		2010	
Aggregate fair value of net derivative payables	\$	16,937	\$	19,777
Collateral posted		11,429		14,629

The following table shows the impact of a single-notch and two-notch ratings downgrade to JPMorgan Chase & Co. and its subsidiaries, primarily JPMorgan Chase Bank, National Association ("JPMorgan Chase Bank, N.A.") at December 31, 2011 and 2010, related to derivative contracts with contingent collateral or termination features that may be triggered upon a downgrade.

Liquidity impact of derivative downgrade triggers

December 31, (in millions)	2011		2010	
	Single-notch downgrade	Two-notch downgrade	Single-notch downgrade	Two-notch downgrade
Amount of additional collateral to be posted	\$ 1,460	\$ 2,054	\$ 1,904	\$ 3,462
Amount required to settle contracts with termination triggers	1,054	1,923	430	994

The following tables show the carrying value of derivative receivables and payables after netting adjustments and adjustments for collateral held and transferred as of December 31, 2011 and 2010.

Impact of netting adjustments on derivative receivables and payables

December 31, (in millions)	Derivative receivables		Derivative payables	
	2011	2010	2011	2010
Gross derivative fair value	\$ 1,884,499	\$ 1,529,412	\$ 1,837,256	\$ 1,485,109
Netting adjustment – offsetting receivables/payables	(1,710,525)	(1,376,969)	(1,710,523)	(1,376,969)
Netting adjustment – cash collateral received/paid	(81,497)	(71,962)	(51,756)	(38,921)
Carrying value on Consolidated Balance Sheets	\$ 92,477	\$ 80,481	\$ 74,977	\$ 69,219

Total derivative collateral

December 31, (in millions)	Collateral held		Collateral transferred	
	2011	2010	2011	2010
Netting adjustment for cash collateral	\$ 81,497	\$ 71,962	\$ 51,756	\$ 38,921
Liquid securities and other cash collateral	21,807	16,486	19,439	10,899
Additional liquid securities and cash collateral	17,615	18,048	10,824	8,435
Total collateral for derivative transactions	\$ 120,919	\$ 106,496	\$ 82,019	\$ 58,255

- As permitted under U.S. GAAP, the Firm has elected to net cash collateral received and paid together with the related derivative receivables and derivative payables when a legally enforceable master netting agreement exists.
- Represents cash collateral received and paid that is not subject to a legally enforceable master netting agreement, and liquid securities collateral held and transferred.
- Represents liquid securities and cash collateral held and transferred at the initiation of derivative transactions, which is available as security against potential exposure that could arise should the fair value of the transactions move, as well as collateral held and transferred related to contracts that have a non-daily call frequency for collateral to be posted, and collateral that the Firm or a counterparty has agreed to return but has not yet settled as of the reporting date. These amounts were not netted against the derivative receivables and payables in the tables above, because, at an individual counterparty level, the collateral exceeded the fair value exposure at both December 31, 2011 and 2010.

Credit derivatives

Credit derivatives are financial instruments whose value is derived from the credit risk associated with the debt of a third-party issuer (the reference entity) and which allow one party (the protection purchaser) to transfer that risk to another party (the protection seller). Credit derivatives expose the protection purchaser to the creditworthiness of the protection seller, as the protection seller is required to make payments under the contract when the reference entity experiences a credit event, such as a bankruptcy, a failure to pay its obligation or a restructuring. The seller of credit protection receives a premium for providing protection but has the risk that the underlying instrument referenced in the contract will be subject to a credit event.

The Firm is both a purchaser and seller of protection in the credit derivatives market and uses these derivatives for two primary purposes. First, in its capacity as a market-maker in the dealer/client business, the Firm actively risk manages a portfolio of credit derivatives by purchasing and selling credit protection, predominantly on corporate debt obligations, to meet the needs of customers. As a seller of protection, the Firm's exposure to a given reference entity may be offset partially, or entirely, with a contract to purchase protection from another counterparty on the same or similar reference entity. Second, the Firm uses credit derivatives to mitigate credit risk associated with its overall derivative receivables and traditional commercial credit lending exposures (loans and unfunded commitments) as well as to manage its exposure to residential and commercial mortgages. In accomplishing the above, the Firm uses different types of credit derivatives. Following is a summary of various types of credit derivatives.

Credit default swaps

Credit derivatives may reference the credit of either a single reference entity ("single-name") or a broad-based index. The Firm purchases and sells protection on both single-name and index-reference obligations. Single-name CDS and index CDS contracts are OTC derivative contracts. Single-name CDS are used to manage the default risk of a single reference entity, while index CDS contracts are used to manage the credit risk associated with the broader credit markets or credit market segments. Like the S&P 500 and other market indices, a CDS index comprises a portfolio of CDS across many reference entities. New series of CDS indices are periodically established with a new underlying portfolio of reference entities to reflect changes in the credit markets. If one of the reference entities in the index experiences a credit event, then the reference entity that defaulted is removed from the index. CDS can also be referenced against specific portfolios of reference names or against customized exposure levels based on specific client demands: for example, to provide protection against the first \$1 million of realized credit losses in a \$10 million portfolio of exposure. Such structures are commonly known as tranche CDS.

For both single-name CDS contracts and index CDS contracts, upon the occurrence of a credit event, under the terms of a CDS contract neither party to the CDS contract has recourse to the reference entity. The protection purchaser has recourse to the protection seller for the difference between the face value of the CDS contract and the fair value of the reference obligation at the time of settling the credit derivative contract, also known as the recovery value. The protection purchaser does not need to hold the debt instrument of the underlying reference entity in order to receive amounts due under the CDS contract when a credit event occurs.

Credit-related notes

A credit-related note is a funded credit derivative where the issuer of the credit-related note purchases from the note investor credit protection on a referenced entity. Under the contract, the investor pays the issuer the par value of the note at the inception of the transaction, and in return, the issuer pays periodic payments to the investor, based on the credit risk of the referenced entity. The issuer also repays the investor the par value of the note at maturity unless the reference entity experiences a specified credit event. If a credit event occurs, the issuer is not obligated to repay the par value of the note, but rather, the issuer pays the investor the difference between the par value of the note and the fair value of the defaulted reference obligation at the time of settlement. Neither party to the credit-related note has recourse to the defaulting reference entity. For a further discussion of credit-related notes, see Note 16 on pages 256–267 of this Annual Report.

The following tables present a summary of the notional amounts of credit derivatives and credit-related notes the Firm sold and purchased as of December 31, 2011 and 2010. Upon a credit event, the Firm as a seller of protection would typically pay out only a percentage of the full notional amount of net protection sold, as the amount actually required to be paid on the contracts takes into account the recovery value of the reference obligation at the time of settlement. The Firm manages the credit risk on contracts to sell protection by purchasing protection with identical or similar underlying reference entities. Other purchased protection referenced in the following tables includes credit derivatives bought on related, but not identical, reference positions (including indices, portfolio coverage and other reference points) as well as protection purchased through credit-related notes.

Notes to consolidated financial statements

The Firm does not use notional amounts of credit derivatives as the primary measure of risk management for such derivatives because the notional amount does not take into account the probability of the occurrence of a credit event, the recovery value of the reference obligation, or related cash instruments and economic hedges, each of which reduces, in the Firm's view, the risks associated with such derivatives.

Total credit derivatives and credit-related notes

December 31, 2011 (in millions)	Maximum payout/Notional amount			
	Protection sold	Protection purchased with identical underlying ^(a)	Net protection (sold)/purchased ^(c)	Other protection purchased ^(d)
Credit derivatives				
Credit default swaps	\$ (2,839,492)	\$ 2,798,207	\$ (41,285)	\$ 29,139
Other credit derivatives ^(b)	(79,711)	4,954	(74,757)	22,292
Total credit derivatives	(2,919,203)	2,803,161	(116,042)	51,431
Credit-related notes	(742)	—	(742)	3,944
Total	\$ (2,919,945)	\$ 2,803,161	\$ (116,784)	\$ 55,375

December 31, 2010 (in millions)	Maximum payout/Notional amount			
	Protection sold	Protection purchased with identical underlying ^(a)	Net protection (sold)/purchased ^(c)	Other protection purchased ^(d)
Credit derivatives				
Credit default swaps	\$ (2,659,240)	\$ 2,652,313	\$ (6,927)	\$ 32,867
Other credit derivatives ^(b)	(93,776)	10,016	(83,760)	24,234
Total credit derivatives	(2,753,016)	2,662,329	(90,687)	57,101
Credit-related notes	(2,008)	—	(2,008)	3,327
Total	\$ (2,755,024)	\$ 2,662,329	\$ (92,695)	\$ 60,428

(a) Primarily consists of total return swaps and credit default swap options.

(b) Represents the total notional amount of protection purchased where the underlying reference instrument is identical to the reference instrument on protection sold; the notional amount of protection purchased for each individual identical underlying reference instrument may be greater or lower than the notional amount of protection sold.

(c) Does not take into account the fair value of the reference obligation at the time of settlement, which would generally reduce the amount the seller of protection pays to the buyer of protection in determining settlement value.

(d) Represents protection purchased by the Firm through single-name and index credit default swaps or credit-related notes.

The following tables summarize the notional and fair value amounts of credit derivatives and credit-related notes as of December 31, 2011 and 2010, where JPMorgan Chase is the seller of protection. The maturity profile is based on the remaining contractual maturity of the credit derivative contracts. The ratings profile is based on the rating of the reference entity on which the credit derivative contract is based. The ratings and maturity profile of credit derivatives and credit-related notes where JPMorgan Chase is the purchaser of protection are comparable to the profile reflected below.

Protection sold – credit derivatives and credit-related notes ratings and maturity profile

December 31, 2011 (in millions)	<1 year	1–5 years	>5 years	Total notional amount	Fair value ^(b)
Risk rating of reference entity					
Investment-grade	\$ (352,215)	\$ (1,262,143)	\$ (345,996)	\$ (1,960,354)	\$ (57,697)
Noninvestment-grade	(241,823)	(589,954)	(127,814)	(959,591)	(85,304)
Total	\$ (594,038)	\$ (1,852,097)	\$ (473,810)	\$ (2,919,945)	\$ (143,001)

December 31, 2010 (in millions)	<1 year	1–5 years	>5 years	Total notional amount	Fair value ^(b)
Risk rating of reference entity					
Investment-grade	\$ (175,618)	\$ (1,194,695)	\$ (336,309)	\$ (1,706,622)	\$ (17,261)
Noninvestment-grade	(148,434)	(702,638)	(197,330)	(1,048,402)	(59,939)
Total	\$ (324,052)	\$ (1,897,333)	\$ (533,639)	\$ (2,755,024)	\$ (77,200)

(a) The ratings scale is based on the Firm's internal ratings, which generally correspond to ratings as defined by S&P and Moody's.

(b) Amounts are shown on a gross basis, before the benefit of legally enforceable master netting agreements and cash collateral received by the Firm.

Note 7 – Noninterest revenue

Investment banking fees

This revenue category includes advisory and equity and debt underwriting fees. Underwriting fees are recognized as revenue when the Firm has rendered all services to the issuer and is entitled to collect the fee from the issuer, as long as there are no other contingencies associated with the fee. Underwriting fees are net of syndicate expense; the Firm recognizes credit arrangement and syndication fees as revenue after satisfying certain retention, timing and yield criteria. Advisory fees are recognized as revenue when the related services have been performed and the fee has been earned.

The following table presents the components of investment banking fees.

Year ended December 31, (in millions)	2011	2010	2009
Underwriting			
Equity	\$ 1,181	\$ 1,589	\$ 2,487
Debt	2,934	3,172	2,739
Total underwriting	4,115	4,761	5,226
Advisory ^(a)	1,796	1,429	1,861
Total investment banking fees	\$ 5,911	\$ 6,190	\$ 7,087

(a) Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. Upon adoption of the guidance, the Firm consolidated its Firm-administered multi-seller conduits. The consolidation of the conduits did not significantly change the Firm's net income as a whole; however, certain advisory fees considered inter-company were eliminated while net interest income and lending-and-deposit-related fees increased.

Principal transactions

Principal transactions revenue consists of trading revenue as well as realized and unrealized gains and losses on private equity investments. Trading revenue is driven by the Firm's client market-making and client driven activities as well as certain risk management activities.

The spread between the price at which the Firm buys and sells financial instruments and physical commodities inventories to and from its clients and other market-makers is recognized as trading revenue. Trading revenue also includes unrealized gains and losses on financial instruments (including those for which the fair value option was elected) and unrealized losses on physical commodities inventories (generally carried at the lower of cost or fair value) that the Firm holds in inventory as a market-maker to meet client needs, or for risk management purposes.

The following table presents principal transactions revenue by major underlying type of risk exposures. This table does not include other types of revenue, such as net interest income on trading assets, which are an integral part of the overall performance of the Firm's client-driven trading activities.

Year ended December 31, (in millions)	2011	2010	2009
Trading revenue by risk exposure			
Interest rate	\$ (873)	\$ (199)	\$ 3,681
Credit	3,393	4,543	546
Foreign exchange	1,154	1,896	2,317
Equity	2,401	2,275	2,056
Commodity ^(a)	2,823	889	1,270
Total trading revenue	8,898	9,404	9,870
Private equity gains/(losses) ^(b)	1,107	1,490	(74)
Principal transactions ^(c)	\$ 10,005	\$ 10,894	\$ 9,796

(a) Includes realized gains and losses and unrealized losses on physical commodities inventories that are generally carried at the lower of cost or fair value, and gains and losses on commodity derivatives and other financial instruments that are carried at fair value through income. Commodity derivatives are frequently used to manage the Firm's risk exposure to its physical commodities inventories.

(b) Includes revenue on private equity investments held in the Private Equity business within Corporate/Private Equity, as well as those held in other business segments.

(c) Principal transactions included DVA related to derivatives and structured liabilities measured at fair value in IB. DVA gains/(losses) were \$1.4 billion, \$509 million, and \$(2.3) billion for the years ended December 31, 2011, 2010 and 2009, respectively.

Lending- and deposit-related fees

This revenue category includes fees from loan commitments, standby letters of credit, financial guarantees, deposit-related fees in lieu of compensating balances, cash management-related activities or transactions, deposit accounts and other loan-servicing activities. These fees are recognized over the period in which the related service is provided.

Asset management, administration and commissions

This revenue category includes fees from investment management and related services, custody, brokerage services, insurance premiums and commissions, and other products. These fees are recognized over the period in which the related service is provided. Performance-based fees, which are earned based on exceeding certain benchmarks or other performance targets, are accrued and recognized at the end of the performance period in which the target is met.

The following table presents components of asset management, administration and commissions.

Year ended December 31, (in millions)	2011	2010	2009
Asset management			
Investment management fees	\$ 6,085	\$ 5,632	\$ 4,997
All other asset management fees	605	496	356
Total asset management fees	6,690	6,128	5,353
Total administration fees	2,171	2,023	1,927
Commission and other fees			
Brokerage commissions	2,753	2,804	2,904
All other commissions and fees	2,480	2,544	2,356
Total commissions and fees	5,233	5,348	5,260
Total asset management, administration and commissions	\$ 14,094	\$ 13,499	\$ 12,540

(a) Includes fees for custody, securities lending, funds services and securities clearance.

Mortgage fees and related income

This revenue category primarily reflects RFS's mortgage production and servicing revenue, including: fees and income derived from mortgages originated with the intent to sell; mortgage sales and servicing including losses related to the repurchase of previously-sold loans; the impact of risk management activities associated with the mortgage pipeline, warehouse loans and MSRs; and revenue related to any residual interests held from mortgage securitizations. This revenue category also includes gains and losses on sales and lower of cost or fair value adjustments for mortgage loans held-for-sale, as well as changes in fair value for mortgage loans originated with the intent to sell and measured at fair value under the fair value option. Changes in the fair value of RFS mortgage servicing rights are reported in mortgage fees and related income. Net interest income from mortgage loans, and securities gains and losses on AFS securities used in mortgage-related risk management activities, are recorded in interest income and securities gains/(losses), respectively. For a further discussion of MSRs, see Note 17 on pages 267–271 of this Annual Report.

Credit card income

This revenue category includes interchange income from credit and debit cards and net fees earned from processing credit card transactions for merchants. Prior to 2010, this revenue category included servicing fees earned in connection with securitization activities; such fees have been eliminated in consolidation since January 1, 2010, when the Firm consolidated its Firm-sponsored credit card securitization trusts (see Note 16 on pages 256–267 of this Annual Report). Credit card income is recognized as earned. Annual fees and direct loan origination costs are deferred and recognized on a straight-line basis over a 12-month period. Expense related to rewards programs is recorded when the rewards are earned by the customer and netted against interchange income.

Credit card revenue sharing agreements

The Firm has contractual agreements with numerous affinity organizations and co-brand partners (collectively, "partners"), which grant the Firm exclusive rights to market to the members or customers of such partners. These partners endorse the credit card programs and provide their mailing lists to the Firm, and they may also conduct marketing activities and provide awards under the various credit card programs. The terms of these agreements generally range from three to 10 years.

The Firm typically makes incentive payments to the partners based on: new account originations; charge volumes; and, the cost of the partners' marketing activities and awards. Payments based on new account originations are accounted for as direct loan origination costs. Payments to partners based on charge volumes are deducted from interchange income as the related revenue is earned. Payments based on marketing efforts undertaken by the partners are expensed by the Firm as incurred and reported as noninterest expense.

Note 8 – Interest income and Interest expense

Interest income and interest expense is recorded in the Consolidated Statements of Income and classified based on the nature of the underlying asset or liability. Interest income and interest expense includes the current-period interest accruals for financial instruments measured at fair value, except for financial instruments containing embedded derivatives that would be separately accounted for in accordance with U.S. GAAP absent the fair value option election; for those instruments, all changes in fair value including any interest elements, are reported in principal transactions revenue. For financial instruments that are not measured at fair value, the related interest is included within interest income or interest expense, as applicable.

Details of interest income and interest expense were as follows.

Year ended December 31, (in millions)	2011	2010	2009
Interest income			
Loans	\$ 37,098	\$ 40,388	\$ 38,704
Securities	9,215	9,540	12,377
Trading assets	11,142	11,007	12,098
Federal funds sold and securities purchased under resale agreements	2,523	1,786	1,750
Securities borrowed	110	175	4
Deposits with banks	599	345	938
Other assets ^(a)	606	541	479
Total interest income ^(b)	61,293	63,782	66,350
Interest expense			
Interest-bearing deposits	3,855	3,424	4,826
Short-term and other liabilities ^{(c)(d)}	2,873	2,364	2,786
Long-term debt ^(e)	6,109	5,848	7,368
Beneficial interests issued by consolidated VIEs	767	1,145	218
Total interest expense ^(b)	13,604	12,781	15,198
Net interest income	47,689	51,001	51,152
Provision for credit losses	7,574	16,639	32,015
Net interest income after provision for credit losses	\$ 40,115	\$ 34,362	\$ 19,137

(a) Predominantly margin loans.

(b) Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. Upon the adoption of the guidance, the Firm consolidated its Firm-sponsored credit card securitization trusts, its Firm-administered multi-seller conduits and certain other consumer loan securitization entities, primarily mortgage-related. The consolidation of these VIEs did not significantly change the Firm's total net income. However, it did affect the classification of items on the Firm's Consolidated Statements of Income; as a result of the adoption of the guidance, certain noninterest revenue was eliminated in consolidation, offset by the recognition of interest income, interest expense, and provision for credit losses.

(c) Includes brokerage customer payables.

(d) Effective January 1, 2011, the long-term portion of advances from FHLBs was reclassified from other borrowed funds to long-term debt. The related interest expense for the prior-year period has also been reclassified to conform with the current presentation.

Note 9 – Pension and other postretirement employee benefit plans

The Firm's defined benefit pension plans and its other postretirement employee benefit ("OPEB") plans (collectively the "Plans") are accounted for in accordance with U.S. GAAP for retirement benefits.

Defined benefit pension plans

The Firm has a qualified noncontributory U.S. defined benefit pension plan that provides benefits to substantially all U.S. employees. The U.S. plan employs a cash balance formula in the form of pay and interest credits to determine the benefits to be provided at retirement, based on eligible compensation and years of service. Employees begin to accrue plan benefits after completing one year of service, and benefits generally vest after three years of service. In November 2009, the Firm announced certain changes to the pay credit schedule and amount of eligible compensation recognized under the U.S. plan effective February 1, 2010. The Firm also offers benefits through defined benefit pension plans to qualifying employees in certain non-U.S. locations based on factors such as eligible compensation, age and/or years of service.

It is the Firm's policy to fund the pension plans in amounts sufficient to meet the requirements under applicable laws. On January 15, 2009, and August 28, 2009, the Firm made discretionary cash contributions to its U.S. defined benefit pension plan of \$1.3 billion and \$1.5 billion, respectively. The Firm does not anticipate any contribution to the U.S. defined benefit pension plan in 2012 at this time. The 2012 contributions to the non-U.S. defined benefit pension plans are expected to be \$49 million of which \$37 million are contractually required.

JPMorgan Chase also has a number of defined benefit pension plans not subject to Title IV of the Employee Retirement Income Security Act. The most significant of these plans is the Excess Retirement Plan, pursuant to which certain employees earn pay and interest credits on compensation amounts above the maximum stipulated by law under a qualified plan. The Firm announced that, effective May 1, 2009, pay credits would no longer be provided on compensation amounts above the maximum stipulated by law. The Excess Retirement Plan had an unfunded projected benefit obligation in the amount of \$272 million and \$266 million, at December 31, 2011 and 2010, respectively.

Defined contribution plans

JPMorgan Chase currently provides two qualified defined contribution plans in the U.S. and other similar arrangements in certain non-U.S. locations, all of which are administered in accordance with applicable local laws and regulations. The most significant of these plans is The JPMorgan Chase 401(k) Savings Plan (the "401(k) Savings Plan"), which covers substantially all U.S. employees. The 401(k) Savings Plan allows employees to make pretax and Roth 401(k) contributions to tax-deferred investment portfolios. The JPMorgan Chase Common Stock Fund, which is an investment option under the 401(k) Savings Plan, is a nonleveraged employee stock ownership plan.

The Firm matched eligible employee contributions up to 5% of benefits-eligible compensation (e.g., base pay) on a per pay period basis through April 30, 2009; commencing May 1, 2009 matching contributions are made annually. Employees begin to receive matching contributions after completing a one-year-of-service requirement. Employees with total annual cash compensation of \$250,000 or more are not eligible for matching contributions. Matching contributions are immediately vested for employees hired before May 1, 2009, and will vest after three years of service for employees hired on or after May 1, 2009. The 401(k) Savings Plan also permits discretionary profit-sharing contributions by participating companies for certain employees, subject to a specified vesting schedule.

Effective August 10, 2009, JPMorgan Chase Bank, N.A. became the sponsor of the WaMu Savings Plan and that plan's assets were merged into the 401(k) Savings Plan effective March 31, 2010.

OPEB plans

JPMorgan Chase offers postretirement medical and life insurance benefits to certain retirees and postretirement medical benefits to qualifying U.S. employees. These benefits vary with length of service and date of hire and provide for limits on the Firm's share of covered medical benefits. The medical and life insurance benefits are both contributory. Postretirement medical benefits also are offered to qualifying U.K. employees.

JPMorgan Chase's U.S. OPEB obligation is funded with corporate-owned life insurance ("COLI") purchased on the lives of eligible employees and retirees. While the Firm owns the COLI policies, COLI proceeds (death benefits, withdrawals and other distributions) may be used only to reimburse the Firm for its net postretirement benefit claim payments and related administrative expense. The U.K. OPEB plan is unfunded.

The following table presents the changes in benefit obligations, plan assets and funded status amounts reported on the Consolidated Balance Sheets for the Firm's U.S. and non-U.S. defined benefit pension and OPEB plans.

214 JPMorgan Chase & Co./2011 Annual Report

Notes to consolidated financial statements

The estimated pretax amounts that will be amortized from AOCI into net periodic benefit cost in 2012 are as follows.

(in millions)	Defined benefit pension plans		OPEB plans	
	U.S.	Non-U.S.	U.S.	Non-U.S.
Net loss	\$ 287	\$ 36	\$ 7	\$ —
Prior service cost/(credit)	(41)	(1)	(1)	—
Total	\$ 246	\$ 35	\$ 6	\$ —

The following table presents the actual rate of return on plan assets for the U.S. and non-U.S. defined benefit pension and OPEB plans.

Year ended December 31,	U.S.			Non-U.S.		
	2011	2010	2009	2011	2010	2009
Actual rate of return:						
Defined benefit pension plans	0.72%	12.23%	13.78%	(4.29)-13.12%	0.77-10.65%	3.17-22.43%
OPEB plans	5.22%	11.23%	15.93%	NA	NA	NA

Plan assumptions

JPMorgan Chase's expected long-term rate of return for U.S. defined benefit pension and OPEB plan assets is a blended average of the investment advisor's projected long-term (10 years or more) returns for the various asset classes, weighted by the asset allocation. Returns on asset classes are developed using a forward-looking approach and are not strictly based on historical returns. Equity returns are generally developed as the sum of inflation, expected real earnings growth and expected long-term dividend yield. Bond returns are generally developed as the sum of inflation, real bond yield and risk spread (as appropriate), adjusted for the expected effect on returns from changing yields. Other asset-class returns are derived from their relationship to the equity and bond markets. Consideration is also given to current market conditions and the short-term portfolio mix of each plan; as a result, in 2011 the Firm generally maintained the same expected return on assets as in the prior year.

For the U.K. defined benefit pension plans, which represent the most significant of the non-U.S. defined benefit pension plans, procedures similar to those in the U.S. are used to develop the expected long-term rate of return on plan

assets, taking into consideration local market conditions and the specific allocation of plan assets. The expected long-term rate of return on U.K. plan assets is an average of projected long-term returns for each asset class. The return on equities has been selected by reference to the yield on long-term U.K. government bonds plus an equity risk premium above the risk-free rate. The expected return on "AA" rated long-term corporate bonds is based on an implied yield for similar bonds.

The discount rate used in determining the benefit obligation under the U.S. defined benefit pension and OPEB plans was selected by reference to the yields on portfolios of bonds with maturity dates and coupons that closely match each of the plan's projected cash flows; such portfolios are derived from a broad-based universe of high-quality corporate bonds as of the measurement date. In years in which these hypothetical bond portfolios generate excess cash, such excess is assumed to be reinvested at the one-year forward rates implied by the Citigroup Pension Discount Curve published as of the measurement date. The discount rate for the U.K. defined benefit pension and OPEB plans represents a rate implied from the yield curve of the year-end iBoxx £ corporate "AA" 15-year-plus bond index.

The following tables present the weighted-average annualized actuarial assumptions for the projected and accumulated postretirement benefit obligations, and the components of net periodic benefit costs, for the Firm's significant U.S. and non-U.S. defined benefit pension and OPEB plans, as of and for the periods indicated.

Weighted-average assumptions used to determine benefit obligations

December 31,	U.S.		Non-U.S.	
	2011	2010	2011	2010
Discount rate:				
Defined benefit pension plans	4.60%	5.50%	1.50-4.80%	1.60-5.50%
OPEB plans	4.70	5.50	—	—
Rate of compensation increase	4.00	4.00	2.75-4.20	3.00-4.50
Health care cost trend rate:				
Assumed for next year	7.00	7.00	—	—
Ultimate	5.00	5.00	—	—
Year when rate will reach ultimate	2017	2017	—	—

Weighted-average assumptions used to determine net periodic benefit costs

Year ended December 31,	U.S.			Non-U.S.		
	2011	2010	2009	2011	2010	2009
Discount rate:						
Defined benefit pension plans	5.50%	6.00%	6.65%	1.60-5.50%	2.00-5.70%	2.00-6.20%
OPEB plans	5.50	6.00	6.70	—	—	—
Expected long-term rate of return on plan assets:						
Defined benefit pension plans	7.50	7.50	7.50	2.40-5.40	2.40-6.20	2.50-6.90
OPEB plans	6.25	7.00	7.00	NA	NA	NA
Rate of compensation increase	4.00	4.00	4.00	3.00-4.50	3.00-4.50	3.00-4.00
Health care cost trend rate:						
Assumed for next year	7.00	7.75	8.50	—	—	—
Ultimate	5.00	5.00	5.00	—	—	—
Year when rate will reach ultimate	2017	2014	2014	—	—	—

The following table presents the effect of a one-percentage-point change in the assumed health care cost trend rate on JPMorgan Chase's total service and interest cost and accumulated postretirement benefit obligation.

Year ended December 31, 2011(in millions)	1-Percentage point increase	1-Percentage point decrease
Effect on total service and interest cost	\$ 1	\$(1)
Effect on accumulated postretirement benefit obligation	27	(24)

At December 31, 2011, the Firm decreased the discount rates used to determine its benefit obligations for the U.S. defined benefit pension and OPEB plans in light of current market interest rates, which will result in an increase in expense of approximately \$47 million for 2012. The 2012 expected long-term rate of return on U.S. defined benefit pension plan assets and U.S. OPEB plan assets are 7.50% and 6.25%, respectively, unchanged from 2011. For 2012, the initial health care benefit obligation trend assumption will be set at 7.00%, and the ultimate health care trend assumption and year to reach ultimate rate will remain at 5.00% and 2017, respectively, unchanged from 2011. As of December 31, 2011, the assumed rate of compensation increase remained at 4.00%. The 2012 interest crediting rate assumption will be set at 5.00%, as compared to 5.25% in 2011.

JPMorgan Chase's U.S. defined benefit pension and OPEB plan expense is sensitive to the expected long-term rate of return on plan assets and the discount rate. With all other assumptions held constant, a 25-basis point decline in the expected long-term rate of return on U.S. plan assets would result in an increase of approximately an aggregate \$29 million in 2012 U.S. defined benefit pension and OPEB plan expense. A 25-basis point decline in the discount rate for the U.S. plans would result in an increase in 2012 U.S. defined benefit pension and OPEB plan expense of approximately an aggregate \$17 million and an increase in the related benefit obligations of approximately an aggregate \$192 million. A 25-basis point increase in the interest crediting rate for the U.S. defined benefit pension plan would result in an increase in 2012 U.S. defined

benefit pension expense of approximately \$19 million and an increase in the related projected benefit obligations of approximately \$82 million. A 25-basis point decline in the discount rates for the non-U.S. plans would result in an increase in the 2012 non-U.S. defined benefit pension plan expense of approximately \$11 million.

Investment strategy and asset allocation

The Firm's U.S. defined benefit pension plan assets are held in trust and are invested in a well-diversified portfolio of equity and fixed income securities, real estate, cash and cash equivalents, and alternative investments (e.g., hedge funds, private equity, real estate and real assets). Non-U.S. defined benefit pension plan assets are held in various trusts and are also invested in well-diversified portfolios of equity, fixed income and other securities. Assets of the Firm's COLI policies, which are used to partially fund the U.S. OPEB plan, are held in separate accounts with an insurance company and are invested in equity and fixed income index funds.

The investment policy for the Firm's U.S. defined benefit pension plan assets is to optimize the risk-return relationship as appropriate to the needs and goals using a global portfolio of various asset classes diversified by market segment, economic sector, and issuer. Assets are managed by a combination of internal and external investment managers. Periodically the Firm performs a comprehensive analysis on the U.S. defined benefit pension plan asset allocations, incorporating projected asset and liability data, which focuses on the short-and long-term impact of the asset allocation on cumulative pension expense, economic cost, present value of contributions and funded status. Currently, approved asset allocation ranges are: U.S. equity 15% to 35%, international equity 15% to 25%, debt securities 10% to 30%, hedge funds 10% to 30%, and real estate, real assets and private equity 5% to 20%. Asset allocations are not managed to a specific target but seek to shift asset class allocations within these stated ranges. Investment strategies incorporate the economic outlook, anticipated implications of the macroeconomic environment on the various asset classes/managers, and maintaining an appropriate level of liquidity for the plan.

Notes to consolidated financial statements

The Firm regularly reviews the asset allocations and all factors that continuously impact the portfolio, which is rebalanced when deemed necessary.

For the U.K. defined benefit pension plans, which represent the most significant of the non-U.S. defined benefit pension plans, the assets are invested to maximize returns subject to an appropriate level of risk relative to the plans' liabilities. In order to reduce the volatility in returns relative to the plan's liability profiles, the U.K. defined benefit pension plans' largest asset allocations are to debt securities of appropriate durations. Other assets, mainly equity securities, are then invested for capital appreciation, to provide long-term investment growth. Similar to the U.S. defined benefit pension plan, asset allocations for the U.K. plans are reviewed and rebalanced on a regular basis.

Investments held by the Plans include financial instruments which are exposed to various risks such as interest rate, market and credit risks. Exposure to a concentration of credit risk is mitigated by the broad diversification of both U.S. and non-U.S. investment instruments. Additionally, the investments in each of the common/collective trust funds and registered investment companies are further diversified into various financial instruments. As of December 31, 2011, assets held by the Firm's U.S. and non-U.S. defined benefit pension and OPEB plans do not include JPMorgan Chase common stock, except in connection with investments in third-party stock-index funds. The plans hold investments in funds that are sponsored or managed by affiliates of JPMorgan Chase in the amount of \$1.6 billion and \$1.7 billion for U.S. plans and \$194 million and \$155 million for non-U.S. plans, as of December 31, 2011 and 2010, respectively.

The following table presents the weighted-average asset allocation of the fair values of total plan assets at December 31 for the years indicated, as well as the respective approved range/target allocation by asset category, for the Firm's U.S. and non-U.S. defined benefit pension and OPEB plans.

December 31, Asset category	Defined benefit pension plans						OPEB plans		
	U.S.			Non-U.S.					
	Target Allocation	% of plan assets 2011	2010	Target Allocation	% of plan assets 2011	2010	Target Allocation	% of plan assets 2011	2010
Debt securities ^(a)	10–30%	20%	29%	72%	74%	71%	50%	50%	50%
Equity securities	25–60	39	40	27	25	28	50	50	50
Real estate	5–20	5	4	—	—	—	—	—	—
Alternatives ^(b)	15–50	36	27	1	1	1	—	—	—
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%

(a) Debt securities primarily include corporate debt, U.S. federal, state, local and non-U.S. government, and mortgage-backed securities.

(b) Alternatives primarily include limited partnerships.

(c) Represents the U.S. OPEB plan only, as the U.K. OPEB plan is unfunded.

Fair value measurement of the plans' assets and liabilities

For information on fair value measurements, including descriptions of level 1, 2, and 3 of the fair value hierarchy and the valuation methods employed by the Firm, see Note 3 on pages 184–198 of this Annual Report.

Pension and OPEB plan assets and liabilities measured at fair value

December 31, 2011 (in millions)	U.S. defined benefit pension plans				Non-U.S. defined benefit pension plans			
	Level 1	Level 2	Level 3	Total fair value	Level 1	Level 2	Level 3	Total fair value
Cash and cash equivalents	\$ 117	\$ —	\$ —	\$ 117	\$ 72	\$ —	\$ —	\$ 72
Equity securities:								
Capital equipment	607	7	—	614	69	12	—	81
Consumer goods	657	—	—	657	64	30	—	94
Banks and finance companies	301	2	—	303	83	13	—	96
Business services	332	—	—	332	48	10	—	58
Energy	173	—	—	173	52	10	—	62
Materials	161	—	1	162	35	6	—	41
Real Estate	11	—	—	11	1	—	—	1
Other	766	274	—	1,040	160	5	—	165
Total equity securities	3,008	283	1	3,292	512	86	—	598
Common/collective trust funds	401	1,125	202	1,728	138	170	—	308
Limited partnerships ⁽¹⁾								
Hedge funds	—	933	1,039	1,972	—	—	—	—
Private equity	—	—	1,367	1,367	—	—	—	—
Real estate	—	—	306	306	—	—	—	—
Real assets ⁽²⁾	—	—	264	264	—	—	—	—
Total limited partnerships	—	933	2,976	3,909	—	—	—	—
Corporate debt securities ⁽³⁾	—	544	2	546	—	958	—	958
U.S. federal, state, local and non-U.S. government debt securities	—	328	—	328	—	904	—	904
Mortgage-backed securities	122	36	—	158	17	—	—	17
Derivative receivables	1	2	—	3	—	7	—	7
Other ⁽⁴⁾	102	60	427	589	74	65	—	139
Total assets measured at fair value ⁽⁵⁾	\$ 3,751	\$ 3,311	\$ 3,608	\$ 10,670	\$ 813	\$ 2,190	\$ —	\$ 3,003
Derivative payables	—	(3)	—	(3)	—	(1)	—	(1)
Total liabilities measured at fair value	\$ —	\$ (3)	\$ —	\$ (3) ⁽⁶⁾	\$ —	\$ (1)	\$ —	\$ (1)

Notes to consolidated financial statements

December 31, 2010 (in millions)	U.S. defined benefit pension plans				Non-U.S. defined benefit pension plans			
	Level 1	Level 2	Level 3	Total fair value	Level 1	Level 2	Level 3	Total fair value
Cash and cash equivalents	\$ —	\$ —	\$ —	\$ —	\$ 81	\$ —	\$ —	\$ 81
Equity securities:								
Capital equipment	748	9	—	757	68	13	—	81
Consumer goods	712	—	—	712	75	21	—	96
Banks and finance companies	414	1	—	415	113	9	—	122
Business services	444	—	—	444	53	10	—	63
Energy	195	—	—	195	59	6	—	65
Materials	205	—	—	205	50	13	—	63
Real estate	21	—	—	21	1	—	—	1
Other	857	6	—	863	194	16	—	210
Total equity securities	3,596	16	—	3,612	613	88	—	701
Common/collective trust funds ^(a)	436	1,263	194	1,893	46	180	—	226
Limited partnerships ^(b)								
Hedge funds	—	959	1,160	2,119	—	—	—	—
Private equity	—	—	1,232	1,232	—	—	—	—
Real estate	—	—	304	304	—	—	—	—
Real assets ^(c)	—	—	—	—	—	—	—	—
Total limited partnerships	—	959	2,696	3,655	—	—	—	—
Corporate debt securities ^(d)	—	424	1	425	—	718	—	718
U.S. federal, state, local and non-U.S. government debt securities	—	453	—	453	—	864	—	864
Mortgage-backed securities	188	55	—	243	1	—	—	1
Derivative receivables	2	194	—	196	—	3	—	3
Other ^(f)	218	58	387	663	18	51	—	69
Total assets measured at fair value ^(e)	\$ 4,440	\$ 3,422	\$ 3,278	\$ 11,140	\$ 759	\$ 1,904	\$ —	\$ 2,663
Derivative payables	—	(177)	—	(177)	—	(25)	—	(25)
Total liabilities measured at fair value	\$ —	\$ (177)	\$ —	\$ (177) ⁽ⁱ⁾	\$ —	\$ (25)	\$ —	\$ (25)

- (a) At December 31, 2011 and 2010, common/collective trust funds generally include commingled funds that primarily included 23% and 22%, respectively, of short-term investment funds; 19% and 21%, respectively, of equity (index) investments; and 19% and 16%, respectively, of international investments.
- (b) The prior period has been revised to consider redemption notification periods, in determining the classification of investments within the fair value hierarchy.
- (c) Unfunded commitments to purchase limited partnership investments for the Plans were \$1.2 billion and \$1.1 billion for 2011 and 2010, respectively.
- (d) Real assets include investments in productive assets such as agriculture, energy rights, mining and timber properties and exclude raw land to be developed for real estate purposes.
- (e) Corporate debt securities include debt securities of U.S. and non-U.S. corporations.
- (f) Other consists of exchange traded funds and participating and non-participating annuity contracts. Exchange traded funds are primarily classified with level 1 of the fair value hierarchy given they are valued using market observable prices. Participating and non-participating annuity contracts are classified within level 3 of the fair value hierarchy due to lack of market mechanisms for transferring each policy and surrender restrictions.
- (g) At December 31, 2011 and 2010, the fair value of investments valued at NAV were \$3.9 billion and \$4.1 billion, respectively, which were classified within the valuation hierarchy as follows: \$0.4 billion and \$0.5 billion in level 1, \$2.1 billion and \$2.2 billion in level 2 and \$1.4 billion and \$1.4 billion in level 3.
- (h) At December 31, 2011 and 2010, excluded U.S. defined benefit pension plan receivables for investments sold and dividends and interest receivables of \$50 million and \$52 million, respectively; and excluded non-U.S. defined benefit pension plan receivables for dividends and interest receivables of \$56 million and \$9 million, respectively.
- (i) At December 31, 2011 and 2010, excluded \$241 million and \$149 million, respectively, of U.S. defined benefit pension plan payables for investments purchased; and \$4 million and \$38 million, respectively, of other liabilities; and excluded non-U.S. defined benefit pension plan payables for investments purchased of \$69 million at December 31, 2011.

The Firm's OPEB plan was partially funded with COLI policies of \$1.4 billion, at December 31, 2011 and 2010, respectively, which were classified in level 3 of the valuation hierarchy.

Changes in level 3 fair value measurements using significant unobservable inputs

Year ended December 31, 2011 (in millions)	Fair value, January 1, 2011	Actual return on plan assets		Purchases, sales and settlements, net	Transfers in and/or out of level 3	Fair value, December 31, 2011
		Realized gains/(losses)	Unrealized gains/(losses)			
U.S. defined benefit pension plans						
Equities	\$ —	\$ —	\$ —	\$ —	\$ 1	\$ 1
Common/collective trust funds	194	35	1	(28)	—	202
Limited partnerships:						
Hedge funds	1,160	(16)	27	(76)	(56)	1,039
Private equity	1,232	56	2	77	—	1,367
Real estate	304	8	40	14	(60)	306
Real assets	—	5	(7)	150	116	264
Total limited partnerships	2,696	53	62	165	—	2,976
Corporate debt securities	1	—	—	1	—	2
Other	387	—	41	(1)	—	427
Total U.S. plans	\$ 3,278	\$ 88	\$ 104	\$ 137	\$ 1	\$ 3,608
Non-U.S. defined benefit pension plans						
Other	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —
Total non-U.S. plans	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —
OPEB plans						
COLI	\$ 1,381	\$ —	\$ 70	\$ (24)	\$ —	\$ 1,427
Total OPEB plans	\$ 1,381	\$ —	\$ 70	\$ (24)	\$ —	\$ 1,427

Year ended December 31, 2010 (in millions)	Fair value, January 1, 2010	Actual return on plan assets		Purchases, sales and settlements, net	Transfers in and/or out of level 3	Fair value, December 31, 2010
		Realized gains/(losses)	Unrealized gains/(losses)			
U.S. defined benefit pension plans						
Equities	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —
Common/collective trust funds	284	—	(90)	—	—	194
Limited partnerships:						
Hedge funds	680	(1)	14	388	79	1,160
Private equity	874	3	108	235	12	1,232
Real estate	196	3	16	89	—	304
Real assets	—	—	—	—	—	—
Total limited partnerships	1,750	5	138	712	91	2,696
Corporate debt securities	—	—	—	—	1	1
Other	334	—	53	—	—	387
Total U.S. plans	\$ 2,368	\$ 5	\$ 101	\$ 712	\$ 92	\$ 3,278
Non-U.S. defined benefit pension plans						
Other	\$ 13	\$ —	\$ (1)	\$ (12)	\$ —	\$ —
Total non-U.S. plans	\$ 13	\$ —	\$ (1)	\$ (12)	\$ —	\$ —
OPEB plans						
COLI	\$ 1,269	\$ —	\$ 137	\$ (25)	\$ —	\$ 1,381
Total OPEB plans	\$ 1,269	\$ —	\$ 137	\$ (25)	\$ —	\$ 1,381

Notes to consolidated financial statements

Year ended December 31, 2009 (in millions)	Fair value, January 1, 2009	Actual return on plan assets		Purchases, sales and settlements, net	Transfers in and/or out of level 3	Fair value, December 31, 2009
		Realized gains/(losses)	Unrealized gains/(losses)			
U.S. defined benefit pension plans						
Equities	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —
Common/collective trust funds	340	—	(56)	—	—	284
Limited partnerships:						
Hedge funds	553	—	136	(9)	—	680
Private equity	810	—	(1)	80	(15)	874
Real estate	203	—	(107)	100	—	196
Real assets	—	—	—	—	—	—
Total limited partnerships	1,566	—	28	171	(15)	1,750
Corporate debt securities	—	—	—	—	—	—
Other	315	—	19	—	—	334
Total U.S. plans	\$ 2,221	\$ —	\$ (9)	\$ 171	\$ (15)	\$ 2,368
Non-U.S. defined benefit pension plans						
Other	\$ 14	\$ —	\$ (1)	\$ —	\$ —	\$ 13
Total non-U.S. plans	\$ 14	\$ —	\$ (1)	\$ —	\$ —	\$ 13
OPEB plans						
COLI	\$ 1,126	\$ —	\$ 172	\$ (29)	\$ —	\$ 1,269
Total OPEB plans	\$ 1,126	\$ —	\$ 172	\$ (29)	\$ —	\$ 1,269

(a) The prior period has been revised to consider redemption notification periods in determining the classification of investments within the fair value hierarchy.

Estimated future benefit payments

The following table presents benefit payments expected to be paid, which include the effect of expected future service, for the years indicated. The OPEB medical and life insurance payments are net of expected retiree contributions.

Year ended December 31, (in millions)	U.S. defined benefit pension plans	Non-U.S. defined benefit pension plans	OPEB before Medicare Part D subsidy	Medicare Part D subsidy
2012	\$ 1,038	\$ 95	\$ 96	\$ 11
2013	1,035	99	95	12
2014	610	101	94	13
2015	610	110	92	14
2016	613	116	90	14
Years 2017–2021	3,084	658	404	80

Note 10 – Employee stock-based incentives

Employee stock-based awards

In 2011, 2010 and 2009, JPMorgan Chase granted long-term stock-based awards to certain key employees under the 2005 Long-Term Incentive Plan (the “2005 Plan”). The 2005 Plan became effective on May 17, 2005, and was last amended in May 2011. Under the terms of the amended 2005 plan, as of December 31, 2011, 318 million shares of common stock are available for issuance through May 2015. The amended 2005 Plan is the only active plan under which the Firm is currently granting stock-based incentive awards. In the following discussion, the 2005 Plan, plus prior Firm plans and plans assumed as the result of acquisitions, are referred to collectively as the “LTI Plans,” and such plans constitute the Firm’s stock-based incentive plans.

Restricted stock units (“RSUs”) are awarded at no cost to the recipient upon their grant. RSUs are generally granted annually and generally vest at a rate of 50% after two

years and 50% after three years and convert into shares of common stock at the vesting date. In addition, RSUs typically include full-career eligibility provisions, which allow employees to continue to vest upon voluntary termination, subject to post-employment and other restrictions based on age or service-related requirements. All of these awards are subject to forfeiture until vested and contain clawback provisions that may result in cancellation prior to vesting under certain specified circumstances. RSUs entitle the recipient to receive cash payments equivalent to any dividends paid on the underlying common stock during the period the RSUs are outstanding and, as such, are considered participating securities as discussed in Note 24 on page 277 of this Annual Report.

Under the LTI Plans, stock options and stock appreciation rights (“SARs”) have generally been granted with an exercise price equal to the fair value of JPMorgan Chase’s common stock on the grant date. The Firm typically awards SARs to certain key employees once per year; the Firm also

periodically grants employee stock options and SARs to individual employees. The 2011, 2010 and 2009 grants of SARs to key employees vest ratably over five years (i.e., 20% per year) and contain clawback provisions similar to RSUs. The 2011 and 2010 grants of SARs contain full-career eligibility provisions; the 2009 grants of SARs do not include any full-career eligibility provisions. SARs generally expire 10 years after the grant date.

The Firm separately recognizes compensation expense for each tranche of each award as if it were a separate award with its own vesting date. Generally, for each tranche granted, compensation expense is recognized on a straight-line basis from the grant date until the vesting date of the respective tranche, provided that the employees will not become full-career eligible during the vesting period. For awards with full-career eligibility provisions and awards granted with no future substantive service requirement, the Firm accrues the estimated value of awards expected to be awarded to employees as of the grant date without giving consideration to the impact of post-employment restrictions. For each tranche granted to employees who will become full-career eligible during the vesting period, compensation expense is recognized on a straight-line basis from the grant date until the earlier of the employee's full-career eligibility date or the vesting date of the respective tranche.

The Firm's policy for issuing shares upon settlement of employee stock-based incentive awards is to issue either new shares of common stock or treasury shares. During 2011, 2010 and 2009, the Firm settled all of its employee stock-based awards by issuing treasury shares.

In January 2008, the Firm awarded to its Chairman and Chief Executive Officer up to 2 million SARs. The terms of this award are distinct from, and more restrictive than, other equity grants regularly awarded by the Firm. The SARs, which have a 10-year term, will become exercisable no earlier than January 22, 2013, and have an exercise price of \$39.83. The number of SARs that will become exercisable (ranging from none to the full 2 million) and their exercise date or dates may be determined by the Board of Directors based on an annual assessment of the performance of both the CEO and JPMorgan Chase. The Firm recognizes this award ratably over an assumed five-year service period, subject to a requirement to recognize changes in the fair value of the award through the grant date. The Firm recognized \$(4) million, \$4 million and \$9 million in compensation expense in 2011, 2010 and 2009, respectively, for this award.

RSUs, employee stock options and SARs activity

Compensation expense for RSUs is measured based on the number of shares granted multiplied by the stock price at the grant date, and for employee stock options and SARs, is measured at the grant date using the Black-Scholes valuation model. Compensation expense for these awards is recognized in net income as described previously. The following table summarizes JPMorgan Chase's RSUs, employee stock options and SARs activity for 2011.

Year ended December 31, 2011 (in thousands, except weighted-average data, and where otherwise stated)	RSUs		Options/SARs			
	Number of shares	Weighted-average grant date fair value	Number of awards	Weighted-average exercise price	Weighted-average remaining contractual life (in years)	Aggregate intrinsic value
Outstanding, January 1	234,121	\$ 30.45	234,527	\$ 43.33		
Granted	59,697	44.05	15,300	44.27		
Exercised or vested	(121,699)	26.95	(15,409)	32.27		
Forfeited	(5,488)	37.05	(4,168)	39.56		
Canceled	NA	NA	(74,489)	51.77		
Outstanding, December 31	166,631	\$ 37.65	155,761	\$ 40.58	4.6	\$ 419,887
Exercisable, December 31	NA	NA	106,335	41.89	3.1	260,309

The total fair value of RSUs that vested during the years ended December 31, 2011, 2010 and 2009, was \$5.4 billion, \$2.3 billion and \$1.3 billion, respectively. The weighted-average grant date per share fair value of stock options and SARs granted during the years ended December 31, 2011, 2010 and 2009, was \$13.04, \$12.27 and \$8.24, respectively. The total intrinsic value of options exercised during the years ended December 31, 2011, 2010 and 2009, was \$191 million, \$154 million and \$154 million, respectively.

Notes to consolidated financial statements

Compensation expense

The Firm recognized the following noncash compensation expense related to its various employee stock-based incentive plans in its Consolidated Statements of Income.

Year ended December 31, (in millions)	2011	2010	2009
Cost of prior grants of RSUs and SARs that are amortized over their applicable vesting periods	\$ 1,986	\$ 2,479	\$ 2,510
Accrual of estimated costs of RSUs and SARs to be granted in future periods including those to full-career eligible employees	689	772	845
Total noncash compensation expense related to employee stock-based incentive plans	\$ 2,675	\$ 3,251	\$ 3,355

At December 31, 2011, approximately \$1.3 billion (pretax) of compensation cost related to unvested awards had not yet been charged to net income. That cost is expected to be amortized into compensation expense over a weighted-average period of 1.0 year. The Firm does not capitalize any compensation cost related to share-based compensation awards to employees.

Cash flows and tax benefits

Income tax benefits related to stock-based incentive arrangements recognized in the Firm's Consolidated Statements of Income for the years ended December 31, 2011, 2010 and 2009, were \$1.0 billion, \$1.3 billion and \$1.3 billion, respectively.

The following table sets forth the cash received from the exercise of stock options under all stock-based incentive arrangements, and the actual income tax benefit realized related to tax deductions from the exercise of the stock options.

Year ended December 31, (in millions)	2011	2010	2009
Cash received for options exercised	\$ 354	\$ 205	\$ 437
Tax benefit realized	31	14	11

(a) The tax benefit realized from dividends or dividend equivalents paid on equity-classified share-based payment awards that are charged to retained earnings are recorded as an increase to additional paid-in capital and included in the pool of excess tax benefits available to absorb tax deficiencies on share-based payment awards.

Valuation assumptions

The following table presents the assumptions used to value employee stock options and SARs granted during the years ended December 31, 2011, 2010 and 2009, under the Black-Scholes valuation model.

Year ended December 31,	2011	2010	2009
Weighted-average annualized valuation assumptions			
Risk-free interest rate	2.58%	3.89%	2.33%
Expected dividend yield	2.20	3.13	3.40
Expected common stock price volatility	34	37	56
Expected life (in years)	6.5	6.4	6.6

(a) In 2011, the expected dividend yield was determined using forward-looking assumptions. In 2010 and 2009 the expected dividend yield was determined using historical dividend yields.

The expected volatility assumption is derived from the implied volatility of JPMorgan Chase's stock options. The expected life assumption is an estimate of the length of time that an employee might hold an option or SAR before it is exercised or canceled, and the assumption is based on the Firm's historical experience.

Note 11 – Noninterest expense

The following table presents the components of noninterest expense.

Year ended December 31, (in millions)	2011	2010	2009
Compensation expense	\$ 29,037	\$ 28,124	\$ 26,928
Noncompensation expense:			
Occupancy expense	3,895	3,681	3,666
Technology, communications and equipment expense	4,947	4,684	4,624
Professional and outside services	7,482	6,767	6,232
Marketing	3,143	2,446	1,777
Other expense ^(c)	13,559	14,558	7,594
Amortization of intangibles	848	936	1,050
Total noncompensation expense	33,874	33,072	24,943
Merger costs	—	—	481 ^(d)
Total noninterest expense	\$ 62,911	\$ 61,196	\$ 52,352

(a) Expense for 2010 includes a payroll tax expense related to the United Kingdom ("U.K.") Bank Payroll Tax on certain compensation awarded from December 9, 2009, to April 5, 2010, to relevant banking employees.

(b) Included litigation expense of \$4.9 billion, \$7.4 billion and \$161 million for the years ended December 31, 2011, 2010 and 2009, respectively.

(c) Included foreclosed property expense of \$718 million, \$1.0 billion and \$1.4 billion for the years ended December 31, 2011, 2010 and 2009, respectively.

(d) Total merger-related costs for the year ended December 31, 2009, were comprised of \$247 million in compensation costs, \$12 million in occupancy costs, and \$222 million in technology and communications and other costs.

Note 12 – Securities

Securities are primarily classified as AFS or trading. Trading securities are discussed in Note 3 on pages 184–198 of this Annual Report. Securities are classified primarily as AFS when used to manage the Firm's exposure to interest rate movements or used for longer-term strategic purposes. AFS securities are carried at fair value on the Consolidated Balance Sheets. Unrealized gains and losses, after any applicable hedge accounting adjustments, are reported as net increases or decreases to accumulated other comprehensive income/(loss). The specific identification method is used to determine realized gains and losses on AFS securities, which are included in securities gains/(losses) on the Consolidated Statements of Income.

Other-than-temporary impairment

AFS debt and equity securities in unrealized loss positions are analyzed as part of the Firm's ongoing assessment of other-than-temporary impairment ("OTTI"). For most types of debt securities, the Firm considers a decline in fair value to be other-than-temporary when the Firm does not expect to recover the entire amortized cost basis of the security. For beneficial interests in securitizations that are rated below "AA" at their acquisition, or that can be contractually prepaid or otherwise settled in such a way that the Firm would not recover substantially all of its recorded investment, the Firm considers an OTTI to have occurred when there is an adverse change in expected cash flows. For AFS equity securities, the Firm considers a decline in fair value to be other-than-temporary if it is probable that the Firm will not recover its amortized cost basis.

Potential OTTI is considered using a variety of factors, including the length of time and extent to which the market value has been less than cost; adverse conditions specifically related to the industry, geographic area or financial condition of the issuer or underlying collateral of a security; payment structure of the security; changes to the rating of the security by a rating agency; the volatility of the fair value changes; and the Firm's intent and ability to hold the security until recovery.

For debt securities, the Firm recognizes OTTI losses in earnings if the Firm has the intent to sell the debt security, or if it is more likely than not that the Firm will be required to sell the debt security before recovery of its amortized cost basis. In these circumstances the impairment loss is equal to the full difference between the amortized cost basis and the fair value of the securities. When the Firm has the intent and ability to hold AFS debt securities in an unrealized loss position, it evaluates the expected cash flows to be received and determines if a credit loss exists. In the event of a credit loss, only the amount of impairment associated with the credit loss is recognized in income. Amounts relating to factors other than credit losses are recorded in OCI.

The Firm's cash flow evaluations take into account the factors noted above and expectations of relevant market and economic data as of the end of the reporting period. For securities issued in a securitization, the Firm estimates cash flows considering underlying loan-level data and structural features of the securitization, such as subordination, excess spread, overcollateralization or other forms of credit enhancement, and compares the losses projected for the underlying collateral ("pool losses") against the level of credit enhancement in the securitization structure to determine whether these features are sufficient to absorb the pool losses, or whether a credit loss exists. The Firm also performs other analyses to support its cash flow projections, such as first-loss analyses or stress scenarios.

For equity securities, OTTI losses are recognized in earnings if the Firm intends to sell the security. In other cases the Firm considers the relevant factors noted above, as well as the Firm's intent and ability to retain its investment for a period of time sufficient to allow for any anticipated recovery in market value, and whether evidence exists to support a realizable value equal to or greater than the carrying value. Any impairment loss on an equity security is equal to the full difference between the amortized cost basis and the fair value of the security.

Realized gains and losses

The following table presents realized gains and losses and credit losses that were recognized in income from AFS securities.

Year ended December 31, (in millions)	2011	2010	2009
Realized gains	\$ 1,811	\$ 3,382	\$ 2,268
Realized losses	(142)	(317)	(580)
Net realized gains ^(a)	1,669	3,065	1,688
Credit losses included in securities gains	(76)	(100)	(578)
Net securities gains	\$ 1,593	\$ 2,965	\$ 1,110

- (a) Proceeds from securities sold were within approximately 4% of amortized cost in 2011, and within approximately 3% of amortized cost in 2010 and 2009.
- (b) Includes other-than-temporary impairment losses recognized in income on certain prime mortgage-backed securities for the year ended December 31, 2011; certain prime mortgage-backed securities and obligations of U.S. states and municipalities for the year ended December 31, 2010; and certain prime and subprime mortgage-backed securities and obligations of U.S. states and municipalities for the year ended December 31, 2009.

Notes to consolidated financial statements

The amortized costs and estimated fair values of AFS and held-to-maturity ("HTM") securities were as follows for the dates indicated.

December 31, (in millions)	2011				2010			
	Amortized cost	Gross unrealized gains	Gross unrealized losses	Fair value	Amortized cost	Gross unrealized gains	Gross unrealized losses	Fair value
Available-for-sale debt securities								
Mortgage-backed securities:								
U.S. government agencies	\$ 101,968	\$ 5,141	\$ 2	\$ 107,107	\$ 117,364	\$ 3,159	\$ 297	\$ 120,226
Residential:								
Prime and Alt-A	2,170	54	218 ^(c)	2,006	2,173	81	250 ^(c)	2,004
Subprime	1	—	—	1	1	—	—	1
Non-U.S.	66,067	170	687	65,550	47,089	290	409	46,970
Commercial	10,632	650	53	11,229	5,169	502	17	5,654
Total mortgage-backed securities	180,838	6,015	960	185,893	171,796	4,032	973	174,855
U.S. Treasury and government agencies	8,184	169	2	8,351	11,258	118	28	11,348
Obligations of U.S. states and municipalities	15,404	1,184	48	16,540	11,732	165	338	11,559
Certificates of deposit	3,017	—	—	3,017	3,648	1	2	3,647
Non-U.S. government debt securities	44,944	402	81	45,265	20,614	191	28	20,777
Corporate debt securities	63,607	216	1,647	62,176	61,717	495	419	61,793
Asset-backed securities:								
Credit card receivables	4,506	149	—	4,655	7,278	335	5	7,608
Collateralized loan obligations	24,474	553	166	24,861	13,336	472	210	13,598
Other	11,273	102	57	11,318	8,968	130	16	9,082
Total available-for-sale debt securities	356,247	8,790	2,961 ^(c)	362,076	310,347	5,939	2,019 ^(c)	314,267
Available-for-sale equity securities	2,693	14	2	2,705	1,894	163	6	2,051
Total available-for-sale securities	\$ 358,940	\$ 8,804	\$ 2,963 ^(c)	\$ 364,781	\$ 312,241	\$ 6,102	\$ 2,025 ^(c)	\$ 316,318
Total held-to-maturity securities	\$ 12	\$ 1	\$ —	\$ 13	\$ 18	\$ 2	\$ —	\$ 20

(a) Includes total U.S. government-sponsored enterprise obligations with fair values of \$89.3 billion and \$94.2 billion at December 31, 2011 and 2010, respectively, which were predominantly mortgage-related.

(b) Consists primarily of bank debt including sovereign government-guaranteed bank debt.

(c) Includes a total of \$91 million and \$133 million (pretax) of unrealized losses related to prime mortgage-backed securities for which credit losses have been recognized in income at December 31, 2011 and 2010, respectively. These unrealized losses are not credit-related and remain reported in AOCI.

Securities impairment

The following tables present the fair value and gross unrealized losses for AFS securities by aging category at December 31, 2011 and 2010.

December 31, 2011 (in millions)	Securities with gross unrealized losses					
	Less than 12 months		12 months or more		Total fair value	Total gross unrealized losses
	Fair value	Gross unrealized losses	Fair value	Gross unrealized losses		
Available-for-sale debt securities						
Mortgage-backed securities:						
U.S. government agencies	\$ 2,724	\$ 2	\$ —	\$ —	\$ 2,724	\$ 2
Residential:						
Prime and Alt-A	649	12	970	206	1,619	218
Subprime	—	—	—	—	—	—
Non-U.S.	30,500	266	25,176	421	55,676	687
Commercial	837	53	—	—	837	53
Total mortgage-backed securities	34,710	333	26,146	627	60,856	960
U.S. Treasury and government agencies	3,369	2	—	—	3,369	2
Obligations of U.S. states and municipalities	147	42	40	6	187	48
Certificates of deposit	—	—	—	—	—	—
Non-U.S. government debt securities	11,901	66	1,286	15	13,187	81
Corporate debt securities	22,230	901	9,585	746	31,815	1,647
Asset-backed securities:						
Credit card receivables	—	—	—	—	—	—
Collateralized loan obligations	5,610	49	3,913	117	9,523	166
Other	4,735	40	1,185	17	5,920	57
Total available-for-sale debt securities	82,702	1,433	42,155	1,528	124,857	2,961
Available-for-sale equity securities	338	2	—	—	338	2
Total securities with gross unrealized losses	\$ 83,040	\$ 1,435	\$ 42,155	\$ 1,528	\$ 125,195	\$ 2,963

December 31, 2010 (in millions)	Securities with gross unrealized losses					
	Less than 12 months		12 months or more		Total fair value	Total gross unrealized losses
	Fair value	Gross unrealized losses	Fair value	Gross unrealized losses		
Available-for-sale debt securities						
Mortgage-backed securities:						
U.S. government agencies	\$ 14,039	\$ 297	\$ —	\$ —	\$ 14,039	\$ 297
Residential:						
Prime and Alt-A	—	—	1,193	250	1,193	250
Subprime	—	—	—	—	—	—
Non-U.S.	35,166	379	1,080	30	36,246	409
Commercial	548	14	11	3	559	17
Total mortgage-backed securities	49,753	690	2,284	283	52,037	973
U.S. Treasury and government agencies	921	28	—	—	921	28
Obligations of U.S. states and municipalities	6,890	330	20	8	6,910	338
Certificates of deposit	1,771	2	—	—	1,771	2
Non-U.S. government debt securities	6,960	28	—	—	6,960	28
Corporate debt securities	18,783	418	90	1	18,873	419
Asset-backed securities:						
Credit card receivables	—	—	345	5	345	5
Collateralized loan obligations	460	10	6,321	200	6,781	210
Other	2,615	9	32	7	2,647	16
Total available-for-sale debt securities	88,153	1,515	9,092	504	97,245	2,019
Available-for-sale equity securities	—	—	2	6	2	6
Total securities with gross unrealized losses	\$ 88,153	\$ 1,515	\$ 9,094	\$ 510	\$ 97,247	\$ 2,025

Notes to consolidated financial statements

Other-than-temporary impairment

The following table presents credit losses that are included in the securities gains and losses table above.

Year ended December 31, (in millions)	2011	2010	2009
Debt securities the Firm does not intend to sell that have credit losses			
Total other-than-temporary impairment losses ^a	\$ (27)	\$ (94)	\$ (946)
Losses recorded in/(reclassified from) other comprehensive income	(49)	(6)	368
Total credit losses recognized in income ^{b)(c)}	\$ (76)	\$ (100)	\$ (578)

- (a) For initial OTTI, represents the excess of the amortized cost over the fair value of AFS debt securities. For subsequent impairments of the same security, represents additional declines in fair value subsequent to previously recorded OTTI, if applicable.
- (b) Represents the credit loss component on certain prime mortgage-backed securities for 2011; certain prime mortgage-backed securities and obligations of U.S. states and municipalities for 2010; and certain prime and subprime mortgage-backed securities and obligations of U.S. states and municipalities for 2009 that the Firm does not intend to sell. Subsequent credit losses may be recorded on securities without a corresponding further decline in fair value if there has been a decline in expected cash flows.
- (c) Excluded from this table are OTTI losses of \$7 million that were recognized in income in 2009, related to subprime mortgage-backed debt securities the Firm intended to sell. These securities were sold in 2009, resulting in the recognition of a recovery of \$1 million.

Changes in the credit loss component of credit-impaired debt securities

The following table presents a rollforward for the years ended December 31, 2011, 2010 and 2009, of the credit loss component of OTTI losses that have been recognized in income, related to debt securities that the Firm does not intend to sell.

Year ended December 31, (in millions)	2011	2010	2009
Balance, beginning of period	\$ 632	\$ 578	\$ —
Additions:			
Newly credit-impaired securities	4	—	578
Increase in losses on previously credit-impaired securities	—	94	—
Losses reclassified from other comprehensive income on previously credit-impaired securities	72	6	—
Reductions:			
Sales of credit-impaired securities	—	(31)	—
Impact of new accounting guidance related to VIEs	—	(15)	—
Balance, end of period	\$ 708	\$ 632	\$ 578

Gross unrealized losses

Gross unrealized losses have generally increased since December 31, 2010, including those that have been in an

unrealized loss position or 3or 3\$.on of a reco1 0 0 1 227.25 av A.vomeb35 -26 Td4n.a(ended ((n3tiar)!\$ S Q7)ses3i-21 iTd unr-7uslyn R9.tls\$cludd [(C)-2 (—)axa 7u0.4r227t* [6li-2.899xa d 1 0 0 0 71 0t3 M 3yn R95(ea)scie13 0 mtl(n R9.tls\$cludd

enhancement in the form of subordination was approximately 10% of the outstanding principal balance of securitized mortgage loans, compared with expected lifetime losses of 1% of the outstanding principal. In assessing potential credit losses, assumptions included recovery rates of 60%, default rates of 0.25% to 0.5% and constant prepayment rates of 15% to 20%. The unrealized loss is considered temporary, based on management's assessment that the estimated future cash flows together with the credit enhancement levels for those securities remain sufficient to support the Firm's investment.

Corporate debt securities

As of December 31, 2011, gross unrealized losses related to corporate debt securities were \$1.6 billion, of which \$746 million related to securities that have been in an unrealized loss position for 12 months or more.

Substantially all of the corporate debt securities are rated investment-grade, including those in an unrealized loss position. Various factors were considered in assessing whether the Firm expects to recover the amortized cost of corporate debt securities including, but not limited to, the strength of issuer credit ratings, the financial condition of guarantors and the length of time and the extent to which a security's fair value has been less than its amortized cost. The fair values of securities in an unrealized loss position were on average within approximately 4% of amortized cost. Based on management's assessment, the Firm expects to recover the entire amortized cost basis of all corporate debt securities that were in an unrealized loss position as of December 31, 2011.

Asset-backed securities – Collateralized loan obligations
As of December 31, 2011, gross unrealized losses related to CLOs were \$166 million, of which \$117 million related to securities that were in an unrealized loss position for 12 months or more. Overall, losses have decreased since December 31, 2010, mainly as a result of lower default forecasts and spread tightening across various asset classes. Substantially all of these securities are rated "AAA," "AA" or "A" and have an average credit enhancement of 30%. The key assumptions considered in analyzing potential credit losses were underlying loan and debt security defaults and loss severity. Based on current default trends for the collateral underlying the securities, the Firm assumed initial collateral default rates of 2% and 4% beginning in 2012 and thereafter. Further, loss severities were assumed to be 48% for loans and 82% for debt securities. Losses on collateral were estimated to occur approximately 18 months after default. The unrealized loss is considered temporary, based on management's assessment that the estimated future cash flows together with the credit enhancement levels for those securities remain sufficient to support the Firm's investment.

Notes to consolidated financial statements

Contractual maturities and yields

The following table presents the amortized cost and estimated fair value at December 31, 2011, of JPMorgan Chase's AFS HTM securities by contractual maturity.

By remaining maturity December 31, 2011 (in millions)	Due in one year or less	Due after one year through five years	Due after five years through 10 years	Due after 10 years ^(c)	Total
Available-for-sale debt securities					
Mortgage-backed securities					
Amortized cost	\$ 15	\$ 3,666	\$ 3,932	\$ 173,225	\$ 180,838
Fair value	15	3,653	4,073	178,152	185,893
Average yield ^(b)	5.04%	3.20%	3.08%	3.64%	3.62%
U.S. Treasury and government agencies					
Amortized cost	\$ 4,949	\$ 2,984	\$ —	\$ 251	\$ 8,184
Fair value	4,952	3,099	—	300	8,351
Average yield ^(b)	0.58%	2.20%	—%	3.89%	1.27%
Obligations of U.S. states and municipalities					
Amortized cost	\$ 61	\$ 306	\$ 1,132	\$ 13,905	\$ 15,404
Fair value	62	326	1,206	14,946	16,540
Average yield ^(b)	3.10%	3.66%	3.59%	4.84%	4.72%
Certificates of deposit					
Amortized cost	\$ 3,017	\$ —	\$ —	\$ —	\$ 3,017
Fair value	3,017	—	—	—	3,017
Average yield ^(b)	4.33%	—%	—%	—%	4.33%
Non-U.S. government debt securities					
Amortized cost	\$ 20,863	\$ 15,967	\$ 7,524	\$ 590	\$ 44,944
Fair value	20,861	16,106	7,700	598	45,265
Average yield ^(b)	1.27%	2.06%	2.86%	4.94%	1.87%
Corporate debt securities					
Amortized cost	\$ 22,019	\$ 30,171	\$ 11,398	\$ 19	\$ 63,607
Fair value	22,091	29,291	10,776	18	62,176
Average yield ^(b)	2.05%	3.09%	4.45%	5.42%	2.97%
Asset-backed securities					
Amortized cost	\$ 2	\$ 5,965	\$ 17,951	\$ 16,335	\$ 40,253
Fair value	2	6,102	18,287	16,443	40,834
Average yield ^(b)	2.28%	2.88%	2.02%	2.51%	2.35%
Total available-for-sale debt securities					
Amortized cost	\$ 50,926	\$ 59,059	\$ 41,937	\$ 204,325	\$ 356,247
Fair value	51,000	58,577	42,042	210,457	362,076
Average yield ^(b)	1.73%	2.75%	2.97%	3.64%	3.14%
Available-for-sale equity securities					
Amortized cost	\$ —	\$ —	\$ —	\$ 2,693	\$ 2,693
Fair value	—	—	—	2,705	2,705
Average yield ^(b)	—%	—%	—%	0.38%	0.38%
Total available-for-sale securities					
Amortized cost	\$ 50,926	\$ 59,059	\$ 41,937	\$ 207,018	\$ 358,940
Fair value	51,000	58,577	42,042	213,162	364,781
Average yield ^(b)	1.73%	2.75%	2.97%	3.60%	3.12%
Total held-to-maturity securities					
Amortized cost	\$ —	\$ 8	\$ 3	\$ 1	\$ 12
Fair value	—	9	3	1	13
Average yield ^(b)	—%	6.90%	6.76%	6.48%	6.84%

- (a) U.S. government agencies and U.S. government-sponsored enterprises were the only issuers whose securities exceeded 10% of JPMorgan Chase's total stockholders' equity at December 31, 2011.
- (b) Average yield is computed using the effective yield of each security owned at the end of the period, weighted based on the amortized cost of each security. The effective yield considers the contractual coupon, amortization of premiums and accretion of discounts, and the effect of related hedging derivatives. Taxable-equivalent amounts are used where applicable.
- (c) Includes securities with no stated maturity. Substantially all of the Firm's residential mortgage-backed securities and collateralized mortgage obligations are due in 10 years or more, based on contractual maturity. The estimated duration, which reflects anticipated future prepayments based on a consensus of dealers in the market, is approximately three years for agency residential mortgage-backed securities, two years for agency residential collateralized mortgage obligations and four years for nonagency residential collateralized mortgage obligations.

Note 13 – Securities financing activities

JPMorgan Chase enters into resale agreements, repurchase agreements, securities borrowed transactions and securities loaned transactions (collectively, “securities financing agreements”) primarily to finance the Firm’s inventory positions, acquire securities to cover short positions, accommodate customers’ financing needs, and settle other securities obligations.

Securities financing agreements are treated as collateralized financings on the Firm’s Consolidated Balance Sheets. Resale and repurchase agreements are generally carried at the amounts at which the securities will be subsequently sold or repurchased, plus accrued interest. Securities borrowed and securities loaned transactions are generally carried at the amount of cash collateral advanced or received. Where appropriate under applicable accounting guidance, resale and repurchase agreements with the same counterparty are reported on a net basis. Fees received and paid in connection with securities financing agreements are recorded in interest income and interest expense, respectively.

The Firm has elected the fair value option for certain securities financing agreements. For further information regarding the fair value option, see Note 4 on pages 198–200 of this Annual Report. The securities financing agreements for which the fair value option has been elected are reported within securities purchased under resale agreements; securities loaned or sold under repurchase agreements; and securities borrowed on the Consolidated Balance Sheets. Generally, for agreements carried at fair value, current-period interest accruals are recorded within interest income and interest expense, with changes in fair value reported in principal transactions revenue. However, for financial instruments containing embedded derivatives that would be separately accounted for in accordance with accounting guidance for hybrid instruments, all changes in fair value, including any interest elements, are reported in principal transactions revenue.

The following table details the Firm’s securities financing agreements, all of which are accounted for as collateralized financings during the periods presented.

December 31, (in millions)	2011	2010
Securities purchased under resale agreements ^(a)	\$ 235,000	\$ 222,302
Securities borrowed ^(b)	142,462	123,587
Securities sold under repurchase agreements ^(c)	\$ 197,789	\$ 262,722
Securities loaned	14,214	10,592

- (a) At December 31, 2011 and 2010, included resale agreements of \$24.9 billion and \$20.3 billion, respectively, accounted for at fair value.
- (b) At December 31, 2011 and 2010, included securities borrowed of \$15.3 billion and \$14.0 billion, respectively, accounted for at fair value.
- (c) At December 31, 2011 and 2010, included repurchase agreements of \$9.5 billion and \$4.1 billion, respectively, accounted for at fair value.

The amounts reported in the table above were reduced by \$115.7 billion and \$112.7 billion at December 31, 2011 and 2010, respectively, as a result of agreements in effect that meet the specified conditions for net presentation under applicable accounting guidance.

JPMorgan Chase’s policy is to take possession, where possible, of securities purchased under resale agreements and of securities borrowed. The Firm monitors the value of the underlying securities (primarily G7 government securities, U.S. agency securities and agency MBS, and equities) that it has received from its counterparties and either requests additional collateral or returns a portion of the collateral when appropriate in light of the market value of the underlying securities. Margin levels are established initially based upon the counterparty and type of collateral and monitored on an ongoing basis to protect against declines in collateral value in the event of default. JPMorgan Chase typically enters into master netting agreements and other collateral arrangements with its resale agreement and securities borrowed counterparties, which provide for the right to liquidate the purchased or borrowed securities in the event of a customer default. As a result of the Firm’s credit risk mitigation practices described above on resale and securities borrowed agreements, the Firm did not hold any reserves for credit impairment on these agreements as of December 31, 2011 and 2010.

For further information regarding assets pledged and collateral received in securities financing agreements, see Note 30 on page 289 of this Annual Report.

Note 14 – Loans

Loan accounting framework

The accounting for a loan depends on management’s strategy for the loan, and on whether the loan was credit-impaired at the date of acquisition. The Firm accounts for loans based on the following categories:

- Originated or purchased loans held-for-investment (i.e., “retained”), other than purchased credit-impaired (“PCI”) loans
- Loans held-for-sale
- Loans at fair value
- PCI loans held-for-investment

The following provides a detailed accounting discussion of these loan categories:

Loans held-for-investment (other than PCI loans)

Originated or purchased loans held-for-investment, other than PCI loans, are measured at the principal amount outstanding, net of the following: allowance for loan losses; net charge-offs; interest applied to principal (for loans accounted for on the cost recovery method); unamortized discounts and premiums; and net deferred loan fees or costs.

Interest income

Interest income on performing loans held-for-investment, other than PCI loans, is accrued and recognized as interest

income at the contractual rate of interest. Purchase price discounts or premiums, as well as net deferred loan fees or costs, are amortized into interest income over the life of the loan to produce a level rate of return.

Nonaccrual loans

Nonaccrual loans are those on which the accrual of interest has been suspended. Loans (other than credit card loans and certain consumer loans insured by U.S. government agencies) are placed on nonaccrual status and considered nonperforming when full payment of principal and interest is in doubt, which for consumer loans, excluding credit card, is generally determined when principal or interest is 90 days or more past due and collateral, if any, is insufficient to cover principal and interest. A loan is determined to be past due when the minimum payment is not received from the borrower by the contractually specified due date or for certain loans (e.g., residential real estate loans), when a monthly payment is due and unpaid for 30 days or more. All interest accrued but not collected is reversed against interest income at the date a loan is placed on nonaccrual status. In addition, the amortization of deferred amounts is suspended. In certain cases, interest income on nonaccrual loans may be recognized to the extent cash is received (i.e., cash basis) when the recorded loan balance is deemed fully collectible; however, if there is doubt regarding the ultimate collectibility of the recorded loan balance, all interest cash receipts are applied to reduce the carrying value of the loan (the cost recovery method).

A loan may be returned to accrual status when repayment is reasonably assured and there has been demonstrated performance under the terms of the loan or, if applicable, the terms of the restructured loan.

As permitted by regulatory guidance, credit card loans are generally exempt from being placed on nonaccrual status; accordingly, interest and fees related to credit card loans continue to accrue until the loan is charged off or paid in full. However, the Firm separately establishes an allowance for the estimated uncollectible portion of billed and accrued interest and fee income on credit card loans.

Allowance for loan losses

The allowance for loan losses represents the estimated probable losses on held-for-investment loans. Changes in the allowance for loan losses are recorded in the provision for credit losses on the Firm's Consolidated Statements of Income. See Note 15 on pages 252–255 for further information on the Firm's accounting policies for the allowance for loan losses.

Charge-offs

Wholesale loans and risk-rated business banking and auto loans are charged off against the allowance for loan losses when it is highly certain that a loss has been realized. This determination includes many factors, including the prioritization of the Firm's claim in bankruptcy, expectations of the workout/restructuring of the loan and valuation of the borrower's equity.

Credit card loans are charged off by the end of the month in which the account becomes 180 days past due, or within 60 days from receiving notification about a specified event (e.g., bankruptcy of the borrower), whichever is earlier.

Consumer loans, other than risk-rated business banking and auto loans and PCI loans, are generally charged off to the allowance for loan losses upon reaching specified stages of delinquency, in accordance with the Federal Financial Institutions Examination Council ("FFIEC") policy. Residential mortgage loans and scored business banking loans are generally charged down to estimated net realizable value (the fair value of collateral less costs to sell) at no later than 180 days past due.

Collateral-dependent loans are charged down to estimated net realizable value when deemed impaired (for example, upon modification in a troubled debt restructuring). A loan is considered to be collateral-dependent when repayment of the loan is expected to be provided solely by the underlying collateral, rather than by cash flows from the borrower's operations, income or other resources.

When a loan is charged down to the estimated net realizable value, the determination of the fair value of the collateral depends on the type of collateral (e.g., securities, real estate). In cases where the collateral is in the form of liquid securities, the fair value is based on quoted market prices or broker quotes. For illiquid securities or other financial assets, the fair value of the collateral is estimated using a discounted cash flow model.

For residential real estate loans, collateral values are based upon external valuation sources. When it becomes likely that a borrower is either unable or unwilling to pay, the Firm obtains a broker's price opinion of the home based on an exterior-only valuation ("exterior opinions"), which is then updated at least every six months thereafter. As soon as practicable after taking physical possession of the property through foreclosure, the Firm obtains an appraisal based on an inspection that includes the interior of the home ("interior appraisals"). Exterior opinions and interior appraisals are discounted based upon the Firm's experience with actual liquidation values as compared to the estimated values provided by exterior opinions and interior appraisals, considering state- and product-specific factors.

For commercial real estate loans, collateral values are generally based on appraisals from internal and external valuation sources. Collateral values are typically updated every six to twelve months, either by obtaining a new appraisal or by performing an internal analysis, in accordance with the Firm's policies. The Firm also considers both borrower- and market-specific factors, which may result in obtaining appraisal updates or broker price opinions at more frequent intervals.

Loans held-for-sale

Held-for-sale loans are measured at the lower of cost or fair value, with valuation changes recorded in noninterest revenue. For wholesale loans, the valuation is performed on an individual loan basis. For consumer loans, the valuation is performed on a portfolio basis.

Interest income on loans held-for-sale is accrued and recognized based on the contractual rate of interest.

Loan origination fees or costs and purchase price discounts or premiums are deferred in a contra loan account until the related loan is sold. The deferred fees and discounts or premiums are an adjustment to the basis of the loan and therefore are included in the periodic determination of the lower of cost or fair value adjustments and/or the gain or losses recognized at the time of sale.

Held-for-sale loans are subject to the nonaccrual policies described above.

Because held-for-sale loans are recognized at the lower of cost or fair value, the Firm's allowance for loan losses and charge-off policies do not apply to these loans.

Loans at fair value

Loans used in a trading strategy or risk managed on a fair value basis are measured at fair value, with changes in fair value recorded in noninterest revenue.

For these loans, the earned current contractual interest payment is recognized in interest income. Changes in fair value are recognized in noninterest revenue. Loan origination fees are recognized upfront in noninterest revenue. Loan origination costs are recognized in the associated expense category as incurred.

Because these loans are recognized at fair value, the Firm's nonaccrual, allowance for loan losses, and charge-off policies do not apply to these loans.

See Note 4 on pages 198–200 of this Annual Report for further information on the Firm's elections of fair value accounting under the fair value option. See Note 3 and Note 4 on pages 184–198 and 198–200 of this Annual Report for further information on loans carried at fair value and classified as trading assets.

PCI loans

PCI loans held-for-investment are initially measured at fair value. PCI loans have evidence of credit deterioration since the loan's origination date and therefore it is probable, at acquisition, that all contractually required payments will not be collected. Because PCI loans are initially measured at fair value, which includes an estimate of future credit losses, no allowance for loan losses related to PCI loans is recorded at the acquisition date. See page 247 of this Note for information on accounting for PCI loans subsequent to their acquisition.

Loan classification changes

Loans in the held-for-investment portfolio that management decides to sell are transferred to the held-for-sale portfolio at the lower of cost or fair value on the date of transfer. Credit-related losses are charged against the allowance for loan losses; losses due to changes in interest rates or foreign currency exchange rates are recognized in noninterest revenue.

In the event that management decides to retain a loan in the held-for-sale portfolio, the loan is transferred to the held-for-investment portfolio at the lower of cost or fair

value on the date of transfer. These loans are subsequently assessed for impairment based on the Firm's allowance methodology. For a further discussion of the methodologies used in establishing the Firm's allowance for loan losses, see Note 15 on pages 252–255 of this Annual Report.

Loan modifications

The Firm seeks to modify certain loans in conjunction with its loss-mitigation activities. Through the modification, JPMorgan Chase grants one or more concessions to a borrower who is experiencing financial difficulty in order to minimize the Firm's economic loss, avoid foreclosure or repossession of the collateral, and to ultimately maximize payments received by the Firm from the borrower. The concessions granted vary by program and by borrower-specific characteristics, and may include interest rate reductions, term extensions, payment deferrals, or the acceptance of equity or other assets in lieu of payments. In certain limited circumstances, loan modifications include principal forgiveness.

Such modifications are accounted for and reported as troubled debt restructurings ("TDRs"). A loan that has been modified in a TDR is generally considered to be impaired until it matures, is repaid, or is otherwise liquidated, regardless of whether the borrower performs under the modified terms. In certain limited cases, the effective interest rate applicable to the modified loan is at or above the current market rate at the time of the restructuring. In such circumstances, and assuming that the loan subsequently performs under its modified terms and the Firm expects to collect all contractual principal and interest cash flows, the loan is disclosed as impaired and as a TDR only during the year of the modification; in subsequent years, the loan is not disclosed as an impaired loan or as a TDR so long as repayment of the restructured loan under its modified terms is reasonably assured.

Loans, except for credit card loans, modified in a TDR are generally placed on nonaccrual status, although in many cases such loans were already on nonaccrual status prior to modification. These loans may be returned to performing status (resuming the accrual of interest) if the following criteria are met: (a) the borrower has performed under the modified terms for a minimum of six months and/or six payments, and (b) the Firm has an expectation that repayment of the modified loan is reasonably assured based on, for example, the borrower's debt capacity and level of future earnings, collateral values, LTV ratios, and other current market considerations. In certain limited and well-defined circumstances in which the loan is current at the modification date, such loans are not placed on nonaccrual status at the time of modification.

Because loans modified in TDRs are considered to be impaired, these loans are evaluated for an asset-specific allowance, which considers the expected re-default rates for the modified loans and is determined based on the same methodology used to estimate the Firm's asset-specific allowance component. A loan modified in a TDR remains subject to the asset-specific allowance methodology

Notes to consolidated financial statements

throughout its remaining life, regardless of whether the loan is performing and has been returned to accrual status. For further discussion of the methodology used to estimate the Firm's asset-specific allowance, see Note 15 on pages 252–255 of this Annual Report.

Foreclosed property

The Firm acquires property from borrowers through loan restructurings, workouts, and foreclosures. Property acquired may include real property (e.g., residential real estate, land, buildings, and fixtures) and commercial and

personal property (e.g., aircraft, railcars, and ships).

At the time JPMorgan Chase takes physical possession, the property is recorded in other assets on the Consolidated Balance Sheets at fair value less estimated costs to sell. Each quarter the fair value of the acquired property is reviewed and adjusted, if necessary. Subsequent changes to fair value are charged/credited to noninterest revenue. Operating expense, such as real estate taxes and maintenance, are charged to other expense.

Loan portfolio

The Firm's loan portfolio is divided into three portfolio segments, which are the same segments used by the Firm to determine the allowance for loan losses: Wholesale; Consumer, excluding credit card; and Credit card. Within each portfolio segment, the Firm monitors and assesses the credit risk in the following classes of loans, based on the risk characteristics of each loan class.

Wholesale ^(a)	Consumer, excluding credit card ^(b)	Credit card
- Commercial and industrial - Real estate - Financial institutions - Government agencies - Other	<u>Residential real estate – excluding PCI</u> - Home equity – senior lien - Home equity – junior lien - Prime mortgage, including option ARMs - Subprime mortgage <u>Other consumer loans</u> - Auto^(c) - Business banking - Student and other <u>Residential real estate – PCI</u> - Home equity - Prime mortgage - Subprime mortgage - Option ARMs	- Chase, excluding accounts originated by Washington Mutual - Accounts originated by Washington Mutual

(a) Includes loans reported in IB, Commercial Banking ("CB"), Treasury & Securities Services ("TSS"), Asset Management ("AM"), and Corporate/Private Equity segments.

(b) Includes loans reported in RFS, auto and student loans reported in Card Services & Auto ("Card"), and residential real estate loans reported in the Corporate/Private Equity and AM segment.

(c) Includes auto and business banking risk-rated loans that apply the wholesale methodology for determining the allowance for loan losses; these loans are managed by Card and RFS, respectively, and therefore, for consistency in presentation, are included with the other consumer loan classes.

The following table summarizes the Firm's loan balances by portfolio segment.

December 31, 2011 (in millions)	Wholesale	Consumer, excluding credit card	Credit card	Total
Retained	\$ 278,395	\$ 308,427	\$ 132,175	\$ 718,997 ^(a)
Held-for-sale	2,524	—	102	2,626
At fair value	2,097	—	—	2,097
Total	\$ 283,016	\$ 308,427	\$ 132,277	\$ 723,720

December 31, 2010 (in millions)	Wholesale	Consumer, excluding credit card	Credit card	Total
Retained	\$ 222,510	\$ 327,464	\$ 135,524	\$ 685,498 ^(a)
Held-for-sale	3,147	154	2,152	5,453
At fair value	1,976	—	—	1,976
Total	\$ 227,633	\$ 327,618	\$ 137,676	\$ 692,927

(a) Loans (other than PCI loans and those for which the fair value option has been selected) are presented net of unearned income, unamortized discounts and premiums, and net deferred loan costs of \$2.7 billion and \$1.9 billion at December 31, 2011 and 2010, respectively.

The following table provides information about the carrying value of retained loans purchased, retained loans sold and retained loans reclassified to held-for-sale during the periods indicated. These tables exclude loans recorded at fair value. On an ongoing basis, the Firm manages its exposure to credit risk. Selling loans is one way that the Firm reduces its credit exposures.

Year ended December 31, 2011 (in millions)	Wholesale	Consumer, excluding credit card	Credit card	Total
Purchases	\$ 906	\$ 7,525	\$ —	\$ 8,431
Sales	3,289	1,384	—	4,673
Retained loans reclassified to held-for-sale	538	—	2,006	2,544

The following table provides information about gains/(losses) on loan sales by portfolio segment.

Year ended December 31, (in millions)	2011	2010	2009
Net gains/(losses) on sales of loans (including lower of cost or fair value adjustments)			
Wholesale	\$ 121	\$ 215	\$ 291
Consumer, excluding credit card	131	265	127
Credit card	(24)	(16)	21
Total net gains/(losses) on sales of loans (including lower of cost or fair value adjustments)	\$ 228	\$ 464	\$ 439

(a) Excludes sales related to loans accounted for at fair value.

Wholesale loan portfolio

Wholesale loans include loans made to a variety of customers from large corporate and institutional clients to certain high-net worth individuals.

The primary credit quality indicator for wholesale loans is the risk rating assigned each loan. Risk ratings are used to identify the credit quality of loans and differentiate risk within the portfolio. Risk ratings on loans consider the probability of default ("PD") and the loss given default ("LGD"). PD is the likelihood that a loan will not be repaid at default. The LGD is the estimated loss on the loan that would be realized upon the default of the borrower and takes into consideration collateral and structural support for each credit facility.

Management considers several factors to determine an appropriate risk rating, including the obligor's debt capacity and financial flexibility, the level of the obligor's earnings, the amount and sources for repayment, the level and nature of contingencies, management strength, and the industry and geography in which the obligor operates. Risk ratings generally represent ratings profiles similar to those defined

by S&P and Moody's. Investment grade ratings range from "AAA/Aaa" to "BBB-/Baa3." Noninvestment grade ratings are classified as noncriticized ("BB+/Ba1 and B-/B3") and criticized ("CCC+"/"Caa1 and below"), and the criticized portion is further subdivided into performing and nonaccrual loans, representing management's assessment of the collectibility of principal and interest. Criticized loans have a higher probability of default than noncriticized loans.

Risk ratings are reviewed on a regular and ongoing basis by Credit Risk Management and are adjusted as necessary for updated information affecting the obligor's ability to fulfill its obligations.

As noted above, the risk rating of a loan considers the industry in which the obligor conducts its operations. As part of the overall credit risk management framework, the Firm focuses on the management and diversification of its industry and client exposures, with particular attention paid to industries with actual or potential credit concern. See Note 5 on page 201 in this Annual Report for further detail on industry concentrations.

Notes to consolidated financial statements

The table below provides information by class of receivable for the retained loans in the Wholesale portfolio segment.

As of or for the year ended December 31, (in millions, except ratios)	Commercial and industrial		Real estate	
	2011	2010	2011	2010
Loans by risk ratings				
Investment grade	\$ 52,428	\$ 31,697	\$ 33,920	\$ 28,504
Noninvestment grade:				
Noncriticized	38,644	30,874	15,972	16,425
Criticized performing	2,254	2,371	3,906	5,769
Criticized nonaccrual	889	1,634	886	2,937
Total noninvestment grade	41,787	34,879	20,764	25,131
Total retained loans	\$ 94,215	\$ 66,576	\$ 54,684	\$ 53,635
% of total criticized to total retained loans	3.34%	6.02%	8.76%	16.23%
% of nonaccrual loans to total retained loans	0.94	2.45	1.62	5.48
Loans by geographic distribution ^(a)				
Total non-U.S.	\$ 30,813	\$ 17,731	\$ 1,497	\$ 1,963
Total U.S.	63,402	48,845	53,187	51,672
Total retained loans	\$ 94,215	\$ 66,576	\$ 54,684	\$ 53,635
Net charge-offs	\$ 124	\$ 403	\$ 256	\$ 862
% of net charge-offs to end-of-period retained loans	0.13%	0.61%	0.47%	1.61%
Loan delinquency ^(b)				
Current and less than 30 days past due and still accruing	\$ 93,060	\$ 64,501	\$ 53,387	\$ 50,299
30–89 days past due and still accruing	266	434	327	290
90 or more days past due and still accruing	—	7	84	109
Criticized nonaccrual	889	1,634	886	2,937
Total retained loans	\$ 94,215	\$ 66,576	\$ 54,684	\$ 53,635

(a) The U.S. and non-U.S. distribution is determined based predominantly on the domicile of the borrower.

(b) The credit quality of wholesale loans is assessed primarily through ongoing review and monitoring of an obligor's ability to meet contractual obligations rather than relying on the past due status, which is generally a lagging indicator of credit quality. For a discussion of more significant risk factors, see pages 235 of this Note.

(c) Represents loans that are considered well-collateralized and therefore still accruing interest.

(d) Other primarily includes loans to SPEs and loans to private banking clients. See Note 1 on pages 182–183 of this Annual Report for additional information on SPEs.

The following table presents additional information on the real estate class of loans within the Wholesale portfolio segment for the periods indicated. The real estate class primarily consists of secured commercial loans mainly to borrowers for multifamily and commercial lessor properties. Multifamily lending specifically finances apartment buildings. Commercial lessors receive financing specifically for real estate leased to retail, office and industrial tenants. Commercial construction and development loans represent financing for the construction of apartments, office and professional buildings and malls. Other real estate loans include lodging, real estate investment trusts ("REITs"), single-family, homebuilders and other real estate.

December 31, (in millions, except ratios)	Multifamily		Commercial lessors	
	2011	2010	2011	2010
Real estate retained loans	\$ 32,524	\$ 30,604	\$ 14,444	\$ 15,796
Criticized exposure	2,451	3,798	1,662	3,593
% of criticized exposure to total real estate retained loans	7.54%	12.41%	11.51%	22.75%
Criticized nonaccrual	\$ 412	\$ 1,016	\$ 284	\$ 1,549
% of criticized nonaccrual to total real estate retained loans	1.27%	3.32%	1.97%	9.81%

(table continued from previous page)

Financial institutions		Government agencies		Other ^(d)		Total retained loans	
2011	2010	2011	2010	2011	2010	2011	2010
\$ 28,804	\$ 22,525	\$ 7,421	\$ 6,871	\$ 74,497	\$ 56,450	\$ 197,070	\$ 146,047
9,132	8,480	378	382	7,583	6,012	71,709	62,173
246	317	4	3	808	320	7,218	8,780
37	136	16	22	570	781	2,398	5,510
9,415	8,933	398	407	8,961	7,113	81,325	76,463
\$ 38,219	\$ 31,458	\$ 7,819	\$ 7,278	\$ 83,458	\$ 63,563	\$ 278,395	\$ 222,510
0.74 %	1.44%	0.26%	0.34%	1.65%	1.73%	3.45%	6.42%
0.10	0.43	0.20	0.30	0.68	1.23	0.86	2.48
\$ 29,996	\$ 19,756	\$ 583	\$ 870	\$ 32,275	\$ 25,831	\$ 95,164	\$ 66,151
8,223	11,702	7,236	6,408	51,183	37,732	183,231	156,359
\$ 38,219	\$ 31,458	\$ 7,819	\$ 7,278	\$ 83,458	\$ 63,563	\$ 278,395	\$ 222,510
\$ (137)	\$ 72	\$ —	\$ 2	\$ 197	\$ 388	\$ 440	\$ 1,727
(0.36)%	0.23%	—%	0.03%	0.24%	0.61%	0.16%	0.78%
\$ 38,129	\$ 31,289	\$ 7,780	\$ 7,222	\$ 81,802	\$ 61,837	\$ 274,158	\$ 215,148
51	31	23	34	1,072	704	1,739	1,493
2	2	—	—	14	241	100	359
37	136	16	22	570	781	2,398	5,510
\$ 38,219	\$ 31,458	\$ 7,819	\$ 7,278	\$ 83,458	\$ 63,563	\$ 278,395	\$ 222,510

(table continued from previous page)

Commercial construction and development		Other		Total real estate loans	
2011	2010	2011	2010	2011	2010
\$ 3,148	\$ 3,395	\$ 4,568	\$ 3,840	\$ 54,684	\$ 53,635
297	619	382	696	4,792	8,706
9.43%	18.23%	8.36%	18.13%	8.76%	16.23%
\$ 69	\$ 174	\$ 121	\$ 198	\$ 886	\$ 2,937
2.19%	5.13%	2.65%	5.16%	1.62%	5.48%

Notes to consolidated financial statements

Wholesale impaired loans and loan modifications

Wholesale impaired loans include loans that have been placed on nonaccrual status and/or that have been modified in a TDR. All impaired loans are evaluated for an asset-specific allowance as described on pages 233–234 of this Note.

The table below set forth information about the Firm's wholesale impaired loans.

December 31, (in millions)	Commercial and industrial		Real estate		Financial institutions		Government agencies		Other		Total retained loans	
	2011	2010	2011	2010	2011	2010	2011	2010	2011	2010	2011	2010
Impaired loans												
With an allowance	\$ 828	\$ 1,512	\$ 621	\$ 2,510	\$ 21	\$ 127	\$ 16	\$ 22	\$ 473	\$ 697	\$ 1,959	\$ 4,868
Without an allowance ^(a)	177	157	292	445	18	8	—	—	103	8	590	618
Total impaired loans	\$ 1,005	\$ 1,669	\$ 913	\$ 2,955	\$ 39	\$ 135	\$ 16	\$ 22	\$ 576	\$ 705	\$ 2,549	\$ 5,486
Allowance for loan losses related to impaired loans	\$ 276	\$ 435	\$ 148	\$ 825	\$ 5	\$ 61	\$ 10	\$ 14	\$ 77	\$ 239	\$ 516	\$ 1,574
Unpaid principal balance of impaired loans ^(b)	1,705	2,453	1,124	3,487	63	244	17	30	1,008	1,046	3,917	7,260

- (a) When the discounted cash flows, collateral value or market price equals or exceeds the recorded investment in the loan, then the loan does not require an allowance. This typically occurs when the impaired loans have been partially charged-off and/or there have been interest payments received and applied to the loan balance.
- (b) Represents the contractual amount of principal owed at December 31, 2011 and 2010. The unpaid principal balance differs from the impaired loan balances due to various factors, including charge-offs; interest payments received and applied to the carrying value; net deferred loan fees or costs; and unamortized discount or premiums on purchased loans.

The following table presents the Firm's average impaired loans for the years ended 2011, 2010 and 2009.

Year ended December 31, (in millions)	2011	2010	2009
Commercial and industrial	\$ 1,309	\$ 1,655	\$ 1,767
Real estate	1,813	3,101	2,420
Financial institutions	84	304	685
Government agencies	20	5	4
Other	634	884	468
Total ^(a)	\$ 3,860	\$ 5,949	\$ 5,344

- (a) The related interest income on accruing impaired loans and interest income recognized on a cash basis were not material for the years ended December 31, 2011, 2010 and 2009.

Loan modifications

Certain loan modifications are considered to be TDRs as they provide various concessions to borrowers who are experiencing financial difficulty. All TDRs are reported as impaired loans in the tables above. The following table provides information about the Firm's wholesale loans that have been modified in TDRs as of the dates presented.

December 31, (in millions)	Commercial and industrial		Real estate		Financial institutions		Government agencies		Other		Total retained loans	
	2011	2010	2011	2010	2011	2010	2011	2010	2011	2010	2011	2010
Loans modified in troubled debt restructurings	\$ 531	\$ 212	\$ 176	\$ 907	\$ 2	\$ 1	\$ 16	\$ 22	\$ 25	\$ 1	\$ 750	\$ 1,143
TDRs on nonaccrual status	415	163	128	831	—	1	16	22	19	1	578	1,018
Additional commitments to lend to borrowers whose loans have been modified in TDRs	147	1	—	—	—	—	—	—	—	—	147	1

TDR activity rollforward

The following table reconciles the beginning and ending balances of wholesale loans modified in TDRs for the period presented and provides information regarding the nature and extent of modifications during the period.

Year ended December 31, 2011 (in millions)	Commercial and industrial	Real estate	Other ^(b)	Total
Beginning balance of TDRs	\$ 212	\$ 907	\$ 24	\$ 1,143
New TDRs	665	113	32	810
Increases to existing TDRs	96	16	—	112
Charge-offs post-modification	(30)	(146)	—	(176)
Sales and other ^(a)	(412)	(714)	(13)	(1,139)
Ending balance of TDRs	\$ 531	\$ 176	\$ 43	\$ 750

- (a) Sales and other are predominantly sales and paydowns, but may include performing loans restructured at market rates that are no longer reported as TDRs.
- (b) Includes loans to Financial institutions, Government agencies and Other.

Financial effects of modifications and redefaults

Loans modified as TDRs during the year ended December 31, 2011, are predominantly term or payment extensions and, to a lesser extent, deferrals of principal and/or interest on commercial and industrial and real estate loans. The average term extension granted on loans with term or payment extensions was 3.3 years for the year ended December 31, 2011. The weighted-average remaining term for all loans modified during the year ended December 31, 2011 was 4.5 years. Wholesale TDR loans that redefaulted within one year of the modification were \$96 million during the year ended December 31, 2011. A payment default is deemed to occur when the borrower has not made a loan payment by its scheduled due date after giving effect to any contractual grace period.

Consumer, excluding credit card loan portfolio

Consumer loans, excluding credit card loans, consist primarily of residential mortgages, home equity loans and lines of credit, auto loans, business banking loans, and student and other loans, with a primary focus on serving the prime consumer credit market. The portfolio also includes home equity loans secured by junior liens and mortgage loans with interest-only payment options to predominantly prime borrowers, as well as certain payment-option loans originated by Washington Mutual that may result in negative amortization.

The table below provides information about consumer retained loans by class, excluding the Credit card loan portfolio segment.

December 31, (in millions)	2011	2010
Residential real estate – excluding PCI		
Home equity:		
Senior lien	\$ 21,765	\$ 24,376
Junior lien	56,035	64,009
Mortgages:		
Prime, including option ARMs	76,196	74,539
Subprime	9,664	11,287
Other consumer loans		
Auto	47,426	48,367
Business banking	17,652	16,812
Student and other	14,143	15,311
Residential real estate – PCI		
Home equity	22,697	24,459
Prime mortgage	15,180	17,322
Subprime mortgage	4,976	5,398
Option ARMs	22,693	25,584
Total retained loans	\$ 308,427	\$ 327,464

Delinquency rates are a primary credit quality indicator for consumer loans. Loans that are more than 30 days past due provide an early warning of borrowers that may be experiencing financial difficulties and/or who may be unable or unwilling to repay the loan. As the loan continues to age, it becomes more clear that the borrower is likely

either unable or unwilling to pay. In the case of residential real estate loans, late-stage delinquencies (greater than 150 days past due) are a strong indicator of loans that will ultimately result in a short sale or foreclosure. In addition to delinquency rates, other credit quality indicators for consumer loans vary based on the class of loan, as follows:

- For residential real estate loans, including both non-PCI and PCI portfolios, the current estimated LTV ratio, or the combined LTV ratio in the case of loans with a junior lien, is an indicator of the potential loss severity in the event of default. Additionally, LTV or combined LTV can provide insight into a borrower's continued willingness to pay, as the delinquency rate of high-LTV loans tends to be greater than that for loans where the borrower has equity in the collateral. The geographic distribution of the loan collateral also provides insight as to the credit quality of the portfolio, as factors such as the regional economy, home price changes and specific events such as hurricanes, earthquakes, etc., will affect credit quality. The borrower's current or "refreshed" FICO score is a secondary credit-quality indicator for certain loans, as FICO scores are an indication of the borrower's credit payment history. Thus, a loan to a borrower with a low FICO score (660 or below) is considered to be of higher risk than a loan to a borrower with a high FICO score. Further, a loan to a borrower with a high LTV ratio and a low FICO score is at greater risk of default than a loan to a borrower that has both a high LTV ratio and a high FICO score.
- For auto, scored business banking and student loans, geographic distribution is an indicator of the credit performance of the portfolio. Similar to residential real estate loans, geographic distribution provides insights into the portfolio performance based on regional economic activity and events.
- Risk-rated business banking and auto loans are similar to wholesale loans in that the primary credit quality indicators are the risk rating that is assigned to the loan and whether the loans are considered to be criticized and/or nonaccrual. Risk ratings are reviewed on a regular and ongoing basis by Credit Risk Management and are adjusted as necessary for updated information affecting borrowers' ability to fulfill their obligations. Consistent with other classes of consumer loans, the geographic distribution of the portfolio provides insights into portfolio performance based on regional economic activity and events.

Residential real estate – excluding PCI loans

The following tables provide information by class for residential real estate – excluding PCI retained loans in the Consumer, excluding credit card, portfolio segment.

The following factors should be considered in analyzing certain credit statistics applicable to the Firm's residential real estate – excluding PCI loans portfolio: (i) junior lien home equity loans may be fully charged off when the loan becomes 180 days past due, the borrower is either unable

Notes to consolidated financial statements

or unwilling to repay the loan, and the value of the collateral does not support the repayment of the loan, resulting in relatively high charge-off rates for this product class; and (ii) the lengthening of loss-mitigation timelines

may result in higher delinquency rates for loans carried at estimated collateral value that remain on the Firm's Consolidated Balance Sheets.

Residential real estate – excluding PCI loans

December 31, (in millions, except ratios)	Home equity			
	Senior lien		Junior lien	
	2011	2010	2011	2010
Loan delinquency ^(a)				
Current and less than 30 days past due	\$ 20,992	\$ 23,615	\$ 54,533	\$ 62,315
30–149 days past due	405	414	1,272	1,508
150 or more days past due	368	347	230	186
Total retained loans	\$ 21,765	\$ 24,376	\$ 56,035	\$ 64,009
% of 30+ days past due to total retained loans	3.55%	3.12%	2.68%	2.65%
90 or more days past due and still accruing	\$ —	\$ —	\$ —	\$ —
90 or more days past due and government guaranteed ^(b)	—	—	—	—
Nonaccrual loans	495	479	792	784
Current estimated LTV ratios ^{(d)(e)(f)}				
Greater than 125% and refreshed FICO scores:				
Equal to or greater than 660	\$ 341	\$ 363	\$ 6,463	\$ 6,928
Less than 660	160	196	2,037	2,495
101% to 125% and refreshed FICO scores:				
Equal to or greater than 660	663	619	8,775	9,403
Less than 660	241	249	2,510	2,873
80% to 100% and refreshed FICO scores:				
Equal to or greater than 660	1,850	1,900	11,433	13,333
Less than 660	601	657	2,616	3,155
Less than 80% and refreshed FICO scores:				
Equal to or greater than 660	15,350	17,474	19,326	22,527
Less than 660	2,559	2,918	2,875	3,295
U.S. government-guaranteed	—	—	—	—
Total retained loans	\$ 21,765	\$ 24,376	\$ 56,035	\$ 64,009
Geographic region				
California	\$ 3,066	\$ 3,348	\$ 12,851	\$ 14,656
New York	3,023	3,272	10,979	12,278
Florida	992	1,088	3,006	3,470
Illinois	1,495	1,635	3,785	4,248
Texas	3,027	3,594	1,859	2,239
New Jersey	687	732	3,238	3,617
Arizona	1,339	1,481	2,552	2,979
Washington	714	776	1,895	2,142
Ohio	1,747	2,010	1,328	1,568
Michigan	1,044	1,176	1,400	1,618
All other ^(g)	4,631	5,264	13,142	15,194
Total retained loans	\$ 21,765	\$ 24,376	\$ 56,035	\$ 64,009

(a) Individual delinquency classifications included mortgage loans insured by U.S. government agencies as follows: current and less than 30 days past due includes \$3.0 billion and \$2.5 billion; 30–149 days past due includes \$2.3 billion and \$2.5 billion; and 150 or more days past due includes \$10.3 billion and \$7.9 billion at December 31, 2011 and 2010, respectively.

(b) These balances, which are 90 days or more past due but insured by U.S. government agencies, are excluded from nonaccrual loans. In predominately all cases, 100% of the principal balance of the loans is insured and interest is guaranteed at a specified reimbursement rate subject to meeting agreed servicing guidelines. These amounts are excluded from nonaccrual loans because reimbursement of insured and guaranteed amounts is proceeding normally. At December 31, 2011 and 2010, these balances included \$7.0 billion and \$2.8 billion, respectively, of loans that are no longer accruing interest because interest has been curtailed by the U.S. government agencies although, in predominately all cases, 100% of the principal is still insured. For the remaining balance, interest is being accrued at the guaranteed reimbursement rate.

(c) Represents the aggregate unpaid principal balance of loans divided by the estimated current property value. Current property values are estimated, at a minimum, quarterly based on home valuation models using nationally recognized home price index valuation estimates incorporating actual data to the extent available and forecasted data where actual data is not available. These property values do not represent actual appraised loan level collateral values; as such, the resulting ratios are necessarily imprecise and should be viewed as estimates.

(d) Junior lien represents combined LTV, which considers all available lien positions related to the property. All other products are presented without consideration of subordinate liens on the property.

(e) Refreshed FICO scores represent each borrower's most recent credit score, which is obtained by the Firm at least on a quarterly basis.

(f) For senior lien home equity loans, prior-period amounts have been revised to conform with the current-period presentation.

(g) At December 31, 2011 and 2010, included mortgage loans insured by U.S. government agencies of \$15.6 billion and \$12.9 billion, respectively.

(h) At December 31, 2011 and 2010, excluded mortgage loans insured by U.S. government agencies of \$12.6 billion and \$10.3 billion, respectively. These amounts were excluded as reimbursement of insured amounts is proceeding normally.

(table continued from previous page)

Mortgages							
Prime, including option ARMs				Subprime		Total residential real estate – excluding PCI	
2011		2010		2011	2010	2011	2010
\$	59,855	\$	59,223	\$	7,585	\$	142,965
	3,475		4,052		820		5,972
	12,866		11,264		1,259		14,723
\$	76,196	\$	74,539	\$	9,664	\$	163,660
	4.96% ^(h)		6.68% ^(h)		21.51%		4.97% ^(h)
\$	—	\$	—	\$	—	\$	—
	11,516		9,417		—		11,516
	3,462		4,320		1,781		6,530
					2,210		7,793
\$	3,168	\$	3,039	\$	367	\$	10,339
	1,416		1,595		1,061		4,674
	4,626		4,733		506		14,570
	1,636		1,775		1,284		5,671
	9,343		10,720		817		23,443
	2,349		2,786		1,556		7,122
	33,849		32,385		1,906		70,431
	4,225		4,557		2,167		11,826
	15,584		12,949		—		15,584
\$	76,196	\$	74,539	\$	9,664	\$	163,660
\$	18,029	\$	19,278	\$	1,463	\$	35,409
	10,200		9,587		1,217		25,419
	4,565		4,840		1,206		9,769
	3,922		3,765		391		9,593
	2,851		2,569		300		8,037
	2,042		2,026		461		6,428
	1,194		1,320		199		5,284
	1,878		2,056		209		4,696
	441		462		234		3,750
	909		963		246		3,599
	30,165		27,673		3,738		51,676
\$	76,196	\$	74,539	\$	9,664	\$	163,660

Notes to consolidated financial statements

The following table represents the Firm's delinquency statistics for junior lien home equity loans as of December 31, 2011 and 2010.

December 31, 2011 (in millions, except ratios)	Delinquencies			Total loans	Total 30+ day delinquency rate
	30–89 days past due	90–149 days past due	150+ days past due		
HELOCs:					
Within the revolving period	\$ 606	\$ 314	\$ 173	\$ 47,760	2.29%
Within the required amortization period	45	19	15	1,636	4.83
HELOANs	188	100	42	6,639	4.97
Total	\$ 839	\$ 433	\$ 230	\$ 56,035	2.68%

December 31, 2010 (in millions, except ratios)	Delinquencies			Total loans	Total 30+ day delinquency rate
	30–89 days past due	90–149 days past due	150+ days past due		
HELOCs:					
Within the revolving period	\$ 665	\$ 384	\$ 145	\$ 54,434	2.19%
Within the required amortization period	41	19	10	1,177	5.95
HELOANs	250	149	31	8,398	5.12
Total	\$ 956	\$ 552	\$ 186	\$ 64,009	2.65%

(a) In general, HELOCs are open-ended, revolving loans for a 10-year period, after which time the HELOC converts to a loan with a 20-year amortization period.

(b) The Firm manages the risk of HELOCs during their revolving period by closing or reducing the undrawn line to the extent permitted by law when borrowers are experiencing financial difficulty or when the collateral does not support the loan amount.

Home equity lines of credit ("HELOCs") within the required amortization period and home equity loans ("HELOANs") have higher delinquency rates than do HELOCs within the revolving period. That is primarily because the fully-amortizing payment required for those products is higher than the minimum payment options available for HELOCs within the revolving period. The higher delinquency rates associated with amortizing HELOCs and HELOANs are factored into the loss estimates produced by the Firm's delinquency roll-rate methodology, which estimates defaults based on the current delinquency status of a portfolio.

Impaired loans

The table below sets forth information about the Firm's residential real estate impaired loans, excluding PCI. These loans are considered to be impaired as they have been modified in a TDR. All impaired loans are evaluated for an asset-specific allowance as described in Note 15 on pages 252–255 of this Annual Report.

December 31, (in millions)	Home equity		
	Senior lien		Junior lien
	2011	2010	2011
Impaired loans			
With an allowance	\$ 319	\$ 211	\$ 622
Without an allowance	16	15	35
Total impaired loans ^(b)	\$ 335	\$ 226	\$ 657
Allowance for loan losses related to impaired loans	\$ 80	\$ 77	\$ 141
Unpaid principal balance of impaired loans ^(b)	433	265	
Impaired loans on nonaccrual status	77	38	35

The following table presents average impaired loans and the related interest income reported by the Firm.

Year ended December 31, (in millions)	Average impaired loans			Interest income on impaired loans ^(a)			Interest income on impaired loans on a cash basis		
	2011	2010	2009	2011	2010	2009	2011	2010	2009
Home equity									
Senior lien	\$ 287	\$ 207	\$ 142	\$ 10	\$ 15	\$ 7	\$ 1	\$ 1	\$ 1
Junior lien	521	266	187	18	10	9	2	1	1
Mortgages									
Prime, including option ARMs	3,859	1,530	496	147	70	34	14	14	8
Subprime	3,083	2,539	1,948	148	121	98	16	19	6
Total residential real estate – excluding PCI	7,750	4,542	2,773	323	216	148	33	35	16

(a) Generally, interest income on loans modified in a TDR is recognized on a cash basis until such time as the borrower has made a minimum of six payments under the new terms. As of December 31, 2011 and 2010, \$886 million and \$580 million, respectively, of loans were TDRs for which the borrowers had not yet made six payments under their modified terms.

Loan modifications

The Firm is participating in the U.S. Treasury's Making Home Affordable ("MHA") programs and is continuing to expand its other loss-mitigation efforts for financially distressed borrowers who do not qualify for the U.S. Treasury's programs. The MHA programs include the Home Affordable Modification Program ("HAMP") and the Second Lien Modification Program ("2MP"). The Firm's other loss-mitigation programs for troubled borrowers who do not qualify for HAMP include the traditional modification programs offered by the GSEs and Ginnie Mae, as well as the Firm's proprietary modification programs, which include concessions similar to those offered under HAMP and 2MP but with expanded eligibility criteria. In addition, the Firm has offered specific targeted modification programs to higher risk borrowers, many of whom were current on their mortgages prior to modification.

In order to be offered a permanent modification under HAMP, a borrower must successfully make three payments under the new terms during a trial modification period. The Firm also offers one proprietary modification program that is similar to HAMP and that includes a comparable trial modification period. Borrowers who do not successfully complete the trial modification period do not qualify to

have their loans permanently modified under that particular program; however, in certain cases, the Firm considers whether the borrower might qualify for a different loan modification program.

Permanent modifications of residential real estate loans, excluding PCI loans, are generally accounted for and reported as TDRs. In addition, in the fourth quarter of 2011, the Firm began to characterize as TDRs loans to borrowers who have been approved for a trial modification either under HAMP or under the proprietary program noted above, even though such loans have not yet been permanently modified. Regardless of whether the borrower successfully completes the trial modification, such loans will continue to be reported as TDRs until charged-off, repaid or otherwise liquidated. The Firm previously considered the risk characteristics of loans in a trial modification in determining its formula-based allowance for loan losses. As a result, the recharacterization of trial modifications as TDRs during the fourth quarter of 2011 did not have a significant impact on the Firm's allowance for loan losses.

There were no additional commitments to lend to borrowers whose residential real estate loans, excluding PCI loans, have been modified in TDRs.

TDR activity rollforward

The following tables reconcile the beginning and ending balances of residential real estate loans, excluding PCI loans, modified in TDRs for the periods presented.

Year ended December 31, 2011 (in millions)	Home equity		Mortgages		Total residential real estate – (excluding PCI)
	Senior lien	Junior lien	Prime, including option ARMs	Subprime	
Beginning balance of TDRs	\$ 226	\$ 283	\$ 2,084	\$ 2,751	\$ 5,344
New TDRs	138	518	3,268	883	4,807
Charge-offs post-modification	(15)	(78)	(119)	(234)	(446)
Foreclosures and other liquidations (e.g., short sales)	—	(11)	(108)	(82)	(201)
Principal payments and other	(14)	(55)	(248)	(99)	(416)
Ending balance of TDRs	\$ 335	\$ 657	\$ 4,877	\$ 3,219	\$ 9,088
Permanent modifications	\$ 285	\$ 634	\$ 4,601	\$ 3,029	\$ 8,549
Trial modifications	\$ 50	\$ 23	\$ 276	\$ 190	\$ 539

(a) Includes all loans to borrowers who were approved for trial modification on or after January 1, 2011, as well as all loans permanently modified during the year ended December 31, 2011. In the event that a trial modification is reported as a new TDR, any subsequent permanent modification of that same loan is not reported as a new TDR.

(b) Includes charge-offs on unsuccessful trial modifications.

Nature and extent of modifications

MHA, as well as the Firm's proprietary modification programs, generally provide various concessions to financially troubled borrowers including, but not limited to, interest rate reductions, term or payment extensions and deferral of principal and/or interest payments that would otherwise have been required under the terms of the original agreement. The following table provides information about how residential real estate loans, excluding PCI loans, were permanently modified during the period presented.

Year ended December 31, 2011	Home equity		Mortgages		Total residential real estate – (excluding PCI)
	Senior lien	Junior lien	Prime, including option ARMs	Subprime	
Number of loans approved for a trial modification, but not permanently modified	654	778	898	1,730	4,060
Number of loans permanently modified	1,006	9,142	9,579	4,972	24,699
Permanent concession granted ^(a)					
Interest rate reduction	80%	95%	53%	80%	75%
Term or payment extension	88	81	71	72	75
Principal and/or interest deferred	10	21	17	19	19
Principal forgiveness	7	20	2	13	11
Other ^(c)	29	7	68	26	35

(a) As a percentage of the number of loans permanently modified. The sum of the percentages exceeds 100% because predominantly all of the permanent modifications include more than one type of concession.

(b) Except for the "Other" category, the percentages representing the various types of concessions granted are estimated to be materially consistent with those related to loans approved for trial modification.

(c) Represents variable interest rate to fixed interest rate modifications. To date, these concessions have solely related to permanent modifications.

Financial effects of modifications and redefaults

The following table provides information about the financial effects of the various concessions granted in permanent modifications of residential real estate loans, excluding PCI, and also about redefaults of certain loans modified in TDRs for the period presented.

Year ended December 31, 2011 (in millions, except weighted-average data and number of loans)	Home equity		Mortgages		Total residential real estate – (excluding PCI)
	Senior lien	Junior lien	Prime, including option ARMs	Subprime	
Weighted-average interest rate of loans with interest rate reductions – before TDR	7.25%	5.46%	5.98%	8.25%	6.44%
Weighted-average interest rate of loans with interest rate reductions – after TDR	3.51	1.49	3.34	3.46	3.09
Weighted-average remaining contractual term (in years) of loans with term or payment extensions – before TDR	18	21	25	23	24
Weighted-average remaining contractual term (in years) of loans with term or payment extensions – after TDR	30	34	35	34	35
Charge-offs recognized upon permanent modification	\$ 1	\$ 117	\$ 61	\$ 19	\$ 198
Principal deferred ^(a)	4	35	167	61	267
Principal forgiven ^(a)	1	62	20	46	129
Number of loans that redefaulted within one year of permanent modification ^(c)	222	1,310	1,142	1,989	4,663
Balance of loans that redefaulted within one year of permanent modification ^(c)	\$ 18	\$ 52	\$ 340	\$ 281	\$ 691
Cumulative permanent modification redefault rates	21%	14%	13%	28%	18%

(a) Represents information about loans that have been permanently modified. The financial effects of such concessions related to loans approved for trial modification are estimated to be materially consistent with the financial effects presented above.

(b) Represents information about loans that have been permanently modified. Principal deferred and principal forgiven related to loans approved for trial modification totaled \$125 million for the year ended December 31, 2011.

(c) Represents loans permanently modified in TDRs that experienced a payment default in the period presented, and for which the payment default occurred within one year of the modification. The dollar amounts presented represent the balance of such loans at the end of the reporting period in which they defaulted. For residential real estate loans modified in TDRs, payment default is deemed to occur when the loan becomes two contractual payments past due. In the event that a modified loan redefaults, it is probable that the loan will ultimately be liquidated through foreclosure or another similar type of liquidation transaction. Defaults of loans modified within the last 12 months may not be representative of ultimate redefault levels.

(d) Based upon permanent modifications completed after October 1, 2009, that are seasoned more than six months.

Approximately 85% of the trial modifications approved on or after July 1, 2010 (the approximate date on which substantial revisions were made to the HAMP program), that are seasoned more than six months have been successfully converted to permanent modifications.

At December 31, 2011, the weighted-average estimated remaining lives of residential real estate loans, excluding PCI loans permanently modified in TDRs were 7.0 years, 6.9 years, 9.0 years and 6.7 years for senior lien home equity, junior lien home equity, prime mortgage, including option ARMs, and subprime mortgage, respectively. The estimated remaining lives of these loans reflect estimated prepayments, both voluntary and involuntary (i.e., foreclosures and other forced liquidations).

Other consumer loans

The tables below provide information for other consumer retained loan classes, including auto, business banking and student loans.

December 31, (in millions, except ratios)	Auto		Business banking		Student and other		Total other consumer	
	2011	2010	2011	2010	2011	2010	2011	2010
Loan delinquency^(a)								
Current and less than 30 days past due	\$46,891	\$47,778	\$17,173	\$16,240	\$12,905	\$13,998	\$76,969	\$78,016
30–119 days past due	528	579	326	351	777	795	1,631	1,725
120 or more days past due	7	10	153	221	461	518	621	749
Total retained loans	\$47,426	\$48,367	\$17,652	\$16,812	\$14,143	\$15,311	\$79,221	\$80,490
% of 30+ days past due to total retained loans	1.13%	1.22%	2.71%	3.40%	1.76% ^(d)	1.61% ^(d)	1.59% ^(d)	1.75% ^(d)
90 or more days past due and still accruing ^(b)	\$ —	\$ —	\$ —	\$ —	\$ 551	\$ 625	\$ 551	\$ 625
Nonaccrual loans	118	141	694	832	69	67	881	1,040
Geographic region								
California	\$ 4,413	\$ 4,307	\$ 1,342	\$ 851	\$ 1,261	\$ 1,330	\$ 7,016	\$ 6,488
New York	3,616	3,875	2,792	2,877	1,401	1,305	7,809	8,057
Florida	1,881	1,923	313	220	658	722	2,852	2,865
Illinois	2,496	2,608	1,364	1,320	851	940	4,711	4,868
Texas	4,467	4,505	2,680	2,550	1,053	1,273	8,200	8,328
New Jersey	1,829	1,842	376	422	460	502	2,665	2,766
Arizona	1,495	1,499	1,165	1,218	316	387	2,976	3,104
Washington	735	716	160	115	249	279	1,144	1,110
Ohio	2,633	2,961	1,541	1,647	880	1,010	5,054	5,618
Michigan	2,282	2,434	1,389	1,401	637	729	4,308	4,564
All other	21,579	21,697	4,530	4,191	6,377	6,834	32,486	32,722
Total retained loans	\$47,426	\$48,367	\$17,652	\$16,812	\$14,143	\$15,311	\$79,221	\$80,490
Loans by risk ratings^(c)								
Noncriticized	\$ 6,775	\$ 5,803	\$11,749	\$10,351	NA	NA	\$18,524	\$16,154
Criticized performing	166	265	817	982	NA	NA	983	1,247
Criticized nonaccrual	3	12	524	574	NA	NA	527	586

- (a) Loans insured by U.S. government agencies under the Federal Family Education Loan Program ("FFELP") are included in the delinquency classification presented based on their payment status. Prior-period amounts have been revised to conform with the current-period presentation.
- (b) These amounts represent student loans, which are insured by U.S. government agencies under the FFELP. These amounts were accruing as reimbursement of insured amounts is proceeding normally.
- (c) For risk-rated business banking and auto loans, the primary credit quality indicator is the risk rating of the loan, including whether the loans are considered to be criticized and/or nonaccrual.
- (d) December 31, 2011 and 2010, excluded loans 30 days or more past due and still accruing, which are insured by U.S. government agencies under the FFELP, of \$989 million and \$1.1 billion, respectively. These amounts were excluded as reimbursement of insured amounts is proceeding normally.

Notes to consolidated financial statements

Other consumer impaired loans and loan modifications

The tables below set forth information about the Firm's other consumer impaired loans, including risk-rated business banking and auto loans that have been placed on nonaccrual status, and loans that have been modified in TDRs.

December 31, (in millions)	Auto		Business banking		Total other consumer	
	2011	2010	2011	2010	2011	2010
Impaired loans						
With an allowance	\$ 88	\$ 102	\$ 713	\$ 774	\$ 801	\$ 876
Without an allowance ^(c)	3	—	—	—	3	—
Total impaired loans	\$ 91	\$ 102	\$ 713	\$ 774	\$ 804	\$ 876
Allowance for loan losses related to impaired loans	\$ 12	\$ 16	\$ 225	\$ 248	\$ 237	\$ 264
Unpaid principal balance of impaired loans	126	132	822	899	948	1,031
Impaired loans on nonaccrual status	41	50	551	647	592	697

- (a) When discounted cash flows, collateral value or market price equals or exceeds the recorded investment in the loan, then the loan does not require an allowance. This typically occurs when the impaired loans have been partially charged off and/or there have been interest payments received and applied to the loan balance.
- (b) Represents the contractual amount of principal owed at December 31, 2011 and 2010. The unpaid principal balance differs from the impaired loan balances due to various factors, including charge-offs; interest payments received and applied to the principal balance; net deferred loan fees or costs; and unamortized discounts or premiums on purchased loans.
- (c) There were no impaired student and other loans at December 31, 2011 and 2010.

The following table presents average impaired loans for the periods presented.

Year ended December 31, (in millions)	Average impaired loans		
	2011	2010	2009
Auto	\$ 92	\$ 120	\$ 100
Business banking	760	682	396
Total other consumer ^(a)	\$ 852	\$ 802	\$ 496

- (a) There were no impaired student and other loans for the years ended 2011, 2010 and 2009.
- (b) The related interest income on impaired loans, including those on a cash basis, was not material for the years ended 2011, 2010 and 2009.

Loan modifications

The following table provides information about the Firm's other consumer loans modified in TDRs. All of these TDRs are reported as impaired loans in the tables above.

December 31, (in millions)	Auto		Business banking		Total other consumer	
	2011	2010	2011	2010	2011	2010
Loans modified in troubled debt restructurings ^(b)	\$ 88	\$ 91	\$ 415	\$ 395	\$ 503	\$ 486
TDRs on nonaccrual status	38	39	253	268	291	307

- (a) These modifications generally provided interest rate concessions to the borrower or deferral of principal repayments.
- (b) Additional commitments to lend to borrowers whose loans have been modified in TDRs as of December 31, 2011 and 2010, were immaterial.
- (c) There were no student and other loans modified in TDRs at December 31, 2011 and 2010.

TDR activity rollforward

The following table reconciles the beginning and ending balances of other consumer loans modified in TDRs for the period presented.

Year ended December 31, 2011						
(in millions)	Auto		Business banking		Total other consumer	
Beginning balance of TDRs	\$	91	\$	395	\$	486
New TDRs		54		195		249
Charge-offs		(5)		(11)		(16)
Foreclosures and other liquidations		—		(3)		(3)
Principal payments and other		(52)		(161)		(213)
Ending balance of TDRs	\$	88	\$	415	\$	503

Financial effects of modifications and redefaults

For auto loans, TDRs typically occur in connection with the bankruptcy of the borrower. In these cases, the loan is

modified with a revised repayment plan that typically incorporates interest rate reductions and, to a lesser extent, principal forgiveness.

For business banking loans, concessions are dependent on individual borrower circumstances and can be of a short-term nature for borrowers who need temporary relief or longer term for borrowers experiencing more fundamental financial difficulties. Concessions are predominantly term or payment extensions, but also may include interest rate reductions.

For the year ended December 31, 2011, the interest rates on auto loans modified in TDRs were reduced on average from 12.45% to 5.70%, and the interest rates on business banking loans modified in TDRs were reduced on average from 7.55% to 5.52%. For business banking loans, the weighted-average remaining term of all loans modified in TDRs during the year ended December 31, 2011, increased from 1.4 years to 2.6 years. For all periods presented, principal forgiveness related to auto loans was immaterial.

The balance of business banking loans modified in TDRs that experienced a payment default during the year ended December 31, 2011, and for which the payment default occurred within one year of the modification, was \$80 million; the corresponding balance of redefaulted auto loans modified in TDRs was insignificant. A payment default is deemed to occur as follows: (1) for scored auto and business banking loans, when the loan is two payments past due; and (2) for risk-rated business banking loans and auto loans, when the borrower has not made a loan payment by its scheduled due date after giving effect to the contractual grace period, if any.

Purchased credit-impaired loans

PCI loans are initially recorded at fair value at acquisition; PCI loans acquired in the same fiscal quarter may be aggregated into one or more pools, provided that the loans have common risk characteristics. A pool is then accounted for as a single asset with a single composite interest rate and an aggregate expectation of cash flows. With respect to the Washington Mutual transaction, all of the consumer loans were aggregated into pools of loans with common risk characteristics.

On a quarterly basis, the Firm estimates the total cash flows (both principal and interest) expected to be collected over the remaining life of each pool. These estimates incorporate assumptions regarding default rates, loss severities, the amounts and timing of prepayments and other factors that reflect then-current market conditions. Probable decreases in expected cash flows (i.e., increased credit losses) trigger the recognition of impairment, which is then measured as the present value of the expected principal loss plus any related foregone interest cash flows, discounted at the pool's effective interest rate. Impairments are recognized through the provision for credit losses and an increase in the allowance for loan losses. Probable and significant increases in expected cash flows (e.g., decreased credit losses, the net benefit of modifications) would first reverse any previously recorded allowance for loan losses with any remaining increases recognized prospectively as a yield adjustment over the remaining estimated lives of the underlying loans. The impacts of (i) pre-payments, (ii)

changes in variable interest rates, and (iii) any other changes in the timing of expected cash flows are recognized prospectively as adjustments to interest income. Disposals of loans — which may include sales of loans, receipt of payments in full by the borrower, or foreclosure — result in removal of the loans from the PCI portfolio.

The Firm continues to modify certain PCI loans. The impact of these modifications is incorporated into the Firm's quarterly assessment of whether a probable and significant change in expected cash flows has occurred, and the loans continue to be accounted for and reported as PCI loans. In evaluating the effect of modifications on expected cash flows, the Firm incorporates the effect of any foregone interest and also considers the potential for redefault. The Firm develops product-specific probability of default estimates, which are used to compute expected credit losses. In developing these probabilities of default, the Firm considers the relationship between the credit quality characteristics of the underlying loans and certain assumptions about home prices and unemployment based upon industry-wide data. The Firm also considers its own historical loss experience to date based on actual redefaulted PCI modified loans.

The excess of cash flows expected to be collected over the carrying value of the underlying loans is referred to as the accretable yield. This amount is not reported on the Firm's Consolidated Balance Sheets but is accreted into interest income at a level rate of return over the remaining estimated lives of the underlying pools of loans.

If the timing and/or amounts of expected cash flows on PCI loans were determined not to be reasonably estimable, no interest would be accreted and the loans would be reported as nonaccrual loans; however, since the timing and amounts of expected cash flows for the Firm's PCI consumer loans are reasonably estimable, interest is being accreted and the loans are being reported as performing loans.

Charge-offs are not recorded on PCI loans until actual losses exceed the estimated losses that were recorded as purchase accounting adjustments at acquisition date. To date, no charge-offs have been recorded for these consumer loans.

The PCI portfolio affects the Firm's results of operations primarily through: (i) contribution to net interest margin; (ii) expense related to defaults and servicing resulting from the liquidation of the loans; and (iii) any provision for loan losses. The PCI loans acquired in the Washington Mutual transaction were funded based on the interest rate characteristics of the loans. For example, variable-rate loans were funded with variable-rate liabilities and fixed-rate loans were funded with fixed-rate liabilities with a similar maturity profile. A net spread will be earned on the declining balance of the portfolio, which is estimated as of December 31, 2011, to have a remaining weighted-average life of 7.5 years.

Notes to consolidated financial statements

Residential real estate – PCI loans

The table below sets forth information about the Firm's consumer, excluding credit card, PCI loans.

December 31, (in millions, except ratios)	Home equity		Prime mortgage		Subprime mortgage		Option ARMs		Total PCI	
	2011	2010	2011	2010	2011	2010	2011	2010	2011	2010
Carrying value ^(a)	\$22,697	\$24,459	\$15,180	\$17,322	\$ 4,976	\$ 5,398	\$22,693	\$25,584	\$65,546	\$72,763
Related allowance for loan losses	1,908	1,583	1,929	1,766	380	98	1,494	1,494	5,711	4,941
Loan delinquency (based on unpaid principal balance)										
Current and less than 30 days past due	\$22,682	\$25,783	\$12,148	\$13,035	\$ 4,388	\$ 4,312	\$17,919	\$18,672	\$57,137	\$61,802
30–149 days past due	1,130	1,348	912	1,468	782	1,020	1,467	2,215	4,291	6,051
150 or more days past due	1,252	1,181	3,000	4,425	2,059	2,710	6,753	9,904	13,064	18,220
Total loans	\$25,064	\$28,312	\$16,060	\$18,928	\$ 7,229	\$ 8,042	\$26,139	\$30,791	\$74,492	\$86,073
% of 30+ days past due to total loans	9.50%	8.93%	24.36%	31.13%	39.30%	46.38%	31.45%	39.36%	23.30%	28.20%
Current estimated LTV ratios (based on unpaid principal balance) ^{(b)(d)(e)}										
Greater than 125% and refreshed FICO scores:										
Equal to or greater than 660	\$ 5,915	\$ 6,289	\$ 2,313	\$ 2,400	\$ 473	\$ 432	\$ 2,509	\$ 2,681	\$11,210	\$11,802
Less than 660	3,299	4,043	2,319	2,744	1,939	2,129	4,608	6,330	12,165	15,246
101% to 125% and refreshed FICO scores:										
Equal to or greater than 660	5,393	6,053	3,328	3,815	434	424	3,959	4,292	13,114	14,584
Less than 660	2,304	2,696	2,314	3,011	1,510	1,663	3,884	5,005	10,012	12,375
80% to 100% and refreshed FICO scores:										
Equal to or greater than 660	3,482	3,995	1,629	1,970	372	374	3,740	4,152	9,223	10,491
Less than 660	1,264	1,482	1,457	1,857	1,197	1,477	3,035	3,551	6,953	8,367
Lower than 80% and refreshed FICO scores:										
Equal to or greater than 660	2,409	2,641	1,276	1,443	198	186	2,189	2,281	6,072	6,551
Less than 660	998	1,113	1,424	1,688	1,106	1,357	2,215	2,499	5,743	6,657
Total unpaid principal balance	\$25,064	\$28,312	\$16,060	\$18,928	\$ 7,229	\$ 8,042	\$26,139	\$30,791	\$74,492	\$86,073
Geographic region (based on unpaid principal balance)										
California	\$15,091	\$17,012	\$ 9,121	\$10,891	\$ 1,661	\$ 1,971	\$13,565	\$16,130	\$39,438	\$46,004
New York	1,179	1,316	1,018	1,111	709	736	1,548	1,703	4,454	4,866
Florida	2,307	2,595	1,265	1,519	812	906	3,201	3,916	7,585	8,936
Illinois	558	627	511	562	411	438	702	760	2,182	2,387
Texas	455	525	168	194	405	435	140	155	1,168	1,309
New Jersey	471	540	445	486	297	316	969	1,064	2,182	2,406
Arizona	468	539	254	359	126	165	362	528	1,210	1,591
Washington	1,368	1,535	388	451	160	178	649	745	2,565	2,909
Ohio	32	38	79	91	114	122	111	131	336	382
Michigan	81	95	239	279	187	214	268	345	775	933
All other	3,054	3,490	2,572	2,985	2,347	2,561	4,624	5,314	12,597	14,350
Total unpaid principal balance	\$25,064	\$28,312	\$16,060	\$18,928	\$ 7,229	\$ 8,042	\$26,139	\$30,791	\$74,492	\$86,073

- (a) Carrying value includes the effect of fair value adjustments that were applied to the consumer PCI portfolio at the date of acquisition.
- (b) Management concluded as part of the Firm's regular assessment of the PCI loan pools that it was probable that higher expected credit losses would result in a decrease in expected cash flows. As a result, an allowance for loan losses for impairment of these pools has been recognized.
- (c) Represents the aggregate unpaid principal balance of loans divided by the estimated current property value. Current property values are estimated, at a minimum, quarterly, based on home valuation models using nationally recognized home price index valuation estimates incorporating actual data to the extent available and forecasted data where actual data is not available. These property values do not represent actual appraised loan level collateral values; as such, the resulting ratios are necessarily imprecise and should be viewed as estimates. Current estimated combined LTV for junior lien home equity loans considers all available lien positions related to the property.
- (d) Refreshed FICO scores represent each borrower's most recent credit score obtained by the Firm. The Firm obtains refreshed FICO scores at least quarterly.
- (e) For home equity loans, prior-period amounts have been revised to conform with the current-period presentation.

Approximately 20% of the PCI home equity portfolio are senior lien loans; the remaining balance are junior lien HELOANS or HELOCs. The following table represents delinquency statistics for PCI junior lien home equity loans based on unpaid principal balance as of December 31, 2011 and 2010.

December 31, 2011 (in millions, except ratios)	Delinquencies			Total loans	Total 30+ day delinquency rate
	30–89 days past due	90–149 days past due	150+ days past due		
HELOCs:					
Within the revolving period	\$ 500	\$ 296	\$ 543	\$ 18,246	7.34%
Within the required amortization period	16	11	5	400	8.00
HELOANS	53	29	44	1,327	9.50
Total	\$ 569	\$ 336	\$ 592	\$ 19,973	7.50%

December 31, 2010 (in millions, except ratios)	Delinquencies			Total loans	Total 30+ day delinquency rate
	30–89 days past due	90–149 days past due	150+ days past due		
HELOCs:					
Within the revolving period	\$ 601	\$ 404	\$ 428	\$ 21,172	6.77%
Within the required amortization period	1	—	1	37	5.41
HELOANS	79	49	46	1,573	11.06
Total	\$ 681	\$ 453	\$ 475	\$ 22,782	7.06%

- (a) In general, HELOCs are open-ended, revolving loans for a 10-year period, after which time the HELOC converts to a loan with a 20-year amortization period.
- (b) Substantially all undrawn HELOCs within the revolving period have been closed.
- (c) Predominantly all of these loans have been modified to provide a more affordable payment to the borrower.

The table below sets forth the accretable yield activity for the Firm's PCI consumer loans for the years ended December 31, 2011, 2010 and 2009, and represents the Firm's estimate of gross interest income expected to be earned over the remaining life of the PCI loan portfolios. This table excludes the cost to fund the PCI portfolios, and therefore does not represent net interest income expected to be earned on these portfolios.

Year ended December 31, (in millions, except ratios)	Total PCI		
	2011	2010	2009
Beginning balance	\$ 19,097	\$ 25,544	\$ 32,619
Accretion into interest income	(2,767)	(3,232)	(4,363)
Changes in interest rates on variable-rate loans	(573)	(819)	(4,849)
Other changes in expected cash flows	3,315	(2,396)	2,137
Balance at December 31	\$ 19,072	\$ 19,097	\$ 25,544
Accretable yield percentage	4.33%	4.35%	5.14%

- (a) Other changes in expected cash flows may vary from period to period as the Firm continues to refine its cash flow model and periodically updates model assumptions. For the year ended December 31, 2011, other changes in expected cash flows were largely driven by the impact of modifications, but also related to changes in prepayment assumptions. For the years ended December 31, 2010 and 2009, other changes in expected cash flows were principally driven by changes in prepayment assumptions, as well as reclassification to the nonaccretable difference. Changes to prepayment assumptions change the expected remaining life of the portfolio, which drives changes in expected future interest cash collections. Such changes do not have a significant impact on the accretable yield percentage.

The factors that most significantly affect estimates of gross cash flows expected to be collected, and accordingly the accretable yield balance, include: (i) changes in the benchmark interest rate indices for variable-rate products such as option ARM and home equity loans; and (ii) changes in prepayment assumptions.

Since the date of acquisition, the decrease in the accretable yield percentage has been primarily related to a decrease in interest rates on variable-rate loans and, to a lesser extent, extended loan liquidation periods. Certain events, such as extended loan liquidation periods, affect the timing of expected cash flows but not the amount of cash expected to

be received (i.e., the accretable yield balance). Extended loan liquidation periods reduce the accretable yield percentage because the same accretable yield balance is recognized against a higher-than-expected loan balance over a longer-than-expected period of time.

Credit card loan portfolio

The Credit card portfolio segment includes credit card loans originated and purchased by the Firm, including those acquired in the Washington Mutual transaction.

Delinquency rates are the primary credit quality indicator for credit card loans as they provide an early warning that borrowers may be experiencing difficulties (30-days past due), as well as information on those borrowers that have been delinquent for a longer period of time (90-days past due). In addition to delinquency rates, the geographic distribution of the loans provides insight as to the credit quality of the portfolio based on the regional economy.

The borrower's credit score is another general indicator of credit quality. Because the borrower's credit score tends to

be a lagging indicator of credit quality, the Firm does not use credit scores as a primary indicator of credit quality. However, the distribution of such scores provides a general indicator of credit quality trends within the portfolio. Refreshed FICO score information for a statistically significant random sample of the credit card portfolio is indicated in the table below, as FICO is considered to be the industry benchmark for credit scores.

The Firm generally originates new card accounts to prime consumer borrowers. However, certain cardholders' refreshed FICO scores may change over time, depending on the performance of the cardholder and changes in credit score technology.

The table below sets forth information about the Firm's credit card loans.

As of or for the year ended December 31, (in millions, except ratios)	Chase, excluding Washington Mutual portfolio ^(b)		Washington Mutual portfolio ^(b)		Total credit card ^(a)	
	2011	2010	2011	2010	2011	2010
Net charge-offs	\$ 5,668	\$ 11,191	\$ 1,257	\$ 2,846	\$ 6,925	\$ 14,037
% of net charge-offs to retained loans	4.91%	8.73%	10.49%	17.73%	5.44%	9.73%
Loan delinquency						
Current and less than 30 days past due and still accruing	\$ 118,054	\$ 117,248	\$ 10,410	\$ 12,670	\$ 128,464	\$ 129,918
30–89 days past due and still accruing	1,509	2,092	299	459	1,808	2,551
90 or more days past due and still accruing	1,558	2,449	344	604	1,902	3,053
Nonaccrual loans	1	2	—	—	1	2
Total retained loans	\$ 121,122	\$ 121,791	\$ 11,053	\$ 13,733	\$ 132,175	\$ 135,524
Loan delinquency ratios						
% of 30+ days past due to total retained loans	2.53%	3.73%	5.82%	7.74%	2.81%	4.14%
% of 90+ days past due to total retained loans	1.29	2.01	3.11	4.40	1.44	2.25
Credit card loans by geographic region						
California	\$ 15,479	\$ 15,454	\$ 2,119	\$ 2,650	\$ 17,598	\$ 18,104
New York	9,755	9,540	839	1,032	10,594	10,572
Texas	9,418	9,217	821	1,006	10,239	10,223
Florida	6,658	6,724	925	1,165	7,583	7,889
Illinois	7,108	7,077	440	542	7,548	7,619
New Jersey	5,208	5,070	396	494	5,604	5,564
Ohio	4,882	5,035	320	401	5,202	5,436
Pennsylvania	4,434	4,521	345	424	4,779	4,945
Michigan	3,777	3,956	217	273	3,994	4,229
Virginia	3,061	3,020	237	295	3,298	3,315
Georgia	2,737	2,834	315	398	3,052	3,232
Washington	2,081	2,053	359	438	2,440	2,491
All other	46,524	47,290	3,720	4,615	50,244	51,905
Total retained loans	\$ 121,122	\$ 121,791	\$ 11,053	\$ 13,733	\$ 132,175	\$ 135,524
Percentage of portfolio based on carrying value with estimated refreshed FICO scores						
Equal to or greater than 660	83.3%	80.6%	62.6%	56.4%	81.4%	77.9%
Less than 660	16.7	19.4	37.4	43.6	18.6	22.1

(a) Refreshed FICO scores are estimated based on a statistically significant random sample of credit card accounts in the credit card portfolio for the period shown. The Firm obtains refreshed FICO scores at least quarterly.

(b) Includes billed finance charges and fees net of an allowance for uncollectible amounts.

Credit card impaired loans and loan modifications

The table below sets forth information about the Firm's impaired credit card loans. All of these loans are considered to be impaired as they have been modified in TDRs.

December 31, (in millions)	Chase, excluding Washington Mutual portfolio		Washington Mutual portfolio		Total credit card	
	2011	2010	2011	2010	2011	2010
Impaired loans with an allowance ^{(a)(b)}						
Credit card loans with modified payment terms	\$ 4,959	\$ 6,685	\$ 1,116	\$ 1,570	\$ 6,075	\$ 8,255
Modified credit card loans that have reverted to pre-modification payment terms	930	1,439	209	311	1,139	1,750
Total impaired loans	\$ 5,889	\$ 8,124	\$ 1,325	\$ 1,881	\$ 7,214	\$ 10,005
Allowance for loan losses related to impaired loans	\$ 2,195	\$ 3,175	\$ 532	\$ 894	\$ 2,727	\$ 4,069

(a) The carrying value and the unpaid principal balance are the same for credit card impaired loans.

(b) There were no impaired loans without an allowance.

(c) Represents credit card loans outstanding to borrowers enrolled in a credit card modification program as of the date presented.

(d) Represents credit card loans that were modified in TDRs but that have subsequently reverted back to the loans' pre-modification payment terms. At December 31, 2011 and 2010, \$762 million and \$1.2 billion, respectively, of loans have reverted back to the pre-modification payment terms of the loans due to noncompliance with the terms of the modified loans. Based on the Firm's historical experience a substantial portion of these loans is expected to be charged-off in accordance with the Firm's standard charge-off policy. The remaining \$377 million and \$590 million at December 31, 2011 and 2010, respectively, of these loans are to borrowers who have successfully completed a short-term modification program. The Firm continues to report these loans as TDRs since the borrowers' credit lines remain closed.

The following table presents average balances of impaired credit card loans and interest income recognized on those loans.

Year ended December 31, (in millions)	Average impaired loans			Interest income on impaired loans		
	2011	2010	2009	2011	2010	2009
Chase, excluding Washington Mutual portfolio	\$ 6,914	\$ 8,747	\$ 3,059	\$ 360	\$ 479	\$ 181
Washington Mutual portfolio	1,585	1,983	991	103	126	70
Total credit card	\$ 8,499	\$ 10,730	\$ 4,050	\$ 463	\$ 605	\$ 251

Loan modifications

JPMorgan Chase may offer one of a number of loan modification programs to credit card borrowers who are experiencing financial difficulty. The Firm has short-term programs for borrowers who may be in need of temporary relief, and long-term programs for borrowers who are experiencing a more fundamental level of financial difficulties. Most of the credit card loans have been modified under long-term programs. Modifications under long-term programs involve placing the customer on a fixed payment plan, generally for 60 months. Modifications under all short- and long-term programs typically include reducing the interest rate on the credit card. Certain borrowers enrolled in a short-term modification program may be given the option to re-enroll in a long-term program. Substantially all modifications are considered to be TDRs.

If the cardholder does not comply with the modified payment terms, then the credit card loan agreement reverts back to its pre-modification payment terms. Assuming that the cardholder does not begin to perform in accordance with those payment terms, the loan continues to age and will ultimately be charged-off in accordance with the Firm's standard charge-off policy. In addition, if a borrower successfully completes a short-term modification program, then the loan reverts back to its pre-modification payment terms. However, in most cases, the Firm does not reinstate the borrower's line of credit.

Notes to consolidated financial statements

The following tables provide information regarding the nature and extent of modifications of credit card loans for the period presented.

Year ended December 31, 2011 (in millions)	Chase, excluding Washington Mutual portfolio		Washington Mutual portfolio		Total credit card	
	Short-term programs	Long-term programs	Short-term programs	Long-term programs	Short-term programs	Long-term programs
New enrollments	\$ 141	\$ 2,075	\$ 26	\$ 448	\$ 167	\$ 2,523

Financial effects of modifications and redefaults

The following tables provide information about the financial effects of the concessions granted on credit card loans modified in TDRs and redefaults for the period presented.

Year ended December 31, 2011 (in millions, except weighted-average data)	Chase, excluding Washington Mutual portfolio		Washington Mutual portfolio		Total credit card	
Weighted-average interest rate of loans – before TDR		14.91%		21.38%		16.05%
Weighted-average interest rate of loans – after TDR		5.04		6.39		5.28
Loans that redefaulted within one year of modification ^(a)	\$	559	\$	128	\$	687

(a) Represents loans modified in TDRs that experienced a payment default in the period presented, and for which the payment default occurred within one year of the modification. The amounts presented represent the balance of such loans as of the end of the quarter in which they defaulted.

For credit card loans modified in TDRs, payment default is deemed to have occurred when the loans become two payments past due. At the time of default, a loan is removed from the modification program and reverts back to its pre-modification terms. Based on historical experience, a substantial portion of these loans are expected to be charged-off in accordance with the Firm's standard charge-off policy. Also based on historical experience, the estimated weighted-average ultimate default rate for modified credit card loans was 35.47% at December 31, 2011, and 36.45% at December 31, 2010.

Note 15 – Allowance for credit losses

JPMorgan Chase's allowance for loan losses covers the wholesale and consumer, including credit card, loan portfolios, and represents management's estimate of probable credit losses inherent in the Firm's loan portfolio. The allowance for loan losses includes an asset-specific component, a formula-based component and a component related to PCI loans, as described below. Management also estimates an allowance for wholesale and consumer lending-related commitments using methodologies similar to those used to estimate the allowance on the underlying loans. During 2011, the Firm did not make any significant changes to the methodologies or policies used to determine its allowance for credit losses; such policies are described in the following paragraphs.

The asset-specific component of the allowance relates to loans considered to be impaired, which includes loans that have been modified in TDRs as well as risk-rated loans that have been placed on nonaccrual status. To determine the asset-specific component of the allowance, larger loans are evaluated individually, while smaller loans are evaluated as pools using historical loss experience for the respective class of assets. Risk-rated loans (primarily wholesale loans) are segmented by risk rating, while scored loans (i.e.,

consumer loans) are pooled by product type.

The Firm generally measures the asset-specific allowance as the difference between the recorded investment in the loan and the present value of the cash flows expected to be collected, discounted at the loan's original effective interest rate. Subsequent changes in impairment are reported as an adjustment to the provision for loan losses. In certain cases, the asset-specific allowance is determined using an observable market price, and the allowance is measured as the difference between the recorded investment in the loan and the loan's fair value. Impaired collateral-dependent loans are charged down to the fair value of collateral less costs to sell and therefore may not be subject to an asset-specific reserve as for other impaired loans. See Note 14 on pages 231–252 of this Annual Report for more information about charge-offs and collateral-dependent loans.

The asset-specific component of the allowance for impaired loans that have been modified in TDRs incorporates the effects of foregone interest, if any, in the present value calculation and also incorporates the effect of the modification on the loan's expected cash flows, which considers the potential for redefault. For wholesale loans modified in TDRs, expected losses incorporate redefaults based on management's expectation of the borrower's ability to repay under the modified terms. For residential real estate loans modified in TDRs, the Firm develops product-specific probability of default estimates, which are applied at a loan level to compute expected losses. In developing these probabilities of default, the Firm considers the relationship between the credit quality characteristics of the underlying loans and certain assumptions about home prices and unemployment, based upon industry-wide data. The Firm also considers its own historical loss experience to date based on actual redefaulted modified loans. For credit card loans modified in TDRs, expected losses incorporate projected redefaults based on the Firm's

historical experience by type of modification program.

The formula-based component is based on a statistical calculation to provide for probable principal losses inherent in performing risk-rated loans and all consumer loans, except for any loans restructured in TDRs and PCI loans. See Note 14 on pages 231–252 of this Annual Report for more information on PCI loans.

For risk-rated loans, the statistical calculation is the product of an estimated probability of default (“PD”) and an estimated loss given default (“LGD”). These factors are differentiated by risk rating and expected maturity. In assessing the risk rating of a particular loan, among the factors considered are the obligor’s debt capacity and financial flexibility, the level of the obligor’s earnings, the amount and sources for repayment, the level and nature of contingencies, management strength, and the industry and geography in which the obligor operates. These factors are based on an evaluation of historical and current information, and involve subjective assessment and interpretation. Emphasizing one factor over another or considering additional factors could impact the risk rating assigned by the Firm to that loan. PD estimates are based on observable external through-the-cycle data, using credit-rating agency default statistics. LGD estimates are based on the Firm’s history of actual credit losses over more than one credit cycle.

For scored loans, the statistical calculation is performed on pools of loans with similar risk characteristics (e.g., product type) and generally computed by applying expected loss factors to outstanding principal balances over an estimated loss emergence period. The loss emergence period represents the time period between the date at which the loss is estimated to have been incurred and the ultimate realization of that loss (through a charge-off). Estimated loss emergence periods may vary by product and may change over time; management applies judgment in estimating loss emergence periods, using available credit information and trends.

Loss factors are statistically derived and sensitive to changes in delinquency status, credit scores, collateral values and other risk factors. The Firm uses a number of different forecasting models to estimate both the PD and the loss severity, including delinquency roll rate models and credit loss severity models. In developing PD and loss severity assumptions, the Firm also considers known and anticipated changes in the economic environment, including changes in home prices, unemployment rates and other risk indicators.

A nationally recognized home price index measure is used to estimate both the PD and the loss severity on residential real estate loans at the metropolitan statistical areas (“MSA”) level. Loss severity estimates are regularly

validated by comparison to actual losses recognized on defaulted loans, market-specific real estate appraisals and property sales activity. The economic impact of potential modifications of residential real estate loans is not included in the statistical calculation because of the uncertainty regarding the type and results of such modifications.

Management applies judgment within an established framework to adjust the results of applying the statistical calculation described above. The determination of the appropriate adjustment is based on management’s view of uncertainties that have occurred but that are not yet reflected in the loss factors and that relate to current macroeconomic and political conditions, the quality of underwriting standards and other relevant internal and external factors affecting the credit quality of the portfolio. In addition, for the risk-rated portfolios, any adjustments made to the statistical calculation also consider concentrated and deteriorating industries. For the scored loan portfolios, adjustments to the statistical calculation are accomplished in part by analyzing the historical loss experience for each major product segment. Factors related to unemployment, home prices, borrower behavior and lien position, the estimated effects of the mortgage foreclosure-related settlement with federal and state officials and uncertainties regarding the ultimate success of loan modifications are incorporated into the calculation, as appropriate. For junior lien products, management considers the delinquency and/or modification status of any senior liens in determining the adjustment.

Management establishes an asset-specific allowance for lending-related commitments that are considered impaired and computes a formula-based allowance for performing wholesale and consumer lending-related commitments. These are computed using a methodology similar to that used for the wholesale loan portfolio, modified for expected maturities and probabilities of drawdown.

Determining the appropriateness of the allowance is complex and requires judgment by management about the effect of matters that are inherently uncertain. Subsequent evaluations of the loan portfolio, in light of the factors then prevailing, may result in significant changes in the allowances for loan losses and lending-related commitments in future periods.

At least quarterly, the allowance for credit losses is reviewed by the Chief Risk Officer, the Chief Financial Officer and the Controller of the Firm and discussed with the Risk Policy and Audit Committees of the Board of Directors of the Firm. As of December 31, 2011, JPMorgan Chase deemed the allowance for credit losses to be appropriate (i.e., sufficient to absorb probable credit losses that are inherent in the portfolio).

Notes to consolidated financial statements

Allowance for credit losses and loans and lending-related commitments by impairment methodology

The table below summarizes information about the allowance for loan losses, loans by impairment methodology, the allowance for lending-related commitments and lending-related commitments by impairment methodology.

Year ended December 31, (in millions)	2011			
	Wholesale	Consumer, excluding credit card	Credit card	Total
Allowance for loan losses				
Beginning balance at January 1,	\$ 4,761	\$ 16,471	\$ 11,034	\$ 32,266
Cumulative effect of change in accounting principles	—	—	—	—
Gross charge-offs	916	5,419	8,168	14,503
Gross recoveries	(476)	(547)	(1,243)	(2,266)
Net charge-offs	440	4,872	6,925	12,237
Provision for loan losses	17	4,670	2,925	7,612
Other	(22)	25	(35)	(32)
Ending balance at December 31,	\$ 4,316	\$ 16,294	\$ 6,999	\$ 27,609
Allowance for loan losses by impairment methodology				
Asset-specific ^(b)	\$ 516	\$ 828	\$ 2,727	\$ 4,071
Formula-based	3,800	9,755	4,272	17,827
PCI	—	5,711	—	5,711
Total allowance for loan losses	\$ 4,316	\$ 16,294	\$ 6,999	\$ 27,609
Loans by impairment methodology				
Asset-specific	\$ 2,549	\$ 9,892	\$ 7,214	\$ 19,655
Formula-based	275,825	232,989	124,961	633,775
PCI	21	65,546	—	65,567
Total retained loans	\$ 278,395	\$ 308,427	\$ 132,175	\$ 718,997
Impaired collateral-dependent loans				
Net charge-offs ^(c)	\$ 128	\$ 110	\$ —	\$ 238
Loans measured at fair value of collateral less cost to sell	833	830 ^(d)	—	1,663
Allowance for lending-related commitments				
Beginning balance at January 1,	\$ 711	\$ 6	\$ —	\$ 717
Cumulative effect of change in accounting principles	—	—	—	—
Provision for lending-related commitments	(40)	2	—	(38)
Other	(5)	(1)	—	(6)
Ending balance at December 31,	\$ 666	\$ 7	\$ —	\$ 673
Allowance for lending-related commitments by impairment methodology				
Asset-specific	\$ 150	\$ —	\$ —	\$ 150
Formula-based	516	7	—	523
Total allowance for lending-related commitments	\$ 666	\$ 7	\$ —	\$ 673
Lending-related commitments by impairment methodology				
Asset-specific	\$ 865	\$ —	\$ —	\$ 865
Formula-based	381,874	62,307	530,616	974,797
Total lending-related commitments	\$ 382,739	\$ 62,307	\$ 530,616	\$ 975,662

(a) Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. Upon adoption of the guidance, the Firm consolidated its Firm-sponsored credit card securitization trusts, its Firm-administered multi-seller conduits and certain other consumer loan securitization entities, primarily mortgage-related. As a result, \$7.4 billion, \$14 million and \$127 million, respectively, of allowance for loan losses were recorded on-balance sheet with the consolidation of these entities. For further discussion, see Note 16 on pages 256–267 of this Annual Report.

(b) Includes risk-rated loans that have been placed on nonaccrual status and loans that have been modified in a TDR.

(c) Prior periods have been revised to conform with the current presentation.

(d) Includes collateral-dependent residential mortgage loans that are charged off to the fair value of the underlying collateral less cost to sell. These loans considered collateral-dependent under regulatory guidance because they involve modifications where an interest-only period is provided or a significant portion of principal is deferred.

(table continued from previous page)

2010

Consumer,	
Wholesale	
\$	7,145
	14
	1,989
	(262)
	1,727
	(673)
	2
\$	4,761
\$	1,574
	3,187
	—
\$	4,761
\$	5,486
	216,980
	44
\$	222,510
\$	636
	1,269
\$	927
	(18)
	(177)
	(21)
\$	711
\$	180
	531
\$	711
\$	1,005
	345,074
\$	346,079

Significant Firm-sponsored variable interest entities

Credit card securitizations

The Card business securitizes originated and purchased credit card loans, primarily through the Chase Issuance Trust (the "Trust"). The Firm's continuing involvement in credit card securitizations includes servicing the receivables, retaining an undivided seller's interest in the receivables, retaining certain senior and subordinated securities and maintaining escrow accounts.

The Firm is considered to be the primary beneficiary of these Firm-sponsored credit card securitization trusts based on the Firm's ability to direct the activities of these VIEs through its servicing responsibilities and other duties, including making decisions as to the receivables that are transferred into those trusts and as to any related modifications and workouts. Additionally, the nature and extent of the Firm's other continuing involvement with the trusts, as indicated above, obligates the Firm to absorb losses and gives the Firm the right to receive certain benefits from these VIEs that could potentially be significant.

Effective January 1, 2010, the Firm consolidated the assets and liabilities of the Firm-sponsored credit card securitization trusts as a result of the implementation of VIE consolidation accounting guidance. See the table on page 264 of this Note for more information on the consolidation of credit card securitizations.

The underlying securitized credit card receivables and other assets are available only for payment of the beneficial interests issued by the securitization trusts; they are not available to pay the Firm's other obligations or the claims of the Firm's other creditors.

The agreements with the credit card securitization trusts require the Firm to maintain a minimum undivided interest in the credit card trusts (which generally ranges from 4% to 12%). As of December 31, 2011 and 2010, the Firm held undivided interests in Firm-sponsored credit card securitization trusts of \$13.7 billion and \$17.2 billion, respectively. The Firm maintained an average undivided interest in principal receivables owned by those trusts of approximately 22% and 19% for the years ended December 31, 2011 and 2010, respectively. The Firm also retained \$541 million and \$1.1 billion of senior securities and \$3.0 billion and \$3.2 billion of subordinated securities in certain of its credit card securitization trusts as of December 31, 2011 and 2010, respectively. The Firm's undivided interests in the credit card trusts and securities retained are eliminated in consolidation.

Firm-sponsored mortgage and other securitization trusts

The Firm securitizes (or has securitized) originated and purchased residential mortgages, commercial mortgages and other consumer loans (including automobile and student loans) primarily in its IB and RFS businesses. Depending on the particular transaction, as well as the respective business involved, the Firm may act as the servicer of the loans and/or retain certain beneficial interests in the securitization trusts.

Notes to consolidated financial statements

The following table presents the total unpaid principal amount of assets held in Firm-sponsored securitization entities in which the Firm has continuing involvement, including those that are consolidated or not consolidated by the Firm. Continuing involvement includes servicing the loans; holding senior interests or subordinated interests; recourse or guarantee arrangements; and derivative transactions. In certain instances, the Firm's only continuing involvement is servicing the loans. See Securitization activity on pages 264–265 of this Note for further information regarding the Firm's cash flows with and interests retained in nonconsolidated VIEs.

December 31, 2011 ^(f) (in billions)	Principal amount outstanding			JPMorgan Chase interest in securitized assets in nonconsolidated VIEs ^(f)		
	Total assets held by securitization VIEs	Assets held in consolidated securitization VIEs	Assets held in nonconsolidated securitization VIEs with continuing involvement	Trading assets	AFS securities	Total interests held by JPMorgan Chase
Securitization-related						
Residential mortgage:						
Prime ^(b)	\$ 129.5	\$ 2.4	\$ 101.0	\$ 0.6	\$ —	\$ 0.6
Subprime	38.3	0.2	35.8	—	—	—
Option ARMs	31.1	—	31.1	—	—	—
Commercial and other	139.3	—	93.3	1.7	2.0	3.7
Student	4.1	4.1	—	—	—	—
Total	\$ 342.3	\$ 6.7	\$ 261.2	\$ 2.3	\$ 2.0	\$ 4.3

December 31, 2010 ^(f) (in billions)	Principal amount outstanding			JPMorgan Chase interest in securitized assets in nonconsolidated VIEs ^(f)		
	Total assets held by securitization VIEs	Assets held in consolidated securitization VIEs	Assets held in nonconsolidated securitization VIEs with continuing involvement	Trading assets	AFS securities	Total interests held by JPMorgan Chase
Securitization-related						
Residential mortgage:						
Prime ^(b)	\$ 153.1	\$ 2.2	\$ 143.8	\$ 0.7	\$ —	\$ 0.7
Subprime	44.0	1.6	40.7	—	—	—
Option ARMs	36.1	0.3	35.8	—	—	—
Commercial and other	153.4	—	106.2	2.0	0.9	2.9
Student	4.5	4.5	—	—	—	—
Total	\$ 391.1	\$ 8.6	\$ 326.5	\$ 2.7	\$ 0.9	\$ 3.6

(a) Excludes U.S. government agency securitizations. See page 265 of this Note for information on the Firm's loan sales to U.S. government agencies.

(b) Includes Alt-A loans.

(c) Consists of securities backed by commercial loans (predominantly real estate) and non-mortgage-related consumer receivables purchased from third parties. The Firm generally does not retain a residual interest in its sponsored commercial mortgage securitization transactions.

(d) The table above excludes the following: retained servicing (see Note 17 on pages 267–271 of this Annual Report for a discussion of MSRs); securities retained from loans sales to U.S. government agencies; interest rate and foreign exchange derivatives primarily used to manage interest rate and foreign exchange risks of securitization entities (See Note 6 on pages 202–210 of this Annual Report for further information on derivatives); senior and subordinated securities of \$110 million and \$8 million, respectively, at December 31, 2011, and \$182 million and \$18 million, respectively, at December 31, 2010, which the Firm purchased in connection with IB's secondary market-making activities.

(e) Includes interests held in re-securitization transactions.

(f) As of December 31, 2011 and 2010, 68% and 66%, respectively, of the Firm's retained securitization interests, which are carried at fair value, were risk rated "A" or better, on an S&P-equivalent basis. The retained interests in prime residential mortgages consisted of \$136 million and \$157 million of investment-grade and \$427 million and \$552 million of noninvestment-grade retained interests at December 31, 2011 and 2010, respectively. The retained interests in commercial and other securitizations trusts consisted of \$3.4 billion and \$2.6 billion of investment-grade and \$283 million and \$250 million of noninvestment-grade retained interests at December 31, 2011 and 2010, respectively.

Residential mortgage

The Firm securitizes residential mortgage loans originated by RFS, as well as residential mortgage loans purchased from third parties by either RFS or IB. RFS generally retains servicing for all residential mortgage loans originated or purchased by RFS, and for certain mortgage loans purchased by IB. For securitizations serviced by RFS, the Firm has the power to direct the significant activities of the VIE because it is responsible for decisions related to loan modifications and workouts. RFS may retain an interest upon securitization.

In addition, IB engages in underwriting and trading activities involving securities issued by Firm-sponsored securitization trusts. As a result, IB at times retains senior and/or subordinated interests (including residual interests) in residential mortgage securitizations upon securitization, and/or reacquires positions in the secondary market in the normal course of business. In certain instances, as a result of the positions retained or reacquired by IB or held by RFS, when considered together with the servicing arrangements entered into by RFS, the Firm is deemed to be the primary beneficiary of certain securitization trusts. See the table on page 264 of this Note for more information on the consolidated residential mortgage securitizations.

The Firm does not consolidate a mortgage securitization (Firm-sponsored or third-party-sponsored) when it is not the servicer (and therefore does not have the power to direct the most significant activities of the trust) or does not hold a beneficial interest in the trust that could potentially be significant to the trust. At December 31, 2011 and 2010, the Firm did not consolidate the assets of certain Firm-sponsored residential mortgage securitization VIEs, in which the Firm had continuing involvement, primarily due to the fact that the Firm did not hold an interest in these trusts that could potentially be significant to the trusts. See the table on page 258 of this Note for further information on interests held in nonconsolidated securitizations.

Commercial mortgages and other consumer securitizations IB originates and securitizes commercial mortgage loans, and engages in underwriting and trading activities involving the securities issued by securitization trusts. IB may retain unsold senior and/or subordinated interests in commercial mortgage securitizations at the time of securitization but, generally, the Firm does not service commercial loan securitizations. For commercial mortgage securitizations the power to direct the significant activities of the VIE generally is held by the servicer or investors in a specified class of securities ("controlling class"). See the table on page 264 of this Note for more information on the consolidated commercial mortgage securitizations, and the table on page 258 of this Note for further information on interests held in nonconsolidated securitizations.

The Firm also securitizes automobile and student loans. The Firm retains servicing responsibilities for all originated and certain purchased student and automobile loans and has the power to direct the activities of these VIEs through these servicing responsibilities. See the table on page 264

of this Note for more information on the consolidated student loan securitizations, and the table on page 258 of this Note for further information on interests held in nonconsolidated securitizations.

Re-securitizations

The Firm engages in certain re-securitization transactions in which debt securities are transferred to a VIE in exchange for new beneficial interests. These transfers occur in connection with both agency (Fannie Mae, Freddie Mac and Ginnie Mae) and nonagency (private-label) sponsored VIEs, which may be backed by either residential or commercial mortgages. The Firm's consolidation analysis is largely dependent on the Firm's role and interest in the re-securitization trusts. During the years ended December 31, 2011, 2010 and 2009, the Firm transferred \$24.9 billion, \$33.9 billion and \$19.1 billion, respectively, of securities to agency VIEs, and \$381 million, \$1.3 billion and \$4.0 billion, respectively, of securities to private-label VIEs.

Most re-securitizations with which the Firm is involved are client-driven transactions in which a specific client or group of clients are seeking a specific return or risk profile. For these transactions, the Firm has concluded that the decision-making power of the entity is shared between the Firm and its client(s), considering the joint effort and decisions in establishing the re-securitization trust and its assets, as well as the significant economic interest the client holds in the re-securitization trust; therefore the Firm does not consolidate the re-securitization VIE.

In more limited circumstances, the Firm creates a re-securitization trust independently and not in conjunction with specific clients. In these circumstances, the Firm is deemed to have the unilateral ability to direct the most significant activities of the re-securitization trust because of the decisions made during the establishment and design of the trust; therefore, the Firm consolidates the re-securitization VIE if the Firm holds an interest that could potentially be significant.

Additionally, the Firm may invest in beneficial interests of third-party securitizations and generally purchases these interests in the secondary market. In these circumstances, the Firm does not have the unilateral ability to direct the most significant activities of the re-securitization trust, either because it wasn't involved in the initial design of the trust, or the Firm is involved with an independent third party sponsor and demonstrates shared power over the creation of the trust; therefore, the Firm does not consolidate the re-securitization VIE.

As of December 31, 2011 and 2010, the Firm did not consolidate any agency re-securitizations. As of December 31, 2011 and 2010, the Firm consolidated \$348 million and \$477 million, respectively, of assets, and \$139 million and \$230 million, respectively, of liabilities of private-label re-securitizations. See the table on page 264 of this Note for more information on the consolidated re-securitization transactions.

As of December 31, 2011 and 2010, total assets of nonconsolidated Firm-sponsored private-label re-securitization entities were \$3.3 billion and \$3.6 billion, respectively. At December 31, 2011 and 2010, the Firm held approximately \$3.6 billion and \$3.5 billion, respectively, of interests in nonconsolidated agency re-securitization entities, and \$14 million and \$46 million, respectively, of senior and subordinated interests in nonconsolidated private-label re-securitization entities. See the table on page 258 of this Note for further information on interests held in nonconsolidated securitizations.

Multi-seller conduits

Multi-seller conduit entities are separate bankruptcy remote entities that purchase interests in, and make loans secured by, pools of receivables and other financial assets pursuant to agreements with customers of the Firm. The conduits fund their purchases and loans through the issuance of highly rated commercial paper. The primary source of repayment of the commercial paper is the cash flows from the pools of assets. In most instances, the assets are structured with deal-specific credit enhancements provided to the conduits by the customers (i.e., sellers) or other third parties. Deal-specific credit enhancements are generally structured to cover a multiple of historical losses expected on the pool of assets, and are typically in the form of overcollateralization provided by the seller. The deal-specific credit enhancements mitigate the Firm's potential losses on its agreements with the conduits.

To ensure timely repayment of the commercial paper, each asset pool financed by the conduits has a minimum 100% deal-specific liquidity facility associated with it provided by JPMorgan Chase Bank, N.A. JPMorgan Chase Bank, N.A., also provides the multi-seller conduit vehicles with uncommitted program-wide liquidity facilities and program-wide credit enhancement in the form of standby letters of credit. The amount of program-wide credit enhancement required varies by conduit and ranges between 5% and 10% of the commercial paper that is outstanding.

The Firm consolidates its Firm-administered multi-seller conduits, as the Firm has both the power to direct the significant activities of the conduits and a potentially significant economic interest in the conduits. As administrative agent and in its role in structuring transactions, the Firm makes decisions regarding asset types and credit quality, and manages the commercial paper funding needs of the conduits. The Firm's interests that could potentially be significant to the VIEs include the fees received as administrative agent and liquidity and program-wide credit enhancement provider, as well as the potential exposure to the liquidity and credit enhancement facilities provided to the conduits. See page 264 of this Note for further information on consolidated VIE assets and liabilities.

In the normal course of business, JPMorgan Chase trades and invests in commercial paper, including commercial paper issued by the Firm-administered multi-seller conduits. The Firm held \$11.3 billion of the commercial

paper issued by the Firm-administered multi-seller conduits at December 31, 2011, which was eliminated in consolidation. The Firm did not hold commercial paper issued by the Firm-administered multi-seller conduits at December 31, 2010. The Firm's investments were not driven by market illiquidity and the Firm is not obligated under any agreement to purchase the commercial paper issued by the Firm-administered multi-seller conduits.

Deal-specific liquidity facilities, program-wide liquidity and credit enhancement provided by the Firm have been eliminated in consolidation. The Firm provides lending-related commitments to certain clients of the Firm-administered multi-seller conduits. The unfunded portion of these commitments was \$10.8 billion and \$10.0 billion at December 31, 2011 and 2010 respectively, which are reported as off-balance sheet lending-related commitments. For more information on off-balance sheet lending-related commitments, see Note 29 on pages 2889 of this Annual Report.

VIEs associated with investor intermediation activities As a financial intermediary, the Firm creates certain types of VIEs and also structures transactions with these VIEs, typically using derivatives, to meet investor needs. The Firm may also provide liquidity and other support. The risks inherent in the derivative instruments or liquidity commitments are managed similarly to other credit, market or liquidity risks to which the Firm is exposed. The principal types of VIEs for which the Firm is engaged in on behalf of clients are municipal bond vehicles, credit-related note vehicles and asset swap vehicles.

Municipal bond vehicles

The Firm has created a series of trusts that provide short-term investors with qualifying tax-exempt investments, and that allow investors in tax-exempt securities to finance their investments at short-term tax-exempt rates. In a typical transaction, the vehicle purchases fixed-rate longer-term highly rated municipal bonds and funds the purchase by issuing two types of securities: (1) puttable floating-rate certificates and (2) inverse floating-rate residual interests ("residual interests"). The maturity of each of the puttable floating-rate certificates and the residual interests is equal to the life of the vehicle, while the maturity of the underlying municipal bonds is typically longer. Holders of the puttable floating-rate certificates may "put," or tender, the certificates if the remarketing agent cannot successfully remarket the floating-rate certificates to another investor. A liquidity facility conditionally obligates the liquidity provider to fund the purchase of the tendered floating-rate certificates. Upon termination of the vehicle, proceeds from the sale of the underlying municipal bonds would first repay any funded liquidity facility or outstanding floating-rate certificates and the remaining amount, if any, would be paid to the residual interests. If the proceeds from the sale of the underlying municipal bonds are not sufficient to repay the liquidity facility, in certain transactions the liquidity provider has recourse to the residual interest holders for reimbursement. Certain residual interest holders may be required to post collateral with the Firm, as liquidity

provider, to support such reimbursement obligations should the market value of the municipal bonds decline.

JPMorgan Chase Bank, N.A. often serves as the sole liquidity provider, and J.P. Morgan Securities LLC as remarketing agent, of the puttable floating-rate certificates. The liquidity provider's obligation to perform is conditional and is limited by certain termination events, which include bankruptcy or failure to pay by the municipal bond issuer or credit enhancement provider, an event of taxability on the municipal bonds or the immediate downgrade of the municipal bond to below investment grade. In addition, the Firm's exposure as liquidity provider is further limited by the high credit quality of the underlying municipal bonds, the excess collateralization in the vehicle or in certain transactions the reimbursement agreements with the residual interest holders. However, a downgrade of JPMorgan Chase Bank, N.A.'s short-term rating does not affect the Firm's obligation under the liquidity facility.

The long-term credit ratings of the puttable floating rate certificates are directly related to the credit ratings of the underlying municipal bonds, to the credit rating of any insurer of the underlying municipal bond, and the Firm's short-term credit rating as liquidity provider. A downgrade in any of these ratings would affect the rating of the puttable floating-rate certificates and could cause demand

for these certificates by investors to decline or disappear.

As remarketing agent, the Firm may hold puttable floating-rate certificates of the municipal bond vehicles. At December 31, 2011 and 2010, respectively, the Firm held \$637 million and \$248 million of these certificates on its Consolidated Balance Sheets. The largest amount held by the Firm at any time during 2011 was \$1.1 billion, or 7.6%, of the municipal bond vehicles' aggregate outstanding puttable floating-rate certificates. The Firm did not have and continues not to have any intent to protect any residual interest holder from potential losses on any of the municipal bond holdings.

The Firm consolidates municipal bond vehicles if it owns the residual interest. The residual interest generally allows the owner to make decisions that significantly impact the economic performance of the municipal bond vehicle, primarily by directing the sale of the municipal bonds owned by the vehicle. In addition, the residual interest owners have the right to receive benefits and bear losses that could potentially be significant to the municipal bond vehicle. The Firm does not consolidate municipal bond vehicles if it does not own the residual interests, since the Firm does not have the power to make decisions that significantly impact the economic performance of the municipal bond vehicle.

The Firm's exposure to nonconsolidated municipal bond VIEs at December 31, 2011 and 2010, including the ratings profile of the VIEs' assets, was as follows.

December 31, (in billions)	Fair value of assets held by VIEs	Liquidity facilities ^(a)	Excess/(deficit) ^(b)	Maximum exposure
Nonconsolidated municipal bond vehicles				
2011	\$ 13.5	\$ 7.9	\$ 5.6	\$ 7.9
2010	13.7	8.8	4.9	8.8

December 31, (in billions, except where otherwise noted)	Ratings profile of VIE assets						Fair value of assets held by VIEs	Wt. avg. expected life of assets (years)
	Investment-grade				Noninvestment-grade			
	AAA to AAA-	AA+ to AA-	A+ to A-	BBB+ to BBB-	BB+ and below			
2011	\$ 1.5	\$ 11.2	\$ 0.7	\$ —	\$ 0.1		\$ 13.5	6.6
2010	1.9	11.2	0.6	—	—		13.7	15.5

(a) The Firm may serve as credit enhancement provider to municipal bond vehicles in which it serves as liquidity provider. The Firm provided insurance on underlying municipal bonds, in the form of letters of credit, of \$10 million at December 31, 2010. The Firm did not provide insurance on underlying municipal bonds at December 31, 2011.

(b) Represents the excess/(deficit) of the fair values of municipal bond assets available to repay the liquidity facilities, if drawn.

(c) The ratings scale is based on the Firm's internal risk ratings and is presented on an S&P-equivalent basis.

Credit-related note and asset swap vehicles

Credit-related note vehicles

The Firm structures transactions with credit-related note vehicles in which the VIE purchases highly rated assets, such as asset-backed securities, and enters into a credit derivative contract with the Firm to obtain exposure to a referenced credit which the VIE otherwise does not hold. The VIE then issues credit-linked notes ("CLNs") with maturities predominantly ranging from one to 10 years in order to transfer the risk of the referenced credit to the

VIE's investors. Clients and investors often prefer using a CLN vehicle since the CLNs issued by the VIE generally carry a higher credit rating than such notes would if issued directly by JPMorgan Chase. As a derivative counterparty in a credit-related note structure, the Firm has a senior claim on the collateral of the VIE and reports such derivatives on its Consolidated Balance Sheets at fair value. The collateral purchased by such VIEs is largely investment-grade, with a significant amount being rated "AAA." The Firm divides its credit-related note structures broadly into two types: static and managed.

In a static credit-related note structure, the CLNs and associated credit derivative contract either reference a single credit (e.g., a multi-national corporation), or all or part of a fixed portfolio of credits. In a managed credit-related note structure, the CLNs and associated credit derivative generally reference all or part of an actively managed portfolio of credits. An agreement exists between a portfolio manager and the VIE that gives the portfolio manager the ability to substitute each referenced credit in the portfolio for an alternative credit. The Firm does not act as portfolio manager; its involvement with the VIE is generally limited to being a derivative counterparty. As a net buyer of credit protection, in both static and managed credit-related note structures, the Firm pays a premium to the VIE in return for the receipt of a payment (up to the notional of the derivative) if one or more of the credits within the portfolio defaults, or if the losses resulting from the default of reference credits exceed specified levels. The Firm does not provide any additional contractual financial support to the VIE. In addition, the Firm has not historically provided any financial support to the CLN vehicles over and above its contractual obligations. Since each CLN is established to the specifications of the investors, the investors have the power over the activities of that VIE that most significantly affect the performance of the CLN. Accordingly, the Firm does not generally consolidate these credit-related note entities. Furthermore, the Firm does not have a variable interest that could potentially be significant. As a derivative counterparty, the Firm has a senior claim on the collateral of the VIE and reports such derivatives on its Consolidated Balance Sheets at fair value. Substantially all of the assets purchased by such VIEs are investment-grade.

Asset swap vehicles

The Firm structures and executes transactions with asset swap vehicles on behalf of investors. In such transactions, the VIE purchases a specific asset or assets and then enters into a derivative with the Firm in order to tailor the interest rate or foreign exchange currency risk, or both, according to investors' requirements. Generally, the assets are held by the VIE to maturity, and the tenor of the derivatives would match the maturity of the assets. Investors typically invest in the notes issued by such VIEs in order to obtain exposure to the credit risk of the specific assets, as well as exposure to foreign exchange and interest rate risk that is tailored to their specific needs. The derivative transaction between the Firm and the VIE may include currency swaps to hedge assets held by the VIE denominated in foreign currency into the investors' local currency or interest rate swaps to hedge the interest rate risk of assets held by the VIE; to add additional interest rate exposure into the VIE in order to increase the return on the issued notes; or to convert an interest-bearing asset into a zero-coupon bond.

The Firm's exposure to asset swap vehicles is generally limited to its rights and obligations under the interest rate and/or foreign exchange derivative contracts. The Firm historically has not provided any financial support to the asset swap vehicles over and above its contractual obligations. The Firm does not generally consolidate these asset swap vehicles, since the Firm does not have the power to direct the significant activities of these entities and does not have a variable interest that could potentially be significant. As a derivative counterparty, the Firm has a senior claim on the collateral of the VIE and reports such derivatives on its Consolidated Balance Sheets at fair value. Substantially all of the assets purchased by such VIEs are investment-grade.

Exposure to nonconsolidated credit-related note and asset swap VIEs at December 31, 2011 and 2010, was as follows.

December 31, 2011 (in billions)	Net derivative receivables	Total exposure ^(a)	Par value of collateral held by VIEs ^(b)
Credit-related notes			
Static structure	\$ 1.0	\$ 1.0	\$ 9.1
Managed structure	2.7	2.7	7.7
Total credit-related notes	3.7	3.7	16.8
Asset swaps	0.6	0.6	8.6
Total	\$ 4.3	\$ 4.3	\$ 25.4

December 31, 2010 (in billions)	Net derivative receivables	Total exposure ^(a)	Par value of collateral held by VIEs ^(b)
Credit-related notes			
Static structure	\$ 1.0	\$ 1.0	\$ 9.5
Managed structure	2.8	2.8	10.7
Total credit-related notes	3.8	3.8	20.2
Asset swaps	0.3	0.3	7.6
Total	\$ 4.1	\$ 4.1	\$ 27.8

(a) On-balance sheet exposure that includes net derivative receivables and trading assets – debt and equity instruments. At both December 31, 2011 and 2010, the amount of trading assets issued by nonconsolidated credit-related note and asset swap vehicles that were held by the Firm were immaterial.

(b) The Firm's maximum exposure arises through the derivatives executed with the VIEs; the exposure varies over time with changes in the fair value of the derivatives. The Firm does not have the power to direct the activities of the VIE that most significantly impact the VIE's economic performance. The Firm's interests in the VIE include investments classified as AFS securities that had fair values of \$2.9 billion and \$3.1 billion at December 31, 2011 and 2010, respectively, and other interests which are classified as loans and have a fair value of approximately \$1.0 billion and \$1.0 billion at December 31, 2011 and 2010, respectively. For more information on AFS securities and loans, see Notes 12 and 14 on pages 225–230 and 231–252, respectively, of this Annual Report.

The Firm consolidated credit-related note vehicles with collateral fair values of \$231 million and \$394 million, at December 31, 2011 and 2010, respectively. The Firm consolidated these vehicles, because in its role as secondary market-maker, it held positions in these entities that provided the Firm with control of certain vehicles. The Firm did not consolidate any asset swap vehicles at December 31, 2011 and 2010.

VIEs sponsored by third parties

Investment in a third-party credit card securitization trust
The Firm holds two interests in a third-party-sponsored VIE, which is a credit card securitization trust that owns credit card receivables issued by a national retailer. The Firm is not the primary beneficiary of the trust as the Firm does not have the power to direct the activities of the VIE that most significantly impact the VIE's economic performance. The Firm's interests in the VIE include investments classified as AFS securities that had fair values of \$2.9 billion and \$3.1 billion at December 31, 2011 and 2010, respectively, and other interests which are classified as loans and have a fair value of approximately \$1.0 billion and \$1.0 billion at December 31, 2011 and 2010, respectively. For more information on AFS securities and loans, see Notes 12 and 14 on pages 225–230 and 231–252, respectively, of this Annual Report.

VIE used in FRBNY transaction

In conjunction with the Bear Stearns merger, in June 2008, the Federal Reserve Bank of New York ("FRBNY") took control, through an LLC formed for this purpose, of a portfolio of \$30.0 billion in assets, based on the value of the portfolio as of March 14, 2008. The assets of the LLC

were funded by a \$28.85 billion term loan from the FRBNY and a \$1.15 billion subordinated loan from JPMorgan Chase. The JPMorgan Chase loan is subordinated to the FRBNY loan and will bear the first \$1.15 billion of any losses of the portfolio. Any remaining assets in the portfolio after repayment of the FRBNY loan, repayment of the JPMorgan Chase loan and the expense of the LLC will be for the account of the FRBNY. The extent to which the FRBNY and JPMorgan Chase loans will be repaid will depend on the value of the assets in the portfolio and the liquidation strategy directed by the FRBNY. The Firm does not consolidate the LLC, as it does not have the power to direct the activities of the VIE that most significantly impact the VIE's economic performance.

Other VIEs sponsored by third parties

The Firm enters into transactions with VIEs structured by other parties. These include, for example, acting as a derivative counterparty, liquidity provider, investor, underwriter, placement agent, trustee or custodian. These transactions are conducted at arm's-length, and individual credit decisions are based on the analysis of the specific VIE, taking into consideration the quality of the underlying assets. Where the Firm does not have the power to direct the activities of the VIE that most significantly impact the VIE's economic performance, or a variable interest that could potentially be significant, the Firm records and reports these positions on its Consolidated Balance Sheets similarly to the way it would record and report positions in respect of any other third-party transaction.

Notes to consolidated financial statements

Consolidated VIE assets and liabilities

The following table presents information on assets and liabilities related to VIEs consolidated by the Firm as of December 31, 2011 and 2010.

December 31, 2011 (in billions)	Assets				Liabilities		
	Trading assets – debt and equity instruments	Loans	Other ^(c)	Total assets ^(d)	Beneficial interests in VIE assets ^(e)	Other ^(f)	Total liabilities
VIE program type							
Firm-sponsored credit card trusts	\$ —	\$ 50.7	\$ 0.8	\$ 51.5	\$ 32.5	\$ —	\$ 32.5
Firm-administered multi-seller conduits	—	29.7	0.2	29.9	18.7	—	18.7
Mortgage securitization entities	1.4	2.3	—	3.7	2.3	1.3	3.6
Other ^(b)	10.7	4.1	1.6	16.4	12.5	0.2	12.7
Total	\$ 12.1	\$ 86.8	\$ 2.6	\$ 101.5	\$ 66.0	\$ 1.5	\$ 67.5

December 31, 2010 (in billions)	Assets				Liabilities		
	Trading assets – debt and equity instruments	Loans	Other ^(c)	Total assets ^(d)	Beneficial interests in VIE assets ^(e)	Other ^(f)	Total liabilities
VIE program type							
Firm-sponsored credit card trusts	\$ —	\$ 67.2	\$ 1.3	\$ 68.5	\$ 44.3	\$ —	\$ 44.3
Firm-administered multi-seller conduits	—	21.1	0.6	21.7	21.6	0.1	21.7
Mortgage securitization entities	1.8	2.9	—	4.7	2.4	1.6	4.0
Other ^(b)	8.0	4.4	1.6	14.0	9.3	0.3	9.6
Total	\$ 9.8	\$ 95.6	\$ 3.5	\$ 108.9	\$ 77.6	\$ 2.0	\$ 79.6

(a) Includes residential and commercial mortgage securitizations as well as re-securitizations.

(b) Primarily comprises student loan securitization entities and municipal bond entities. The Firm consolidated \$4.1 billion and \$4.5 billion of student loan securitization entities as of December 31, 2011 and 2010, respectively, and \$9.3 billion and \$4.6 billion of municipal bond vehicles as of December 31, 2011 and 2010, respectively.

(c) Includes assets classified as cash, derivative receivables, AFS securities, and other assets within the Consolidated Balance Sheets.

(d) The assets of the consolidated VIEs included in the program types above are used to settle the liabilities of those entities. The difference between total assets and total liabilities recognized for consolidated VIEs represents the Firm's interest in the consolidated VIEs for each program type.

(e) The interest-bearing beneficial interest liabilities issued by consolidated VIEs are classified in the line item on the Consolidated Balance Sheets titled, "Beneficial interests issued by consolidated variable interest entities." The holders of these beneficial interests do not have recourse to the general credit of JPMorgan Chase. Included in beneficial interests in VIE assets are long-term beneficial interests of \$39.7 billion and \$52.6 billion at December 31, 2011 and 2010, respectively. The maturities of the long-term beneficial interests as of December 31, 2011, were as follows: \$13.5 billion under one year, \$17.8 billion between one and five years, and \$8.4 billion over five years, all respectively.

(f) Includes liabilities classified as accounts payable and other liabilities in the Consolidated Balance Sheets.

Supplemental information on loan securitizations

The Firm securitizes and sells a variety of loans, including residential mortgage, credit card, automobile, student and commercial (primarily related to real estate) loans, as well as debt securities. The primary purposes of these securitization transactions are to satisfy investor demand and to generate liquidity for the Firm.

For loan securitizations in which the Firm is not required to consolidate the trust, the Firm records the transfer of the loan receivable to the trust as a sale when the accounting criteria for a sale are met. Those criteria are: (1) the transferred financial assets are legally isolated from the Firm's creditors; (2) the transferee or beneficial interest holder can pledge or exchange the transferred financial assets; and (3) the Firm does not maintain effective control over the transferred financial assets (e.g., the Firm cannot repurchase the transferred assets before their maturity and it does not have the ability to unilaterally cause the holder to return the transferred assets).

For loan securitizations accounted for as a sale, the Firm recognizes a gain or loss based on the difference between the value of proceeds received (including cash, beneficial interests, or servicing assets received) and the carrying value of the assets sold. Gains and losses on securitizations are reported in noninterest revenue.

Securitization activity

The following tables provide information related to the Firm's securitization activities for the years ended December 31, 2011, 2010 and 2009, related to assets held in JPMorgan Chase-sponsored securitization entities that were not consolidated by the Firm, and sale accounting was achieved based on the accounting rules in effect at the time of the securitization.

For the year ended December 31, 2009, there were no mortgage loans that were securitized, except for commercial and other, and there were no cash flows from the Firm to the SPEs related to recourse arrangements.

Effective January 1, 2010, all of the Firm-sponsored credit card securitization trusts and predominantly all of the Firm-sponsored student loan and auto securitization trusts were consolidated as a result of the accounting guidance related to VIEs and, accordingly, are not included in the securitization activity tables below for the years ended December 31, 2011 and 2010.

Prior to January 1, 2010, the Firm did not consolidate its credit card, residential and commercial mortgage, automobile, and certain student loan securitizations based on the accounting guidance in effect at that time. The Firm recorded only its retained interests in the entities on its Consolidated Balance Sheets.

Year ended December 31, (in millions, except rates)	2011		2010		2009		
	Residential mortgage ^{(d)(e)}	Commercial and other ^(f)	Residential mortgage ^{(d)(e)}	Commercial and other ^(f)	Residential mortgage ^{(d)(e)}	Commercial and other ^(f)	Credit card
Principal securitized	\$ —	\$ 5,961	\$ 35	\$ 2,237	\$ —	\$ 500	\$ 26,538
Pretax gains	—	— ^(g)	—	— ^(g)	—	— ^(g)	22
All cash flows during the period:							
Proceeds from new securitizations	\$ —	\$ 6,142	\$ 36	\$ 2,369	\$ —	\$ 542	\$ 26,538
Servicing fees collected	755	4	968	4	1,111	18	1,251
Other cash flows received	—	—	—	—	11	—	5,000
Proceeds from collections reinvested in revolving securitizations	—	—	—	—	—	—	161,428
Purchases of previously transferred financial assets (or the underlying collateral) ^(b)	772	—	321	—	165	249	—
Cash flows received on the interests that continue to be held by the Firm	235	178	319	143	538	120	261
Key assumptions used to measure retained interests originated during the year (rates per annum)							
Prepayment rate ^(a)		—%		100%		100%	16.7%
		CPY		CPY		CPY	PPR
Weighted-average life (in years)		1.7		7.1		9.0	0.5
Expected credit losses		—%		—%		—%	8.9%
Discount rate		3.5		7.7		10.7	16.0

- (a) Proceeds from residential and commercial mortgage securitizations are received in the form of securities. During 2011, \$4.0 billion and \$2.1 billion of commercial mortgage securitizations were classified in levels 2 and 3 of the fair value hierarchy, respectively. During 2010, \$2.2 billion and \$172 million of residential and commercial mortgage securitizations were classified in levels 2 and 3 of the fair value hierarchy, respectively. During 2009, \$380 million and \$162 million of residential and commercial mortgage securitizations were classified in levels 2 and 3 of the fair value hierarchy, respectively; and \$12.8 billion of proceeds from credit card securitizations were received as securities and were classified in level 2 of the fair value hierarchy.
- (b) Includes cash paid by the Firm to reacquire assets from off-balance sheet, nonconsolidated entities – for example, loan repurchases due to representation and warranties and servicer clean-up calls.
- (c) CPY: constant prepayment yield; PPR: principal payment rate.
- (d) Includes prime, Alt-A, subprime, option ARMS, and re-securitizations. Excludes sales for which the Firm did not securitize the loan (including loans sold to Ginnie Mae, Fannie Mae and Freddie Mac).
- (e) There were no retained interests held in the residential mortgage securitization completed in 2010. There were no residential mortgage securitizations in 2011 and 2009.
- (f) Includes commercial, student loan and automobile loan securitizations.
- (g) The Firm elected the fair value option for loans pending securitization. The carrying value of these loans accounted for at fair value approximated the proceeds received from securitization.

Loans sold to agencies and other third-party-sponsored securitization entities

In addition to the amounts reported in the securitization activity tables above, the Firm, in the normal course of business, sells originated and purchased mortgage loans on a nonrecourse basis, predominantly to Ginnie Mae, Fannie Mae and Freddie Mac (the "Agencies"). These loans are sold primarily for the purpose of securitization by the Agencies, which also provide credit enhancement of the loans through certain guarantee provisions. The Firm does not consolidate these securitization vehicles as it is not the primary beneficiary. For a limited number of loan sales, the Firm is obligated to share a portion of the credit risk associated with the sold loans with the purchaser. See Note 29 on pages 283–289 of this Annual Report for additional

information about the Firm's loans sales- and securitization-related indemnifications.

The following table summarizes the activities related to loans sold to U.S. government-sponsored agencies and third-party-sponsored securitization entities.

Year ended December 31, (in millions)	2011	2010	2009
Carrying value of loans sold ^(a)	\$ 150,632	\$ 156,615	\$ 154,571
Proceeds received from loan sales as cash	2,864	3,887	1,702
Proceeds from loans sales as securities ^(c)	145,340	149,786	149,343
Total proceeds received from loan sales	\$ 148,204	\$ 153,673	\$ 151,045
Gains on loan sales	133	212	89

(a) Predominantly to U.S. government agencies.

Notes to consolidated financial statements

- (b) MSRs were excluded from the above table. See Note 17 on pages 267–271 of this Annual Report for further information on originated MSRs.
- (c) Predominantly includes securities from U.S. government agencies that are generally sold shortly after receipt.

Options to repurchase delinquent loans

In addition to the Firm's obligation to repurchase certain loans due to material breaches of representations and warranties as discussed in Note 29 on pages 283–289 of this Annual Report, the Firm also has the option to repurchase delinquent loans that it services for Ginnie Mae, as well as for other U.S. government agencies in certain arrangements. The Firm typically elects to repurchase delinquent loans from Ginnie Mae as it continues to service them and/or manage the foreclosure process in accordance with the applicable requirements, and such loans continue

to be insured or guaranteed. When the Firm's repurchase option becomes exercisable, such loans must be reported on the Consolidated Balance Sheets as a loan with a corresponding liability. As of December 31, 2011 and 2010, the Firm had recorded on its Consolidated Balance Sheets \$15.7 billion and \$13.0 billion, respectively, of loans that either had been repurchased or for which the Firm had an option to repurchase. Predominately all of the amounts presented above relate to loans that have been repurchased from Ginnie Mae. Additionally, real estate owned resulting from voluntary repurchases of loans was \$1.0 billion and \$1.9 billion as of December 31, 2011 and 2010, respectively. Substantially all of these loans and real estate owned are insured or guaranteed by U.S. government agencies, and where applicable, reimbursement is proceeding normally. For additional information, refer to Note 14 on pages 231–252 of this Annual Report.

JPMorgan Chase's interest in securitized assets held at fair value

The following table outlines the key economic assumptions used to determine the fair value, as of December 31, 2011 and 2010, of certain of the Firm's retained interests in nonconsolidated VIEs (other than MSRs), that are valued using modeling techniques. The table also outlines the sensitivities of those fair values to immediate 10% and 20% adverse changes in assumptions used to determine fair value. For a discussion of MSRs, see Note 17 on pages 267–271 of this Annual Report.

December 31, (in millions, except rates and where otherwise noted)	Commercial and other	
	2011	2010
JPMorgan Chase interests in securitized assets	\$ 3,663	\$ 2,906
Weighted-average life (in years)	3.0	3.3
Weighted-average constant prepayment rate	—%	—%
	CPR	CPR
Impact of 10% adverse change	\$ —	\$ —
Impact of 20% adverse change	—	—
Weighted-average loss assumption	0.2%	2.1%
Impact of 10% adverse change	\$ (61)	\$ (76)
Impact of 20% adverse change	(119)	(151)
Weighted-average discount rate	28.2%	16.4%
Impact of 10% adverse change	\$ (75)	\$ (69)
Impact of 20% adverse change	(136)	(134)

- (a) The Firm's interests in prime mortgage securitizations were \$555 million and \$708 million, as of December 31, 2011 and 2010, respectively. These include retained interests in Alt-A loans and re-securitization transactions. The Firm's interests in subprime mortgage securitizations were \$31 million and \$14 million, as of December 31, 2011 and 2010, respectively. Additionally, the Firm had interests in option ARM mortgage securitizations of \$23 million and \$29 million at December 31, 2011 and 2010, respectively.
- (b) Includes certain investments acquired in the secondary market but predominantly held for investment purposes.
- (c) CPR: constant prepayment rate.

The sensitivity analysis in the preceding table is hypothetical. Changes in fair value based on a 10% or 20% variation in assumptions generally cannot be extrapolated easily, because the relationship of the change in the assumptions to the change in fair value may not be linear. Also, in the table, the effect that a change in a particular assumption may have on the fair value is calculated without changing any other assumption. In reality, changes in one factor may result in changes in another, which might counteract or magnify the sensitivities. The above sensitivities also do not reflect risk management practices the Firm may undertake to mitigate such risks.

Loan delinquencies and liquidation losses

The table below includes information about delinquencies, liquidation losses and components of nonconsolidated securitized financial assets in which the Firm has continuing involvement as of December 31, 2011 and 2010.

As of or for the year ended December 31, (in millions)	Securitized assets		90 days past due		Liquidation losses	
	2011	2010	2011	2010	2011	2010
Securitized loans ^(a)						
Residential mortgage:						
Prime mortgage ^(b)	\$ 101,004	\$ 143,764	\$ 24,285	\$ 33,093	\$ 5,650	\$ 6,257
Subprime mortgage	35,755	40,721	14,293	15,456	3,086	3,598
Option ARMs	31,075	35,786	9,999	10,788	1,907	2,305
Commercial and other	93,336	106,245	4,836	5,791	1,101	618
Total loans securitized ^(c)	\$ 261,170	\$ 326,516	\$ 53,413	\$ 65,128	\$ 11,744	\$ 12,778

(a) Total assets held in securitization-related SPEs were \$342.3 billion and \$391.1 billion, respectively, at December 31, 2011 and 2010. The \$261.2 billion and \$326.5 billion, respectively, of loans securitized at December 31, 2011 and 2010, excludes: \$74.4 billion and \$56.0 billion, respectively, of securitized loans in which the Firm has no continuing involvement, and \$6.7 billion and \$8.6 billion, respectively, of loan securitizations consolidated on the Firm's Consolidated Balance Sheets at December 31, 2011 and 2010.

(b) Includes Alt-A loans.

(c) Includes securitized loans that were previously recorded at fair value and classified as trading assets.

Implementation of change in consolidation accounting guidance for VIEs

On January 1, 2010, the Firm implemented consolidation accounting guidance related to VIEs. The following table summarizes the incremental impact at adoption of the new guidance.

(in millions, except ratios)	U.S. GAAP assets	U.S. GAAP liabilities	Stockholders' equity	Tier 1 capital
As of December 31, 2009	\$ 2,031,989	\$ 1,866,624	\$ 165,365	11.10%
Impact of new accounting guidance for consolidation of VIEs				
Credit card	60,901	65,353	(4,452)	(0.30)
Multi-seller conduits	17,724	17,744	(20)	—
Mortgage & other	9,059	9,107	(48)	(0.04)
Total impact of new guidance	87,684	92,204	(4,520)	(0.34)
Beginning balance as of January 1, 2010	\$ 2,119,673	\$ 1,958,828	\$ 160,845	10.76%

Note 17 – Goodwill and other intangible assets

Goodwill and other intangible assets consist of the following.

December 31, (in millions)	2011	2010	2009
Goodwill	\$48,188	\$48,854	\$48,357
Mortgage servicing rights	7,223	13,649	15,531
Other intangible assets:			
Purchased credit card relationships	\$ 602	\$ 897	\$ 1,246
Other credit card-related intangibles	488	593	691
Core deposit intangibles	594	879	1,207
Other intangibles	1,523	1,670	1,477
Total other intangible assets	\$ 3,207	\$ 4,039	\$ 4,621

Goodwill

Goodwill is recorded upon completion of a business combination as the difference between the purchase price and the fair value of the net assets acquired. Subsequent to initial recognition, goodwill is not amortized but is tested for impairment during the fourth quarter of each fiscal year, or more often if events or circumstances, such as adverse changes in the business climate, indicate there may be impairment.

The goodwill associated with each business combination is allocated to the related reporting units, which are

determined based on how the Firm's businesses are managed and how they are reviewed by the Firm's Operating Committee. The following table presents goodwill attributed to the business segments.

December 31, (in millions)	2011	2010	2009
Investment Bank	\$ 5,276	\$ 5,278	\$ 4,959
Retail Financial Services	16,489	16,496	16,514
Card Services & Auto	14,507	14,522	14,451
Commercial Banking	2,864	2,866	2,868
Treasury & Securities Services	1,668	1,680	1,667
Asset Management	7,007	7,635	7,521
Corporate/Private Equity	377	377	377
Total goodwill	\$ 48,188	\$ 48,854	\$ 48,357

The following table presents changes in the carrying amount of goodwill.

Year ended December 31, (in millions)	2011	2010	2009
Balance at beginning of period	\$48,854	\$48,357	\$48,027
Changes during the period from:			
Business combinations	97	556	271
Dispositions	(685)	(19)	—
Other ^(b)	(78)	(40)	59
Balance at December 31 ^(a)	\$48,188	\$48,854	\$48,357

- (a) Reflects gross goodwill balances as the Firm has not recognized any impairment losses to date.
- (b) Includes foreign currency translation adjustments and other tax-related adjustments.

The net reduction in goodwill was predominantly due to AM's sale of its investment in an asset manager.

Impairment testing

Goodwill was not impaired at December 31, 2011 or 2010, nor was any goodwill written off due to impairment during 2011, 2010 or 2009.

The goodwill impairment test is performed in two steps. In the first step, the current fair value of each reporting unit is compared with its carrying value, including goodwill. If the fair value is in excess of the carrying value (including goodwill), then the reporting unit's goodwill is considered not to be impaired. If the fair value is less than the carrying value (including goodwill), then a second step is performed. In the second step, the implied current fair value of the reporting unit's goodwill is determined by comparing the fair value of the reporting unit (as determined in step one) to the fair value of the net assets of the reporting unit, as if the reporting unit were being acquired in a business combination. The resulting implied current fair value of goodwill is then compared with the carrying value of the reporting unit's goodwill. If the carrying value of the goodwill exceeds its implied current fair value, then an impairment charge is recognized for the excess. If the carrying value of goodwill is less than its implied current fair value, then no goodwill impairment is recognized.

The primary method the Firm uses to estimate the fair value of its reporting units is the income approach. The models project cash flows for the forecast period and use the perpetuity growth method to calculate terminal values. These cash flows and terminal values are then discounted using an appropriate discount rate. Projections of cash flows are based on the reporting units' earnings forecasts, which include the estimated effects of regulatory and legislative changes (including, but not limited to the Dodd-Frank Wall Street Reform and Consumer Protection Act (the "Dodd-Frank Act"), the CARD Act, and limitations on non-sufficient funds and overdraft fees), and which are reviewed with the Operating Committee of the Firm. The discount rate used for each reporting unit represents an estimate of the cost of equity for that reporting unit and is determined considering the Firm's overall estimated cost of equity (estimated using the Capital Asset Pricing Model), as adjusted for the risk characteristics specific to each reporting unit (for example, for higher levels of risk or uncertainty associated with the business or management's forecasts and assumptions). To assess the reasonableness of the discount rates used for each reporting unit management compares the discount rate to the estimated cost of equity for publicly traded institutions with similar businesses and risk characteristics. In addition, the weighted average cost of equity (aggregating the various reporting units) is compared with the Firm's overall estimated cost of equity to ensure reasonableness.

The valuations derived from the discounted cash flow models are then compared with market-based trading and transaction multiples for relevant competitors. Trading and transaction comparables are used as general indicators to assess the general reasonableness of the estimated fair values, although precise conclusions generally cannot be drawn due to the differences that naturally exist between the Firm's businesses and competitor institutions. Management also takes into consideration a comparison between the aggregate fair value of the Firm's reporting units and JPMorgan Chase's market capitalization. In evaluating this comparison, management considers several factors, including (a) a control premium that would exist in a market transaction, (b) factors related to the level of execution risk that would exist at the firmwide level that do not exist at the reporting unit level and (c) short-term market volatility and other factors that do not directly affect the value of individual reporting units.

While no impairment of goodwill was recognized, the Firm's consumer lending businesses in RFS and Card remain at an elevated risk of goodwill impairment due to their exposure to U.S. consumer credit risk and the effects of economic, regulatory and legislative changes. The valuation of these businesses is particularly dependent upon economic conditions (including new unemployment claims and home prices), regulatory and legislative changes (for example, those related to residential mortgage servicing, foreclosure and loss mitigation activities, and those that may affect consumer credit card use), and the amount of equity capital required. In addition, the earnings or estimated cost of equity of the Firm's capital markets businesses could also be affected by regulatory or legislative changes. The assumptions used in the discounted cash flow valuation models were determined using management's best estimates. The cost of equity reflected the related risks and uncertainties, and was evaluated in comparison to relevant market peers. Deterioration in these assumptions could cause the estimated fair values of these reporting units and their associated goodwill to decline, which may result in a material impairment charge to earnings in a future period related to some portion of the associated goodwill.

Mortgage servicing rights

Mortgage servicing rights represent the fair value of expected future cash flows for performing servicing activities for others. The fair value considers estimated future servicing fees and ancillary revenue, offset by estimated costs to service the loans, and generally declines over time as net servicing cash flows are received, effectively amortizing the MSR asset against contractual servicing and ancillary fee income. MSRs are either purchased from third parties or recognized upon sale or securitization of mortgage loans if servicing is retained.

As permitted by U.S. GAAP, the Firm elected to account for its MSRs at fair value. The Firm treats its MSRs as a single class of servicing assets based on the availability of market inputs used to measure the fair value of its MSR asset and its treatment of MSRs as one aggregate pool for risk

management purposes. The Firm estimates the fair value of MSRs using an option-adjusted spread (“OAS”) model, which projects MSR cash flows over multiple interest rate scenarios in conjunction with the Firm’s prepayment model, and then discounts these cash flows at risk-adjusted rates. The model considers portfolio characteristics, contractually specified servicing fees, prepayment assumptions, delinquency rates, late charges, other ancillary revenue and costs to service, and other economic factors. The Firm compares fair value estimates and assumptions to observable market data where available, and also considers recent market activity and actual portfolio experience.

The fair value of MSRs is sensitive to changes in interest rates, including their effect on prepayment speeds. MSRs typically decrease in value when interest rates decline because declining interest rates tend to increase prepayments and therefore reduce the expected life of the net servicing cash flows that comprise the MSR asset. Conversely, securities (e.g., mortgage-backed securities), principal-only certificates and certain derivatives (i.e., those for which the Firm receives fixed-rate interest payments) increase in value when interest rates decline. JPMorgan Chase uses combinations of derivatives and securities to manage changes in the fair value of MSRs. The intent is to offset any interest-rate related changes in the fair value of MSRs with changes in the fair value of the related risk management instruments.

The following table summarizes MSR activity for the years ended December 31, 2011, 2010 and 2009.

Year ended December 31, (in millions, except where otherwise noted)	2011	2010	2009
Fair value at beginning of period	\$ 13,649	\$ 15,531	\$ 9,403
MSR activity			
Originations of MSRs	2,570	3,153	3,615
Purchase of MSRs	33	26	2
Disposition of MSRs	—	(407)	(10)
Changes due to modeled amortization	(1,910)	(2,386)	(3,286)
Net additions and amortization	693	386	321
Changes due to market interest rates	(5,392)	(2,224)	5,844
Other changes in valuation due to inputs and assumptions	(1,727)	(44)	(37)
Total change in fair value of MSRs	(7,119)	(2,268)	5,807
Fair value at December 31	\$ 7,223	\$ 13,649	\$ 15,531
Change in unrealized gains/(losses) included in income related to MSRs held at December 31	\$ (7,119)	\$ (2,268)	\$ 5,807
Contractual service fees, late fees and other ancillary fees included in income	\$ 3,977	\$ 4,484	\$ 4,818
Third-party mortgage loans serviced at December 31 (in billions)	\$ 910	\$ 976	\$ 1,091
Servicer advances at December 31 (in billions) ^(d)	\$ 11.1	\$ 9.9	\$ 7.7

- (a) Represents the aggregate impact of changes in model inputs and assumptions such as costs to service, home prices, mortgage spreads, ancillary income, and assumptions used to derive prepayment speeds, as well as changes to the valuation models themselves.
- (b) Includes changes related to commercial real estate of \$(9) million,

\$ (1) million and \$(4) million for the years ended December 31, 2011, 2010 and 2009, respectively.

- (c) Includes \$31 million, \$40 million and \$41 million related to commercial real estate at December 31, 2011, 2010 and 2009, respectively.
- (d) Represents amounts the Firm pays as the servicer (e.g., scheduled principal and interest to a trust, taxes and insurance), which will generally be reimbursed within a short period of time after the advance from future cash flows from the trust or the underlying loans. The Firm’s credit risk associated with these advances is minimal because reimbursement of the advances is senior to all cash payments to investors. In addition, the Firm maintains the right to stop payment if the collateral is insufficient to cover the advance.

During the year ended December 31, 2011, the fair value of the MSR decreased by \$6.4 billion. This decrease was predominately due to a decline in market interest rates, which resulted in a loss of \$5.4 billion. These losses were offset by gains of \$5.6 billion on derivatives used to hedge the MSR asset; these derivatives are recognized on the Consolidated Balance Sheets separately from the MSR asset. Also contributing to the decline in fair value of the MSR asset was a \$1.7 billion decrease related to revised cost to service and ancillary income assumptions incorporated in the MSR valuation. The increased cost to service assumptions reflect the estimated impact of higher servicing costs to enhance servicing processes, particularly loan modification and foreclosure procedures, including costs to comply with Consent Orders entered into with banking regulators. The increase in the cost to service assumption contemplates significant and prolonged increases in staffing levels in the core and default servicing functions. The decreased ancillary income assumption is similarly related to a reassessment of business practices in consideration of the Consent Orders and the existing industry-wide regulatory environment, which is broadly affecting market participants.

Also in the fourth quarter of 2011, the Firm revised its OAS assumption and updated its proprietary prepayment model; these changes had generally offsetting effects. The Firm’s OAS assumption is based upon capital and return requirements that the Firm believes a market participant would consider, taking into account factors such as the pending Basel III capital rules. Consequently, the OAS assumption for the Firm’s portfolio increased by approximately 400 basis points and decreased the fair value of the MSR asset by approximately \$1.2 billion.

Since 2009, the Firm has continued to refine its proprietary prepayment model based on a number of market-related factors, including a downward trend in home prices, a general tightening of credit underwriting standards and the associated impact on refinancing activity. In the fourth quarter of 2011, the Firm further enhanced its proprietary prepayment model to incorporate: (i) the impact of the Home Affordable Refinance Program (“HARP”) 2.0, and (ii) assumptions that will limit modeled refinancings due to the combined influences of relatively strict underwriting standards and reduced levels of expected home price appreciation. In the aggregate, these refinements increased the fair value of the MSR asset by approximately \$1.2 billion.

The decrease in the fair value of the MSR results in a lower asset value that will amortize in future periods against contractual and ancillary fee income received in future periods. While there is expected to be higher levels of noninterest expense associated with higher servicing costs in those future periods, there will also be less MSR amortization, which will have the effect of increasing mortgage fees and related income. The amortization of the MSR is reflected in the tables above under "Changes due to modeled amortization."

The following table presents the components of mortgage fees and related income (including the impact of MSR risk management activities) for the years ended December 31, 2011, 2010 and 2009.

Year ended December 31, (in millions)	2011	2010	2009
RFS mortgage fees and related income			
Net production revenue:			
Production revenue	\$3,395	\$3,440	\$2,115
Repurchase losses	(1,347)	(2,912)	(1,612)
Net production revenue	2,048	528	503
Net mortgage servicing revenue			
Operating revenue:			
Loan servicing revenue	4,134	4,575	4,942
Changes in MSR asset fair value due to modeled amortization	(1,904)	(2,384)	(3,279)
Total operating revenue	2,230	2,191	1,663
Risk management:			
Changes in MSR asset fair value due to market interest rates	(5,390)	(2,224)	5,804
Other changes in MSR asset fair value due to inputs or assumptions in model ^(a)	(1,727)	(44)	—
Derivative valuation adjustments and other	5,553	3,404	(4,176)
Total risk management	(1,564)	1,136	1,628
Total RFS net mortgage servicing revenue	666	3,327	3,291
All other	7	15	(116)
Mortgage fees and related income	\$2,721	\$3,870	\$3,678

(a) Represents the aggregate impact of changes in model inputs and assumptions such as costs to service, home prices, mortgage spreads, ancillary income, and assumptions used to derive prepayment speeds, as well as changes to the valuation models themselves.

The table below outlines the key economic assumptions used to determine the fair value of the Firm's MSRs at December 31, 2011 and 2010; and it outlines the sensitivities of those fair values to immediate adverse changes in those assumptions, as defined below.

Year ended December 31, (in millions, except rates)	2011	2010
Weighted-average prepayment speed assumption ("CPR")	18.07%	11.29%
Impact on fair value of 10% adverse change	\$ (585)	\$ (809)
Impact on fair value of 20% adverse change	(1,118)	(1,568)
Weighted-average option adjusted spread	7.83%	3.94%
Impact on fair value of 100 basis points adverse change	\$ (269)	\$ (578)
Impact on fair value of 200 basis points adverse change	(518)	(1,109)

CPR: Constant prepayment rate.

The sensitivity analysis in the preceding table is hypothetical and should be used with caution. Changes in fair value based on variation in assumptions generally cannot be easily extrapolated, because the relationship of the change in the assumptions to the change in fair value are often highly inter-related and may not be linear. In this table, the effect that a change in a particular assumption may have on the fair value is calculated without changing any other assumption. In reality, changes in one factor may result in changes in another, which would either magnify or counteract the impact of the initial change.

Other intangible assets

Other intangible assets are recorded at their fair value upon completion of a business combination or certain other transactions, and generally represent the value of customer relationships or arrangements. Subsequently, the Firm's intangible assets with finite lives, including core deposit intangibles, purchased credit card relationships, and other intangible assets, are amortized over their useful lives in a manner that best reflects the economic benefits of the intangible asset. The \$832 million decrease in other intangible assets during 2011, was due to \$848 million in amortization.

The components of credit card relationships, core deposits and other intangible assets were as follows.

December 31, (in millions)	December 31, 2011			December 31, 2010		
	Gross amount	Accumulated amortization ^(a)	Net carrying value	Gross amount	Accumulated amortization	Net carrying value
Purchased credit card relationships	\$ 3,826	\$ 3,224	\$ 602	\$ 5,789	\$ 4,892	\$ 897
Other credit card-related intangibles	844	356	488	907	314	593
Core deposit intangibles	4,133	3,539	594	4,280	3,401	879
Other intangibles	2,467	944	1,523	2,515	845	1,670

(a) The decrease in the gross amount and accumulated amortization from December 31, 2010, was due to the removal of fully amortized assets.

In addition to the finite lived intangible assets in the previous table, the Firm has intangible assets of approximately \$600 million consisting primarily of asset management advisory contracts, which were determined to have an indefinite life and are not amortized.

Amortization expense

The following table presents amortization expense related to credit card relationships, core deposits and other intangible assets.

December 31, (in millions)	2011	2010	2009
Purchased credit card relationships	\$ 295	\$ 355	\$ 421
Other credit card-related intangibles	106	111	94
Core deposit intangibles	285	328	390
Other intangibles	162	142	145
Total amortization expense	\$ 848	\$ 936	\$ 1,050

Future amortization expense

The following table presents estimated future amortization expense related to credit card relationships, core deposits and other intangible assets at December 31, 2011.

For the year ended December 31, (in millions)	Purchased credit card relationships	Other credit card-related intangibles	Core deposit intangibles	Other intangibles	Total
2012	\$ 253	\$ 106	\$ 240	\$ 147	\$ 746
2013	212	103	195	140	650
2014	109	102	103	122	436
2015	23	94	26	105	248
2016	4	34	14	98	150

Impairment testing

The Firm's intangible assets are tested for impairment annually or more often if events or changes in circumstances indicate that the asset might be impaired.

The impairment test for a finite-lived intangible asset compares the undiscounted cash flows associated with the use or disposition of the intangible asset to its carrying value. If the sum of the undiscounted cash flows exceeds its carrying value, then no impairment charge is recorded. If the sum of the undiscounted cash flows is less than its carrying value, then an impairment charge is recognized to the extent the carrying amount of the asset exceeds its fair value.

The impairment test for indefinite-lived intangible assets compares the fair value of the intangible asset to its carrying amount. If the carrying value exceeds the fair value, then an impairment charge is recognized for the difference.

Note 18 – Premises and equipment

Premises and equipment, including leasehold improvements, are carried at cost less accumulated depreciation and amortization. JPMorgan Chase computes depreciation using the straight-line method over the estimated useful life of an asset. For leasehold improvements, the Firm uses the straight-line method computed over the lesser of the remaining term of the leased facility or the estimated useful life of the leased asset. JPMorgan Chase has recorded immaterial asset retirement obligations related to asbestos remediation in those cases where it has sufficient information to estimate the obligations' fair value.

JPMorgan Chase capitalizes certain costs associated with the acquisition or development of internal-use software. Once the software is ready for its intended use, these costs are amortized on a straight-line basis over the software's expected useful life and reviewed for impairment on an ongoing basis.

At December 31, 2011 and 2010, time deposits in denominations of \$100,000 or more were as follows.

December 31, (in millions)	2011	2010
U.S. offices	\$ 57,802	\$ 59,653
Non-U.S. offices	50,614	44,544
Total	\$108,416	\$104,197

At December 31, 2011, the maturities of interest-bearing time deposits were as follows.

December 31, 2011 (in millions)	U.S.	Non-U.S.	Total
2012	\$ 68,345	\$ 67,107	\$ 135,452
2013	7,222	1,086	8,308
2014	1,947	219	2,166
2015	2,051	22	2,073
2016	2,532	102	2,634
After 5 years	641	57	698
Total	\$ 82,738	\$ 68,593	\$ 151,331

Note 19 – Deposits

At December 31, 2011 and 2010, noninterest-bearing and interest-bearing deposits were as follows.

December 31, (in millions)	2011	2010
U.S. offices		
Noninterest-bearing	\$ 346,670	\$ 228,555
Interest-bearing		
Demand ^(a)	47,075	33,368
Savings ^(b)	375,051	334,632
Time (included \$3,861 and \$2,733 at fair value) ^(c)	82,738	87,237
Total interest-bearing deposits	504,864	455,237
Total deposits in U.S. offices	851,534	683,792
Non-U.S. offices		
Noninterest-bearing	18,790	10,917
Interest-bearing		
Demand	188,202	174,417
Savings	687	607
Time (included \$1,072 and \$1,636 at fair value) ^(c)	68,593	60,636
Total interest-bearing deposits	257,482	235,660
Total deposits in non-U.S. offices	276,272	246,577
Total deposits	\$ 1,127,806	\$ 930,369

- (a) Includes Negotiable Order of Withdrawal ("NOW") accounts, and certain trust accounts.
 (b) Includes Money Market Deposit Accounts ("MMDAs").
 (c) Includes structured notes classified as deposits for which the fair value option has been elected. For further discussion, see Note 4 on pages 198–200 of this Annual Report.

Note 20 – Accounts payable and other liabilities

The following table details the components of accounts payable and other liabilities.

December 31, (in millions)	2011	2010
Brokerage payables	\$ 121,353	\$ 95,359
Accounts payable and other liabilities	81,542	74,971
Total	\$ 202,895	\$ 170,330

- (a) Includes payables to customers, brokers, dealers and clearing organizations, and securities fails.
 (b) Includes \$51 million and \$236 million accounted for at fair value at December 31, 2011 and 2010, respectively.

Note 21 – Long-term debt

JPMorgan Chase issues long-term debt denominated in various currencies, although predominantly U.S. dollars, with both fixed and variable interest rates. Included in senior and subordinated debt below are various equity-linked or other indexed instruments, which the Firm has elected to measure at fair value. Changes in fair value are recorded in principal transaction revenue in the Consolidated Statements of Income. The following table is a summary of long-term debt carrying values (including unamortized original issue discount, valuation adjustments and fair value adjustments, where applicable) by remaining contractual maturity as of December 31, 2011.

By remaining maturity at December 31, (in millions, except rates)		2011				2010 Total
		Under 1 year	1-5 years	After 5 years	Total	
Parent company						
Senior debt:	Fixed rate ^(a)	\$ 17,142	\$ 40,060	\$ 39,276	\$ 96,478	\$ 98,787
	Variable rate ^(b)	24,186	25,684	5,909	55,779	59,027
	Interest rates ^(c)	0.32-7.00%	0.60-7.00%	0.41-7.25%	0.32-7.25%	0.24-7.25%
Subordinated debt:	Fixed rate	\$ 1,005	\$ 8,919	\$ 9,243	\$ 19,167	\$ 22,000
	Variable rate	118	1,827	9	1,954	1,996
	Interest rates ^(c)	6.63-6.63%	1.09-5.75%	2.16-8.53%	1.09-8.53%	1.37-8.53%
Subtotal		\$ 42,451	\$ 76,490	\$ 54,437	\$ 173,378	\$ 181,810
Subsidiaries						
FHLB advances:	Fixed rate	\$ 18	\$ 4,548	\$ 172	\$ 4,738	\$ 7,324
	Variable rate	5,500	6,822	763	13,085	15,660
	Interest rates ^(c)	0.32-0.44%	0.32-2.04%	0.41-0.44%	0.32-2.04%	0.21-4.05%
Senior debt:	Fixed rate	\$ 699	\$ 2,963	\$ 2,884	\$ 6,546	\$ 5,228
	Variable rate	6,465	17,327	4,465	28,257	30,545
	Interest rates ^(c)	0.33-0.57%	0.13-4.28%	4.00-14.21%	0.13-14.21%	0.21-14.21%
Subordinated debt:	Fixed rate	\$ —	\$ 1,672	\$ 7,083	\$ 8,755	\$ 8,605
	Variable rate	—	1,150	—	1,150	1,150
	Interest rates ^(c)	—%	0.87-5.88%	4.38-8.25%	0.87-8.25%	0.63-8.25%
Subtotal		\$ 12,682	\$ 34,482	\$ 15,367	\$ 62,531	\$ 68,512
Junior subordinated debt:	Fixed rate	\$ —	\$ —	\$ 15,784	\$ 15,784	\$ 15,249
	Variable rate	—	—	5,082	5,082	5,082
	Interest rates ^(c)	—%	—%	0.93-8.75%	0.93-8.75%	0.79-8.75%
Subtotal		\$ —	\$ —	\$ 20,866	\$ 20,866	\$ 20,331
Total long-term debt ^{(e)(f)(g)}		\$ 55,133	\$ 110,972	\$ 90,670	\$ 256,775	\$ 270,653
Long-term beneficial interests:						
	Fixed rate	\$ 2,012	\$ 2,474	\$ 1,775	\$ 6,261	\$ 9,795
	Variable rate	11,474	15,306	6,693	33,473	42,759
	Interest rates	0.06-11.00%	0.06-5.63%	0.02-9.19%	0.02-11.00%	0.05-11.00%
Total long-term beneficial interests ^(h)		\$ 13,486	\$ 17,780	\$ 8,468	\$ 39,734	\$ 52,554

(a) Included \$8.4 billion and \$18.5 billion as of December 31, 2011 and 2010, respectively, guaranteed by the FDIC under the Temporary Liquidity Guarantee ("TLG") Program.

(b) Included \$11.9 billion and \$17.9 billion as of December 31, 2011 and 2010, respectively, guaranteed by the FDIC under the TLG Program.

(c) The interest rates shown are the range of contractual rates in effect at year-end, including non-U.S. dollar fixed- and variable-rate issuances, which excludes the effects of the associated derivative instruments used in hedge accounting relationships, if applicable. The use of these derivative instruments modifies the Firm's exposure to the contractual interest rates disclosed in the table above. Including the effects of the hedge accounting derivatives, the range of modified rates in effect at December 31, 2011, for total long-term debt was (0.37)% to 14.21%, versus the contractual range of 0.13% to 14.21% presented in the table above. The interest rate ranges shown exclude structured notes accounted for at fair value.

(d) Effective January 1, 2011, \$23.0 billion of long-term advances from FHLBs were reclassified from other borrowed funds to long-term debt. The prior-year period has been revised to conform with the current presentation.

(e) Included long-term debt of \$23.8 billion and \$31.3 billion secured by assets totaling \$89.4 billion and \$92.0 billion at December 31, 2011 and 2010, respectively. The amount of long-term debt secured by assets does not include amounts related to hybrid instruments.

(f) Included \$34.7 billion and \$38.8 billion of outstanding structured notes accounted for at fair value at December 31, 2011 and 2010, respectively.

(g) Included \$2.1 billion and \$879 million of outstanding zero-coupon notes at December 31, 2011 and 2010, respectively. The aggregate principal amount of these notes at their respective maturities was \$5.0 billion and \$2.7 billion, respectively.

(h) Included on the Consolidated Balance Sheets in beneficial interests issued by consolidated VIEs. Also included \$1.3 billion and \$1.5 billion of outstanding structured notes accounted for at fair value at December 31, 2011 and 2010, respectively. Excluded short-term commercial paper and other short-term beneficial interests of \$26.2 billion and \$25.1 billion at December 31, 2011 and 2010, respectively.

Notes to consolidated financial statements

- (i) At December 31, 2011, long-term debt in the aggregate of \$28.6 billion was redeemable at the option of JPMorgan Chase, in whole or in part, prior to maturity, based on the terms specified in the respective notes.
- (j) The aggregate carrying values of debt that matures in each of the five years subsequent to 2011 is \$55.1 billion in 2012, \$34.9 billion in 2013, \$30.4 billion in 2014, \$21.6 billion in 2015 and \$24.1 billion in 2016.

The weighted-average contractual interest rates for total long-term debt excluding structured notes accounted for at fair value were 3.57% and 3.50% as of December 31, 2011 and 2010, respectively. In order to modify exposure to interest rate and currency exchange rate movements, JPMorgan Chase utilizes derivative instruments, primarily interest rate and cross-currency interest rate swaps, in conjunction with some of its debt issues. The use of these instruments modifies the Firm's interest expense on the associated debt. The modified weighted-average interest rates for total long-term debt, including the effects of related derivative instruments, were 2.67% and 2.36% as of December 31, 2011 and 2010, respectively.

The Firm commenced its participation in the TLG Program in December 2008. The TLG Program was available to, among others, all U.S. depository institutions insured by the FDIC and all U.S. bank holding companies, unless they opted out or the FDIC terminated their participation. Under the TLG Program, the FDIC guaranteed through the earlier of maturity or December 31, 2012, certain senior unsecured debt issued through October 31, 2009, in return for a fee to be paid based on the amount and maturity of the debt. Under the TLG Program, the FDIC would pay the unpaid principal and interest on an FDIC-guaranteed debt instrument upon the failure of the participating entity to make a timely payment of principal or interest in accordance with the terms of the instrument.

The Parent Company has guaranteed certain long-term debt of its subsidiaries, including both long-term debt and structured notes sold as part of the Firm's market-making

activities. These guarantees rank on parity with all of the Firm's other unsecured and unsubordinated indebtedness. Guaranteed liabilities were \$3.0 billion and \$3.7 billion at December 31, 2011 and 2010, respectively.

The Firm's unsecured debt does not contain requirements that would call for an acceleration of payments, maturities or changes in the structure of the existing debt, provide any limitations on future borrowings or require additional collateral, based on unfavorable changes in the Firm's credit ratings, financial ratios, earnings or stock price.

Junior subordinated deferrable interest debentures held by trusts that issued guaranteed capital debt securities At December 31, 2011, the Firm had established 26 wholly-owned Delaware statutory business trusts ("issuer trusts") that had issued guaranteed capital debt securities.

The junior subordinated deferrable interest debentures issued by the Firm to the issuer trusts, totaling \$20.9 billion and \$20.3 billion at December 31, 2011 and 2010, respectively, were reflected in the Firm's Consolidated Balance Sheets in long-term debt, and in the table on the preceding page under the caption "Junior subordinated debt" (i.e., trust preferred capital debt securities). The Firm also records the common capital securities issued by the issuer trusts in other assets in its Consolidated Balance Sheets at December 31, 2011 and 2010. The debentures issued to the issuer trusts by the Firm, less the common capital securities of the issuer trusts, qualified as Tier 1 capital as of December 31, 2011.

The following is a summary of the outstanding trust preferred capital debt securities, including unamortized original issue discount, issued by each trust, and the junior subordinated deferrable interest debenture issued to each trust, as of December 31, 2011.

December 31, 2011 (in millions)	Amount of trust preferred capital debt securities issued by trust ^(a)	Principal amount of debenture issued to trust ^(b)	Issue date	Stated maturity of trust preferred capital securities and debentures	Earliest redemption date	Interest rate of trust preferred capital securities and debentures	Interest payment/ distribution dates
Bank One Capital III	\$474	\$765	2000	2030	Any time	8.75%	Semiannually
Bank One Capital VI	525	552	2001	2031	Any time	7.20%	Quarterly
Chase Capital II	482	497	1997	2027	Any time	LIBOR + 0.50%	Quarterly
Chase Capital III	295	305	1997	2027	Any time	LIBOR + 0.55%	Quarterly
Chase Capital VI	241	249	1998	2028	Any time	LIBOR + 0.625%	Quarterly
First Chicago NBD Capital I	249	256	1997	2027	Any time	LIBOR + 0.55%	Quarterly
J.P. Morgan Chase Capital X	1,000	1,016	2002	2032	Any time	7.00%	Quarterly
J.P. Morgan Chase Capital XI	1,075	1,009	2003	2033	Any time	5.88%	Quarterly
J.P. Morgan Chase Capital XII	400	391	2003	2033	Any time	6.25%	Quarterly
JPMorgan Chase Capital XIII	465	480	2004	2034	2014	LIBOR + 0.95%	Quarterly
JPMorgan Chase Capital XIV	600	587	2004	2034	Any time	6.20%	Quarterly
JPMorgan Chase Capital XV	93	132	2005	2035	Any time	5.88%	Semiannually
JPMorgan Chase Capital XVI	500	493	2005	2035	Any time	6.35%	Quarterly
JPMorgan Chase Capital XVII	496	720	2005	2035	Any time	5.85%	Semiannually
JPMorgan Chase Capital XVIII	748	749	2006	2036	Any time	6.95%	Semiannually
JPMorgan Chase Capital XIX	563	564	2006	2036	Any time	6.63%	Quarterly
JPMorgan Chase Capital XX	905	907	2006	2036	Any time	6.55%	Semiannually
JPMorgan Chase Capital XXI	836	837	2007	2037	2012	LIBOR + 0.95%	Quarterly
JPMorgan Chase Capital XXII	911	912	2007	2037	Any time	6.45%	Semiannually
JPMorgan Chase Capital XXIII	643	643	2007	2047	2012	LIBOR + 1.00%	Quarterly
JPMorgan Chase Capital XXIV	700	700	2007	2047	2012	6.88%	Quarterly
JPMorgan Chase Capital XXV	1,493	2,292	2007	2037	2037	6.80%	Semiannually
JPMorgan Chase Capital XXVI	1,815	1,815	2008	2048	2013	8.00%	Quarterly
JPMorgan Chase Capital XXVII	995	995	2009	2039	2039	7.00%	Semiannually
JPMorgan Chase Capital XXVIII	1,500	1,500	2009	2039	2014	7.20%	Quarterly
JPMorgan Chase Capital XXIX	1,500	1,500	2010	2040	2015	6.70%	Quarterly
Total	\$19,504	\$20,866					

- (a) Represents the amount of trust preferred capital debt securities issued to the public by each trust, including unamortized original issue discount.
- (b) Represents the principal amount of JPMorgan Chase debentures issued to each trust, including unamortized original-issue discount. The principal amount of debentures issued to the trusts includes the impact of hedging and purchase accounting fair value adjustments that were recorded on the Firm's Consolidated Financial Statements.

Note 22 – Preferred stock

At December 31, 2011 and 2010, JPMorgan Chase was authorized to issue 200 million shares of preferred stock, in one or more series, with a par value of \$1 per share.

In the event of a liquidation or dissolution of the Firm, JPMorgan Chase's preferred stock then outstanding takes precedence over the Firm's common stock for the payment of dividends and the distribution of assets.

Dividends on the Fixed-to-Floating Rate Non-Cumulative Perpetual Preferred Stock, Series I shares are payable semiannually at a fixed annual dividend rate of 7.90% through April 2018, and then become payable quarterly at an annual dividend rate of three-month LIBOR plus 3.47%. Dividends on the 8.625% Non-Cumulative Preferred Stock, Series J are payable quarterly.

On August 20, 2010, the Firm redeemed all of the outstanding shares of its 6.15% Cumulative Preferred Stock, Series E; 5.72% Cumulative Preferred Stock, Series F; and 5.49% Cumulative Preferred Stock, Series G at their stated redemption value. On June 17, 2009, the Firm redeemed all outstanding shares of the Fixed Rate Cumulative Perpetual Preferred Stock, Series K ("Series K Preferred Stock") and repaid the full \$25.0 billion principal amount together with accrued but unpaid dividends. The following is a summary of JPMorgan Chase's preferred stock outstanding as of December 31, 2011 and 2010.

December 31,		Fixed-to-Floating Rate Non-Cumulative Perpetual Preferred Stock, Series I	8.625% Non-Cumulative Perpetual Preferred Stock, Series J	Total preferred stock
Contractual rate in effect at December 31, 2011		7.900%	8.625%	
Shares ^(a)				
	2011	600,000	180,000	780,000
	2010	600,000	180,000	780,000
Carrying value (in millions)				
	2011	\$ 6,000	\$ 1,800	\$ 7,800
	2010	6,000	1,800	7,800
Earliest redemption date		4/30/2018	9/1/2013	
Share value and redemption price per share ^(b)		\$ 10,000	\$ 10,000	

(a) Represented by depositary shares.

(b) The redemption price includes the amount shown in the table plus any accrued but unpaid dividends.

Dividend and stock repurchase restrictions

Prior to the redemption of the Series K Preferred Stock on June 17, 2009, the Firm was subject to certain restrictions regarding the declaration of dividends and share repurchases. As a result of the redemption of the Series K Preferred Stock, JPMorgan Chase is no longer subject to any of these restrictions.

Note 23 – Common stock

At December 31, 2011 and 2010, JPMorgan Chase was authorized to issue 9.0 billion shares of common stock with a par value of \$1 per share. On June 5, 2009, the Firm issued \$5.8 billion, or 163 million new shares, of its common stock at \$35.25 per share.

Common shares issued (newly issued or distributed from treasury) by JPMorgan Chase during the years ended December 31, 2011, 2010 and 2009 were as follows.

Year ended December 31, (in millions)	2011	2010	2009
Issued – balance at January 1	4,104.9	4,104.9	3,941.6
New open market issuances	—	—	163.3
Total issued – balance at December 31	4,104.9	4,104.9	4,104.9
Treasury – balance at January 1	(194.6)	(162.9)	(208.8)
Purchase of treasury stock	(226.9)	(77.9)	—
Share repurchases related to employee stock-based awards	(0.1)	(0.1)	(1.1)
Issued from treasury:			
Employee benefits and compensation plans	88.3	45.3	45.7
Employee stock purchase plans	1.1	1.0	1.3
Total issued from treasury	89.4	46.3	47.0
Total treasury – balance at December 31	(332.2)	(194.6)	(162.9)
Outstanding	3,772.7	3,910.3	3,942.0

(a) Participants in the Firm's stock-based incentive plans may have shares withheld to cover income taxes.

Pursuant to the U.S. Treasury's Capital Purchase Program, the Firm issued to the U.S. Treasury a Warrant to purchase up to 88,401,697 shares of the Firm's common stock, at an exercise price of \$42.42 per share, subject to certain antidilution and other adjustments. The U.S. Treasury exchanged the Warrant for 88,401,697 warrants, each of which was a warrant to purchase a share of the Firm's common stock at an exercise price of \$42.42 per share and, on December 11, 2009, sold the warrants in a secondary public offering for \$950 million. The warrants are exercisable, in whole or in part, at any time and from time to time until October 28, 2018. As part of its common equity repurchase program discussed below, the Firm repurchased 10,167,698 warrants during 2011, with 78,233,999 warrants remaining outstanding at December 31, 2011. The repurchase of the warrants resulted in a \$122 million adjustment to capital surplus.

On March 18, 2011, the Board of Directors approved a \$15.0 billion common equity (i.e., common stock and warrants) repurchase program, of which \$8.95 billion was authorized for repurchase in 2011. The \$15.0 billion repurchase program superseded a \$10.0 billion repurchase program approved in 2007. During 2011 and 2010, the Firm repurchased (on a trade-date basis) an aggregate of 240 million and 78 million shares of common stock and warrants, for \$8.95 billion and \$3.0 billion, at an average price per unit of \$37.35 and \$38.49, respectively. The Firm

did not repurchase any of the warrants during 2010, and did not repurchase any shares of its common stock or warrants during 2009. For additional information regarding repurchases of the Firm's equity securities, see Part II, Item 5: Market for registrant's common equity, related stockholder matters and issuer purchases of equity securities, on pages 18–20 of JPMorgan Chase's 2011 Form 10-K.

The Firm may, from time to time, enter into written trading plans under Rule 10b5-1 of the Securities Exchange Act of 1934 to facilitate repurchases in accordance with the repurchase program. A Rule 10b5-1 repurchase plan allows the Firm to repurchase its equity during periods when it would not otherwise be repurchasing common equity – for example, during internal trading “black-out periods.” All purchases under a Rule 10b5-1 plan must be made according to a predefined plan established when the Firm is not aware of material nonpublic information.

As of December 31, 2011, approximately 408 million unissued shares of common stock were reserved for issuance under various employee incentive, compensation, option and stock purchase plans, director compensation plans, and the warrants sold by the U.S. Treasury as discussed above.

Note 24 – Earnings per share

Earnings per share (“EPS”) is calculated under the two-class method under which all earnings (distributed and undistributed) are allocated to each class of common stock and participating securities based on their respective rights to receive dividends. JPMorgan Chase grants restricted stock and RSUs to certain employees under its stock-based compensation programs, which entitle recipients to receive nonforfeitable dividends during the vesting period on a basis equivalent to the dividends paid to holders of common stock; these unvested awards meet the definition of participating securities. Options issued under employee benefit plans that have an antidilutive effect are excluded from the computation of diluted EPS.

The following table presents the calculation of basic and diluted EPS for the years ended December 31, 2011, 2010 and 2009.

Year ended December 31, (in millions, except per share amounts)	2011	2010	2009
Basic earnings per share			
Income before extraordinary gain	\$18,976	\$17,370	\$11,652
Extraordinary gain	—	—	76
Net income	\$18,976	\$17,370	\$11,728
Less: Preferred stock dividends	629	642	1,327
Less: Accelerated amortization from redemption of preferred stock issued to the U.S. Treasury	—	—	1,112 ^(c)
Net income applicable to common equity	18,347	16,728	9,289 ^(c)
Less: Dividends and undistributed earnings allocated to participating securities	779	964	515
Net income applicable to common stockholders	\$17,568	\$15,764	\$ 8,774
Total weighted-average basic shares outstanding	3,900.4	3,956.3	3,862.8
Per share			
Income before extraordinary gain	\$ 4.50	\$ 3.98	\$ 2.25 ^(c)
Extraordinary gain	—	—	0.02
Net income	\$ 4.50	\$ 3.98	\$ 2.27 ^(c)

Year ended December 31, (in millions, except per share amounts)	2011	2010	2009
Diluted earnings per share			
Net income applicable to common stockholders	\$17,568	\$15,764	\$ 8,774
Total weighted-average basic shares outstanding	3,900.4	3,956.3	3,862.8
Add: Employee stock options, SARs and warrants	19.9	20.6	16.9
Total weighted-average diluted shares outstanding ^(a)	3,920.3	3,976.9	3,879.7
Per share			
Income before extraordinary gain	\$ 4.48	\$ 3.96	\$ 2.24 ^(c)
Extraordinary gain	—	—	0.02
Net income per share	\$ 4.48	\$ 3.96	\$ 2.26 ^(c)

- (a) Excluded from the computation of diluted EPS (due to the antidilutive effect) were options issued under employee benefit plans and the warrants originally issued in 2008 under the U.S. Treasury's Capital Purchase Program to purchase shares of the Firm's common stock. The aggregate number of shares issuable upon the exercise of such options and warrants was 133 million, 233 million and 266 million for the full years ended December 31, 2011, 2010 and 2009 respectively.
- (b) Participating securities were included in the calculation of diluted EPS using the two-class method, as this computation was more dilutive than the calculation using the treasury stock method.
- (c) The calculation of basic and diluted EPS and net income applicable to common equity for full year 2009 includes a one-time, noncash reduction of \$1.1 billion, or \$0.27 per share, resulting from repayment of the U.S. Troubled Asset Relief Program (“TARP”) preferred capital.

Note 25 – Accumulated other comprehensive income/(loss)

AOCI includes the after-tax change in unrealized gains and losses on AFS securities, foreign currency translation adjustments (including the impact of related derivatives), cash flow hedging activities, and net loss and prior service costs/(credit) related to the Firm's defined benefit pension and OPEB plans.

As of or for the year ended December 31, (in millions)	Unrealized gains/ (losses) on AFS securities ^(b)	Translation adjustments, net of hedges	Cash flow hedges	Net loss and prior service costs/(credit) of defined benefit pension and OPEB plans	Accumulated other comprehensive income/(loss)
Balance at December 31, 2008	\$ (2,101)	\$ (598)	\$ (202)	\$ (2,786)	\$ (5,687)
Net change	4,133 ^(c)	582	383	498	5,596
Balance at December 31, 2009	\$ 2,032 ^(d)	\$ (16)	\$ 181	\$ (2,288)	\$ (91)
Cumulative effect of changes in accounting principles ^(a)	(144)	—	—	—	(144)
Net change	610 ^(e)	269	25	332	1,236
Balance at December 31, 2010	\$ 2,498 ^(d)	\$ 253	\$ 206	\$ (1,956)	\$ 1,001
Net change	1,067 ^(f)	(279)	(155)	(690)	(57)
Balance at December 31, 2011	\$ 3,565 ^(d)	\$ (26)	\$ 51	\$ (2,646)	\$ 944

- (a) Reflects the effect of the adoption of accounting guidance related to the consolidation of VIEs, and to embedded credit derivatives in beneficial interest securitized financial assets. AOCI decreased by \$129 million due to the adoption of the accounting guidance related to VIEs, as a result of the reversal of the fair value adjustments taken on retained AFS securities that were eliminated in consolidation; for further discussion see Note 16 on pages 256–267 of this Annual Report. AOCI decreased by \$15 million due to the adoption of the new guidance related to credit derivatives embedded in certain of the Firm's AFS securities; for further discussion see Note 6 on pages 202–210 of this Annual Report.
- (b) Represents the after-tax difference between the fair value and amortized cost of securities accounted for as AFS.
- (c) The net change during 2009 was due primarily to overall market spread and market liquidity improvement as well as changes in the composition of investments.
- (d) Included after-tax unrealized losses not related to credit on debt securities for which credit losses have been recognized in income of \$(56) million, \$(8) million and \$(226) million at December 31, 2011, 2010 and 2009, respectively.
- (e) The net change during 2010 was due primarily to the narrowing of spreads on commercial and non-agency MBS as well as on collateralized loan obligations; also reflects increased market value on pass-through MBS due to narrowing of spreads and other market factors.
- (f) The net change for 2011 was due primarily to increased market value on agency MBS and municipal securities, partially offset by the widening of spreads on non-U.S. corporate debt and the realization of gains due to portfolio repositioning.

The following table presents the before- and after-tax changes in the components of other comprehensive income/(loss).

Year ended December 31, (in millions)	2011			2010			2009		
	Before tax	Tax effect	After tax	Before tax	Tax effect	After tax	Before tax	Tax effect	After tax
Unrealized gains/(losses) on AFS securities:									
Net unrealized gains/(losses) arising during the period	\$ 3,361	\$(1,322)	\$ 2,039	\$ 3,982	\$(1,540)	\$ 2,442	\$ 7,870	\$(3,029)	\$ 4,841
Reclassification adjustment for realized (gains)/ losses included in net income	(1,593)	621	(972)	(2,982)	1,150	(1,832)	(1,152)	444	(708)
Net change	1,768	(701)	1,067	1,000	(390)	610	6,718	(2,585)	4,133
Translation adjustments:									
Translation	(672)	255	(417)	402	(139)	263	1,139	(398)	741
Hedges	226	(88)	138	11	(5)	6	(259)	100	(159)
Net change	(446)	167	(279)	413	(144)	269	880	(298)	582
Cash flow hedges:									
Net unrealized gains/(losses) arising during the period	50	(19)	31	247	(96)	151	767	(308)	459
Reclassification adjustment for realized (gains)/ losses included in net income	(301)	115	(186)	(206)	80	(126)	(124)	48	(76)
Net change	(251)	96	(155)	41	(16)	25	643	(260)	383
Net loss and prior service cost/(credit) of defined benefit pension and OPEB plans:									
Net gains/(losses) and prior service credits arising during the period	(1,291)	502	(789)	294	(96)	198	494	(200)	294
Reclassification adjustment for net loss and prior service credits included in net income	162	(63)	99	224	(90)	134	337	(133)	204
Net change	(1,129)	439	(690)	518	(186)	332	831	(333)	498
Total other comprehensive income/(loss)	\$ (58)	\$ 1	\$ (57)	\$ 1,972	\$ (736)	\$ 1,236	\$ 9,072	\$(3,476)	\$ 5,596

Note 26 – Income taxes

JPMorgan Chase and its eligible subsidiaries file a consolidated U.S. federal income tax return. JPMorgan Chase uses the asset and liability method to provide income taxes on all transactions recorded in the Consolidated Financial Statements. This method requires that income taxes reflect the expected future tax consequences of temporary differences between the carrying amounts of assets or liabilities for book and tax purposes. Accordingly, a deferred tax asset or liability for each temporary difference is determined based on the tax rates that the Firm expects to be in effect when the underlying items of income and expense are realized. JPMorgan Chase's expense for income taxes includes the current and deferred portions of that expense. A valuation allowance is established to reduce deferred tax assets to the amount the Firm expects to realize.

Due to the inherent complexities arising from the nature of the Firm's businesses, and from conducting business and being taxed in a substantial number of jurisdictions, significant judgments and estimates are required to be made. Agreement of tax liabilities between JPMorgan Chase and the many tax jurisdictions in which the Firm files tax returns may not be finalized for several years. Thus, the Firm's final tax-related assets and liabilities may ultimately be different from those currently reported.

The components of income tax expense/(benefit) included in the Consolidated Statements of Income were as follows for each of the years ended December 31, 2011, 2010, and 2009.

Year ended December 31, (in millions)	2011	2010	2009
Current income tax expense			
U.S. federal	\$ 3,719	\$ 4,001	\$ 4,698
Non-U.S.	1,183	2,712	2,368
U.S. state and local	1,178	1,744	971
Total current income tax expense	6,080	8,457	8,037
Deferred income tax expense/ (benefit)			
U.S. federal	2,109	(753)	(2,867)
Non-U.S.	102	169	(454)
U.S. state and local	(518)	(384)	(301)
Total deferred income tax expense/(benefit)	1,693	(968)	(3,622)
Total income tax expense	\$ 7,773	\$ 7,489	\$ 4,415

Total income tax expense includes \$76 million, \$485 million and \$280 million of tax benefits recorded in 2011, 2010, and 2009, respectively, as a result of tax audit resolutions.

The preceding table does not reflect the tax effect of certain items that are recorded each period directly in stockholders' equity and certain tax benefits associated with the Firm's employee stock-based compensation plans. The tax effect of all items recorded directly to stockholders' equity resulted in an increase of \$927 million in 2011, an

increase of \$1.8 billion in 2010, and a decrease of \$3.7 billion in 2009.

U.S. federal income taxes have not been provided on the undistributed earnings of certain non-U.S. subsidiaries, to the extent that such earnings have been reinvested abroad for an indefinite period of time. Based on JPMorgan Chase's ongoing review of the business requirements and capital needs of its non-U.S. subsidiaries, combined with the formation of specific strategies and steps taken to fulfill these requirements and needs, the Firm has determined that the undistributed earnings of certain of its subsidiaries would be indefinitely reinvested to fund current and future growth of the related businesses. As management does not intend to use the earnings of these subsidiaries as a source of funding for its U.S. operations, such earnings will not be distributed to the U.S. in the foreseeable future. For 2011, pretax earnings of approximately \$2.6 billion were generated and will be indefinitely reinvested in these subsidiaries. At December 31, 2011, the cumulative amount of undistributed pretax earnings in these subsidiaries approximated \$21.8 billion. If the Firm were to record a deferred tax liability associated with these undistributed earnings, the amount would be approximately \$4.9 billion at December 31, 2011.

Tax expense applicable to securities gains and losses for the years 2011, 2010 and 2009 was \$617 million, \$1.1 billion, and \$427 million, respectively.

A reconciliation of the applicable statutory U.S. income tax rate to the effective tax rate for each of the years ended December 31, 2011, 2010 and 2009, is presented in the following table.

Year ended December 31,	2011	2010	2009
Statutory U.S. federal tax rate	35.0%	35.0%	35.0%
Increase/(decrease) in tax rate resulting from:			
U.S. state and local income taxes, net of U.S. federal income tax benefit	1.6	3.6	2.7
Tax-exempt income	(2.1)	(2.4)	(3.9)
Non-U.S. subsidiary earnings	(2.3)	(2.2)	(1.7)
Business tax credits	(4.0)	(3.7)	(5.5)
Other, net	0.9	(0.2)	0.9
Effective tax rate	29.1%	30.1%	27.5%

(a) Includes earnings deemed to be reinvested indefinitely in non-U.S. subsidiaries.

Deferred income tax expense/(benefit) results from differences between assets and liabilities measured for financial reporting purposes versus income tax return purposes. Deferred tax assets are recognized if, in management's judgment, their realizability is determined to be more likely than not. If a deferred tax asset is determined to be unrealizable, a valuation allowance is established. The significant components of deferred tax assets and liabilities are reflected in the following table as of December 31, 2011 and 2010.

Notes to consolidated financial statements

December 31, (in millions)	2011	2010
Deferred tax assets		
Allowance for loan losses	\$ 10,689	\$ 12,287
Employee benefits	4,570	4,279
Accrued expenses and other	9,186	7,850
Non-U.S. operations	2,943	956
Tax attribute carryforwards	1,547	2,348
Gross deferred tax assets	\$ 28,935	\$ 27,720
Valuation allowance	(1,303)	(1,784)
Deferred tax assets, net of valuation allowance	\$ 27,632	\$ 25,936
Deferred tax liabilities		
Depreciation and amortization	\$ 6,358	\$ 4,823
Leasing transactions	2,569	2,160
Non-U.S. operations	2,790	1,136
Other, net ^(a)	1,139	1,497
Gross deferred tax liabilities	\$ 12,856	\$ 9,616
Net deferred tax assets	\$ 14,776	\$ 16,320

(a) The prior-year period has been revised to conform with the current presentation.

JPMorgan Chase has recorded deferred tax assets of \$1.5 billion at December 31, 2011, in connection with U.S. federal, state and local, and non-U.S. subsidiary net operating loss carryforwards. At December 31, 2011, the U.S. federal net operating loss carryforwards were approximately \$4.1 billion; the state and local net operating loss carryforward was approximately \$642 million; and the non-U.S. subsidiary net operating loss carryforward was \$116 million. If not utilized, the U.S. federal net operating loss carryforwards and the state and local net operating loss carryforward will expire between 2027 and 2030. The non-U.S. subsidiary net operating loss carryforward has an unlimited carryforward period. A valuation allowance has been recorded for losses associated with non-U.S. subsidiaries and certain portfolio investments, and certain state and local tax benefits. During 2011, the valuation allowance decreased by \$481 million predominantly related to the realization of state and local tax benefits.

At December 31, 2011, 2010 and 2009, JPMorgan Chase's unrecognized tax benefits, excluding related interest expense and penalties, were \$7.2 billion, \$7.8 billion and \$6.6 billion, respectively, of which \$4.0 billion, \$3.8 billion and \$3.5 billion, respectively, if recognized, would reduce the annual effective tax rate. As JPMorgan Chase is presently under audit by a number of taxing authorities, it is reasonably possible that significant changes in the gross balance of unrecognized tax benefits may occur within the next 12 months. JPMorgan Chase does not expect that any changes over the next twelve months in its gross balance of unrecognized tax benefits caused by such audits would result in a significant change in its annual effective tax rate. The following table presents a reconciliation of the beginning and ending amount of unrecognized tax benefits for the years ended December 31, 2011, 2010 and 2009.

Unrecognized tax benefits

Year ended December 31, (in millions)	2011	2010	2009
Balance at January 1,	\$ 7,767	\$ 6,608	\$ 5,894
Increases based on tax positions related to the current period	516	813	584
Decreases based on tax positions related to the current period	(110)	(24)	(6)
Increases based on tax positions related to prior periods	496	1,681	703
Decreases based on tax positions related to prior periods	(1,433)	(1,198)	(322)
Decreases related to settlements with taxing authorities	(16)	(74)	(203)
Decreases related to a lapse of applicable statute of limitations	(31)	(39)	(42)
Balance at December 31,	\$ 7,189	\$ 7,767	\$ 6,608

After-tax interest expense/(benefit) and penalties related to income tax liabilities recognized in income tax expense were \$184 million, \$(54) million and \$101 million in 2011, 2010 and 2009, respectively.

At December 31, 2011 and 2010, in addition to the liability for unrecognized tax benefits, the Firm had accrued \$1.7 billion and \$1.6 billion, respectively, for income tax-related interest and penalties.

JPMorgan Chase is continually under examination by the Internal Revenue Service, by taxing authorities throughout the world, and by many states throughout the U.S. The following table summarizes the status of significant income tax examinations of JPMorgan Chase and its consolidated subsidiaries as of December 31, 2011.

December 31, 2011	Periods under examination	Status
JPMorgan Chase – U.S.1993 – 2002		Refund claims under review
JPMorgan Chase – U.S.2003 – 2005 ^(a)		Field examination completed, JPMorgan Chase intends to file refund claims
Bank One – U.S.	2000 – 2004	Refund claims under review
Bear Stearns – U.S.	2003 – 2005	In appeals process
Bear Stearns – U.S.	2006 – 2008	Field examination
JPMorgan Chase – United Kingdom	2006 – 2010	Field examination
JPMorgan Chase – New York State and City	2005 – 2007	Field examination
JPMorgan Chase – California	2006 – 2008	Field examination

(a) JPMorgan Chase anticipates that the IRS will commence in 2012 an examination of the years 2006 through 2008.

The following table presents the U.S. and non-U.S. components of income before income tax expense and extraordinary gain for the years ended December 31, 2011, 2010 and 2009.

Year ended December 31, (in millions)	2011	2010	2009
U.S.	\$ 16,336	\$ 16,568	\$ 6,263
Non-U.S. ^(a)	10,413	8,291	9,804
Income before income tax and extraordinary gain	\$ 26,749	\$ 24,859	\$ 16,067

(a) For purposes of this table, non-U.S. income is defined as income generated from operations located outside the U.S.

Note 27 – Restrictions on cash and intercompany funds transfers

The business of JPMorgan Chase Bank, National Association (“JPMorgan Chase Bank, N.A.”) is subject to examination and regulation by the Office of the Comptroller of the Currency (“OCC”). The Bank is a member of the U.S. Federal Reserve System, and its deposits in the U.S. are insured by the FDIC.

The Board of Governors of the Federal Reserve System (the “Federal Reserve”) requires depository institutions to maintain cash reserves with a Federal Reserve Bank. The average amount of reserve balances deposited by the Firm’s bank subsidiaries with various Federal Reserve Banks was approximately \$4.4 billion and \$803 million in 2011 and 2010, respectively.

Restrictions imposed by U.S. federal law prohibit JPMorgan Chase and certain of its affiliates from borrowing from banking subsidiaries unless the loans are secured in specified amounts. Such secured loans to the Firm or to other affiliates are generally limited to 10% of the banking subsidiary’s total capital, as determined by the risk-based capital guidelines; the aggregate amount of all such loans is limited to 20% of the banking subsidiary’s total capital.

The principal sources of JPMorgan Chase’s income (on a parent company-only basis) are dividends and interest from JPMorgan Chase Bank, N.A., and the other banking and nonbanking subsidiaries of JPMorgan Chase. In addition to

dividend restrictions set forth in statutes and regulations, the Federal Reserve, the OCC and the FDIC have authority under the Financial Institutions Supervisory Act to prohibit or to limit the payment of dividends by the banking organizations they supervise, including JPMorgan Chase and its subsidiaries that are banks or bank holding companies, if, in the banking regulator’s opinion, payment of a dividend would constitute an unsafe or unsound practice in light of the financial condition of the banking organization.

At January 1, 2012, JPMorgan Chase’s banking subsidiaries could pay, in the aggregate, \$7.4 billion in dividends to their respective bank holding companies without the prior approval of their relevant banking regulators. The capacity to pay dividends in 2012 will be supplemented by the banking subsidiaries’ earnings during the year.

In compliance with rules and regulations established by U.S. and non-U.S. regulators, as of December 31, 2011 and 2010, cash in the amount of \$25.4 billion and \$25.0 billion, respectively, and securities with a fair value of \$23.4 billion and \$9.7 billion, respectively, were segregated in special bank accounts for the benefit of securities and futures brokerage customers. In addition, as of December 31, 2011 and 2010, the Firm had other restricted cash of \$4.2 billion and \$2.7 billion, respectively, primarily representing cash reserves held at non-U.S. central banks and held for other general purposes.

Note 28 – Regulatory capital

The Federal Reserve establishes capital requirements, including well-capitalized standards for the consolidated financial holding company. The OCC establishes similar capital requirements and standards for the Firm’s national banks, including JPMorgan Chase Bank, N.A., and Chase Bank USA, N.A.

There are two categories of risk-based capital: Tier 1 capital and Tier 2 capital. Tier 1 capital consists of common stockholders’ equity, perpetual preferred stock, noncontrolling interests in subsidiaries and trust preferred capital debt securities, less goodwill and certain other adjustments. Tier 2 capital consists of preferred stock not qualifying as Tier 1 capital, subordinated long-term debt and other instruments qualifying as Tier 2 capital, and the aggregate allowance for credit losses up to a certain percentage of risk-weighted assets. Total capital is Tier 1 capital plus Tier 2 capital. Under the risk-based capital guidelines of the Federal Reserve, JPMorgan Chase is required to maintain minimum ratios of Tier 1 and Total capital to risk-weighted assets, as well as minimum leverage ratios (which are defined as Tier 1 capital divided by adjusted quarterly average assets). Failure to meet these minimum requirements could cause the Federal Reserve to take action. Banking subsidiaries also are subject to these capital requirements by their respective primary regulators. As of December 31, 2011 and 2010, JPMorgan Chase and all of its banking subsidiaries were well-capitalized and met all capital requirements to which each was subject.

Notes to consolidated financial statements

The following table presents the regulatory capital, assets and risk-based capital ratios for JPMorgan Chase and its significant banking subsidiaries at December 31, 2011 and 2010. These amounts are determined in accordance with regulations issued by the Federal Reserve and/or OCC.

December 31, (in millions, except ratios)	JPMorgan Chase & Co.		JPMorgan Chase Bank, N.A.		Chase Bank USA, N.A.		Well-capitalized ratios ^(f)	Minimum capital ratios ^(f)
	2011	2010	2011	2010	2011	2010		
Regulatory capital								
Tier 1 ^(a)	\$ 150,384	\$ 142,450	\$ 98,426	\$ 91,764	\$ 11,903	\$ 12,966		
Total	188,088	182,216	136,017	130,444	15,448	16,659		
Assets								
Risk-weighted ^{(b)(c)}	\$1,221,198	\$1,174,978	\$1,042,898	\$ 965,897	\$107,421	\$116,992		
Adjusted average ^(d)	2,202,087	2,024,515	1,789,194	1,611,486	106,312	117,368		
Capital ratios								
Tier 1 ^(a)	12.3%	12.1%	9.4%	9.5%	11.1%	11.1%	6.0%	4.0%
Total	15.4	15.5	13.0	13.5	14.4	14.2	10.0	8.0
Tier 1 leverage	6.8	7.0	5.5	5.7	11.2	11.0	5.0 ^(g)	3.0 ^(h)

- (a) At December 31, 2011, for JPMorgan Chase and JPMorgan Chase Bank, N.A., trust preferred capital debt securities were \$19.6 billion and \$600 million, respectively. If these securities were excluded from the calculation at December 31, 2011, Tier 1 capital would be \$130.8 billion and \$97.8 billion, respectively, and the Tier 1 capital ratio would be 10.7% and 9.4%, respectively. At December 31, 2011, Chase Bank USA, N.A. had no trust preferred capital debt securities.
- (b) Risk-weighted assets consist of on- and off-balance sheet assets that are assigned to one of several broad risk categories and weighted by factors representing their risk and potential for default. On-balance sheet assets are risk-weighted based on the perceived credit risk associated with the obligor or counterparty, the nature of any collateral, and the guarantor, if any. Off-balance sheet assets such as lending-related commitments, guarantees, derivatives and other applicable off-balance sheet positions are risk-weighted by multiplying the contractual amount by the appropriate credit conversion factor to determine the on-balance sheet credit-equivalent amount, which is then risk-weighted based on the same factors used for on-balance sheet assets. Risk-weighted assets also incorporate a measure for the market risk related to applicable trading assets—debt and equity instruments, and foreign exchange and commodity derivatives. The resulting risk-weighted values for each of the risk categories are then aggregated to determine total risk-weighted assets.
- (c) Includes off-balance sheet risk-weighted assets at December 31, 2011, of \$301.1 billion, \$291.0 billion and \$38 million, and at December 31, 2010, of \$282.9 billion, \$274.2 billion and \$31 million, for JPMorgan Chase, JPMorgan Chase Bank, N.A. and Chase Bank USA, N.A., respectively.
- (d) Adjusted average assets, for purposes of calculating the leverage ratio, include total quarterly average assets adjusted for unrealized gains/(losses) securities, less deductions for disallowed goodwill and other intangible assets, investments in certain subsidiaries, and the total adjusted carrying value of nonfinancial equity investments that are subject to deductions from Tier 1 capital.
- (e) Asset and capital amounts for JPMorgan Chase's banking subsidiaries reflect intercompany transactions; whereas the respective amounts for JPMorgan Chase reflect the elimination of intercompany transactions.
- (f) As defined by the regulations issued by the Federal Reserve, OCC and FDIC.
- (g) Represents requirements for banking subsidiaries pursuant to regulations issued under the FDIC Improvement Act. There is no Tier 1 leverage component in the definition of a well-capitalized bank holding company.
- (h) The minimum Tier 1 leverage ratio for bank holding companies and banks is 3% or 4%, depending on factors specified in regulations issued by the Federal Reserve and OCC.

Note: Rating agencies allow measures of capital to be adjusted upward for deferred tax liabilities, which have resulted from both nontaxable business combinations and from tax-deductible goodwill. The Firm had deferred tax liabilities resulting from nontaxable business combinations totaling \$414 million and \$647 million at December 31, 2011 and 2010, respectively; and deferred tax liabilities resulting from tax-deductible goodwill of \$2.3 billion and \$1.9 billion at December 31, 2011 and 2010, respectively.

A reconciliation of the Firm's Total stockholders' equity to Tier 1 capital and Total qualifying capital is presented in the table below.

December 31, (in millions)	2011	2010
Tier 1 capital		
Total stockholders' equity	\$ 183,573	\$ 176,106
Effect of certain items in accumulated other comprehensive income/(loss) excluded from Tier 1 capital	(970)	(748)
Qualifying hybrid securities and noncontrolling interests ^(a)	19,668	19,887
Less: Goodwill	45,873	46,915
Fair value DVA on derivative and structured note liabilities related to the Firm's credit quality	2,150	1,261
Investments in certain subsidiaries and other	993	1,032
Other intangible assets ^(b)	2,871	3,587
Total Tier 1 capital	150,384	142,450
Tier 2 capital		
Long-term debt and other instruments qualifying as Tier 2	22,275	25,018
Qualifying allowance for credit losses	15,504	14,959
Adjustment for investments in certain subsidiaries and other	(75)	(211)
Total Tier 2 capital	37,704	39,766
Total qualifying capital	\$ 188,088	\$ 182,216

(a) Primarily includes trust preferred capital debt securities of certain business trusts.

(b) Goodwill and other intangible assets are net of any associated deferred tax liabilities.

Note 29 – Off-balance sheet lending-related financial instruments, guarantees, and other commitments

JPMorgan Chase provides lending-related financial instruments (e.g., commitments and guarantees) to meet the financing needs of its customers. The contractual amount of these financial instruments represents the maximum possible credit risk to the Firm should the counterparty draw upon the commitment or the Firm be required to fulfill its obligation under the guarantee, and should the counterparty subsequently fail to perform according to the terms of the contract. Most of these commitments and guarantees expire without being drawn or a default occurring. As a result, the total contractual amount of these instruments is not, in the Firm's view, representative of its actual future credit exposure or funding requirements.

To provide for the risk of loss inherent in wholesale and consumer (excluding credit card) contracts, an allowance for credit losses on lending-related commitments is maintained. See Note 15 on pages 252–255 of this Annual Report for further discussion regarding the allowance for credit losses on lending-related commitments. The following table summarizes the contractual amounts and carrying values of off-balance sheet lending-related financial instruments, guarantees and other commitments at December 31, 2011 and 2010. The amounts in the table below for credit card and home equity lending-related commitments represent the total available credit for these products. The Firm has not experienced, and does not anticipate, that all available lines of credit for these products will be utilized at the same time. The Firm can reduce or cancel credit card lines of credit by providing the borrower notice or, in some cases, without notice as permitted by law. The Firm may reduce or close home equity lines of credit when there are significant decreases in the value of the underlying property, or when there has been a demonstrable decline in the creditworthiness of the borrower. Also, the Firm typically closes credit card lines when the borrower is 60 days or more past due.

Notes to consolidated financial statements

Off-balance sheet lending-related financial instruments, guarantees and other commitments

	Contractual amount					Carrying value ^e	
	2011					2011	2010
By remaining maturity at December 31, (in millions)	Expires in 1 year or less	Expires after 1 year through 3 years	Expires after 3 years through 5 years	Expires after 5 years	Total	Total	
Lending-related							
Consumer, excluding credit card:							
Home equity – senior lien	\$ 933	\$ 4,780	\$ 4,870	\$ 5,959	\$ 16,542	\$ 17,662	\$ —
Home equity – junior lien	2,096	8,964	8,075	7,273	26,408	30,948	—
Prime mortgage	1,500	—	—	—	1,500	1,266	—
Subprime mortgage	—	—	—	—	—	—	—
Auto	6,431	97	149	17	6,694	5,246	1
Business banking	9,480	430	63	326	10,299	9,702	6
Student and other	82	169	127	486	864	579	—
Total consumer, excluding credit card	20,522	14,440	13,284	14,061	62,307	65,403	7
Credit card	530,616	—	—	—	530,616	547,227	—
Total consumer	551,138	14,440	13,284	14,061	592,923	612,630	7
Wholesale:							
Other unfunded commitments to extend credit	61,083	61,628	87,830	4,710	215,251	199,859	347
Standby letters of credit and other financial guarantees ^{(b)(c)(d)}	27,982	34,671	36,448	2,798	101,899	94,837	696
Unused advised lines of credit	46,695	11,324	327	1,857	60,203	44,720	—
Other letters of credit ^(d)	4,218	1,020	148	—	5,386	6,663	2
Total wholesale	139,978	108,643	124,753	9,365	382,739	346,079	1,045
Total lending-related	\$ 691,116	\$ 123,083	\$ 138,037	\$ 23,426	\$ 975,662	\$ 958,709	\$ 1,052
Other guarantees and commitments							
Securities lending indemnifications	\$ 186,077	\$ —	\$ —	\$ —	\$ 186,077	\$ 181,717	NA
Derivatives qualifying as guarantees	2,998	5,117	31,097	36,381	75,593	87,768	\$ 457
Unsettled reverse repurchase and securities borrowing agreements	39,939	—	—	—	39,939	39,927	—
Loan sale and securitization-related indemnifications:							
Mortgage repurchase liability ^(f)	NA	NA	NA	NA	NA	NA	3,557
Loans sold with recourse	NA	NA	NA	NA	10,397	10,982	148
Other guarantees and commitments	1,030	279	299	4,713	6,321	6,492	(5)

- (a) At December 31, 2011 and 2010, reflects the contractual amount net of risk participations totaling \$1.1 billion and \$542 million, respectively, for other unfunded commitments to extend credit; \$19.8 billion and \$22.4 billion, respectively, for standby letters of credit and other financial guarantees; and \$974 million and \$1.1 billion, respectively, for other letters of credit. In regulatory filings with the Federal Reserve these commitments are shown gross risk participations.
- (b) At December 31, 2011 and 2010, included credit enhancements and bond and commercial paper liquidity commitments to U.S. states and municipalities, hospitals and other not-for-profit entities of \$48.6 billion and \$43.4 billion, respectively. These commitments also include liquidity facilities to nonconsolidated municipal bond VIEs; for further information, see Note 16 on pages 256–267 of this Annual Report.
- (c) At December 31, 2011 and 2010, included unissued standby letters of credit commitments of \$44.1 billion and \$41.6 billion, respectively.
- (d) At December 31, 2011 and 2010, JPMorgan Chase held collateral relating to \$41.5 billion and \$37.8 billion, respectively, of standby letters of credit; and \$1.3 billion and \$2.1 billion, respectively, of other letters of credit.
- (e) At December 31, 2011 and 2010, collateral held by the Firm in support of securities lending indemnification agreements was \$186.3 billion and \$185.0 billion, respectively. Securities lending collateral comprises primarily cash and securities issued by governments that are members of the Organisation for Economic Co-operation and Development (“OECD”) and U.S. government agencies.
- (f) Represents notional amounts of derivatives qualifying as guarantees.
- (g) Represents the estimated mortgage repurchase liability related to indemnifications for breaches of representations and warranties in loan sale and securitization agreements. For additional information, see Loan sale and securitization-related indemnifications on pages 286–287 of this Note.
- (h) At December 31, 2011 and 2010, included unfunded commitments of \$789 million and \$1.0 billion, respectively, to third-party private equity funds; and \$1.5 billion and \$1.4 billion, respectively, to other equity investments. These commitments included \$820 million and \$1.0 billion, respectively, related to investments that are generally fair valued at net asset value as discussed in Note 3 on pages 184–198 of this Annual Report. In addition, at December 31, 2011 and 2010, included letters of credit hedged by derivative transactions and managed on a market risk basis of \$3.9 billion and \$3.8 billion, respectively.
- (i) For lending-related products, the carrying value represents the allowance for lending-related commitments and the guarantee liability; for derivative-related products, the carrying value represents the fair value. For all other products the carrying value represents the valuation reserve.

Other unfunded commitments to extend credit
Other unfunded commitments to extend credit generally comprise commitments for working capital and general corporate purposes, as well as extensions of credit to support commercial paper facilities and bond financings in the event that those obligations cannot be remarketed to new investors.

Also included in other unfunded commitments to extend credit are commitments to noninvestment-grade counterparties in connection with leveraged and acquisition finance activities, which were \$6.1 billion and \$5.9 billion at December 31, 2011 and 2010, respectively. For further information, see Note 3 and Note 4 on pages 184–198 and 198–200 respectively, of this Annual Report.

Guarantees

U.S. GAAP requires that a guarantor recognize, at the inception of a guarantee, a liability in an amount equal to the fair value of the obligation undertaken in issuing the guarantee. U.S. GAAP defines a guarantee as a contract that contingently requires the guarantor to pay a guaranteed party based upon: (a) changes in an underlying asset, liability or equity security of the guaranteed party; or (b) a third party's failure to perform under a specified agreement. The Firm considers the following off-balance sheet lending-related arrangements to be guarantees under U.S. GAAP: standby letters of credit and financial guarantees, securities lending indemnifications, certain indemnification agreements included within third-party contractual arrangements and certain derivative contracts.

As required by U.S. GAAP, the Firm initially records guarantees at the inception date fair value of the obligation assumed (e.g., the amount of consideration received or the net present value of the premium receivable). For certain types of guarantees, the Firm records this fair value amount in other liabilities with an offsetting entry recorded in cash (for premiums received), or other assets (for premiums

receivable). Any premium receivable recorded in other assets is reduced as cash is received under the contract, and the fair value of the liability recorded at inception is amortized into income as lending and deposit-related fees over the life of the guarantee contract. For indemnifications provided in sales agreements, a portion of the sale proceeds is allocated to the guarantee, which adjusts the gain or loss that would otherwise result from the transaction. For these indemnifications, the initial liability is amortized to income as the Firm's risk is reduced (i.e., over time or when the indemnification expires). Any contingent liability that exists as a result of issuing the guarantee or indemnification is recognized when it becomes probable and reasonably estimable. The contingent portion of the liability is not recognized if the estimated amount is less than the carrying amount of the liability recognized at inception (adjusted for any amortization). The recorded amounts of the liabilities related to guarantees and indemnifications at December 31, 2011 and 2010, excluding the allowance for credit losses on lending-related commitments, are discussed below.

Standby letters of credit and other financial guarantees
Standby letters of credit ("SBLC") and other financial guarantees are conditional lending commitments issued by the Firm to guarantee the performance of a customer to a third party under certain arrangements, such as commercial paper facilities, bond financings, acquisition financings, trade and similar transactions. The carrying values of standby and other letters of credit were \$698 million and \$707 million at December 31, 2011 and 2010, respectively, which were classified in accounts payable and other liabilities on the Consolidated Balance Sheets; these carrying values included \$319 million and \$347 million, respectively, for the allowance for lending-related commitments, and \$379 million and \$360 million, respectively, for the guarantee liability and corresponding asset.

The following table summarizes the types of facilities under which standby letters of credit and other letters of credit arrangements are outstanding by the ratings profiles of the Firm's customers, as of December 31, 2011 and 2010.

Standby letters of credit, other financial guarantees and other letters of credit

December 31, (in millions)	2011		2010	
	Standby letters of credit and other financial guarantees	Other letters of credit	Standby letters of credit and other financial guarantees	Other letters of credit
Investment-grade ^(a)	\$ 78,884	\$ 4,105	\$ 70,236	\$ 5,289
Noninvestment-grade ^(b)	23,015	1,281	24,601	1,374
Total contractual amount ^(b)	\$ 101,899 ^(c)	\$ 5,386	\$ 94,837 ^(c)	\$ 6,663
Allowance for lending-related commitments	\$ 317	\$ 2	\$ 345	\$ 2
Commitments with collateral	41,529	1,264	37,815	2,127

(a) The ratings scale is based on the Firm's internal ratings which generally correspond to ratings as defined by S&P and Moody's.

(b) At December 31, 2011 and 2010, reflects the contractual amount net of risk participations totaling \$19.8 billion and \$22.4 billion, respectively, for standby letters of credit and other financial guarantees; and \$974 million and \$1.1 billion, respectively, for other letters of credit. In regulatory filings with the Federal Reserve these commitments are shown gross of risk participations.

(c) At December 31, 2011 and 2010, included unissued standby letters of credit commitments of \$44.1 billion and \$41.6 billion, respectively.

Advised lines of credit

An advised line of credit is a revolving credit line which specifies the maximum amount the Firm may make available to an obligor, on a nonbinding basis. The borrower receives written or oral advice of this facility. The Firm may cancel this facility at any time by providing the borrower notice or, in some cases, without notice as permitted by law.

Securities lending indemnifications

Through the Firm's securities lending program, customers' securities, via custodial and non-custodial arrangements, may be lent to third parties. As part of this program, the Firm provides an indemnification in the lending agreements which protects the lender against the failure of the third-party borrower to return the lent securities in the event the Firm did not obtain sufficient collateral. To minimize its liability under these indemnification agreements, the Firm obtains cash or other highly liquid collateral with a market value exceeding 100% of the value of the securities on loan from the borrower. Collateral is marked to market daily to help assure that collateralization is adequate. Additional collateral is called from the borrower if a shortfall exists, or collateral may be released to the borrower in the event of overcollateralization. If a borrower defaults, the Firm would use the collateral held to purchase replacement securities in the market or to credit the lending customer with the cash equivalent thereof.

Derivatives qualifying as guarantees

In addition to the contracts described above, the Firm transacts certain derivative contracts that have the characteristics of a guarantee under U.S. GAAP. These contracts include written put options that require the Firm to purchase assets upon exercise by the option holder at a specified price by a specified date in the future. The Firm may enter into written put option contracts in order to meet client needs, or for other trading purposes. The terms of written put options are typically five years or less.

Derivative guarantees also include contracts such as stable value derivatives that require the Firm to make a payment of the difference between the market value and the book value of a counterparty's reference portfolio of assets in the event that market value is less than book value and certain other conditions have been met. Stable value derivatives, commonly referred to as "stable value wraps", are transacted in order to allow investors to realize investment returns with less volatility than an unprotected portfolio and are typically longer-term or may have no stated maturity, but allow the Firm to terminate the contract under certain conditions.

Derivative guarantees are recorded on the Consolidated Balance Sheets at fair value in trading assets and trading liabilities. The total notional value of the derivatives that the Firm deems to be guarantees was \$75.6 billion and \$87.8 billion at December 31, 2011 and 2010, respectively. The notional amount generally represents the Firm's maximum exposure to derivatives qualifying as guarantees. However, exposure to certain stable value contracts is contractually limited to a substantially lower percentage of the notional amount; the notional amount on

these stable value contracts was \$26.1 billion and \$25.9 billion and the maximum exposure to loss was \$2.8 billion and \$2.7 billion, at December 31, 2011 and 2010, respectively. The fair values of the contracts reflect the probability of whether the Firm will be required to perform under the contract. The fair value related to derivatives that the Firm deems to be guarantees were derivative payables of \$555 million and \$390 million and derivative receivables of \$98 million and \$96 million at December 31, 2011 and 2010, respectively. The Firm reduces exposures to these contracts by entering into offsetting transactions, or by entering into contracts that hedge the market risk related to the derivative guarantees. In addition to derivative contracts that meet the characteristics of a guarantee, the Firm is both a purchaser and seller of credit protection in the credit derivatives market. For a further discussion of credit derivatives, see Note 6 on pages 202–210 of this Annual Report.

Unsettled reverse repurchase and securities borrowing agreements

In the normal course of business, the Firm enters into reverse repurchase agreements and securities borrowing agreements that settle at a future date. At settlement, these commitments require that the Firm advance cash to and accept securities from the counterparty. These agreements generally do not meet the definition of a derivative, and therefore, are not recorded on the Consolidated Balance Sheets until settlement date. At December 31, 2011 and 2010, the amount of commitments related to forward starting reverse repurchase agreements and securities borrowing agreements were \$14.4 billion and \$14.4 billion, respectively. Commitments related to unsettled reverse repurchase agreements and securities borrowing agreements with regular way settlement periods were \$25.5 billion and \$25.5 billion at December 31, 2011 and 2010, respectively.

Loan sales- and securitization-related indemnifications Mortgage repurchase liability

In connection with the Firm's loan sale and securitization activities with the GSEs and other loan sale and private-label securitization transactions, as described in Note 16 on pages 256–267 of this Annual Report, the Firm has made representations and warranties that the loans sold meet certain requirements. The Firm may be, and has been, required to repurchase loans and/or indemnify the GSEs and other investors for losses due to material breaches of these representations and warranties. Although there have been both generalized allegations, as well as specific demands that the Firm should repurchase loans sold or deposited into private-label securitizations, and the Firm experienced an increase in the number of requests for loan files ("file requests") in the latter part of 2011, loan-level repurchase demands and repurchases from private-label securitizations have been limited to date. Generally, the maximum amount of future payments the Firm would be required to make for breaches of these representations and warranties would be equal to the unpaid principal balance of such loans that are deemed to have defects that were

sold to purchasers (including securitization-related SPEs) plus, in certain circumstances, accrued and unpaid interest on such loans and certain expense.

Subsequent to the Firm's acquisition of certain assets and liabilities of Washington Mutual from the FDIC in September 2008, the Firm resolved and/or limited certain current and future repurchase demands for loans sold to the GSEs by Washington Mutual, although it remains the Firm's position that such obligations remain with the FDIC receivership. The Firm will continue to evaluate and may pay (subject to reserving its rights for indemnification by the FDIC) certain future repurchase demands related to individual loans, subject to certain limitations, and has considered such potential repurchase demands in its repurchase liability.

To estimate the Firm's mortgage repurchase liability arising from breaches of representations and warranties, the Firm considers:

- (i) the level of outstanding unresolved repurchase demands,
- (ii) estimated probable future repurchase demands considering information about file requests, delinquent and liquidated loans, resolved and unresolved mortgage insurance rescission notices and the Firm's historical experience,
- (iii) the potential ability of the Firm to cure the defects identified in the repurchase demands ("cure rate"),
- (iv) the estimated severity of loss upon repurchase of the loan or collateral, make-whole settlement, or indemnification,
- (v) the Firm's potential ability to recover its losses from third-party originators, and
- (vi) the terms of agreements with certain mortgage insurers and other parties.

Based on these factors, the Firm has recognized a mortgage repurchase liability of \$3.6 billion and \$3.3 billion, as of December 31, 2011 and 2010, respectively, which is reported in accounts payable and other liabilities net of probable recoveries from third-party correspondents of \$577 million and \$517 million at December 31, 2011 and 2010, respectively.

Substantially all of the estimates and assumptions underlying the Firm's established methodology for computing its recorded mortgage repurchase liability — including factors such as the amount of probable future demands from purchasers, trustees or investors, the ability of the Firm to cure identified defects, the severity of loss upon repurchase or foreclosure, and recoveries from third parties — require application of a significant level of management judgment. Estimating the mortgage repurchase liability is further complicated by historical data and uncertainty surrounding numerous external factors, including: (i) macro-economic factors and (ii) the level of future demands, which is dependent, in part, on actions taken by third parties such as the GSEs, mortgage insurers, trustees and investors.

While the Firm uses the best information available to it in estimating its mortgage repurchase liability, the estimation process is inherently uncertain and imprecise and, accordingly, losses in excess of the amounts accrued as of December 31, 2011, are reasonably possible. The Firm believes the estimate of the range of reasonably possible losses, in excess of its established repurchase liability, is from \$0 to approximately \$2 billion at December 31, 2011. This estimated range of reasonably possible loss considers the Firm's GSE-related exposure based on an assumed peak to trough decline in home prices of 44%, which is an additional 9 percentage point decline in home prices beyond the Firm's current assumptions which were derived from a nationally recognized home price index. Although the Firm does not consider a further decline in home prices of this magnitude likely to occur, such a decline could increase the level of loan delinquencies, thereby potentially increasing the repurchase demand rate from the GSEs and increasing loss severity on repurchased loans, each of which could affect the Firm's mortgage repurchase liability. Claims related to private-label securitizations have, thus far, generally manifested themselves through threatened or pending litigation, which the Firm has considered with other litigation matters as discussed in Note 31 on pages 290–299 of this Annual Report. Actual repurchase losses could vary significantly from the Firm's recorded mortgage repurchase liability or this estimate of reasonably possible additional losses, depending on the outcome of various factors, including those considered above.

The following table summarizes the change in the mortgage repurchase liability for each of the periods presented.

Summary of changes in mortgage repurchase liability

Year ended December 31, (in millions)	2011	2010	2009
Repurchase liability at beginning of period	\$ 3,285	\$ 1,705	\$ 1,093
Realized losses	(1,263)	(1,423)	(1,253) ^(d)
Provision for repurchase losses	1,535	3,003	1,865
Repurchase liability at end of period	\$ 3,557 ^(c)	\$ 3,285	\$ 1,705

- (a) Mortgage repurchase liabilities associated with pending or threatened litigation are not reported in this table because the Firm separately evaluates its exposure to such repurchases in establishing its litigation reserves.
- (b) Includes principal losses and accrued interest on repurchased loans, "make-whole" settlements, settlements with claimants, and certain related expense. For the years ended December 31, 2011, 2010 and 2009, make-whole settlements were and \$640 million, \$632 million and \$277 million, respectively.
- (c) Includes \$173 million at December 31, 2011, related to future demands on loans sold by Washington Mutual to the GSEs.
- (d) Includes the Firm's resolution of certain current and future repurchase demands for certain loans sold by Washington Mutual.

Loans sold with recourse

The Firm provides servicing for mortgages and certain commercial lending products on both a recourse and nonrecourse basis. In nonrecourse servicing, the principal credit risk to the Firm is the cost of temporary servicing

advances of funds (i.e., normal servicing advances). In recourse servicing, the servicer agrees to share credit risk with the owner of the mortgage loans, such as Fannie Mae or Freddie Mac or a private investor, insurer or guarantor. Losses on recourse servicing predominantly occur when foreclosure sales proceeds of the property underlying a defaulted loan are less than the sum of the outstanding principal balance, plus accrued interest on the loan and the cost of holding and disposing of the underlying property. The Firm's securitizations are predominantly nonrecourse, thereby effectively transferring the risk of future credit losses to the purchaser of the mortgage-backed securities issued by the trust. At December 31, 2011 and 2010, the unpaid principal balance of loans sold with recourse totaled \$10.4 billion and \$11.0 billion, respectively. The carrying value of the related liability that the Firm has recorded, which is representative of the Firm's view of the likelihood it will have to perform under its recourse obligations, was \$148 million and \$153 million at December 31, 2011 and 2010, respectively.

Other off-balance sheet arrangements

Indemnification agreements – general

In connection with issuing securities to investors, the Firm may enter into contractual arrangements with third parties that require the Firm to make a payment to them in the event of a change in tax law or an adverse interpretation of tax law. In certain cases, the contract also may include a termination clause, which would allow the Firm to settle the contract at its fair value in lieu of making a payment under the indemnification clause. The Firm may also enter into indemnification clauses in connection with the licensing of software to clients ("software licensees") or when it sells a business or assets to a third party ("third-party purchasers"), pursuant to which it indemnifies software licensees for claims of liability or damages that may occur subsequent to the licensing of the software, or third-party purchasers for losses they may incur due to actions taken by the Firm prior to the sale of the business or assets. It is difficult to estimate the Firm's maximum exposure under these indemnification arrangements, since this would require an assessment of future changes in tax law and future claims that may be made against the Firm that have not yet occurred. However, based on historical experience, management expects the risk of loss to be remote.

Credit card charge-backs

Chase Paymentech Solutions, Card's merchant services business and a subsidiary of JPMorgan Chase Bank, N.A., is a global leader in payment processing and merchant acquiring.

Under the rules of Visa USA, Inc., and MasterCard International, JPMorgan Chase Bank, N.A., is liable primarily for the amount of each processed credit card sales transaction that is the subject of a dispute between a cardmember and a merchant. If a dispute is resolved in the cardmember's favor, Chase Paymentech will (through the cardmember's issuing bank) credit or refund the amount to the cardmember and will charge back the transaction to the

merchant. If Chase Paymentech is unable to collect the amount from the merchant, Chase Paymentech will bear the loss for the amount credited or refunded to the cardmember. Chase Paymentech mitigates this risk by withholding future settlements, retaining cash reserve accounts or by obtaining other security. However, in the unlikely event that: (1) a merchant ceases operations and is unable to deliver products, services or a refund; (2) Chase Paymentech does not have sufficient collateral from the merchant to provide customer refunds; and (3) Chase Paymentech does not have sufficient financial resources to provide customer refunds, JPMorgan Chase Bank, N.A., would be liable for the amount of the transaction. For the year ended December 31, 2011, Chase Paymentech incurred aggregate credit losses of \$13 million on \$553.7 billion of aggregate volume processed, and at December 31, 2011, it held \$204 million of collateral. For the year ended December 31, 2010, Chase Paymentech incurred aggregate credit losses of \$12 million on \$469.3 billion of aggregate volume processed, and at December 31, 2010, it held \$189 million of collateral. For the year ended December 31, 2009, Chase Paymentech incurred aggregate credit losses of \$11 million on \$409.7 billion of aggregate volume processed, and at December 31, 2009, it held \$213 million of collateral. The Firm believes that, based on historical experience and the collateral held by Chase Paymentech, the fair value of the Firm's charge back-related obligations, which are representative of the payment or performance risk to the Firm, is immaterial.

Exchange and clearinghouse guarantees

The Firm is a member of several securities and futures exchanges and clearinghouses, both in the U.S. and other countries. Membership in some of these organizations requires the Firm to pay a pro rata share of the losses incurred by the organization as a result of the default of another member. Such obligations vary with different organizations. These obligations may be limited to members who dealt with the defaulting member or to the amount (or a multiple of the amount) of the Firm's contribution to a member's guarantee fund, or, in a few cases, the obligation may be unlimited. It is difficult to estimate the Firm's maximum exposure under these membership agreements, since this would require an assessment of future claims that may be made against the Firm that have not yet occurred. However, based on historical experience, management expects the risk of loss to be remote.

The Firm clears transactions on behalf of its clients through various clearinghouses, and the Firm stands behind the performance of its clients on such trades. The Firm mitigates its exposure to loss in the event of a client default by requiring that clients provide appropriate amounts of margin at the inception and throughout the life of the transaction, and can cease the provision of clearing services if clients do not adhere to their obligations under the clearing agreement. It is difficult to estimate the Firm's maximum exposure under such transactions, as this would

require an assessment of transactions that clients may execute in the future. However, based upon historical experience, management believes it is unlikely that the Firm will have to make any material payments under these arrangements and the risk of loss is expected to be remote.

Guarantees of subsidiaries

In the normal course of business, JPMorgan Chase & Co. ("Parent Company") may provide counterparties with guarantees of certain of the trading and other obligations of its subsidiaries on a contract-by-contract basis, as negotiated with the Firm's counterparties. The obligations of the subsidiaries are included on the Firm's Consolidated Balance Sheets, or are reflected as off-balance sheet commitments; therefore, the Parent Company has not recognized a separate liability for these guarantees. The Firm believes that the occurrence of any event that would trigger payments by the Parent Company under these guarantees is remote.

The Parent Company has guaranteed certain debt of its subsidiaries, including both long-term debt and structured notes sold as part of the Firm's market-making activities. These guarantees are not included in the table on page 284 of this Note. For additional information, see Note 21 on pages 273–275 of this Annual Report.

Note 30 – Commitments, pledged assets and collateral

Lease commitments

At December 31, 2011, JPMorgan Chase and its subsidiaries were obligated under a number of noncancelable operating leases for premises and equipment used primarily for banking purposes, and for energy-related tolling service agreements. Certain leases contain renewal options or escalation clauses providing for increased rental payments based on maintenance, utility and tax increases, or they require the Firm to perform restoration work on leased premises. No lease agreement imposes restrictions on the Firm's ability to pay dividends, engage in debt or equity financing transactions or enter into further lease agreements.

The following table presents required future minimum rental payments under operating leases with noncancelable lease terms that expire after December 31, 2011.

Year ended December 31, (in millions)	
2012	\$ 1,753
2013	1,758
2014	1,577
2015	1,438
2016	1,300
After 2016	7,188
Total minimum payments required ^(a)	15,014
Less: Sublease rentals under noncancelable subleases	(1,542)
Net minimum payment required	\$ 13,472

(a) Lease restoration obligations are accrued in accordance with U.S. GAAP, and are not reported as a required minimum lease payment.

Total rental expense was as follows.

Year ended December 31, (in millions)	2011	2010	2009
Gross rental expense	\$ 2,228	\$ 2,212	\$ 1,884
Sublease rental income	(403)	(545)	(172)
Net rental expense	\$ 1,825	\$ 1,667	\$ 1,712

Pledged assets

At December 31, 2011, assets were pledged to collateralize repurchase agreements, other securities financing agreements, derivative transactions and for other purposes, including to secure borrowings and public deposits. Certain of these pledged assets may be sold or repledged by the secured parties and are identified as financial instruments owned (pledged to various parties) on the Consolidated Balance Sheets. In addition, at December 31, 2011 and 2010, the Firm had pledged \$270.3 billion and \$288.7 billion, respectively, of financial instruments it owns that may not be sold or repledged by the secured parties. Total assets pledged do not include assets of consolidated VIEs; these assets are used to settle the liabilities of those entities. The significant components of the Firm's pledged assets were as follows.

December 31, (in billions)	2011	2010
Securities	\$ 134.8	\$ 112.1
Loans	198.6	214.8
Trading assets and other	122.8	123.2
Total assets pledged ^(a)	\$ 456.2	\$ 450.1

(a) Total assets pledged do not include assets of consolidated VIEs; these assets are used to settle the liabilities of those entities. See Note 16 on pages 256–267 of this Annual Report for additional information on assets and liabilities of consolidated VIEs.

Collateral

At December 31, 2011 and 2010, the Firm had accepted assets as collateral that it could sell or repledge, deliver or otherwise use with a fair value of approximately \$742.1 billion and \$655.0 billion, respectively. This collateral was generally obtained under resale agreements, securities borrowing agreements, customer margin loans and derivative agreements. Of the collateral received, approximately \$515.8 billion and \$521.3 billion, respectively, were sold or repledged, generally as collateral under repurchase agreements, securities lending agreements or to cover short sales and to collateralize deposits and derivative agreements.

Note 31 – Litigation

Contingencies

As of December 31, 2011, the Firm and its subsidiaries are defendants or putative defendants in numerous legal proceedings, including private, civil litigations and regulatory/government investigations. The litigations range from individual actions involving a single plaintiff to class action lawsuits with potentially millions of class members. Investigations involve both formal and informal proceedings, by both governmental agencies and self-regulatory organizations. These legal proceedings are at varying stages of adjudication, arbitration or investigation, and involve each of the Firm's lines of business and geographies and a wide variety of claims (including common law tort and contract claims and statutory antitrust, securities and consumer protection claims), some of which present novel legal theories.

The Firm believes the estimate of the aggregate range of reasonably possible losses, in excess of reserves established, for its legal proceedings is from \$0 to approximately \$5.1 billion at December 31, 2011. This estimated aggregate range of reasonably possible losses is based upon currently available information for those proceedings in which the Firm is involved, taking into account the Firm's best estimate of such losses for those cases for which such estimate can be made. For certain cases, the Firm does not believe that an estimate can currently be made. The Firm's estimate involves significant judgment, given the varying stages of the proceedings (including the fact that many are currently in preliminary stages), the existence in many such proceedings of multiple defendants (including the Firm) whose share of liability has yet to be determined, the numerous yet-unresolved issues in many of the proceedings (including issues regarding class certification and the scope of many of the claims) and the attendant uncertainty of the various potential outcomes of such proceedings. Accordingly, the Firm's estimate will change from time to time, and actual losses may be more than the current estimate.

Set forth below are descriptions of the Firm's material legal proceedings.

Auction-Rate Securities Investigations and Litigation. Beginning in March 2008, several regulatory authorities initiated investigations of a number of industry participants, including the Firm, concerning possible state and federal securities law violations in connection with the sale of auction-rate securities. The market for many such securities had frozen and a significant number of auctions for those securities began to fail in February 2008.

The Firm, on behalf of itself and affiliates, agreed to a settlement in principle with the New York Attorney General's Office which provided, among other things, that the Firm would offer to purchase at par certain auction-rate securities purchased from J.P. Morgan Securities LLC, Chase Investment Services Corp. and Bear, Stearns & Co. Inc. by individual investors, charities and small- to medium-sized businesses. The Firm also agreed to a substantively similar

settlement in principle with the Office of Financial Regulation for the State of Florida and the North American Securities Administrators Association ("NASAA") Task Force, which agreed to recommend approval of the settlement to all remaining states, Puerto Rico and the U.S. Virgin Islands. The Firm has finalized the settlement agreements with the New York Attorney General's Office and the Office of Financial Regulation for the State of Florida. The settlement agreements provide for the payment of penalties totaling \$25 million to all states. The Firm is currently in the process of finalizing consent agreements with NASAA's member states; more than 45 of these consent agreements have been finalized to date.

The Firm also faces a number of civil actions relating to the Firm's sales of auction-rate securities, including a putative securities class action in the United States District Court for the Southern District of New York that seeks unspecified damages, and individual arbitrations and lawsuits in various forums brought by institutional and individual investors that, together, seek damages totaling approximately \$50 million. The actions generally allege that the Firm and other firms manipulated the market for auction-rate securities by placing bids at auctions that affected these securities' clearing rates or otherwise supported the auctions without properly disclosing these activities. Some actions also allege that the Firm misrepresented that auction-rate securities were short-term instruments. The lawsuits are being coordinated before the federal District Court in New York.

Additionally, the Firm was named in two putative antitrust class actions. The actions allege that the Firm, along with numerous other financial institution defendants, colluded to maintain and stabilize the auction-rate securities market and then to withdraw their support for the auction-rate securities market. In January 2010, the District Court dismissed both actions. An appeal is pending in the United States Court of Appeals for the Second Circuit.

Bear Stearns Hedge Fund Matter. The Bear Stearns Companies LLC (formerly The Bear Stearns Companies Inc.) ("Bear Stearns"), certain current or former subsidiaries of Bear Stearns, including Bear Stearns Asset Management, Inc. ("BSAM") and Bear, Stearns & Co. Inc., and certain individuals formerly employed by Bear Stearns are named defendants (collectively the "Bear Stearns defendants") in multiple civil actions and arbitrations relating to alleged losses resulting from the failure of the Bear Stearns High Grade Structured Credit Strategies Master Fund, Ltd. (the "High Grade Fund") and the Bear Stearns High Grade Structured Credit Strategies Enhanced Leverage Master Fund, Ltd. (the "Enhanced Leverage Fund") (collectively, the "Funds"). BSAM served as investment manager for both of the Funds, which were organized such that there were U.S. and Cayman Islands "feeder funds" that invested substantially all their assets, directly or indirectly, in the Funds. The Funds are in liquidation.

There are currently three civil actions pending in the United States District Court for the Southern District of New York

relating to the Funds. One of these actions involves a derivative lawsuit brought on behalf of purchasers of partnership interests in the U.S. feeder fund to the Enhanced Leverage Fund, alleging that the Bear Stearns defendants mismanaged the Funds. This action seeks, among other things, unspecified compensatory damages based on alleged investor losses. The parties have reached an agreement to settle this derivative action, pursuant to which BSAM would pay a maximum of approximately \$18 million. BSAM has reserved the right not to proceed with this settlement if plaintiff is unable to secure the participation of investors whose net contributions meet a prescribed percentage of the aggregate net contributions to this feeder fund. The court has preliminarily approved the settlement, which remains subject to final court approval. (A separate derivative action, also alleging that the Bear Stearns defendants mismanaged the Funds, was brought on behalf of purchasers of partnership interests in the U.S. feeder fund to the High Grade Fund, and was dismissed following a Court-approved settlement with similar terms, pursuant to which BSAM paid approximately \$19 million). The second pending action, brought by the Joint Voluntary Liquidators of the Cayman Islands feeder funds, makes allegations similar to those asserted in the derivative lawsuits related to the U.S. feeder funds, alleges net losses of approximately \$700 million and seeks compensatory and punitive damages. The parties presently are engaged in discovery.

The third action was brought by Bank of America and Banc of America Securities LLC (together "BoFA") alleging breach of contract and fraud in connection with a \$4 billion securitization in May 2007 known as a "CDO-squared," for which BSAM served as collateral manager. This securitization was composed of certain collateralized debt obligation holdings that were purchased by BoFA from the Funds. BoFA alleges that it incurred losses in excess of \$3 billion and seeks damages in an amount to be determined, although the amount of damages that BoFA seeks may be substantially less than its alleged losses. Discovery is ongoing.

Bear Stearns Shareholder Litigation and Related Matters. Various shareholders of Bear Stearns have commenced purported class actions against Bear Stearns and certain of its former officers and/or directors on behalf of all persons who purchased or otherwise acquired common stock of Bear Stearns between December 14, 2006, and March 14, 2008 (the "Class Period"). During the Class Period, Bear Stearns had between 115 million and 120 million common shares outstanding, and the price per share of those securities declined from a high of \$172.61 to a low of \$30 at the end of the period. The actions, originally commenced in several federal courts, allege that the defendants issued materially false and misleading statements regarding Bear Stearns' business and financial results and that, as a result of those false statements, Bear Stearns' common stock traded at artificially inflated prices during the Class Period. In addition, several individual shareholders of Bear Stearns have also commenced or threatened to commence their

own arbitration proceedings and lawsuits asserting claims similar to those in the putative class actions. Certain of these matters have been dismissed or settled.

Separately, an agreement in principle has been reached to resolve a class action brought under the Employee Retirement Income Security Act ("ERISA") against Bear Stearns and certain of its former officers and/or directors on behalf of participants in the Bear Stearns Employee Stock Ownership Plan for alleged breaches of fiduciary duties in connection with the management of that Plan. Under the settlement, which remains subject to final documentation and court approval, the class will receive \$10 million.

Bear Stearns, former members of Bear Stearns' Board of Directors and certain of Bear Stearns' former executive officers have also been named as defendants in a shareholder derivative and class action suit which is pending in the United States District Court for the Southern District of New York. Plaintiffs assert claims for breach of fiduciary duty, violations of federal securities laws, waste of corporate assets and gross mismanagement, unjust enrichment, abuse of control and indemnification and contribution in connection with the losses sustained by Bear Stearns as a result of its purchases of subprime loans and certain repurchases of its own common stock. Certain individual defendants are also alleged to have sold their holdings of Bear Stearns common stock while in possession of material nonpublic information. Plaintiffs seek compensatory damages in an unspecified amount. The District Court dismissed the action, and plaintiffs have appealed.

City of Milan Litigation and Criminal Investigation. In January 2009, the City of Milan, Italy (the "City") issued civil proceedings against (among others) JPMorgan Chase Bank, N.A. and J.P. Morgan Securities Ltd. (together, "JPMorgan Chase") in the District Court of Milan. The proceedings relate to (a) a bond issue by the City in June 2005 (the "Bond"), and (b) an associated swap transaction, which was subsequently restructured on a number of occasions between 2005 and 2007 (the "Swap"). The City seeks damages and/or other remedies against JPMorgan Chase (among others) on the grounds of alleged "fraudulent and deceitful acts" and alleged breach of advisory obligations in connection with the Swap and the Bond, together with related swap transactions with other counterparties. The civil proceedings have been stayed pending the determination of an application by JPMorgan Chase to the Supreme Court in Rome challenging jurisdiction, which was heard in November 2011.

In March 2010, a criminal judge directed four current and former JPMorgan Chase personnel and JPMorgan Chase Bank, N.A. (as well as other individuals and three other banks) to go forward to a full trial that started in May 2010. Although the Firm is not charged with any crime and does not face criminal liability, if one or more of its employees were found guilty, the Firm could be subject to administrative sanctions, including restrictions on its ability

to conduct business in Italy and monetary penalties. Hearings have continued on a weekly basis since May 2010.

Enron LitigationJPMorgan Chase and certain of its officers and directors are involved in several lawsuits seeking damages arising out of the Firm's banking relationships with Enron Corp. and its subsidiaries ("Enron"). A number of actions and other proceedings against the Firm previously were resolved, including a class action lawsuit captioned *Newby v. Enron Corp.* and adversary proceedings brought by Enron's bankruptcy estate. The remaining Enron-related actions include an individual action by an Enron investor, an action by an Enron counterparty and a purported class action filed on behalf of JPMorgan Chase employees who participated in the Firm's 401(k) plan asserting claims under ERISA for alleged breaches of fiduciary duties by JPMorgan Chase, its directors and named officers. The class action has been dismissed, and is on appeal to the United States Court of Appeals for the Second Circuit. Motions to dismiss are pending in the other two actions.

Interchange LitigationA group of merchants has filed a series of putative class action complaints in several federal courts. The complaints allege that Visa and MasterCard, as well as certain other banks and their respective bank holding companies, conspired to set the price of credit and debit card interchange fees, enacted respective association rules in violation of antitrust laws, and engaged in tying/bundling and exclusive dealing. The complaint seeks unspecified damages and injunctive relief based on the theory that interchange fees would be lower or eliminated but for the challenged conduct. Based on publicly available estimates, Visa and MasterCard branded payment cards generated approximately \$40 billion of interchange fees industry-wide in 2010. All cases have been consolidated in the United States District Court for the Eastern District of New York for pretrial proceedings. The Court has dismissed all claims relating to periods prior to January 2004. The Court has not yet ruled on motions relating to the remainder of the case or plaintiffs' class certification motion. Fact and expert discovery have closed.

In addition to the consolidated class action complaint, plaintiffs filed supplemental complaints challenging the initial public offerings ("IPOs") of MasterCard and Visa (the "IPO Complaints"). With respect to the MasterCard IPO, plaintiffs allege that the offering violated Section 7 of the Clayton Act and Section 1 of the Sherman Act and that the offering was a fraudulent conveyance. With respect to the Visa IPO, plaintiffs are challenging the Visa IPO on antitrust theories parallel to those articulated in the MasterCard IPO pleading. Defendants have filed motions to dismiss the IPO Complaints. The Court has not yet ruled on those motions.

The parties also have filed motions seeking summary judgment as to various claims in the complaints. Oral argument on these summary judgment motions was heard in November 2011.

Investment Management LitigationFour cases have been filed claiming that investment portfolios managed by J.P. Morgan Investment Management Inc. ("JPMorgan

Investment Management") were inappropriately invested in securities backed by subprime residential real estate collateral. Plaintiffs claim that JPMorgan Investment Management and related defendants are liable for losses of more than \$1 billion in market value of these securities. The first case was filed by NM Homes One, Inc. in federal District Court in New York. Following rulings on motions addressed to the pleadings, plaintiff's claims for breach of contract, breach of fiduciary duty, negligence and gross negligence survive, and discovery is proceeding. In the second case, filed by Assured Guaranty (U.K.) in New York state court, discovery is proceeding on plaintiff's claims for breach of contract, breach of fiduciary duty and gross negligence. In the third case, filed by Ambac Assurance UK Limited in New York state court, the lower court granted JPMorgan Investment Management's motion to dismiss. The New York State Appellate Division reversed the lower court's decision and discovery is proceeding. The fourth case, filed by CMMF LLP in New York state court, asserts claims under New York law for breach of fiduciary duty, gross negligence, breach of contract and negligent misrepresentation. The lower court denied in part defendants' motion to dismiss and discovery is proceeding.

Lehman Brothers Bankruptcy ProceedingsIn May 2010, Lehman Brothers Holdings Inc. ("LBHI") and its Official Committee of Unsecured Creditors (the "Committee") filed a complaint (and later an amended complaint) against JPMorgan Chase Bank, N.A. in the United States Bankruptcy Court for the Southern District of New York that asserts both federal bankruptcy law and state common law claims, and seeks, among other relief, to recover \$8.6 billion in collateral that was transferred to JPMorgan Chase Bank, N.A. in the weeks preceding LBHI's bankruptcy. The amended complaint also seeks unspecified damages on the grounds that JPMorgan Chase Bank, N.A.'s collateral requests hastened LBHI's demise. The Firm has moved to dismiss plaintiffs' amended complaint in its entirety, and has also moved to transfer the litigation from the Bankruptcy Court to the United States District Court for the Southern District of New York. Neither motion has yet been decided, but following argument on the motion to transfer the litigation, the District Court directed the Bankruptcy Court to decide the motion to dismiss while the District Court is considering the transfer motion. The Firm also filed counterclaims against LBHI alleging that LBHI fraudulently induced the Firm to make large clearing advances to Lehman against inappropriate collateral, which left the Firm with more than \$25 billion in claims (the "Clearing Claims") against the estate of Lehman Brothers Inc. ("LBI"), LBHI's broker-dealer subsidiary. These claims have been paid in full, subject to the outcome of the litigation. Discovery is underway with a trial scheduled for 2012. In August 2011, LBHI and the Committee filed an objection to the deficiency claims asserted by JPMorgan Chase Bank, N.A. against LBHI with respect to the Clearing Claims, principally on the grounds that the Firm had not conducted the sale of the securities collateral held for such claims in a commercially reasonable manner. The Firm has received and is in various

stages of responding to regulatory investigations regarding Lehman.

LIBOR Investigations and LitigationJPMorgan Chase has received various subpoenas and requests for documents and, in some cases, interviews, from the United States Department of Justice, United States Commodity Futures Trading Commission, United States Securities and Exchange Commission, European Commission, United Kingdom Financial Services Authority, Canadian Competition Bureau and Swiss Competition Commission. The documents and information sought all relate to the process by which rates were submitted to the British Bankers Association ("BBA") in connection with the setting of the BBA's London Interbank Offered Rate ("LIBOR"), principally in 2007 and 2008. The inquiries from some of the regulators also relate to similar processes by which EURIBOR rates are submitted to the European Banking Federation and TIBOR rates are submitted to the Japanese Bankers' Association during similar time periods. The Firm is cooperating with these inquiries.

In addition, the Firm has been named as a defendant along with other banks in a series of individual and class actions filed in various U.S. federal courts alleging that since 2006 the defendants either individually suppressed the LIBOR rate artificially or colluded in submitting rates for LIBOR that were artificially low. Plaintiffs allege that they transacted in U.S. dollar LIBOR-based derivatives or other financial instruments whose values are impacted by changes in U.S. dollar LIBOR, and assert a variety of claims including antitrust claims seeking treble damages. All cases have been consolidated for pre-trial purposes in the United States District Court for the Southern District of New York. In November 2011, the District Court entered an Order appointing interim lead counsel for the two proposed classes: (i) plaintiffs who allegedly purchased U.S. dollar LIBOR-based financial instruments directly from the defendants in the over-the-counter market, and (ii) plaintiffs who allegedly purchased U.S. dollar LIBOR-based financial instruments on an exchange.

Madoff LitigationJPMorgan Chase & Co., JPMorgan Chase Bank, N.A., J.P. Morgan Securities LLC, and J.P. Morgan Securities Ltd. have been named as defendants in a lawsuit brought by the trustee (the "Trustee") for the liquidation of Bernard L. Madoff Investment Securities LLC ("Madoff"). The Trustee has served an amended complaint in which he has asserted 28 causes of action against JPMorgan Chase, 20 of which seek to avoid certain transfers (direct or indirect) made to JPMorgan Chase that are alleged to have been preferential or fraudulent under the federal Bankruptcy Code and the New York Debtor and Creditor Law. The remaining causes of action involve claims for, among other things, aiding and abetting fraud, aiding and abetting breach of fiduciary duty, conversion, contribution and unjust enrichment. The complaint generally alleges that JPMorgan Chase, as Madoff's long-time bank, facilitated the maintenance of Madoff's Ponzi scheme and overlooked signs of wrongdoing in order to obtain profits and fees. The complaint asserts common law claims that purport to seek

approximately \$19 billion in damages, together with bankruptcy law claims to recover approximately \$425 million in transfers that JPMorgan Chase allegedly received directly or indirectly from Bernard Madoff's brokerage firm. By order dated October 31, 2011, the United States District Court for the Southern District of New York granted JPMorgan Chase's motion to dismiss the common law claims asserted by the Trustee, and returned the remaining claims to the Bankruptcy Court for further proceedings. The Trustee has appealed this decision.

Separately, J.P. Morgan Trust Company (Cayman) Limited, JPMorgan (Suisse) SA, J.P. Morgan Securities Ltd., Bear Stearns Alternative Assets International Ltd. and J.P. Morgan Clearing Corp. have been named as defendants in lawsuits presently pending in Bankruptcy Court in New York arising out of the liquidation proceedings of Fairfield Sentry Limited and Fairfield Sigma Limited (together, "Fairfield"), so-called Madoff feeder funds. These actions are based on theories of mistake and restitution and seek to recover payments made to defendants by the funds totaling approximately \$150 million. Pursuant to an agreement with the Trustee, the liquidators of Fairfield have voluntarily dismissed their action against J.P. Morgan Securities Ltd. without prejudice to refile. The other actions remain outstanding. The Bankruptcy Court has stayed these actions. In addition, a purported class action was brought against JPMorgan Chase in the United States District Court for the Southern District of New York, as is a motion by separate potential class plaintiffs to add claims against JPMorgan Chase, JPMorgan Chase Bank, N.A., J.P. Morgan Securities LLC and J.P. Morgan Securities Ltd. to an already-pending purported class action in the same court. The allegations in these complaints largely track those raised by the Trustee. The Court dismissed these complaints and plaintiffs have appealed.

Finally, JPMorgan Chase is a defendant in five actions pending in New York state court and two purported class actions in federal court in New York. The allegations in all of these actions are essentially identical, and involve claims against the Firm for aiding and abetting fraud, aiding and abetting breach of fiduciary duty, conversion and unjust enrichment. In the state court actions, the Firm's motion to dismiss is pending. The Firm has moved to dismiss the state court actions and intends to move to dismiss the federal actions.

The Firm is also responding to various governmental inquiries concerning the Madoff matter.

MF Global. JPMorgan Chase & Co. has been named as one of several defendants in six putative class action lawsuits brought by customers of MF Global in federal district courts in Montana and New York. The actions allege, among other things, that the Firm aided and abetted MF Global's alleged misuse of customer money and breaches of fiduciary duty and was unjustly enriched by the transfer of \$200 million in customer segregated funds by MF Global.

In addition, J.P. Morgan Securities LLC has been named as one of several defendants in a putative class action filed in

federal district court in New York on behalf of purchasers of MF Global's publicly traded securities including the securities issued pursuant to MF Global's February 2011 and August 2011 convertible note offerings. The complaint, which asserts violations of the Securities Act of 1933 against the underwriter defendants, alleges that the offering documents contained materially false and misleading statements and omissions regarding MF Global's financial position, including its exposure to European sovereign debt. The Firm is also responding to various governmental inquiries concerning MF Global.

Mortgage-Backed Securities and Repurchase Litigation and Regulatory Investigations. JPMorgan Chase and affiliates, Bear Stearns and affiliates and Washington Mutual affiliates have been named as defendants in a number of cases in their various roles as issuer or underwriter in MBS offerings. These cases include purported class action suits, actions by individual purchasers of securities or by trustees for the benefit of purchasers of securities, and actions by monoline insurance companies that guaranteed payments of principal and interest for particular tranches of securities offerings. Although the allegations vary by lawsuit, these cases generally allege that the offering documents for securities issued by dozens of securitization trusts contained material misrepresentations and omissions, including with regard to the underwriting standards pursuant to which the underlying mortgage loans were issued, or assert that various representations or warranties relating to the loans were breached at the time of origination. There are currently pending and tolled investor and monoline claims involving approximately \$120 billion of such securities, a number that decreased significantly in the fourth quarter of 2011 largely due to favorable rulings on standing in the class actions discussed below.

In the actions against the Firm as an MBS issuer (and, in some cases, also as an underwriter of its own MBS offerings), three purported class actions are pending against JPMorgan Chase and Bear Stearns, and/or certain of their affiliates and current and former employees, in the United States District Courts for the Eastern and Southern Districts of New York. Defendants moved to dismiss these actions. In the first of these three actions, the court dismissed claims relating to all but one of the offerings. In the second action, the court dismissed claims as to certain offerings and tranches for lack of standing, but allowed claims to proceed relating to some offerings and certificates including ones raised by newly intervening plaintiffs; both parties have sought leave to appeal these rulings. In the third action, the Firm's motion to dismiss remains pending. In a fourth purported class action pending in the United States District Court for the Western District of Washington, Washington Mutual affiliates, WaMu Asset Acceptance Corp. and WaMu Capital Corp., along with certain former officers or directors of WaMu Asset Acceptance Corp., have been named as defendants. The court there denied plaintiffs' motion for leave to amend their complaint to add JPMorgan Chase Bank, N.A., as a defendant on the theory that it is a successor to Washington Mutual Bank. In October 2011, the

court certified a class of plaintiff investors to pursue the claims asserted, but limited those claims to the 13 tranches of MBS in which a named plaintiff purchased. Discovery is proceeding.

In addition to class actions, the Firm is also a defendant in individual actions brought against certain affiliates of JPMorgan Chase, Bear Stearns and Washington Mutual as issuers (and, in some cases, as underwriters). These actions involve claims by governmental agencies, including the Federal Housing Finance Administration, the National Credit Union Administration and the Federal Home Loan Banks of Pittsburgh, Seattle, San Francisco, Chicago, Indianapolis, Atlanta and Boston, as well as by or to benefit various institutional investors, including Cambridge Place Investment Management, various affiliates of the Allstate Corporation, the Charles Schwab Corporation, Massachusetts Mutual Life Insurance Company, Western & Southern Life Insurance Company, HSH Nordbank, IKB International, S.A., Sealink Funding, Ltd., Landesbank Baden-Wuerttemberg, Stichting Pensioenfonds ABP, Bayerische Landesbank, Union Central Life Insurance Company, Capital Ventures International, John Hancock Life Insurance Company and certain affiliates, Dexia SA/NV and certain affiliates, Deutsche Zentral-Genossenschaftsbank and Asset Management Fund and certain affiliates. These actions are pending in federal and state courts across the country and are at various stages of litigation.

EMC Mortgage LLC (formerly EMC Mortgage Corporation) ("EMC"), an indirect subsidiary of JPMorgan Chase & Co., and certain other JPMorgan Chase entities currently are defendants in four pending actions commenced by bond insurers that guaranteed payments of principal and interest on approximately \$3.5 billion of certain classes of six different MBS offerings sponsored by EMC. One of those actions, commenced by Syncora Guarantee, Inc., is pending in the United States District Court for the Southern District of New York against EMC only. Syncora has also filed two actions in New York state court: the first, against J.P. Morgan Securities LLC, asserts tort claims arising out of the same transaction as its federal complaint; the second asserts various tort and contract claims relating to a separate transaction against J.P. Morgan Securities LLC, JPMorgan Chase Bank, N.A. and Bear Stearns Asset-Backed Securities I LLC. Ambac has filed a similar complaint in New York state court relating to four MBS offerings, which alleges various contract and tort claims against EMC, J.P. Morgan Securities LLC and JPMorgan Chase Bank, N.A. These Ambac and Syncora actions seek unspecified damages and specific performance. In December 2011, Assured Guaranty Corp. dismissed its case filed against EMC with respect to one MBS offering that was pending in the United States District Court for the Southern District of New York.

In actions against the Firm solely as an underwriter of other issuers' MBS offerings, the Firm has contractual rights to indemnification from the issuers, but those indemnity rights may prove effectively unenforceable where the issuers are now defunct, such as affiliates of IndyMac Bancorp.

("IndyMac Trusts") and Thornburg Mortgage ("Thornburg"). The Firm may also be contractually obligated to indemnify underwriters in certain deals it issued. With respect to the IndyMac Trusts, J.P. Morgan Securities LLC, along with numerous other underwriters and individuals, is named as a defendant, both in its own capacity and as successor to Bear Stearns, in a purported class action pending in the United States District Court for the Southern District of New York brought on behalf of purchasers of securities in various IndyMac Trust MBS offerings. The court in that action has dismissed claims as to certain such securitizations, including all offerings in which no named plaintiff purchased securities, and allowed claims as to other offerings to proceed. Plaintiffs' motion to certify a class of investors in certain offerings is pending, and discovery is ongoing. In addition, J.P. Morgan Securities LLC and JPMorgan Chase are named as defendants in an individual action filed by the Federal Home Loan Bank of Pittsburgh in connection with a single offering by an affiliate of IndyMac Bancorp. Discovery in that action is ongoing and defendants moved for partial summary judgment in November 2011. Separately, J.P. Morgan Securities LLC, as successor to Bear, Stearns & Co. Inc., along with other underwriters and certain individuals, are defendants in an action pending in state court in California brought by MBIA Insurance Corp. ("MBIA"). The action relates to certain securities issued by IndyMac trusts in offerings in which Bear Stearns was an underwriter, and as to which MBIA provided guaranty insurance policies. MBIA purports to be subrogated to the rights of the MBS holders, and seeks recovery of sums it has paid and will pay pursuant to those policies. Discovery is ongoing. With respect to Thornburg, a Bear Stearns subsidiary is also a named defendant in a purported class action pending in the United States District Court for the District of New Mexico along with a number of other financial institutions that served as depositors and/or underwriters for three Thornburg MBS offerings. The Court granted in part defendants' motion to dismiss but indicated that plaintiffs could replead. Plaintiffs filed another amended complaint in December 2011, while defendants have asked the court to reconsider its ruling denying in part the defendants' motion to dismiss.

The Firm or its affiliates are defendants in three actions brought by trustees of MBS on behalf of the purchasers of securities. In the first, Wells Fargo, as trustee for a single MBS trust, has filed an action against EMC Mortgage in Delaware state court alleging that EMC breached various representations and warranties and seeking the repurchase of more than 800 mortgage loans by EMC and indemnification for the trustee attorneys' fees and costs. In the second, a trustee for a single MBS trust filed a summons with notice in New York state court against EMC, Bear Stearns & Co. Inc. and JPMorgan Chase & Co., seeking damages for breach of contract. The Firm has not yet been served with the complaint. In the third, the Firm is a defendant in an action commenced by Deutsche Bank National Trust Co., acting as trustee for various MBS trusts. That case is described in more detail below with respect to

the Washington Mutual Litigations.

There is no assurance that the Firm will not be named as a defendant in additional MBS-related litigation, and the Firm has entered into agreements with a number of entities that purchased such securities which toll the statutes of limitations and repose with respect to their claims. In addition, the Firm has received several demands by securitization trustees that threaten litigation, as well as demands by investors directing or threatening to direct trustees to investigate claims or bring litigation, based on purported obligations to repurchase loans out of securitization trusts and alleged servicing deficiencies. These include but are not limited to a demand from a law firm, as counsel to a group of certificateholders who purport to have 25% or more of the voting rights in as many as 191 different trusts sponsored by the Firm with an original principal balance of more than \$174 billion (excluding 52 trusts sponsored by Washington Mutual, with an original principal balance of more than \$58 billion), made to various trustees to investigate potential repurchase and servicing claims.

A shareholder complaint has been filed in New York state court against the Firm and two affiliates, members of the boards of directors thereof and certain employees, asserting claims based on alleged wrongful actions and inactions relating to residential mortgage originations and securitizations. The action seeks an accounting and damages. The defendants have moved to dismiss the action.

In addition to the above-described litigation, the Firm has also received, and responded to, a number of subpoenas and informal requests for information from federal and state authorities concerning mortgage-related matters, including inquiries concerning a number of transactions involving the Firm's origination and purchase of whole loans, underwriting and issuance of MBS, treatment of early payment defaults and potential breaches of securitization representations and warranties, and due diligence in connection with securitizations. In January 2012, the Firm was advised by SEC staff that they are considering recommending to the Commission that civil or administrative actions be pursued arising out of two separate investigations they have been conducting. The first involves potential claims against J.P. Morgan Securities LLC relating to due diligence conducted for two mortgage-backed securitizations and corresponding disclosures. The second involves potential claims against Bear Stearns entities, JPMorgan Chase & Co. and J.P. Morgan Securities LLC relating to settlements of claims against originators involving loans included in a number of Bear Stearns securitizations. In both investigations, the SEC staff has invited the Firm to submit responses to the proposed actions.

Mortgage Foreclosure Investigations and Litigation. JPMorgan Chase and four other firms have agreed to a settlement in principle (the "global settlement") with a number of federal and state government agencies, including the U.S. Department of Justice, the U.S.

Department of Housing and Urban Development, the Consumer Financial Protection Bureau and the State Attorneys General, relating to the servicing and origination of mortgages. The global settlement, which is subject to the execution of a definitive agreement and court approval, calls for the Firm to, among other things: (i) make cash payments of approximately \$1.1 billion (a portion of which will be set aside for payments to borrowers); (ii) provide approximately \$500 million of refinancing relief to certain “underwater” borrowers whose loans are owned by the Firm; and (iii) provide approximately \$3.7 billion of additional relief for certain borrowers, including reductions of principal on first and second liens, payments to assist with short sales, deficiency balance waivers on past foreclosures and short sales, and forbearance assistance for unemployed homeowners. (If the Firm does not meet certain targets for provision of the refinancing or other borrower relief within certain prescribed time periods, the Firm will instead make cash payments.) In addition, under the global settlement the Firm will be required to adhere to certain enhanced mortgage servicing standards.

The global settlement releases the Firm from further claims related to servicing activities, including foreclosures and loss mitigation activities; certain origination activities; and certain bankruptcy-related activities. Not included in the global settlement are any claims arising out of securitization activities, including representations made to investors respecting mortgage-backed securities; criminal claims; and repurchase demands from the GSEs, among other items.

The Firm also entered into agreements in principle with the Federal Reserve and the OCC for the payment of civil money penalties related to conduct that was the subject of consent orders entered into with the banking regulators in April 2011. The Firm's payment obligations under those agreements will be deemed satisfied by the Firm's payments and provisions of relief under the global settlement.

The Attorneys General of Massachusetts and New York have separately filed lawsuits against the Firm, other servicers and a mortgage recording company asserting claims for various alleged wrongdoings relating to mortgage assignments and use of the industry's electronic mortgage registry. The Firm has moved to dismiss the Massachusetts action, and has yet to respond to the New York action.

Five purported class action lawsuits were filed against the Firm relating to its mortgage foreclosure procedures. Two of those suits were dismissed with prejudice. A third suit has been resolved, and its dismissal will be obtained shortly. Additionally, the Firm is defending a purported class action brought against Bank of America involving an EMC loan.

A shareholder derivative action has been filed in New York state court against the Firm's board of directors alleging that the board failed to exercise adequate oversight as to wrongful conduct by the Firm regarding mortgage servicing. The action seeks a declaratory judgment and damages.

Municipal Derivatives Investigations and Litigation. Purported class action lawsuits and individual actions (the “Municipal Derivatives Actions”) have been filed against JPMorgan Chase and Bear Stearns, as well as numerous other providers and brokers, alleging antitrust violations in the reportedly \$100 billion to \$300 billion annual market for financial instruments related to municipal bond offerings referred to collectively as “municipal derivatives.” In July 2011, the Firm settled with federal and state governmental agencies to resolve their investigations into similar alleged conduct. The Municipal Derivatives Actions have been consolidated and/or coordinated in the United States District Court for the Southern District of New York. The court denied in part and granted in part defendants' motions to dismiss the purported class and individual actions, permitting certain claims to proceed against the Firm and others under federal and California state antitrust laws and under the California false claims act. Subsequently, a number of additional individual actions asserting substantially similar claims, including claims under New York and West Virginia state antitrust statutes, were filed against JPMorgan Chase, Bear Stearns and numerous other defendants. These cases are also being coordinated for pretrial purposes in the United States District Court for the Southern District of New York. Discovery is ongoing.

In addition, civil actions have been commenced against the Firm relating to certain Jefferson County, Alabama (the “County”) warrant underwritings and swap transactions. In November 2009, J.P. Morgan Securities LLC settled with the SEC to resolve its investigation into those transactions. Following that settlement, the County and a putative class of sewer rate payers filed complaints against the Firm and several other defendants in Alabama state court. The suits allege that the Firm made payments to certain third parties in exchange for being chosen to underwrite more than \$3 billion in warrants issued by the County and to act as the counterparty for certain swaps executed by the County. The complaints also allege that the Firm concealed these third-party payments and that, but for this concealment, the County would not have entered into the transactions. The Court denied the Firm's motions to dismiss the complaints in both proceedings. The Firm filed mandamus petitions with the Alabama Supreme Court, seeking immediate appellate review of these decisions. The mandamus petition in the County's lawsuit was denied in April 2011. In November and December, 2011, the County filed notices of bankruptcy with the trial court in each of the cases and with the Alabama Supreme Court stating that it was a Chapter 9 Debtor in the U.S. Bankruptcy Court for the Northern District of Alabama and providing notice of the automatic stay. Subsequently, the portion of the sewer rate payer action involving claims against the Firm was removed by certain defendants to the United States District Court for the Northern District of Alabama. In its order finding that removal of this action was proper, the District Court referred the action to the District's Bankruptcy Court, where the action remains pending.

Two insurance companies that guaranteed the payment of principal and interest on warrants issued by the County have filed separate actions against the Firm in New York state court. Their complaints assert that the Firm fraudulently misled them into issuing insurance based upon substantially the same alleged conduct described above and other alleged non-disclosures. One insurer claims that it insured an aggregate principal amount of nearly \$1.2 billion and seeks unspecified damages in excess of \$400 million as well as unspecified punitive damages. The other insurer claims that it insured an aggregate principal amount of more than \$378 million and seeks recovery of \$4 million allegedly paid under the policies to date as well as any future payments and unspecified punitive damages. In December 2010, the court denied the Firm's motions to dismiss each of the complaints. The Firm has filed a cross-claim and a third party claim against the County for indemnity and contribution. The County moved to dismiss, which the court denied in August 2011. In consequence of its November 2011 bankruptcy filing, the County has asserted that these actions are stayed.

Overdraft Fee/Debit Posting Order Litigation JPMorgan Chase Bank, N.A. has been named as a defendant in several purported class actions relating to its practices in posting debit card transactions to customers' deposit accounts. Plaintiffs allege that the Firm improperly re-ordered debit card transactions from the highest amount to the lowest amount before processing these transactions in order to generate unwarranted overdraft fees. Plaintiffs contend that the Firm should have processed such transactions in the chronological order they were authorized. Plaintiffs seek the disgorgement of all overdraft fees paid to the Firm by plaintiffs since approximately 2003 as a result of the re-ordering of debit card transactions. The claims against the Firm have been consolidated with numerous complaints against other national banks in multi-District litigation pending in the United States District Court for the Southern District of Florida. The Firm's motion to compel arbitration of certain plaintiffs' claims was initially denied by the District Court. On appeal, the United States Court of Appeals for the Eleventh Circuit vacated the District Court's order and remanded the case for reconsideration in light of a recent ruling by the United States Supreme Court in an unrelated case addressing the enforcement of an arbitration provision in a consumer product agreement. The Firm has reached an agreement in principle to settle this matter in exchange for the Firm paying \$110 million and agreeing to change certain overdraft fee practices. The settlement is subject to documentation and court approval.

Petters Bankruptcy and Related Matters JPMorgan Chase and certain of its affiliates, including One Equity Partners ("OEP"), have been named as defendants in several actions filed in connection with the receivership and bankruptcy proceedings pertaining to Thomas J. Petters and certain affiliated entities (collectively, "Petters") and the Polaroid Corporation. The principal actions against JPMorgan Chase and its affiliates have been brought by a court-appointed receiver for Petters and the trustees in bankruptcy

proceedings for three Petters entities. These actions generally seek to avoid, on fraudulent transfer and preference grounds, certain purported transfers in connection with (i) the 2005 acquisition by Petters of Polaroid, which at the time was majority-owned by OEP; (ii) two credit facilities that JPMorgan Chase and other financial institutions entered into with Polaroid; and (iii) a credit line and investment accounts held by Petters. The actions collectively seek recovery of approximately \$450 million. Defendants have moved to dismiss the complaints in the actions filed by the Petters bankruptcy trustees.

Securities Lending Litigation JPMorgan Chase Bank, N.A. has been named as a defendant in four putative class actions asserting ERISA and other claims pending in the United States District Court for the Southern District of New York brought by participants in the Firm's securities lending business. A fifth lawsuit was filed in New York state court by an individual participant in the program. Three of the purported class actions, which have been consolidated, relate to investments of approximately \$500 million in medium-term notes of Sigma Finance Inc. ("Sigma"). In August 2010, the Court certified a plaintiff class consisting of all securities lending participants that held Sigma medium-term notes on September 30, 2008, including those that held the notes by virtue of participation in the investment of cash collateral through a collective fund, as well as those that held the notes by virtue of the investment of cash collateral through individual accounts. The Court granted JPMorgan Chase's motion for partial summary judgment as to plaintiffs' duty of loyalty claim, finding that the Firm did not have a conflict of interest when it provided repurchase financing to Sigma while also holding Sigma medium-term notes in securities lending accounts. Trial on the remaining duty of prudence claim is scheduled to begin in February 2012. In December 2011, JPMorgan Chase filed third-party claims for indemnification and contribution against the investment fiduciaries for three unnamed class members that maintained individual securities lending accounts. The parties have reached an agreement in principle to settle this action. The settlement is subject to documentation and court approval.

The fourth putative class action concerns investments of approximately \$500 million in Lehman Brothers medium-term notes. The Firm has moved to dismiss the amended complaint and is awaiting a decision. Discovery is proceeding while the motion is pending. The New York state court action, which is not a class action, concerns the plaintiff's alleged loss of money in both Sigma and Lehman Brothers medium-term notes. The Firm has answered the complaint. Discovery is proceeding.

Service Members Civil Relief Act and Housing and Economic Recovery Act Investigations and Litigation Multiple government officials have conducted inquiries into the Firm's procedures related to the Service Members Civil Relief Act ("SCRA") and the Housing and Economic Recovery Act of 2008 ("HERA"). These inquiries were prompted by the Firm's public statements about its SCRA and HERA

compliance and actions to remedy certain instances in which the Firm mistakenly charged active or recently-active military personnel mortgage interest and fees in excess of that permitted by SCRA and HERA, and in a number of instances, foreclosed on borrowers protected by SCRA and HERA. The Firm has implemented a number of procedural enhancements and controls to strengthen its SCRA and HERA compliance. In addition, an individual borrower filed a nationwide class action in United States District Court for South Carolina against the Firm alleging violations of the SCRA related to home loans. The Firm agreed to pay \$27 million plus attorneys' fees, in addition to reimbursements previously paid by the Firm, to settle the class action. Additional borrowers were subsequently added to the class, and the Firm agreed to pay an additional \$8 million into the settlement fund. The court entered a final order approving the settlement in January 2012.

Washington Mutual Litigation. Subsequent to JPMorgan Chase's acquisition from the FDIC of substantially all of the assets and certain specified liabilities of Washington Mutual Bank ("Washington Mutual Bank") in September 2008, Washington Mutual Bank's parent holding company, Washington Mutual, Inc. ("WMI") and its wholly-owned subsidiary, WMI Investment Corp. (together, the "Debtors"), both commenced voluntary cases under Chapter 11 of Title 11 of the United States Code in the United States Bankruptcy Court for the District of Delaware (the "Bankruptcy Case"). In the Bankruptcy Case, the Debtors have asserted rights and interests in certain assets. The assets in dispute include principally the following: (a) approximately \$4 billion in trust securities contributed by WMI to Washington Mutual Bank (the "Trust Securities"); (b) the right to tax refunds arising from overpayments attributable to operations of Washington Mutual Bank and its subsidiaries; (c) ownership of and other rights in approximately \$4 billion that WMI contends are deposit accounts at Washington Mutual Bank and one of its subsidiaries; and (d) ownership of and rights in various other contracts and other assets (collectively, the "Disputed Assets").

WMI, JPMorgan Chase and the FDIC have since been involved in litigations over these and other claims pending in the Bankruptcy Court for the District of Delaware (the "Bankruptcy Court") and the United States District Court for the District of Columbia.

In May 2010, WMI, JPMorgan Chase and the FDIC announced a global settlement agreement among themselves and significant creditor groups (the "WaMu Global Settlement"). The WaMu Global Settlement is incorporated into WMI's Chapter 11 plan ("the Plan") submitted to the Bankruptcy Court. The WaMu Global Settlement resolves numerous disputes among WMI, JPMorgan Chase, the FDIC in its capacity as receiver for Washington Mutual Bank and the FDIC in its corporate capacity, as well as those of significant creditor groups, including disputes relating to the Disputed Assets. After several amendments to the Plan to address deficiencies

identified by the Bankruptcy Court that were unrelated to the WaMu Global Settlement, in February 2012 the Bankruptcy Court confirmed the Plan, including the WaMu Global Settlement.

Other proceedings related to Washington Mutual's failure are also pending before the Bankruptcy Court. Among other actions, in July 2010, certain holders of the Trust Securities commenced an adversary proceeding in the Bankruptcy Court against JPMorgan Chase, WMI, and other entities seeking, among other relief, a declaratory judgment that WMI and JPMorgan Chase do not have any right, title or interest in the Trust Securities. In early January 2011, the Bankruptcy Court granted summary judgment to JPMorgan Chase and denied summary judgment to the plaintiffs in the Trust Securities adversary proceeding. The plaintiffs have appealed that decision to the United States District Court for the District of Delaware. In connection with the current Plan, these plaintiffs filed a motion seeking a stay of further confirmation proceedings pending their appeal from the Bankruptcy Court's determination that they have no interest in the Trust Securities and are instead owners of WMI preferred equity. In January 2012, the Bankruptcy Court denied their motion, and the District Court denied their motions for a stay pending appeal and mandamus relief.

Other proceedings related to Washington Mutual's failure are pending before the United States District Court for the District of Columbia and include a lawsuit brought by Deutsche Bank National Trust Company, initially against the FDIC, asserting an estimated \$6 billion to \$10 billion in damages based upon alleged breach of various mortgage securitization agreements and alleged violation of certain representations and warranties given by certain WMI subsidiaries in connection with those securitization agreements. The case includes assertions that JPMorgan Chase may have assumed liabilities for the alleged breaches of representations and warranties in the mortgage securitization agreements. The District Court denied as premature motions by the Firm and the FDIC that sought a ruling on whether the FDIC retained liability for Deutsche Bank's claims. Discovery is underway.

In addition, JPMorgan Chase was sued in an action originally filed in state court in Texas (the "Texas Action") by certain holders of WMI common stock and debt of WMI and Washington Mutual Bank who seek unspecified damages alleging that JPMorgan Chase acquired substantially all of the assets of Washington Mutual Bank from the FDIC at a price that was allegedly too low. The Texas Action was transferred to the United States District Court for the District of Columbia, which ultimately granted JPMorgan Chase's and the FDIC's motions to dismiss the complaint, but the United States Court of Appeals for the District of Columbia Circuit reversed the trial court's dismissal and remanded the case for further proceedings. Plaintiffs, which now include only holders of Washington Mutual Bank debt following their voluntary dismissal of claims brought as holders of WMI common stock and debt, have filed an amended complaint alleging that JPMorgan Chase caused

the closure of Washington Mutual Bank and damaged them by causing their bonds issued by Washington Mutual Bank to lose substantially all of their value. JPMorgan Chase and the FDIC have again moved to dismiss this action.

* * *

In addition to the various legal proceedings discussed above, JPMorgan Chase and its subsidiaries are named as defendants or are otherwise involved in a substantial number of other legal proceedings. The Firm believes it has meritorious defenses to the claims asserted against it in its currently outstanding legal proceedings and it intends to defend itself vigorously in all such matters. Additional legal proceedings may be initiated from time to time in the future.

The Firm has established reserves for several hundred of its currently outstanding legal proceedings. The Firm accrues for potential liability arising from such proceedings when it is probable that such liability has been incurred and the amount of the loss can be reasonably estimated. The Firm evaluates its outstanding legal proceedings each quarter to assess its litigation reserves, and makes adjustments in such reserves, upwards or downwards, as appropriate, based on management's best judgment after consultation with counsel. During the years ended December 31, 2011, 2010 and 2009, the Firm incurred \$4.9 billion, \$7.4 billion

and \$161 million, respectively, of litigation expense. There is no assurance that the Firm's litigation reserves will not need to be adjusted in the future.

In view of the inherent difficulty of predicting the outcome of legal proceedings, particularly where the claimants seek very large or indeterminate damages, or where the matters present novel legal theories, involve a large number of parties or are in early stages of discovery, the Firm cannot state with confidence what will be the eventual outcomes of the currently pending matters, the timing of their ultimate resolution or the eventual losses, fines, penalties or impact related to those matters. JPMorgan Chase believes, based upon its current knowledge, after consultation with counsel and after taking into account its current litigation reserves, that the legal proceedings currently pending against it should not have a material adverse effect on the Firm's consolidated financial condition. The Firm notes, however, that in light of the uncertainties involved in such proceedings, there is no assurance the ultimate resolution of these matters will not significantly exceed the reserves it has currently accrued; as a result, the outcome of a particular matter may be material to JPMorgan Chase's operating results for a particular period, depending on, among other factors, the size of the loss or liability imposed and the level of JPMorgan Chase's income for that period.

Note 32 – International operations

The following table presents income statement-related and balance sheet-related information for JPMorgan Chase by major international geographic area. The Firm defines international activities for purposes of this footnote presentation as business transactions that involve clients residing outside of the U.S., and the information presented below is based predominantly on the domicile of the client, the location from which the client relationship is managed or the location of the trading desk. However, many of the Firm's U.S. operations serve international businesses.

As the Firm's operations are highly integrated, estimates and subjective assumptions have been made to apportion revenue and expense between U.S. and international operations. These estimates and assumptions are consistent with the allocations used for the Firm's segment reporting as set forth in Note 33 on pages 300–303 of this Annual Report.

The Firm's long-lived assets for the periods presented are not considered by management to be significant in relation to total assets. The majority of the Firm's long-lived assets are located in the United States.

Notes to consolidated financial statements

As of or for the year ended December 31, (in millions)	Revenue ^(c)	Expenses ^(d)	Income before income tax expense and extraordinary gain	Net income	Total assets
2011					
Europe/Middle East and Africa	\$ 16,212	\$ 9,157	\$ 7,055	\$ 4,844	\$ 566,866
Asia and Pacific	5,992	3,802	2,190	1,380	156,411
Latin America and the Caribbean	2,273	1,711	562	340	51,481
Total international	24,477	14,670	9,807	6,564	774,758
North America ^(a)	72,757	55,815	16,942	12,412	1,491,034
Total	\$ 97,234	\$ 70,485	\$ 26,749	\$ 18,976	\$ 2,265,792
2010^(b)					
Europe/Middle East and Africa	\$ 14,135	\$ 8,777	\$ 5,358	\$ 3,635	\$ 446,547
Asia and Pacific	6,073	3,677	2,396	1,614	151,379
Latin America and the Caribbean	1,750	1,181	569	362	33,192
Total international	21,958	13,635	8,323	5,611	631,118
North America ^(a)	80,736	64,200	16,536	11,759	1,486,487
Total	\$ 102,694	\$ 77,835	\$ 24,859	\$ 17,370	\$ 2,117,605
2009^(b)					
Europe/Middle East and Africa	\$ 16,294	\$ 8,620	\$ 7,674	\$ 5,212	\$ 375,406
Asia and Pacific	5,429	3,528	1,901	1,286	112,798
Latin America and the Caribbean	1,867	1,083	784	463	23,692
Total international	23,590	13,231	10,359	6,961	511,896
North America ^(a)	76,844	71,136	5,708	4,767	1,520,093
Total	\$ 100,434	\$ 84,367	\$ 16,067	\$ 11,728	\$ 2,031,989

(a) Substantially reflects the U.S.

(b) The regional allocation of revenue, expense and net income for 2010 and 2009 has been modified to conform with current allocation methodologies.

(c) Revenue is composed of net interest income and noninterest revenue.

(d) Expense is composed of noninterest expense and the provision for credit losses.

Note 33 – Business segments

The Firm is managed on a line of business basis. There are six major reportable business segments – Investment Bank, Retail Financial Services, Card Services & Auto, Commercial Banking, Treasury & Securities Services and Asset Management, as well as a Corporate/Private Equity segment. The business segments are determined based on the products and services provided, or the type of customer served, and they reflect the manner in which financial information is currently evaluated by management. Results of these lines of business are presented on a managed basis. For a definition of managed basis, see Explanation and Reconciliation of the Firm's use of non-GAAP financial measures, on pages 76–78 of this Annual Report. For a further discussion concerning JPMorgan Chase's business segments, see Business Segment Results on pages 79–80 of this Annual Report.

The following is a description of each of the Firm's business segments:

Investment Bank

J.P. Morgan is one of the world's leading investment banks, with deep client relationships and broad product capabilities. The clients of IB are corporations, financial institutions, governments and institutional investors. The Firm offers a full range of investment banking products and services in all major capital markets, including advising on

corporate strategy and structure, capital-raising in equity and debt markets, sophisticated risk management, market-making in cash securities and derivative instruments, prime brokerage, and research.

Retail Financial Services

RFS serves consumers and businesses through personal service at bank branches and through ATMs, online banking and telephone banking. RFS is organized into Consumer & Business Banking and Mortgage Banking (including Mortgage Production and Servicing, and Real Estate Portfolios). Consumer & Business Banking includes branch banking and business banking activities. Mortgage Production and Servicing includes mortgage origination and servicing activities. Real Estate Portfolios comprises residential mortgages and home equity loans, including the PCI portfolio acquired in the Washington Mutual transaction. Customers can use more than 5,500 bank branches (third largest nationally) and more than 17,200 ATMs (second largest nationally), as well as online and mobile banking around the clock. More than 33,500 branch salespeople assist customers with checking and savings accounts, mortgages, home equity and business loans, and investments across the 23-state footprint from New York and Florida to California. As one of the largest mortgage originators in the U.S., Chase helps customers buy or refinance homes resulting in approximately \$150 billion of

mortgage originations annually. Chase also services more than 8 million mortgages and home equity loans.

Card Services & Auto

Card Services & Auto is one of the nation's largest credit card issuers, with over \$132 billion in credit card loans. Customers have over 65 million open credit card accounts (excluding the commercial card portfolio), and used Chase credit cards to meet over \$343 billion of their spending needs in 2011. Through its Merchant Services business, Chase Paymentech Solutions, Card is a global leader in payment processing and merchant acquiring. Consumers also can obtain loans through more than 17,200 auto dealerships and 2,000 schools and universities nationwide.

Commercial Banking

CB delivers extensive industry knowledge, local expertise and dedicated service to more than 24,000 clients nationally, including corporations, municipalities, financial institutions and not-for-profit entities with annual revenue generally ranging from \$10 million to \$2 billion, and nearly 35,000 real estate investors/owners. CB partners with the Firm's other businesses to provide comprehensive solutions, including lending, treasury services, investment banking and asset management, to meet its clients' domestic and international financial needs.

Treasury & Securities Services

TSS is a global leader in transaction, investment and information services. TSS is one of the world's largest cash management providers and a leading global custodian. Treasury Services ("TS") provides cash management, trade, wholesale card and liquidity products and services to small- and mid-sized companies, multinational corporations, financial institutions and government entities. TS partners with IB, CB, RFS and Asset Management businesses to serve clients firmwide. Certain TS revenue is included in other segments' results. Worldwide Securities Services holds, values, clears and services securities, cash and alternative investments for investors and broker-dealers, and manages depositary receipt programs globally.

Asset Management

AM, with assets under supervision of \$1.9 trillion, is a global leader in investment and wealth management. AM clients include institutions, retail investors and high-net-worth individuals in every major market throughout the world. AM offers global investment management in equities, fixed income, real estate, hedge funds, private equity and

liquidity products, including money-market instruments and bank deposits. AM also provides trust and estate, banking and brokerage services to high-net-worth clients, and retirement services for corporations and individuals. The majority of AM's client assets are in actively managed portfolios.

Corporate/Private Equity

The Corporate/Private Equity sector comprises Private Equity, Treasury, the Chief Investment Office, corporate staff units and expense that is centrally managed. Treasury and the Chief Investment Office manage capital, liquidity, and structural risks of the Firm. The corporate staff units include Central Technology and Operations, Internal Audit, Executive Office, Finance, Human Resources, Marketing & Communications, Legal & Compliance, Corporate Real Estate and General Services, Risk Management, Corporate Responsibility and Strategy & Development. Other centrally managed expense includes the Firm's occupancy and pension-related expense, net of allocations to the business.

Business segment changes

Commencing July 1, 2011, the Firm's business segments have been reorganized as follows:

Auto and Student Lending transferred from the RFS segment and are reported with Card in a single segment. Retail Financial Services continues as a segment, organized in two components: Consumer & Business Banking (formerly Retail Banking) and Mortgage Banking (including Mortgage Production and Servicing, and Real Estate Portfolios).

The business segment information associated with RFS and Card have been revised to reflect the business reorganization retroactive to January 1, 2009.

Effective January 1, 2010, the Firm enhanced its line of business equity framework to better align equity assigned to the lines of business with changes anticipated to occur in each line of business, and to reflect the competitive and regulatory landscape. The lines of business are now capitalized based on the Tier 1 common standard, rather than the Tier 1 capital standard. In addition, effective January 1, 2011, capital allocated to Card was reduced, largely reflecting portfolio runoff and the improving risk profile of the business; and capital allocated to TSS was increased, reflecting growth in the underlying business.

Segment results

The following tables provide a summary of the Firm's segment results for 2011, 2010 and 2009 on a managed basis. Prior to the January 1, 2010, adoption of the accounting guidance related to VIEs, the impact of credit card securitization adjustments had been included in reconciling items; as a result, the total Firm results are on a reported basis. Finally, total net revenue (noninterest revenue and net interest income) for each of the segments is presented on a tax-equivalent basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in the managed results on a basis comparable to taxable investments and securities. This non-GAAP financial measure allows management to assess the comparability of revenue arising from both taxable and tax-exempt sources. The corresponding income tax impact related to tax-exempt items is recorded within income tax expense/(benefit).

Notes to consolidated financial statements

Segment results and reconciliation^(a)

As of or the year ended December 31, (in millions, except ratios)	Investment Bank			Retail Financial Services			Card Services & Auto			Commercial Banking		
	2011	2010	2009	2011	2010	2009	2011	2010	2009	2011	2010	2009
Noninterest revenue	\$ 17,971	\$ 18,253	\$ 18,522	\$ 10,405	\$ 11,227	\$ 11,414	\$ 4,892	\$ 4,278	\$ 3,706	\$ 2,195	\$ 2,200	\$ 1,817
Net interest income	8,303	7,964	9,587	16,133	17,220	18,383	14,249	16,194	19,493	4,223	3,840	3,903
Total net revenue	26,274	26,217	28,109	26,538	28,447	29,797	19,141	20,472	23,199	6,418	6,040	5,720
Provision for credit losses	(286)	(1,200)	2,279	3,999	8,919	14,754	3,621	8,570	19,648	208	297	1,454
Credit allocation income/ (expense) ^(b)	—	—	—	—	—	—	—	—	—	—	—	—
Noninterest expense ^(c)	16,116	17,265	15,401	19,458	16,483	15,512	8,045	7,178	6,617	2,278	2,199	2,176
Income/(loss) before income tax expense/ (benefit) and extraordinary gain	10,444	10,152	10,429	3,081	3,045	(469)	7,475	4,724	(3,066)	3,932	3,544	2,090
Income tax expense/ (benefit)	3,655	3,513	3,530	1,403	1,317	(134)	2,931	1,852	(1,273)	1,565	1,460	819
Income/(loss) before extraordinary gain	6,789	6,639	6,899	1,678	1,728	(335)	4,544	2,872	(1,793)	2,367	2,084	1,271
Extraordinary gain ^(d)	—	—	—	—	—	—	—	—	—	—	—	—
Net income/(loss)	\$ 6,789	\$ 6,639	\$ 6,899	\$ 1,678	\$ 1,728	\$ (335)	\$ 4,544	\$ 2,872	\$ (1,793)	\$ 2,367	\$ 2,084	\$ 1,271
Average common equity	\$ 40,000	\$ 40,000	\$ 33,000	\$ 25,000	\$ 24,600	\$ 22,457	\$ 16,000	\$ 18,400	\$ 17,543	\$ 8,000	\$ 8,000	\$ 8,000
Total assets	776,430	825,150	706,944	274,795	299,950	322,185	208,467	208,793	255,029	158,040	142,646	130,280
Return on average common equity ^(e)	17%	17%	21%	7%	7%	(1)%	28%	16%	(10)%	30%	26%	16%
Overhead ratio	61	66	55	73	58	52	42	35	29	35	36	38

- (a) In addition to analyzing the Firm's results on a reported basis, management reviews the Firm's lines of business results on a "managed basis," which is a non-GAAP financial measure. The Firm's definition of managed basis starts with the reported U.S. GAAP results and includes certain reclassifications discussed below that do not have any impact on net income as reported by the lines of business or by the Firm as a whole.
- (b) IB manages traditional credit exposures related to the Global Corporate Bank ("GCB") on behalf of IB and TSS. Effective January 1, 2011, IB and TSS share the economics related to the Firm's GCB clients. Included within this allocation are net revenue, provision for credit losses and expenses. Prior to 2011, years reflected a reimbursement to IB for a portion of the total costs of managing the credit portfolio. IB recognizes this credit allocation as a component of all other income.
- (c) Includes merger costs, which are reported in the Corporate/Private Equity segment. There were no merger costs in 2011 and 2010. Merger costs attributed to the business segments for 2009 was as follows.

Year ended December 31, (in millions)	2009
Investment Bank	\$ 27
Retail Financial Services	228
Card Services & Auto	40
Commercial Banking	6
Treasury & Securities Services	11
Asset Management	6
Corporate/Private Equity	163

- (d) On September 25, 2008, JPMorgan Chase acquired the banking operations of Washington Mutual from the FDIC for \$1.9 billion. The fair value of the assets acquired exceeded the purchase price, which resulted in negative goodwill. In accordance with U.S. GAAP for business combinations, nonfinancial assets that are not held-for-sale, such as premises and equipment and other intangibles, acquired in the Washington Mutual transaction were written down against that negative goodwill. The negative goodwill that remained after writing down nonfinancial assets was recognized as an extraordinary gain. A preliminary gain of \$1.9 billion was recognized at December 31, 2008. As a result of the final refinement of the purchase price allocation in 2009, the Firm recognized a \$76 million increase in the extraordinary gain. The final total extraordinary gain that resulted from the Washington Mutual transaction was \$2.0 billion.
- (e) Ratio is based on income/(loss) before extraordinary gain for 2009.
- (f) Effective January 1, 2010, the Firm adopted accounting guidance related to VIEs. Prior to the adoption of the new guidance, managed results for credit card securitizations excluded the impact of credit card securitizations on total net revenue, provision for credit losses and average assets, as JPMorgan Chase treated the sold receivables as if they were still on the balance sheet in evaluating the credit performance of the entire managed credit card portfolio, as operations are funded, and decisions are made about allocating resources, such as employees and capital, based on managed information. These adjustments are eliminated in reconciling items to arrive at the Firm's reported U.S. GAAP results. The related securitization adjustments were as follows:

Year ended December 31, (in millions)	2009
Noninterest revenue	\$ (1,494)
Net interest income	7,937
Provision for credit losses	6,443
Total assets	80,882

(table continued from previous page)

Treasury & Securities Services			Asset Management			Corporate/Private Equity			Reconciling Items ^(g)			Total		
2011	2010	2009	2011	2010	2009	2011	2010	2009	2011	2010	2009	2011	2010	2009
\$ 4,544	\$ 4,757	\$ 4,747	\$ 7,895	\$ 7,485	\$ 6,372	\$ 3,638	\$ 5,359	\$ 2,771	\$ (1,995)	\$ (1,866)	\$ (67)	\$ 49,545	\$ 51,693	\$ 49,282
3,158	2,624	2,597	1,648	1,499	1,593	505	2,063	3,863	(530)	(403)	(8,267)	47,689	51,001	51,152
7,702	7,381	7,344	9,543	8,984	7,965	4,143	7,422	6,634	(2,525)	(2,269)	(8,334)	97,234	102,694	100,434
1	(47)	55	67	86	188	(36)	14	80	—	—	(6,443)	7,574	16,639	32,015
8	(121)	(121)	—	—	—	—	—	—	(8)	121	121	—	—	—
5,863	5,604	5,278	7,002	6,112	5,473	4,149	6,355	1,895	—	—	—	62,911	61,196	52,352
1,846	1,703	1,890	2,474	2,786	2,304	30	1,053	4,659	(2,533)	(2,148)	(1,770)	26,749	24,859	16,067
642	624	664	882	1,076	874	(772)	(205)	1,705	(2,533)	(2,148)	(1,770)	7,773	7,489	4,415
1,204	1,079	1,226	1,592	1,710	1,430	802	1,258	2,954	—	—	—	18,976	17,370	11,652
—	—	—	—	—	—	—	—	76	—	—	—	—	—	76
\$ 1,204	\$ 1,079	\$ 1,226	\$ 1,592	\$ 1,710	\$ 1,430	\$ 802	\$ 1,258	\$ 3,030	\$ —	\$ —	\$ —	\$ 18,976	\$ 17,370	\$ 11,728
\$ 7,000	\$ 6,500	\$ 5,000	\$ 6,500	\$ 6,500	\$ 7,000	\$ 70,766	\$ 57,520	\$ 52,903	\$ —	\$ —	\$ —	\$ 173,266	\$ 161,520	\$ 145,903
68,665	45,481	38,054	86,242	68,997	64,502	693,153	526,588	595,877	NA	NA	(80,882)	2,265,792	2,117,605	2,031,989
17%	17%	25%	25%	26%	20%	NM	NM	NM	NM	NM	NM	11%	10%	6%
76	76	72	73	68	69	NM	NM	NM	NM	NM	NM	65	60	52

(g) Segment managed results reflect revenue on a tax-equivalent basis with the corresponding income tax impact recorded within income tax expense/ (benefit). These adjustments are eliminated in reconciling items to arrive at the Firm's reported U.S. GAAP results. Tax-equivalent adjustments for the years ended December 31, 2011, 2010 and 2009 were as follows.

Year ended December 31, (in millions)	2011	2010	2009
Noninterest revenue	\$ 2,003	\$ 1,745	\$ 1,440
Net interest income	530	403	330
Income tax expense	2,533	2,148	1,770

Note 34 – Parent company

Parent company – Statements of income			
Year ended December 31, (in millions)	2011	2010	2009
Income			
Dividends from subsidiaries:			
Bank and bank holding company	\$ 10,852	\$ 16,554	\$ 15,235
Nonbank ^(a)	2,651	932	1,036
Interest income from subsidiaries	1,099	985	1,501
Other interest income	384	294	266
Other income from subsidiaries, primarily fees:			
Bank and bank holding company	809	680	233
Nonbank	92	312	742
Other income/(loss)	(85)	157	844
Total income	15,802	19,914	19,857
Expense			
Interest expense to subsidiaries	1,121	1,263	1,118
Other interest expense	4,447	3,782	4,696
Other noninterest expense	649	540	988
Total expense	6,217	5,585	6,802
Income before income tax benefit and undistributed net income of subsidiaries	9,585	14,329	13,055
Income tax benefit	1,089	511	1,269
Equity in undistributed net income of subsidiaries	8,302	2,530	(2,596)
Net income	\$ 18,976	\$ 17,370	\$ 11,728
Parent company – Balance sheets			
December 31, (in millions)	2011	2010	
Assets			
Cash and due from banks	\$ 132	\$ 96	
Deposits with banking subsidiaries	91,622	80,201	
Trading assets	18,485	16,038	
Available-for-sale securities	3,657	3,176	
Loans	1,880	1,849	
Advances to, and receivables from, subsidiaries:			
Bank and bank holding company	39,888	54,887	
Nonbank	83,138	72,080	
Investments (at equity) in subsidiaries:			
Bank and bank holding company	157,160	150,876	
Nonbank ^(a)	42,231	38,000	
Goodwill and other intangibles	1,027	1,050	
Other assets	15,506	17,171	
Total assets	\$ 454,726	\$ 435,424	
Liabilities and stockholders' equity			
Borrowings from, and payables to, subsidiaries ^(b)	\$ 30,231	\$ 28,332	
Other borrowed funds, primarily commercial paper	59,891	41,874	
Other liabilities	7,653	7,302	
Long-term debt ^(c)	173,378	181,810	
Total liabilities^(c)	271,153	259,318	
Total stockholders' equity	183,573	176,106	
Total liabilities and stockholders' equity	\$ 454,726	\$ 435,424	

- (a) Subsidiaries include trusts that issued guaranteed capital debt securities ("issuer trusts"). The Parent received dividends of \$13 million, \$13 million and \$14 million from the issuer trusts in 2011, 2010 and 2009, respectively. For further discussion on these issuer trusts, see Note 21 on pages 273–275 of this Annual Report.
- (b) At December 31, 2011, long-term debt that contractually matures in 2012 through 2016 totaled \$42.5 billion, \$17.4 billion, \$24.9 billion, \$16.7 billion and \$17.5 billion, respectively.
- (c) For information regarding the Firm's guarantees of its subsidiaries' obligations, see Note 21 and Note 29 on pages 273–275 and 283–289, respectively, of this Annual Report.
- (d) Represents the assumption of Bear Stearns long-term debt by JPMorgan Chase & Co.

Parent company – Statements of cash flows

Year ended December 31, (in millions)	2011	2010	2009
Operating activities			
Net income	\$ 18,976	\$ 17,370	\$ 11,728
Less: Net income of subsidiaries	21,805	20,016	13,675
Parent company net loss	(2,829)	(2,646)	(1,947)
Cash dividends from subsidiaries	13,414	17,432	16,054
Other, net	889	1,685	1,852
Net cash provided by operating activities	11,474	16,471	15,959
Investing activities			
Net change in:			
Deposits with banking subsidiaries	20,866	7,692	(27,342)
Available-for-sale securities:			
Purchases	(1,109)	(1,387)	(1,454)
Proceeds from sales and maturities	886	745	522
Loans, net	153	(90)	209
Advances to subsidiaries, net	(28,105)	8,051	28,808
Investments (at equity) in subsidiaries, net	(1,530)	(871)	(6,582)
Net cash (used in)/provided by investing activities	(8,839)	14,140	(5,839)
Financing activities			
Net change in borrowings from subsidiaries ^(d)	2,827	(2,039)	(4,935)
Net change in other borrowed funds	16,268	(11,843)	1,894
Proceeds from the issuance of long-term debt	33,566	21,610	32,304
Proceeds from the assumption of subsidiaries long-term debt	—	—	15,264
Repayments of long-term debt	(41,747)	(32,893)	(31,964)
Excess tax benefits related to stock-based compensation	867	26	17
Redemption of preferred stock issued to the U.S. Treasury	—	—	(25,000)
Redemption of other preferred stock	—	(352)	—
Proceeds from issuance of common stock	—	—	5,756
Treasury stock and warrants repurchased	(8,863)	(2,999)	—
Dividends paid	(3,895)	(1,486)	(3,422)
All other financing activities, net	(1,622)	(641)	33
Net cash used in financing activities	(2,599)	(30,617)	(10,053)
Net increase/(decrease) in cash and due from banks	36	(6)	67
Cash and due from banks at the beginning of the year, primarily with bank subsidiaries	96	102	35
Cash and due from banks at the end of the year, primarily with bank subsidiaries	\$ 132	\$ 96	\$ 102
Cash interest paid	\$ 5,800	\$ 5,090	\$ 5,629
Cash income taxes paid, net	5,885	7,001	3,124

Selected quarterly financial data (unaudited)

(Table continued on next page)

As of or for the period ended (in millions, except per share, ratio and headcount data)	2011				2010			
	4th quarter	3rd quarter	2nd quarter	1st quarter	4th quarter	3rd quarter	2nd quarter	1st quarter
Selected income statement data								
Noninterest revenue	\$ 9,340	\$ 11,946	\$ 14,943	\$ 13,316	\$ 13,996	\$ 11,322	\$ 12,414	\$ 13,961
Net interest income	12,131	11,817	11,836	11,905	12,102	12,502	12,687	13,710
Total net revenue	21,471	23,763	26,779	25,221	26,098	23,824	25,101	27,671
Total noninterest expense	14,540	15,534	16,842	15,995	16,043	14,398	14,631	16,124
Pre-provision profit ^(a)	6,931	8,229	9,937	9,226	10,055	9,426	10,470	11,547
Provision for credit losses	2,184	2,411	1,810	1,169	3,043	3,223	3,363	7,010
Income before income tax expense	4,747	5,818	8,127	8,057	7,012	6,203	7,107	4,537
Income tax expense	1,019	1,556	2,696	2,502	2,181	1,785	2,312	1,211
Net income	\$ 3,728	\$ 4,262	\$ 5,431	\$ 5,555	\$ 4,831	\$ 4,418	\$ 4,795	\$ 3,326
Per common share data								
Average: Basic	\$ 0.90	\$ 1.02	\$ 1.28	\$ 1.29	\$ 1.13	\$ 1.02	\$ 1.10	\$ 0.75
Diluted	0.90	1.02	1.27	1.28	1.12	1.01	1.09	0.74
Cash dividends declared per share	0.25	0.25	0.25	0.25	0.05	0.05	0.05	0.05
Book value per share	46.59	45.93	44.77	43.34	43.04	42.29	40.99	39.38
Common shares outstanding								
Average: Basic	3,801.9	3,859.6	3,958.4	3,981.6	3,917.0	3,954.3	3,983.5	3,970.5
Diluted	3,811.7	3,872.2	3,983.2	4,014.1	3,935.2	3,971.9	4,005.6	3,994.7
Common shares at period-end	3,772.7	3,798.9	3,910.2	3,986.6	3,910.3	3,925.8	3,975.8	3,975.4
Share price ^(c)								
High	\$ 37.54	\$ 42.55	\$ 47.80	\$ 48.36	\$ 43.12	\$ 41.70	\$ 48.20	\$ 46.05
Low	27.85	28.53	39.24	42.65	36.21	35.16	36.51	37.03
Close	33.25	30.12	40.94	46.10	42.42	38.06	36.61	44.75
Market capitalization	125,442	114,422	160,083	183,783	165,875	149,418	145,554	177,897
Financial ratios								
Return on common equity	8%	9%	12%	13%	11%	10%	12%	8%
Return on tangible common equity	11	13	17	18	16	15	17	12
Return on assets	0.65	0.76	0.99	1.07	0.92	0.86	0.94	0.66
Overhead ratio	68	65	63	63	61	60	58	58
Deposits-to-loans ratio	156	157	152	145	134	131	127	130
Tier 1 capital ratio	12.3	12.1	12.4	12.3	12.1	11.9	12.1	11.5
Total capital ratio	15.4	15.3	15.7	15.6	15.5	15.4	15.8	15.1
Tier 1 leverage ratio	6.8	6.8	7.0	7.2	7.0	7.1	6.9	6.6
Tier 1 common capital ratio ^(b)	10.1	9.9	10.1	10.0	9.8	9.5	9.6	9.1
Selected balance sheet data (period-end)								
Trading assets	\$ 443,963	\$ 461,531	\$ 458,722	\$ 501,148	\$ 489,892	\$ 475,515	\$ 397,508	\$ 426,128
Securities	364,793	339,349	324,741	334,800	316,336	340,168	312,013	344,376
Loans	723,720	696,853	689,736	685,996	692,927	690,531	699,483	713,799
Total assets	2,265,792	2,289,240	2,246,764	2,198,161	2,117,605	2,141,595	2,014,019	2,135,796
Deposits	1,127,806	1,092,708	1,048,685	995,829	930,369	903,138	887,805	925,303
Long-term debt ^(f)	256,775	273,688	279,228	269,616	270,653	271,495	260,442	278,685
Common stockholders' equity	175,773	174,487	175,079	172,798	168,306	166,030	162,968	156,569
Total stockholders' equity	183,573	182,287	182,879	180,598	176,106	173,830	171,120	164,721
Headcount	260,157	256,663	250,095	242,929	239,831	236,810	232,939	226,623

Supplementary information

(Table continued from previous page)

As of or for the period ended (in millions, except ratio data)	2011				2010			
	4th quarter	3rd quarter	2nd quarter	1st quarter	4th quarter	3rd quarter	2nd quarter	1st quarter
Credit quality metrics								
Allowance for credit losses	\$ 28,282	\$ 29,036	\$ 29,146	\$ 30,438	\$ 32,983	\$ 35,034	\$ 36,748	\$ 39,126
Allowance for loan losses to total retained loans	3.84%	4.09%	4.16%	4.40%	4.71%	4.97%	5.15%	5.40%
Allowance for loan losses to retained loans excluding purchased credit-impaired loans	3.35	3.74	3.83	4.10	4.46	5.12	5.34	5.64
Nonperforming assets	\$ 11,036	\$ 12,194	\$ 13,240	\$ 14,986	\$ 16,557	\$ 17,656	\$ 18,156	\$ 19,019
Net charge-offs ^(a)	2,907	2,507	3,103	3,720	5,104	4,945	5,714	7,910
Net charge-off rate ^(b)	1.64%	1.44%	1.83%	2.22%	2.95%	2.84%	3.28%	4.46%

- (a) Pre-provision profit is total net revenue less noninterest expense. The Firm believes that this financial measure is useful in assessing the ability of a lending institution to generate income in excess of its provision for credit losses.
- (b) On March 18, 2011, the Board of Directors increased the Firm's quarterly stock dividend from \$0.05 to \$0.25 per share.
- (c) Share prices shown for JPMorgan Chase's common stock are from the New York Stock Exchange. JPMorgan Chase's common stock is also listed and traded on the London Stock Exchange and the Tokyo Stock Exchange.
- (d) Tier 1 common capital ratio ("Tier 1 common ratio") is Tier 1 common capital ("Tier 1 common") divided by risk-weighted assets. The Firm uses Tier 1 common capital along with the other capital measures to assess and monitor its capital position. For further discussion of Tier 1 common ratio, see Regulatory capital on pages 119–122 of this Annual Report.
- (f) Effective January 1, 2011, the long-term portion of advances from FHLBs was reclassified from other borrowed funds to long-term debt. Prior periods have been revised to conform with the current presentation.
- (g) Excludes the impact of residential real estate PCI loans. For further discussion, see Allowance for credit losses on pages 155–157 of this Annual Report.
- (h) Net charge-offs and net charge-off rates for the fourth quarter of 2010 include the effect of \$632 million of charge-offs related to the estimated net realizable value of the collateral underlying delinquent residential home loans. Because these losses were previously recognized in the provision and allowance for loan losses, this adjustment had no impact on the Firm's net income.

Short-term and other borrowed funds

The following table provides a summary of JPMorgan Chase's short-term and other borrowed funds for the years indicated.

As of or for the year ended December 31, (in millions, except rates)	2011	2010	2009
Federal funds purchased and securities loaned or sold under repurchase agreements:			
Balance at year-end	\$ 213,532	\$ 276,644	\$ 261,413
Average daily balance during the year	256,283	278,603	275,862
Maximum month-end balance	289,835	314,161	310,802
Weighted-average rate at December 31	0.16%	0.18%	0.04%
Weighted-average rate during the year	0.21	(0.07) ^(d)	0.21
Commercial paper:			
Balance at year-end	\$ 51,631	\$ 35,363	\$ 41,794
Average daily balance during the year	42,653	36,000	39,055
Maximum month-end balance	51,631	50,554	53,920
Weighted-average rate at December 31	0.12%	0.21%	0.18%
Weighted-average rate during the year	0.17	0.20	0.28
Other borrowed funds ^(b)			
Balance at year-end	\$ 88,626	\$ 111,272	\$ 97,838
Average daily balance during the year	107,543	104,951	99,785
Maximum month-end balance	127,517	120,437	155,693
Weighted-average rate at December 31	1.60%	5.71%	3.92%
Weighted-average rate during the year	2.50	2.89	2.83
Short-term beneficial interests ^(c)			
Commercial paper and other borrowed funds:			
Balance at year-end	\$ 26,243	\$ 25,095	\$ 4,787
Average daily balance during the year	25,125	21,853	3,275
Maximum month-end balance	26,780	25,095	7,751
Weighted-average rate at December 31	0.18%	0.25%	0.17%
Weighted-average rate during the year	0.23	0.27	0.24

(a) Includes securities sold but not yet purchased.

(b) Effective January 1, 2011, \$23.0 billion of long-term advances from FHLBs were reclassified from other borrowed funds to long-term debt. The prior periods have been revised to conform with the current presentation.

(c) Included on the Consolidated Balance Sheets in beneficial interests issued by consolidated variable interest entities.

(d) Reflects a benefit from the favorable market environments for U.S. dollar-roll financings.

Federal funds purchased represent overnight funds. Securities loaned or sold under repurchase agreements generally mature between one day and three months. Commercial paper generally is issued in amounts not less than \$100,000, and with maturities of 270 days or less. Other borrowed funds consist of demand notes, term federal funds purchased, and various other borrowings that generally have maturities of one year or less.

Glossary of Terms

ACH:Automated Clearing House.

Active mobile customers Retail banking users of all mobile platforms, which include: SMS text, Mobile Browser, iPhone, iPad and Android, who have been active in the past 90 days.

Allowance for loan losses to total loans Represents period-end allowance for loan losses divided by retained loans.

Assets under management Represent assets actively managed by AM on behalf of Private Banking, Institutional and Retail clients. Includes "Committed capital not Called," on which AM earns fees. Excludes assets managed by American Century Companies, Inc., in which the Firm sold its ownership interest on August 31, 2011.

Assets under supervision Represent assets under management as well as custody, brokerage, administration and deposit accounts.

Average managed asset Refers to total assets on the Firm's Consolidated Balance Sheets plus credit card receivables that have been securitized and removed from the Firm's Consolidated Balance Sheets, for periods ended prior to the January 1, 2010, adoption of new accounting guidance requiring the consolidation of the Firm-sponsored credit card securitization trusts.

Beneficial interests issued by consolidated VIEs: Represents the third-party interests issued by VIEs that JPMorgan Chase consolidates where the third-party interest holders do not have recourse to the general credit of JPMorgan Chase. The underlying obligations of the VIEs consist of short-term borrowings, commercial paper and long-term debt.

Benefit obligation:Refers to the projected benefit obligation for pension plans and the accumulated postretirement benefit obligation for OPEB plans.

Client advisors:Investment product specialists, including Private Client Advisors, Financial Advisors, Financial Advisor Associates, Senior Financial Advisors, Independent Financial Advisors and Financial Advisor Associate trainees, who advise clients on investment options, including annuities, mutual funds, stock trading services, etc., sold by the Firm or by third party vendors through retail branches, Chase Private Client branches and other channels.

Client investment managed accounts Assets actively managed by Chase Wealth Management on behalf of clients. The percentage of managed accounts is calculated by dividing managed account assets by total client investment assets.

Contractual credit card charge-off In accordance with the Federal Financial Institutions Examination Council policy, credit card loans are charged off by the end of the month in which the account becomes 180 days past due or within 60

credit bureaus, typically produced from statistical models by Fair Isaac Corporation utilizing data collected by the credit bureaus.

Forward points: Represents the interest rate differential between two currencies, which is either added to or subtracted from the current exchange rate (i.e., “spot rate”) to determine the forward exchange rate.

G7 government bonds: Bonds issued by the government of one of countries in the “Group of Seven” (“G7”) nations. Countries in the G7 are Canada, France, Germany, Italy, Japan, the United Kingdom and the United States.

Global Corporate Bank: TSS and IB formed a joint venture to create the Firm’s Global Corporate Bank. With a team of bankers, the Global Corporate Bank serves multinational clients by providing them access to TSS products and services and certain IB products, including derivatives, foreign exchange and debt. The cost of this effort and the credit that the Firm extends to these clients is shared between TSS and IB.

Headcount-related expenses: Includes salary and benefits (excluding performance-based incentives), and other noncompensation costs related to employees.

Home equity - senior lien: Represents loans where JP Morgan Chase holds the first security interest on the property.

Home equity - junior lien: Represents loans where JP Morgan Chase holds a security interest that is subordinate in rank to other liens.

Interchange income: A fee paid to a credit card issuer in the clearing and settlement of a sales or cash advance transaction.

Interests in purchased receivables: Represents an ownership interest in cash flows of an underlying pool of receivables transferred by a third-party seller into a bankruptcy-remote entity, generally a trust.

Investment-grade: An indication of credit quality based on JPMorgan Chase’s internal risk assessment system. “Investment grade” generally represents a risk profile similar to a rating of a “BBB-”/“Baa3” or better, as defined by independent rating agencies.

ISDA: International Swaps and Derivatives Association.

LLC: Limited Liability Company.

Loan-to-value (“LTV”) ratio: For residential real estate loans, the relationship, expressed as a percentage, between the principal amount of a loan and the appraised value of the collateral (i.e., residential real estate) securing the loan.

Origination date LTV ratio

The LTV ratio at the origination date of the loan. Origination date LTV ratios are calculated based on the actual appraised values of collateral (i.e., loan-level data) at the origination date.

Current estimated LTV ratio

An estimate of the LTV as of a certain date. The current estimated LTV ratios are calculated using estimated

collateral values derived from a nationally recognized home price index measured at the metropolitan statistical area (“MSA”) level. These MSA-level home price indices comprise actual data to the extent available and forecasted data where actual data is not available. As a result, the estimated collateral values used to calculate these ratios do not represent actual appraised loan-level collateral values; as such, the resulting LTV ratios are necessarily imprecise and should therefore be viewed as estimates.

Combined LTV ratio

The LTV ratio considering all lien positions related to the property. Combined LTV ratios are used for junior lien home equity products.

Managed basis: A non-GAAP presentation of financial results that includes reclassifications to present revenue on a fully taxable-equivalent basis. For periods ended prior to the January 1, 2010, adoption of accounting guidance requiring the consolidation of the Firm-sponsored credit card securitization trusts, the Firm’s managed-basis presentation also included certain reclassification adjustments that assumed credit card loans that were securitized remained on the Consolidated Balance Sheets. Management uses this non-GAAP financial measure at the segment level, because it believes this provides information to enable investors to understand the underlying operational performance and trends of the particular business segment and facilitates a comparison of the business segment with the performance of competitors.

Managed credit card portfolio: Refers to credit card receivables on the Firm’s Consolidated Balance Sheets plus credit card receivables that have been securitized and removed from the Firm’s Consolidated Balance Sheets, for periods ended prior to the January 1, 2010, adoption of accounting guidance requiring the consolidation of the Firm-sponsored credit card securitization trusts.

Mark-to-market exposure: A measure, at a point in time, of the value of a derivative or foreign exchange contract in the open market. When the fair value is positive, it indicates the counterparty owes JPMorgan Chase and, therefore, creates credit risk for the Firm. When the fair value is negative, JPMorgan Chase owes the counterparty; in this situation, the Firm has liquidity risk.

Master netting agreement: An agreement between two counterparties who have multiple derivative contracts with each other that provides for the net settlement of all contracts, as well as cash collateral, through a single payment, in a single currency, in the event of default on or termination of any one contract.

Mortgage product types:

Alt-A

Alt-A loans are generally higher in credit quality than subprime loans but have characteristics that would disqualify the borrower from a traditional prime loan. Alt-A lending characteristics may include one or more of the following: (i) limited documentation; (ii) a high combined-loan-to-value (“CLTV”) ratio; (iii) loans secured by non-

Glossary of Terms

owner occupied properties; or (iv) a debt-to-income ratio above normal limits. Perhaps the most important characteristic is limited documentation. A substantial proportion of traditional Alt-A loans are those where a borrower does not provide complete documentation of his or her assets or the amount or source of his or her income.

Option ARMs

The option ARM real estate loan product is an adjustable-rate mortgage loan that provides the borrower with the option each month to make a fully amortizing, interest-only or minimum payment. The minimum payment on an option ARM loan is based on the interest rate charged during the introductory period. This introductory rate is usually significantly below the fully indexed rate. The fully indexed rate is calculated using an index rate plus a margin. Once the introductory period ends, the contractual interest rate charged on the loan increases to the fully indexed rate and adjusts monthly to reflect movements in the index. The minimum payment is typically insufficient to cover interest accrued in the prior month, and any unpaid interest is deferred and added to the principal balance of the loan. Option ARM loans are subject to payment recast, which converts the loan to a variable-rate fully amortizing loan upon meeting specified loan balance and anniversary date triggers.

Prime

Prime mortgage loans generally have low default risk and are made to borrowers with good credit records and a monthly income at least three to four times greater than their monthly housing expense (mortgage payments plus taxes and other debt payments). These borrowers provide full documentation and generally have reliable payment histories.

Subprime

Subprime loans are designed for customers with one or more high risk characteristics, including but not limited to: (i) unreliable or poor payment histories; (ii) a high LTV ratio of greater than 80% (without borrower-paid mortgage insurance); (iii) a high debt-to-income ratio; (iv) an occupancy type for the loan is other than the borrower's primary residence; or (v) a history of delinquencies or late payments on the loan.

MSR risk management revenue Includes changes in the fair value of the MSR asset due to market-based inputs, such as interest rates and volatility, as well as updates to assumptions used in the MSR valuation model; and derivative valuation adjustments and other, which represents changes in the fair value of derivative instruments used to offset the impact of changes in the market-based inputs to the MSR valuation model.

Multi-asset: Any fund or account that allocates assets under management to more than one asset class (e.g., long-term fixed income, equity, cash, real assets, private equity or hedge funds).

NA: Data is not applicable or available for the period presented.

Net charge-off rate: Represents net charge-offs (annualized) divided by average retained loans for the reporting period.

Net yield on interest-earning assets: The average rate for interest-earning assets less the average rate paid for all sources of funds.

NM: Not meaningful.

OPEB: Other postretirement employee benefits.

Overhead ratio: Noninterest expense as a percentage of total net revenue.

Participating securities: Represents unvested stock-based compensation awards containing nonforfeitable rights to dividends or dividend equivalents (collectively, "dividends"), which are included in the earnings-per-share calculation using the two-class method.

Personal bankers: Retail branch office personnel who acquire, retain and expand new and existing customer relationships by assessing customer needs and recommending and selling appropriate banking products and services.

Portfolio activity: Describes changes to the risk profile of existing lending-related exposures and their impact on the allowance for credit losses from changes in customer profiles and inputs used to estimate the allowances.

Pre-provision profit: Total net revenue less noninterest expense. The Firm believes that this financial measure is useful in assessing the ability of a lending institution to generate income in excess of its provision for credit losses.

Pretax margin: Represents income before income tax expense divided by total net revenue, which is, in management's view, a comprehensive measure of pretax performance derived by measuring earnings after all costs are taken into consideration. It is, therefore, another basis that management uses to evaluate the performance of TSS and AM against the performance of their respective competitors.

Purchased credit-impaired ("PCI") loans: Acquired loans deemed to be credit-impaired under the FASB guidance for PCI loans. The guidance allows purchasers to aggregate credit-impaired loans acquired in the same fiscal quarter into one or more pools, provided that the loans have common risk characteristics (e.g., FICO score, geographic location). A pool is then accounted for as a single asset with a single composite interest rate and an aggregate expectation of cash flows. Wholesale loans are determined to be credit-impaired if they meet the definition of an impaired loan under U.S. GAAP at the acquisition date. Consumer loans are determined to be credit-impaired based on specific risk characteristics of the loan, including product type, LTV ratios, FICO scores, and past due status.

Real estate investment trust ("REIT") A special purpose investment vehicle that provides investors with the ability to participate directly in the ownership or financing of real-estate related assets by pooling their capital to purchase and manage income property (i.e., equity REIT) and/or

mortgage loans (i.e., mortgage REIT). REITs can be publicly or privately-held and they also qualify for certain favorable tax considerations.

Reported basis Financial statements prepared under U.S. GAAP, which excludes the impact of taxable-equivalent adjustments.

Retained loans Loans that are held-for-investment excluding loans held-for-sale and loans at fair value.

Risk-weighted assets ("RWA") Risk-weighted assets consist of on- and off-balance sheet assets that are assigned to one of several broad risk categories and weighted by factors representing their risk and potential for default. On-balance sheet assets are risk-weighted based on the perceived credit risk associated with the obligor or counterparty, the nature of any collateral, and the guarantor, if any. Off-balance sheet assets such as lending-related commitments, guarantees, derivatives and other applicable off-balance sheet positions are risk-weighted by multiplying the contractual amount by the appropriate credit conversion factor to determine the on-balance sheet credit equivalent amount, which is then risk-weighted based on the same factors used for on-balance sheet assets. RWA also incorporate a measure for the market risk related to applicable trading assets-debt and equity instruments, and foreign exchange and commodity derivatives. The resulting risk-weighted values for each of the risk categories are then aggregated to determine total RWA.

Sales specialists: Retail branch office and field personnel, including Business Bankers, Relationship Managers and Loan Officers, who specialize in marketing and sales of various business banking products (i.e., business loans, letters of credit, deposit accounts, Chase Paymentech, etc.) and mortgage products to existing and new clients.

Seed capital Initial JPMorgan capital invested in products, such as mutual funds, with the intention of ensuring the fund is of sufficient size to represent a viable offering to clients, enabling pricing of its shares, and allowing the manager to develop a commercially attractive track record. After these goals are achieved, the intent is to remove the Firm's capital from the investment.

Stress testing: A scenario that measures market risk under unlikely but plausible events in abnormal markets.

TARP: Troubled Asset Relief Program.

Taxable-equivalent basis For managed results, total net revenue for each of the business segments and the Firm is presented on a tax-equivalent basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in the managed results on a basis comparable to taxable investments and securities. This non-GAAP financial measure allows management to assess the comparability of revenue arising from both taxable and tax-exempt sources. The corresponding income tax impact related to tax-exempt items is recorded within income tax expense.

Troubled debt restructuring ("TDR") Occurs when the Firm modifies the original terms of a loan agreement by granting a concession to a borrower that is experiencing financial difficulty.

Unaudited: Financial statements and information that have not been subjected to auditing procedures sufficient to permit an independent certified public accountant to express an opinion.

U.S. GAAP Accounting principles generally accepted in the United States of America.

U.S. government-sponsored enterprise obligations: Obligations of agencies originally established or chartered by the U.S. government to serve public purposes as specified by the U.S. Congress; these obligations are not explicitly guaranteed as to the timely payment of principal and interest by the full faith and credit of the U.S. government.

U.S. Treasury U.S. Department of the Treasury.

Value-at-risk ("VaR"): A measure of the dollar amount of potential loss from adverse market moves in an ordinary market environment.

Washington Mutual transaction On September 25, 2008, JPMorgan Chase acquired the banking operations of Washington Mutual Bank ("Washington Mutual") from the FDIC. The Washington Mutual acquisition resulted in negative goodwill, and accordingly, the Firm recorded an extraordinary gain. A preliminary gain of \$1.9 billion was recognized at December 31, 2008. The final total extraordinary gain that resulted from the Washington Mutual transaction was \$2.0 billion.

Rt. Hon. Tony Blair

Chairman of the Council
Quartet Representative and
Former Prime Minister of Great Britain
and Northern Ireland
London, United Kingdom

Khalid A. Al-Falih

President and Chief Executive Officer
Saudi Aramco
Dhahran, Saudi Arabia

Kofi A. Annan

Former UN Secretary-General
Chairman, Kofi Annan Foundation
Geneva, Switzerland

Alberto Baillères

Presidente del Consejo de Administración
Grupo Bal
México D.F., Mexico

Franco Bernabé

Chairman and Chief Executive Officer
Telecom Italia S.p.A.
Rome, Italy

Michael A. Chaney

Chairman
National Australia Bank Limited
Perth, Western Australia

Anatoly B. Chubais

Director General
Russian Corporation of
Nanotechnologies – RUSNANO
Moscow, Russian Federation

Hon. Richard M. Daley

Former Mayor of Chicago
Chicago, Illinois

Martin Feldstein

President Emeritus
National Bureau of Economic
Research, Inc.
Cambridge, Massachusetts

Gao Xi-Qing

Vice Chairman, President and
Chief Investment Officer
China Investment Corporation
Beijing, The People's Republic of China

Jürgen Grossmann

Chief Executive Officer
RWE AG
Essen, Germany

William B. Harrison, Jr.

Former Chairman and
Chief Executive Officer
JPMorgan Chase & Co.
New York, New York

Hon. Carla A. Hills

Chairman and Chief Executive Officer
Hills & Company International
Consultants
Washington, District of Columbia

Franz B. Humer

Chairman
Roche Holding Ltd.
Basel, Switzerland

Hon. Henry A. Kissinger

Chairman
Kissinger Associates, Inc.
New York, New York

Mustafa V. Koç

Chairman of the Board of Directors
Koç Holding A.Ş.
Istanbul, Turkey

Hon. Lee Kuan Yew

Former Minister Mentor,
Republic of Singapore and
Senior Advisor, Government of
Singapore Investment Corporation
Singapore, Republic of Singapore

Gérard Mestrallet

Chairman and Chief Executive Officer
GDF SUEZ
Paris la Défense, France

Akio Mimura

Representative Director and Chairman
Nippon Steel Corporation
Tokyo, Japan

Patrice Motsepe

Executive Chairman
African Rainbow Minerals Limited
Chislehurst, Sandton, South Africa

Michael Pram Rasmussen

Chairman of the Board
A.P. Møller-Maersk Group
Copenhagen, Denmark

Ratan Naval Tata

Chairman
Tata Sons Limited
Mumbai, India

Hon. Tung Chee Hwa GBM

Vice Chairman
National Committee of the Chinese
People's Political
Consultative Conference
Hong Kong, The People's Republic
of China

Álvaro Uribe Vélez

Former President of the Republic
of Colombia
Bogotá DC, Colombia

Cees J.A. van Lede

Former Chairman and Chief Executive
Officer, Board of Management
Akzo Nobel
Arnhem, The Netherlands

Douglas A. Warner III

Former Chairman of the Board
J.P. Morgan Chase & Co. Inc.
New York, New York

John S. Watson

Chairman and Chief Executive Officer
Chevron Corporation
San Ramon, California

Jaime Augusto Zobel de Ayala

Chairman and Chief Executive Officer
Ayala Corporation
Makati City, Philippines

Ex-Officio Members

Jamie Dimon

Chairman and Chief Executive Officer
JPMorgan Chase & Co.
New York, New York

Andrew Crockett

Special Advisor to the Chairman
JPMorgan Chase & Co.
San Francisco, California

Arminio Fraga Neto

Founding Partner and Chief Investment
Officer, Gávea Investimentos, Ltd.
JPMorgan Chase & Co.
Rio de Janeiro, Brazil

Dr. Jacob A. Frenkel

Chairman, International
JPMorgan Chase & Co.
New York, New York

Walter A. Gubert

Vice Chairman
JPMorgan Chase & Co.
Brussels, Belgium

David Mayhew, CBE

Former Chairman, J.P. Morgan Cazenove
JPMorgan Chase & Co.
London, United Kingdom

Regional Advisory Board

Stanley M. Bergman

Chairman and Chief Executive Officer
Henry Schein Inc.

Richard A. Bernstein

Chairman and
Chief Executive Officer
P&E Capital, Inc.

Rod Brayman

President and
Chief Executive Officer
Phoenix Beverages

Christopher B. Combe

Chairman and
Chief Executive Officer
Combe Incorporated

Joseph J. Corasanti

President and
Chief Executive Officer
CONMED Corporation

Roger N. Farah

President and
Chief Operating Officer
Ralph Lauren Corporation

James Fernandez

Executive Vice President and
Chief Operating Officer
Tiffany & Co.

Stanley Fleishman

Chief Executive Officer
Jetro Cash & Carry Enterprise, LLC

Jeffrey S. Flug

President
Union Square Hospitality Group

Martin E. Franklin

Chairman and
Chief Executive Officer
Jarden Corporation

Neil Golub

Chief Executive Officer and
Chairman of the Board
The Golub Corporation

David R. Jaffe

President and
Chief Executive Officer
Ascena Retail Group, Inc.

Tod Johnson

Chairman and
Chief Executive Officer
The NPD Group, Inc.

Richard W. Kunes

Executive Vice President,
Chief Financial Officer
The Estée Lauder Companies

Stephen J. Large

President and
Chief Executive Officer
SI Group, Inc.

Richard S. LeFrak

Chairman and
Chief Executive Officer
LeFrak Organization

Richard B. Leventhal

Chairman and
Chief Executive Officer
Fedway Associates, Inc.

John LiDestri

President and
Chief Executive Officer
LiDestri Foods, Inc.

Leo Liebowitz

Chairman
Getty Realty Corp.

William L. Mack

Founder and Chairman
AREA Property Partners

Peter Markson

Chairman and
Chief Executive Officer
Paris Accessories, Inc.

James F. McCann

Chief Executive Officer
1-800-Flowers.com

John Morphy

Former Senior Vice President,
Chief Financial Officer
and Secretary
Paychex, Inc.

Samuel I. Newhouse III

General Manager
Advance Publications Inc.

William C. Rudin

Vice Chairman and
Chief Executive Officer
Rudin Management Company, Inc.

Steven M. Safyer, M.D.

President and
Chief Executive Officer
Montefiore Medical Center

C. David Sammons

President and
Chief Executive Officer
Subaru Distributors Corporation

John Shalam

Chairman
Audiovox Corporation

Kenneth L. Wallach

Chairman and
Chief Executive Officer
Central National-Gottesman Inc.

Fred Wilpon

Chairman
Sterling Equities, Inc.

Frank Lourenso

Chairman
Regional Advisory Board

Board of Directors

James A. Bell ¹

Retired Executive Vice President
The Boeing Company
(Aerospace)

Crandall C. Bowles ^{1, 4}

Chairman
Springs Industries, Inc.
(Window fashions)

Stephen B. Burke ^{2, 3}

Chief Executive Officer
NBCUniversal, LLC
Executive Vice President
Comcast Corporation
(Television and entertainment)

David M. Cote ^{4, 5}

Chairman and
Chief Executive Officer
Honeywell International Inc.
(Diversified technology and manufacturing)

James S. Crown ⁵

President
Henry Crown and Company
(Diversified investments)

Jamie Dimon

Chairman and
Chief Executive Officer
JPMorgan Chase & Co.

Ellen V. Futter ^{4, 5}

President and Trustee
American Museum of
Natural History
(Museum)

William H. Gray, III ^{1, 4}

Chairman
Gray Global Strategies, Inc.
(Consulting and advisory)

Laban P. Jackson, Jr. ¹

Chairman and
Chief Executive Officer
Clear Creek Properties, Inc.
(Real estate development)

David C. Novak ^{2, 3}

Chairman and
Chief Executive Officer
Yum! Brands, Inc.
(Franchised restaurants)

Lee R. Raymond ^{2, 3}

Retired Chairman and
Chief Executive Officer
Exxon Mobil Corporation
(Oil and gas)

William C. Weldon ^{2, 3}

Chairman and
Chief Executive Officer
Johnson & Johnson
(Health care products)

Member of:

- 1 Audit Committee
- 2 Compensation & Management Development Committee
- 3 Corporate Governance & Nominating Committee
- 4 Public Responsibility Committee
- 5 Risk Policy Committee

Operating Committee

Jamie Dimon

Chairman and
Chief Executive Officer

Frank J. Bisignano

Chief Administrative Officer and
Mortgage Banking CEO

Douglas L. Braunstein

Chief Financial Officer

Michael J. Cavanagh

Treasury & Securities Services CEO

Stephen M. Cutler

General Counsel

John L. Donnelly

Head of Human Resources

Ina R. Drew

Chief Investment Officer

Mary Callahan Erdoes

Asset Management CEO

John J. Hogan

Chief Risk Officer

Todd Maclin

Consumer & Business
Banking CEO

Douglas B. Petno

Commercial Banking CEO

Gordon A. Smith

Card, Merchant Services &
Auto Finance CEO

Jes Staley

Investment Bank CEO

Barry L. Zubrow

Head of Corporate & Regulatory Affairs

Executive Committee

Gaby A. Abdelnour
Asia Pacific

Nicolas Aguzin
Latin America

Peter K. Barker
California

Paul T. Bateman
Asset Management

Philip F. Bleser
Global Corporate Bank

Peter J. Bocian
Corporate Services and Finance

Phyllis J. Campbell
Pacific Northwest

Richard M. Cashin
One Equity Partners

Guy Chiarello
Chief Information Officer

Paul H. Compton
Investment Bank

Kimberly B. Davis
Global Philanthropy

Craig M. Delany
Mortgage Banking

Klaus Diederichs
Investment Bank

Phil Di Iorio
Asset Management

Joseph M. Evangelisti
Corporate Communications

Dr. Jacob A. Frenkel
JPMorgan Chase International

Martha J. Gallo
Chief Compliance Officer

George C. Gatch
Asset Management

Scott Geller
Business Banking

Walter A. Gubert
Europe, Middle East and Africa

Gregory L. Guyett
Global Corporate Bank

Carlos M. Hernandez
Investment Bank

Thomas J. Higgins
Chief Administrative Office

Robert C. Holmes
Commercial Banking

Dinkar Jetley
Worldwide Securities Services

Catherine M. Keating
Asset Management

James B. Lee, Jr.
Investment Bank

Achilles O. Macris
Chief Investment Office

Mel R. Martinez
Florida, Mexico, Central America and Caribbean;
JPMorgan Chase Foundation

Blythe S. Masters
Investment Bank

Donald H. McCree, III
Treasury Services

Ryan McInerney
Consumer Banking

Michael O'Brien
Asset Management

Sandra E. O'Connor
Finance

Daniel E. Pinto
Investment Bank and EMEA

Louis Rauchenberger
Treasury & Securities Services

Richard Sabo
Chief Investment Office

Emilio Saracho
Investment Bank

Peter L. Scher
Corporate Responsibility

Eileen M. Serra
Card Services

Marc Sheinbaum
Auto Finance and Student Loans

Larry D. Slaughter
Investment Bank

Barry Sommers
Chase Wealth Management

Jack M. Stephenson
Mobile, E-commerce and Payments

Glenn Tilton
Midwest

Irene Tse
Chief Investment Office

Lauren M. Tyler
Audit

Jeffrey H. Urwin
Investment Bank

William S. Wallace
Card Services

Shannon S. Warren
Finance

Kevin P. Watters
Mortgage Banking

Kevin D. Willsey
Investment Bank

Douglas Wurth
Asset Management

Matthew E. Zames
Investment Bank

Other Corporate Officers

Anthony J. Horan
Secretary

Sarah M. Youngwood
Investor Relations

Regional Chief Executive Officers

Asia Pacific

Gaby A. Abdelnour

Europe, Middle East, Africa

Daniel E. Pinto

Latin America

Nicolas Aguzin

Senior Country Officers

Asia Pacific

Australia/New Zealand
Robert C. Priestley

China
Zili Shao

Hong Kong
Michael Fung

India
Kalpana Morparia

Indonesia
Haryanto Budiman

Japan
Christopher L. Harvey

Korea
Steve Lim

Malaysia
Steve R. Clayton

Philippines
Roberto L. Panlilio

Singapore
Philip Lee

Taiwan
Carl K. Chien

Thailand
M.L. Chayotid Kridakon

Vietnam
Dominic Price

Europe

Benelux/Nordics
John S. Gilbert

Central & Eastern Europe
Walter Schuster

France
Isabelle Seillier

Germany/Austria/Switzerland
Karl Altenburg

Iberia
Enrique Casanueva

Italy
Francesco Rossi Ferrini

Portugal
Guillermo Baygual

Russia/Ukraine
Jeffrey R. Costello

Middle East, North Africa

Middle East/North Africa
Sjoerd Leenart

Bahrain
Ali Moosa

Israel
Roy Navon

Nigeria
Tosin T. Adewuyi

Saudi Arabia
Abdulaziz Al Helaissi

Turkey
Emre Derman

UAE
Ehsun A. Zaidi

Sub-Saharan Africa

Sub-Saharan Africa
John J. Coulter

Latin America

Andean Region
Lisandro Miguens

Argentina
Facundo D. Gomes Minujin

Brazil
Claudio F. Berquo

Chile
Vicente Monge

Mexico
Eduardo F. Cepeda

North America

Canada
David E. Rawlings

JPMorgan Chase & Co.

Corporate headquarters

270 Park Avenue
New York, NY 10017-2070
Telephone: 212-270-6000
jpmorganchase.com

Principal subsidiaries

JPMorgan Chase Bank,
National Association
Chase Bank USA,
National Association
J.P. Morgan Securities LLC
J.P. Morgan Securities Ltd.

Annual Report on Form 10-K

The Annual Report on Form 10-K of JPMorgan Chase & Co. as filed with the U.S. Securities and Exchange Commission will be made available without charge upon request to:

Office of the Secretary
JPMorgan Chase & Co.
270 Park Avenue
New York, NY 10017-2070

Stock listing

New York Stock Exchange
London Stock Exchange
Tokyo Stock Exchange

The New York Stock Exchange ticker symbol for the common stock of JPMorgan Chase & Co. is JPM.

Financial information about JPMorgan Chase & Co. can be accessed by visiting the Investor Relations website at jpmorganchase.com. Additional questions should be addressed to:

Investor Relations
JPMorgan Chase & Co.
270 Park Avenue
New York, NY 10017-2070
Telephone: 212-270-6000

Directors

To contact any of the Board members or committee chairs, the Presiding Director or the non-management directors as a group, please mail correspondence to:

JPMorgan Chase & Co.
Attention (Board member(s))
Office of the Secretary
270 Park Avenue
New York, NY 10017-2070

The Corporate Governance Principles of the Board, the charters of the principal Board committees, the Code of Conduct, the Code of Ethics for Finance Professionals and other governance information can be accessed by visiting our website at jpmorganchase.com and clicking on "Governance" under the "About us" tab.

Transfer agent and registrar

Computershare Shareowner Services LLC
480 Washington Boulevard
Jersey City, NJ 07310-2053
Telephone: 800-758-4651
computershare.com

Investor Services Program

JPMorgan Chase & Co.'s Investor Services Program offers a variety of convenient, low-cost services to make it easier to reinvest dividends and buy and sell shares of JPMorgan Chase & Co. common stock. A brochure and enrollment materials may be obtained by contacting the Program Administrator, Computershare Shareowner Services LLC, by calling 800-758-4651, by writing to the address indicated above or by visiting its website at bnymellon.com/shareowner/equityaccess.

Direct deposit of dividends

For information about direct deposit of dividends, please contact Computershare Shareowner Services LLC.

Stockholder inquiries

Contact Computershare Shareowner Services LLC:

By telephone:

Within the United States, Canada and Puerto Rico: 800-758-4651 (toll free)

From all other locations:
201-680-6578 (collect)

TDD service for the hearing impaired within the United States, Canada and Puerto Rico: 800-231-5469 (toll free)

All other locations:
201-680-6610 (collect)

By mail:

Computershare Shareowner Services LLC
480 Washington Boulevard
Jersey City, NJ 07310-2053

Duplicate mailings

If you receive duplicate mailings because you have more than one account listing and you wish to consolidate your accounts, please write to Computershare Shareowner Services LLC at the address above.

Independent registered public accounting firm

PricewaterhouseCoopers LLP
300 Madison Avenue
New York, NY 10017-6204

"JPMorgan Chase," "J.P. Morgan," "Chase," the Octagon Symbol and other words or symbols in this report that identify JPMorgan Chase services are service marks of JPMorgan Chase & Co. Other words or symbols in this report that identify other parties' goods and services may be trademarks or service marks of those other parties.

As of the beginning of 2009, JPMorgan Chase & Co. has distributed shareholder information under the U.S. Securities and Exchange Commission "Notice and Access" rule. As a result, the firm prints 700,000 fewer Annual Reports and Proxy Statements, which saves on an annual basis approximately 6,400 trees and 800 metric tons of CO₂ emissions.

This Annual Report is printed on paper made from well-managed forests and other controlled sources. The paper is independently certified by BVQI to the Forest Stewardship Council (FSC) standards. The paper contains a minimum of 20% post-consumer waste recycled fibers.



